

A Complete Solution Guide to Principles of Mathematical Analysis

KIT-WING YU



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by Kit-Wing Yu, PhD

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Preface

Professor Walter Rudin^a is the author of the classical and famous textbooks: *Principles of Mathematical Analysis*, *Real and Complex Analysis*, and *Functional Analysis*. (People commonly call them “*Baby Rudin*”, “*Papa Rudin*” and “*Grandpa Rudin*” respectively.) Undoubtedly, they have produced important and extensive impacts to the study of mathematical analysis at university level since their publications.

In my memory, *Principles of Mathematical Analysis* was the standard textbook when I was a year 2 undergraduate mathematics student many years ago. I read Chapters 1-7 plus 11 again when I prepared my Ph.D. qualifying examination. In my personal experience, I think that chapters in these books are well-organized and expositions of theorems are clear, precise and well-written. I believe that you will agree with me after reading them.

Although *Principles of Mathematical Analysis* stopped to have any update after 1976, many instructors nowadays still like to select this book as the standard textbook or one of the main reference books in their analysis courses. I guess the “beauty” of this book that I described above is one of the reasons for this kind of choice. Furthermore, many people believe that the best way to study mathematics is by working through examples and exercises. Thus the excellent and well-designed exercises provided in the book is another main factor that attracts us. This can be justified if you type some keywords in google, then you will see many people are discussing and finding solutions of those exercises in different platforms. For example, the Math Stack Exchange:

<https://math.stackexchange.com/>

Actually, Professor Roger Cooke wrote “*Solutions Manual to Walter Rudin’s Principles of Mathematical Analysis*” in 1976, see

<https://minds.wisconsin.edu/handle/1793/67009>

However, a glance at the files will reveal that the readability of the manual is a bit low because of the lacks of equation numbers and supporting illustrations. Besides, the most important point is that it seems that some solutions are *incomplete*!

Therefore, these give “birth” to this book: I decide to write the solutions myself. In other words, this book is **not** a collection of solutions from others, but rather I use my **own words** and **own ways** to solve and prove the problems. In fact, I have written a *complete* solution guide so that it helps every mathematics student and instructor to understand the ideas and applications of the theorems in Rudin’s book.

As a mathematics instructor at a college, I understand that the growth of a mathematics student depends largely on how hard he/she does exercises. When your instructor asks you to do some exercises from Rudin, you are not suggested to read my solutions unless you have tried your best to prove them yourselves.

^ahttps://en.wikipedia.org/wiki/Walter_Rudin.

The features of this book are as follows:

- It covers all the 285 exercises with *detailed* and *complete* solutions. As a matter of fact, my solutions show every detail, every step and every theorem that I applied. That's why my book has over 390 pages!
- There are 55 illustrations and 3 tables for explaining the mathematical concepts or ideas used behind the questions or theorems.
- Sections in each chapter are added so as to increase the readability of the exercises.
- Different colors are used frequently in order to highlight or explain problems, lemmas, remarks, main points/formulas involved, or show the steps of manipulation in some complicated proofs. (ebook only)
- Necessary lemmas with proofs and references are provided because some questions require additional mathematical concepts which are not covered by Rudin.
- Three appendices are included which further explain and supplement some theories in Chapters 10 and 11.

Since the solutions are written solely by me, you may find typos or mistakes. If you really find such a mistake, please send your valuable comments or opinions to

`kitwing@hotmail.com`.

Then I will post the updated errata on my new website

<https://sites.google.com/view/yukitwing/>.

Finally, if the sales of this book are good (I hope so), then I will start to write a solution guide of the book *Real and Complex Analysis* soon.

Kit Wing Yu
February 2018

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CHAPTER 1

The Real and Complex Number Systems

Unless the contrary is explicitly stated, all numbers that are mentioned in these exercises are understood to be real.

1.1 Problems on rational numbers and fields

Problem 1.1

Rudin Chapter 1 Exercise 1.

Proof. Assume that $r + x$ was rational. Then it follows from Definition 1.12(A1), (A4) and (A5) that

$$x = (r + x) - x$$

is also rational, a contradiction. Similarly, if rx was rational, then it follows from Definition 1.12(M1), M(4) and M(5) that

$$x = \frac{rx}{r}$$

is also rational, a contradiction. This ends the proof of the problem. ■

Problem 1.2

Rudin Chapter 1 Exercise 2.

Proof. Assume that $\sqrt{12}$ was rational so that $\sqrt{12} = \frac{m}{n}$, where m and n are co-prime integers. Then we have $m^2 = 12n^2$ and thus m is divisible by 3. Let $m = 3k$ for some integer k . Then we have $m^2 = 9k^2$ and this shows that

$$k^2 = \frac{4n^2}{3},$$

so n is divisible by 3. This contradicts the fact that m and n are co-prime, completing the proof of the problem. ■

Problem 1.3

Rudin Chapter 1 Exercise 3.

Proof. Since $x \neq 0$, there exists $\frac{1}{x} \in F$ such that $x \cdot \frac{1}{x} = 1$.

- (a) Therefore, it follows from Definition 1.12(M2), (M3), (M4) and (M5) that $xy = xz$ implies that $y = z$.

- (b) Similarly, it follows from Definition 1.12(M2), (M3), (M4) and (M5) that $xy = x$ implies that $y = 1$.
- (c) Similarly, it follows from Definition 1.12(M2), (M3), (M4) and (M5) that $xy = 1$ implies that $y = \frac{1}{x}$.
- (d) Since $\frac{1}{x} \in F$, we have $\frac{1}{\frac{1}{x}} \in F$ such that $\frac{1}{x} \cdot \frac{1}{\frac{1}{x}} = 1$. Now we have $\frac{1}{x} \cdot \frac{1}{\frac{1}{x}} = \frac{1}{x} \cdot x (= 1)$, then Proposition 1.15(a) implies that $\frac{1}{\frac{1}{x}} = x$.

This completes the proof of the problem. ■

1.2 Properties of supremums and infimums

Problem 1.4

Rudin Chapter 1 Exercise 4.

Proof. Since $E \subset S$, the definitions give $\alpha \leq x$ and $x \leq \beta$ for all $x \in E$. Thus Definition 1.5(ii) implies that $\alpha \leq \beta$. This completes the proof of the problem. ■

Problem 1.5

Rudin Chapter 1 Exercise 5.

Proof. Theorem 1.19 says that $\mathbb{R}$ is an ordered set with the least-upper-bound property. Since A is a non-empty subset of $\mathbb{R}$ and A is bounded below, $\inf A$ exists in $\mathbb{R}$ by Definition 1.10. Furthermore, $-A$ is a non-empty subset of $\mathbb{R}$. Let y be a lower bound of A , i.e. $y \leq x$ for all $x \in A$. Then we have $-x \leq -y$ for all $x \in A$. Thus $-y$ is an upper bound of $-A$ and $\sup(-A)$ exists in $\mathbb{R}$ by Definition 1.10.

Let $\alpha = \inf A$ and $\beta = \sup(-A)$. By definition, we have $y \leq \beta$ for all $y \in -A$, where $y = -x$ for some $x \in A$. It implies that $x = -y \geq -\beta$ for all $x \in A$, so $-\beta$ is a lower bound of A and then $-\beta \leq \alpha$. Similarly, we have $\alpha \leq x$ for all $x \in A$ so that $-\alpha \geq -x$ for all $x \in A$. It implies that $-\alpha$ is an upper bound of $-A$, so $\beta \leq -\alpha$ and then $-\beta \geq \alpha$. Hence we have $\alpha = -\beta$, i.e. $\inf A = -\sup(-A)$. This completes the proof of the problem. ■

1.3 An index law and the logarithm

Problem 1.6

Rudin Chapter 1 Exercise 6.

Proof.

- (a) Since $b^m > 0$ and $n \in \mathbb{N}$, Theorem 1.21 implies that there exists one and only one real y such that $y^n = b^m$. Similarly, there exists one and only one real z such that $z^q = b^p$. We have

$$y^{nq} = (y^n)^q = (b^m)^q = b^{mq} = b^{pn} = (b^p)^n = (z^q)^n = z^{qn}$$

which implies that $y = z$, i.e., $(b^m)^{\frac{1}{n}} = (b^p)^{\frac{1}{q}}$.

- (b) Let $b^r = b^{\frac{m}{n}}$ and $b^s = b^{\frac{p}{q}}$. Without loss of generality, we may assume that n and q are positive. Then the corollary of Theorem 1.21 implies that

$$b^{r+s} = b^{\frac{mq+np}{nq}} = (b^{mq+np})^{\frac{1}{nq}} = (b^{mq} \times b^{np})^{\frac{1}{nq}} = (b^{mq})^{\frac{1}{nq}} \times (b^{np})^{\frac{1}{nq}} = b^{\frac{m}{n}} \times b^{\frac{p}{q}} = b^r \times b^s.$$

- (c) By definition, $B(r) = \{b^t \mid t \in \mathbb{Q}, t \leq r\}$, where $r \in \mathbb{Q}$. It is clear that $b^r \in B(r)$, so it is a non-empty subset of $\mathbb{R}$. Since $b > 1$, we have $b^t \leq b^r$ for all $t \leq r$ so that b^r is an upper bound of $B(r)$. Therefore, Theorem 1.19 and Definition 1.10 show that $\sup B(r)$ exists in $\mathbb{R}$.

Now we show that $b^r = \sup B(r)$. If $0 < \gamma < b^r$, then γ is obviously not an upper bound of $B(r)$ because $b^r \in B(r)$. By Definition 1.8, we have $b^r = \sup B(r)$.

- (d) By part (c), we know that b^x, b^y and b^{x+y} are all well-defined in $\mathbb{R}$. By definition, we have

$$B(x) = \{b^r \mid r \in \mathbb{Q}, r \leq x\}, \quad B(y) = \{b^s \mid s \in \mathbb{Q}, s \leq y\},$$

$$B(x+y) = \{b^t \mid t \in \mathbb{Q}, t \leq x+y\}.$$

Before continuing the proof, we need to show several results:

Lemma 1.1

For every real x and y , we define

$$B(x, y) = B(x) \times B(y) = \{b^r \times b^s \mid r, s \in \mathbb{Q}, r \leq x, s \leq y\}.$$

Then we have $b^x \times b^y = \sup B(x, y)$.

Proof of Lemma 1.1. By definition, b^x and b^y are upper bounds of $B(x)$ and $B(y)$ respectively, so we have $b^r \leq b^x$ and $b^s \leq b^y$ for every $b^r \in B(x)$ and $b^s \in B(y)$. Therefore, we have $b^r \times b^s \leq b^x \times b^y$ for every $b^r \times b^s \in B(x, y)$. In other words, $b^x \times b^y$ is an upper bound of $B(x, y)$.

Let $0 < \alpha < b^x \times b^y$. Then we have $\frac{\alpha}{b^x} < b^y$. We define the number $p = \frac{1}{2}(\frac{\alpha}{b^x} + b^y)$. It is obvious from this definition that $\frac{\alpha}{b^x} < p < b^y$. By $\frac{\alpha}{b^x} < p$, we have $\frac{\alpha}{p} < b^x$ and so there exists $b^r \in B(x)$ such that

$$\frac{\alpha}{p} < b^r. \quad (1.1)$$

Similarly, the inequality $p < b^y$ implies that there exists $b^s \in B(y)$ such that

$$p < b^s. \quad (1.2)$$

Now inequalities (1.1) and (1.2) show that

$$\alpha < b^r \times b^s$$

for some $b^r \times b^s \in B(x, y)$. Hence α is not an upper bound of $B(x, y)$ and we have

$$b^x \times b^y = \sup B(x, y),$$

completing the proof of the lemma. ■

Lemma 1.2

Let S be a set of positive real numbers and bounded above and $S^{-1} = \{x^{-1} \mid x \in S\}$. Then we have $\sup S = \frac{1}{\inf S^{-1}}$.

Proof of Lemma 1.2. Suppose that α is an upper bound of S , i.e., $0 < x \leq \alpha$ for all $x \in S$. Then we have $0 < \alpha^{-1} \leq x^{-1}$ for all $x^{-1} \in S^{-1}$. Hence the result follows directly from the definitions of the least upper bound and the greatest lower bound. ■

Lemma 1.3

For every real x , we have $b^{-x} = \frac{1}{b^x}$.

Proof of Lemma 1.3. We have two facts:

- **Fact 1.** If $b > 1$, then $b^{\frac{1}{n}} > 1$ for every positive integer n . Otherwise, $0 < b^{\frac{1}{n}} < 1$ implies that $0 < b = (b^{\frac{1}{n}})^n < 1^n = 1$ by Theorem 1.21, a contradiction.
- **Fact 2.** If m and n are positive integers such that $n > m$, then $b^{\frac{1}{m}} > b^{\frac{1}{n}}$. Otherwise, it follows from **Fact 1** that $1 < b^{\frac{1}{m}} < b^{\frac{1}{n}}$ and so it implies that

$$b = (b^{\frac{1}{m}})^m < (b^{\frac{1}{n}})^m < (b^{\frac{1}{n}})^n = b,$$

a contradiction.

Let r and s be rational. Define $A(x) = \{b^s \mid s \in \mathbb{Q}, s \geq x\}$. We next want to prove that $\sup B(x) = \inf A(x)$. In fact, it is clear that $\sup B(x) \leq \inf A(x)$ by definitions. Suppose that $D = \inf A(x) - \sup B(x)$. Assume that $D > 0$. By **Fact 2** above, $b^{\frac{1}{n}} - 1$ is decreasing as n is increasing, so there exists a positive integer N such that

$$b^x(b^{\frac{1}{N}} - 1) < D$$

for all $n \geq N$. By Theorem 1.20(b), we see that there exist $r, s \in \mathbb{Q}$ such that $x - \frac{1}{2n} < s < x$ and $x < r < x + \frac{1}{2n}$. Thus we have $r - s < x + \frac{1}{2n} - (x - \frac{1}{2n}) = \frac{1}{n}$ and this implies that

$$D < b^r - b^s = b^s(b^{r-s} - 1) \leq b^x(b^{r-s} - 1) < D,$$

a contradiction. Hence we must have $\sup B(x) = \inf A(x)$.

Define $B'(-x) = \{b^{-t} \mid t \in \mathbb{Q}, t \leq -x\}$. Let $s = -t$. Then we have $b^{-t} = b^s$ and the inequality $t \leq -x$ is equivalent to $s \geq x$. Thus we have $B'(-x) = A(x)$. Now it follows from Lemma 1.2 and the above facts that

$$b^{-x} = \sup B(-x) = \frac{1}{\inf B'(-x)} = \frac{1}{\inf A(x)} = \frac{1}{\sup B(x)} = \frac{1}{b^x},$$

completing the proof of the lemma. ■

We can continue our proof of the problem. Let b^r and b^s be any elements of $B(x)$ and $B(y)$ respectively. Then we have r and s are rational numbers such that $r \leq x$ and $s \leq y$. Since the sum $t = r + s$ is also rational and $t \leq x + y$, part (b) implies that $b^r \times b^s = b^{r+s} \leq b^{x+y}$. Thus b^{x+y} is an upper bound of the set $B(x, y)$ and Lemma 1.1 implies that

$$b^x \times b^y \leq b^{x+y}.$$

Assume that $b^x b^y < b^{x+y}$. We deduce from Lemma 1.3 that

$$b^y = (b^{-x} b^x) b^y = b^{-x} (b^x b^y) < b^{-x} b^{x+y} \leq b^{-x+(x+y)} = b^y,$$

a contradiction. Hence we have the desired result that $b^x b^y = b^{x+y}$.

This completes the proof of the problem. ■

Problem 1.7

Rudin Chapter 1 Exercise 7.

Proof.

- (a) Since $b > 1$, it is obvious that

$$b^n - 1 = (b - 1)(b^{n-1} + b^{n-2} + \cdots + 1) \geq (b - 1)(1 + 1 + \cdots + 1) = n(b - 1)$$

for any positive integer n .

- (b) It is obvious that $b^{\frac{1}{n}} > 1$; otherwise, we have $b = (b^{\frac{1}{n}})^n < 1$ which is impossible. The result follows by replacing b by the real number $b^{\frac{1}{n}}$ in part (a) and Problem 1.6(a).
- (c) If $t > 1$ and $n > \frac{b-1}{t-1}$, then part (b) implies that

$$b - 1 \geq n(b^{\frac{1}{n}} - 1) > \frac{b-1}{t-1} \times (b^{\frac{1}{n}} - 1)$$

and so $b^{\frac{1}{n}} < t$.

- (d) Let w be a number such that $b^w < y$. Let $t = y \cdot b^{-w}$. It is easily to check that $t > 1$. If n is sufficiently large enough, then we have $n > \frac{b-1}{t-1}$. Hence it follows from parts (c) and (b) that $b^{\frac{1}{n}} < t = y \cdot b^{-w}$ and thus $b^{w+\frac{1}{n}} < y$ for sufficiently large n .
- (e) Let w be a number such that $b^w > y$. Let $t = y^{-1} \cdot b^w$. It is obvious that $t > 1$. If n is sufficiently large enough, then we have $n > \frac{b-1}{t-1}$. Hence it follows from part (c) and then part (b) that

$$n(y^{-1}b^w - 1) > b - 1 \geq n(b^{\frac{1}{n}} - 1)$$

and thus $b^{w-\frac{1}{n}} > y$ for sufficiently large n .

- (f) We have $A = \{w \in \mathbb{R} \mid b^w < y\}$. Since x is the least upper bound of A , we have $w \leq x$ for all $w \in A$. If $b^x < y$, then part (d) implies that $b^{x+\frac{1}{n}} < y$ for sufficiently large n and so $x + \frac{1}{n} \in A$. Therefore, we have $x + \frac{1}{n} \leq x$ and then $\frac{1}{n} \leq 0$, a contradiction. Similarly, if $b^x > y$, then $x \notin A$ and so $w < x$ for all $w \in A$. Now part (e) implies that $b^{x-\frac{1}{n}} > y$ for sufficiently large n , so we have

$$w < x - \frac{1}{n} < x$$

for some sufficiently large n . This means that $x - \frac{1}{n}$ is an upper bound of A , contradicting to the fact that x is the least upper bound of A . Hence, we must have

$$b^x = y.$$

- (g) The uniqueness of x follows from the uniqueness of the least upper bound of the set A . We finish the proof of the problem. ■

1.4 Properties of the complex field

Problem 1.8

Rudin Chapter 1 Exercise 8.

Proof. Assume that $i > 0$. Then we have $i \cdot i > i \cdot 0$ which implies that $-1 > 0$, a contradiction. Similarly, the case $i < 0$ is impossible. ■

Problem 1.9

Rudin Chapter 1 Exercise 9.

Proof. We check Definition 1.5 in this case. Let $z = a + bi, w = c + di \in \mathbb{C}$. If $a \neq c$, then we have either $z < w$ or $z > w$. If $a = c$ and $b \neq d$, then we have either $z < w$ or $z > w$. If $a = c$ and $b = d$, then we have $z = w$. Therefore, this relation satisfies Definition 1.5(i).

Let $z = a + bi, w = c + di$ and $q = e + fi$ be complex numbers such that $z < w$ and $w < q$. Since $z < w$, we have either $a < c$ or $a = c$ and $b < d$. Similarly, since $w < q$, we have either $c < e$ or $c = e$ and $d < f$. Combining the above two results, we get either $a < e$ or $a = e$ and $b < f$. This proves Definition 1.5(ii). Hence this turns $\mathbb{C}$ into an ordered set.

Let $S = \{z = a + bi \mid b \in \mathbb{R}, a < 0\} \subset \mathbb{C}$. Since $-1 \in S$, it is clear that S is not empty. We will show that $\sup S$ does not exist in $\mathbb{C}$. Assume that $w = c + di$ was the least upper bound of S for some $c, d \in \mathbb{R}$. That is, $z \leq w$ for all $z \in S$. If $c > 0$, then the definition of the dictionary order implies that $z < \zeta = 0 + bi < w$ for all $z \in S$, contradicting to the fact that $w = \sup S$. If $c < 0$, then we have $c < \frac{c}{2}$ so that $\zeta = \frac{c}{2} + di \in S$ and $w < \zeta$, contradicting to the fact that $w = \sup S$. Hence this order set does not have the least-upper-bound property, completing the proof of the problem. ■

Problem 1.10

Rudin Chapter 1 Exercise 10.

Proof. If $v \geq 0$, then we have

$$z^2 = a^2 + 2abi - b^2 = u + 2\left(\frac{|w|^2 - u^2}{4}\right)^{\frac{1}{2}}i = u + (|w|^2 - u^2)^{\frac{1}{2}}i = u + (v^2)^{\frac{1}{2}}i = u + vi = w.$$

If $v \leq 0$, then we have

$$\bar{z}^2 = a^2 - 2abi - b^2 = u - (|w|^2 - u^2)^{\frac{1}{2}}i = u - (v^2)^{\frac{1}{2}}i = u - (-v)i = u + vi = w.$$

By the above results, we see that every non-zero complex number w has two complex square roots $z, \bar{z}$ which are defined as in the question. However, when $w = 0$, then $u = v = 0$ which imply that $a = b = 0$ and thus $z = \bar{z} = 0$. Hence, the complex number 0 has only one complex square root which is 0 itself, completing the proof of the problem. ■

Problem 1.11

Rudin Chapter 1 Exercise 11.

Proof. Let $|z| = r > 0$. We define $z = rw$, where $w = \frac{z}{r}$. Then it is easily to see that this expression satisfies the required conditions. Assume $r_1 w_1 = r_2 w_2$, where $r_1 > 0, r_2 > 0$ and $|w_1| = |w_2| = 1$. Then we have $\frac{r_1}{r_2} = \frac{w_1}{w_2}$ which leads to

$$\frac{r_1}{r_2} = \left| \frac{w_1}{w_2} \right| = \frac{|w_1|}{|w_2|} = 1.$$

Hence we have $r_1 = r_2$ and then $w_1 = w_2$. This completes the proof of the problem. ■

Problem 1.12

Rudin Chapter 1 Exercise 12.

Proof. This result follows from induction and Theorem 1.33(e). ■

Problem 1.13

Rudin Chapter 1 Exercise 13.

Proof. Since

$$|x| = |x - y + y| \leq |x - y| + |y|$$

by Theorem 1.33(e), we have $|x| - |y| \leq |x - y|$. Similarly, since

$$|y| = |y - x + x| \leq |y - x| + |x|$$

by Theorem 1.33(e), we have $-|x - y| \leq |x| - |y|$. Hence these two results together imply that the desired result, completing the proof of the problem. ■

Problem 1.14

Rudin Chapter 1 Exercise 14.

Proof. It follows from Definition 1.32 that

$$|1 + z|^2 + |1 - z|^2 = (1 + z)(1 + \bar{z}) + (1 - z)(1 - \bar{z}) = 2(1 + |z|^2)$$

holds. This completes the proof of the problem. ■

Problem 1.15

Rudin Chapter 1 Exercise 15.

Proof. By the proof of the Schwarz inequality (Theorem 1.35), the equality holds if and only if each term in the sum $\sum_{j=1}^n |Ba_j - Cb_j|^2$ is zero, i.e.,

$$\begin{aligned} |Ba_j - Cb_j|^2 &= 0 \\ (Ba_j - Cb_j)(\overline{Ba_j - Cb_j}) &= 0 \\ B^2|a_j|^2 + C^2|b_j|^2 - 2BC\operatorname{Re}(a_j\bar{b}_j) &= 0. \end{aligned} \tag{1.3}$$

By Theorem 1.33(d), we have $\operatorname{Re}(a_j\bar{b}_j) \leq |a_j||b_j|$ and so the relation (1.3) implies that

$$0 \geq B^2|a_j|^2 + C^2|b_j|^2 - 2BC|a_j||b_j| = (B|a_j| - C|b_j|)^2.$$

Hence we have the equality holds if and only if $B|a_j| = C|b_j|$ if and only if $\frac{|a_j|}{|b_j|}$ is a constant for $j = 1, 2, \dots, n$. This completes the proof of the problem. ■

1.5 Properties of Euclidean spaces

Problem 1.16

Rudin Chapter 1 Exercise 16.

Proof. Let $\mathbf{m}$ be the mid-point of $\mathbf{x}$ and $\mathbf{y}$ and $\mathbf{u} = \mathbf{z} - \mathbf{m} = (u_1, u_2, \dots, u_k)$. Geometrically, these conditions say that the three vectors $\mathbf{x}, \mathbf{y}$ and $\mathbf{u}$ form an isosceles triangle with sides r, r and d . In other words, $\mathbf{u}$ must satisfy the equations

$$\mathbf{u} \cdot (\mathbf{x} - \mathbf{y}) = 0 \quad \text{and} \quad |\mathbf{u}|^2 = r^2 - \frac{d^2}{4}. \tag{1.4}$$

- (a) Since $|\mathbf{x} - \mathbf{y}| = d > 0$, we have from Theorem 1.37(b) that $\mathbf{x}$ and $\mathbf{y}$ are distinct and then we may assume without loss of generality that $x_1 \neq y_1$. If $u_2, u_3, \dots, u_k$ are arbitrary, then we define

$$u_1 = \frac{-1}{x_1 - y_1} [u_2(x_2 - y_2) + \dots + u_k(x_k - y_k)].$$

Then it is clear that this vector $\mathbf{u} = (u_1, u_2, \dots, u_k)$ satisfies the equation $\mathbf{u} \cdot (\mathbf{x} - \mathbf{y}) = 0$. Since $u_2, u_3, \dots, u_k$ are arbitrary, there are infinitely many u_1 and then $\mathbf{u}$.

Next, suppose that $\mathbf{u}$ is a vector satisfying the equation $\mathbf{u} \cdot (\mathbf{x} - \mathbf{y}) = 0$. Since $2r > d$, we must have $r^2 - \frac{d^2}{4} > 0$. If $|\mathbf{u}|^2 \neq r^2 - \frac{d^2}{4}$, then we consider the vector $\bar{\mathbf{u}}$ defined by

$$\bar{\mathbf{u}} = \frac{\sqrt{r^2 - \frac{d^2}{4}}}{|\mathbf{u}|} \mathbf{u}.$$

Thus it is easy to check that $\bar{\mathbf{u}}$ satisfies both equations (1.4). In fact, this proves that there are infinitely many $\mathbf{u}$ satisfying both equations (1.4) and hence there are infinitely many $\mathbf{z}$ such that

$$|\mathbf{z} - \mathbf{x}| = |\mathbf{z} - \mathbf{y}| = r.$$

- (b) If $2r = d$, then we have $r^2 - \frac{d^2}{4} = 0$ and so $|\mathbf{u}|^2 = 0$. It follows from Theorem 1.37(b) that $\mathbf{u} = \mathbf{0}$, i.e., $\mathbf{z} = \mathbf{m}$. This proves the uniqueness of such $\mathbf{z}$.
- (c) If $2r < d$, then we have $r^2 - \frac{d^2}{4} < 0$ and it is clear that there is no $\mathbf{u}$ such that $|\mathbf{u}|^2 < 0$. Hence there is no such $\mathbf{z}$ in this case.

When $k = 1$ or 2 , the results will be different from those when $k \geq 3$ and the analysis is given as follows:

- **Case (i):** $k = 2$. If $2r > d$, then

$$\bar{\mathbf{u}} = \frac{\sqrt{r^2 - \frac{d^2}{4}}}{|\mathbf{u}|} \mathbf{u}$$

are the only vectors satisfying the equations (1.4), where

$$\mathbf{u} = \pm \left(-\frac{x_2 - y_2}{x_1 - y_1} u_2, u_2 \right).$$

If $2r = d$, then $\mathbf{u} = \mathbf{0}$ is the only vector such that $\mathbf{u} \cdot (\mathbf{x} - \mathbf{y}) = 0$ and $|\mathbf{u}|^2 = 0$. If $2r < d$, then there is no such $\mathbf{u}$. Hence, there are two such $\mathbf{z}$ if $2r > d$, exactly one such $\mathbf{z}$ if $2r = d$ and no such $\mathbf{z}$ if $2r < d$.

- **Case (ii):** $k = 1$. Then there is no point u satisfying both

$$u(x_1 - y_1) = 0 \quad \text{and} \quad |u|^2 = r^2 - \frac{d^2}{4}$$

if $2r \neq d$. However, $u = 0$ is the only point such that $u(x_1 - y_1) = 0$ and $|u|^2 = 0$ if $2r = d$. Hence, there is no such z if $2r \neq d$ and only one such z if $2r = d$.

This completes the proof of the problem. ■

Problem 1.17

Rudin Chapter 1 Exercise 17.

Proof. Let $\mathbf{x} = (x_1, x_2, \dots, x_k)$ and $\mathbf{y} = (y_1, y_2, \dots, y_k)$. Then we have

$$|\mathbf{x} + \mathbf{y}|^2 + |\mathbf{x} - \mathbf{y}|^2 = \sum_{j=1}^k [(x_j + y_j)^2 + (x_j - y_j)^2] = 2 \sum_{j=1}^k x_j^2 + 2 \sum_{j=1}^k y_j^2 = 2|\mathbf{x}|^2 + 2|\mathbf{y}|^2.$$

Suppose that $\mathbf{x}$ and $\mathbf{y}$ are sides of a parallelogram. Then $\mathbf{x} + \mathbf{y}$ and $\mathbf{x} - \mathbf{y}$ are the diagonals of the parallelogram and the above result can be interpreted as follows: the sum of the squares of the lengths of the diagonals (left-hand side) is double to the sum of the squares of the lengths of the sides (right-hand side). This completes the proof of the problem. ■

Problem 1.18

Rudin Chapter 1 Exercise 18.

Proof. We define $\mathbf{x} = (x_1, x_2, \dots, x_k)$, where $x_j \in \mathbb{R}$ for $j = 1, 2, \dots, k$. If $x_1 = x_2 = \dots = x_k = 0$, then the element $\mathbf{y} = (1, 0, \dots, 0)$ satisfies the requirements that $\mathbf{y} \neq \mathbf{0}$ and $\mathbf{x} \cdot \mathbf{y} = 0$. Otherwise, without loss of generality, we may assume that $x_1 \neq 0$. If we define

$$\mathbf{y} = \left(-\frac{x_2 + x_3 + \dots + x_k}{x_1}, 1, \dots, 1 \right),$$

then we still have $\mathbf{y} \neq \mathbf{0}$ and $\mathbf{x} \cdot \mathbf{y} = 0$. Let x be a non-zero real number. If $y \neq 0$, then Proposition 1.16(b) implies that $x \cdot y \neq 0$. Hence this is not true if $k = 1$, finishing the proof of the problem. ■

Problem 1.19

Rudin Chapter 1 Exercise 19.

Proof. Let $3\mathbf{c} = 4\mathbf{b} - \mathbf{a}$ and $3r = 2|\mathbf{b} - \mathbf{a}|$. Since $|\mathbf{x}|^2 = \mathbf{x} \cdot \mathbf{x}$ (Definition 1.36), we have the following relations:

$$\begin{aligned} |\mathbf{x} - \mathbf{a}| = 2|\mathbf{x} - \mathbf{b}| &\Leftrightarrow |\mathbf{x}|^2 - 2\mathbf{a} \cdot \mathbf{x} + |\mathbf{a}|^2 = 4|\mathbf{x}|^2 - 8\mathbf{b} \cdot \mathbf{x} + 4|\mathbf{b}|^2 \\ &\Leftrightarrow 3|\mathbf{x}|^2 - 8\mathbf{b} \cdot \mathbf{x} + 2\mathbf{a} \cdot \mathbf{x} + 4|\mathbf{b}|^2 - |\mathbf{a}|^2 = 0 \\ &\Leftrightarrow 9|\mathbf{x}|^2 - 24\mathbf{b} \cdot \mathbf{x} + 6\mathbf{a} \cdot \mathbf{x} + 12|\mathbf{b}|^2 - 3|\mathbf{a}|^2 = 0 \\ &\Leftrightarrow (3\mathbf{x} - 4\mathbf{b} + \mathbf{a}) \cdot (3\mathbf{x} - 4\mathbf{b} + \mathbf{a}) = 4(\mathbf{b} - \mathbf{a})(\mathbf{b} - \mathbf{a}) \\ &\Leftrightarrow \left(\mathbf{x} - \frac{4}{3}\mathbf{b} + \frac{1}{3}\mathbf{a} \right) \cdot \left(\mathbf{x} - \frac{4}{3}\mathbf{b} + \frac{1}{3}\mathbf{a} \right) = \frac{4}{9}(\mathbf{b} - \mathbf{a})(\mathbf{b} - \mathbf{a}) \\ &\Leftrightarrow \left| \mathbf{x} - \frac{4}{3}\mathbf{b} + \frac{1}{3}\mathbf{a} \right| = \frac{2}{3}|\mathbf{b} - \mathbf{a}| \\ &\Leftrightarrow |\mathbf{x} - \mathbf{c}| = r, \end{aligned}$$

completing the proof of the problem. ■

1.6 A supplement to the proof of Theorem 1.19

Problem 1.20

Rudin Chapter 1 Exercise 20.

Proof. Let us recall the first two properties now:

$$(I) \quad \alpha \neq \emptyset \text{ and } \alpha \neq \mathbb{Q}.$$

(II) If $p \in \alpha$ and $q \in \mathbb{Q}$ such that $q < p$, then $q \in \alpha$.

Suppose that “(III) If $p \in \alpha$, then $p < r$ for some $r \in \alpha$ ” is deleted in the definition of a cut, see p. 17. Then it is easy to see that **Step 2** and **Step 3** are still satisfied. For **Step 4**, we still have the definition of addition of cuts: if $\alpha, \beta \in R$, then $\alpha + \beta = \{r + s \mid r \in \alpha, s \in \beta\}$. We define

$$0^* = \{p \in \mathbb{Q} \mid p \leq 0\}.$$

It is clear that 0^* satisfies (I) and (II), but it has the maximum element 0. It is also obvious that the addition so defined satisfies axioms (A1) to (A3).

Let $\alpha \in R$. If $r \in \alpha$ and $s \in 0^*$, then either $r + s < r$ or $r + s = r$ which imply that $r + s \in \alpha$, i.e., $\alpha + 0^* \subseteq \alpha$. For any $p \in \alpha$, we always have $p + 0 = p$ so that $p \in \alpha + 0^*$, i.e., $\alpha \subseteq \alpha + 0^*$. Hence we have $\alpha = \alpha + 0^*$ and the addition satisfies (A4).

Let $\alpha = \{p \in \mathbb{Q} \mid p < 0\} \in R$. Assume that there was $\beta \in R$ such that $\alpha + \beta = 0^*$. Since $\alpha \neq \varnothing$, we have $p \in \alpha$ and then β contains an element $q \in \mathbb{Q}$ such that $p + q = 0$. Since p is a negative rational, q must be a positive rational. By Theorem 1.20(b), we have $0 < q_1 < q$ for some $q_1 \in \mathbb{Q}$. By (II), we have $q_1 \in \beta$. However, we have $p + q_1 > 0$ so that $p + q_1 \notin 0^*$ by definition. This contradicts the assumption and hence we have the fact that the addition does not satisfy (A5). This completes the proof of the problem. ■

CHAPTER 2

Basic Topology

2.1 The empty set and properties of algebraic numbers

Problem 2.1

Rudin Chapter 2 Exercise 1.

Proof. Assume that there was a set A such that $\emptyset$ is not a subset of it. Then $\emptyset$ contains an element which is not in A . However, $\emptyset$ has no element by definition. Hence no such element exists and $\emptyset$ must be a subset of every set. This completes the proof of the problem. ■

Problem 2.2

Rudin Chapter 2 Exercise 2.

Proof. We don't use the hint to prove the result. For each positive integer n , we let P_n be the set of all polynomials of degree less than or equal to n with integer coefficients. Then it is easy to check that the mapping $f : P_n \rightarrow \mathbb{Z}^{n+1}$ defined by

$$f(a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n) = (a_0, a_1, \dots, a_n)$$

is bijective. By Example 2.5, it is clear that $\mathbb{Z}$ is countable. Thus it follows from Theorem 2.13 that $\mathbb{Z}^{n+1}$ is countable and thus P_n is also countable. For each $p(z) \in P_n$, we let $B_{p(z)}$ be the set of all roots of $p(z)$. Since a polynomial $p(z) \in P_n$ has *at most* n (distinct) roots, $B_{p(z)}$ is a finite set. By the corollary to Theorem 2.12, the set

$$S = \bigcup_{p(z) \in P_n} B_{p(z)}$$

is at most countable. Since every positive integer is an algebraic number (consider the polynomial $z - n$), the set of all algebraic numbers A is infinite. Since A is a subset of S , we have A is countable, completing the proof of the problem. ■

Problem 2.3

Rudin Chapter 2 Exercise 3.

Proof. Assume that all real numbers were algebraic. Then Problem 2.2 implies that $\mathbb{R}$ is countable which contradicts the corollary of Theorem 2.43. Hence there exist real numbers which are not algebraic. This completes the proof of the problem. ■

2.2 The uncountability of irrational numbers

Problem 2.4

Rudin Chapter 2 Exercise 4.

Proof. The set of all irrational real numbers $\mathbb{R} \setminus \mathbb{Q}$ is uncountable. Otherwise the corollary of Theorem 2.12 implies that

$$\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q}$$

is countable, a contradiction to the corollary of Theorem 2.43. This completes the proof of the problem. ■

2.3 Limit points, open sets and closed sets

Problem 2.5

Rudin Chapter 2 Exercise 5.

Proof. We let

$$E_0 = \left\{ \frac{1}{n} \mid n = 1, 2, \dots \right\}, \quad E_1 = \left\{ 2 + \frac{1}{n} \mid n = 1, 2, \dots \right\}, \quad E_2 = \left\{ 4 + \frac{1}{n} \mid n = 1, 2, \dots \right\}$$

and

$$E = E_0 \cup E_1 \cup E_2.$$

It is clear that E_0, E_1 and E_2 are bounded. Furthermore, by Example 2.21(e), we know that each of E_0, E_1 and E_2 has exactly one limit point, namely 0, 2 and 4 respectively. Hence the E satisfies the required conditions. This completes the proof of the problem. ■

Problem 2.6

Rudin Chapter 2 Exercise 6.

Proof. Let $p \in (E')'$ and $N_r(p)$ be a neighborhood of p for some $r > 0$. Since p is a limit point of E' , Definition 2.18(b) implies that there exists a point $q \neq p$ in $N_r(p)$ such that $q \in E'$. By the definition of E' , q is a limit point of E and thus there exists a point $s \neq q$ in $N_h(q)$ such that $s \in E$ for every $h > 0$. If we take $h = \frac{1}{2} \min(d(p, q), r - d(p, q))$, then we have

$$N_h(q) \subset N_r(p).$$

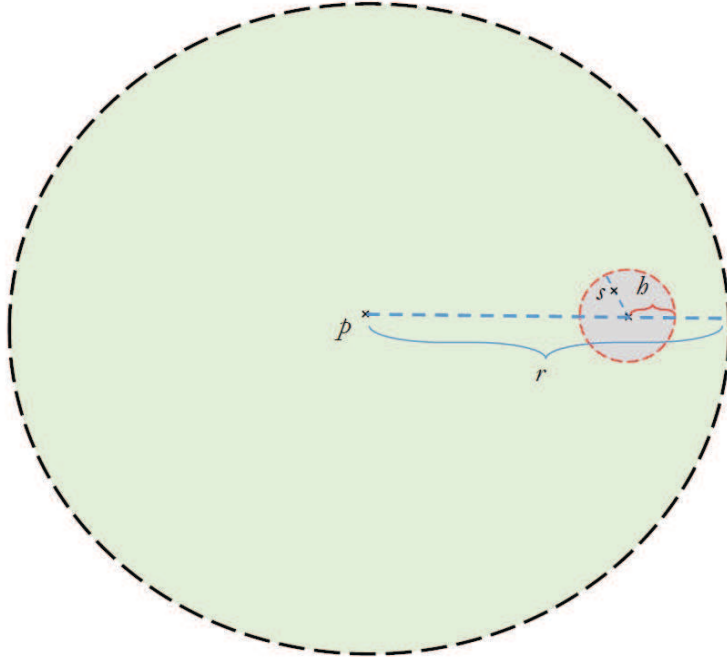
Furthermore, we must have $s \neq p$. Otherwise, we have $p \in N_h(q)$. If $h = \frac{1}{2}d(p, q)$, then this fact implies that

$$d(p, q) < h = \frac{1}{2}d(p, q),$$

a contradiction. If $h = \frac{1}{2}(r - d(p, q))$, then we have $r - d(p, q) \leq d(p, q)$, but $d(p, q) < h = \frac{1}{2}(r - d(p, q))$ which implies that

$$2d(p, q) < r - d(p, q),$$

a contradiction again. Thus what we have shown is that every neighborhood $N_r(p)$ of p contains a point $s \neq p$ such that $s \in E$. By Definition 2.18(b), p is a limit point of E and hence $p \in E'$ and E' is closed by Definition 2.18(d). See Figure 2.1 for the neighborhoods $N_h(q)$ and $N_r(p)$ below.

Figure 2.1: The neighborhoods $N_h(q)$ and $N_r(p)$.

We first show that $(\overline{E})' \subseteq E'$. Suppose that $p \in (\overline{E})'$. Then p is a limit point of $\overline{E} = E' \cup E$, so for every $r > 0$, there exists $q \in N_r(p)$ and $q \neq p$ such that $q \in \overline{E} = E' \cup E$. If $q \in E$, then p is already a limit point of E and thus $p \in E'$. If $q \in E'$, then the argument in the previous paragraph can be applied to obtain that $p \in E'$. So we have $(\overline{E})' \subseteq E'$. Conversely, suppose that $p \in E'$ and $N_r(p)$ is a neighborhood of p for some $r > 0$. Since p is a limit point of E , Definition 2.18(b) implies that there exists a point $q \neq p$ in $N_r(p)$ such that $q \in E$. Recall that $\overline{E} = E \cup E'$, we must have $q \in \overline{E}$ and thus p is a limit point of $\overline{E}$. Hence $E' \subseteq (\overline{E})'$ and then $E' = (\overline{E})'$.

The sets E and E' may have different limit points. For example, we consider the set $E = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ whose only limit point is 0, i.e., $E' = \{0\}$. Since E' has only one element, Theorem 2.20 implies that $(E')' = \emptyset$. This completes the proof of the problem. ■

Problem 2.7

Rudin Chapter 2 Exercise 7.

Proof.

- (a) If $x \in \overline{B_n}$, then $x \in B_n$ or $x \in B'_n$. If $x \in B_n$, then $x \in A_i$ for some $i \in \{1, 2, \dots, n\}$, so $x \in \overline{A_i}$. Therefore we have

$$x \in \bigcup_{i=1}^n \overline{A_i}.$$

If $x \in B'_n$, then x is a limit point of B_n . We claim that $x \in A'_i$ for some $i \in \{1, 2, \dots, n\}$. Assume that $x \notin A'_i$ for all $i \in \{1, 2, \dots, n\}$. Then for each $i \in \{1, 2, \dots, n\}$, there exists a neighborhood $N_{r_i}(x)$ of x for some $r_i > 0$ such that $N_{r_i}(x) \cap A_i = \emptyset$. Let $r = \min_{1 \leq i \leq n} \{r_i\} > 0$.^a Then we must

have $N_r(x) \cap A_i = \emptyset$ for all $i \in \{1, 2, \dots, n\}$, so $N_r(x) \cap B_n = \emptyset$ by Remarks 2.11, contradicting the fact that x is a limit point of B_n . Hence we must have $x \in A'_i$ for some $i \in \{1, 2, \dots, n\}$ and

^aWe can define this r because there are only finitely many subsets, namely $A_1, A_2, \dots, A_n$, of a metric space.

then

$$\overline{B_n} \subseteq \bigcup_{i=1}^n \overline{A_i}. \quad (2.1)$$

Conversely, if $x \in \bigcup_{i=1}^n \overline{A_i}$, then $x \in \overline{A_i}$ and thus $x \in A_i$ or $x \in A'_i$ for some $i \in \{1, 2, \dots, n\}$. If $x \in A_i$, then we have $x \in B_n \subseteq \overline{B_n}$. If $x \in A'_i$, then x is a limit point of A_i . Thus there exists $y \in N_r(x)$ for every $r > 0$ and $y \neq x$ such that $y \in A_i$. Since $A_i \subseteq B_n$ by definition, x is also a limit point of B_n , i.e., $x \in B'_n \subseteq \overline{B_n}$. Both cases together imply that

$$\bigcup_{i=1}^n \overline{A_i} \subseteq \overline{B_n}. \quad (2.2)$$

Hence the above set relations (2.1) and (2.2) imply that

$$\overline{B_n} = \bigcup_{i=1}^n \overline{A_i},$$

for $n = 1, 2, \dots$

- (b) The result follows from a similar argument of part (a). However, the inclusion may be proper. For example, we consider $A_i = \{\frac{1}{i}\}$ for all $i \in \mathbb{N}$, so we have $B = \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$. By the corollary of Theorem 2.20, we have $A'_i = \emptyset$ and thus $\overline{A_i} = \{\frac{1}{i}\}$ for all $i \in \mathbb{N}$. Thus we have

$$\bigcup_{i=1}^{\infty} \overline{A_i} = \{1, \frac{1}{2}, \frac{1}{3}, \dots\},$$

but $\overline{B} = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$. Hence we have $\overline{B} \supset \bigcup_{i=1}^{\infty} \overline{A_i}$.

This completes the proof of the problem. ■

Problem 2.8

Rudin Chapter 2 Exercise 8.

Proof. Let $p = (p_1, p_2) \in E$. Since E is open in $\mathbb{R}^2$, we have $N_s(p) \subseteq E$ for some $s > 0$. In other words, there exists $q = (q_1, q_2) \in N_s(p)$ such that $(q_1, q_2) \neq (p_1, p_2)$. Let $r > 0$. If $s \leq r$, then it is obvious that $(q_1, q_2) \in N_s(p) \subseteq N_r(p)$ in this case. If $r < s$, then we consider the point $p' = (p_1 + \frac{1}{2}r, p_2)$. Since

$$0 < \sqrt{(p_1 + \frac{1}{2}r - p_1)^2 + (p_2 - p_2)^2} = \frac{1}{2}r < r,$$

we have $p' \in N_r(p)$ and $p' \neq p$. Hence p is a limit point of E .

However, the case is not true for closed sets. For example, let $E = \{(1, 0), (2, 0)\}$. Then E is closed by Example 2.21(c) but $E' = \emptyset$ by the corollary of Theorem 2.20. This completes the proof of the problem. ■

Problem 2.9

Rudin Chapter 2 Exercise 9.

Proof.

- (a) Let $p \in E^\circ$. To show that p is an interior point of E° , it is equivalent to show that there exists a neighborhood N of p such that $N \subseteq E^\circ$ and this is equivalent to show that every point of such N is an interior point of E . By definition, p is an interior point of E , so there is a neighborhood N_p of p such that $N_p \subseteq E$. Now for each $q \in N_p$, since N_p is an open set by Theorem 2.19, there exists a neighborhood N_q of q such that

$$N_q \subseteq N_p \subseteq E.$$

Thus every point of N_p is an interior point of E by Definition 2.18(e). Hence we have p is an interior point of E° , i.e., E° is always open.

- (b) By Definition 2.18(e), we always have $E^\circ \subseteq E$. Now it follows from Definition 2.18(f) that E is open if and only if every point of E is an interior point of E if and only if $E \subseteq E^\circ$. Hence we have $E = E^\circ$ if and only if E is open.

- (c) Let $p \in G$. Since G is open, there exists a neighborhood N of p such that

$$N \subseteq G \subseteq E.$$

Hence p is also an interior point of E , i.e., $p \in E^\circ$ and then $G \subseteq E^\circ$.

- (d) If $x \in (E^\circ)^c$, then $x \notin E^\circ$. This means x is not an interior point of E . If $x \notin E$, then $x \in E^c \subseteq \overline{E^c}$. If $x \in E$, then Definition 2.18(e) implies that every neighborhood N of x must satisfy $N \cap E^c \neq \emptyset$. Hence x is a limit point of E^c by Definition 2.18(b) and then $x \in (E^c)' \subseteq \overline{E^c}$. Hence we have $(E^\circ)^c \subseteq \overline{E^c}$.

Conversely, if $x \in \overline{E^c}$, then we have $x \in E^c$ or $x \in (E^c)'$. If $x \in E^c$, then $x \notin E$. Since $E^\circ \subseteq E$, we have $x \notin E^\circ$ and thus $x \in (E^\circ)^c$. If $x \in (E^c)'$, then x is a limit point of E^c . If $x \notin (E^\circ)^c$, then $x \in E^\circ$. By definition, x is an interior point of E so that there exists a neighborhood N of x such that $N \subseteq E$ which gives $N \cap E^c = \emptyset$, a contradiction to the fact that x is a limit point of E^c . Hence we have $x \in (E^\circ)^c$ and then $\overline{E^c} \subseteq (E^\circ)^c$.

Combining the two parts, we must have $(E^\circ)^c = \overline{E^c}$.

- (e) It is not always true. For example, take $E = (-1, 0) \cup (0, 1)$ in $\mathbb{R}$. Then it is easy to see that $E^\circ = E$ by part (b) but $\overline{E} = [-1, 1]$ so that $\overline{E^\circ} = (-1, 1)$. Hence we have the fact that $E^\circ \neq \overline{E^\circ}$.
- (f) It is not always true. For example, take $E = \{1\}$ in $\mathbb{R}$. Then the corollary of Theorem 2.20 yields that E is closed so that $\overline{E} = E$. However, we have $E^\circ = \emptyset$ by definition so that $\overline{E^\circ} = \emptyset$. Hence we have the fact that $\overline{E^\circ} \neq \overline{E}$.

This completes the proof of the problem. ■

2.4 Some metrics

Problem 2.10

Rudin Chapter 2 Exercise 10.

Proof. It is clear that Definition 2.15(a) and (b) are true for this d . Since X is an infinite set, there exists $r \in X$ such that $r \neq p$ and $r \neq q$. Thus for $r \in X$, we have the following cases:

- **Case (i):** $p = q = r$. Then $d(p, q) = d(p, r) = d(r, q) = 0$ so that $d(p, q) = d(p, r) + d(r, q)$.
- **Case (ii):** $p = q$ but $r \neq p$. Then $d(p, q) = 0$ and $d(p, r) = d(r, q) = 1$ so that

$$d(p, q) < d(p, r) + d(r, q).$$

- **Case (iii):** $p \neq q$ and $r = p$. Then $d(p, q) = d(r, q) = 1$ and $d(p, r) = 0$ so that

$$d(p, q) = d(p, r) + d(r, q).$$

- **Case (iv):** $p \neq q$ and $r = q$. Then $d(p, q) = d(p, r) = 1$ and $d(r, q) = 0$ so that

$$d(p, q) = d(p, r) + d(r, q).$$

- **Case (v):** $p \neq q, r \neq p$ and $r \neq q$. Then $d(p, q) = d(p, r) = d(r, q) = 1$ so that

$$d(p, q) < d(p, r) + d(r, q).$$

Hence Definition 2.15(c) is also true for this d and then it is a metric.

Let $x \in X$. We consider its neighborhood

$$N_r(x) = \{y \in X \mid d(x, y) < r\}.$$

Since $N_r(x) \subset \{x\}$ for any $0 < r \leq 1$, the set $\{x\}$ is open in X for every $x \in X$. By Theorem 2.24(a), every subset of X is open in X . Next, it follows from Theorem 2.23 that every subset of X is also closed in X . It is clear that every finite subset of X is compact. Assume $K \subseteq X$ was compact and infinite. For $x \in K$, since $G_x = \{x\}$ is open in X and

$$K \subseteq \bigcup_{x \in K} G_x,$$

the collection $\{G_x\}$ is an open cover of K . Since K is compact, there exists a positive integer n such that

$$K \subseteq \bigcup_{i=1}^n G_i = \{x_1, \dots, x_n\}$$

which contradicts the hypothesis that K is infinite. Hence K is compact in X if and only if K is a finite subset of X . This completes the proof of the problem. ■

Problem 2.11

Rudin Chapter 2 Exercise 11.

Proof. We answer the questions one by one:

- **For d_1 :** It is clear that $d_1(3, 1) = 4, d_1(3, 2) = 1$ and $d_1(2, 1) = 1$, so we have $d_1(3, 1) > d_1(3, 2) + d_1(2, 1)$. Hence d_1 is not a metric.
- **For d_2 :** It is easy to check that the function d_2 satisfies Definition 2.15(a) and (b). For any non-negative real numbers a and b , we have $a + b \leq a + 2\sqrt{ab} + b$. This certainly implies that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and so for any $x, y, z \in \mathbb{R}$,

$$d_2(x, y) = \sqrt{|x - y|} \leq \sqrt{|x - z| + |z - y|} \leq \sqrt{|x - z|} + \sqrt{|z - y|} = d_2(x, z) + d_2(z, y).$$

Hence d_2 is a metric.

- **For d_3 and d_4 :** The function d_3 is not a metric because $d_3(1, -1) = |1^2 - (-1)^2| = 0$. Similarly, the function d_4 is not a metric too because $d_4(1, 1) = |1 - 2| = 1$.
- **For d_5 :** It is clear that d_5 satisfies Definition 2.15(a) and (b). To show that d_5 also satisfies Definition 2.15(c), we need a lemma first:

Lemma 2.1

Suppose that a, b and c are non-negative real numbers. If $a \leq b + c$, then we have

$$\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c}.$$

Proof of Lemma 2.1. Since $0 \leq a \leq b + c$, we have $1 \leq 1 + a \leq 1 + b + c$ and then

$$\frac{a}{1+a} = \frac{1+a-1}{1+a} = 1 - \frac{1}{1+a} \leq 1 - \frac{1}{1+b+c} = \frac{b}{1+b+c} + \frac{c}{1+b+c} \leq \frac{b}{1+b} + \frac{c}{1+c},$$

completing the proof of the lemma. ■

Now if we put $a = |x - y|$, $b = |x - z|$ and $c = |y - z|$ into Lemma 2.1, we immediately have the result that

$$d_5(x, y) \leq d_5(x, z) + d_5(z, y).$$

Hence d_5 is a metric.

Now we end the proof of the problem. ■

2.5 Compact sets

Problem 2.12

Rudin Chapter 2 Exercise 12.

Proof. Let $K = \{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\}$ and $\{G_\alpha\}$ be a collection of open subsets of $\mathbb{R}$ such that $K \subseteq \bigcup_\alpha G_\alpha$. Then we must have $0 \in G_{\alpha_1}$ for some α_1 . Since G_{α_1} is open in $\mathbb{R}$, 0 is an interior point of G_{α_1} by Definition 2.18(f). Thus there exists an interval (a, b) , where $a < 0 < b$, such that $(a, b) \subseteq G_{\alpha_1}$. By Theorem 1.20(a) (the Archimedean property), there exists a positive integer N such that $Nb > 1$, i.e., $b > \frac{1}{N}$. Therefore, we have $\frac{1}{n} \in (a, b) \subseteq G_{\alpha_1}$ for all positive integers $n \geq N$.

We rewrite $K = \{1, \frac{1}{2}, \dots, \frac{1}{N-1}\} \cup \{0, \frac{1}{N}, \frac{1}{N+1}, \dots\}$. Now it follows from the previous paragraph that

$$\left\{0, \frac{1}{N}, \frac{1}{N+1}, \dots\right\} \subseteq G_{\alpha_1}.$$

In addition, since $\{1, \frac{1}{2}, \dots, \frac{1}{N-1}\}$ is a finite set, there are finitely many $G_{\alpha_2}, G_{\alpha_3}, \dots, G_{\alpha_m}$ such that

$$\left\{1, \frac{1}{2}, \dots, \frac{1}{N-1}\right\} \subseteq G_{\alpha_2} \cup G_{\alpha_3} \cup \dots \cup G_{\alpha_m}.$$

Hence we have

$$K \subseteq \bigcup_{i=1}^m G_{\alpha_i},$$

i.e., K is compact by Definition 2.32, completing the proof of the problem. ■

Problem 2.13

Rudin Chapter 2 Exercise 13.

Proof. We define

$$K_0 = \left\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\right\} \quad \text{and} \quad K_n = \left\{\frac{1}{n} + \frac{1}{m} \mid m = n, n+1, \dots\right\},$$

where $n = 1, 2, \dots$. We also define $K = \bigcup_{n=0}^{\infty} K_n$. By definition, K_0 and each K_n have limit point 0 and $\frac{1}{n}$ respectively. Therefore we have $\{0, 1, \frac{1}{2}, \frac{1}{3}, \dots\} \subseteq K'$.

We claim that $K' = K_0$. Let $p \in \mathbb{R}$ be a limit point of K . If $p < 0$, then we define $\delta = \frac{|p|}{2}$ and thus $(p - \delta, p + \delta) \cap K = \emptyset$. If $p > 2$, then we define $\delta = \frac{p-2}{2}$ and thus $(p - \delta, p + \delta) \cap K = \emptyset$. If $1 < p \leq 2$,

then we define $\delta = \frac{1}{2} \min(p-1, 2-p)$ and thus the set $(p-\delta, p+\delta) \cap K$ contain only finitely many points of K . Hence we have shown that

$$K' \subseteq [0, 1].$$

Next, we suppose that $p \in [0, 1] \setminus K_0$. By this assumption, there exists a positive integer k such that $\frac{1}{k+1} < p < \frac{1}{k}$. Since $\frac{1}{n} + \frac{1}{n} \geq \frac{1}{n} + \frac{1}{m}$ and $\frac{1}{n} + \frac{1}{n} \geq \frac{1}{m} + \frac{1}{m}$ for all $m \geq n$, $\frac{2}{k+m}$ is the maximum of the set $K_{k+m} \cup K_{k+m+1} \cup \dots$. Define $\delta = \frac{1}{2} \min(p - \frac{1}{k+1}, \frac{1}{k} - p)$, so $(p-\delta, p+\delta) \subset (\frac{1}{k+1}, \frac{1}{k})$. If $\delta = \frac{1}{2}(\frac{1}{k} - p)$, then we have

$$p - \delta = \frac{3p}{2} - \frac{1}{2k} > \frac{1}{2} \left(\frac{3}{k+1} - \frac{1}{k} \right) = \frac{2k-1}{2(k+1)^2} > \frac{2}{k+m}$$

for all $m > \frac{4(k+1)^2}{2k-1} - k$. In this case, the interval $(p-\delta, p+\delta)$ contains only finitely many points of $K_1 \cup K_2 \cup \dots \cup K_{m-1}$. Similarly, if $\delta = \frac{1}{2}(p - \frac{1}{k+1})$, then we have

$$p - \delta = \frac{p}{2} + \frac{1}{2(k+1)} > \frac{1}{k+1} > \frac{2}{k+m}$$

for all $m > k+2$. In this case, the interval $(p-\delta, p+\delta)$ contains only finitely many points of $K_1 \cup \dots \cup K_{m-1}$. Now both cases show that the interval $(p-\delta, p+\delta)$ can only contain finitely many points of K and thus it is not a limit point of K by Theorem 2.20. Therefore we have $K' = K_0 \subset K$ so that K is closed.

Since $|x| \leq 2$ for all $x \in K$, the set K is a bounded set. Hence it follows from Theorem 2.41 (the Heine-Borel theorem) that K is compact, completing the proof of the problem. ■

Problem 2.14

Rudin Chapter 2 Exercise 14.

Proof. For each $n = 2, 3, \dots$, we consider the interval $G_n = (\frac{1}{n}, 1)$. If $x \in (0, 1)$, then it follows from Theorem 1.20(a) (the Archimedean property) that there exists a positive integer n such that $nx > 1$, i.e., $x \in G_n$. Furthermore, we have

$$(0, 1) \subseteq \bigcup_{i=2}^{\infty} G_n,$$

i.e., $\{G_2, G_3, \dots\}$ is an open cover of the segment $(0, 1)$.

Assume that $\{G_{n_1}, G_{n_2}, \dots, G_{n_k}\}$ was a finite subcover of $(0, 1)$, where $n_1, n_2, \dots, n_k$ are positive integers and $2 \leq n_1 < n_2 < \dots < n_k$. By definition, we have $G_{n_1} \subseteq G_{n_2} \subseteq \dots \subseteq G_{n_k}$ and so

$$(0, 1) \subseteq \bigcup_{i=1}^k G_{n_i} \subseteq G_{n_k},$$

contradicting to the fact that $\frac{1}{2n_k} \in (0, 1)$ but $\frac{1}{2n_k} \notin (\frac{1}{n_k}, 1)$. Hence $\{G_2, G_3, \dots\}$ does not have a finite subcover for $(0, 1)$. This completes the proof of the problem. ■

2.6 Further topological properties of $\mathbb{R}$

Problem 2.15

Rudin Chapter 2 Exercise 15.

Proof. We take the metric space to be the real number line $\mathbb{R}$.

- **Example 1.** Let $E_n = [n, \infty)$, where $n = 1, 2, 3, \dots$. It follows from the corollary of Theorem 2.23 that each E_n is closed in $\mathbb{R}$. Furthermore, if $1 \leq n_1 < n_2 < \dots < n_k$, then we have

$$\bigcap_{i=1}^k E_{n_i} = [n_k, \infty) \neq \emptyset,$$

but $\bigcap_{n=1}^{\infty} E_n = \emptyset$ because if x is a real number such that $x \in \bigcap_{n=1}^{\infty} E_n$, then $x > n$ for all $n \in \mathbb{N}$, a contradiction.

- **Example 2.** Let $F_n = (0, \frac{1}{n})$, where $n = 1, 2, 3, \dots$. It is clear that each F_n is bounded. If $1 \leq n_1 < n_2 < \dots < n_k$, then we have

$$\bigcap_{i=1}^k F_{n_i} = \left(0, \frac{1}{n_k}\right) \neq \emptyset,$$

but $\bigcap_{n=1}^{\infty} F_n = \emptyset$ because if x is a real number such that $x \in \bigcap_{n=1}^{\infty} F_n$, then $x < \frac{1}{n}$ for all $n \in \mathbb{N}$, a contradiction.

We note that the above examples satisfy the condition that $E_{n+1} \subseteq E_n$ and $F_{n+1} \subseteq F_n$ for $n = 1, 2, \dots$. Hence they show that Theorem 2.36 and its corollary become false if the word “compact” is replaced by “closed” or by “bounded”. This completes the proof of the problem. ■

Problem 2.16

Rudin Chapter 2 Exercise 16.

Proof. We have $E = \{p \in \mathbb{Q} \mid 2 < p^2 < 3\} = (-\sqrt{3}, -\sqrt{2}) \cup (\sqrt{2}, \sqrt{3})$. It is easy to check the boundedness of E from Definition 2.18(i) because we have $0 \in \mathbb{Q}$ and $|p| < 10$ for all $p \in E$. Let $p \in \mathbb{Q}$ be a limit point of E . Thus for every $r > 0$, there exists $q \in E$ and $q \neq p$ such that $|p - q| < r$. In particular, if we take $r = \frac{1}{n}$, where n is a large positive integer, then we have

$$0 < q - \frac{1}{n} < p < q + \frac{1}{n}.$$

Since $2 < q^2 < 3$, we have $2 < q^2 - \delta$ and $q^2 + \delta < 3$ for some $\delta > 0$. Now we choose $n > \max\{\frac{3q}{\delta}, \frac{1}{q}\}$. We can deduce from this that

$$p^2 < \left(q + \frac{1}{n}\right)^2 = q^2 + \frac{2q}{n} + \frac{1}{n^2} < q^2 + \frac{3q}{n} < q^2 + \delta < 3 \quad (2.3)$$

and

$$p^2 > \left(q - \frac{1}{n}\right)^2 = q^2 - \frac{2q}{n} + \frac{1}{n^2} > q^2 - \frac{2q}{n} > q^2 - \delta > 2. \quad (2.4)$$

Thus the inequalities (2.5) and (2.6) imply that $p \in E$ and hence E is closed by Definition 2.18(d).

Assume that E was compact in $\mathbb{Q}$. For each $n = 1, 2, \dots$, we let the sets $V_n = \{p \in \mathbb{Q} \mid 2 - \frac{1}{n} < p^2 < 3\}$, $V_n^- = \left(-\sqrt{3}, -\sqrt{2 - \frac{1}{n}}\right)$ and $V_n^+ = \left(\sqrt{2 - \frac{1}{n}}, \sqrt{3}\right)$. Then it is easy to see that

$$V_n = V_n^- \cup V_n^+. \quad (2.5)$$

If $p \in E$, then either $-\sqrt{3} < p < -\sqrt{2}$ or $\sqrt{2} < p < \sqrt{3}$ which implies that either

$$-\sqrt{3} < p < -\sqrt{2 - \frac{1}{n}} \quad \text{or} \quad \sqrt{2 - \frac{1}{n}} < p < \sqrt{3}$$

for some $n \in \mathbb{N}$. Thus we have $p \in V_n^- \cup V_n^+ = V_n$ for some $n \in \mathbb{N}$ so that

$$E \subseteq \bigcup_{n=1}^{\infty} V_n. \quad (2.6)$$

Let x_- and x_+ be the mid-points of the intervals $\left(-\sqrt{3}, -\sqrt{2 - \frac{1}{n}}\right)$ and $\left(\sqrt{2 - \frac{1}{n}}, \sqrt{3}\right)$ respectively. If $q \in V_n$, then we have either $q \in V_n^-$ or $q \in V_n^+$. If $q \in V_n^-$, then there exists a positive real number $h > 0$

such that $d(x_-, q) = r - h$, where $r = \left| \sqrt{3} - \sqrt{2 - \frac{1}{n}} \right|$. For all s such that $d(q, s) < h$, we have from Definition 2.15(c) that

$$d(x_-, s) \leq d(x_-, q) + d(q, s) < r - h + h = r$$

so that $s \in V_n^-$, i.e., $N_h(q) \subseteq V_n^-$. By similar argument, we also have $N_h(q) \subseteq V_n^+$ if $q \in V_n^+$. Hence it follows from the definition (2.5) that q is an interior point of V_n (i.e., V_n is open in $\mathbb{Q}$) and then the relation (2.6) implies that $\{V_n\}$ is an open cover of E . Since E is compact, we have

$$E \subseteq V_{n_1} \cup V_{n_2} \cup \cdots \cup V_{n_k},$$

where $n_1, n_2, \dots, n_k$ are positive integers and $n_1 < n_2 < \cdots < n_k$. Since $V_{n+1} \subseteq V_n$ for all positive integers n , we have $E \subseteq V_{n_1}$, a contradiction.

The argument of proving the openness of the set V_n in the previous paragraph can also be applied to show that E is open in $\mathbb{Q}$. Hence, this completes the proof of the problem. ■

Problem 2.17

Rudin Chapter 2 Exercise 17.

Proof. By definition, the element of E must be in the representation $0.x_1x_2x_3\dots$, where $x_n \in \{4, 7\}$ for all $n = 1, 2, \dots$. We answer the questions one by one:

- **Is E countable?** By a similar argument as the proof of Theorem 2.14, we can show that E is uncountable.
- **Is E dense in $[0, 1]$?** Let $x = 0.1$ and $y = 0.2$. By definition, we have $x < y < z$ for all $z \in E$. Hence E is not dense in $[0, 1]$.
- **Is E compact?** Since $0.4 \leq x < 0.8$ and $[0, 1]$ is bounded, E is bounded. Let x be a limit point of E . It is clear that $0.4 < x < 0.8$. Assume that $x \notin E$. If we let x_n be the first digit in x which is not 4 or 7 and

$$h = 0.\underbrace{00\dots 0}_n 1,$$

then it is not difficult to see that the n th or $(n+1)$ th digit of every number in the interval $(x-h, x+h)$ is not 4 or 7. This implies that $(x-h, x+h) \cap E = \emptyset$ and x is not a limit point of E , a contradiction. Hence we have $x \in E$ so that E is closed and then compact by Theorem 2.41 (the Heine-Borel theorem).

- **Is E perfect?** We know that E is closed, i.e., every limit point of E is a point of E . Let $x = 0.x_1x_2x_3\dots$ and $N_h(x)$ be an element of E and a neighborhood of E respectively, where $h > 0$. Let $0 < h < 1$, i.e., $h = 0.h_1h_2\dots$ ^b Let h_n be the first non-zero digit in h . Let $y = 0.y_1y_2\dots$ be the number such that if the $(n+1)$ th digit of x is 4 (resp. 7), then the $(n+1)$ th digit of y is 7 (resp. 4) and all other digits of y are kept the same as those of x . It follows from this definition that $y \neq x$ and

$$|x - y| = 0.\underbrace{0\dots 0}_n 3 < h$$

so that $y \in N_h(x)$. It is obvious that $y \in E$ and so x is a limit point of E . Hence E is perfect by Definition 2.18(h).

This completes the proof of the problem. ■

Problem 2.18

Rudin Chapter 2 Exercise 18.

^bWe note that the argument of proving the result when $0 < h < 1$ can be applied completely to prove the case when $h \geq 1$.

Proof. By Problem 2.17, we know that E is a nonempty perfect set. However, E may contain a rational number, e.g. $\frac{47}{99} = 0.47474747 \dots \in E$. In fact, we have the following result about $\mathbb{Q}$:

Lemma 2.2

Every rational number in $(0, 1)$ has a terminating or repeating decimal expansion.

Proof of Lemma 2.2. Let $\frac{p}{q}$ be a rational number, where p and q are positive integers and $p < q$. To find the first decimal digit of $\frac{p}{q}$, we consider $10p \div q$ and the division algorithm (i.e., given two integers a and b , with $b \neq 0$, there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < |b|$) gives $10p = q_1q + p_1$, where q_1 and p_1 are unique integers such that $0 \leq p_1 < q$. To find the second decimal digit of $\frac{p}{q}$, we consider $10p_1 \div q$ and the division algorithm again gives $10p_1 = q_2q + p_2$, where q_2 and p_2 are unique integers such that $0 \leq p_2 < q$. We repeat this process to get the sequence of integers $\{p_1, p_2, \dots\}$ such that $0 \leq p_1, p_2, \dots < q$. Since there are only finite number of integers between 0 and q , this sequence will terminate or repeat. ■

Now we can go back to the proof of the problem. We let $F = a + E = \{a + x \mid x \in E\}$, where $a = 0.1010010001 \dots$. Since F is just a translation of the perfect set E , F is also perfect. The chosen constant a ensures that elements in F will not have terminating or repeating decimal expansion and thus are irrational by Lemma 2.2. Hence F is the desired set. This completes the proof of the problem. ■

2.7 Properties of connected sets

Problem 2.19

Rudin Chapter 2 Exercise 19.

Proof.

- (a) Since A and B are closed, we have $\overline{A} = A$ and $\overline{B} = B$ by Theorem 2.27(b). Since A and B are disjoint, we always have $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Hence they are separated.
- (b) Let A and B be disjoint open sets. Without loss of generality, we may assume that $\overline{A} \cap B \neq \emptyset$. Let $x \in \overline{A} \cap B$. Since $A \cap B = \emptyset$ and $\overline{A} = A \cup A'$, we must have $x \in A'$. Since $x \in B$ and B is open, there is a neighborhood N of x such that $N \subseteq B$. Since x is a limit point of A , there exists $y \in N$ and $y \neq x$ such that $y \in A$. Hence $y \in A \cap B$, a contradiction.
- (c) Fix $p \in X$ and $\delta > 0$. By definition, we have $A = \{q \in X \mid d(p, q) < \delta\}$ and $B = \{q \in X \mid d(p, q) > \delta\}$ which are obviously disjoint open sets by Theorem 2.19. By the result of part (b), A and B are separated.
- (d) Let X be a connected metric space with at least two (distinct) points. Let the two points be a and b . Assume that X was countable, i.e., $X = \{a, b, x_3, x_4, \dots\}$. We define $r = d(a, b)$. Since a and b are distinct, it follows from Definition 2.15(a) that $r > 0$. Since $[0, r] = (0, r) \cup \{0, r\}$, $(0, r)$ must be uncountable by the corollary of Theorem 2.43. Therefore there exists $\delta \in (0, r)$ such that

$$d(a, x) \neq \delta$$

for all $x \in X$. Now if we define $A = \{x \in X \mid d(a, x) < \delta\}$ and $B = \{x \in X \mid d(a, x) > \delta\}$, then they are nonempty ($a \in A$ and $b \in B$) and the result of (c) shows that A and B are separated. However,

$$X = A \cup B$$

so that X is not connected by Definition 2.45. This contradiction proves the problem. We complete the proof of the problem. ■

Problem 2.20*Rudin Chapter 2 Exercise 20.*

Proof. Let E be a connected set. Assume that $\overline{E} = A \cup B$, where $\overline{A} \cap B = A \cap \overline{B} = \emptyset$. Since $A \cap E \subseteq A$ and $B \cap E \subseteq B$, we have $\overline{A \cap E} \subseteq \overline{A}$ and $\overline{B \cap E} \subseteq \overline{B}$. Since $A \subseteq \overline{A}$, we have $A \cap B \subseteq \overline{A} \cap B = \emptyset$. Since $E \subseteq \overline{E} = A \cup B$, we have

$$E = E \cap \overline{E} = E \cap (A \cup B) = (E \cap A) \cup (E \cap B).$$

Since $(\overline{A \cap E}) \cap (B \cap E) \subseteq \overline{A} \cap B = \emptyset$ and $(\overline{B \cap E}) \cap (A \cap E) \subseteq \overline{B} \cap A = \emptyset$, we have E is disconnected which is a contradiction. Hence $\overline{E}$ must be connected.

However, the interior of a connected set may not be connected. We prove the following lemma first:

Lemma 2.3

Suppose that A and B are connected sets and $A \cap B \neq \emptyset$. Then the set $E = A \cup B$ is also connected.

Proof of Lemma 2.3. Assume that E was not connected. By definition, there are separated sets U and V such that $U \cup V = E = A \cup B$. Let $U_A = U \cap A$ and $V_A = V \cap A$. Since U and V are separated, we have

$$\overline{U_A} \cap V_A \subseteq (\overline{U} \cap \overline{A}) \cap (V \cap A) = (\overline{U} \cap V) \cap (\overline{A} \cap A) = \emptyset \cap A = \emptyset$$

and similarly $U_A \cap \overline{V_A} = \emptyset$. Since $U_A \cup V_A = A$, A is not connected which contradicts the hypothesis. Therefore we have either

$$U_A = \emptyset \quad \text{or} \quad V_A = \emptyset. \quad (2.7)$$

Similarly, the sets $U_B = U \cap B$ and $V_B = V \cap B$ will imply that B is not connected, a contradiction again. Therefore we have either

$$U_B = \emptyset \quad \text{or} \quad V_B = \emptyset. \quad (2.8)$$

Now we verify that the cases (2.7) and (2.8) will induce a contradiction. For examples, if $U_A = \emptyset$ and $U_B = \emptyset$, then $U \cap A = \emptyset$ and $U \cap B = \emptyset$ but they imply that

$$U = U \cap E = U \cap (A \cup B) = (U \cap A) \cup (U \cap B) = \emptyset$$

which contradicts $U \neq \emptyset$; if $U_A = \emptyset$ and $V_B = \emptyset$, then $A \cap U = \emptyset$ and $B \cap V = \emptyset$ which give $(A \cap B) \cap U = \emptyset$ and $(A \cap B) \cap V = \emptyset$ respectively. Thus these imply that

$$\emptyset = [(A \cap B) \cap U] \cup [(A \cap B) \cap V] = (A \cap B) \cap (U \cup V) = (A \cap B) \cap E = A \cap B$$

which contradicts the fact that $A \cap B \neq \emptyset$. Other cases can be done similarly, so we have the desired result that E is connected. ■

Let's go back to the proof of the problem. For example, we consider the disks

$$A = \{\mathbf{x} \in \mathbb{R}^2 \mid |\mathbf{x}| \leq 1\} \quad \text{and} \quad B = \{\mathbf{x} \in \mathbb{R}^2 \mid |\mathbf{x} - (2, 0)| \leq 1\}.$$

Define $E = A \cup B$. By Definition 2.17, both A and B are convex so that we obtain from Problem 2.21(c) that A and B are connected. It is clear that $\mathbf{x} = (1, 0)$ is the (only) common point of A and B . Hence it follows from Lemma 2.3 that E is connected. However, it is easy to check from the definition that

$$E^\circ = A^\circ \cup B^\circ,$$

where $A^\circ = \{\mathbf{x} \in \mathbb{R}^2 \mid |\mathbf{x}| < 1\}$ and $B^\circ = \{\mathbf{x} \in \mathbb{R}^2 \mid |\mathbf{x} - 2| < 1\}$. Since A° and B° are clearly separated, E° is not connected. We end the proof of the problem. ■

Problem 2.21*Rudin Chapter 2 Exercise 21.**Proof.*

- (a) By definition, we have $A_0 = \{t \in \mathbb{R} \mid \mathbf{p}(t) \in A\}$ and $B_0 = \{t \in \mathbb{R} \mid \mathbf{p}(t) \in B\}$. If $t \in \overline{A_0} \cap B_0$, then $t \in \overline{A_0}$ and $t \in B_0$. Since $t \in B_0$, we have $\mathbf{p}(t) \in B$. Since $t \in \overline{A_0}$, we have $t \in A_0$ or $t \in A'_0$. If $t \in A_0$, then $\mathbf{p}(t) \in A$. Thus $\mathbf{p}(t) \in A \cap B \subseteq \overline{A} \cap B$ and then A and B are not separated, a contradiction.

Suppose that $t \in A'_0$. Then t is a limit point of A_0 . Therefore we have $s \in N_h(t)$ and $s \neq t$ such that $s \in A_0$ for every $h > 0$. Since $s \in N_h(t)$, we have $|s - t| < h$ and thus

$$|\mathbf{p}(s) - \mathbf{p}(t)| = |(-s + t)\mathbf{a} + (s - t)\mathbf{b}| \leq |s - t|(|\mathbf{a}| + |\mathbf{b}|) < h(|\mathbf{a}| + |\mathbf{b}|).$$

In fact, this implies that $\mathbf{p}(N_h(t)) = N_{h(|\mathbf{a}|+|\mathbf{b}|)}(\mathbf{p}(t))$ so that

$$\mathbf{p}(s) \in N_{h(|\mathbf{a}|+|\mathbf{b}|)}(\mathbf{p}(t)).$$

If $\mathbf{p}(s) = \mathbf{p}(t)$, then we have $\mathbf{a} = \mathbf{b}$ which contradicts the fact that A and B are separated. Thus we have $\mathbf{p}(s) \neq \mathbf{p}(t)$. Recall that $s \in A_0$ so that $\mathbf{p}(s) \in A$. Hence we have actually shown that $\mathbf{p}(t)$ is a limit point of A , i.e., $\mathbf{p}(t) \in \overline{A}$ which gives $\mathbf{p}(t) \in \overline{A} \cap B$ and then A and B are not separated, a contradiction.

- (b) Assume that $\mathbf{p}(t) \in A \cup B$ for all $t \in (0, 1)$. Since $\mathbf{p}(0) = \mathbf{a}$ and $\mathbf{p}(1) = \mathbf{b}$, we have $\mathbf{p}(t) \in A \cup B$ for all $t \in [0, 1]$ and thus $[0, 1] \subseteq A_0 \cup B_0$ by definition. Let $C_0 = [0, 1] \cap A_0$ and $D_0 = [0, 1] \cap B_0$. Then C_0 and D_0 are nonempty because $\mathbf{p}(0) = \mathbf{a} \in A$ and $\mathbf{p}(1) = \mathbf{b} \in B$. Furthermore, since A_0 and B_0 are separated, we have

$$\overline{C_0} \cap D_0 = \emptyset \quad \text{and} \quad C_0 \cap \overline{D_0} = \emptyset.$$

Therefore $[0, 1]$ is not connected which contradicts Theorem 2.47. Hence we have $\mathbf{p}(t_0) \notin A \cup B$ for some $t_0 \in (0, 1)$.

- (c) By Definition 2.17, a set $S \subseteq \mathbb{R}^k$ is said to be convex if, for all $\mathbf{x}, \mathbf{y} \in S$ and all $t \in (0, 1)$, we have $(1 - t)\mathbf{x} + t\mathbf{y} \in S$, see Figure 2.2.

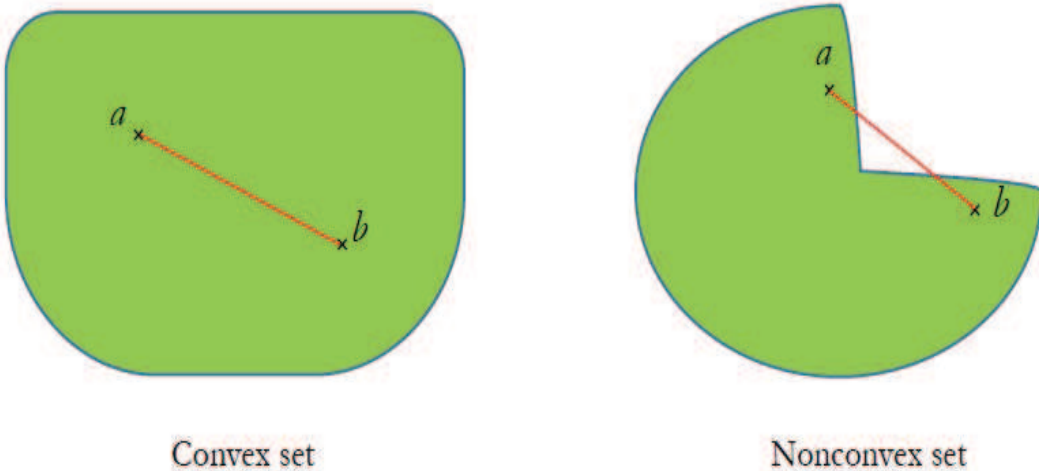


Figure 2.2: Convex sets and nonconvex sets.

Assume that S was not connected. Then we have $S = A \cup B$, where A and B are separated. Pick $\mathbf{a} \in A$ and $\mathbf{b} \in B$. By part (b), there exists $t_0 \in (0, 1)$ such that $\mathbf{p}(t_0) \notin A \cup B = S$, so S is not convex which is contrary to our hypothesis. Hence S must be connected.

This completes the proof of the problem. ■

2.8 Separable metric spaces and bases and a special case of Baire's theorem

Problem 2.22

Rudin Chapter 2 Exercise 22.

Proof. Let $\mathbb{Q}^k = \{(q_1, q_2, \dots, q_k) \mid q_1, q_2, \dots, q_k \in \mathbb{Q}\}$. Since $\mathbb{Q}$ is countable, Theorem 2.13 shows that $\mathbb{Q}^k$ is countable. To prove that $\mathbb{R}^k$ is separable, we must show that every nonempty open subset of $\mathbb{R}^k$ contains at least one element in $\mathbb{Q}^k$.^c Let S be an open set of $\mathbb{R}^k$ and $\mathbf{p} = (p_1, p_2, \dots, p_k) \in S$. By Definition 2.18(f), there exists $h > 0$ such that $B(\mathbf{p}, h) \subseteq S$. By definition, $B(\mathbf{p}, h) = \{\mathbf{x} \in \mathbb{R}^k \mid d(\mathbf{x}, \mathbf{p}) < h\}$. If we take $\mathbf{x} = (p_1 - \frac{h}{2k}, p_2 - \frac{h}{2k}, \dots, p_k - \frac{h}{2k})$, then we have

$$d(\mathbf{x}, \mathbf{p}) = \sum_{i=1}^k \left| p_i - \frac{h}{2k} - p_i \right| = \frac{h}{2} < h$$

so that $\mathbf{x} \in B(\mathbf{p}, h)$. Since $p_i - \frac{h}{2k} < p_i$, where $i = 1, 2, \dots, k$, Theorem 1.20(b) shows that there exist $q_i \in \mathbb{Q}$ such that $p_i - \frac{h}{2k} < q_i < p_i$, where $i = 1, 2, \dots, k$. Let $\mathbf{q} = (q_1, q_2, \dots, q_k)$. By Definition 2.15(c), we have

$$d(\mathbf{q}, \mathbf{p}) \leq d(\mathbf{q}, \mathbf{x}) + d(\mathbf{x}, \mathbf{p}) < \frac{h}{2} + \frac{h}{2} = h.$$

Hence we have $\mathbf{q} \in B(\mathbf{p}, h) \subseteq S$ and $\mathbb{Q}^k$ is dense in $\mathbb{R}^k$. This ends the proof of the problem. ■

Problem 2.23

Rudin Chapter 2 Exercise 23.

Proof. Let X be a separable metric space. Since X is separable, it contains a countable dense subset. Let $C = \{x_1, x_2, \dots\}$ be a countable dense subset of X . Let $\mathcal{B} = \{N_{q_m}(x_k)\}$ be a collection of subsets of X , where $q_m \in \mathbb{Q}^+$ and $x_k \in C$. (Here $\mathbb{Q}^+$ is the set of all positive rational numbers.) By Theorem 2.19, each $N_{q_m}(x_k)$ is open in X . Since C and $\mathbb{Q}^+$ are countable, the set $\mathcal{B}$ is also countable by Theorem 2.13.

Suppose that $x \in X$ and G is an open subset of X containing x . Since G is open, x is an interior point of G and thus we have $N_h(x) \subseteq G$ for some $h > 0$, see the **black** dotted circle in Figure 2.3.

Since C is dense in X , there exists $x_k \in C$ such that $x_k \in N_{\frac{h}{2}}(x)$, see the **red** dotted circle in Figure 2.3. It is clear from Theorem 1.20(b) that $d(x, x_k) < q_m < \frac{h}{2}$ for some $q_m \in \mathbb{Q}^+$ so that $x \in N_{q_m}(x_k)$. Since $d(x, x_k) < q_m < \frac{h}{2}$, for every $y \in N_{q_m}(x_k)$, we have

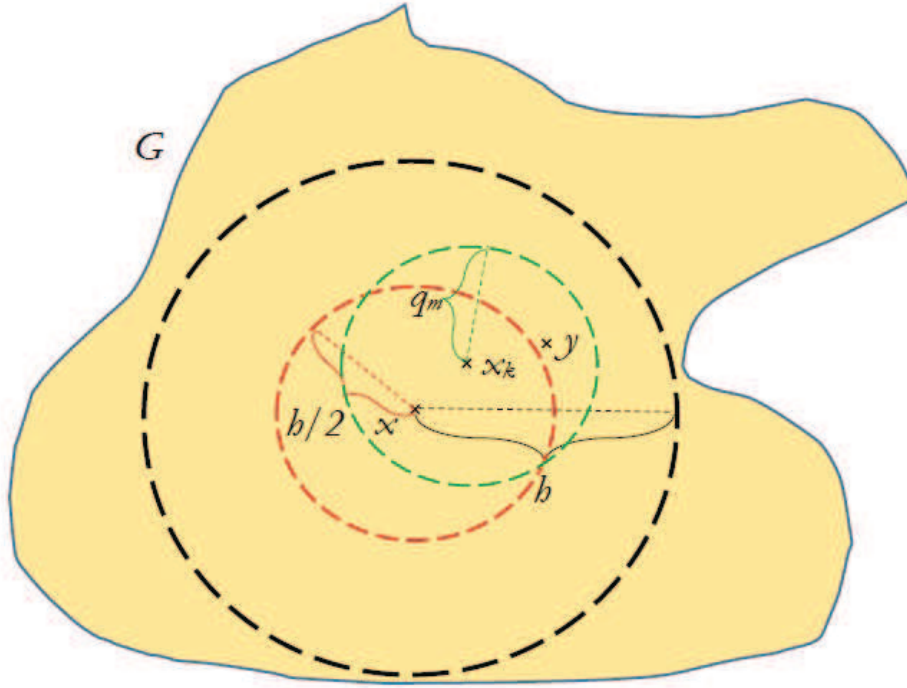
$$d(y, x) \leq d(y, x_k) + d(x_k, x) < q_m + q_m < \frac{h}{2} + \frac{h}{2} = h$$

so that $y \in N_h(x)$, see the **green** dotted circle in Figure 2.3. Hence we have

$$x \in N_{q_m}(x_k) \subseteq N_h(x) \subseteq G,$$

i.e., $\mathcal{B}$ is a countable base for X . This completes the proof of the problem. ■

^cThat is, every point of $\mathbb{R}^k$ is a limit point of $\mathbb{Q}^k$. Since $\mathbb{Q}^k \subseteq \mathbb{R}^k$ and $\mathbb{R}^k$ is closed in $\mathbb{R}^k$, it follows from Theorem 2.27(c) that $\overline{\mathbb{Q}^k} \subseteq \mathbb{R}^k$. By Definition 2.26, $(\mathbb{Q}^k)'$ is the set of all limit points of $\mathbb{Q}^k$. Since every point of $\mathbb{R}^k$ is a limit point of $\mathbb{Q}^k$, we have $\mathbb{R}^k \subseteq (\mathbb{Q}^k)' \subseteq \overline{\mathbb{Q}^k}$. Hence we have $\mathbb{R}^k = \overline{\mathbb{Q}^k}$.

Figure 2.3: The sets $N_h(x)$, $N_{\frac{h}{2}}(x)$ and $N_{q_m}(x_k)$.**Problem 2.24**

Rudin Chapter 2 Exercise 24.

Proof. By definition (see Problem 2.22), we have to prove that X has a countable dense subset. Fix $\delta > 0$, since $X \neq \emptyset$, we can pick $x_1 \in X$. Then we can choose $x_2 \in X$ such that $d(x_1, x_2) \geq \delta$. Otherwise, we have $d(x_1, x) < \delta$ for all $x \in X$ which means that

$$X \subseteq N_\delta(x_1).$$

In this case, we replace δ by $\frac{\delta}{2}$ and choose $x_2 \in X$ such that $d(x_1, x_2) \geq \frac{\delta}{2}$. Otherwise, we have $X \subseteq N_{\frac{\delta}{2}}(x_1)$. Then the process can be repeated but it must stop after a finite number of steps because

$$X \subseteq N_{\frac{\delta}{2^n}}(x_1)$$

for all nonnegative integers n . However, this implies that $X = \emptyset$, a contradiction.

Let j be a positive integer. Having chosen $x_1, \dots, x_j \in X$, choose $x_{j+1} \in X$, if possible, so that $d(x_i, x_{j+1}) \geq \delta$ for $i = 1, \dots, j$. In other words, $x_1, x_2, \dots, x_{j+1}$ are *distinct* elements. This process must stop after a finite number of steps. Otherwise, we have the infinite subset $E = \{x_1, x_2, \dots\}$ of X and the hypothesis guarantees that E has a limit point $x \in X$. By Theorem 2.20, every neighborhood of x contains infinitely many points of E . Therefore we have $x_j, x_{j+1}, \dots \in N_{\frac{\delta}{2}}(x)$ for some positive integer j so that $d(x_i, x) < \frac{\delta}{2}$ for all $i = j, j+1, \dots$. By this, we have

$$d(x_j, x_{j+1}) \leq d(x_j, x) + d(x, x_{j+1}) < \frac{\delta}{2} + \frac{\delta}{2} = \delta,$$

a contradiction.

Therefore X can be covered by finitely many neighborhoods of radius δ . For each $n = 1, 2, \dots$, we take $\delta = \frac{1}{n}$ and consider the set $E_n = \{x_{n1}, x_{n2}, \dots, x_{nm_n}\}$, where $x_{n1}, x_{n2}, \dots, x_{nm_n}$ are the centers of such neighborhoods for some positive integer m_n , i.e.,

$$X \subseteq N_{\frac{1}{n}}(x_{n1}) \cup N_{\frac{1}{n}}(x_{n2}) \cup \dots \cup N_{\frac{1}{n}}(x_{nm_n}). \quad (2.9)$$

Since the set E_n is at most countable (in fact, it is finite) for each n , the corollary of Theorem 2.12 shows that

$$E = \bigcup_{n=1}^{\infty} E_n$$

is countable.

We claim that E is a countable dense subset of the metric space X . In other words, we have to verify that every nonempty open subset V of X contains at least one element of E . Let $x \in V \subseteq X$. Since x is an interior point of V , we have $N_h(x) \subseteq V$ for some $h > 0$. By Theorem 1.20(a) (the Archimedean property), there exists a positive integer n such that $\frac{1}{n} < h$. Fix this n and we follow from the relation (2.9) that

$$x \in N_{\frac{1}{n}}(x_{nk})$$

for some $k \in \{1, 2, \dots, m_n\}$. By definition, we have $d(x, x_{nk}) < \frac{1}{n} < h$ so that

$$x_{nk} \in N_h(x) \subseteq V.$$

Hence the claim is true, i.e., E is a countable dense subset of X . This completes the proof of the problem. ■

Problem 2.25

Rudin Chapter 2 Exercise 25.

Proof. Since K is compact, it follows from Theorem 2.37 that every infinite subset E of K has a limit point in K . Since K is a metric space, we have K is separable by Problem 2.24. Hence we deduce from Problem 2.23 that K has a countable base, completing the proof of the problem. ■

Problem 2.26

Rudin Chapter 2 Exercise 26.

Proof. By Problem 2.24, X is separable. Then by Problem 2.23, it has a countable base. It follows that every open cover of X has a *countable* subcover $\{G_n\}, n = 1, 2, 3, \dots$. If no finite subcollection of $\{G_n\}$ covers X , then the complement

$$F_n = X \setminus (G_1 \cup G_2 \cup \dots \cup G_n)$$

is nonempty for each n . However, $\bigcap F_n$ is an empty set. Otherwise, $x \in \bigcap F_n$ implies that $x \in F_n$ for every n and then

$$x \notin G_1 \cup G_2 \cup \dots \cup G_n$$

for each n . This contradicts the fact that $\{G_n\}$ is an open cover of X .

Let $E = \{x_1, x_2, \dots\}$, where $x_i \in F_i$ and $i = 1, 2, \dots$. We note that $F_{n+1} \subseteq F_n$, so it may happen that $x_{n+1} = x_n$. Assume that E was finite, i.e., $E = \{x_{n_1}, \dots, x_{n_k}\}$. Since $\{G_n\}$ is an open cover of X , we must have $x_{n_i} \in G_{m_i}$, where $m_i \in \mathbb{N}$ for $i = 1, 2, \dots, k$.^d Let $m = \max(m_1, m_2, \dots, m_k)$. Since $x_m \in F_m$ by definition of E , we have

$$x_m \notin G_{m_1} \cup G_{m_2} \cup \dots \cup G_{m_k}. \quad (2.10)$$

However, since $x_m = x_{n_i}$ for some $i = 1, 2, \dots, k$, we must have $x_m \in G_{m_i}$ which contradicts the hypothesis (2.10). Therefore E must be an infinite set and it follows from the hypothesis that E has a limit point x in X .

Again we recall that X is covered by $G_1, G_2, \dots$ so that $x \in G_n$ for some n . Since G_n is open, there is a $\delta > 0$ such that $N_\delta(x) \subseteq G_n$. Let m be a positive integer greater than n . If $x_m \in N_\delta(x)$, then we have $x_m \in G_n$ and so

$$x_m \in G_1 \cup \dots \cup G_n \cup \dots \cup G_m$$

which implies that $x_m \notin F_m$, a contradiction. Therefore x is not a limit point of E which is a contradiction. Hence we have the desired result that X is compact. This completes the proof of the problem. ■

^dWe don't require that $n_i = m_i$ in general.

Problem 2.27*Rudin Chapter 2 Exercise 27.*

Proof. Suppose that $E \subseteq \mathbb{R}^k$, E is uncountable, and let P be the set of all condensation points of E .

- **At most countably many points of E are not in P .** This statement is equivalent to the statement that

$$P^c \cap E \text{ is at most countable.}^e$$

By Problem 2.22, we know that $\mathbb{R}^k$ is separable and then Problem 2.23 implies that it has a countable base. Let $\{V_n\}$ be a countable base of $\mathbb{R}^k$, let W be the union of those V_n for which $E \cap V_n$ is at most countable.

Let $p \in P$. Assume that $p \in W$. Since W is open by Theorem 2.24(a), we have $p \in V_k \subseteq W$ for some positive integer k because $\{V_n\}$ is a base of X (see Problem 2.23). Since V_k is open, we have $N_h(p) \subseteq V_k$ for some $h > 0$. By definition of W , $E \cap V_k$ is at most countable and so is $E \cap N_h(p)$, but this contradicts the fact that $N_h(p) \cap E$ is uncountable. Hence we have $p \in W^c$ so that

$$P \subseteq W^c.$$

If $p \in W^c$, then $p \notin W$. Assume that $N_h(p)$ was a neighborhood of p such that $N_h(p) \cap E$ has at most countably many points for some $h > 0$. Since $\{V_n\}$ is a base of X and $N_h(p)$ is open, there exists V_k such that $p \in V_k \subseteq N_h(p)$ for some positive integer k . Since $N_h(p) \cap E$ is at most countable, $V_k \cap E$ is at most countable too and this implies that

$$p \in V_k \subseteq W,$$

a contradiction. Hence $N_h(p) \cap E$ must be uncountable for all $h > 0$ so that $p \in P$, i.e., $W^c \subseteq P$.

Now the above two paragraphs show our desired result that

$$P = W^c. \quad (2.11)$$

- **P is perfect.** Since W is open, $P = W^c$ is closed by the relation (2.11) and Theorem 2.23.^e Next, we let $x \in P$ and $N_h(x)$ be a neighborhood of x , where $h > 0$. Assume that $N_h(x) \cap P = \{x\}$. This means that if $y \in N_h(x) \setminus \{x\}$, then $y \in P^c$ and it is easy to get from the relation (2.11) that $y \in W$. Thus we have $N_h(x) \setminus \{x\} \subseteq W$. Since $N_h(x) = (N_h(x) \setminus \{x\}) \cup \{x\}$, we have $N_h(x) \subseteq W \cup \{x\}$ and so

$$N_h(x) \cap E \subseteq (W \cup \{x\}) \cap E \subseteq (W \cap E) \cup \{x\}. \quad (2.12)$$

By definition, W is the union of those V_n for which $E \cap V_n$ is at most countable. By the corollary of Theorem 2.12, we have $W \cap E$ is at most countable and then (2.12) implies that $N_h(x) \cap E$ is at most countable too. This implies that $x \notin P$, a contradiction. Hence there exists $y \in N_h(x)$ and $y \neq x$ such that $y \in P$ for every $h > 0$. In other words, x is a limit point of P and we obtain from Definition 2.18(h) that P is perfect.

This completes the proof of the problem. ■

Problem 2.28*Rudin Chapter 2 Exercise 28.*

^eWe can prove that P is closed directly without using the relation (2.11): Let p be a limit point of P . Then for every $r > 0$, the neighborhood $N_r(p)$ of p contains a point $q \neq p$ such that $q \in P$. By definition, q is a condensation point of E , so the neighborhood $N_h(q)$ contains uncountably many points of E , where $h = \frac{1}{2} \min(d(p, q), r - d(p, q))$. Since we have $N_h(q) \subset N_r(p)$ (see Figure 2.1 for clarification), p is also a condensation point of E . That is $p \in P$ and so P is closed.

Proof. Let E be a closed set in a separable metric space X . If E is at most countable, then we have $E = \emptyset \cup E$. By Theorem 2.23, the empty set is closed. Since $\emptyset$ contains no point, it contains no isolated point. Thus the empty set must be perfect and we are done in this case.

Suppose that E is uncountable. If we read the proof of Problem 2.27 in detail, then it can be seen that the proof *does not* depend on the metric space in which P is embedded. In fact, what the proof requires is that the space X has a countable base and this is automatically satisfied because of Problem 2.23. Therefore the set P of all condensation points of E is perfect. By definition, a condensation point of E must be a limit point of E so that $P \subseteq E'$. Since E is closed, we have

$$P \subseteq E$$

by Theorem 2.27(b). Let $F = E \setminus P$. By Problem 2.27 again, the set F must be at most countably many points. Hence we have

$$E = P \cup F$$

and this completes the proof of the problem. ■

Problem 2.29

Rudin Chapter 2 Exercise 29.

Proof. Let G be an open set in $\mathbb{R}$ and $x \in G$. We shall construct the segments with the required properties. The idea of the construction is that for each $x \in G$, we have to find the *maximal* segment containing x and show that such segment is a subset of G . Next, we have to show that two maximal segments are disjoint. The construction is divided into several steps.

- **Step 1:** Let $E_x = \{y \in \mathbb{R} \mid (x, y) \subseteq G\}$ and $F_x = \{y \in \mathbb{R} \mid (y, x) \subseteq G\}$. Since G is open, we have $x \in (x - h, x + h) \subseteq G$ for some $h > 0$ so that $x + h \in E_x$, $x - h \in F_x$ and thus E_x and F_x are nonempty. We define $a_x = \inf F_x$, $b_x = \sup E_x$ and the segment

$$I_x = (a_x, b_x).^f \quad (2.13)$$

- **Step 2:** We show that $I_x \subseteq G$. By definition, we have $x \in I_x$. Let $p \in I_x \setminus \{x\}$. Then we have either $a_x < p < x$ or $x < p < b_x$. Suppose that $a_x < p < x$. Then it follows from Theorem 1.20(b) (or by Problem 2.22) that

$$a_x < q < p$$

for some $q \in \mathbb{Q}$. Since a_x is the greatest lower bound of F_x , we must have $y < q$ for some $y \in F_x$ so that

$$(q, x) \subset (y, x) \subseteq G. \quad (2.14)$$

Since $q < p < x$, we obtain from the relation (2.14) that $p \in G$ so that $(a_x, x) \subseteq G$. The other side is similar and we omit the details here. By these, we have proven **Step 2** that $I_x \subseteq G$. For each $x \in G$, we have $x \in I_x \subseteq G$ which implies that

$$G = \bigcup_{x \in G} I_x.$$

- **Step 3:** The segment defined in (2.13) is the *maximal* segment containing x . To show this property, we must have

$$a_x \notin G.$$

Otherwise, we have $(a_x - h, a_x + h) \subseteq G$ for some $h > 0$ as G is an open set. This implies that $(a_x - h, x) \subseteq G$ so that $a_x - h \in F_x$ and then we have $a_x < a_x - h$, but a_x is the greatest lower bound of F_x . This is clearly a contradiction. Hence we have $a_x \notin G$. Similarly, we must have $b_x \notin G$. Now if I_x was not the maximal segment containing x , then we have

$$I_x \subseteq (a, b) \subseteq G,$$

where $a < a_x$ or $b_x < b$. These imply that $a_x \in G$ or $b_x \in G$, but both lead to a contradiction.

^fHere a_x and b_x can be possibly $-\infty$ and $+\infty$ respectively.

- **Step 4:** If $r, s \in G$, then we have either

$$I_r = I_s \quad \text{or} \quad I_r \cap I_s = \emptyset.$$

Let $p \in I_r \cap I_s$. Then it follows from Theorem 2.24(a) that $I_r \cup I_s$ is an open set containing the point p . By **Step 3** above, we have $I_r = I_r \cup I_s$ and $I_s = I_r \cup I_s$ which imply that $I_r = I_s$.

We have shown from **Step 1** to **Step 4** that the open set G is a union of disjoint segments. By Theorem 1.20(b), each I_x contains at least one rational number. Since $\mathbb{Q}$ is countable, the union of disjoint segments is at most countable. This finishes the proof of the problem. ■

Problem 2.30

Rudin Chapter 2 Exercise 30.

Proof. Assume that the interior of every F_n was empty. That is, $F_n^\circ = \emptyset$ for every n . Then we have $\bigcup_1^\infty F_n^\circ = \emptyset$ and so $(\bigcup_1^\infty F_n^\circ)^c = \mathbb{R}^k$. By Theorem 2.22, we have

$$\bigcap_1^\infty (F_n^\circ)^c = \left(\bigcup_1^\infty F_n^\circ \right)^c = \mathbb{R}^k. \quad (2.15)$$

By Problem 2.9(d), we have $(F_n^\circ)^c = \overline{F_n^c}$. By this and the relation (2.15), we have $\bigcap_1^\infty \overline{F_n^c} = \mathbb{R}^k$ which implies that

$$\mathbb{R}^k = \overline{F_n^c} \quad (2.16)$$

for every n . For every n , since F_n is closed, F_n^c is open and thus this and the relation (2.16) deduce that each F_n^c is a dense open subset of $\mathbb{R}^k$.

As suggested by Rudin, we can “imitate” the proof of Theorem 2.43 to obtain the required result. When we read the proof of Theorem 2.43 closely, we see that the core idea of it is to construct a shrinking sequence of nonempty compact sets K_n so that the corollary of Theorem 2.36 can be applied to obtain a contradiction. Now we follow this idea to *construct* such a shrinking sequence of nonempty compact sets in the following paragraph, see also Figure 2.4 for the idea of constructing the shrinking sequence.

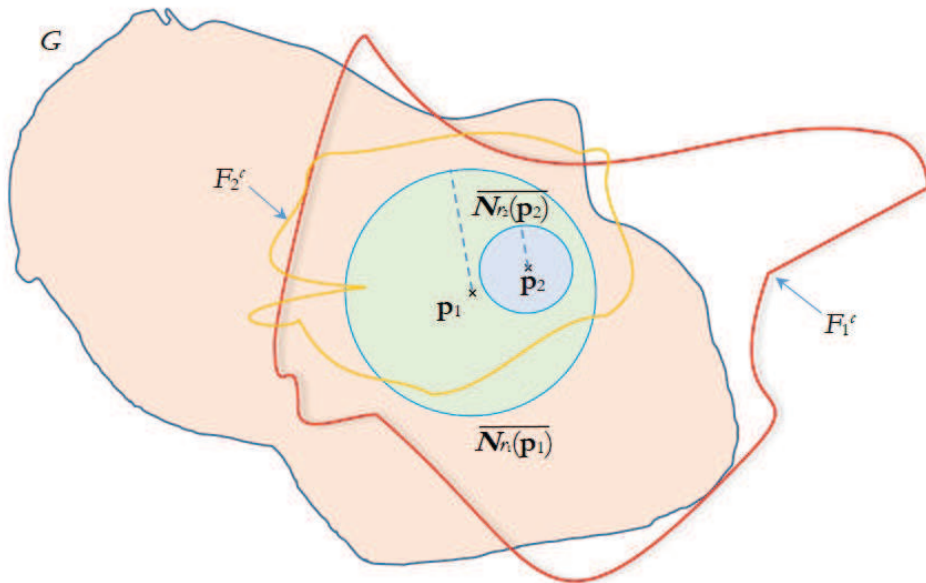


Figure 2.4: The construction of the shrinking sequence.

Let G be an open set of $\mathbb{R}^k$.^g Since F_1^c is a dense subset of $\mathbb{R}^k$, there exists $\mathbf{p} \in F_1^c$ such that $\mathbf{p} \in G$, see Problem 2.22 and its footnote. Thus we have

$$G_1 = F_1^c \cap G \neq \emptyset.$$

Since F_1^c is open in $\mathbb{R}^k$, G_1 must be open in $\mathbb{R}^k$ by Theorem 2.24(c). Let $\mathbf{p}_1 \in G_1$.^h Since G_1 is open, we have $N_{r_1}(\mathbf{p}_1) \subseteq G_1$ for some $r_1 > 0$. Without loss of generality, we may assume that

$$\overline{N_{r_1}(\mathbf{p}_1)} \subseteq G_1.$$

Since F_2^c is a dense subset of $\mathbb{R}^k$, the set $G_2 = F_2^c \cap N_{r_1}(\mathbf{p}_1)$ is a nonempty open subset of $\mathbb{R}^k$. Let $\mathbf{p}_2 \in G_2$. Then we can choose $r_2 > 0$ small enough such that

$$\overline{N_{r_2}(\mathbf{p}_2)} \subseteq G_2.$$

By definition, we have

$$\overline{N_{r_2}(\mathbf{p}_2)} \subseteq G_2 = F_2^c \cap N_{r_1}(\mathbf{p}_1) \subset \overline{N_{r_1}(\mathbf{p}_1)}.$$

Now we can continue this process to obtain the following shrinking sequence

$$\cdots \subset \overline{N_{r_3}(\mathbf{p}_3)} \subset \overline{N_{r_2}(\mathbf{p}_2)} \subset \overline{N_{r_1}(\mathbf{p}_1)}. \quad (2.17)$$

By Theorem 2.27(a), each $\overline{N_{r_n}(\mathbf{p}_n)}$ is closed. It is clear that each $\overline{N_{r_n}(\mathbf{p}_n)}$ is bounded, so Theorem 2.41 (the Heine-Borel theorem) implies that each member in (2.17) is compact. Therefore the corollary of Theorem 2.36 shows that $\bigcap_1^\infty \overline{N_{r_n}(\mathbf{p}_n)}$ is nonempty. Since it is true that $\overline{N_{r_n}(\mathbf{p}_n)} \subseteq F_n^c$ for every n , we have

$$\bigcap_1^\infty \overline{N_{r_n}(\mathbf{p}_n)} \subseteq \bigcap_1^\infty F_n^c$$

which says that the set $\bigcap_1^\infty F_n^c$ is nonempty. However, this and Theorem 2.22 lead the result that

$$\left(\bigcup_1^\infty F_n \right)^c \neq \emptyset.$$

In other words, we have $\bigcup_1^\infty F_n \neq \mathbb{R}^k$ which contradicts our hypothesis. Hence there is at least one F_n having a nonempty interior. This completes the proof of the problem. ■

^gThe construction of the required shrinking sequence can be “seen” from Figure 2.4.

^hHere we don’t assume that $\mathbf{p} = \mathbf{p}_1$.

CHAPTER 3

Numerical Sequences and Series

3.1 Problems on sequences

Problem 3.1

Rudin Chapter 3 Exercise 1.

Proof. Suppose that $\{s_n\}$ converges to s . By Definition 3.1, for every $\varepsilon > 0$, there is an integer N such that $n \geq N$ implies that $|s_n - s| < \varepsilon$. By Problem 1.13, we have

$$||s_n| - |s|| \leq |s_n - s| < \varepsilon$$

for all $n \geq N$. Hence the sequence $\{|s_n|\}$ converges to $|s|$. However, the converse is not true. For example, we let $s_n = (-1)^n$ so that $\{s_n\}$ is divergent, but $|s_n| = 1 \rightarrow 1$. This completes the proof of the problem. ■

Problem 3.2

Rudin Chapter 3 Exercise 2.

Proof. We have

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 + n} - n) = \lim_{n \rightarrow \infty} \frac{n^2 + n - n^2}{\sqrt{n^2 + n} + n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} = \frac{1}{2}.$$

This completes the proof of the problem. ■

Problem 3.3

Rudin Chapter 3 Exercise 3.

Proof. It follows from $s_1 = \sqrt{2} < 2$ and the induction that

$$s_{k+1} = \sqrt{2 + \sqrt{s_k}} < \sqrt{2 + 2} < 2$$

when $s_k < 2$. Hence we have $s_n < 2$ for all $n = 1, 2, 3, \dots$. By similar argument, we can show that $s_n > 0$ for all $n = 1, 2, 3, \dots$. Therefore we have

$$0 < s_n < 2$$

for all $n = 1, 2, 3, \dots$ so that $\{s_n\}$ is bounded. Since

$$\sqrt{s_n} - 2 < \sqrt{2} - 2 \quad \text{and} \quad s_{n+1} - s_n = \sqrt{2 + \sqrt{s_n}} - s_n = \frac{-(\sqrt{s_n} - 2)(\sqrt{s_n} + 1)}{\sqrt{2 + \sqrt{s_n}} + s_n},$$

we have $s_{n+1} - s_n > 0$ so that $\{s_n\}$ is strictly increasing. By Theorem 3.14 (Monotone Convergence Theorem), we have $\{s_n\}$ converges. This completes the proof of the problem. ■

Problem 3.4

Rudin Chapter 3 Exercise 4.

Proof. By checking the first few terms, we see that

$$s_1 = 0, s_2 = 0, s_3 = \frac{1}{2}, s_4 = \frac{1}{4}, s_5 = \frac{3}{4}, s_6 = \frac{3}{8}, s_7 = \frac{7}{8}, \dots$$

Therefore we can show by induction that

$$s_{2m} = \frac{2^{m-1} - 1}{2^m} \quad \text{and} \quad s_{2m+1} = \frac{2^m - 1}{2^m},$$

where $m = 1, 2, \dots$. By Definition 3.16, we have $E = \{\frac{1}{2}, 1\}$. Hence we have

$$\limsup_{n \rightarrow \infty} s_n = 1 \quad \text{and} \quad \liminf_{n \rightarrow \infty} s_n = \frac{1}{2},$$

completing the proof of the problem. ■

Problem 3.5

Rudin Chapter 3 Exercise 5.

Proof. Let $s = \limsup_{n \rightarrow \infty} (a_n + b_n)$, $a = \limsup_{n \rightarrow \infty} a_n$ and $b = \limsup_{n \rightarrow \infty} b_n$. Suppose that the sum on the right is not of the form $\infty - \infty$.

- **Case (i):** When $a = +\infty$ or $b = +\infty$. The inequality holds trivially in this case.
- **Case (ii):** When $a < +\infty$ and $b < +\infty$. By Theorem 3.17(a), there exists a subsequence $\{a_{n_k} + b_{n_k}\}$ such that $s = \lim_{k \rightarrow \infty} (a_{n_k} + b_{n_k})$.
 - If any one of $\{a_{n_k}\}$ and $\{b_{n_k}\}$ converges, say $\{a_{n_k}\}$, then the equation $b_{n_k} = (a_{n_k} + b_{n_k}) - a_{n_k}$ implies that $\{b_{n_k}\}$ also converges and it follows from Theorem 3.3(a) and Definition 3.16 that

$$s = \lim_{k \rightarrow \infty} (a_{n_k} + b_{n_k}) = \lim_{k \rightarrow \infty} a_{n_k} + \lim_{k \rightarrow \infty} b_{n_k} \leq a + b.$$

- If both $\{a_{n_k}\}$ and $\{b_{n_k}\}$ diverge but $\{a_{n_k} + b_{n_k}\}$ converge^a, then we consider

$$\alpha = \limsup_{k \rightarrow \infty} a_{n_k} \quad \text{and} \quad \beta = \limsup_{k \rightarrow \infty} b_{n_k}.$$

By Definition 3.16, we have

$$\limsup_{k \rightarrow \infty} a_{n_k} \leq \limsup_{n \rightarrow \infty} a_n \quad \text{and} \quad \limsup_{k \rightarrow \infty} b_{n_k} \leq \limsup_{n \rightarrow \infty} b_n$$

^aFor example, $a_{n_k} = (-1)^{k+1}$ and $b_{n_k} = (-1)^k$ so that $a_{n_k} + b_{n_k} = 0$ for all positive integers k .

so that α and β are finite. By Theorem 3.17(a) again, there exists a subsequence $\{a_{n_{k_j}}\}$ of the *divergent* sequence $\{a_{n_k}\}$ such that

$$\lim_{j \rightarrow \infty} a_{n_{k_j}} = \alpha.$$

We note that $\{b_{n_k}\}$ *diverges*, so it may happen that its subsequence also diverges. However, we recall from Definition 3.5 that a sequence $\{p_n\}$ converges to p if and only if every subsequence of $\{p_n\}$ converges to p . Since $s = \lim_{k \rightarrow \infty} (a_{n_k} + b_{n_k})$, the subsequence $\{a_{n_{k_j}} + b_{n_{k_j}}\}$ converges to this s too. Thus it follows from Theorem 3.3(a) that

$$\lim_{j \rightarrow \infty} b_{n_{k_j}} = \lim_{j \rightarrow \infty} [(a_{n_{k_j}} + b_{n_{k_j}}) - a_{n_{k_j}}] = \lim_{j \rightarrow \infty} (a_{n_{k_j}} + b_{n_{k_j}}) - \lim_{j \rightarrow \infty} a_{n_{k_j}} = s - \alpha.$$

By this, we know that the subsequence $\{b_{n_{k_j}}\}$ of the *divergent* sequence $\{b_{n_k}\}$ is convergent. Hence we obtain from the definition of s and Theorem 3.3(a) that

$$\begin{aligned} \limsup_{n \rightarrow \infty} (a_n + b_n) &= \lim_{k \rightarrow \infty} (a_{n_k} + b_{n_k}) \\ &= \lim_{j \rightarrow \infty} (a_{n_{k_j}} + b_{n_{k_j}}) \\ &= \lim_{j \rightarrow \infty} a_{n_{k_j}} + \lim_{j \rightarrow \infty} b_{n_{k_j}} \\ &\leq \alpha + \beta \\ &\leq a + b \\ &= \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n. \end{aligned}$$

This completes the proof of the problem. ■

3.2 Problems on series

Problem 3.6

Rudin Chapter 3 Exercise 6.

Proof.

(a) We have

$$a_n = \sqrt{n+1} - \sqrt{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \geq \frac{1}{2\sqrt{n+1}} = \frac{1}{2 \cdot (n+1)^{\frac{1}{2}}} \geq 0.$$

By Theorem 3.28, $\sum \frac{1}{(n+1)^{\frac{1}{2}}}$ diverges. By Theorem 3.25 (Comparison Test), $\sum a_n$ diverges.

(b) We have

$$a_n = \frac{\sqrt{n+1} - \sqrt{n}}{n} = \frac{(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n})}{n(\sqrt{n+1} + \sqrt{n})} = \frac{1}{n(\sqrt{n+1} + \sqrt{n})} \leq \frac{1}{2n\sqrt{n}} = \frac{1}{2n^{\frac{3}{2}}}.$$

Since $\sum \frac{1}{n^{\frac{3}{2}}}$ converges by Theorem 3.28, the comparison test shows that $\sum a_n$ converges.

(c) Since $\frac{3}{2} > 1$, we have $(\frac{3}{2})^n \geq (\frac{3}{2})^2 > 2$ for $n \geq 2$. Then it can be shown by induction that

$$n < \left(\frac{3}{2}\right)^n$$

for $n \geq 2$. By this, we have $\sqrt[n]{n} - 1 < \frac{1}{2}$ which gives $a_n < \frac{1}{2^n}$ for $n \geq 2$. Since $\sum \frac{1}{2^n}$ converges by Theorem 3.26, it follows from Theorem 3.25 (Comparison Test) that

$$\sum_{n=2}^{\infty} a_n$$

converges. Hence $\sum a_n$ converges because $a_1 = 0$.

(d) The convergence of $\sum a_n$ depends on the value of $|z|$.

It is clear that if $|z| \leq 1$, then the triangle inequality implies that $|1 + z^n| \leq 1 + |z|^n$ so that

$$|a_n| = \left| \frac{1}{1 + z^n} \right| \geq \frac{1}{1 + |z|^n} \geq \frac{1}{2} \neq 0.$$

By Theorem 3.23, we acquire $\sum a_n$ diverges if $|z| \leq 1$.

On the other hand, if $|z| > 1$, then the triangle inequality implies that $|z^n| \leq |1 + z^n| + 1$ so that

$$|a_n| = \left| \frac{1}{1 + z^n} \right| \leq \frac{1}{|z|^n - 1}.$$

Since $|z| > 1$, we have $|z| = 1 + \delta$ for some $\delta > 0$. Let N be the least positive integer such that^b

$$N \geq \frac{\log 2}{\log(1 + \delta)}.$$

Then for all positive integers n such that $n > N$, we have $|z|^n = (1 + \delta)^n > 2$ so that

$$|a_n| \leq \frac{1}{|z|^n - 1} \leq \frac{2}{|z|^n}.$$

Hence it follows from Theorem 3.25 (Comparison Test) that $\sum a_n$ converges.

This completes the proof of the problem. ■

Problem 3.7

Rudin Chapter 3 Exercise 7.

Proof. Suppose that $a_n \geq 0$ for all positive integers n and $\sum a_n$ converges. Since $0 \leq (\sqrt{a_n} - \frac{1}{n})^2$, we have

$$\frac{\sqrt{a_n}}{n} \leq \frac{1}{2} \left(a_n + \frac{1}{n^2} \right).$$

By Theorem 3.28, $\sum \frac{1}{n^2}$ converges. By the assumption and Theorem 3.25 (Comparison Test), we see that

$$\sum \frac{\sqrt{a_n}}{n}$$

converges. This completes the proof of the problem. ■

Problem 3.8

Rudin Chapter 3 Exercise 8.

^bThe base of the logarithm here is supposed to be 10.

Proof. Suppose that $\{b_n\}$ is increasing.^c Since $\{b_n\}$ is bounded, there exists a positive number M such that $-M \leq b_n \leq M$ for all positive integers n . Therefore we have $-M|a_n| \leq |a_n|b_n \leq M|a_n|$ and then

$$|a_n b_n| \leq M|a_n|$$

for all positive integers n . Since $\sum a_n$ converges, $\sum M|a_n|$ also converges and it follows from Theorem 3.25 (Comparison Test) that $\sum a_n b_n$ converges. This completes the proof of the problem.^d ■

Problem 3.9

Rudin Chapter 3 Exercise 9.

Proof.

- (a) Since $\left|\frac{(n+1)^3}{n^3}\right| = \left(1 + \frac{1}{n}\right)^3$, we have

$$\alpha = \limsup_{n \rightarrow \infty} \left|\frac{(n+1)^3}{n^3}\right| = \limsup_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^3 = 1.$$

Hence we have $R = 1$.

- (b) Since $\left|\frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}}\right| = \frac{2}{n+1}$, we have

$$\alpha = \limsup_{n \rightarrow \infty} \left|\frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}}\right| = \limsup_{n \rightarrow \infty} \frac{2}{n+1} = 0.$$

Hence we have $R = \infty$.

- (c) Since $\left|\frac{\frac{2^{n+1}}{(n+1)^2}}{\frac{2^n}{n^2}}\right| = 2\left(\frac{n}{n+1}\right)^2$, we have

$$\alpha = \limsup_{n \rightarrow \infty} \left|\frac{\frac{2^{n+1}}{(n+1)^2}}{\frac{2^n}{n^2}}\right| = \limsup_{n \rightarrow \infty} 2\left(\frac{n}{n+1}\right)^2 = 2.$$

Hence we have $R = \frac{1}{2}$.

- (d) Since $\left|\frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}}\right| = \frac{1}{3}\left(\frac{n+1}{n}\right)^3$, we have

$$\alpha = \limsup_{n \rightarrow \infty} \left|\frac{\frac{(n+1)^3}{3^{n+1}}}{\frac{n^3}{3^n}}\right| = \limsup_{n \rightarrow \infty} \frac{1}{3}\left(\frac{n+1}{n}\right)^3 = \frac{1}{3}.$$

Hence we have $R = 3$.

We complete the proof of the problem. ■

Problem 3.10

Rudin Chapter 3 Exercise 10.

Proof. Suppose that $\{a_{n_k}\}$ is the subsequence of $\{a_n\}$ such that $a_{n_k} \neq 0$, where n_k are positive integers such that $n_1 < n_2 < \dots$. The subsequence is infinite by the given condition. Since $|a_{n_k}| \geq 1$, we have $\sqrt[n_k]{|a_{n_k}|} \geq 1$ so that

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \geq 1.$$

Hence we have $R = \frac{1}{\alpha} \leq 1$ which is the desired result. ■

^cThe case for decreasing sequences is similar, so we omit the details here.

^dThis result is well-known as Abel's Test.

Problem 3.11*Rudin Chapter 3 Exercise 11.**Proof.*

- (a) If the sequence
- $\{a_n\}$
- is not bounded, then

$$\frac{a_n}{1+a_n} = \frac{1}{1+\frac{1}{a_n}} \rightarrow 1.$$

It follows from Theorem 3.23 that $\sum \frac{a_n}{1+a_n}$ diverges. If the sequence $\{a_n\}$ is bounded, then $a_n > 0$ implies that there is a positive real number M such that $a_n < M$ for all positive integers n and thus $1+a_n \leq 1+M$. Therefore we have

$$\frac{a_n}{1+a_n} \geq \frac{a_n}{1+M} > 0$$

and Theorem 3.25 (Comparison Test) yields that

$$\sum \frac{a_n}{1+a_n}$$

diverges.

- (b) Since $a_n > 0$, we have $s_{N+k} \geq s_{N+j}$ for any fixed positive integer N and for all $j = 1, 2, \dots, k$. Therefore we have $\frac{1}{s_{N+k}} \leq \frac{1}{s_{N+j}}$ for $j = 1, 2, \dots, k$. Hence we have

$$\begin{aligned} \frac{a_{N+1}}{s_{N+1}} + \dots + \frac{a_{N+k}}{s_{N+k}} &\geq \frac{a_{N+1}}{s_{N+k}} + \frac{a_{N+2}}{s_{N+k}} + \dots + \frac{a_{N+k}}{s_{N+k}} \\ &= \frac{1}{s_{N+k}}(a_{N+1} + a_{N+2} + \dots + a_{N+k}) \\ &= \frac{1}{s_{N+k}}(s_{N+k} - s_N) \\ &= 1 - \frac{s_N}{s_{N+k}}. \end{aligned} \tag{3.1}$$

Since $\{s_k\}$ diverges, $\{s_{N+k}\}$ diverges. Since $a_n > 0$ for all $n \in \mathbb{N}$, we have

$$\lim_{k \rightarrow \infty} s_{N+k} = +\infty. \tag{3.2}$$

Combining the inequality (3.1) and the limit (3.2), we have

$$\lim_{k \rightarrow \infty} \left(\frac{a_{N+1}}{s_{N+1}} + \dots + \frac{a_{N+k}}{s_{N+k}} \right) \geq \lim_{k \rightarrow \infty} \left(1 - \frac{s_N}{s_{N+k}} \right) = 1 - s_N \lim_{k \rightarrow \infty} \frac{1}{s_{N+k}} = 1. \tag{3.3}$$

Assume that $\sum \frac{a_n}{s_n}$ was convergent. By Theorem 3.22, there exists an integer N such that

$$\frac{a_{N+1}}{s_{N+1}} + \dots + \frac{a_m}{s_m} \leq \frac{1}{2}$$

if $m \geq N+1$, but this contradicts the result (3.3) if we take $m \rightarrow \infty$ in the above inequality. Hence

$$\sum_{n=1}^{\infty} \frac{a_n}{s_n}$$

diverges.

(c) Since $s_n \geq s_{n-1}$ for every positive integer $n \geq 2$, we have

$$\frac{1}{s_{n-1}} - \frac{1}{s_n} = \frac{s_n - s_{n-1}}{s_{n-1}s_n} \geq \frac{a_n}{s_n^2}. \quad (3.4)$$

Since $a_n > 0$, we have $s_n > 0$. This and the inequality (3.4) together imply that

$$\sum_{k=1}^n \frac{a_k}{s_k^2} = \frac{a_1}{s_1^2} + \sum_{k=2}^n \frac{a_k}{s_k^2} \leq \frac{1}{a_1} + \sum_{k=2}^n \left(\frac{1}{s_{k-1}} - \frac{1}{s_k} \right) = \frac{1}{a_1} + \frac{1}{s_1} - \frac{1}{s_n} < \frac{2}{a_1}.$$

Hence the partial sums of the series $\{\sum \frac{a_n}{s_n^2}\}$ are bounded and it follows from Theorem 3.24 that

$$\sum \frac{a_n}{s_n^2}$$

converges.

(d) Since $a_n > 0$, we have $n^2 a_n < 1 + n^2 a_n$ and thus

$$\frac{a_n}{1 + n^2 a_n} \leq \frac{1}{n^2}.$$

By Theorem 3.28, $\sum \frac{1}{n^2}$ converges. By Theorem 3.25 (Comparison Test), the series

$$\sum \frac{a_n}{1 + n^2 a_n}$$

converges.

However, the convergence of the series

$$\sum \frac{a_n}{1 + n a_n}$$

depends on the choice of the sequence $\{a_n\}$.

– If $a_n = 1$, then we have $\sum a_n$ diverges and it follows from Theorem 3.28 that the series

$$\sum \frac{1}{1 + n}$$

diverges.

– To construct a sequence $\{a_n\}$ so that $\sum a_n$ diverges but $\sum \frac{a_n}{1 + n a_n}$ converges, we note that the terms

$$\frac{1}{n} \quad \text{and} \quad \frac{a_n}{1 + n a_n}$$

are of the same magnitude. Therefore, if there are “too many” terms $\frac{a_n}{1 + n a_n}$ in the series, then the series must be divergent because $\sum \frac{1}{n}$ diverges.

This observation motivates the construction of such sequence $\{a_n\}$ as follows: Define $\{a_n\}$ by

$$a_n = \begin{cases} 2^k, & \text{if } n = 2^k, \text{ where } k = 0, 1, 2, \dots; \\ 0, & \text{otherwise.} \end{cases}$$

For examples, $a_1 = 1$, $a_2 = 2$, $a_3 = 0$, $a_4 = 4$, $a_5 = a_6 = a_7 = 0$, $a_8 = 8$ and

$$a_9 = \dots = a_{15} = 0, a_{16} = 16, \dots$$

It is clear that

$$\sum_{n=1}^{\infty} a_n = \sum_{k=0}^{\infty} 2^k \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{a_n}{1 + n a_n} = \sum_{k=0}^{\infty} \frac{2^k}{1 + 2^{2k}}.$$

By Theorem 3.26, the series

$$\sum 2^k$$

diverges. Since $\frac{2^k}{1+2^{2k}} \leq \frac{1}{2^k}$ and $\sum \frac{1}{2^k}$ is convergent by Theorem 3.26, we obtain from Theorem 3.25 (Comparison Test) that the series

$$\sum \frac{2^k}{1+2^{2k}}$$

converges.

This completes the proof of the problem. ■

Problem 3.12

Rudin Chapter 3 Exercise 12.

Proof.

- (a) If $m < n$, then $r_n < r_m$ so that the sequence $\{r_n\}$ is strictly decreasing and $\frac{1}{r_m} < \frac{1}{r_n}$. It follows from this that

$$\begin{aligned} \frac{a_m}{r_m} + \frac{a_{m+1}}{r_{m+1}} + \cdots + \frac{a_n}{r_n} &> \frac{a_m}{r_m} + \frac{a_{m+1}}{r_m} + \cdots + \frac{a_n}{r_m} \\ &= \frac{r_m - r_{n+1}}{r_m} \\ &= 1 - \frac{r_{n+1}}{r_m} \\ &> 1 - \frac{r_n}{r_m}. \end{aligned} \tag{3.5}$$

We use similar argument as in the proof of Problem 3.11(b). Assume that $\sum \frac{a_n}{r_n}$ was convergent. Since $a_n > 0$ for every $n \in \mathbb{N}$, it is easy to see that $r_n > 0$ for all $n \in \mathbb{N}$. By Theorem 3.22, there is an integer N such that

$$\frac{a_m}{r_m} + \cdots + \frac{a_n}{r_n} \leq \frac{1}{2} \tag{3.6}$$

if $n > m \geq N$. Therefore, by putting the inequality (3.6) into the inequality (3.5), we obtain that

$$1 - \frac{r_n}{r_N} \leq \frac{1}{2} \tag{3.7}$$

if $n > N$. Recall that $\{r_n\}$ is strictly decreasing and $r_n > 0$ for all $n \in \mathbb{N}$. Therefore, we apply Theorem 3.14 (Monotone Convergence Theorem) to $\{r_n\}$ to get the result that

$$\lim_{n \rightarrow \infty} r_n = 0.$$

Hence we deduce from this and the inequality (3.7) that

$$1 = \lim_{n \rightarrow \infty} \left(1 - \frac{r_n}{r_N}\right) \leq \frac{1}{2},$$

a contradiction and so the required result follows.

- (b) Since

$$\sqrt{r_n} - \sqrt{r_{n+1}} = \frac{(\sqrt{r_n} - \sqrt{r_{n+1}})(\sqrt{r_n} + \sqrt{r_{n+1}})}{\sqrt{r_n} + \sqrt{r_{n+1}}} = \frac{r_n - r_{n+1}}{\sqrt{r_n} + \sqrt{r_{n+1}}} > \frac{a_n}{2\sqrt{r_n}},$$

the desired inequality follows. Since

$$\sum_{k=1}^n \frac{a_k}{\sqrt{r_k}} < 2(\sqrt{r_1} - \sqrt{r_{n+1}}) < 2\sqrt{r_1},$$

it follows from Theorem 3.24 that $\sum \frac{a_n}{\sqrt{r_n}}$ converges.

This completes the proof of the problem. ■

Problem 3.13*Rudin Chapter 3 Exercise 13.*

Proof. Let $\sum a_n$ and $\sum b_n$ be two absolutely convergent series. Let $\sum c_n$ be the Cauchy product of the two series, where $c_n = \sum_{k=0}^n a_k b_{n-k}$. We have to show that $\sum |c_n|$ converges and we follow part of the idea of the proof of Theorem 3.50.

Since $\sum a_n$ and $\sum b_n$ converge absolutely, we let $A = \sum |a_n|$ and $B = \sum |b_n|$. Put $A_n = \sum_{k=0}^n |a_k|$, $B_n = \sum_{k=0}^n |b_k|$ and $C_n = \sum_{k=0}^n |c_k|$. Now for all $n \geq 0$, $|a_n|$ and $|b_n|$ are nonnegative terms of A_n and B_n respectively. This implies that $A_n \leq A$ and $B_n \leq B$ for all $n \geq 0$ and then we have

$$\begin{aligned} |C_n| &= \sum_{k=0}^n |c_k| \\ &\leq |a_0||b_0| + (|a_0||b_1| + |a_1||b_0|) + \cdots + (|a_0||b_n| + |a_1||b_{n-1}| + \cdots + |a_n||b_0|) \\ &= |a_0|B_n + |a_1|B_{n-1} + \cdots + |a_n|B_0 \\ &\leq |a_0|B + |a_1|B + \cdots + |a_n|B \\ &= A_n B \\ &\leq AB. \end{aligned}$$

Hence $\{C_n\}$ is a bounded sequence (bounded by AB) and we deduce from Theorem 3.24 that $\sum |c_n|$ converges. We finish the proof of the problem. ■

Problem 3.14*Rudin Chapter 3 Exercise 14.*

Proof.

- (a) Given $\epsilon > 0$, there exists a positive integer N such that $|s_n - s| < \frac{\epsilon}{2}$ for all $n \geq N$. Fix this N , we have

$$\begin{aligned} |\sigma_n - s| &= \left| \frac{(s_0 - s) + \cdots + (s_{N-1} - s) + (s_N - s) + \cdots + (s_n - s)}{n+1} \right| \\ &\leq \frac{1}{n+1} (|s_0 - s| + \cdots + |s_{N-1} - s|) + \frac{1}{n+1} \underbrace{(|s_N - s| + \cdots + |s_n - s|)}_{(n-N+1) \text{ terms}} \\ &\leq \frac{1}{n+1} (|s_0 - s| + \cdots + |s_{N-1} - s|) + \frac{n-N+1}{n+1} \cdot \frac{\epsilon}{2} \\ &< \frac{1}{n+1} (|s_0 - s| + \cdots + |s_{N-1} - s|) + \frac{\epsilon}{2}. \end{aligned} \tag{3.8}$$

Let $M = \max(|s_0 - s|, \dots, |s_{N-1} - s|)$ and N' be the least integer such that $N' > \frac{2M}{\epsilon} - 2$. Since N is fixed, M and then N' are fixed too. Then for all $n \geq N' + 1$, we have $\frac{M}{n+1} < \frac{\epsilon}{2}$ so that inequality (3.8) implies that

$$|\sigma_n - s| < \epsilon$$

for all $n \geq N' + 1$. This shows that $\lim \sigma_n = s$.

- (b) Let $s_n = (-1)^n$. Then $\{s_n\}$ is obviously divergent. Since $\sigma_n = 0$ if n is odd and $\sigma_n = \frac{1}{n+1}$ if n is even, we have $\lim_{n \rightarrow \infty} \sigma_n = 0$.

(c) Such sequence $\{s_n\}$ must satisfy two conditions:

- **Condition (1):** It must contain a divergent subsequence.
- **Condition (2):** The growth of the sum $s_0 + s_1 + \cdots + s_n = O(n^{\frac{1}{k}})$ for some $k \geq 2$ as $n \rightarrow \infty$ ^e so that

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1} \rightarrow 0$$

as $n \rightarrow \infty$.

Now we define the sequence $\{s_n\}$ by

$$s_n = \begin{cases} n^{\frac{1}{3}} = k, & \text{if } n = k^3, \text{ where } k = 1, 2, \dots; \\ \frac{1}{n^2}, & \text{otherwise.} \end{cases}$$

This sequence $\{s_n\}$ satisfies **Condition (1)** because

$$s_{k^3} = k \rightarrow \infty$$

as $k \rightarrow \infty$ so that $\limsup s_n = \infty$. Next, for any positive integer n , let k be the largest positive integer such that $k^3 \leq n < (k+1)^3$. Then we have

$$0 \leq s_0 + s_1 + \cdots + s_n \leq \sum_{m=1}^k m + \sum_{m=1}^{(k+1)^3} \frac{1}{m^2} = \frac{k(k+1)}{2} + \sum_{m=1}^{(k+1)^3} \frac{1}{m^2}$$

so that

$$0 \leq \sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1} \leq \frac{1}{k^3+1} \left[\frac{k(k+1)}{2} + \sum_{m=1}^{(k+1)^3} \frac{1}{m^2} \right]. \quad (3.9)$$

By Theorem 3.28, $\sum \frac{1}{m^2}$ converges. Since the right-hand side of the inequality (3.9) tends to 0 as $k \rightarrow \infty$, it follows from the remark^f above Theorem 3.20 that

$$\lim_{n \rightarrow \infty} \sigma_n = 0.$$

That is, this sequence $\{s_n\}$ satisfies **Condition (2)**. Hence this prove part (c).

(d) It is clear that the equation is true for $n = 1$. Assume that it is also true for $n = k$, where k is a positive integer. Then for $n = k+1$ we have

$$\begin{aligned} \frac{1}{k+2} \sum_{j=1}^{k+1} j a_j &= \frac{1}{k+2} \left[\sum_{j=1}^k j a_j + (k+1) a_{k+1} \right] \\ &= \frac{1}{k+2} [(k+1)(s_k - \sigma_k) + (k+1) a_{k+1}] \\ &= \frac{1}{k+2} [(k+1) s_{k+1} - (k+1) \sigma_k] \\ &= \frac{1}{k+2} [(k+1) s_{k+1} + s_{k+1} - (k+2) \sigma_{k+1}] \\ &= s_{k+1} - \sigma_{k+1}. \end{aligned}$$

Hence the induction shows that the expression is true for all positive integers.

Since $\lim(n a_n) = 0$, we obtain from part (a) that

$$\lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{k=1}^n k a_k = 0.$$

^eLet $f(n)$ and $g(n)$ be two functions defined on $\mathbb{N}$. One writes $f(n) = O(g(n))$ as $n \rightarrow \infty$ if and only if there is a positive constant M such that for all sufficiently large values of n , we have $|f(n)| \leq M|g(n)|$.

^fThat is, if $0 \leq x_n \leq s_n$ for $n \geq N$, where N is some fixed number, and if $s_n \rightarrow 0$, then $x_n \rightarrow 0$.

Let $\lim \sigma_n = \sigma$. Then it follows from Theorem 3.3(a) that

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(\sigma_n + \frac{1}{n+1} \sum_{k=1}^n k a_k \right) = \sigma + 0 = \sigma.$$

Hence $\{s_n\}$ converges to σ .

(e) If $m < n$, then we have

$$\begin{aligned} & (m+1)(\sigma_n - \sigma_m) + \sum_{i=m+1}^n (s_n - s_i) \\ &= (m+1) \left(\frac{s_0 + s_1 + \cdots + s_n}{n+1} - \frac{s_0 + s_1 + \cdots + s_m}{m+1} \right) + [(s_n - s_{m+1}) + \cdots + (s_n - s_n)] \\ &= \frac{m+1}{n+1} (s_0 + s_1 + \cdots + s_n) + (n-m)s_n - (s_0 + s_1 + \cdots + s_m) \\ &= \frac{m+1-n-1}{n+1} (s_0 + s_1 + \cdots + s_n) + (n-m)s_n \\ &= (n-m)(s_n - \sigma_n) \end{aligned}$$

which yields the desired formula. Since $i \geq m+1$, we have $n-i \leq n-m-1$ and $\frac{1}{i+1} \leq \frac{1}{m+2}$ so that

$$\begin{aligned} |s_n - s_i| &= |s_n - s_{n-1} + s_{n-1} - s_{n-2} + \cdots + s_{i+1} - s_i| \\ &\leq |s_n - s_{n-1}| + |s_{n-1} - s_{n-2}| + \cdots + |s_{i+1} - s_i| \\ &\leq \underbrace{\frac{M}{n} + \frac{M}{n-1} + \cdots + \frac{M}{i+1}}_{(n-i) \text{ terms}} \\ &\leq \frac{M}{i+1} + \cdots + \frac{M}{i+1} \\ &= \frac{(n-i)}{i+1} M \\ &\leq \frac{(n-m-1)}{m+2} M. \end{aligned}$$

Fix $\epsilon > 0$. Since $(m+m\epsilon+\epsilon+1+\epsilon) - (m+m\epsilon+\epsilon) = 1+\epsilon > 1$, there exists an integer n such that $m+m\epsilon+\epsilon \leq n < m+m\epsilon+\epsilon+1+\epsilon$ and the inequalities are equivalent to

$$m \leq \frac{n-\epsilon}{1+\epsilon} < m+1.$$

It is easy to see that the above inequalities imply that

$$\frac{m+1}{n-m} \leq \frac{1}{\epsilon} \quad \text{and} \quad \frac{n-m-1}{m+2} < \epsilon.$$

Since $|s_n - s_i| \leq \frac{n-m-1}{m+2} M$, we have $|s_n - s_i| < M\epsilon$ and hence

$$\limsup_{n \rightarrow \infty} |s_n - \sigma| \leq M\epsilon.$$

Since ϵ was arbitrary, $\lim s_n = \sigma$.

This completes the proof of the problem. ■

Problem 3.15

Rudin Chapter 3 Exercise 15.

Proof. We prove the theorems one by one:

- **Proof of generalized Theorem 3.22:** Let $\mathbf{a} = (a_1, a_2, \dots, a_k)$. We further let, for each positive integer n , $\mathbf{a}_n = (a_{n1}, a_{n2}, \dots, a_{nk})$, where $a_{n1}, \dots, a_{nk} \in \mathbb{R}$. Now $\sum_{i=1}^n \mathbf{a}_i$ converges to $\mathbf{a} \in \mathbb{R}^k$ if and only if $\sum_{i=1}^n a_{ij}$ converges to a_j for each $j = 1, 2, \dots, k$.^g By Theorem 3.22, we have $\sum_{i=1}^n a_{ij}$ converges to a_j if and only if for every ϵ there is an integer N_j such that

$$\left| \sum_{i=n}^m a_{ij} \right| < \frac{\epsilon}{k} \quad (3.10)$$

if $m \geq n \geq N_j$.

If $\sum_{i=1}^n \mathbf{a}_i$ converges, then it follows from the inequality (3.10) and

$$\left| \sum_{i=n}^m \mathbf{a}_i \right| \leq \left| \sum_{i=n}^m a_{i1} \right| + \left| \sum_{i=n}^m a_{i2} \right| + \dots + \left| \sum_{i=n}^m a_{ik} \right|$$

that

$$\left| \sum_{i=n}^m \mathbf{a}_i \right| < \frac{\epsilon}{k} + \dots + \frac{\epsilon}{k} \leq \epsilon \quad (3.11)$$

for $m \geq n \geq N = \max(N_1, \dots, N_k)$.

Conversely, if the inequality (3.11) holds for $m \geq n \geq N$, then since

$$\left| \sum_{i=n}^m a_{ij} \right| \leq \left| \sum_{i=n}^m \mathbf{a}_i \right|,$$

for each $j = 1, 2, \dots, k$, we have

$$\left| \sum_{i=n}^m a_{ij} \right| < \epsilon$$

for $m \geq n \geq N$. By Theorem 3.22 again, $\sum_{i=1}^n a_{ij}$ converges to a_j for each $j = 1, 2, \dots, k$ so that

$\sum_{i=1}^n \mathbf{a}_i$ converges to $\mathbf{a}$.

- **Proof of generalized Theorem 3.23:** We take $m = n$ in the inequality (3.11), then it becomes $|\mathbf{a}_n| < \epsilon$ for all $n \geq N$. Since $\mathbf{a}_n = (a_{n1}, a_{n2}, \dots, a_{nk})$, we have

$$\sqrt{a_{n1}^2 + a_{n2}^2 + \dots + a_{nk}^2} < \epsilon$$

for all $n \geq N$. By Definition 3.1, we have

$$\lim_{n \rightarrow \infty} (a_{n1}^2 + a_{n2}^2 + \dots + a_{nk}^2) = 0.$$

Since $a_{nj}^2 \geq 0$ for $1 \leq j \leq k$, we have $\lim_{n \rightarrow \infty} a_{nj} = 0$ for $1 \leq j \leq k$, i.e.,

$$\lim_{n \rightarrow \infty} \mathbf{a}_n = \mathbf{0}.$$

^gThis holds because we have $\left| \sum_{i=1}^n a_{ij} - a_j \right| \leq \left| \sum_{i=1}^n \mathbf{a}_i - \mathbf{a} \right| \leq \left| \sum_{i=1}^n a_{i1} - a_1 \right| + \dots + \left| \sum_{i=1}^n a_{ik} - a_k \right|$, where $j = 1, 2, \dots, k$.

- **Proof of generalized Theorem 3.25(a):** Given $\epsilon > 0$, there exists $N \geq N_0$ such that $m \geq n \geq N$ implies

$$\sum_{k=n}^m c_k \leq \epsilon$$

by the Cauchy criterion. It follows from Theorem 1.37(e) that

$$\left| \sum_{k=n}^m \mathbf{a}_k \right| \leq \sum_{k=n}^m |\mathbf{a}_k| \leq \sum_{k=n}^m c_k < \epsilon$$

for all $m \geq n \geq N$. By the generalized Theorem 3.22, we have $\sum \mathbf{a}_k$ converges.

- **Proof of generalized Theorem 3.33:** If $\alpha < 1$, then we can choose β so that $\alpha < \beta < 1$ and an integer N such that $\sqrt[n]{|\mathbf{a}_n|} < \beta$ for $n \geq N$. That is, for all $n \geq N$, we have

$$|\mathbf{a}_n| < \beta^n.$$

Since $0 < \beta < 1$, $\sum \beta^n$ converges by Theorem 3.26. Hence it follows from the generalized Theorem 3.25(a) that $\sum |\mathbf{a}_n|$ converges.

If $\alpha > 1$, then we obtain from Theorem 3.17(a) that there is a sequence $\{n_k\}$ such that

$$\sqrt[n_k]{|\mathbf{a}_{n_k}|} \rightarrow \alpha.$$

Hence $|\mathbf{a}_n| > 1$ for infinitely many values of n which contradicts the generalized Theorem 3.23.

- **Proof of generalized Theorem 3.34:** If part (a) holds, we can find $\beta < 1$ and an integer N such that

$$\left| \frac{\mathbf{a}_{n+1}}{\mathbf{a}_n} \right| < \beta$$

for $n \geq N$. In particular, we have

$$|\mathbf{a}_{N+p}| < \beta^p |\mathbf{a}_N|, \quad (3.12)$$

where p is a positive integer. It follows from the inequality (3.12) that

$$|\mathbf{a}_n| < |\mathbf{a}_N| \beta^{-N} \cdot \beta^n$$

for $n \geq N$. Hence part (a) follows from the generalized Theorem 3.25(a) because $\sum \beta^n$ converges.

If $|\mathbf{a}_{n+1}| \geq |\mathbf{a}_n|$ for $n \geq n_0$, it is easily seen that the condition $\mathbf{a}_n \rightarrow \mathbf{0}$ does not hold, and part (b) follows.

- **Proof of generalized Theorem 3.42:** For every integer n , let $\mathbf{a}_n = (a_{n1}, \dots, a_{nk})$. Since the partial sums $\mathbf{A}_n$ of $\sum \mathbf{a}_n$ form a bounded sequence, we choose a positive constant M such that $|\mathbf{A}_n| \leq M$ for all n . Let A_{nj} be the partial sum of the series $\sum a_{nj}$ for $1 \leq j \leq k$. Then we have

$$|A_{nj}| \leq \sqrt{A_{n1}^2 + A_{n2}^2 + \dots + A_{nk}^2} = |\mathbf{A}_n| \leq M \quad (3.13)$$

for $1 \leq j \leq k$.

Given $\epsilon > 0$, we follow from the facts $b_0 \geq b_1 \geq b_2 \geq \dots$ and $\lim_{n \rightarrow \infty} b_n = 0$ that there is an integer N such that

$$b_N \leq \frac{\epsilon}{2\sqrt{k}M}. \quad (3.14)$$

Let $q \geq p \geq N$. Now we have $b_n - b_{n+1} \geq 0$ for all nonnegative integers n . It follows from this, inequalities (3.13), (3.14) and the facts that $(a+b)^2 \leq (|a| + |b|)^2$ for every $a, b \in \mathbb{R}$, we have

$$\left| \sum_{n=p}^q \mathbf{a}_n b_n \right|$$

$$\begin{aligned}
&= \left| \left(\sum_{n=p}^q a_{n1} b_n, \dots, \sum_{n=p}^q a_{nk} b_n \right) \right| \\
&= \left| \left(\sum_{n=p}^{q-1} A_{n1}(b_n - b_{n+1}) + A_{q1}b_q - A_{(p-1)1}b_p, \dots, \sum_{n=p}^{q-1} A_{nk}(b_n - b_{n+1}) + A_{qk}b_q - A_{(p-1)k}b_p \right) \right| \\
&= \sqrt{\sum_{j=1}^k \left[\underbrace{\sum_{n=p}^{q-1} A_{nj}(b_n - b_{n+1})}_{\text{This is } a.} + \underbrace{A_{qj}b_q - A_{(p-1)j}b_p}_{\text{This is } b.} \right]^2} \\
&\leq \sqrt{\sum_{j=1}^k \left[\left| \sum_{n=p}^{q-1} A_{nj}(b_n - b_{n+1}) \right| + |A_{qj}b_q - A_{(p-1)j}b_p| \right]^2} \\
&\leq \sqrt{\sum_{j=1}^k M^2 \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + b_q + b_p \right]^2} \\
&= \sqrt{\sum_{j=1}^k (4M^2 b_p^2)} \\
&\leq 2Mb_N \sqrt{k} \\
&\leq \epsilon.
\end{aligned}$$

Now the convergence of the series $\sum \mathbf{a}_n b_n$ follows immediately from the generalized Theorem 3.22.

- **Proof of generalized Theorem 3.45:** The assertion follows from the inequality

$$\left| \sum_{k=n}^m \mathbf{a}_k \right| \leq \sum_{k=n}^m |\mathbf{a}_k|$$

plus the generalized Theorem 3.22.

- **Proof of generalized Theorem 3.47:** Let $\mathbf{A}_n = \sum_{k=0}^n \mathbf{a}_k$ and $\mathbf{B}_n = \sum_{k=0}^n \mathbf{b}_k$. Then we acquire

$$\mathbf{A}_n + \mathbf{B}_n = \sum_{k=0}^n (\mathbf{a}_k + \mathbf{b}_k). \quad (3.15)$$

Since $\lim_{n \rightarrow \infty} \mathbf{A}_n = \mathbf{A}$ and $\lim_{n \rightarrow \infty} \mathbf{B}_n = \mathbf{B}$, we see from the expression (3.15) that

$$\lim_{n \rightarrow \infty} (\mathbf{A}_n + \mathbf{B}_n) = \mathbf{A} + \mathbf{B}.$$

The proof of the second assertion is similar.

- **Proof of generalized Theorem 3.55:** Let $\sum \mathbf{a}'_n$ be a rearrangement with partial sums $\mathbf{s}'_n$. Given $\epsilon > 0$, there exists an integer N such that $m \geq n \geq N$ implies that

$$\sum_{i=n}^m |\mathbf{a}_i| < \epsilon. \quad (3.16)$$

Now choose a positive integer p such that the integers $1, 2, \dots, N$ are all contained in the set $k_1, k_2, \dots, k_p$ (here we use the notation of Definition 3.52). Then if $n > p$, the vectors $\mathbf{a}_1, \dots, \mathbf{a}_N$ will cancel in the difference $\mathbf{s}_n - \mathbf{s}'_n$ so that the inequality (3.16) implies that

$$|\mathbf{s}_n - \mathbf{s}'_n| < \epsilon.$$

Hence $\{\mathbf{s}'_n\}$ converges to the same sum as $\{\mathbf{s}_n\}$.

This completes the proof of the problem. ■

3.3 Recursion formulas of sequences

Problem 3.16*Rudin Chapter 3 Exercise 16.**Proof.*

- (a) Since $\alpha > 0$, it can be shown by induction that $x_n > \sqrt{\alpha} > 0$ for all positive integers n .^h By this, we have

$$x_{n+1} - x_n = \frac{x_n^2 + \alpha}{2x_n} - x_n = -\left(\frac{-\alpha + x_n^2}{2x_n}\right) \leq 0.$$

Thus $\{x_n\}$ decreases monotonically. Since $\sqrt{\alpha} < x_n \leq x_1$ for all positive integers n , the sequence $\{x_n\}$ is bounded and Theorem 3.14 (Monotone Convergence Theorem) implies that $\{x_n\}$ converges. Let $x = \lim x_n$. Then Theorem 3.3 implies that

$$\begin{aligned} \lim_{n \rightarrow \infty} x_{n+1} &= \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \\ x &= \frac{1}{2} \left(x + \frac{\alpha}{x} \right) \\ x &= \pm \sqrt{\alpha}. \end{aligned}$$

Since $x_n > 0$ for all positive integers n , we have $x > 0$ and hence $x = \sqrt{\alpha}$, as desired.

- (b) We have

$$\epsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \frac{(x_n - \sqrt{\alpha})^2}{x_n} = \frac{\epsilon_n^2}{2x_n}.$$

As shown in part (a) that $x_n > \sqrt{\alpha} > 0$ for all positive integers n . Thus if $\beta = 2\sqrt{\alpha}$, then we have

$$\epsilon_{n+1} = \frac{\epsilon_n^2}{2x_n} < \frac{\epsilon_n^2}{2\sqrt{\alpha}} = \frac{\epsilon_n^2}{\beta}.$$

By this, it is clear that $\epsilon_2 < \beta \left(\frac{\epsilon_1}{\beta} \right)^2$. Assume that $\epsilon_{k+1} < \beta \left(\frac{\epsilon_1}{\beta} \right)^{2^k}$ for some positive integer k . For $n = k + 1$, we have

$$\epsilon_{k+2} < \frac{\epsilon_{k+1}^2}{\beta} < \frac{1}{\beta} \left[\beta \left(\frac{\epsilon_1}{\beta} \right)^{2^k} \right]^2 = \beta \left(\frac{\epsilon_1}{\beta} \right)^{2^{k+1}}.$$

Hence it follows from induction that

$$\epsilon_{n+1} < \beta \left(\frac{\epsilon_1}{\beta} \right)^{2^n}$$

for $n = 1, 2, \dots$

- (c) Since $25 < 27$, we have $5 < 3\sqrt{3}$ and so $\frac{2-\sqrt{3}}{2\sqrt{3}} < \frac{1}{10}$. Since $\epsilon = x_1 - \sqrt{3} = 2 - \sqrt{3}$ and $\beta = 2\sqrt{3}$, we have

$$\frac{\epsilon_1}{\beta} < \frac{1}{10}.$$

By part (b), we have

$$\epsilon_5 < 2\sqrt{3} \cdot 10^{-16} < 4 \cdot 10^{-16} \quad \text{and} \quad \epsilon_6 < 2\sqrt{3} \cdot 10^{-32} < 4 \cdot 10^{-32}.$$

We complete the proof of the problem. ■

^hNote that $x_{n+1}^2 - \alpha = \frac{1}{4} \left(x_n - \frac{\alpha}{x_n} \right)^2$ for every positive integer n .

Problem 3.17

Rudin Chapter 3 Exercise 17.

Proof. We need two results to prove (a) and (b). Lemma 3.1 is about the magnitude between x_n and $\sqrt{\alpha}$. In Lemma 3.2, we obtain the equations of $x_{2n+1} - x_{2n-1}$ and $x_{2n} - x_{2n-2}$.

Lemma 3.1

For every positive integer n , we have

$$x_{2n-1} > \sqrt{\alpha} \quad \text{and} \quad x_{2n} < \sqrt{\alpha}.$$

Proof of Lemma 3.1. It is clear from the definition and induction that $x_n > 0$ for all positive integers n . Since $\alpha > 1$, we have $\sqrt{\alpha} - 1 > 0$ and then

$$x_{n+1} - \sqrt{\alpha} = \frac{\sqrt{\alpha} - 1}{1 + x_n}(\sqrt{\alpha} - x_n) \quad (3.17)$$

for all positive integers n . Therefore the expression (3.17) implies that

$$\begin{aligned} x_{2n+1} - \sqrt{\alpha} &= \frac{(\sqrt{\alpha} - 1)^2}{(1 + x_{2n})(1 + x_{2n-1})}(x_{2n-1} - \sqrt{\alpha}), \\ x_{2n+2} - \sqrt{\alpha} &= \frac{(\sqrt{\alpha} - 1)^2}{(1 + x_{2n+1})(1 + x_{2n})}(x_{2n} - \sqrt{\alpha}). \end{aligned} \quad (3.18)$$

for all positive integers n . Since $x_1 > \alpha$ and $x_2 < \alpha$, we can show by induction and the expressions (3.18) that

$$x_{2n-1} > \sqrt{\alpha} \quad \text{and} \quad x_{2n} < \sqrt{\alpha}$$

for all positive integers n . ■

Lemma 3.2

For every positive integer n , we have

$$x_{2n+1} - x_{2n-1} = \frac{2(\alpha - x_{2n-1}^2)}{1 + \alpha + 2x_{2n-1}} \quad \text{and} \quad x_{2n+2} - x_{2n} = \frac{2(\alpha - x_{2n}^2)}{1 + \alpha + 2x_{2n}}. \quad (3.19)$$

Proof of Lemma 3.2. We note that

$$\begin{aligned} x_{n+1} - x_{n-1} &= \frac{\alpha + x_n}{1 + x_n} - x_{n-1} \\ &= \frac{\alpha + \frac{\alpha + x_{n-1}}{1 + x_{n-1}}}{1 + \frac{\alpha + x_{n-1}}{1 + x_{n-1}}} - x_{n-1} \\ &= \frac{\alpha(1 + x_{n-1}) + \alpha + x_{n-1}}{1 + x_{n-1} + \alpha + x_{n-1}} - x_{n-1} \\ &= \frac{2(\alpha - x_{n-1}^2)}{1 + \alpha + 2x_{n-1}}. \end{aligned}$$

Hence it is easily seen that the expressions (3.19) follows from this. ■

It is time to return to the proof of Problem 3.17.

- (a) Let n be a positive integer. By Lemma 3.1, we have $x_{2n-1}^2 - \alpha > 0$. Hence it follows from this and Lemma 3.2 that $x_{2n+1} < x_{2n-1}$.
- (b) Similarly, Lemma 3.1 implies that $x_{2n}^2 - \alpha < 0$ and we obtain from Lemma 3.2 that $x_{2n+1} > x_{2n}$.
- (c) By Lemma 3.1 and part (a), $\{x_{2n-1}\}$ is monotonically decreasing, so Theorem 3.14 (Monotone Convergence Theorem) shows that $\{x_{2n-1}\}$ converges. Similarly, since Lemma 3.1 and part (b) imply that $\{x_{2n}\}$ is monotonically increasing, we obtain from Theorem 3.14 (Monotone Convergence Theorem) that $\{x_{2n}\}$ converges. Furthermore, it can be deduced easily from the expressions (3.19) that

$$\lim_{n \rightarrow \infty} x_{2n-1} = \lim_{n \rightarrow \infty} x_{2n} = \sqrt{\alpha}. \quad (3.20)$$

By the triangle inequality, we have

$$|x_m - x_n| \leq |x_m - \sqrt{\alpha}| + |\sqrt{\alpha} - x_n| \quad (3.21)$$

for positive integers m and n . Hence it follows from the limits (3.20) and the inequality (3.21) that $\{x_n\}$ is a Cauchy sequence. By Theorem 3.11(c), the sequence $\{x_n\}$ converges. Let $\lim x_n = x$. Then we have

$$\begin{aligned} \lim_{n \rightarrow \infty} x_{n+1} &= \lim_{n \rightarrow \infty} \left(\frac{\alpha + x_n}{1 + x_n} \right) \\ x &= \frac{\alpha + x}{1 + x} \\ x &= \pm \sqrt{\alpha}. \end{aligned}$$

Since $x_n > 0$ for every positive integer n , we have $x > 0$ and hence $x = \sqrt{\alpha}$.

- (d) Put $\epsilon_n = |x_n - \sqrt{\alpha}|$. We know from the expression (3.17) that

$$\epsilon_{n+1} = \frac{|1 - \sqrt{\alpha}|}{1 + x_n} \epsilon_n = \frac{(1 - \sqrt{\alpha})^2}{(1 + x_n)(1 + x_{n-1})} \epsilon_{n-1}.$$

By the definition of x_n , we have $(1 + x_{n+1})(1 + x_n) = 1 + \alpha + 2x_n > 1 + \alpha$ so that

$$\epsilon_{n+1} = \frac{(1 - \sqrt{\alpha})^2}{(1 + x_n)(1 + x_{n-1})} \epsilon_{n-1} < \frac{(1 - \sqrt{\alpha})^2}{1 + \alpha} \epsilon_{n-1} = \frac{\sqrt{\alpha} - 1}{\sqrt{\alpha} + 1} \epsilon_{n-1}$$

which implies that

$$\epsilon_{2n+1} < \left(\frac{\sqrt{\alpha} - 1}{\sqrt{\alpha} + 1} \right)^n \epsilon_1 \quad \text{and} \quad \epsilon_{2n+2} < \left(\frac{\sqrt{\alpha} - 1}{\sqrt{\alpha} + 1} \right)^n \epsilon_2, \quad (3.22)$$

where n is a positive integer.

To compare the rapidity of convergence of the process with the one described in Problem 3.16, we take the same example that $\alpha = 3$ and $x_1 = 2$. Then we have $\frac{\sqrt{3}-1}{\sqrt{3}+1} < \frac{3}{10}$, $\epsilon_1 = |2 - \sqrt{3}| < \frac{3}{10}$ and $\epsilon_2 = |\frac{5}{3} - \sqrt{3}| < \frac{1}{10}$. By these and the inequalities (3.22), we have

$$\epsilon_{2n+1} < \left(\frac{3}{10} \right)^{n+1} \quad \text{and} \quad \epsilon_{2n+2} < \frac{1}{10} \cdot \left(\frac{3}{10} \right)^n,$$

where n is a positive integer. Put $n = 2$ in the above estimates, we have

$$\epsilon_5 < \left(\frac{3}{10} \right)^3 \quad \text{and} \quad \epsilon_6 < \frac{1}{10} \cdot \left(\frac{3}{10} \right)^2.$$

Now it is easy to see that the rapidity of convergence in Problem 3.16 is much faster than the one in this problem.

This completes the analysis of the problem. ■

Problem 3.18

Rudin Chapter 3 Exercise 18.

Proof. Firstly, it is trivial to check that $p \geq 2$ because if $p = 1$, then $x_{n+1} = \alpha$ which is meaningless. Secondly, the recursion formula here reduces to the recursion formula in Problem 3.16 when $p = 2$. Thirdly, if $\lim_{n \rightarrow \infty} x_n$ exists and let it be x , then it can be shown from the recursion formula that $x = \sqrt[p]{\alpha}$.

To prove that $\{x_n\}$ converges, we show that $x_n > \sqrt[p]{\alpha}$ for all positive integers n and $\{x_n\}$ is monotonically decreasing in the following two lemmata:

Lemma 3.3

Suppose that $\alpha > 0$ and $x_1 > \sqrt[p]{\alpha}$. For every positive integer n , we have $x_n > \sqrt[p]{\alpha}$.

Proof of Lemma 3.3. Assume that $x_k > \sqrt[p]{\alpha}$ for some positive integer k . By the binomial theorem, if $0 < x < 1$, then we have

$$(1 - x)^p = 1 - px + \frac{p(p-1)}{2}x^2 - \cdots > 1 - px. \quad (3.23)$$

Let $y = 1 - x$. The inequality (3.23) becomes

$$y^p > 1 - p(1 - y). \quad (3.24)$$

Next, we put $y = \frac{\sqrt[p]{\alpha}}{x_k}$ into the inequality (3.24) to get

$$\begin{aligned} \frac{\alpha}{x_k^p} &> 1 - p\left(1 - \frac{\sqrt[p]{\alpha}}{x_k}\right) \\ p\left(1 - \frac{\sqrt[p]{\alpha}}{x_k}\right) &> 1 - \frac{\alpha}{x_k^p} \\ x_k - \sqrt[p]{\alpha} &> \frac{1}{p}x_k - \frac{\alpha}{px_k^{p-1}} \\ \frac{p-1}{p}x_k + \frac{\alpha}{px_k^{p-1}} &> \sqrt[p]{\alpha} \\ x_{k+1} &> \sqrt[p]{\alpha}, \end{aligned}$$

completing the proof of the lemma. ■

Lemma 3.4

The sequence $\{x_n\}$ is monotonically decreasing.

Proof of Lemma 3.4. It is clear from the definition of x_n that

$$x_{n+1} - x_n = -\frac{1}{p}x_n + \frac{\alpha}{px_n^{p-1}} = \frac{1}{px_n^{p-1}}(\alpha - x_n^p)$$

for every positive integer n . By Lemma 3.3, we have $\alpha - x_n^p < 0$ so that

$$x_{n+1} - x_n < 0$$

for every positive integer n . In other words, the sequence $\{x_n\}$ is monotonically decreasing. ■

Now we can continue our proof of Problem 3.18. By Lemmae 3.3 and 3.4, the sequence $\{x_n\}$ is monotonically decreasing and bounded below by $\sqrt[m]{\alpha}$. Hence it follows from Theorem 3.14 (Monotone Convergence Theorem) that $\{x_n\}$ converges and the analysis preceding Lemma 3.3 gives the limit of it must be $\sqrt[m]{\alpha}$. This completes the proof of the problem. $\blacksquare$

3.4 A representation of the Cantor set

Problem 3.19

Rudin Chapter 3 Exercise 19.

Proof. Let $E = \left\{x(a) \mid x(a) = \sum_{n=1}^{\infty} \frac{\alpha_n}{3^n}, \alpha_n \in \{0, 2\}\right\}$. Recall that the Cantor set P is defined by

$$P = [0, 1] \setminus \bigcup_{m=1}^{\infty} \bigcup_{k=0}^{3^{m-1}-1} \left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right),$$

see equation (2.24) on [21, p. 42], see Figure 3.1.ⁱ



Figure 3.1: The Cantor set.

In other words, $x \notin P$ if and only if $x \in \left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right)$ for some $m = 1, 2, \dots$ and $k = 0, 1, \dots, 3^{m-1} - 1$.

Suppose that $b = \{\beta_n\}$ and $x(b) = \sum_{n=1}^{\infty} \frac{\beta_n}{3^n}$, where $\beta_n \in \{0, 1, 2\}$. We want to show that $x(b) \notin P$ if and only if $\beta_n = 1$ for some positive integer n . Then the previous paragraph says that $x(b) \notin P$ if and only if

$$\sum_{n=1}^{\infty} \frac{\beta_n}{3^n} \in \left(\frac{3k+1}{3^m}, \frac{3k+2}{3^m}\right) \quad (3.25)$$

for some positive integer m and $k = 0, 1, \dots, 3^{m-1} - 1$. Fix this m and it is obvious that the relation (3.25) is equivalent to

$$\sum_{n=1}^{\infty} \frac{\beta_n}{3^{n-m}} \in (3k+1, 3k+2). \quad (3.26)$$

Since

$$\sum_{n=1}^{\infty} \frac{\beta_n}{3^{n-m}} = \sum_{n=1}^{m-1} \beta_n 3^{m-n} + \beta_m + \sum_{n=m+1}^{\infty} \frac{\beta_n}{3^{n-m}}$$

and each $m-n$ is a positive integer for $n = 1, 2, \dots, m-1$, we have each 3^{m-n} is divisible by 3 and then we have

$$\sum_{n=1}^{m-1} \beta_n 3^{m-n} = 3N \quad (3.27)$$

ⁱThe figure can be found in https://en.wikipedia.org/wiki/Cantor_set.

for some positive integer N . Let $\gamma_m = \beta_m + \sum_{n=m+1}^{\infty} \frac{\beta_n}{3^{n-m}}$, so we have from the relation (3.26) and the expression (3.27) that

$$3N + \gamma_m \in (3k + 1, 3k + 2). \quad (3.28)$$

Since $0 \leq \beta_n \leq 2$ for all n , we get from Theorem 3.26 that

$$0 \leq \gamma_m \leq 2 + \sum_{n=m+1}^{\infty} \frac{2}{3^{n-m}} = 2 \left(1 + \frac{1}{3} + \frac{1}{3^2} + \cdots \right) = 3.$$

If $\gamma_m = 0$ or 3 , then $3N + \gamma_m = 3N$ or $3N + 3$ which contradicts the relation (3.28). In fact, the bounds of γ_m force that $N = k$ and $1 < \gamma_m < 2$. Since

$$0 \leq \sum_{n=m+1}^{\infty} \frac{\beta_n}{3^{n-m}} \leq 1,$$

we must have $0 < \beta_m < 2$ for this fixed positive integer m . However, we acquire $\beta_m \in \{0, 1, 2\}$ which implies that $\beta_m = 1$ for a positive integer m .

Hence we have shown that $x(b) \notin P$ if and only if $\beta_n = 1$ for some positive integer n and this means that

$$E = P$$

as required. We finish the proof of the problem. ■

3.5 Cauchy sequences and the completions of metric spaces

Problem 3.20

Rudin Chapter 3 Exercise 20.

Proof. Given $\epsilon > 0$, there is an integer N_1 such that $d(p_{n_i}, p) < \frac{\epsilon}{2}$ for $n_i \geq N_1$. Since $\{p_n\}$ is a Cauchy sequence, there is an integer N_2 such that $d(p_n, p_m) < \frac{\epsilon}{2}$ for $m, n \geq N_2$. Put $N = \max(N_1, N_2)$. Then for all $n \geq N$, we have

$$d(p_n, p) \leq d(p_n, p_{n_i}) + d(p_{n_i}, p) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence the full sequence $\{p_n\}$ converges to p , completing the proof of the problem. ■

Problem 3.21

Rudin Chapter 3 Exercise 21.

Proof. Let $E = \bigcap_{n=1}^{\infty} E_n$. Since each E_n is nonempty, we can construct a sequence $\{p_n\}$, where $p_n \in E_n$. Since $E_m \subseteq E_n$ if $m \geq n$, we have

$$p_m \in E_n \quad (3.29)$$

if $m \geq n$.

We first show that the sequence $\{p_n\}$ is convergent. Given that $\epsilon > 0$. Since each E_n is bounded, we know from Definition 3.9 that each $\text{diam } E_n$ is well-defined. Besides, since $\text{diam } E_n \rightarrow 0$ as $n \rightarrow \infty$, there is an integer N such that

$$\text{diam } E_n < \epsilon$$

for all $n \geq N$. In particular, we take $n = N$ and we obtain from Definition 3.9 that $S_n = \{d(p, q) \mid p, q \in E_n\}$ and

$$d(p, q) \leq \sup S_N = \text{diam } E_N < \epsilon \quad (3.30)$$

for $p, q \in E_N$.^j Since $E_n \supseteq E_{n+1}$, we have $E_m \subseteq E_n \subseteq E_N$ so that

$$p_m, p_n \in E_N \quad (3.31)$$

for any integers m, n with $m \geq n \geq N$. Therefore it follows from the inequality (3.30) and the relation (3.31) that $d(p_n, p_m) < \epsilon$ for all $m \geq n \geq N$ which shows that $\{p_n\}$ is a Cauchy sequence by Definition 3.8. Since X is a complete metric space, Definition 3.12 ensures that the sequence $\{p_n\}$ converges to a point $p \in X$.

Next, we prove that $E = \{p\}$. To this end, we first show that $p \in E$. Define A to be the subset of $\mathbb{N}$ such that $n \in A$ if and only if $p_n = p$. We also define B to be the complement of A so that $n \in B$ if and only if $p_n \neq p$. By definition, we know that $n \in A$ if and only if $p \in E_n$. (However, $n \in B$ does not imply that $p \notin E_n$.) Now there are two cases for consideration:

- **Case (i): B is finite.** Then we have $B = \{n_1, n_2, \dots, n_k\}$, where $n_1 < n_2 < \dots < n_k$. This implies that $p_n = p$ for all $n \geq n_k + 1$ and thus $p \in E_n$ for all $n \geq n_k + 1$. Since

$$E_{n_{k+1}} \subseteq E_{n_k} \subseteq E_{n_{k-1}} \subseteq \dots \subseteq E_{n_1},$$

we have $p \in E_n$ for all $n \in B$. Hence we have $p \in E$ in this case.

- **Case (ii): B is countable.** We want to show that p is a limit point of each E_n , so we fix the integer n first. Since $p_m \rightarrow p$ as $m \rightarrow \infty$, we get from Theorem 3.2(a) that if $N_r(p)$ is any neighborhood of p , then there is an integer N such that

$$p_m \in N_r(p)$$

for all $m \geq N$. Since B is countable, we can pick an element $m \in B$ with $m \geq n$ and $m \geq N$ such that the relation (3.29) yields $p_m \in E_n$. By definition, we have $p_m \neq p$ and hence it follows from Definition 2.18(b) that p is a limit point of each E_n . Since E_n is closed for each positive integer n , we have $p \in E_n$ for each positive integer n and thus $p \in E$ in this case too.

Now we have shown that $p \in E$ and we shall prove the uniqueness of p . Assume that $p' \in E$ but $p' \neq p$. Then we have $d(p, p') > 0$ and $p, p' \in E_n$ for every positive integer n . By Theorem 1.20(b), there is $q \in \mathbb{Q}$ which does not depend on n such that $0 < q < d(p, p')$. Therefore Definition 3.9 implies that

$$\text{diam } E_n \geq q > 0$$

for every positive integer n , a contradiction. Hence we must have $p = p'$, i.e., $E = \{p\}$, completing the proof of the problem. ■

Problem 3.22

Rudin Chapter 3 Exercise 22.

Proof. Basically, we follow the idea of proof of Problem 2.30. Let G be an open set of X . Since G_1 is a dense subset of X , there exists $p \in G_1$ such that $p \in G$. Thus the set

$$F_1 = G_1 \cap G$$

is nonempty. Since G_1 is open in X , F_1 must be open in X by Theorem 2.24(c). Let $p_1 \in F_1$. (Here we don't assume that $p = p_1$.) Since F_1 is an open subset, we have

$$E_1 = N_{r_1}(p_1) \subseteq F_1 \subseteq G_1$$

for some $r_1 > 0$. Without loss of generality, we may assume that

$$\overline{E_1} = \overline{N_{r_1}(p_1)} \subseteq F_1 \subseteq G_1.$$

^jIn fact, the inequality (3.30) holds for all $n \geq N$ with E_N replaced by E_n .

Since G_2 is a dense subset of X , the set

$$F_2 = G_2 \cap E_1$$

is a nonempty open subset of X . Let $p_2 \in F_2$. Then we can choose $r_2 > 0$ small enough such that

$$\overline{E_2} = \overline{N_{r_2}(p_2)} \subseteq F_2 \subseteq G_2.$$

By definition, we have

$$\overline{E_2} = \overline{N_{r_2}(p_2)} \subseteq F_2 = G_2 \cap E_1 = G_2 \cap N_{r_1}(p_1) \subset \overline{N_{r_1}(p_1)} = \overline{E_1}.$$

Now we can continue this process to obtain the following shrinking sequence

$$\cdots \subset \overline{E_3} \subset \overline{E_2} \subset \overline{E_1}. \quad (3.32)$$

Since each E_n is actually a neighborhood, $\overline{E_n}$ is closed and bounded. Furthermore, the sequence (3.32) implies that $\{r_n\}$ is a strictly decreasing sequence of positive real numbers (here 0 is the greatest lower bound of $\{r_n\}$) and it follows from Theorem 3.14 (Monotone Convergence Theorem) that $\lim_{n \rightarrow \infty} r_n = 0$. Now it is clear that, for each positive integer n ,

$$\text{diam } \overline{E_n} = \sup\{d(p, q) \mid p, q \in \overline{E_n}\} = 2r_n,$$

so we must have

$$\lim_{n \rightarrow \infty} \text{diam } \overline{E_n} = \lim_{n \rightarrow \infty} (2r_n) = 0.$$

Since X is a nonempty complete metric space, Problem 3.21 implies that

$$\bigcap_{n=1}^{\infty} \overline{E_n} = \{p\}$$

for some $p \in X$. Since $p \in \overline{E_n} \subseteq G_n$ for each positive integer n , we have $p \in \bigcap_{n=1}^{\infty} G_n$, completing the proof of the problem. ■

Problem 3.23

Rudin Chapter 3 Exercise 23.

Proof. We follow the hint. For any positive integers m and n , we have

$$d(p_n, q_n) \leq d(p_n, p_m) + d(p_m, q_m) + d(q_m, q_n) \quad \text{or} \quad d(p_m, q_m) \leq d(p_n, p_m) + d(p_n, q_n) + d(q_m, q_n) \quad (3.33)$$

which implies that

$$|d(p_n, q_n) - d(p_m, q_m)| \leq d(p_n, p_m) + d(q_m, q_n).$$

Since $\{p_n\}$ and $\{q_n\}$ are Cauchy sequences in a metric space X , there is an integer N such that

$$d(p_n, p_m) < \frac{\epsilon}{2} \quad \text{and} \quad d(q_n, q_m) < \frac{\epsilon}{2} \quad (3.34)$$

for all $m, n \geq N$. It follows from the inequalities (3.33) and (3.34) that

$$|d(p_n, q_n) - d(p_m, q_m)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for $m, n \geq N$. Therefore the sequence $\{d(p_n, q_n)\}$ is a Cauchy sequence in $\mathbb{R}$. By Theorem 3.11(c), the sequence $\{d(p_n, q_n)\}$ is convergent. This ends the proof of the problem. ■

Problem 3.24

Rudin Chapter 3 Exercise 24.

Proof.

(a) We check that this satisfies Definition 2.3.

- **Reflexive:** It is clear that $\{p_n\} \sim \{p_n\}$ because $d(p_n, p_n) = 0$ by definition.
- **Symmetric:** Since $d(p_n, q_n) = d(q_n, p_n)$, we have $\{p_n\} \sim \{q_n\}$ implies $\{q_n\} \sim \{p_n\}$.
- **Transitive:** Suppose that $\{p_n\}, \{q_n\}$ and $\{r_n\}$ are Cauchy sequences such that $\{p_n\} \sim \{q_n\}$ and $\{q_n\} \sim \{r_n\}$. Then we have $\lim_{n \rightarrow \infty} d(p_n, q_n) = 0$ and $\lim_{n \rightarrow \infty} d(q_n, r_n) = 0$. The triangle inequality gives

$$d(p_n, r_n) \leq d(p_n, q_n) + d(q_n, r_n)$$

which yields $\lim_{n \rightarrow \infty} d(p_n, r_n) = 0$. Thus we have $\{p_n\} \sim \{r_n\}$.

Hence this is an equivalence relation.

(b) Let $\{p_n\}, \{p'_n\} \in P$ and $\{q_n\}, \{q'_n\} \in Q$. By the triangle inequality, we have

$$d(p'_n, q'_n) \leq d(p'_n, p_n) + d(p_n, q_n) + d(q_n, q'_n)$$

which gives

$$\lim_{n \rightarrow \infty} d(p'_n, q'_n) \leq \lim_{n \rightarrow \infty} d(p'_n, p_n) + \lim_{n \rightarrow \infty} d(p_n, q_n) + \lim_{n \rightarrow \infty} d(q_n, q'_n).$$

Since $\{p_n\} \sim \{p'_n\}$ and $\{q_n\} \sim \{q'_n\}$, we have $\lim_{n \rightarrow \infty} d(p'_n, p_n) = \lim_{n \rightarrow \infty} d(q'_n, q_n) = 0$. Therefore, we have

$$\lim_{n \rightarrow \infty} d(p'_n, q'_n) \leq \lim_{n \rightarrow \infty} d(p_n, q_n). \quad (3.35)$$

Now the roles of p_n, q_n and p'_n, q'_n can be exchanged in the inequality (3.35), so we have

$$\lim_{n \rightarrow \infty} d(p_n, q_n) \leq \lim_{n \rightarrow \infty} d(p'_n, q'_n).$$

Hence we have $\lim_{n \rightarrow \infty} d(p_n, q_n) = \lim_{n \rightarrow \infty} d(p'_n, q'_n)$, i.e., $\Delta(P, Q)$ is unchanged.

Next, we check Δ is a metric in X^* .

- **Condition 1:** If $P \neq Q$, then we have $\{p_n\} \not\sim \{q_n\}$ for all $\{p_n\} \in P$ and $\{q_n\} \in Q$. By the definition from part (a), $\{p_n\}$ and $\{q_n\}$ are *not* equivalent so that $\lim_{n \rightarrow \infty} d(p_n, q_n) > 0$. By the definition of Δ , we have

$$\Delta(P, Q) > 0.$$

Since $d(p_n, p_n) = 0$ for every $\{p_n\} \in P$, we have

$$\Delta(P, P) = \lim_{n \rightarrow \infty} d(p_n, p_n) = 0.$$

- **Condition 2:** Since $d(p_n, q_n) = d(q_n, p_n)$ for every $\{p_n\} \in P$ and $\{q_n\} \in Q$, we have

$$\Delta(P, Q) = \Delta(Q, P).$$

- **Condition 3:** Now it is easily seen that the inequality

$$\Delta(P, Q) \leq \Delta(P, R) + \Delta(R, Q)$$

follows directly from the triangle inequality $d(p_n, q_n) \leq d(p_n, r_n) + d(r_n, q_n)$ for every $\{p_n\} \in P$, $\{q_n\} \in Q$ and $\{r_n\} \in R$.

(c) Let $\{P_n\}$ be a Cauchy sequence in X^* with the metric Δ . To show that $\{P_n\}$ converges in X^* , it is equivalent to showing that *there exists* a $P \in X^*$ such that $\Delta(P_n, P) \rightarrow 0$ as $n \rightarrow \infty$.

- **Existence of P .** For each positive integer n , recall that elements of each equivalence class P_n are Cauchy sequences in X with the metric d , so we can choose

$$\{p_{n1}, p_{n2}, p_{n3}, \dots\} \in P_n.$$

Given that $\epsilon > 0$. Since $\{P_n\}$ is a Cauchy sequence in X^* , there exists an integer $N_\epsilon^{(1)}$ such that

$$\Delta(P_r, P_s) < \frac{\epsilon}{4}$$

for every $r, s \geq N_\epsilon^{(1)}$. By the definition of Δ , there exists a positive integer $N_\epsilon^{(2)}$ such that

$$d(p_{rt}, p_{st}) < \frac{\epsilon}{4} \quad (3.36)$$

for $t \geq N_\epsilon^{(2)}$.

Now for each positive integer k , the sequence $\{p_{k1}, p_{k2}, \dots\} \in P_k$ is a Cauchy sequence in X , so there exists an integer N_k such that

$$d(p_{kn}, p_{km}) < \frac{1}{2^k} \quad (3.37)$$

if $m, n \geq N_k$.

Lemma 3.5

Suppose that $p_k = p_{kN_k}$ for each positive integer k . We define the sequence $\{p_1, p_2, \dots\}$ which is a Cauchy sequence in X .

Proof of Lemma 3.5. We note that the integer N_k depends solely on k (not the arbitrary constant ϵ), so the sequence $\{p_1, p_2, \dots\}$ is well-defined. Let $N_\epsilon^{(3)} = \max(N_\epsilon^{(1)}, \lceil 2 + \frac{\log \frac{1}{\epsilon}}{\log 2} \rceil)$ and $n = \max(N_r, N_s, N_\epsilon^{(2)})$. (We denote $\lceil x \rceil$ to be the function which gives the smallest integer greater than or equal to x .)

If r and s are *fixed* positive integers such that $r, s \geq N_\epsilon^{(3)}$, then we have $\frac{1}{2^r} \leq \frac{\epsilon}{4}$, $\frac{1}{2^s} \leq \frac{\epsilon}{4}$ and it follows from the inequalities (3.36), (3.37) and the triangle inequality that

$$\begin{aligned} d(p_r, p_s) &= d(p_{rN_r}, p_{sN_s}) \\ &\leq d(p_{rN_r}, p_{rn}) + d(p_{rn}, p_{sn}) + d(p_{sn}, p_{sN_s}) \\ &< \frac{1}{2^r} + \frac{1}{2^s} + d(p_{rn}, p_{sn}) \\ &< \frac{1}{2^r} + \frac{1}{2^s} + \frac{\epsilon}{4} \\ &< \frac{\epsilon}{4} + \frac{\epsilon}{4} + \frac{\epsilon}{4} \\ &< \epsilon. \end{aligned}$$

Hence $\{p_1, p_2, \dots\}$ is a Cauchy sequence in X , completing the proof of Lemma 3.5. ■

By Lemma 3.5, we can define P to be the equivalent class containing the Cauchy sequence $\{p_1, p_2, \dots\}$ and it is obvious that $P \in X^*$.

- **Convergence of $\{P_m\}$.** We want to show that

$$\Delta(P_m, P) \rightarrow 0$$

as $m \rightarrow \infty$. Given that $\epsilon > 0$. By the proof of Lemma 3.5, we see that if $m, t \geq N_\epsilon^{(3)}$, then we have

$$d(p_m, p_t) < \frac{3\epsilon}{4}. \quad (3.38)$$

We put $m \geq N_\epsilon^{(3)}$. Since $N_\epsilon^{(3)} = \max(N_\epsilon^{(1)}, \lceil 2 + \frac{\log \frac{1}{\epsilon}}{\log 2} \rceil)$, we have $\frac{1}{2^m} \leq \frac{\epsilon}{4}$. Then we obtain from the triangle inequality and the inequality (3.37) that

$$\begin{aligned} d(p_{mt}, p_t) &\leq d(p_{mt}, p_m) + d(p_m, p_t) \\ &= d(p_{mt}, p_{mN_m}) + d(p_m, p_t) \\ &< \frac{1}{2^m} + d(p_m, p_t) \end{aligned} \quad (3.39)$$

if $t \geq N_m$.

Let $N_\epsilon^{(4)} = \max(N_m, N_\epsilon^{(3)})$. Now for every $t \geq N_\epsilon^{(4)}$, since $m \geq N_\epsilon^{(3)}$ and $t \geq N_\epsilon^{(4)} \geq N_\epsilon^{(3)}$, it follows from the inequality (3.38) that

$$d(p_m, p_t) < \frac{3\epsilon}{4}.$$

Recall that this chosen m gives $\frac{1}{2^m} \leq \frac{\epsilon}{4}$ and $t \geq N_\epsilon^{(4)} \geq N_m$, so we obtain from these and the inequality (3.39) that

$$d(p_{mt}, p_t) < \frac{1}{2^m} + d(p_m, p_t) < \frac{\epsilon}{4} + \frac{3\epsilon}{4} = \epsilon \quad (3.40)$$

if $m \geq N_\epsilon^{(3)}$ and $t \geq N_\epsilon^{(4)}$. Therefore we can easily see from the inequality (3.40) and the definition $\Delta(P_m, P) = \lim_{t \rightarrow \infty} d(p_{mt}, p_t)$ that $\Delta(P_m, P) \rightarrow 0$ as $m \rightarrow \infty$.

Hence we have shown that a Cauchy sequence $\{P_n\}$ in X^* converges to $P \in X^*$. By Definition 3.12, X^* is complete.

- (d) The sequence $\{p, p, \dots\}$ is the required Cauchy sequence. Since $\{p, p, \dots\} \in P_p$ and $\{q, q, \dots\} \in P_q$ for all $p, q \in X$, we get from the part (b) that

$$\Delta(P_p, P_q) = \lim_{n \rightarrow \infty} d(p, q) = d(p, q).$$

In other words, this shows that $\varphi : X \rightarrow X^*$ defined by $\varphi(p) = P_p$ is an isometry.

- (e) To prove that $\varphi(X)$ is dense in X^* , we must show that every neighborhood of $P \in X^*$ contains $\varphi(p) \in \varphi(X)$. To this end, let $P \in X^*$ and $\{p_n\} \in P$. Given that $\epsilon > 0$. Since $\{p_n\}$ is a Cauchy sequence in X , there exists a positive integer N such that $d(p_m, p_n) < \frac{\epsilon}{2}$ for all $m, n \geq N$. By part (d), we have

$$\varphi(p_N) = P_{p_N}.$$

Recall that P_{p_N} contains the Cauchy sequence all of whose terms are p_N , so it follows from this and part (b) that

$$\Delta(P, \varphi(p_N)) = \Delta(P, P_{p_N}) = \lim_{n \rightarrow \infty} d(p_n, p_N) \leq \frac{\epsilon}{2} < \epsilon.$$

In other words, it means that the neighborhood of P with radius ϵ contains an element $\varphi(p_N) \in \varphi(X)$. Hence $\varphi(X)$ is dense in X^* .

Suppose that X is complete. Let $P \in X^*$ and $\{p_n\} \in P$. Since $\{p_n\}$ is a Cauchy sequence in the complete metric space X , we have $p_n \rightarrow p$ for some $p \in X$. Thus we have

$$\Delta(P, P_p) = \lim_{n \rightarrow \infty} d(p_n, p) = 0.$$

Hence we have $P = P_p$ and then $\varphi(X) = X^*$.

This completes the proof of the problem. ■

Problem 3.25

Rudin Chapter 3 Exercise 25.

Proof. Now we have $X = \mathbb{Q}$. By Problem 3.24(e), $\mathbb{Q}$ is dense in $\mathbb{Q}^*$. By the footnote **c** in Chapter 2, we have

$$\mathbb{Q}^* = \overline{\mathbb{Q}} = \mathbb{R},$$

finishing the proof of the problem. ■

CHAPTER 4

Continuity

4.1 Properties of continuous functions

Problem 4.1

Rudin Chapter 4 Exercise 1.

Proof. The answer is no! For example, let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(0) = 1$ and $f(x) = 0$ for $x \in \mathbb{R} \setminus \{0\}$. Then it is easy to check that

$$\lim_{h \rightarrow 0} [f(x+h) - f(x-h)] = 0$$

for every $x \in \mathbb{R}$. However, f is not continuous at 0. To see this, we pick $\epsilon = \frac{1}{2}$ and for every $\delta > 0$, those $x \in \mathbb{R}$ with $0 < |x - 0| < \delta$ always imply that

$$|f(x) - f(0)| = |0 - 1| = 1 > \frac{1}{2}.$$

By Definition 4.5, f is not continuous at 0. We finish the proof of the problem. ■

Problem 4.2

Rudin Chapter 4 Exercise 2.

Proof. Since $x \in E$ implies $f(x) \in f(E) \subseteq \overline{f(E)}$, we have

$$E \subseteq f^{-1}(\overline{f(E)}).$$

By Theorem 2.27(a), $\overline{f(E)}$ is closed in Y . Thus the corollary of Theorem 4.8 ensures that $f^{-1}(\overline{f(E)})$ is closed in X . By Theorem 2.27(c), we must have $\overline{E} \subseteq f^{-1}(\overline{f(E)})$ and hence we have

$$f(\overline{E}) \subseteq \overline{f(E)}$$

for every set $E \subseteq X$.

To prove the second assertion, we consider the example $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x^2}.$$

Take $E = \mathbb{Z}$, so $\overline{E} = \mathbb{Z}$ and $f(\overline{E}) = f(\mathbb{Z}) = \{1, \frac{1}{2^2}, \frac{1}{3^2}, \dots\}$, but $\overline{f(E)} = \overline{f(\mathbb{Z})} = \{0\} \cup \{1, \frac{1}{2^2}, \frac{1}{3^2}, \dots\}$. Hence we have the desired result that

$$f(\overline{E}) \subsetneq \overline{f(E)},$$

completing the proof of the problem. ■

Problem 4.3*Rudin Chapter 4 Exercise 3.**Proof.* By definition, we have

$$Z(f) = \{p \in X \mid f(p) = 0\} = f^{-1}(\{0\}).$$

Since $\{0\}$ is a closed set in $\mathbb{R}$, the corollary of Theorem 4.8 implies that $Z(f)$ is closed in X . We complete the proof of the problem. ■

Problem 4.4*Rudin Chapter 4 Exercise 4.*

Proof. By definition, the statement “ $f(E)$ is dense in $f(X)$ ” is equivalent to the statement “every point of $f(X)$ is a limit point of $f(E)$ ” which is also equivalent to $f(X) \subseteq (f(E))' \subseteq \overline{f(E)}$. Since it is clear that $f(E) \subseteq f(X)$, the last statement is equivalent to

$$\overline{f(E)} = f(X). \quad (4.1)$$

The direction $\overline{f(E)} \subseteq f(X)$ is obvious. For the other direction, we note from Problem 4.2 that

$$f(\overline{E}) \subseteq \overline{f(E)}$$

for every set $E \subseteq X$. If E is a dense subset of X , then we have $\overline{E} = X$ and this implies that $f(X) \subseteq \overline{f(E)}$ which is the relation (4.1).

Let $p \in X$. Thus it is a limit point of E . Since E is dense in X , there exists a sequence $\{p_n\}$ in E such that $\lim_{n \rightarrow \infty} p_n = p$. Since f and g are continuous at p , it follows from Theorem 4.2 that

$$\lim_{n \rightarrow \infty} f(p_n) = f(p) \quad \text{and} \quad \lim_{n \rightarrow \infty} g(p_n) = g(p).$$

Since $f(p_n) = g(p_n)$ for all positive integers n , we have

$$f(p) = \lim_{n \rightarrow \infty} f(p_n) = \lim_{n \rightarrow \infty} g(p_n) = g(p),$$

completing the proof of the problem. ■

4.2 The extension, the graph and the restriction of a continuous function**Problem 4.5***Rudin Chapter 4 Exercise 5.*

Proof. We have $f : E \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and E is closed in $\mathbb{R}$. By Theorem 2.23, E^c is open in $\mathbb{R}$ and Problem 2.29 tells us that E^c is the union of an at most countable collection of disjoint segments, possibly including $(-\infty, a)$ or (b, ∞) for some $a, b \in \mathbb{R}$.^a We define $g : \mathbb{R} \rightarrow \mathbb{R}$ to be the function such that

$$g(x) = f(x)$$

for all $x \in E$. To define g on E^c , we let (a_n, b_n) be one of the disjoint segments of E^c , where $a_n < b_n$. There are two cases:

^aFor examples, if $E = [0, 1]$, then we have $E^c = (-\infty, 0) \cup (1, \infty)$; if $E = (-\infty, 0] \cup [1, +\infty)$, then we have $E^c = (0, 1)$.

- **Case (i):** E^c does not contain $(-\infty, a)$ or (b, ∞) . We define g on $[a_n, b_n]$ by

$$g(x) = \begin{cases} f(a_n), & \text{if } x = a_n; \\ m(x - a_n) + f(a_n) & \text{if } a_n < x < b_n; \\ f(b_n), & \text{if } x = b_n, \end{cases}$$

where

$$m = \frac{f(b_n) - f(a_n)}{b_n - a_n}.$$

In other words, the graph of g is a straight line connecting $(a_n, f(a_n))$ and $(b_n, f(b_n))$ with slope m , see Figure 4.1 for an illustration.

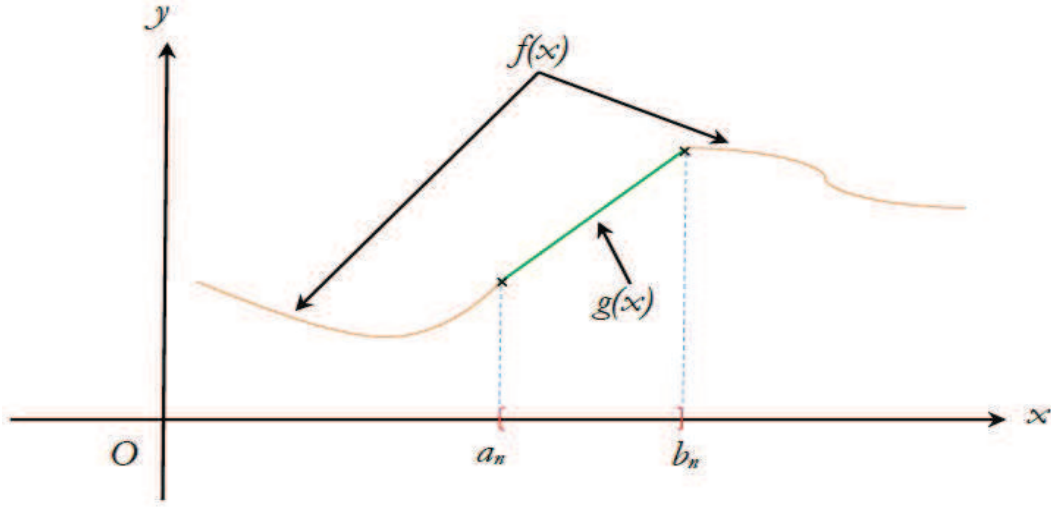


Figure 4.1: The graph of g on $[a_n, b_n]$.

- **Case (ii):** E^c contains $(-\infty, a)$ or (b, ∞) . We consider the case $(-\infty, a)$ only because the case $(b, +\infty)$ can be done similarly. We define g on $(-\infty, a]$ by

$$g(x) = f(a)$$

for all $x \in (-\infty, a]$. In other words, the graph of g is a constant function.

Since a straight line is continuous on its domain, the only points of uncertainty of continuity of $g(x)$ are at the endpoints $x = a_n$ and $x = b_n$. Given $\epsilon > 0$ small enough and suppose that $m \neq 0$. Since f is continuous at a_n , there exists a $\delta > 0$ such that

$$|f(x) - f(a_n)| < \epsilon$$

if $|x - a_n| < \delta$. Let $\delta_1 = \min(\delta, \frac{\epsilon}{|m|})$.^b Then, for $|x - a_n| < \delta_1$, we have

$$\begin{aligned} |g(x) - g(a_n)| &= \begin{cases} |m(x - a_n) + f(a_n) - f(a_n)|, & \text{if } a_n \leq x < a_n + \delta_1; \\ |f(x) - f(a_n)|, & \text{if } a_n - \delta_1 < x < a_n, \end{cases} \\ &= \begin{cases} |m| \cdot |x - a_n|, & \text{if } a_n \leq x < a_n + \delta_1; \\ |f(x) - f(a_n)|, & \text{if } a_n - \delta_1 < x < a_n, \end{cases} \\ &< \begin{cases} \epsilon, & \text{if } a_n \leq x < a_n + \delta_1; \\ \epsilon, & \text{if } a_n - \delta_1 < x < a_n. \end{cases} \end{aligned}$$

^bWe note that m can be negative, so we take its absolute value to make $\frac{\epsilon}{|m|}$ positive.

If $m = 0$, then we take $\delta_1 = \min(\epsilon, \delta)$ in the above analysis and we get

$$\begin{aligned} |g(x) - g(a_n)| &= \begin{cases} |f(a_n) - f(a_n)|, & \text{if } a_n \leq x < a_n + \delta_1; \\ |f(x) - f(a_n)|, & \text{if } a_n - \delta_1 < x < a_n, \end{cases} \\ &= \begin{cases} 0, & \text{if } a_n \leq x < a_n + \delta_1; \\ |f(x) - f(a_n)|, & \text{if } a_n - \delta_1 < x < a_n, \end{cases} \\ &< \begin{cases} \epsilon, & \text{if } a_n \leq x < a_n + \delta_1; \\ \epsilon, & \text{if } a_n - \delta_1 < x < a_n. \end{cases} \end{aligned}$$

Hence we have

$$|g(x) - g(a_n)| < \epsilon$$

if $|x - a_n| < \delta_1$ and then g is continuous at $x = a_n$ by Definition 4.5. Since the continuity of g in the case for the endpoint $x = b_n$ and the case for E^c containing $(-\infty, a)$ or $(b, +\infty)$ can be done similarly as above, we skip the details here.

Note that the word “closed” cannot be omitted in the above result. We modify Example 4.27(c) to be

$$f(x) = \begin{cases} -x - 2, & \text{if } x < 0; \\ x + 2, & \text{if } x > 0. \end{cases}$$

Then f has a *simple discontinuity* at $x = 0$ and is continuous at every other point of the open set $(-\infty, 0) \cup (0, +\infty)$. Since $f(0+) = 2$ and $f(0-) = -2$, this makes f impossible to have a continuous extension.

Suppose that $E \subseteq \mathbb{R}$ is a closed set and $\mathbf{f} : E \rightarrow \mathbb{R}^k$ is a vector-valued function defined by

$$\mathbf{f}(x) = (f_1(x), f_2(x), \dots, f_k(x)),$$

where each $f_i : E \rightarrow \mathbb{R}$ is a real function defined on E . By Theorem 4.10(a), $\mathbf{f}$ is continuous on E if and only if each of the functions $f_1, \dots, f_k$ is continuous on E . By the previous analysis, each f_i has a continuous extension g_i . Therefore, the vector-valued function $\mathbf{g} : \mathbb{R} \rightarrow \mathbb{R}^k$ defined by

$$\mathbf{g}(x) = (g_1(x), g_2(x), \dots, g_k(x))$$

is continuous on $\mathbb{R}$ by Theorem 4.10(a). For each $1 \leq i \leq k$, we have $g_i(x) = f_i(x)$ for all $x \in E$ and this implies that

$$\mathbf{g}(x) = \mathbf{f}(x)$$

for all $x \in E$. This completes the proof of the problem. ■

Problem 4.6

Rudin Chapter 4 Exercise 6.

Proof. Let $f : E \rightarrow Y$, E and Y are metric spaces. By definition, since $E \times f(E) = \{(x, f(y)) \mid x, y \in E\}$, we have

$$\text{graph}(f) = \{(x, f(x)) \mid x \in E\} \subseteq E \times f(E) \subseteq E \times Y.$$

Before we start to prove the result, there are a few points to note. We know that E and Y may *not* be subsets of $\mathbb{R}^k$ for some positive integer k , so we *cannot* apply any result relevant to $\mathbb{R}^k$ (e.g. Theorems 2.41, 4.10, 4.15 and etc.) directly in this problem. Instead, the strategy we use here is that we *define* a metric on the set $E \times Y$ and we consider the mapping $g : E \rightarrow E \times f(E)$ defined by

$$g(x) = (x, f(x)).$$

If we can show that g is continuous, then $g(E) = \text{graph}(f)$ is compact by Theorem 4.14.

The first step is to define the metric in $E \times Y$ induced by the metrics d_E and d_Y . Let $p_1, p_2 \in E \times Y$. Thus we have $p_1 = (x_1, y_1)$ and $p_2 = (x_2, y_2)$ for some $x_1, x_2 \in E$ and $y_1, y_2 \in Y$. We define

$$d_{E \times Y}(p_1, p_2) = d_E(x_1, x_2) + d_Y(y_1, y_2), \quad (4.2)$$

where d_E and d_Y are the metrics in the spaces E and Y respectively. We must check Definition 2.15:

- Since $p_1 = p_2$ if and only if $x_1 = x_2$ and $y_1 = y_2$ if and only if $d_E(x_1, x_2) = 0$ and $d_Y(y_1, y_2) = 0$, we have $d_{E \times Y}(p_1, p_2) = 0$ if and only if $p_1 = p_2$.
- Since $d_E(x_1, x_2) = d_E(x_2, x_1)$ and $d_Y(y_1, y_2) = d_Y(y_2, y_1)$, we have

$$d_{E \times Y}(p_1, p_2) = d_{E \times Y}(p_2, p_1).$$

- For any $p_1, p_2, p_3 \in E \times Y$, we have

$$\begin{aligned} d_{E \times Y}(p_1, p_2) &= d_E(x_1, x_2) + d_Y(y_1, y_2) \\ &\leq d_E(x_1, x_3) + d_E(x_3, x_2) + d_Y(y_1, y_3) + d_Y(y_3, y_2) \\ &= d_{E \times Y}(p_1, p_3) + d_{E \times Y}(p_3, p_2). \end{aligned}$$

These show that the definition (4.2) is indeed a metric in $E \times Y$. By Example 2.16, we know that every subset Y of a metric space X is a metric in its own right, with the *same* distance function. Hence the expression (4.2) is also a metric in $E \times f(E)$.

The second step is to show that g is continuous on E . Let p be a point of E . Given that $\epsilon > 0$. Since f is continuous at E , it is continuous at p by Definition 4.5. Thus there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \frac{\epsilon}{2} \quad (4.3)$$

if $d_E(x, p) < \delta$ and $x \in E$. We let $\delta' = \min(\frac{\epsilon}{2}, \delta)$. Then we still have the inequality (4.3) if $d_E(x, p) < \delta'$ and $x \in E$. Hence it follows from the definition (4.2) and the inequality (4.3) that

$$d_{E \times Y}(g(x), g(p)) = d_E(x, p) + d_Y(f(x), f(p)) < \delta' + \frac{\epsilon}{2} \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} < \epsilon$$

if $d_E(x, p) < \delta'$ and $x \in E$. Since p is arbitrary, g is continuous on E by Definition 4.5 and then $g(E) = \text{graph}(f)$ is compact by Theorem 4.14.

Conversely, suppose that $\text{graph}(f)$ is compact and $p \in E$. By Theorem 2.34, we know that $\text{graph}(f)$ is a closed subset of $E \times f(E)$. By Theorem 4.6, f is continuous at p if and only if

$$\lim_{x \rightarrow p} f(x) = f(p).$$

The latter statement is equivalent to the fact that

$$\lim_{n \rightarrow \infty} f(p_n) = f(p) \quad (4.4)$$

for every sequence $\{p_n\}$ in E such that $p_n \neq p$ and $\lim_{n \rightarrow \infty} p_n = p$ by Theorem 4.2.

Let $\{p_n\}$ be a sequence in E such that $p_n \neq p$ and $\lim_{n \rightarrow \infty} p_n = p$ and let $q_n = (p_n, f(p_n))$. Then it is clear that $\{q_n\}$ is a sequence in $\text{graph}(f)$. Since $\text{graph}(f)$ is a compact metric space, we obtain from Theorem 3.6(a) that a subsequence

$$\{q_{n_k} = (p_{n_k}, f(p_{n_k}))\}$$

converges to a point $q = (q_1, f(q_1)) \in \text{graph}(f)$. Since $\lim_{n \rightarrow \infty} p_n = p$ if and only if every subsequence of $\{p_n\}$ converges to p (Definition 3.5) and the uniqueness of the limit of a sequence (Theorem 3.2(b)), we must have $q_1 = p$ and then

$$\lim_{k \rightarrow \infty} f(p_{n_k}) = f(q_1) = f(p). \quad (4.5)$$

In fact, the limit (4.5) is true for every subsequence $\{p_{n_j}\}$ converging to p . Otherwise, there was a subsequence $\{p_{n_j}\}$ converging to p but $\lim_{j \rightarrow \infty} f(p_{n_j}) = r \neq f(p)$.^c Then this implies that

$$\lim_{j \rightarrow \infty} q_{n_j} = (p, r) \notin \text{graph}(f)$$

which means the set $\text{graph}(f)$ does not contain its limit point (p, r) , but this certainly contradicts the fact that $\text{graph}(f)$ is closed (Definition 2.18(b)). Therefore the limit (4.5) is true for every subsequence $\{p_{n_j}\}$ converging to p so that Definition 3.5 induces the fact (4.4). Hence, since p is arbitrary, we have f is continuous on E , finishing the proof of the problem. ■

^cIf $r = f(p')$ for some $p' \in E$, then it is easily proven that $p' = p$. So it is reasonable to assume that $r \notin f(E)$.

Problem 4.7*Rudin Chapter 4 Exercise 7.*

Proof. We show the result steps by steps as follows:

- **f is bounded on $\mathbb{R}^2$:** By the A.M. $\geq$ G.M., we have $\frac{x^2+y^4}{2} \geq \sqrt{x^2y^4} = |x|y^2$ so that

$$|f(x, y)| \leq \left| \frac{xy^2}{x^2 + y^4} \right| = \frac{|x|y^2}{x^2 + y^4} \leq \frac{1}{2}$$

for every $(x, y) \in \mathbb{R}^2$. Hence f is bounded on $\mathbb{R}^2$.

- **g is unbounded in every neighborhood of $(0, 0)$:** If we take $x = y^3$, then we have

$$g(x, y) = g(y^3, y) = \frac{y^5}{y^6 + y^6} = \frac{1}{2y}$$

so that $g(x, y) \rightarrow \infty$ as $y \rightarrow 0$. Hence g is unbounded in every neighborhood of $(0, 0)$.

- **f is not continuous at $(0, 0)$:** If we take $x = y^2$, then we have $f(x, y) = f(y^2, y) = \frac{1}{2}$ so that

$$|f(x, y) - f(0, 0)| = \frac{1}{2}$$

for every y . Therefore, f does not satisfy Definition 4.5 when $\epsilon < \frac{1}{2}$ which means that f is not continuous at $(0, 0)$.

- **The restrictions of both f and g to every straight line in $\mathbb{R}^2$ are continuous:**

- **Case (i): The straight line does not pass through the origin.** We have $y = mx + c$ for some $m, c \in \mathbb{R}$ and $c \neq 0$. Let $L : \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{(0, 0)\}$ be defined by

$$L(x) = (x, mx + c)$$

and $E = \{(x, mx + c) \mid x \in \mathbb{R}\}$ be the graph of L . Now we have $f_E = f \circ L : \mathbb{R} \rightarrow \mathbb{R}$. Since the mapping L is obviously continuous on $\mathbb{R}$ by Theorem 4.10(a) and f is continuous on $\mathbb{R}^2 \setminus \{(0, 0)\}$, it follows from Theorem 4.7 that f_E is continuous on $\mathbb{R}$. The case for the continuity of g_E is similar to that of f_E , so we omit its details here.

- **Case (ii): The straight line passes through the origin.** We have $y = mx$ for some $m \in \mathbb{R}$. Let $L' : \mathbb{R} \rightarrow \mathbb{R}^2$ be defined by

$$L'(x) = (x, mx)$$

and $E' = \{(x, mx) \mid x \in \mathbb{R}\}$ be the graph of L' . It is clear that f is continuous at every point except at the origin, so we only need to check the continuity of $f_{E'} = f \circ L' : \mathbb{R} \rightarrow \mathbb{R}$ at the origin. By definition, we have

$$f(x, mx) = \frac{x(mx)^2}{x^2 + (mx)^4} = \frac{m^2x}{1 + m^4x^2}.$$

If $m = 0$, then we have $f(x, 0) = 0$ so that $f_{E'}$ is continuous at the origin because

$$|f(x, 0) - f(0, 0)| = 0$$

for every $x \in \mathbb{R}$. Without loss of generality, we may assume that $m \neq 0$. Given that $\epsilon > 0$. We let $\delta = \frac{\epsilon}{m^2}$. If $|x| < \delta$, then we have

$$|f(x, mx) - f(0, 0)| = \left| \frac{m^2x}{1 + m^4x^2} \right| \leq m^2|x| < m^2\delta = \epsilon.$$

By Definition 4.5, $f_{E'}$ is continuous at the origin. The case for the continuity of $g_{E'}$ is similar to that of $f_{E'}$, so we omit its details here.

We complete the proof of the analysis. ■

4.3 Problems on uniformly continuous functions

Problem 4.8*Rudin Chapter 4 Exercise 8.*

Proof. Since E is bounded in $\mathbb{R}$, Theorem 1.19 guarantees that $\inf E$ and $\sup E$ exist in $\mathbb{R}$. Let them be a and b respectively so that $E \subseteq [a, b]$. Since f is uniformly continuous on E , we take $\epsilon = 1$ there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < 1$$

for all $x, y \in E$ and $|x - y| < \delta$. Let $\Delta = \frac{2(b-a)}{\delta}$ and N be the least positive integer such that $N \geq \frac{2(b-a)}{\delta}$. Then we define the intervals

$$I_n = \left[a + (n-1)\frac{\delta}{2}, a + n\frac{\delta}{2} \right],$$

where $n = 1, 2, \dots, N$. By definition, we have

$$I_1 = \left[a, a + \frac{\delta}{2} \right] \quad \text{and} \quad I_N = \left[a + (N-1)\frac{\delta}{2}, a + N\frac{\delta}{2} \right],$$

where $a + N\frac{\delta}{2} \geq b$. See Figure 4.2.

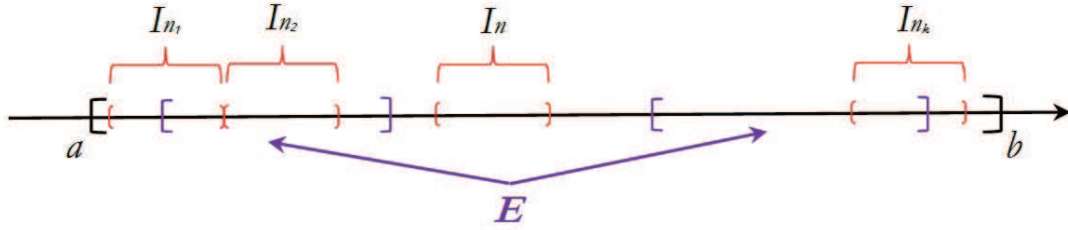


Figure 4.2: The sets E and I_{n_i} .

Since the width of each I_n is $\frac{\delta}{2}$, we have

$$|f(x) - f(y)| < 1 \tag{4.6}$$

for all $x, y \in I_n \cap E$.^d In the following discussion, we suppose that $n_1, n_2, \dots, n_k$ are positive integers such that $1 \leq n_1 < n_2 < \dots < n_k \leq N$ and $I_{n_i} \cap E \neq \emptyset$ for $i = 1, 2, \dots, k$. Therefore, we can pick and fix an element $x_{n_i} \in I_{n_i} \cap E$ and it follows from the triangle inequality and the inequality (4.6) that

$$|f(x)| \leq |f(x) - f(x_{n_i})| + |f(x_{n_i})| < 1 + |f(x_{n_i})| \tag{4.7}$$

for all $x \in I_{n_i} \cap E$, where $i = 1, 2, \dots, k$. Let $M = \max(|f(x_{n_1})|, |f(x_{n_2})|, \dots, |f(x_{n_k})|)$. It is clear that

$$(I_{n_1} \cap E) \cup (I_{n_2} \cap E) \cup \dots \cup (I_{n_k} \cap E) = E.$$

If $x \in E$, then we have $x \in I_{n_i} \cap E$ for some $i = 1, 2, \dots, k$ and it follows from the inequality (4.7) that

$$|f(x)| < 1 + M.$$

Hence f is bounded on E .

We know that $f(x) = x$ is a real uniformly continuous function on $\mathbb{R}$ (we just need to take $\delta = \epsilon$ in Definition 4.18). However, it is obviously not bounded on $\mathbb{R}$, completing the proof of the problem. ■

^dNote that it may happen that $I_n \cap E = \emptyset$ for some $n \in \{1, 2, \dots, N\}$.

Problem 4.9*Rudin Chapter 4 Exercise 9.*

Proof. Let us restate the two requirements first:

- **Statement 1:** To every $\epsilon > 0$ there exists a $\delta > 0$ such that $\text{diam } f(E) < \epsilon$ for all $E \subseteq X$ with $\text{diam } E < \delta$.
- **Statement 2:** To every $\epsilon > 0$ there exists a $\delta > 0$ such that $d_Y(f(p), f(q)) < \epsilon$ for all $p, q \in E \subseteq X$ with $d_X(p, q) < \delta$.

Recall from Definition 3.9 that if E is a nonempty subset of a metric space X ,

$$S_E = \{d_X(p, q) \mid p, q \in E\}$$

and

$$S_{f(E)} = \{d_Y(P, Q) \mid P, Q \in f(E)\} = \{d_Y(f(p), f(q)) \mid p, q \in E\}.$$

Then we have

$$\text{diam } E = \sup S_E \quad \text{and} \quad \text{diam } f(E) = \sup S_{f(E)}. \quad (4.8)$$

Suppose that **Statement 1** is true. Let $p, q \in E$. If $d_X(p, q) < \delta$, then Definition 1.7 implies that δ is *greater than* any element of S_E and so the first expression (4.8) implies that $\text{diam } E < \delta$. Since **Statement 1** is true, we have $\text{diam } f(E) < \epsilon$. By the second expression (4.8) again, the latter inequality implies that $d_Y(f(p), f(q)) < \epsilon$. This shows **Statement 2** is true. Next, we suppose that **Statement 2** is true. If $\text{diam } E < \delta$, then the first expression (4.8) implies that $d_X(p, q) < \delta$ for all $p, q \in E$ and thus $d_Y(f(p), f(q)) < \epsilon$ for all $p, q \in E$. By this and the second expression (4.8), we have

$$\text{diam } f(E) < \epsilon.$$

Therefore, **Statement 1** is true. Hence **Statement 1** and **Statement 2** are equivalent, completing the proof of the problem. ■

Problem 4.10*Rudin Chapter 4 Exercise 10.*

Proof. Assume that f was not uniformly continuous on X . Then it follows from Definition 4.18 that there exists a $\epsilon > 0$ such that *for every* $\delta > 0$, we have

$$d_Y(f(p), f(q)) \geq \epsilon$$

for some $p, q \in X$ with $d_X(p, q) < \delta$. For each positive integer n , if we take $\delta = \frac{1}{n}$, then there are sequences $\{p_n\}, \{q_n\}$ in X such that $d_X(p_n, q_n) < \frac{1}{n}$ but

$$d_Y(f(p_n), f(q_n)) \geq \epsilon. \quad (4.9)$$

Furthermore, we know from the inequality (4.9) that $p_n \neq q_n$ for all positive integers n . Since X is a compact metric space, Theorem 3.6(a)^e implies that a subsequence of $\{p_n\}$, namely $\{p_{n_k}\}$, converges to $p \in X$. By the triangle inequality, we have

$$d_X(q_{n_k}, p) \leq d_X(q_{n_k}, p_{n_k}) + d_X(p_{n_k}, p) < \frac{1}{n_k} + d_X(p_{n_k}, p). \quad (4.10)$$

Thus this inequality (4.10) implies that the subsequence $\{q_{n_k}\}$ of $\{q_n\}$ also converges to the point p . Since p is a point of the domain of f , f is continuous at p . Therefore we can get from Theorems 4.2(a) and 4.6 that there exists integers N_p and N_q such that

$$d_Y(f(p_{n_k}), p) < \frac{\epsilon}{4} \quad \text{and} \quad d_Y(f(q_{n_k}), p) < \frac{\epsilon}{4}$$

^eWe can apply Theorem 2.37 here instead of Theorem 3.6(a).

for all $k \geq N_p$ and $k \geq N_q$ respectively. Hence if $k \geq N = \max(N_p, N_q)$, then we get from the inequality (4.9) that

$$\epsilon \leq d_Y(f(p_{n_k}), f(q_{n_k})) \leq d_Y(f(p_{n_k}), f(p)) + d_Y(f(p), f(q_{n_k})) < \frac{\epsilon}{4} + \frac{\epsilon}{4} = \frac{\epsilon}{2},$$

a contradiction. Hence this proves Theorem 4.19 that f is uniformly continuous on the compact metric space X . This completes the proof of the problem. ■

Problem 4.11

Rudin Chapter 4 Exercise 11.

Proof. Given that $\epsilon > 0$. By Definition 4.18, there exists a $\delta > 0$ such that $d_Y(f(p), f(q)) < \epsilon$ for all $p, q \in X$ and $d_X(p, q) < \delta$. Since $\{x_n\}$ is a Cauchy sequence in X , there is an integer N such that $d_X(x_n, x_m) < \delta$ if $m, n \geq N$. Hence we have

$$d_Y(f(x_n), f(x_m)) < \epsilon$$

if $m, n \geq N$, i.e., $\{f(x_n)\}$ is a Cauchy sequence by Definition 3.8.

Alternative proof of Problem 4.13: Let $p \in X \setminus E$. Since E is a dense subset of X , there is a sequence $\{p_n\}$ in E such that

$$\lim_{n \rightarrow \infty} p_n = p.$$

By Theorem 3.11(a), $\{p_n\}$ is a Cauchy sequence in X . Since f is uniformly continuous on E , our Problem 4.11 implies that $\{f(p_n)\}$ is also a Cauchy sequence in $\mathbb{R}$ and we follow from Theorem 3.11(c) that $\{f(p_n)\}$ converges to $P \in \mathbb{R}$. Suppose that there is another sequence $\{q_n\}$ in E which converges to p . By the previous analysis, we know that $\{f(q_n)\}$ converges to $Q \in \mathbb{R}$. Now we have the following result about P and Q :

Lemma 4.1

We have $P = Q$.

Proof of Lemma 4.1. Given that $\delta > 0$. Since $p_n \rightarrow p$ and $q_n \rightarrow p$ as $n \rightarrow \infty$, there exist positive integers N_1 and N_2 such that

$$d_X(p_n, p) < \frac{\delta}{2} \quad \text{and} \quad d_X(q_n, p) < \frac{\delta}{2}$$

if $n \geq N_1$ and $n \geq N_2$ respectively. Let $N = \max(N_1, N_2)$. Then the triangle inequality implies that

$$d_X(p_n, q_n) \leq d_X(p_n, p) + d_X(q_n, p) < \frac{\delta}{2} + \frac{\delta}{2} = \delta$$

for $n \geq N$.

Since f is uniformly continuous on E , for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(a) - f(b)| < \epsilon$$

for all $a, b \in E$ and $d_X(a, b) < \delta$. Thus if we take $a = p_n$ and $b = q_n$, then there exists a positive integer N such that $d_X(p_n, q_n) < \delta$ for all $n \geq N$. Therefore we have for all $n \geq N$,

$$|f(p_n) - f(q_n)| < \epsilon. \tag{4.11}$$

By Definition 3.1,

$$\lim_{n \rightarrow \infty} [f(p_n) - f(q_n)] = 0$$

and it follows from Theorem 3.3(a) that $P = Q$. This completes the proof of the lemma. ■

We can continue the proof of the problem. We define $g : X \rightarrow \mathbb{R}$ by

$$g(p) = \begin{cases} f(p), & \text{if } p \in E; \\ \lim_{n \rightarrow \infty} f(p_n), & \text{if } p \in X \setminus E, \{p_n\} \subseteq E \text{ and } \lim_{n \rightarrow \infty} p_n = p. \end{cases} \quad (4.12)$$

By Lemma 4.1, the definition (4.12) is well-defined. It remains to show that g is continuous at every $p \in X$. At the first glance, one may think that it is *not* necessary to consider points in E because f is uniformly continuous on E . However, the definition only says that

$$|f(x) - f(p)| < \frac{\epsilon}{3} \quad (4.13)$$

is true for all $x, p \in E$ (not $x, p \in X$) satisfying $d_X(x, p) < \delta$. If x or p is a point of X , then the validity of the inequality (4.13) is *not* clear to us. Therefore, we have to check that g satisfies Definition 4.5 on X . Given that $\epsilon > 0$.

- **Case (i):** $p \in E$. In this case, $g(p) = f(p)$. Thus it is easy to see that there exists a $\delta > 0$ such that

$$|g(x) - g(p)| = |f(x) - f(p)| < \frac{\epsilon}{3}$$

for all $x \in E$ and $d_X(x, p) < \delta$. Thus, without loss of generality, we may assume that $x \in X \setminus E$. Let $d_X(x, p) < \delta$. Since E is dense in X , we have $\{x_n\} \subseteq E$ such that $\lim_{n \rightarrow \infty} x_n = x$. Since we have $g(x) = \lim_{n \rightarrow \infty} f(x_n)$, there exists a positive integer N such that

$$d_X(x_N, x) + d_X(x, p) < \delta \quad \text{and} \quad |g(x) - f(x_N)| < \frac{2\epsilon}{3}. \quad (4.14)$$

Now it follows from the first inequality in (4.14) that

$$d_X(x_N, p) < \delta. \quad (4.15)$$

Again, we follow from the inequality (4.13), the second inequality in (4.14) and the inequality (4.15) that

$$|g(x) - g(p)| = |g(x) - f(p)| \leq |g(x) - f(x_N)| + |f(x_N) - f(p)| < \frac{2\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

for all $x \in X \setminus E$ and $d_X(x, p) < \delta$.

- **Case (ii):** $p \in X \setminus E$. Similarly, we let $d_X(x, p) < \delta$. Basically, we apply the argument used in **Case (i)**. Since E is dense in X , we can find $\{x_n\}, \{p_n\} \subseteq E$ such that

$$\lim_{n \rightarrow \infty} x_n = x \quad \text{and} \quad \lim_{n \rightarrow \infty} p_n = p.$$

Furthermore, since we have

$$g(x) = \lim_{n \rightarrow \infty} f(x_n) \quad \text{and} \quad g(p) = \lim_{n \rightarrow \infty} f(p_n),$$

there exists a positive integer M such that

$$2d_X(p_M, p) + d_X(x, p) < \delta \quad \text{and} \quad |g(p) - g(p_M)| < \frac{\epsilon}{3} \quad (4.16)$$

and

$$2d_X(x_M, x) + d_X(x, p) < \delta \quad \text{and} \quad |g(x) - g(x_M)| < \frac{\epsilon}{3}. \quad (4.17)$$

Now it follows from the left-hand inequalities in (4.16) and (4.17) that

$$d_X(x_M, p_M) \leq d_X(x_M, x) + d_X(x, p) + d(p, p_M) < \delta. \quad (4.18)$$

Again, we follow from the inequality (4.13), the right-hand inequalities in (4.16) and (4.17) as well as the inequality (4.18) that

$$|g(x) - g(p)| \leq |g(x) - g(x_M)| + |g(x_M) - g(p_M)| + |g(p_M) - g(p)|$$

$$\begin{aligned}
&< \frac{\epsilon}{3} + |f(x_M) - f(p_M)| + \frac{\epsilon}{3} \\
&< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\
&= \epsilon
\end{aligned}$$

for all $x \in X \setminus E$ and $d_X(x, p) < \delta$.

Since ϵ is arbitrary, we know from Definition 4.5 that g is continuous on X . Hence g is *the* continuous extension of f and this extension is unique because of Problem 4.4. This completes the proof of the problem ■

Problem 4.12

Rudin Chapter 4 Exercise 12.

Proof. The statement can be stated as follows: “Suppose that X, Y, Z are metric spaces, $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ and $h : X \rightarrow Z$ is a mapping defined by

$$h(x) = g(f(x))$$

for all $x \in X$. If f is uniformly continuous on X and g is uniformly continuous on Y , then h is uniformly continuous on X .”

We follow the idea of proof in [21, Theorem 4.7, p. 86]. Given that $\epsilon > 0$. Since g is uniformly continuous on Y , there exists a $\eta > 0$ such that

$$d_Z(g(f(p)), g(f(q))) < \epsilon \quad (4.19)$$

for all $f(p), f(q) \in Y$ and $d_Y(f(p), f(q)) < \eta$. Since f is uniformly continuous on X , there exists a $\delta > 0$ such that

$$d_Y(f(p), f(q)) < \eta \quad (4.20)$$

for all $p, q \in X$ and $d_X(p, q) < \delta$. It follows from the inequalities (4.19) and (4.20) that

$$d_Z(h(p), h(q)) = d_Z(g(f(p)), g(f(q))) < \epsilon$$

for all $p, q \in X$ and $d_X(p, q) < \delta$. This completes the proof of the problem. ■

Problem 4.13

Rudin Chapter 4 Exercise 13.

Proof. Let $f : E \rightarrow \mathbb{R}$. For each $p \in X \setminus E$ and each positive integer n , let

$$V_n(p) = \left\{ q \in E \mid d(p, q) < \frac{1}{n} \right\}.$$

We want to apply Problem 3.21 to $\{\overline{f(V_n(p))}\}$, so we check the conditions as follows:

- Since $f(p) \in f(V_n(p)) \subseteq \overline{f(V_n(p))}$ for each positive integer n , we have

$$\overline{f(V_n(p))} \neq \emptyset.$$

- Since f is uniformly continuous on E , it is also a real uniformly continuous function on each bounded^f $V_n(p)$. By Problem 4.8, f is bounded on $V_n(p)$. Therefore, Definition 4.13 implies that

$$\overline{f(V_n(p))}$$

is bounded.

^fSee Definition 2.18(i).

- By Problem 4.9, we have $\text{diam } f(V_n(p)) \rightarrow 0$ as $n \rightarrow \infty$. By Theorem 3.10(a),

$$\text{diam } \overline{f(V_n(p))} = \text{diam } f(V_n(p))$$

so that $\text{diam } \overline{f(V_n(p))} \rightarrow 0$ as $n \rightarrow \infty$.

- It is clear that $\overline{f(V_n(p))}$ is closed in $\mathbb{R}$ for every positive integer n . Since $V_{n+1}(p) \subseteq V_n(p)$, we have

$$f(V_{n+1}(p)) \subseteq f(V_n(p))$$

and thus

$$\overline{f(V_{n+1}(p))} \subseteq \overline{f(V_n(p))}.$$

Now what we have shown is that $\{\overline{f(V_n(p))}\}$ is a sequence of nonempty, closed and bounded sets in $\mathbb{R}$,

$$\overline{f(V_{n+1}(p))} \subseteq \overline{f(V_n(p))} \quad \text{and} \quad \lim_{n \rightarrow \infty} \text{diam } \overline{f(V_n(p))} = 0.$$

Since $\mathbb{R}$ is complete by Theorem 3.11(c), Problem 3.21 guarantees that

$$\bigcap_{n=1}^{\infty} \overline{f(V_n(p))}$$

consists of exactly one point. Let this point be $g(p)$. By similar argument as in the proof of Problem 4.11, it can be shown that the function g so defined is the continuous extension of the function f . Therefore we won't repeat the proof here.

Next, we answer the questions one by one as follows:

- **Could the range space $\mathbb{R}$ be replaced by $\mathbb{R}^k$?** Since Problem 3.21 is valid for any complete metric space and $\mathbb{R}^k$ is complete by Theorem 3.11(c), the range space $\mathbb{R}$ can be replaced by $\mathbb{R}^k$.
- **Could the range space $\mathbb{R}$ be replaced by any compact metric space?** Since a compact metric space must be complete by Theorem 3.11(b), the range space $\mathbb{R}$ can be replaced by any compact metric space.
- **Could the range space $\mathbb{R}$ be replaced by any complete metric space?** The answer is affirmative.
- **Could the range space $\mathbb{R}$ be replaced by any metric space?** The answer is no! For example, we consider $f : \mathbb{Q} \rightarrow \mathbb{Q} \subset \mathbb{R}$ defined by

$$f(x) = 2x.$$

It is obvious that f is a uniformly continuous real function on $\mathbb{Q}$. (Take $\delta = \frac{\varepsilon}{2}$ in Definition 4.18.) Assume that $g : \mathbb{R} \rightarrow \mathbb{Q}$ was a continuous extension of f from $\mathbb{Q}$ to $\mathbb{R}$. Then Theorem 4.22 yields that $g(\mathbb{R}) \subseteq \mathbb{Q}$ is connected which is a contradiction because any subset of $\mathbb{Q}$ cannot be connected by Theorem 2.47.

This completes the proof of the problem. ■

4.4 Further properties of continuous functions

Problem 4.14

Rudin Chapter 4 Exercise 14.

Proof. Let $g(x) = x - f(x)$. If $g(0) = 0$ or $g(1) = 0$, then we are done. Suppose that $g(0) \neq 0$ and $g(1) \neq 0$. Then we must have

$$f(0) > 0 \quad \text{and} \quad f(1) < 1$$

which imply that

$$g(0) = 0 - f(0) < 0 \quad \text{and} \quad g(1) = 1 - f(1) > 0.$$

By Theorem 4.23, there exists $c \in (0, 1)$ such that $g(c) = 0$, i.e., $f(c) = c$. This completes the proof of the problem. ■

Problem 4.15*Rudin Chapter 4 Exercise 15.*

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous open mapping. Assume that f was not monotonic. In other words, there exists $a < c < b$ such that either

$$f(a) > f(c) \quad \text{and} \quad f(c) < f(b) \quad (4.21)$$

or

$$f(a) < f(c) \quad \text{and} \quad f(c) > f(b). \quad (4.22)$$

If inequalities (4.21) hold, then we consider the *restriction of f* (see Problem 4.7) on the compact set $[a, b]$. By Theorem 4.16, there exists $p, q \in [a, b]$ such that f attains its maximum and minimum at p and q respectively. Certainly, we have $q \in (a, b)$ because of the inequalities (4.21).

- **Case (i):** $p \in (a, b)$. By definition, we have

$$f(q) \leq f(x) \leq f(p) \quad (4.23)$$

for all $x \in (a, b)$ so that

$$f((a, b)) \subseteq [f(q), f(p)]. \quad (4.24)$$

By Theorem 2.47, (a, b) is connected. Thus Theorem 4.22 implies that $f((a, b))$ is a connected subset of $\mathbb{R}$. Since $f(p), f(q) \in f((a, b))$ and the inequalities (4.23), Theorem 2.47 again implies that

$$[f(q), f(p)] \subseteq f((a, b)). \quad (4.25)$$

Now we obtain from the set relations (4.24) and (4.25) that

$$f((a, b)) = [f(q), f(p)].$$

Since f is an open map and (a, b) is an open set in $\mathbb{R}$, $f((a, b)) = [f(q), f(p)]$ is open in $\mathbb{R}$ which is a contradiction.

- **Case (ii):** $p \notin (a, b)$. Then f attains its maximum *only* at the end points a or b . However, by applying similar argument as obtaining the set relations (4.24) and (4.25), we can show that either

$$f((a, b)) = [f(q), f(a)) \quad \text{or} \quad f((a, b)) = [f(q), f(b)).$$

Since both $[f(q), f(a))$ and $[f(q), f(b))$ are not open, it contradicts the hypothesis that f is an open map.

Therefore, inequalities (4.21) cannot hold. Similarly, inequalities (4.22) cannot hold. Hence, f must be monotonic. This completes the proof of the problem. ■

4.5 Discontinuous functions**Problem 4.16***Rudin Chapter 4 Exercise 16.*

Proof. Both the functions $f(x) = [x]$ and $g(x) = (x)$ have simple discontinuities at all integers. To see this, let $n \in \mathbb{Z}$, then we have

$$f(n+) = \lim_{x \rightarrow n+} f(x) = n, \quad f(n-) = \lim_{x \rightarrow n-} f(x) = n - 1$$

and

$$g(n+) = \lim_{x \rightarrow n+} g(x) = 0, \quad g(n-) = \lim_{x \rightarrow n-} g(x) = 1.$$

Let us show that f and g are continuous on $\mathbb{R} \setminus \mathbb{Z}$. Let $x \in \mathbb{R} \setminus \mathbb{Z}$. Then we have $x \in (k-1, k)$ for some $k \in \mathbb{Z}$ so that

$$f(x) = [x] = k - 1 \quad \text{and} \quad g(x) = (x) = x - [x] = x - k + 1.$$

In other words, $f(x)$ is constant on each $(k-1, k)$, so f is continuous on each $(k-1, k)$. Similarly, since $g(x)$ is a polynomial of x on each $(k-1, k)$, g is continuous on each $(k-1, k)$. See Figure 4.3 for the graphs of the functions $[x]$ (the blue line) and (x) (the red line):

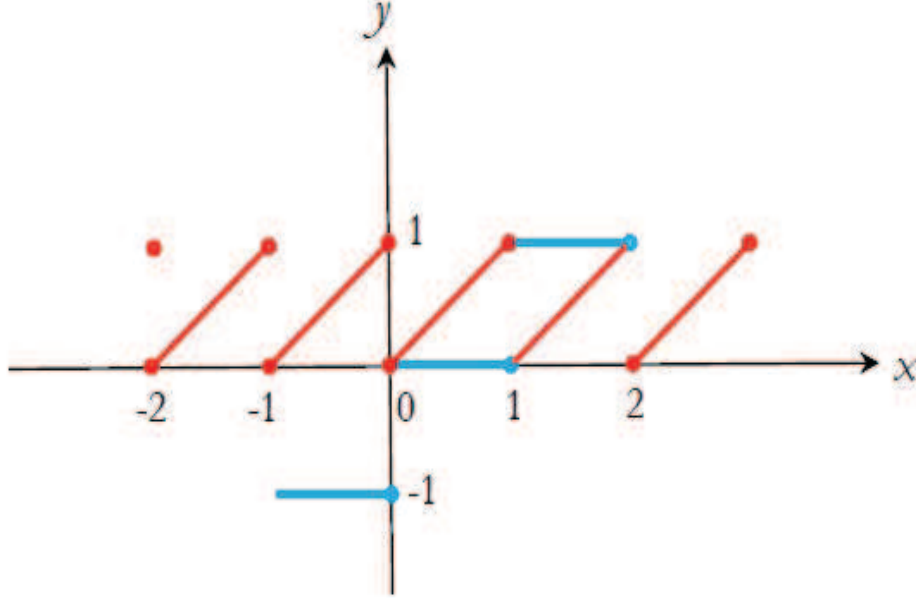


Figure 4.3: The graphs of $[x]$ and (x) .

This completes the proof of the problem. ■

Problem 4.17

Rudin Chapter 4 Exercise 17.

Proof. By Definition 4.26, there are two ways in which a function can have a simple discontinuity, so we consider the cases separately.

- **Case (i):** $f(x+) \neq f(x-)$. Let $E_+ = \{x \mid f(x-) < f(x+)\} \subseteq (a, b)$. Since $f(x-) < f(x+)$, Theorem 1.20(b) implies the existence of a rational number p such that

$$f(x-) < p < f(x+).$$

Therefore this p satisfies condition (a). Fix this p , Theorem 1.20(b) again implies the existence of a $\delta > 0$ such that $f(x-) + \delta < p$. By Definition 4.25, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(t) - f(x-)| < \epsilon$$

for all $a < t < x$ and $x - t < \delta$. Since the ϵ can be chosen such as $f(x-) + \epsilon < p$, we have $f(t) < p$ for all t such that $x - \delta < t < x$. Since there exists a rational number q such that $x - \delta < q < t$, condition (b) is satisfied. Similarly, there exists a rational number r satisfying condition (c). Now we have shown that for each $x \in E_+$, we have associated a triple (p, q, r) of rational numbers satisfying conditions (a) to (c). By Theorem 2.13, the set of all such triples is countable.

Suppose that the triple (p, q, r) is associated with x and y in E_+ . Assume that $x < y$. Then we have $x < t_0 < y$. Since $x < t < r < b$ implies $f(t) > p$, we have $f(t_0) > p$. However, since $a < q < t < y$

implies $f(t) < p$, we have $f(t_0) < p$. Thus a contradiction occurs and we cannot have $x < y$. Now the inequality $y < x$ can be shown to be impossible by similar argument. Therefore, we have $x = y$. In other words, each triple is associated with *at most* one point of E_+ . Hence the set E_+ is at most countable.

If we define $E_- = \{x \mid f(x+) < f(x-)\} \subseteq (a, b)$, then the above analysis can also be applied to show that E_- is also at most countable.

- **Case (ii):** $f(x+) = f(x-) \neq f(x)$. Let $F_+ = \{x \mid f(x+) = f(x-) < f(x)\} \subseteq (a, b)$. By applying the argument in **Case (i)**, it can be shown that each point $x \in F_+$ is associated a triple (p, q, r) of rational numbers such that

(1) $f(x-) = f(x+) < p < f(x)$ and $a < q < t < x$ or $x < t < r < b$ implies that

$$f(t) < p,$$

(2) each such triple (p, q, r) is associated with *at most* one point of F_+ . By Theorem 2.13, F_+ is at most countable.

Next if we define $F_- = \{x \mid f(x+) = f(x-) > f(x)\} \subseteq (a, b)$, then it is very clear from the above analysis that it is also at most countable.

Since the set of points at which f has a simple discontinuity, namely G , is the union of E_+ , E_- , F_+ and F_- and each set is at most countable, G is also at most countable by Theorem 2.13, completing the proof of the problem. ■

Problem 4.18

Rudin Chapter 4 Exercise 18.

Proof. The function in question is called Thomae's function, named after Carl Johannes Thomae. Read the website https://en.wikipedia.org/wiki/Thomae%27s_function for more details.

Let α be irrational. We want to show that for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(\alpha)| = |f(x)| < \epsilon$$

for all $x \in \mathbb{R}$ with $|x - \alpha| < \delta$. Given that $\epsilon > 0$. By Theorem 1.20(a) (the Archimedean property), we let n be the least positive integer such that $\frac{1}{n} < \epsilon$. Now it is clear from the definition that $f(x) = 0$ if x is irrational and so the crux idea of the proof is that the δ is constructed in order that all rational numbers in the interval $(\alpha - \delta, \alpha + \delta)$ cannot have denominators $\leq n$.

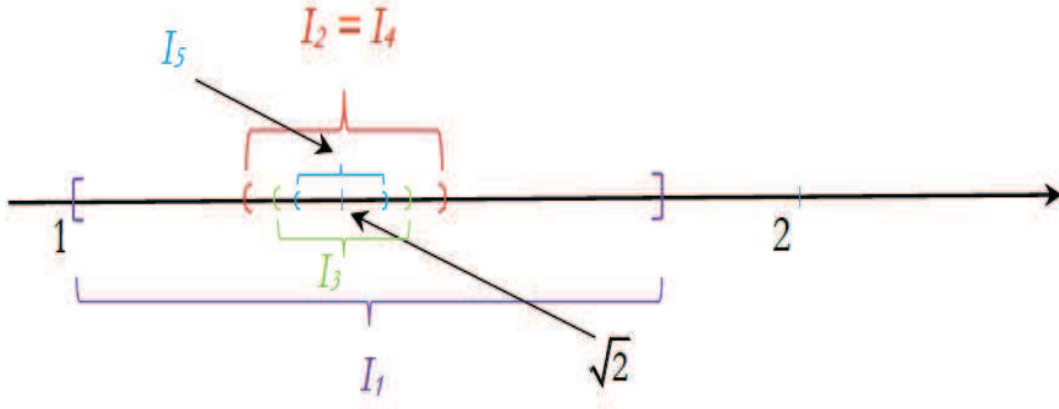
The construction of the number δ is as follows: Let k be a positive integer $\leq n$. Since α is irrational, $k\alpha$ is irrational too. By applying the proof of Theorem 1.20(b), we can find an integer m_k such that $m_k < k\alpha < m_k + 1$ which implies that

$$\alpha \in I_k = \left(\frac{m_k}{k}, \frac{m_k + 1}{k} \right).$$

Since the width of I_k is less than $\frac{1}{k}$, the interval I_k *does not* contain a rational number with k as its denominator. For this k , we define $\delta_k = \min(|\frac{m_k}{k} - \alpha|, |\alpha - \frac{m_k + 1}{k}|)$. Since $(\alpha - \delta_k, \alpha + \delta_k) \subset I_k$, $(\alpha - \delta_k, \alpha + \delta_k)$ *does not* contain a rational number with k as its denominator too. Next, we define $\delta = \min(\delta_1, \delta_2, \dots, \delta_n)$ and we consider the interval $(\alpha - \delta, \alpha + \delta)$. By the above analysis, this interval *does not* contain a rational number whose denominator $\leq n$. If $x \in (\alpha - \delta, \alpha + \delta)$ and $x = \frac{p}{q}$, then we must have $q \geq n + 1$ and thus

$$|f(x)| = \frac{1}{q} \leq \frac{1}{n+1} < \frac{1}{n} < \epsilon,$$

as required. By definition, f is continuous at every irrational number, i.e., $\lim_{x \rightarrow \alpha} f(x) = f(\alpha) = 0$. See Figure 4.4 for an example that $\alpha = \sqrt{2}$ and $n = 5$.

Figure 4.4: An example for $\alpha = \sqrt{2}$ and $n = 5$.

Let $x = \frac{m}{n}$. To prove that f has a simple discontinuity at x , we check Definitions 4.25 and 4.26 directly. We need the following result:

Lemma 4.2

The set $\mathbb{R} \setminus \mathbb{Q}$ is also dense in $\mathbb{R}$.

Proof of Lemma 4.2. Let $x, y \in \mathbb{R}$ and $x < y$. Suppose first that $0 \notin (x, y)$. We consider $\frac{x}{\sqrt{2}} < \frac{y}{\sqrt{2}}$. By Theorem 1.20(b), we have

$$\frac{x}{\sqrt{2}} < r < \frac{y}{\sqrt{2}}$$

for some $r \in \mathbb{Q} \setminus \{0\}$ and this means that $x < r\sqrt{2} < y$. By Problem 1.1, $r\sqrt{2}$ is irrational. Next, we suppose that $x < 0 < y$. Then we consider $\frac{x}{\sqrt{2}} < 0$. By Theorem 1.20(b) again, we have

$$\frac{x}{\sqrt{2}} < r' < 0$$

for some $r' \in \mathbb{Q} \setminus \{0\}$ which implies that $x < r'\sqrt{2} < 0 < y$. This completes the proof of the lemma. ■

We can continue our proof of the problem. By Lemma 4.2, there exists an irrational $\alpha_k \in (x, x + \frac{1}{k})$ for each positive integer k . In other words, we have $\lim_{k \rightarrow \infty} \alpha_k = x$ and then

$$\lim_{k \rightarrow \infty} f(\alpha_k) = 0.$$

By this, it is reasonable to claim that $f(x+) = 0$. To prove this, we need the following lemma:

Lemma 4.3

Let $\{r_k = \frac{p_k}{q_k}\}$ be a sequence of rational numbers such that $r_k > \frac{m}{n}$ for all k and $r_k \rightarrow \frac{m}{n}$. Then we have

$$q_k \rightarrow +\infty \quad \text{as } k \rightarrow \infty.$$

Proof of Lemma 4.3. Assume that the set $\{q_k\}$ was bounded. Then there is a positive integer N such that $1 \leq q_k \leq N$ for all k . If $\{p_k\}$ is bounded, then $\{r_k\}$ is a finite sequence which contradicts the hypothesis that $r_k \rightarrow \frac{m}{n}$ as $k \rightarrow \infty$. Thus we have $p_k \rightarrow +\infty$ as $k \rightarrow \infty$. However, this means that $r_k \rightarrow +\infty$ as $k \rightarrow \infty$, a contradiction again. Hence we have

$$q_k \rightarrow +\infty \quad \text{as } k \rightarrow \infty.$$

■

Now we return to the proof of the problem. By the definition of f , we have $\{f(r_k)\} = \{\frac{1}{q_k}\}$. By Lemma 4.3, we have $\frac{1}{q_k} \rightarrow 0$ as $k \rightarrow \infty$. In other words, we have

$$\lim_{k \rightarrow \infty} f(r_k) = 0.$$

By Definition 4.25, we have $f(x+) = 0$. Similarly, we also have $f(x-) = 0$. Since $f(x) = f(\frac{m}{n}) = \frac{1}{n}$, we have

$$f(x+) = f(x-) \neq f(x).$$

By Definition 4.26, f has a simple discontinuity at every rational point. This completes the proof of the problem. ■

Problem 4.19

Rudin Chapter 4 Exercise 19.

Proof. For every rational r , let $E_r = \{x \in \mathbb{R} \mid f(x) = r\}$. By the hypothesis, E_r is closed in $\mathbb{R}$. Assume that f was not continuous at x_0 . By Theorems 4.2 and 4.6, there exists a sequence $\{x_n\}$ in $\mathbb{R}$ such that $x_n \rightarrow x_0$ and $x_n \neq x_0$ for all n but $f(x_n) \not\rightarrow f(x_0)$. Thus there exists a subsequence $\{x_{n_k}\}$ of $\{x_n\}$ such that

$$|f(x_{n_k}) - f(x_0)| \geq \epsilon$$

for some $\epsilon > 0$ and for all k . For the convenience of the discussion, we may rename this subsequence as $\{x_n\}$ so that

$$f(x_n) - f(x_0) \geq \epsilon$$

for all n , i.e., $f(x_n) \geq f(x_0) + \epsilon$ for all n . By Theorem 1.20(b), there is a rational r such that

$$f(x_n) \geq f(x_0) + \epsilon > r > f(x_0)$$

for all n . By the assumption, we have $f(t_n) = r$ for some t_n between x_0 and x_n . Since $x_n \rightarrow x_0$ as $n \rightarrow \infty$, we must have $t_n \rightarrow x_0$ as $n \rightarrow \infty$. By this and the fact that $t_n \in E_r$ for all n , we know that x_0 is a limit point of E_r . Since E_r is closed, $x_0 \in E_r$ and thus $f(x_0) = r$ which contradicts our choice of r . Hence f is continuous, completing the proof of the analysis. ■

4.6 The distance function ρ_E

Problem 4.20

Rudin Chapter 4 Exercise 20.

Proof.

(a) By definition, it is clear that $\rho_E(x) \geq 0$.

Suppose that $x \in \overline{E}$. Then $x \in E$ or $x \in E'$. If $x \in E$, then it is clear that $\rho_E(x) = 0$. If $x \in E'$, then x is a limit point of E . Thus for every $N_{\frac{1}{n}}(x)$, there is a point $z_n \in N_{\frac{1}{n}}(x)$ but $z_n \neq x$ such that $z_n \in E$. By the choice of $N_{\frac{1}{n}}(x)$, we have $d(x, z_n) < \frac{1}{n}$ which gives

$$0 \leq \rho_E(x) < \frac{1}{n}.$$

By the remark preceding Theorem 3.20, we must have $\rho_E(x) = 0$.

Conversely, we suppose that $\rho_E(x) = 0$. If $x \in E$, then we are done. Now without loss of generality, we may assume that $x \notin E$. We claim that for every positive integer n , there exists a $z \in E$ such that $d(x, z) < \frac{1}{n}$. Otherwise, there is a positive integer N such that

$$\frac{1}{N} \leq d(x, z)$$

for all $z \in E$ but this means that $\frac{1}{N}$ is a lower bound of the set $A_x = \{d(x, z) \mid z \in E\}$. Thus we have

$$\rho_E(x) \geq \frac{1}{N} > 0,$$

a contradiction. This proves our claim. In addition, it is easy to see that $z \neq x$ because $x \notin E$.

Next, we let $N_\delta(x)$ be a neighborhood of x for some $\delta > 0$. Then there exists a positive integer n such that $\frac{1}{n} < \delta$ so that

$$N_{\frac{1}{n}}(x) \subset N_\delta(x).$$

The preceding analysis makes sure that there exists a $z \in N_\delta(x)$ such that $z \neq x$ and $z \in E$. Hence, it follows from Definition 2.18(b) that x is a limit point of E . Hence we have $x \in E' \subseteq \overline{E}$, completing the proof of part (a).

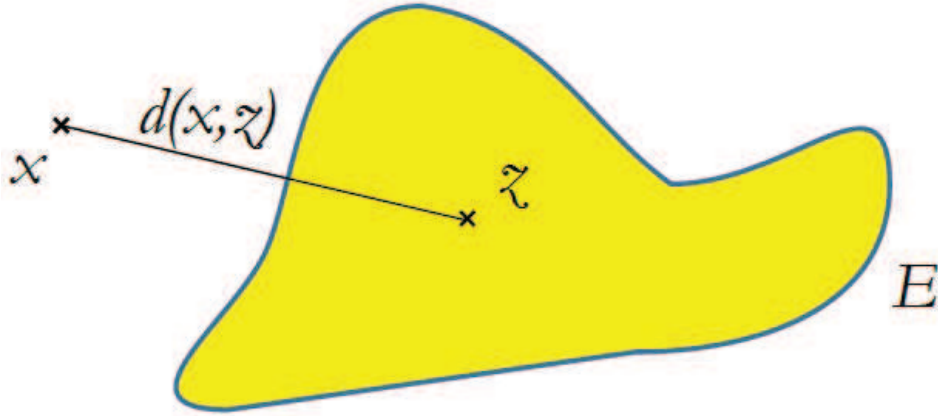


Figure 4.5: The distance from $x \in X$ to E .

- (b) Since $\rho_E(x) \leq d(x, z) \leq d(x, y) + d(y, z)$ for all $x, y \in X$, we have

$$\rho_E(x) \leq d(x, y) + \rho_E(y). \quad (4.26)$$

Similarly, we have $\rho_E(y) \leq d(y, z) \leq d(y, x) + d(x, z)$ for all $x, y \in X$ which implies that

$$\rho_E(y) \leq d(x, y) + \rho_E(x). \quad (4.27)$$

Combining inequalities (4.26) and (4.27), we have

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y)$$

for all $x \in X, y \in X$. Given $\epsilon > 0$. Let $\delta = \frac{\epsilon}{2}$. Then for all $x, y \in X$ with $d(x, y) < \delta$, we have

$$|\rho_E(x) - \rho_E(y)| \leq d(x, y) < \delta = \frac{\epsilon}{2} < \epsilon.$$

By Definition 4.18, ρ_E is a uniformly continuous function on X . This finishes the proof of the problem. ■

Problem 4.21*Rudin Chapter 4 Exercise 21.*

Proof. We define the function $\rho_F : K \rightarrow \mathbb{R}$ by

$$\rho_F(x) = \inf_{z \in F} d(x, z),$$

where $x \in K$. Since F is closed, we have $\overline{F} = F$ by Theorem 2.27(b). Since $F \cap K = \emptyset$, we have

$$\overline{F} \cap K = \emptyset$$

and Problem 4.20(a) implies that $\rho_F(x) \neq 0$ for all $x \in K$. Therefore, we follow from Problem 4.20(b) that ρ_F is a positive (uniformly) continuous function on K . By Theorem 4.16, there exists an $a \in K$ such that $\rho_F(a) = \min_{x \in K} \rho_F(x)$. Since ρ_F is positive, we must have $\rho_F(a) > 0$ so that there exists a $\delta > 0$ such that

$$0 < \delta < \rho_F(a) \leq \rho_F(q) \leq d(p, q)$$

for all $p \in F$ and $q \in K$.

Suppose that $K = \{1, 2, 3, \dots\}$ and $F = \{1 + \frac{1}{2}, 2 + \frac{1}{3}, \dots\}$. Then both K and F are closed as well as $K \cap F = \emptyset$, but they are not compact. Since $n \in K$ and $n + \frac{1}{n+1} \in F$ for every $n \in \mathbb{N}$, we have

$$d\left(n, n + \frac{1}{n+1}\right) = \frac{1}{n} \rightarrow 0$$

as $n \rightarrow \infty$. Thus our conclusion fails in this case and this ends the proof of the problem. ■

Problem 4.22*Rudin Chapter 4 Exercise 22.*

Proof. By definition, we have $\rho_A \geq 0$ and $\rho_B \geq 0$ on X . We claim that

$$\rho_A(x) + \rho_B(x) > 0 \tag{4.28}$$

for all $x \in X$. Since A and B are closed, it follows from Problem 4.20(a) that $\rho_A(x) = 0$ if and only if $x \in A$ and $\rho_B(x) = 0$ if and only if $x \in B$. Since $A \cap B = \emptyset$, there is no point $x \in X$ such that

$$\rho_A(x) = \rho_B(x) = 0$$

which yields the result (4.28). By Problem 4.20(b), both ρ_A and ρ_B are continuous functions on X . By Theorem 4.9, we see immediately that f is a continuous function on X .

To find the range of f , we note that $0 \leq \rho_A(x) \leq \rho_A(x) + \rho_B(x)$. Combining this and the inequality (4.28), we have

$$0 \leq \frac{\rho_A(x)}{\rho_A(x) + \rho_B(x)} \leq 1.$$

Since $f(a) = 0$ if $a \in A$ and $f(b) = 1$ if $b \in B$, we have

$$f(X) = [0, 1].$$

Since $f(p) = 0$ if and only if $\rho_A(p) = 0$ and A is closed, we deduce from Problem 4.20(a) that

$$\rho_A(p) = 0 \quad \text{if and only if} \quad p \in A.$$

Hence $f(p) = 0$ precisely on A . The result that $f(p) = 1$ precisely on B can be proven similarly, so we omit the details here.

Now the above result can be applied to obtain a converse of Problem 4.3: Let $A \subseteq X$ be closed. Then $A = Z(f)$ for some real continuous function f . To see why it is true, we consider two cases:

- **Case (i):** $A = X$. Then the function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = 0$ for all $x \in X$ satisfies the required conditions.
- **Case (ii):** $A \subset X$. Then we take any point $y \in X \setminus A$. Thus $B = \{y\}$ is closed and $A \cap B = \emptyset$. So the f defined in this problem is the desired continuous real function with the property that $Z(f) = A$.

Let $V = f^{-1}([0, \frac{1}{2}))$ and $W = f^{-1}((\frac{1}{2}, 1])$. It is obvious that V and W are disjoint because if $x \in V \cap W$, then $f(x) \in [0, \frac{1}{2})$ and $f(x) \in (\frac{1}{2}, 1]$ which is impossible. Since

$$\left[0, \frac{1}{2}\right) = [0, 1] \cap \left(-\frac{1}{2}, \frac{1}{2}\right) \quad \text{and} \quad \left(\frac{1}{2}, 1\right] = [0, 1] \cap \left(\frac{1}{2}, \frac{3}{2}\right),$$

Theorem 2.30 implies that both $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$ are open sets in $[0, 1]$. By Theorem 4.8, we know that V and W are disjoint open sets in X . Finally, if $p \in A$, then $f(p) = 0$ so that

$$p \in V,$$

i.e., $A \subseteq V$. Similarly, if $p \in B$, then $f(p) = 1$ so that $p \in W$, i.e., $B \subseteq W$. This completes the proof of the problem. ■

4.7 Convex functions

Problem 4.23

Rudin Chapter 4 Exercise 23.

Proof. Before we prove the results, let's look at the graph of a convex function first:

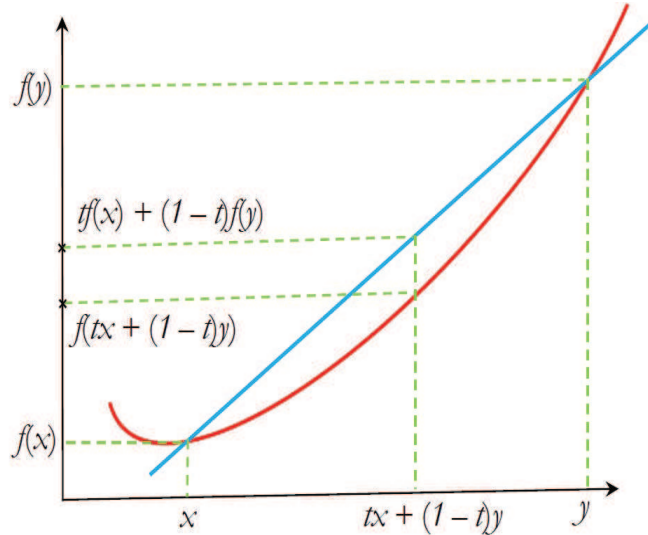


Figure 4.6: The graph of a convex function f .

From Figure 4.6, we see that the graph of a convex function is *below* the straight line connecting the points $(x, f(x))$ and $(y, f(y))$. Furthermore, the graph indicates that the equality holds if and only if $x = y$. Now we are going to prove the assertions one by one.

- **f is continuous in (a, b) .** Suppose that $p, q \in (a, b)$ and $p < q$. We first show that

Lemma 4.4

The function f is bounded in $[p, q]$.

Proof of Lemma 4.4. Let $0 < r < \min(\frac{q-p}{4}, \frac{p-a}{4}, \frac{b-q}{4})$, $M_{p,q} = \max(f(p), f(q))$ and $t \in (p, q)$. It is obvious that $[p+r, q-r] \subset (a, b)$. If $\lambda = \frac{t-p}{q-p}$, then we have $0 < \lambda < 1$ and $\lambda q + (1-\lambda)p = t$. Therefore, the definition implies that

$$\begin{aligned} f(t) &= f(\lambda q + (1-\lambda)p) \\ &\leq \lambda f(q) + (1-\lambda)f(p) \\ &\leq \lambda M_{p,q} + (1-\lambda)M_{p,q} \\ &= M_{p,q} \end{aligned}$$

for all $t \in (p, q)$. In other words, f is bounded above in $[p, q]$.

Since $a < p-r < p < q$, if $t \in (p, q)$ and $\lambda = \frac{r}{t-p+r}$, then $0 < \lambda < 1$ and we have $p = (1-\lambda)(p-r) + \lambda t$. By the definition of a convex function, we have

$$f(p) = f((1-\lambda)(p-r) + \lambda t) \leq (1-\lambda)f(p-r) + \lambda f(t)$$

so that

$$\begin{aligned} f(t) &\geq \frac{1}{\lambda}f(p) - \frac{1-\lambda}{\lambda}f(p-r) \\ &= \frac{t-p+r}{r}f(p) - \frac{t-p}{r}f(p-r) \\ &> f(p) - \frac{q-p}{r}f(p-r) \end{aligned}$$

for all $t \in (p, q)$. In other words, f is bounded below in $[p, q]$. Hence there is a positive number $N_{p,q}$ such that $|f(x)| \leq N_{p,q}$ for all $x \in [p, q]$, completing the proof of the lemma. ■

Let's return to the proof of the problem. Given that $\epsilon > 0$ and x is a *fixed* number in the interval (a, b) . Then Theorem 1.20(b) ensures that *there exist* $p, q \in (a, b)$ such that $a < p < x < q < b$. Suppose that $0 < \kappa < \frac{1}{2} \min(x-p, q-x)$. Then we have

$$a < p < p + \kappa < x < q - \kappa < q < b.$$

See Figure 4.7 for the positions of the points $p, p + \kappa, q - \kappa$ and q .



Figure 4.7: The positions of the points $p, p + \kappa, q - \kappa$ and q .

Case (i): Let y be a real number such that $p < p + \kappa < x < y < q - \kappa < q$. If $\lambda = \frac{x-p}{y-p}$, then it is easy to check that $0 < \lambda < 1$ and $x = (1-\lambda)p + \lambda y$. Thus we have $y - p \geq \kappa$ and Lemma 4.4 implies that

$$\begin{aligned} f(x) &\leq (1 - \lambda)f(p) + \lambda f(y) \\ &= f((1 - \lambda)p + \lambda y) \end{aligned} \quad (4.29)$$

$$\begin{aligned} f(x) - f(y) &\leq (1 - \lambda)[f(p) - f(y)] \\ &= \frac{y - x}{y - p}[f(p) - f(y)] \\ &< \frac{2N_{p,q}}{\kappa}(y - x) \end{aligned} \quad (4.30)$$

for some positive number $N_{p,q}$. Similarly, if $\lambda = \frac{y-x}{q-x}$, then it is easy to check that $0 < \lambda < 1$ and $y = (1 - \lambda)x + \lambda q$. Thus we have $q - x > \kappa$ and Lemma 4.4 implies that

$$\begin{aligned} f(y) &\leq (1 - \lambda)f(x) + \lambda f(q) \\ &= f((1 - \lambda)x + \lambda q) \\ f(y) - f(x) &\leq \lambda[f(q) - f(x)] \\ &= \frac{y - x}{q - x}[f(q) - f(x)] \\ &< \frac{2N_{p,q}}{\kappa}(y - x). \end{aligned} \quad (4.31)$$

By combining the inequalities (4.29) and (4.31), we always have

$$|f(y) - f(x)| < \frac{2N_{p,q}}{\kappa}(y - x) \quad (4.32)$$

for all $y \in [p + \kappa, q - \kappa]$ with $x < y$.

Case (ii): Let y be a real number such that $p < p + \kappa < y < x < q - \kappa < q$. Then, instead of the inequalities (4.29) and (4.31), we have

$$f(y) - f(x) < \frac{2N_{p,q}}{\kappa}(x - y) \quad \text{and} \quad f(x) - f(y) < \frac{2N_{p,q}}{\kappa}(x - y) \quad (4.33)$$

for some positive number $N_{p,q}$. Again, by combining the inequalities (4.33), we see that

$$|f(y) - f(x)| < \frac{2N_{p,q}}{\kappa}(x - y) \quad (4.34)$$

for all $y \in [p + \kappa, q - \kappa]$ with $y < x$.

Hence, if we let $\delta = \frac{\kappa}{2N_{p,q}}\epsilon$, then it follows from the inequalities (4.32) and (4.34) that

$$|f(y) - f(x)| < \epsilon$$

for all $y \in [p + \kappa, q - \kappa]$ with $|y - x| < \delta$. By Definition 4.5, f is continuous at x and then on (a, b) .

- **Every increasing convex function of a convex function is convex.** Let $f : (a, b) \rightarrow \mathbb{R}$ be a convex function, $g : f((a, b)) \rightarrow \mathbb{R}$ be an increasing convex function and $h : (a, b) \rightarrow \mathbb{R}$ be defined by $h = g \circ f$.

Further, let $a < x, y < b$ and $0 < \lambda < 1$. We first show that

$$\lambda f(x) + (1 - \lambda)f(y) \in f((a, b)). \quad (4.35)$$

To this end, if $f(x) = f(y)$, then we have the (4.35) holds trivially. So, without loss of generality, we may assume that $f(x) < f(y)$.[§] Then we have

$$f(x) < \lambda f(x) + (1 - \lambda)f(y) < f(y).$$

[§]The case for $f(x) > f(y)$ is similar, so we omit the details here.

Since f is convex, it is continuous on (a, b) . Since (a, b) is connected, Theorem 4.22 implies that $f((a, b))$ is connected. By Theorem 2.47, we must have the (4.35).

Since f is convex, we have $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$. Since g is increasing and convex plus $\lambda f(x) + (1 - \lambda)f(y) \in f((a, b))$, we have

$$\begin{aligned} h(\lambda x + (1 - \lambda)y) &= g(f(\lambda x + (1 - \lambda)y)) \\ &\leq g(\lambda f(x) + (1 - \lambda)f(y)) \\ &\leq \lambda g(f(x)) + (1 - \lambda)g(f(y)) \\ &= \lambda h(x) + (1 - \lambda)h(y). \end{aligned}$$

By definition, h is convex.

- **The validity of inequalities.** Suppose that f is convex in (a, b) and $a < s < t < u < b$. If $\lambda = \frac{u-t}{u-s}$, then we have $0 < \lambda < 1$ and $\lambda s + (1 - \lambda)u = t$. Thus the definition implies that

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\leq \lambda f(x) + (1 - \lambda)f(y) \\ f(t) &\leq \frac{u-t}{u-s}f(s) + \left(1 - \frac{u-t}{u-s}\right)f(u) \\ \frac{f(t) - f(s)}{t-s} &\leq \frac{f(u) - f(s)}{u-s}. \end{aligned} \tag{4.36}$$

Similarly, if we let $\lambda = \frac{t-s}{u-s}$, then we have $0 < \lambda < 1$ and $\lambda u + (1 - \lambda)s = t$. Thus the definition implies that

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\leq \lambda f(x) + (1 - \lambda)f(y) \\ f(t) &\leq \frac{t-s}{u-s}f(u) + \left(1 - \frac{t-s}{u-s}\right)f(s) \\ \frac{f(u) - f(s)}{u-s} &\leq \frac{f(u) - f(t)}{u-t}. \end{aligned} \tag{4.37}$$

Hence, by combining the inequalities (4.36) and (4.37), we have the desired inequalities in the question.

This completes the proof of the problem. ■

Problem 4.24

Rudin Chapter 4 Exercise 24.

Proof. We have a lemma first:

Lemma 4.5

Let f be a continuous real function defined in (a, b) satisfying the hypothesis of the problem. For every $x, y \in (a, b)$ and $\lambda = \frac{m}{2^n}$, where m and n are positive integers such that $0 < m < 2^n$, we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \tag{4.38}$$

Proof of Lemma 4.5. We prove it by induction on n . If $n = 1$, then we see from the hypothesis of the problem that the inequality (4.38) holds for $\lambda = \frac{1}{2}$. Thus it is true for the case $n = 1$. Assume that it is also true for $n = k$ for some positive integer k , i.e.,

$$f\left(\frac{m}{2^k}x + \left(1 - \frac{m}{2^k}\right)y\right) \leq \frac{m}{2^k}f(x) + \left(1 - \frac{m}{2^k}\right)f(y), \quad (4.39)$$

where m is a positive integer such that $0 < m < 2^k$. Let $n = k + 1$ and $0 < m < 2^{k+1}$. We can rewrite

$$\frac{m}{2^{k+1}} = \frac{1}{2} \cdot \frac{2m}{2^{k+1}} = \frac{1}{2} \left(\frac{m-1}{2^{k+1}} + \frac{m+1}{2^{k+1}} \right).$$

If we let $\mu = \frac{m-1}{2^{k+1}}$ and $\nu = \frac{m+1}{2^{k+1}}$, then we have $\lambda = \frac{1}{2}(\mu + \nu)$ and it is clear that

$$\lambda x + (1 - \lambda)y = \frac{1}{2}\{(\mu + \nu)x + [2 - (\mu + \nu)]y\} = \frac{[\mu x + (1 - \mu)y] + [\nu x + (1 - \nu)y]}{2}.$$

By definition, we have $\mu \in [0, 1)$ and $\nu \in (0, 1]$. Now we split the proof into two cases as follows:

- **Cases (i):** $\mu = 0$ or $\nu = 1$. We know that $\mu = 0$ if and only if $m = 1$. In this particular case, we have

$$\lambda x + (1 - \lambda)y = \frac{1}{2^{k+1}}x + \left(1 - \frac{1}{2^{k+1}}\right)y = \frac{1}{2} \cdot \left[\frac{1}{2^k}x + \left(1 - \frac{1}{2^k}\right)y + y \right].$$

Since $\frac{1}{2^k}x + (1 - \frac{1}{2^k})y \in (a, b)$ for all $x, y \in (a, b)$, the hypothesis of the problem and the inequality (4.39) imply that

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= f\left(\frac{\frac{1}{2^k}x + \left(1 - \frac{1}{2^k}\right)y + y}{2}\right) \\ &\leq \frac{1}{2} \left[f\left(\frac{1}{2^k}x + \left(1 - \frac{1}{2^k}\right)y\right) + f(y) \right] \\ &\leq \frac{1}{2} \left[\frac{1}{2^k}f(x) + \left(1 - \frac{1}{2^k}\right)f(y) + f(y) \right] \\ &= \lambda f(x) + (1 - \lambda)f(y). \end{aligned}$$

In other words, the inequality (4.38) also holds in this particular case. Similarly, $\nu = 1$ if and only if $m = 2^{k+1} - 1$. In this particular case, we have

$$\lambda x + (1 - \lambda)y = \frac{2^{k+1} - 1}{2^{k+1}}x + \left(1 - \frac{2^{k+1} - 1}{2^{k+1}}\right)y = \frac{1}{2} \cdot \left[\frac{1}{2^k}y + \left(1 - \frac{1}{2^k}\right)x + x \right].$$

Thus similar argument can be applied to show that the inequality (4.38) also holds in this particular case.

- **Case (ii):** $\mu, \nu \in (0, 1)$. By definition, we have $0 < m - 1 < m + 1 < 2^{k+1}$. Besides, we have $\mu x + (1 - \mu)y \in (a, b)$ and $\nu x + (1 - \nu)y \in (a, b)$ for all $x, y \in (a, b)$. Therefore, it follows from the hypothesis of the problem and the inequality (4.39) that

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &= f\left(\frac{[\mu x + (1 - \mu)y] + [\nu x + (1 - \nu)y]}{2}\right) \\ &\leq \frac{1}{2} \cdot [f(\mu x + (1 - \mu)y) + f(\nu x + (1 - \nu)y)] \\ &\leq \frac{[\mu f(x) + (1 - \mu)f(y)] + [\nu f(x) + (1 - \nu)f(y)]}{2} \\ &= \lambda f(x) + (1 - \lambda)f(y). \end{aligned}$$

Hence the inequality (4.39) is also true for $n = k + 1$ and the induction shows that the inequality (4.39) is true for all positive integers k , where m is a positive integer such that $0 < m < 2^k$. This completes the proof of Lemma 4.5. ■

Now we may continue our proof of the problem. Let us accept the fact that every real number $\alpha \in (0, 1)$ has a unique binary representation in the sense that if $\lambda_n = \frac{a_n}{2^n}$, where $a_n \in \{0, 1\}$, then we have

$$\alpha = \sum_{n=1}^{\infty} \lambda_n.$$

By Theorem 3.26, the series is convergent (absolutely). If we let $\alpha_n = \sum_{k=1}^n \lambda_k$, then we have $\alpha_n \rightarrow \alpha$ as $n \rightarrow \infty$. It is easy to see that α_n is in the form of $\frac{m}{2^n}$, where m is a positive integer such that $0 < m < 2^n$, therefore Lemma 4.5 implies that

$$f(\alpha_n x + (1 - \alpha_n)y) \leq \alpha_n f(x) + (1 - \alpha_n)f(y). \quad (4.40)$$

Since f is continuous on (a, b) , we apply Theorems 3.3, 3.19, 4.2 and 4.6 to the inequality (4.40) to get

$$\begin{aligned} \lim_{n \rightarrow \infty} f(\alpha_n x + (1 - \alpha_n)y) &\leq \lim_{n \rightarrow \infty} [\alpha_n f(x) + (1 - \alpha_n)f(y)] \\ f(\alpha x + (1 - \alpha)y) &\leq \lim_{n \rightarrow \infty} \alpha_n f(x) + \lim_{n \rightarrow \infty} (1 - \alpha_n)f(y) \\ f(\alpha x + (1 - \alpha)y) &\leq \alpha f(x) + (1 - \alpha)f(y). \end{aligned}$$

By Problem 4.23, we see that f is convex. This finishes the proof of the problem. ■

4.8 Other properties of continuous functions

Problem 4.25

Rudin Chapter 4 Exercise 25.

Proof.

- (a) We follow the hint. Take $\mathbf{z} \notin K + C$, put $F = \mathbf{z} - C = \{\mathbf{z} - \mathbf{y} \mid \mathbf{y} \in C\}$. Assume that $K \cap F \neq \emptyset$. Then we have $\mathbf{x} \in K$ and $\mathbf{x} \in F$. By definition, $\mathbf{x} = \mathbf{z} - \mathbf{y}$ for some $\mathbf{y} \in C$ which is equivalent to

$$\mathbf{z} = \mathbf{x} + \mathbf{y} \in K + C,$$

a contradiction. Therefore K and F are disjoint. Since K is compact and C is closed, Problem 4.21 ensures that there exists a $\delta > 0$ such that

$$|\mathbf{p} - \mathbf{q}| > \delta$$

for all $\mathbf{p} \in K$ and $\mathbf{q} \in F$.

Now we consider the open ball $B(\mathbf{z}, \delta) = \{\mathbf{x} \in \mathbb{R}^k \mid |\mathbf{x} - \mathbf{z}| < \delta\}$. Assume that there was $\mathbf{x} \in \mathbb{R}^k$ such that $\mathbf{x} \in B(\mathbf{z}, \delta) \cap (K + C)$. Then we have $|\mathbf{x} - \mathbf{z}| < \delta$ and $\mathbf{x} = \mathbf{p} + \mathbf{y}$ for some $\mathbf{p} \in K$ and $\mathbf{y} \in C$. For this $\mathbf{y}$, we have $\mathbf{q} = \mathbf{z} - \mathbf{y} \in F$ which implies

$$\mathbf{x} = \mathbf{p} + \mathbf{y} = \mathbf{p} + \mathbf{z} - \mathbf{q}.$$

Hence, by the definition of the chosen δ , we have

$$\delta < |\mathbf{p} - \mathbf{q}| = |\mathbf{x} - \mathbf{z}| < \delta,$$

a contradiction.

- (b) We have $C_2 = \{n\alpha \mid n \in C_1\}$. For every $k \in \mathbb{Z}$, the set $(k-1, k)$ is open in $\mathbb{R}$. By Theorem 2.24(a), $\bigcup_{k \in \mathbb{Z}} (k-1, k)$ is open in $\mathbb{R}$. Since $\mathbb{R} \setminus C_1 = \bigcup_{k \in \mathbb{Z}} (k-1, k)$, C_1 is closed in $\mathbb{R}$ by Theorem 2.23. By similar argument, it can be shown that C_2 is closed. By definition, we have $C_1 + C_2 = \{h + k\alpha \mid h, k \in C_1\}$.

Now the mapping $f : C_1 + C_2 \rightarrow C_1 \times C_1$ defined by $f(h + k\alpha) = (h, k)$ shows that $C_1 + C_2$ is equivalent to a subset of $C_1 \times C_1$. Since $C_1 \times C_1$ is countable and $C_1 \subset C_1 + C_2$, it follows from Theorem 2.8 that $C_1 + C_2$ is countable.^b

For a positive integer n , we consider the fractional part $\{n\alpha\} = n\alpha - [n\alpha]$. To prove the density of the set, we need a preliminary result first:

Lemma 4.6

Suppose that α is irrational and $0 < \theta < 1$. For every $\epsilon > 0$, there exists a positive integer k such that $|\{k\alpha\} - \theta| < \epsilon$. Hence, if $h = [k\alpha]$, then we have $|k\alpha - h - \theta| < \epsilon$.

Proof of Lemma 4.6. We note that if $m \neq n$, then $\{m\alpha\} \neq \{n\alpha\}$. Otherwise, we have $m\alpha - [m\alpha] = n\alpha - [n\alpha]$ which implies that

$$\alpha = \frac{[m\alpha] - [n\alpha]}{m - n}$$

is rational, a contradiction. Given that $\epsilon > 0$ and choose θ such that $0 < \theta < 1$. Recall Dirichlet's theorem that for any irrational α , there exist integers h and k such that $|k\alpha - h| < \epsilon$. Now we have either $k\alpha > h$ or $k\alpha < h$. Suppose that $k\alpha > h$. Since $|k\alpha - h| = k\alpha - h = \{k\alpha\} + [k\alpha] - h$, h and $[k\alpha]$ are integers and $\{k\alpha\}$, we have $[k\alpha] = h$ and

$$0 < \{k\alpha\} < \epsilon. \quad (4.41)$$

Now we consider the following sequence $\{k\alpha\}, \{2k\alpha\}, \{3k\alpha\}, \dots$. Since $k\alpha = [k\alpha] + \{k\alpha\}$, we have $mk\alpha = m[k\alpha] + m\{k\alpha\}$ for every integer m . Thus we have

$$m\{k\alpha\} = \{mk\alpha\} \quad \text{if and only if} \quad \{k\alpha\} < \frac{1}{m}. \quad (4.42)$$

By this observation, we let N be the largest integer satisfying $\{k\alpha\} < \frac{1}{N}$. In other words, we always have

$$\frac{1}{N+1} < \{k\alpha\} < \frac{1}{N}. \quad (4.43)$$

By the relation (4.42) and the inequalities (4.43), we have $\{mk\alpha\} = m\{k\alpha\}$ for $m = 1, 2, \dots, N$. Therefore, the numbers $\{k\alpha\}, \{2k\alpha\}, \dots, \{Nk\alpha\}$ form an increasing chain with equal spacing $\{k\alpha\}$. Now it follows from the inequalities (4.43) that

$$1 - \frac{1}{N} < \{Nk\alpha\} < 1$$

and then this and the inequalities (4.41) imply that

$$|1 - \{Nk\alpha\}| < \frac{1}{1+N} < \{k\alpha\} < \epsilon.$$

Hence the numbers $\{k\alpha\}, \{2k\alpha\}, \dots, \{Nk\alpha\}$ subdivide the interval $(0, 1)$ into subintervals of length less than ϵ . Recall that $0 < \theta < 1$, so θ lies in one of these subintervals and the result follows, proving Lemma 4.6. ■

For a general real θ , we note that $\theta = [\theta] + \{\theta\}$, where $0 < \{\theta\} < 1$. Thus Lemma 4.6 implies that

$$|k\alpha - [k\alpha] - \{\theta\}| < \epsilon$$

^bThis result is known as the one-dimensional case of the famous Kronecker's Approximation Theorem. For details, please read [2, §7.4, pp. 148, 149] or <http://mathworld.wolfram.com/KroneckersApproximationTheorem.html>. Here our proof is based on the one presented in [2, §7.4, pp. 148, 149].

or equivalently,

$$|k\alpha - ([k\alpha] - [\theta]) - \theta| < \epsilon. \quad (4.44)$$

If we take $h = [k\alpha] - [\theta]$ in the inequality (4.44), then we have the desired result that $C_1 + C_2$ is dense in $\mathbb{R}$.

This ends the proof of the problem. ■

Problem 4.26

Rudin Chapter 4 Exercise 26.

Proof. We have $f : X \rightarrow Y, g : Y \rightarrow Z$ and $h : X \rightarrow Z$. Since Y is compact and g is continuous on Y , $g(Y)$ is compact by Theorem 4.14. Since $g : Y \rightarrow g(Y)$ is continuous and bijective, we obtain from Theorem 4.17 that $g^{-1} : g(Y) \rightarrow Y$ is a bijective continuous mapping. By Theorem 4.19, g^{-1} is uniformly continuous on $g(Y)$. Since $f(x) = g^{-1}(h(x))$ and h is uniformly continuous on X , it follows from Problem 4.12 that f is also uniformly continuous on X . If h is continuous at $p \in X$, then Theorem 4.7 says that f is also continuous at $p \in X$. This completes the proof of the problem. ■

Differentiation

5.1 Problems on differentiability of a function

Problem 5.1

Rudin Chapter 5 Exercise 1.

Proof. We have $-(x-y)^2 \leq f(x) - f(y) \leq (x-y)^2$. If $t > x$, then $-(t-x) \leq \frac{f(t)-f(x)}{t-x} \leq t-x$. If $t < x$, then we have $-(t-x) \geq \frac{f(t)-f(x)}{t-x} \geq t-x$. By these, we have

$$-|t-x| \leq \frac{f(t)-f(x)}{t-x} \leq |t-x|.$$

By Definition 5.1 and the remark preceding Theorem 3.20, we have

$$f'(x) = \lim_{t \rightarrow x} \frac{f(t)-f(x)}{t-x} = 0$$

for all $x \in \mathbb{R}$. By Theorem 5.11(b), f is constant which completes the proof of the problem. ■

Problem 5.2

Rudin Chapter 5 Exercise 2.

Proof. By Theorem 5.10, we have $f(x_2) - f(x_1) = (x_2 - x_1)f'(x)$ for every pair of numbers x_1, x_2 in (a, b) with $x_1 < x_2$ and for *some* x between x_1 and x_2 . Since $f'(x) > 0$ in (a, b) , we have $f(x_2) - f(x_1) > 0$ which means that f is strictly increasing in (a, b) .

To prove the second assertion, we prove the following result first:

Lemma 5.1

If f is strictly increasing in (a, b) , then f is one-to-one in (a, b) .

Proof of Lemma 5.1. If $x_1, x_2 \in (a, b)$ and $x_1 \neq x_2$, then either $x_1 < x_2$ or $x_1 > x_2$. If $x_1 < x_2$, then since f is strictly increasing in (a, b) , we have $f(x_1) < f(x_2)$. Similarly, if $x_1 > x_2$, then since f is strictly increasing in (a, b) , we have $f(x_1) > f(x_2)$. In both cases, we have $f(x_1) \neq f(x_2)$. By Definition 2.2, f is one-to-one in (a, b) , completing the proof of Lemma 5.1. ■

Suppose that c and d are real numbers such that $a < c < d < b$. By Lemma 5.1, f is one-to-one in $[c, d]$. Since f is differentiable in (a, b) , it is also continuous in (a, b) by Theorem 5.2. In particular, f is continuous on $[c, d]$. Let $E = f([c, d])$. By Theorem 4.17, the inverse function $g : E \rightarrow [c, d]$ of f is well-defined and

$$g(f(x)) = x. \quad (5.1)$$

Besides, g is continuous on E .

Let $t, y \in E$ and $t \neq y$. Then there exist $s, x \in [c, d]$ such that $f(s) = t$ and $f(x) = y$. Now we have from the expression (5.1) that

$$\phi(t) = \frac{g(t) - g(y)}{t - y} = \frac{g(f(s)) - g(f(x))}{f(s) - f(x)} = \frac{s - x}{f(s) - f(x)}. \quad (5.2)$$

Since $t \rightarrow y$ if and only if $s \rightarrow x$ and $f'(x) > 0$ in (a, b) , the expression (5.2) and Theorem 4.4(c) imply that

$$g'(f(x)) = g'(y) = \lim_{t \rightarrow y} \phi(t) = \lim_{t \rightarrow y} \frac{g(t) - g(y)}{t - y} = \lim_{s \rightarrow x} \frac{s - x}{f(s) - f(x)} = \lim_{s \rightarrow x} \frac{1}{\frac{f(s) - f(x)}{s - x}} = \frac{1}{f'(x)},$$

where $x \in [c, d]$. For any $x \in (a, b)$, it is obvious that we can find c, d such that $x \in [c, d] \subset (a, b)$, $g'(f(x)) = \frac{1}{f'(x)}$ is true for every $x \in (a, b)$. This completes the proof of the problem. ■

Problem 5.3

Rudin Chapter 5 Exercise 3.

Proof. Since g is differentiable in $\mathbb{R}$, f is also differentiable in $\mathbb{R}$ by Theorem 5.3(a). In fact, we acquire

$$f'(x) = 1 + \epsilon g'(x).$$

Since $-M \leq g'(x) \leq M$ in $\mathbb{R}$, we have

$$1 - \epsilon M \leq f'(x) \leq 1 + \epsilon M.$$

Thus if ϵ is small enough, then we have $1 - \epsilon M > 0$ so that $f'(x) > 0$ in $\mathbb{R}$. By Problem 5.2, we have f is strictly increasing in $\mathbb{R}$ and then it is one-to-one in $\mathbb{R}$. This completes the proof of the problem. ■

Problem 5.4

Rudin Chapter 5 Exercise 4.

Proof. Let $f(x) = C_0x + C_1\frac{x^2}{2} + \cdots + C_n\frac{x^{n+1}}{n+1}$ be a polynomial defined in $\mathbb{R}$. In particular, f is continuous on $[0, 1]$ and differentiable in $(0, 1)$. By Theorem 5.10, there exists a $\alpha \in (0, 1)$ such that $f(1) - f(0) = f'(\alpha)$. Since $f(1) = f(0) = 0$, the equation

$$f'(x) = C_0 + C_1x + \cdots + C_{n-1}x^{n-1} + C_nx^n = 0$$

has at least one real root in $(0, 1)$ which is the desired result, completing the proof of the problem. ■

Problem 5.5

Rudin Chapter 5 Exercise 5.

Proof. Since f is differentiable in $(0, +\infty)$, it is continuous on $(0, +\infty)$ by Theorem 5.2. In particular, f is continuous on $[x, x+1]$ and differentiable in $(x, x+1)$ for every $x > 0$. By Theorem 5.10 and the definition of g , we have

$$g(x) = f(x+1) - f(x) = (x+1-x)f'(y) = f'(y) \quad (5.3)$$

for some $y \in (x, x+1)$. If $x \rightarrow +\infty$, then we have $y \rightarrow +\infty$ and so $f'(y) \rightarrow 0$. Hence it follows from the expression (5.3) that $g(x) \rightarrow 0$ as $x \rightarrow +\infty$. This finishes the proof of the problem. ■

Problem 5.6

Rudin Chapter 5 Exercise 6.

Proof. By Theorem 5.3 and condition (b), g is differentiable in $(0, +\infty)$ and

$$g'(x) = \frac{xf'(x) - f(x)}{x^2}.$$

We note that

$$g'(x) \geq 0 \quad \text{if and only if} \quad xf'(x) - f(x) \geq 0.$$

For every $x > 0$, conditions (a) and (b) imply that f is continuous in $[0, x]$ and differentiable in $(0, x)$. By Theorem 5.10, we have

$$f(x) - f(0) = (x-0)f'(\xi)$$

for some $\xi \in (0, x)$. By condition (c), we have $f(x) = xf'(\xi)$. By condition (d), we have $f'(\xi) \leq f'(x)$ and thus $f(x) \leq xf'(x)$. Therefore, we have $g'(x) \geq 0$ for every $x > 0$ and Theorem 5.11(a) implies that g is monotonically increasing in $(0, +\infty)$. ■

Problem 5.7

Rudin Chapter 5 Exercise 7.

Proof. Since $f(x), g(x)$ are differentiable, $g'(x) \neq 0$ and $f(x) = g(x) = 0$, it follows from Theorem 4.4(c) that

$$\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{g(t) - g(x)} = \lim_{t \rightarrow x} \frac{\frac{f(t) - f(x)}{t - x}}{\frac{g(t) - g(x)}{t - x}} = \frac{f'(x)}{g'(x)}$$

which is our desired result. This completes the proof of the problem. ■

Problem 5.8

Rudin Chapter 5 Exercise 8.

Proof. Since f' is continuous in $[a, b]$ and $[a, b]$ is compact, Theorem 4.19 implies that f' is uniformly continuous in $[a, b]$. By Definition 4.18, there exists a $\delta > 0$ such that

$$|f'(t) - f'(x)| < \epsilon \quad (5.4)$$

for every $x, t \in [a, b]$ with $|t - x| < \delta$. Suppose that $t < x$. Then we deduce from Theorem 5.10 that there exists a $\xi \in (t, x)$ such that

$$\frac{f(x) - f(t)}{x - t} = f'(\xi). \quad (5.5)$$

Since $0 < |\xi - x| < |t - x| < \delta$, the inequality (5.4) and the expression (5.5) together give

$$\left| \frac{f(x) - f(t)}{x - t} - f'(x) \right| = |f'(\xi) - f'(x)| < \epsilon$$

which is our desired result in the case $t < x$. Since the case for $x < t$ is similar, we omit the details here.

We claim that this also holds for vector-valued functions: $\mathbf{f}'(x)$ is continuous on $[a, b]$ and $\epsilon > 0$. Then there exists $\delta > 0$ such that

$$\left| \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x} - \mathbf{f}'(x) \right| < \epsilon$$

whenever $0 < |t - x| < \delta$ and $x, t \in [a, b]$. To prove this result, suppose that

$$\mathbf{f}(x) = (f_1(x), f_2(x), \dots, f_k(x)),$$

then Remark 5.16 implies that $\mathbf{f}$ is differentiable on $[a, b]$ if and only if each $f_1, \dots, f_k$ is differentiable on $[a, b]$. Furthermore, we apply Theorem 4.10 to get the result that $\mathbf{f}'(x)$ is continuous on $[a, b]$ if and only if each $f'_1, \dots, f'_k$ is continuous on $[a, b]$ and thus they are uniformly continuous on $[a, b]$ by Theorem 4.19. In other words, for each $i = 1, 2, \dots, k$, there exists $\delta_i > 0$ such that

$$\left| \frac{f_i(t) - f_i(x)}{t - x} - f'_i(x) \right| < \frac{\epsilon}{\sqrt{k}} \quad (5.6)$$

whenever $0 < |t - x| < \delta_i$ and $t, x \in [a, b]$. Let $\delta = \min(\delta_1, \dots, \delta_k)$. Hence we follow from the inequalities (5.6) and the proof of Theorem 4.10 that for every $0 < |t - x| < \delta$ and $t, x \in [a, b]$, we have

$$\begin{aligned} \left| \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x} - \mathbf{f}'(x) \right| &= \left| \left(\frac{f_1(t) - f_1(x)}{t - x} - f'_1(x), \dots, \frac{f_k(t) - f_k(x)}{t - x} - f'_k(x) \right) \right| \\ &\leq \sqrt{\sum_{i=1}^k \left| \frac{f_i(t) - f_i(x)}{t - x} - f'_i(x) \right|^2} \\ &< \sqrt{\sum_{i=1}^k \frac{\epsilon^2}{k}} \\ &= \epsilon. \end{aligned}$$

This completes the proof of the problem. ■

Problem 5.9

Rudin Chapter 5 Exercise 9.

Proof. The answer is affirmative. By Definition 5.1, we have to show that the limit

$$\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0}$$

exists. Define the functions

$$F(x) = f(x) - f(0) \quad \text{and} \quad G(x) = x.$$

Then they are real and differentiable in $(0, 1)$, and $G'(x) = 1 \neq 0$ on $(0, 1)$. Since

$$\frac{F'(x)}{G'(x)} = f'(x) \rightarrow 3$$

as $x \rightarrow 0$, we apply Theorem 5.13 to the functions $F(x)$ and $G(x)$ to get $\lim_{x \rightarrow 0} \frac{F(x)}{G(x)} = 3$. As

$$\lim_{x \rightarrow 0} \frac{F(x)}{G(x)} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0},$$

the derivative $f'(0)$ exists and its value must be 3.^a We complete the proof of the problem. ■

^aIf one checks the proof carefully, then it can be seen easily that the number 3 can be replaced by any real number.

Problem 5.10*Rudin Chapter 5 Exercise 10.*

Proof. By the hint, we acquire

$$\frac{f(x)}{g(x)} = \left(\frac{f(x)}{x} - A \right) \cdot \frac{1}{\frac{g(x)}{x}} + A \frac{1}{\frac{g(x)}{x}}. \quad (5.7)$$

Let $A = A_1 + iA_2$ and $B = B_1 + iB_2$, where $A_1, A_2, B_1, B_2 \in \mathbb{R}$. Let $f = f_1 + if_2$ and $g = g_1 + ig_2$, where f_1, f_2, g_1 and g_2 are real functions on $(0, 1)$. By the hypothesis, we have

$$\lim_{x \rightarrow 0} f'_1(x) = A_1, \quad \lim_{x \rightarrow 0} f'_2(x) = A_2, \quad \lim_{x \rightarrow 0} g'_1(x) = B_1 \quad \text{and} \quad \lim_{x \rightarrow 0} g'_2(x) = B_2.$$

Then the formula (5.7) can be rewritten as

$$\frac{f(x)}{g(x)} = \left(\frac{f_1(x)}{x} + i \frac{f_2(x)}{x} - A \right) \cdot \frac{1}{\frac{g_1(x)}{x} + \frac{g_2(x)}{x}} + A \frac{1}{\frac{g_1(x)}{x} + \frac{g_2(x)}{x}}. \quad (5.8)$$

Obviously, we have the following facts:

- **Fact 1.** $f(x) \rightarrow 0$ as $x \rightarrow 0$ if and only if $f_1(x) \rightarrow 0$ and $f_2(x) \rightarrow 0$ as $x \rightarrow 0$;
- **Fact 2.** $f'(x) \rightarrow A$ as $x \rightarrow 0$ if and only if $f'_1(x) \rightarrow A_1$ and $f'_2(x) \rightarrow A_2$ as $x \rightarrow 0$;
- **Fact 3.** $g(x) \rightarrow 0$ as $x \rightarrow 0$ if and only if $g_1(x) \rightarrow 0$ and $g_2(x) \rightarrow 0$ as $x \rightarrow 0$;
- **Fact 4.** $g'(x) \rightarrow B$ as $x \rightarrow 0$ if and only if $g'_1(x) \rightarrow B_1$ and $g'_2(x) \rightarrow B_2$ as $x \rightarrow 0$.

We obtain from Theorem 5.13 that

$$\lim_{x \rightarrow 0} \frac{f_1(x)}{x} = \lim_{x \rightarrow 0} f'_1(x) = A_1 \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{f_2(x)}{x} = \lim_{x \rightarrow 0} f'_2(x) = A_2.$$

Similarly, we have

$$\lim_{x \rightarrow 0} \frac{g_1(x)}{x} = \lim_{x \rightarrow 0} g'_1(x) = B_1 \quad \text{and} \quad \lim_{x \rightarrow 0} \frac{g_2(x)}{x} = \lim_{x \rightarrow 0} g'_2(x) = B_2.$$

Since $B \neq 0$, it is impossible that $B_1 = B_2 = 0$. Hence, it follows from the formula (5.8) that

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = (A_1 + iA_2 - A) \cdot \frac{1}{B_1 + iB_2} + A \cdot \frac{1}{B_1 + iB_2} = 0 \cdot \frac{1}{B} + \frac{A}{B} = \frac{A}{B}.$$

This completes the proof of the problem. ■

Problem 5.11*Rudin Chapter 5 Exercise 11.*

Proof. Let E be the neighborhood of x in which f is defined. Suppose further that h is so small such that $x + h, x - h \in E$. Then we can define

$$F(h) = f(x + h) + f(x - h) - 2f(x) \quad \text{and} \quad G(h) = h^2.$$

Since $f''(x)$ exists, f is continuous at x by Theorem 5.2. It is clear that x is a limit point of E and so Theorem 4.6 implies that

$$\lim_{h \rightarrow 0} f(x + h) = f(x) \quad \text{and} \quad \lim_{h \rightarrow 0} f(x - h) = f(x).$$

Therefore, we have $F(h) \rightarrow 0$ and $G(h) \rightarrow 0$ as $h \rightarrow 0$. By Definition 5.14, since $f''(x)$ exists at x , $f'(t)$ must exist in $(x - \delta, x + \delta)$ for some $\delta > 0$ and then f must be differentiable in $(x - \delta, x + \delta)$. Since our h can be chosen arbitrarily so that

$$(x - h, x + h) \subseteq (x - \delta, x + \delta),$$

we can conclude that $F(h)$ is differentiable as a function of h and Theorem 5.5^b gives

$$F'(h) = f'(x + h) - f'(x - h).$$

By Theorem 5.13, we have

$$\lim_{h \rightarrow 0} \frac{f(x + h) + f(x - h) - 2f(x)}{h^2} = \lim_{h \rightarrow 0} \frac{F(h)}{G(h)} = \lim_{h \rightarrow 0} \frac{F'(h)}{G'(h)} = \lim_{h \rightarrow 0} \frac{f'(x + h) - f'(x - h)}{2h}. \quad (5.9)$$

On the other hand, we follow from Definition 5.1 that

$$\begin{aligned} f''(x) &= \frac{1}{2}(f''(x) + f''(x)) \\ &= \frac{1}{2} \left[\lim_{h \rightarrow 0} \frac{f'(x + h) - f'(x)}{h} + \lim_{h \rightarrow 0} \frac{f'(x) - f'(x - h)}{h} \right] \\ &= \lim_{h \rightarrow 0} \frac{f'(x + h) - f'(x - h)}{2h}. \end{aligned} \quad (5.10)$$

By comparing the limits (5.9) and (5.10), we have the desired result. This finishes the proof of the problem. ■

Problem 5.12

Rudin Chapter 5 Exercise 12.

Proof. We have

$$f(x) = \begin{cases} -x^3, & \text{if } x \leq 0; \\ x^3, & \text{if } x > 0 \end{cases}$$

so that

$$f'(x) = \begin{cases} -3x^2, & \text{if } x < 0; \\ 3x^2, & \text{if } x > 0, \end{cases} \quad \text{and} \quad f''(x) = \begin{cases} -6x, & \text{if } x < 0; \\ 6x, & \text{if } x > 0. \end{cases}$$

To compute $f'(0)$, we consider $f'(0+)$ and $f'(0-)$ so that

$$f'(0+) = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{f(t) - f(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{t^3}{t} = 0 \quad \text{and} \quad f'(0-) = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{f(t) - f(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{-t^3}{t} = 0.$$

Hence we have $f'(0) = f'(0+) = f'(0-) = 0$.

Similarly, to compute $f''(0)$, we consider $f''(0+)$ and $f''(0-)$ so that

$$f''(0+) = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{f'(t) - f'(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{3t^2}{t} = 0 \quad \text{and} \quad f''(0-) = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{f'(t) - f'(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{-3t^2}{t} = 0.$$

Hence we have $f''(0) = f''(0+) = f''(0-) = 0$.

Finally, we consider $f^{(3)}(0+)$ and $f^{(3)}(0-)$ so that

$$f^{(3)}(0+) = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{f''(t) - f''(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t > 0}} \frac{6t}{t} = 6 \quad \text{and} \quad f^{(3)}(0-) = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{f''(t) - f''(0)}{t - 0} = \lim_{\substack{t \rightarrow 0 \\ t < 0}} \frac{-6t}{t} = -6.$$

Since $f^{(3)}(0+) \neq f^{(3)}(0-)$, we conclude that $f^{(3)}(0)$ does not exist, finishing the proof of the problem. ■

^bBy our construction, we know that the function $a(h) = x + h$ is continuous on $[0, \frac{\delta}{2}]$, $a'(h) = 1$ exists on $[0, \frac{\delta}{2}]$, the interval $(x - \delta, x + \delta)$ certainly contains $a([0, \frac{\delta}{2}]) = [x, x + \frac{\delta}{2}]$, and f is differentiable on $[x, x + \frac{\delta}{2}]$. Hence the functions a and f satisfy the conditions of Theorem 5.5.

Problem 5.13*Rudin Chapter 5 Exercise 13.*

Proof. It is unavoidable to make two assumptions here. First of all, by Example 5.6(a) or [21, Eqn. (49), Chap. 8], we know that the function $\sin x$ is differentiable in $\mathbb{R}$. Secondly, when $x < 0$, the function x^a is a complex multi-valued function. In fact, we have $x^a = e^{a \ln x}$, where $\ln x = \ln |x| + i \arg x$ is the complex logarithm function. If we take the principal branch of the function $\ln x$, then the power function x^a is single-valued. Here we assume without proof that x^a is differentiable in $[-1, 1]$ and

$$(x^a)' = ax^{a-1}.$$

- (a) By Theorem 4.9, we know that f is continuous for all $x \neq 0$. Therefore, only the continuity at the point $x = 0$ should be considered.

- **Case (i):** $a > 0$. Then since $-1 \leq |\sin(|x|^{-c})| \leq 1$ for all $x \in [-1, 1]$, we have

$$-|x^a| \leq |f(x)| \leq |x^a|.$$

Since $\lim_{x \rightarrow 0} |x^a| = 0$, we have $\lim_{x \rightarrow 0} |f(x)| = 0$ and thus $\lim_{x \rightarrow 0} f(x) = 0$. Since $f(0) = 0$, we have $\lim_{x \rightarrow 0} f(x) = f(0)$ and so Theorem 4.6 implies that f is continuous at $x = 0$.

- **Case (ii):** $a = 0$. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{(2n\pi + \frac{\pi}{2})^{\frac{1}{c}}}$. Since $c > 0$, we have $\lim_{n \rightarrow \infty} x_n = 0$. Now we have

$$f(x_n) = x_n^0 \sin(|x_n|^{-c}) = \sin\left(2n\pi + \frac{\pi}{2}\right) = 1 \quad (5.11)$$

for all positive integers n so that $\lim_{n \rightarrow \infty} f(x_n) = 1 \neq f(0)$. Thus f is not continuous at $x = 0$ in this case.

- **Case (iii):** $a < 0$. Similarly, we consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{(2n\pi + \frac{\pi}{2})^{\frac{1}{c}}}$. Instead of the expression (5.11), we have

$$f(x_n) = x_n^a \sin(|x_n|^{-c}) = \left(2n\pi + \frac{\pi}{2}\right)^{-\frac{a}{c}} \sin\left(2n\pi + \frac{\pi}{2}\right) = \left(2n\pi + \frac{\pi}{2}\right)^{-\frac{a}{c}} \quad (5.12)$$

for all positive integers n . Since $a < 0$ and $c > 0$, $-\frac{a}{c} > 0$ and then the expression (5.12) implies that $\lim_{n \rightarrow \infty} f(x_n) = \infty \neq f(0)$. Thus f is not continuous at $x = 0$ in this case.

Hence we establish the result that f is continuous if and only if $a > 0$.

- (b) Note that we have

$$\phi(x) = \frac{f(x) - f(0)}{x - 0} = \frac{x^a \sin(|x|^{-c}) - 0}{x - 0} = x^{a-1} \sin(|x|^{-c}) \quad (5.13)$$

and so $f'(0) = \lim_{x \rightarrow 0} \phi(x) = \lim_{x \rightarrow 0} x^{a-1} \sin(|x|^{-c})$.

- **Case (i):** $a \leq 1$. We consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{(n\pi + \frac{\pi}{2})^{\frac{1}{c}}}$. Since $c > 0$, we have $\lim_{n \rightarrow \infty} x_n = 0$. By a similar argument as in part (a), we have

$$\phi(x_n) = x_n^{a-1} \sin(|x_n|^{-c}) = \left(n\pi + \frac{\pi}{2}\right)^{-\frac{a-1}{c}} \sin\left(n\pi + \frac{\pi}{2}\right) = (-1)^n \left(n\pi + \frac{\pi}{2}\right)^{-\frac{a-1}{c}} \quad (5.14)$$

for all positive integers n . Now if $a = 1$, then

$$\phi(x_{2k}) = 1 \quad \text{and} \quad \phi(x_{2k+1}) = -1.$$

By Theorem 4.2, $\lim_{n \rightarrow \infty} \phi(x_n)$ does not exist in this case! If $a < 1$, then $-\frac{a-1}{c} > 0$ and the expression (5.14) implies that $\phi(x_{2k}) \rightarrow +\infty$ and $\phi(x_{2k+1}) \rightarrow -\infty$ as $k \rightarrow \infty$. Thus $\lim_{n \rightarrow \infty} \phi(x_n)$ does not exist in this case! Therefore, we have shown that if $f'(0)$ exists, then $a > 1$.

- **Case (ii):** $a > 1$. Then since $-1 \leq |\sin(|x|^{-c})| \leq 1$ for all $x \in [-1, 1]$, we have

$$0 \leq |\phi(x)| \leq |x|^{a-1}.$$

Since $\lim_{x \rightarrow 0} |x|^{a-1} = 0$, we have $\lim_{x \rightarrow 0} |\phi(x)| = 0$ and thus $\lim_{x \rightarrow 0} \phi(x) = 0$. In other words, we have $f'(0) = 0$ in this case.

- (c) Since x^a and $\sin(|x|^{-c})$ are differentiable in $[-1, 1] \setminus \{0\}$, Theorem 5.3(b) implies that f is differentiable in $[-1, 1] \setminus \{0\}$. Suppose that $a > 1$. By the result of part (b) and Theorem 5.5, we have

$$f'(x) = \begin{cases} x^{a-1}[a \sin(x^{-c}) - cx^{-c} \cos(x^{-c})], & \text{if } 0 < x \leq 1; \\ 0, & \text{if } x = 0; \\ x^{a-1}[a \sin(-x)^{-c} - c(-x)^{-c} \cos(-x)^{-c}], & \text{if } -1 \leq x < 0. \end{cases} \quad (5.15)$$

- **Case (i):** $a \geq 1 + c$. Then we have $a > 1$ and $a - 1 - c \geq 0$. Therefore, we can deduce from the expressions (5.15) that

$$|f'(x)| \leq |ax^{a-1} \sin(|x|^{-c})| + |cx^{a-1-c} \cos(|x|^{-c})| \leq a|x|^{a-1} + c|x|^{a-1-c} \leq a + c$$

for all $x \in [-1, 1]$. This shows that f' is bounded by $a + c$ in this case.

- **Case (ii):** $a < 1 + c$. Consider the sequence $\{x_n\}$ defined by $x_n = (2n\pi)^{-\frac{1}{c}} > 0$. Then we have $\sin(x_n^{-c}) = \sin 2n\pi = 0$, $\cos(x_n^{-c}) = \cos 2n\pi = 1$ and so

$$f'(x_n) = -c(x_n)^{a-1-c} = -c(2n\pi)^{\frac{c+1-a}{c}} \rightarrow -\infty$$

as $n \rightarrow \infty$. Therefore, f' is unbounded on $[-1, 1]$ in this case.

Hence we have shown that f' is bounded if and only if $a \geq 1 + c$.

- (d) By the expressions (5.15), we know that f' is continuous^c for all $x \in [-1, 1] \setminus \{0\}$, so we check whether f' is continuous at $x = 0$ or not.

- **Case (i):** $a > 1 + c$. Then we have $a > 1$ and $a - 1 - c > 0$. Since

$$|f'(x)| \leq |ax^{a-1} \sin(|x|^{-c})| + |cx^{a-1-c} \cos(|x|^{-c})| \leq a|x|^{a-1} + c|x|^{a-1-c},$$

we have $\lim_{x \rightarrow 0} |f'(x)| = 0$ which implies that $\lim_{x \rightarrow 0} f'(x) = 0$. By the expression (5.15), we have $\lim_{x \rightarrow 0} f'(x) = f'(0)$, so f' is continuous on $[-1, 1]$ by Theorem 4.6.

- **Case (ii):** $a < 1 + c$. Then we know from part (c) that f' is unbounded on $[-1, 1]$. Assume that f' was continuous on the compact set $[-1, 1]$. By Theorem 4.15, f' is bounded on $[-1, 1]$, a contradiction. Thus it is not continuous on $[-1, 1]$.

- **Case (iii):** $a = 1 + c$. Then we consider the sequence $\{x_n\}$ defined by $x_n = (2n\pi)^{-\frac{1}{c}} > 0$. Then we have $\sin(x_n^{-c}) = \sin(2n\pi) = 0$, $\cos(x_n^{-c}) = \cos(2n\pi) = 1$ and so

$$f'(x_n) = -c(x_n)^{a-1-c} = -c.$$

Since $f'(0+) = \lim_{n \rightarrow \infty} f'(x_n) = -c \neq 0 = f'(0)$, f' is not continuous at $x = 0$ in this case.

Hence we have shown that f' is continuous if and only if $a > 1 + c$.

- (e) It follows from the expressions (5.15) that

$$\begin{aligned} \varphi(x) &= \frac{f'(x) - f'(0)}{x - 0} \\ &= \frac{x^{a-1}[a \sin(|x|^{-c}) - c|x|^{-c} \cos(|x|^{-c})]}{x - 0} \\ &= x^{a-2}[a \sin(|x|^{-c}) - c|x|^{-c} \cos(|x|^{-c})] \end{aligned}$$

and so $f''(0) = \lim_{x \rightarrow 0} \varphi(x) = \lim_{x \rightarrow 0} x^{a-2}[a \sin(|x|^{-c}) - c|x|^{-c} \cos(|x|^{-c})]$.

^cBy [21, Eqn. (49), Chap. 8] again, we assume that the function $\cos x$ is differentiable in $\mathbb{R}$.

- **Case (i):** $a \leq 2 + c$. We consider the sequences $\{x_n\}$ and $\{y_n\}$ defined by $x_n = \frac{1}{(2n\pi)^{\frac{1}{c}}}$ and $y_n = \frac{1}{[(2n+1)\pi]^{\frac{1}{c}}}$. Since $c > 0$, we have $\lim_{n \rightarrow \infty} x_n = 0$ and $\lim_{n \rightarrow \infty} y_n = 0$. By a similar argument as in part (b), we have

$$\begin{aligned}\varphi(x_n) &= x_n^{a-2} [a \sin(|x_n|^{-c}) - c|x_n|^{-c} \cos(|x_n|^{-c})] \\ &= (2n\pi)^{-\frac{a-2}{c}} [a \sin(2n\pi) - c(2n\pi) \cos(2n\pi)] \\ &= -c(2n\pi)^{\frac{c+2-a}{c}}\end{aligned}\tag{5.16}$$

and

$$\begin{aligned}\varphi(y_n) &= y_n^{a-2} [a \sin(|y_n|^{-c}) - c|y_n|^{-c} \cos(|y_n|^{-c})] \\ &= [(2n+1)\pi]^{-\frac{a-2}{c}} \{a \sin(2n+1)\pi - c[(2n+1)\pi] \cos(2n+1)\pi\} \\ &= c[(2n+1)\pi]^{\frac{c+2-a}{c}}\end{aligned}\tag{5.17}$$

for all positive integers n . If $a = c + 2$, then the expressions (5.16) and (5.17) imply that $\varphi(x_n) = -c$ and $\varphi(y_n) = c$ respectively. Since $\varphi(x_n) \neq \varphi(y_n)$, $f''(0)$ does not exist by Theorem 4.2. If $a < c + 2$, then we obtain from the expression (5.16) that $\lim_{n \rightarrow \infty} \varphi(x_n) = -\infty$.

Thus $f''(0)$ does not exist by Theorem 4.2.

- **Case (ii):** $a > 2 + c$. Then since $-1 \leq |\sin(x^{-c})| \leq 1$ and $-1 \leq |\cos(x^{-c})| \leq 1$ for all $x \in [-1, 1]$, we have

$$0 \leq |\varphi(x)| \leq a|x|^{a-2} + c|x|^{a-2-c}.$$

Since $\lim_{x \rightarrow 0} |x|^{a-1} = 0$ and $\lim_{x \rightarrow 0} |x|^{a-2-c} = 0$, we have $\lim_{x \rightarrow 0} |\varphi(x)| = 0$ and thus $\lim_{x \rightarrow 0} \varphi(x) = 0$. In other words, we have $f''(0) = 0$ in this case.

Hence we have shown that $f''(0)$ exists if and only if $a > 2 + c$.

- (f) Suppose that $a > 2 + c$. By the result of part (e) and the expressions (5.15), we have

$$f''(x) = \begin{cases} (a(a-1)x^{a-2} - c^2x^{a-2-2c}) \sin(x^{-c}) - c(2a-1-c)x^{a-2-c} \cos(x^{-c}), & \text{if } 0 < x \leq 1; \\ 0, & \text{if } x = 0; \\ (a(a-1) - c^2(-x)^{-2c})x^{a-2} \sin(-x)^{-c} - c(2a-1-c)x^{a-2}(-x)^{-c} \cos(-x)^{-c}, & \text{if } -1 \leq x < 0. \end{cases}\tag{5.18}$$

- **Case (i):** $a \geq 2 + 2c$. Then we have $a - 2 \geq 2c > 0$, $a - 2 - c \geq c > 0$ and $a - 2 - 2c \geq 0$. Therefore, we can deduce from the expressions (5.18) that

$$\begin{aligned}|f''(x)| &\leq |[a(a-1)x^{a-2} - c^2x^{a-2-2c}] \sin(|x|^{-c})| + |c(2a-1-c)x^{a-2-c} \cos(|x|^{-c})| \\ &\leq a(a-1)|x|^{a-2} + c^2|x|^{a-2-2c} + c(2a-1-c)|x|^{a-2-c} \\ &\leq a(a-1) + c^2 + c(2a-1-c) \\ &= a(a-1) + c(2a-1)\end{aligned}$$

for all $x \in [-1, 1]$. This shows that f'' is bounded by $a(a-1) + c(2a-1)$ in this case.

- **Case (ii):** $a < 2 + 2c$ and $2a - 1 - c \neq 0$. If $a \geq 2$, then we consider the sequence $\{x_n\}$ defined by $x_n = (2n\pi + \frac{\pi}{2})^{-\frac{1}{c}} > 0$ so that we have

$$\sin(x_n^{-c}) = \sin\left(2n + \frac{1}{2}\right)\pi = 1 \quad \text{and} \quad \cos(x_n^{-c}) = \cos\left(2n + \frac{1}{2}\right)\pi = 0.$$

Thus we have

$$\begin{aligned}f''(x_n) &= [a(a-1)x_n^{a-2} - c^2x_n^{a-2-2c}] \sin(x_n^{-c}) - c(2a-1-c)x_n^{a-2-c} \cos(x_n^{-c}) \\ &= a(a-1)\left(2n\pi + \frac{\pi}{2}\right)^{\frac{2-a}{c}} - c^2\left(2n\pi + \frac{\pi}{2}\right)^{\frac{2+2c-a}{c}}.\end{aligned}$$

Since $\frac{2-a}{c} \leq 0$ and $\frac{2+2c-a}{c} > 0$, we have $f''(x_n) \rightarrow -\infty$ as $n \rightarrow +\infty$.

If $a < 2$, then we consider the sequence $\{y_n\}$ defined by $y_n = (2n\pi)^{-\frac{1}{c}} > 0$ so that we have $\sin(y_n^{-c}) = \sin(2n\pi) = 0$ and $\cos(y_n^{-c}) = \cos(2n\pi) = 1$. Thus we have

$$\begin{aligned} f''(y_n) &= [a(a-1)y_n^{a-2} - c^2 y_n^{a-2-2c}] \sin(y_n^{-c}) - c(2a-1-c)y_n^{a-2-c} \cos(y_n^{-c}) \\ &= -c(2a-1-c)(2n\pi)^{\frac{c+2-a}{c}}. \end{aligned}$$

Since $\frac{c+2-a}{c} > 1$ and $2a-1-c \neq 0$, we have $f''(y_n) \rightarrow -\infty$. Therefore, f'' is unbounded in this case.

– **Case (iii):** $a < 2 + 2c$ and $2a-1-c = 0$. Then we have from the expressions (5.18) that

$$f''(x) = \begin{cases} (a(a-1)x^{a-2} - c^2 x^{a-2-2c}) \sin(x^{-c}), & \text{if } 0 < x \leq 1; \\ 0, & \text{if } x = 0; \\ (a(a-1) - c^2(-x)^{-2c})x^{a-2} \sin(-x)^{-c}, & \text{if } -1 \leq x < 0. \end{cases}$$

If $a \geq 2$, then we can apply the same sequence $\{x_n\}$ as in **Case (ii)** so that we have

$$f''(x_n) = a(a-1) \left(2n\pi + \frac{\pi}{2}\right)^{\frac{2-a}{c}} - c^2 \left(2n\pi + \frac{\pi}{2}\right)^{\frac{2+2c-a}{c}}. \quad (5.19)$$

Since $\frac{2-a}{c} \leq 0$ and $\frac{2+2c-a}{c} > 0$, we have $f''(x_n) \rightarrow -\infty$ as $n \rightarrow +\infty$.

If $a < 2$, then the expression (5.19) can be rewritten as

$$f''(x_n) = \left(2n\pi + \frac{\pi}{2}\right)^{\frac{2-a}{c}} \left[a(a-1) - c^2 \left(2n\pi + \frac{\pi}{2}\right)^2 \right].$$

Since $\frac{2-a}{c} > 0$, we have $f''(x_n) \rightarrow -\infty$ as $n \rightarrow +\infty$. Therefore, f'' is also unbounded in this case.

Hence we have shown that f'' is bounded if and only if $a \geq 2 + 2c$.

(g) By the expressions (5.18), we know that f'' is continuous for all $x \in [-1, 1] \setminus \{0\}$, so we check whether f'' is continuous at $x = 0$ or not.

– **Case (i):** $a > 2 + 2c$. Then we have $a > 2$ and $a-2-c > a-2-2c > 0$. Since

$$\begin{aligned} |f''(x)| &\leq |a(a-1)x^{a-2} - c^2 x^{a-2-2c} \sin(|x|^{-c})| + |c(2a-1-c)x^{a-2-c} \cos(|x|^{-c})| \\ &\leq a(a-1)|x|^{a-2} + c^2|x|^{a-2-2c} + c(2a-1-c)|x|^{a-2-c}, \end{aligned}$$

we have $\lim_{x \rightarrow 0} |f''(x)| = 0$ which implies that $\lim_{x \rightarrow 0} f''(x) = 0$. By the expressions (5.18) again, we have $\lim_{x \rightarrow 0} f''(x) = f''(0)$, so f'' is continuous at 0 by Theorem 4.6.

– **Case (ii):** $a < 2 + 2c$. Then we know from part (f) that f'' is unbounded on $[-1, 1]$. Assume that f'' was continuous on the compact set $[-1, 1]$. By Theorem 4.15, f'' is bounded on $[-1, 1]$, a contradiction. Thus it is not continuous on $[-1, 1]$.

– **Case (iii):** $a = 2 + 2c$. Then we consider the sequence $\{x_n\}$ defined by $x_n = (2n\pi + \frac{\pi}{2})^{-\frac{1}{c}} > 0$. Then we have $\sin(x_n^{-c}) = 1$, $\cos(x_n^{-c}) = 0$ and so

$$f''(x_n) = a(a-1)x_n^{a-2} - c^2 x_n^{a-2-2c} = a(a-1) \left(2n\pi + \frac{\pi}{2}\right)^{-2} - c^2.$$

Since $f''(0+) = \lim_{n \rightarrow \infty} f''(x_n) = -c^2 \neq 0 = f''(0)$, f'' is not continuous at $x = 0$ in this case.

Hence we have shown that f'' is continuous if and only if $a > 2 + 2c$.

This completes the proof of the problem. ■

Problem 5.14

Rudin Chapter 5 Exercise 14.

Proof. We prove the results one by one.

- **f is convex if and only if f' is monotonically increasing.**

– Suppose that f is convex in (a, b) . Then it follows from Problem 4.23 that

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(s)}{u - s} \leq \frac{f(u) - f(t)}{u - t} \quad (5.20)$$

whenever $a < s < t < u < b$. Similarly, we have

$$\frac{f(u) - f(t)}{u - t} \leq \frac{f(v) - f(t)}{v - t} \leq \frac{f(v) - f(u)}{v - u} \quad (5.21)$$

whenever $a < t < u < v < b$.

Combining the inequalities (5.20) and (5.21), we have

$$\frac{f(t) - f(s)}{t - s} \leq \frac{f(u) - f(t)}{u - t} \leq \frac{f(v) - f(u)}{v - u} \quad (5.22)$$

whenever $a < s < t < u < v < b$. Since f is differentiable in (a, b) , we have from Definition 5.1 that $f'(x) = f'(x+) = f'(x-)$ for every $x \in (a, b)$. In particular, we have $f'(s) = f'(s+)$ and $f'(u) = f'(u+)$. Thus these and the inequalities (5.22) together imply that

$$f'(s) = f'(s+) = \lim_{\substack{t \rightarrow s \\ t > s}} \frac{f(t) - f(s)}{t - s} \leq \lim_{\substack{v \rightarrow u \\ v > u}} \frac{f(v) - f(u)}{v - u} = f'(u+) = f'(u)$$

if $a < s < u < b$. Hence it means that f' is monotonically increasing in (a, b) .

– Suppose that f' is monotonically increasing in (a, b) . Let $a < x < b$, $a < y < b$ and $0 < \lambda < 1$, we want to show that

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (5.23)$$

If $t = \lambda x + (1 - \lambda)y$, then it is easily to see that

$$0 < \lambda = \frac{y - t}{y - x} < 1 \quad \text{and} \quad 1 - \lambda = \frac{t - x}{y - x}.$$

After doing some simple algebra, we can show that the inequality (5.23) is *equivalent* to the following inequalities

$$\begin{aligned} f(t) &\leq \frac{y - t}{y - x} f(x) + \frac{t - x}{y - x} f(y) \\ (y - x)f(t) &\leq (y - t)f(x) + (t - x)f(y) \\ [(y - t) + (t - x)]f(t) &\leq (y - t)f(x) + (t - x)f(y) \\ \frac{f(t) - f(x)}{t - x} &\leq \frac{f(y) - f(t)}{y - t}. \end{aligned} \quad (5.24)$$

Since f is differentiable in (a, b) , f is differentiable in $[x, y]$ particularly and then it is continuous on $[x, y]$. By Theorem 5.10, we have

$$f(t) - f(x) = (t - x)f'(\eta) \quad \text{and} \quad f(y) - f(t) = (y - t)f'(\xi), \quad (5.25)$$

where t is *any* real number such that $x < t < y$, η and ξ are *some* real numbers such that $\eta \in (x, t)$ and $\xi \in (t, y)$ respectively. Now we follow from the inequality (5.24) and the expressions (5.25) that

$$f'(\eta) \leq f'(\xi). \quad (5.26)$$

By our assumption, f' is monotonically increasing in (a, b) , so the inequality

$$f'(r) \leq f'(s)$$

must hold for *any* real numbers r, s with $a < r < s < b$. In particular, the inequality (5.26) holds so that the inequality (5.24) and then the inequality (5.23) also hold. By Problem 4.23, f is convex in (a, b) .

- f is convex if and only if $f''(x) \geq 0$ for all $x \in (a, b)$.

– Suppose that f is convex in (a, b) . Then the previous part shows that f' is monotonically increasing in (a, b) . That is, if $a < x < t < b$, then we have $f'(x) \leq f'(t)$ and similarly, if $a < t < x < b$, then we have $f'(t) \leq f'(x)$. In both cases, we always have

$$\frac{f'(t) - f'(x)}{t - x} \geq 0 \quad (5.27)$$

for every $x, t \in (a, b)$. Thus it follows from the inequality (5.27) and Definition 5.1 that

$$f''(x) = \lim_{t \rightarrow x} \frac{f'(t) - f'(x)}{t - x} \geq 0$$

for all $x \in (a, b)$.

– Suppose that $f''(x) \geq 0$ for all $x \in (a, b)$. By Theorem 5.11, f' must be monotonically increasing in (a, b) . Thus the previous part shows that f is convex in (a, b) .

Hence, we complete the proof of the problem. ■

5.2 Applications of Taylor's theorem

Problem 5.15

Rudin Chapter 5 Exercise 15.

Proof. The inequality $M_1^2 \leq 4M_0M_2$ is called Landau's inequality.

We note that the inequality holds trivially if " $M_0 = +\infty$ or $M_2 = +\infty$ " or " $M_1 = 0$ ". Suppose that $M_1 \neq 0, 0 \leq M_0 < +\infty$ and $0 \leq M_2 < +\infty$. Furthermore, if $M_0 = 0$, then we have $|f(x)| \leq 0$ for all $x \in (a, +\infty)$ so that

$$f(x) = f'(x) = f''(x) = 0$$

for all $x \in (a, +\infty)$. Thus we have $M_0 = M_1 = M_2 = 0$ which contradicts to the fact that $M_1 \neq 0$. Next, if $M_2 = 0$, then we have $|f''(x)| \leq 0$ for all $x \in (a, +\infty)$ so that

$$f''(x) = 0$$

for all $x \in (a, +\infty)$. By Theorem 5.11(b), we have $f'(x) = c$ and then $f(x) = cx + d$ for some constants c and d .^d In this case, we have $M_1 = c$ and $M_0 = +\infty$ which contradicts to the fact that $M_0 < +\infty$. Therefore, we may suppose further that

$$0 < M_0 < +\infty \quad \text{and} \quad 0 < M_2 < +\infty.$$

Let $x \in (a, +\infty)$. Then it is true that $[x, x + 2h] \subset (a, +\infty)$ for every $h > 0$. By the hypothesis, f is a twice-differentiable real function in $[x, x + 2h]$. In other words, the function f satisfies the conditions of Theorem 5.15 (Taylor's Theorem). Put $P(t) = f(x) + f'(x)(t - x)$ and so Theorem 5.15 ($\alpha = x$ and $\beta = x + 2h$) implies that

$$f(x + 2h) = P(x + 2h) + \frac{f''(\xi)}{2}(x + 2h - x)^2 = f(x) + f'(x) \cdot (2h) + 2f''(\xi)h^2 \quad (5.28)$$

^dWe apply the theory of Chap. 6 here to obtain that $f(x) = cx + d$.

for some $\xi \in (x, x + 2h)$. Now we rewrite (5.28) to get

$$f'(x) = \frac{1}{2h}[f(x + 2h) - f(x)] - hf''(\xi)$$

and then

$$|f'(x)| \leq \frac{1}{2h}[|f(x + 2h)| + |f(x)|] + h|f''(\xi)| \quad (5.29)$$

for some $\xi \in (x, x + 2h)$. By the definitions of M_0, M_1 and M_2 , the inequality (5.29) implies that

$$|f'(x)| \leq hM_2 + \frac{M_0}{h}, \quad (5.30)$$

where $x \in (a, +\infty)$.

Since h is arbitrary and M_0, M_2 are positive numbers, we may take $h = \sqrt{\frac{M_0}{M_2}} > 0$ in the inequality (5.30) to get $|f'(x)| \leq 2\sqrt{M_0M_2}$ on $(a, +\infty)$. Since it is true for all $x \in (a, +\infty)$, we have $M_1 \leq 2\sqrt{M_0M_2}$ which gives

$$M_1^2 \leq 4M_0M_2 \quad (5.31)$$

as desired.

Consider the given example in the hint, we acquire

$$f'(x) = \begin{cases} 4x & (-1 < x < 0), \\ \frac{4x}{(x^2+1)^2} & (0 < x < \infty), \end{cases} \quad \text{and} \quad f''(x) = \begin{cases} 4 & (-1 < x < 0), \\ \frac{4-12x^2}{(x^2+1)^3} & (0 < x < \infty). \end{cases}$$

Since $f'(x) \rightarrow 0$ and $f''(x) \rightarrow 4$ as $x \rightarrow 0$, the footnote of Problem 5.9 shows that $f'(0) = 0$ and $f''(0) = 4$. Thus we have

$$f'(x) = \begin{cases} 4x & (-1 < x < 0), \\ 0 & (x = 0), \\ \frac{4x}{(x^2+1)^2} & (0 < x < \infty), \end{cases} \quad \text{and} \quad f''(x) = \begin{cases} 4 & (-1 < x < 0), \\ 4 & (x = 0), \\ \frac{4-12x^2}{(x^2+1)^3} & (0 < x < \infty). \end{cases} \quad (5.32)$$

From the definition of f and the expressions (5.32), we have $M_0 = 1, M_1 = 4$ and $M_2 = 4$ which give the equality (5.31).

We claim that the inequality $M_1^2 \leq 4M_0M_2$ also holds for vector-valued functions. Suppose that $\mathbf{f} = (f_1, f_2, \dots, f_n)$, where $f_1, f_2, \dots, f_n$ are twice-differentiable real functions on $(a, +\infty)$. By Remarks 5.16, the vector-valued function $\mathbf{f}$ is also twice-differentiable on $(a, +\infty)$. Let, further, M_0, M_1 and M_2 be the least upper bounds of $|\mathbf{f}(x)|, |\mathbf{f}'(x)|$ and $|\mathbf{f}''(x)|$ on $(a, +\infty)$ respectively. For any $c \in (a, +\infty)$, we consider the function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F(x) = f'_1(c)f_1(x) + f'_2(c)f_2(x) + \dots + f'_n(c)f_n(x).$$

Since each f_k is twice-differentiable real function on $(a, +\infty)$, Theorem 5.3 implies that F is also a twice-differentiable real function on $(a, +\infty)$. Therefore, the previous analysis shows that, instead of the inequality $|f'(x)| \leq 2\sqrt{M_0M_2}$, we have

$$|F'(x)| \leq 2\sqrt{\mathcal{M}_0\mathcal{M}_2}, \quad (5.33)$$

where $\mathcal{M}_0, \mathcal{M}_1$ and $\mathcal{M}_2$ are the least upper bounds of $|F(x)|, |F'(x)|$ and $|F''(x)|$, respectively, on $(a, +\infty)$.

Next, let's recall the Cauchy-Schwarz inequality for vectors: If $\mathbf{u} = (u_1, u_2, \dots, u_n)$ and $\mathbf{v} = (v_1, v_2, \dots, v_n)$, then we have

$$|u_1v_1 + u_2v_2 + \dots + u_nv_n| \leq \sqrt{u_1^2 + u_2^2 + \dots + u_n^2} \cdot \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}. \quad (5.34)$$

Apply the result (5.34) to $|F(x)|$ and $|F''(x)|$ to get

$$\begin{aligned} |F(x)| &= |f'_1(c)f_1(x) + f'_2(c)f_2(x) + \dots + f'_n(c)f_n(x)| \\ &\leq \sqrt{[f'_1(c)]^2 + [f'_2(c)]^2 + \dots + [f'_n(c)]^2} \cdot \sqrt{[f_1(x)]^2 + [f_2(x)]^2 + \dots + [f_n(x)]^2} \end{aligned}$$

$$\leq M_1 \cdot M_0$$

and

$$\begin{aligned} |F''(x)| &= |f'_1(c)f''_1(x) + f'_2(c)f''_2(x) + \cdots + f'_n(c)f''_n(x)| \\ &\leq \sqrt{[f'_1(c)]^2 + [f'_2(c)]^2 + \cdots + [f'_n(c)]^2} \cdot \sqrt{[f''_1(x)]^2 + [f''_2(x)]^2 + \cdots + [f''_n(x)]^2} \\ &\leq M_1 \cdot M_2 \end{aligned}$$

on $(a, +\infty)$. Therefore, it follows from these and the inequality (5.33) that

$$|F'(x)| \leq 2\sqrt{\mathcal{M}_0\mathcal{M}_2} \leq 2\sqrt{M_1 \cdot M_0 \cdot M_1 \cdot M_2} = 2M_1\sqrt{M_0M_2}, \quad (5.35)$$

where $x \in (a, +\infty)$. In particular, we may take $x = c$ in the inequality (5.35) to obtain

$$[f'_1(c)]^2 + [f'_2(c)]^2 + \cdots + [f'_n(c)]^2 \leq 2M_1\sqrt{M_0M_2}. \quad (5.36)$$

Since the right-hand side of the inequality (5.36) is *independent* of the choice of c , we let c run through all values in $(a, +\infty)$ so that $M_1^2 \leq 2M_1\sqrt{M_0M_2}$ which implies that the desired inequality (5.31). Finally, the equality also holds for vector-valued functions if we consider $\mathbf{f}(x) = (f(x), 0, \dots, 0)$, where $f(x)$ is the given example in the hint. This completes the proof of the problem. ■

Problem 5.16

Rudin Chapter 5 Exercise 16.

Proof. By the hypothesis, we have $|f''(x)| \leq K$ on $(0, \infty)$ for some positive K . Let

$$M_0(a) = \sup_{x \in (a, \infty)} |f(x)|, \quad M_1(a) = \sup_{x \in (a, \infty)} |f'(x)| \quad \text{and} \quad M_2(a) = \sup_{x \in (a, \infty)} |f''(x)|.$$

Thus Problem 5.15 implies that

$$M_1^2(a) \leq 4M_2(a)M_0(a) \leq 4KM_0(a). \quad (5.37)$$

Since $f(x) \rightarrow 0$ as $x \rightarrow \infty$, we have $M_0(a) \rightarrow 0$ as $a \rightarrow \infty$. Otherwise, there was a sequence $\{a_n\}$, a positive integer N and a $\epsilon > 0$ such that $a_n \rightarrow \infty$ as $n \rightarrow \infty$ but

$$M_0(a_n) > \epsilon$$

for all $n \geq N$. Since $M_0(a_n)$ is the least upper bound of $|f(x)|$ on (a_n, ∞) , there exist $x_n \in (a_n, \infty)$ such that $|f(x_n)| > \epsilon$ for all $n \geq N$, contradicting to the fact that $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, it follows from the inequality (5.37) that

$$\lim_{a \rightarrow \infty} M_1^2(a) \leq 4K \lim_{a \rightarrow \infty} M_0(a) = 0. \quad (5.38)$$

It is clear that $a \rightarrow \infty$ implies that $x \rightarrow \infty$. Since $|f(x)| \leq M_1(a)$ for all $x \in (a, \infty)$, we follow from this and the inequality (5.38) that

$$\lim_{x \rightarrow \infty} |f'(x)| \leq \lim_{a \rightarrow \infty} M_1(a) = 0.$$

Hence we have $f'(x) \rightarrow 0$ as $x \rightarrow \infty$, completing the proof of the problem. ■

Problem 5.17

Rudin Chapter 5 Exercise 17.

Proof. By Theorem 5.15, we have

$$f(1) = P(1) + \frac{f^{(3)}(s)}{3!}(1-0)^3 \quad \text{and} \quad f(-1) = P(-1) + \frac{f^{(3)}(t)}{3!}(0-1)^3 \quad (5.39)$$

for some $s \in (0, 1)$ and $t \in (-1, 0)$. By the hypothesis, we have

$$P(1) = f(0) + f'(0) \cdot 1 + \frac{f''(0)}{2} \cdot 1^2 = \frac{f''(0)}{2}$$

and

$$P(-1) = f(0) + f'(0) \cdot (-1) + \frac{f''(0)}{2} \cdot (-1)^2 = \frac{f''(0)}{2}.$$

Therefore, the expressions (5.39) imply that

$$1 = \frac{f''(0)}{2} + \frac{f^{(3)}(s)}{6} \quad \text{and} \quad 0 = \frac{f''(0)}{2} - \frac{f^{(3)}(t)}{6}$$

so that $f^{(3)}(s) + f^{(3)}(t) = 6$. Assume that $f^{(3)}(x) < 3$ for all $x \in (-1, 1)$. Then we have $f^{(3)}(s) < 3$ and $f^{(3)}(t) < 3$ so that $f^{(3)}(s) + f^{(3)}(t) < 6$, a contradiction. Hence we have

$$f^{(3)}(x) \geq 3$$

for some $x \in (-1, 1)$. This ends the proof of the problem. ■

Problem 5.18

Rudin Chapter 5 Exercise 18.

Proof. We can prove this version of Taylor's theorem by induction. For $n = 1$, we have

$$P(\beta) + \frac{Q^{(1-1)}(\alpha)}{(1-1)!}(\beta - \alpha)^1 = \sum_{i=0}^0 \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + Q(\alpha)(\beta - \alpha) = f(\alpha) - [f(\alpha) - f(\beta)] = f(\beta).$$

Therefore, the statement is true for $n = 1$. Assume that it is true for $n = k$, i.e.,

$$f(\beta) = P(\beta) + \frac{Q^{(k-1)}(\alpha)}{(k-1)!}(\beta - \alpha)^k = \sum_{i=0}^{k-1} \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + \frac{Q^{(k-1)}(\alpha)}{(k-1)!}(\beta - \alpha)^k. \quad (5.40)$$

For $n = k + 1$, since $f(t) - f(\beta) = (t - \beta)Q(t)$, we have

$$\begin{aligned} f'(t) &= (t - \beta)Q'(t) + Q(t) \\ f''(t) &= (t - \beta)Q''(t) + 2Q'(t) \\ &\vdots \\ f^{(k)}(t) &= (t - \beta)Q^{(k)}(t) + kQ^{(k-1)}(t). \end{aligned} \quad (5.41)$$

Therefore, it follows from the assumption (5.40) and the expression (5.41) that

$$\begin{aligned} P(\beta) + \frac{Q^{(k)}(\alpha)}{k!}(\beta - \alpha)^{k+1} &= \sum_{i=0}^k \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + \frac{Q^{(k)}(\alpha)}{k!}(\beta - \alpha)^{k+1} \\ &= \sum_{i=0}^{k-1} \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + \frac{f^{(k)}(\alpha)}{k!}(\beta - \alpha)^k + \frac{Q^{(k)}(\alpha)}{k!}(\beta - \alpha)^{k+1} \\ &= \sum_{i=0}^{k-1} \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + \left[-\frac{Q^{(k)}(\alpha)}{k!}(\beta - \alpha)^{k+1} + \frac{Q^{(k-1)}(\alpha)}{(k-1)!}(\beta - \alpha)^k \right] \end{aligned}$$

$$\begin{aligned}
& + \frac{Q^{(k)}(\alpha)}{k!}(\beta - \alpha)^{k+1} \\
& = \sum_{i=0}^{k-1} \frac{f^{(i)}(\alpha)}{i!}(\beta - \alpha)^i + \frac{Q^{(k-1)}(\alpha)}{(k-1)!}(\beta - \alpha)^k \\
& = f(\beta).
\end{aligned}$$

Thus the statement is also true for $n = k + 1$. Hence we follow from induction that it is true for all positive integers n , completing the proof of the problem. ■

Problem 5.19

Rudin Chapter 5 Exercise 19.

Proof.

(a) Let $\alpha_n < 0 < \beta_n$. Since $f'(0)$ exists, $|\beta_n - \alpha_n| \geq |\beta_n|$ and $|\beta_n - \alpha_n| \geq |\alpha_n|$, we have

$$\begin{aligned}
|D_n - f'(0)| & = \left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(0) \right| \\
& \leq \left| \frac{f(\beta_n) - f(0) - \beta_n f'(0)}{\beta_n - \alpha_n} \right| + \left| \frac{f(\alpha_n) - f(0) - \alpha_n f'(0)}{\beta_n - \alpha_n} \right| \quad (5.42)
\end{aligned}$$

$$\leq \left| \frac{f(\beta_n) - f(0)}{\beta_n} - f'(0) \right| + \left| \frac{f(\alpha_n) - f(0)}{\alpha_n} - f'(0) \right|. \quad (5.43)$$

Since $f'(0)$ exists, Definition 5.1 and Theorem 4.2 imply that

$$\lim_{n \rightarrow \infty} \frac{f(p_n) - f(0)}{p_n - 0} = f'(0)$$

for every sequence $\{p_n\}$ in $(-1, 1)$ such that $p_n \neq 0$ and $\lim_{n \rightarrow \infty} p_n = 0$. In particular, we have

$$\lim_{n \rightarrow \infty} \frac{f(\alpha_n) - f(0)}{\alpha_n} = \lim_{n \rightarrow \infty} \frac{f(\beta_n) - f(0)}{\beta_n} = f'(0).$$

Hence we deduce from these and the inequality (5.43) that $\lim_{n \rightarrow \infty} D_n = f'(0)$.

(b) Since $0 < \alpha_n < \beta_n$ and $\{\frac{\beta_n}{\beta_n - \alpha_n}\}$ is bounded, we have $\frac{\beta_n}{\beta_n - \alpha_n} \leq M$ for some positive M so that

$$\frac{1}{\beta_n - \alpha_n} \leq \frac{M}{\beta_n} < \frac{M}{\alpha_n}.$$

Therefore, these and the inequality (5.42) imply that

$$\begin{aligned}
|D_n - f'(0)| & < M \left| \frac{f(\beta_n) - f(0) - \beta_n f'(0)}{\beta_n} \right| + M \left| \frac{f(\alpha_n) - f(0) - \alpha_n f'(0)}{\alpha_n} \right| \\
& = M \left| \frac{f(\beta_n) - f(0)}{\beta_n} - f'(0) \right| + M \left| \frac{f(\alpha_n) - f(0)}{\alpha_n} - f'(0) \right|.
\end{aligned}$$

Hence, a similar argument as in part (a) gives the desired result that $\lim_{n \rightarrow \infty} D_n = f'(0)$.

(c) Since f' is continuous in $(-1, 1)$ and $-1 < \alpha_n < \beta_n < 1$, f is continuous on $[\alpha_n, \beta_n]$ and differentiable in (α_n, β_n) . By Theorem 5.10, we have

$$\begin{aligned}
f(\beta_n) - f(\alpha_n) & = (\beta_n - \alpha_n)f'(\xi_n) \\
D_n & = f'(\xi_n), \quad (5.44)
\end{aligned}$$

where $\xi_n \in (\alpha_n, \beta_n)$. Again, by the continuity of f' , we have $\lim_{n \rightarrow \infty} f'(\xi_n) = f'(0)$ and so the expression (5.44) implies that

$$\lim_{n \rightarrow \infty} D_n = f'(0).$$

Recall from Example 5.6(b) that the function $f : (-1, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & (x \neq 0); \\ 0, & (x = 0) \end{cases}$$

is differentiable in $(-1, 1)$, but f' is not continuous at 0, i.e., $\lim_{x \rightarrow 0} f'(x) \neq f'(0) = 0$. Let $\alpha_n = \frac{1}{2n\pi + \frac{\pi}{2}}$ and $\beta_n = \frac{1}{2n\pi}$. Then we have $-1 < \alpha_n < \beta_n < 1$, $\alpha_n \rightarrow 0$ and $\beta_n \rightarrow 0$ as $n \rightarrow \infty$. Since

$$f(\alpha_n) = \alpha_n^2 \sin \frac{1}{\alpha_n} = \frac{\sin(2n\pi + \frac{\pi}{2})}{(2n\pi + \frac{\pi}{2})^2} = \frac{1}{(2n\pi + \frac{\pi}{2})^2} \quad \text{and} \quad f(\beta_n) = \beta_n^2 \sin \frac{1}{\beta_n} = \frac{\sin(2n\pi)}{(2n\pi)^2} = 0,$$

we have

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} = \frac{-1}{(2n\pi + \frac{\pi}{2})^2} \times (4n) \left(2n\pi + \frac{\pi}{2} \right) = \frac{-4n}{2n\pi + \frac{\pi}{2}}$$

so that

$$\lim_{n \rightarrow \infty} D_n = -\frac{2}{\pi} \neq f'(0).$$

This completes the proof of the problem. ■

Problem 5.20

Rudin Chapter 5 Exercise 20.

Proof. When we read the Eqn. (24) of Theorem 5.15, we may modify it to the following form: Suppose $\mathbf{f} = (f_1, f_2, \dots, f_m) : [a, b] \rightarrow \mathbb{R}^m$, where $f_1, f_2, \dots, f_m$ are real functions on $[a, b]$, n is a positive integer, $\mathbf{f}^{(n-1)}$ is continuous on $[a, b]$, $\mathbf{f}^{(n)}$ exists for every $t \in (a, b)$. Let α, β be distinct points of $[a, b]$, and define

$$\mathbf{P}(t) = (P_1(t), P_2(t), \dots, P_m(t)) = \sum_{k=0}^{n-1} \frac{\mathbf{f}^{(k)}(\alpha)}{k!} (t - \alpha)^k, \quad (5.45)$$

where $P_i(t) = \sum_{k=0}^{n-1} \frac{f_i^{(k)}(\alpha)}{k!} (t - \alpha)^k$ for $i = 1, 2, \dots, m$. Then there exists a point $x \in (\alpha, \beta)$ such that

$$|\mathbf{f}(\beta) - \mathbf{P}(\beta)| \leq \left| \frac{\mathbf{f}^{(n)}(x)}{n!} \right| (\beta - \alpha)^n. \quad (5.46)$$

We see that the inequality (5.46) follows immediately from Theorem 5.15 (Taylor's Theorem) if $m = 1$. For the general case, we let

$$\mathbf{z} = \frac{1}{|\mathbf{f}(\beta) - \mathbf{P}(\beta)|} [\mathbf{f}(\beta) - \mathbf{P}(\beta)]$$

which is clearly a unit vector. Let, further, $\varphi(t) = \mathbf{z} \cdot \mathbf{f}(t)$, where $t \in [\alpha, \beta]$. Then it is easy to see that φ is a real-valued continuous function on $[\alpha, \beta]$ which is differentiable in (α, β) . Thus it is the case when $m = 1$, so we deduce from Theorem 5.15 that

$$|\varphi(\beta) - Q(\beta)| \leq \left| \frac{\varphi^{(n)}(x)}{n!} \right| (\beta - \alpha)^n, \quad (5.47)$$

where

$$Q(\beta) = \sum_{k=0}^{n-1} \frac{\varphi^{(k)}(\alpha)}{k!} (\beta - \alpha)^k.$$

Since $\varphi^{(k)}(t) = \mathbf{z} \cdot \mathbf{f}^{(k)}(t)$, where $k = 0, 1, \dots, n$, we have

$$Q(\beta) = \mathbf{z} \cdot \sum_{k=0}^{n-1} \frac{\mathbf{f}^{(k)}(\alpha)}{k!} (\beta - \alpha)^k = \mathbf{z} \cdot \mathbf{P}(\beta). \quad (5.48)$$

Hence, it follows from the inequality (5.47) and the expression (5.48) that

$$\begin{aligned} |\mathbf{z} \cdot [\mathbf{f}(\beta) - \mathbf{P}(\beta)]| &\leq \left| \frac{\mathbf{z} \cdot \mathbf{f}^{(n)}(x)}{n!} \right| (\beta - \alpha)^n \\ \left| \frac{1}{|\mathbf{f}(\beta) - \mathbf{P}(\beta)|} [\mathbf{f}(\beta) - \mathbf{P}(\beta)] \cdot [\mathbf{f}(\beta) - \mathbf{P}(\beta)] \right| &\leq |\mathbf{z}| \left| \frac{\mathbf{f}^{(n)}(x)}{n!} \right| (\beta - \alpha)^n \\ \frac{1}{|\mathbf{f}(\beta) - \mathbf{P}(\beta)|} |\mathbf{f}(\beta) - \mathbf{P}(\beta)|^2 &\leq \left| \frac{\mathbf{f}^{(n)}(x)}{n!} \right| (\beta - \alpha)^n \\ |\mathbf{f}(\beta) - \mathbf{P}(\beta)| &\leq \left| \frac{\mathbf{f}^{(n)}(x)}{n!} \right| (\beta - \alpha)^n \end{aligned}$$

which is exactly our desired inequality (5.46). This finishes the proof of the problem. ■

5.3 Derivatives of higher order and iteration methods

Problem 5.21

Rudin Chapter 5 Exercise 21.

Proof. We prove the last assertion only because it implies all the other cases. We note from Theorem 8.6(b) that the exponential function e^x has derivatives of all orders on $\mathbb{R}$, so we construct the desired function based on this property.

Let E be a non-empty closed subset of $\mathbb{R}$. Then $\overline{E} = \mathbb{R} \setminus E$ is an open subset of $\mathbb{R}$. Recall from Problem 2.29 that every open set in $\mathbb{R}$ is the union of an *at most countable* collection of *disjoint* segments. Thus we suppose that

$$\overline{E} = \bigcup_{k=1}^{\infty} (a_k, b_k),$$

where $(a_1, b_1), (a_2, b_2), \dots$ are disjoint.^e By this definition, we know that all the endpoints a_k and b_k belong to E , i.e., $a_k, b_k \in E$.

We start with the function

$$g(x) = \begin{cases} e^{-\frac{1}{x^2}}, & \text{if } x > 0; \\ 0, & \text{if } x \leq 0. \end{cases} \quad (5.49)$$

By Theorem 8.6(c), we have $g(x) = 0$ if and only if $x \leq 0$. Furthermore, it follows from Theorem 5.5 (Chain Rule) and Theorem 8.6(b) that g is differentiable at all non-zero x . We claim that g is also differentiable at 0. To this end, we follow from the definition (5.49) and Theorem 8.6(f) that

$$g'(0+) = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{g(x) - g(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x > 0}} \frac{e^{-\frac{1}{x^2}}}{x} = 0$$

and

$$g'(0-) = \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{g(x) - g(0)}{x - 0} = \lim_{\substack{x \rightarrow 0 \\ x < 0}} \frac{0}{x} = 0.$$

Since $g'(0+) = g'(0-)$, we obtain our claim. In fact, it can be shown that g has derivatives of all orders on $\mathbb{R}$.

^eIt may happen that $(a_k, b_k) = (-\infty, b_k)$ or (a_k, ∞) for some positive integer k .

Next, we construct functions which have derivatives of all orders in the segments (a_k, b_k) , (a_k, ∞) and $(-\infty, a_k)$ respectively. Firstly, for each segment (a_k, b_k) , we define

$$f_k(x) = g((x - a_k)(b_k - x)).$$

If $x \leq a_k$, then $(x - a_k)(b_k - x) \leq 0$ so that $f_k(x) = 0$. Similarly, if $x \geq b_k$, then $(x - a_k)(b_k - x) \leq 0$ so that $f_k(x) = 0$. By these observations and Theorem 8.6(c), we have

$$f_k(x) = 0 \quad \text{if and only if} \quad x \notin (a_k, b_k).$$

Since g has derivatives of all orders on $\mathbb{R}$, it follows from Theorem 5.5 (Chain Rule) that the function f_k also has derivatives of all orders on $\mathbb{R}$. In fact, the expression of f_k is given as follows:

$$\begin{aligned} f_k(x) &= \begin{cases} g((x - a_k)(b_k - x)), & \text{if } a_k < x < b_k; \\ 0, & \text{if } x \leq a_k \text{ or } x \geq b_k, \end{cases} \\ &= \begin{cases} e^{-\frac{1}{(x-a_k)^2(x-b_k)^2}}, & \text{if } a_k < x < b_k; \\ 0, & \text{if } x \leq a_k \text{ or } x \geq b_k. \end{cases} \end{aligned} \quad (5.50)$$

Secondly, for the segment (a_k, ∞) , we consider the function $f_k(x) = g(x - a_k)$ so that $f_k(x) = 0$ if and only if $x \leq a_k$ and f_k has derivatives of all orders. Finally, for the segment $(-\infty, a_k)$, we consider the function $f_k(x) = g(a_k - x)$ so that $f_k(x) = 0$ if and only if $x \geq a_k$ and f_k has derivatives of all orders.^f

Now it is time to define the function f which has derivative of all orders and $f(x) = 0$ if and only if $x \in E$. In fact, we define

$$f(x) = \begin{cases} f_k(x), & \text{if } x \in (a_k, b_k) \subseteq \overline{E}; \\ 0, & \text{if } x \in E. \end{cases}$$

Hence it is easy to check that $f(x) = 0$ if and only if $x \in E$ and we follow from the definition (5.50) that the expression of f is given by

$$\begin{aligned} f(x) &= \begin{cases} f_k(x), & \text{if } x \in (a_k, b_k) \subseteq \overline{E}; \\ 0, & \text{if } x \in E. \end{cases} \\ &= \begin{cases} e^{-\frac{1}{(x-a_k)^2(x-b_k)^2}}, & \text{if } x \in (a_k, b_k) \subseteq \overline{E}; \\ 0, & \text{if } x \in E. \end{cases} \end{aligned}$$

We finish the proof of the problem. ■

Problem 5.22

Rudin Chapter 5 Exercise 22.

Proof.

- (a) Let $F(x) = f(x) - x$. By Theorem 5.3(a), F is differentiable in $(-\infty, \infty)$. Since $f'(x) \neq 1$ for every real x , we have

$$F'(x) \neq 0$$

for every real x . Assume that $a < b$ were two fixed points of f . Then we have

$$F(a) = F(b) = 0.$$

By Theorem 5.10 (the Mean Value Theorem), we have

$$0 = F(b) - F(a) = (b - a)F'(\xi)$$

for some $\xi \in (a, b)$, but this clearly implies that $F'(\xi) = 0$, a contradiction. Hence f has *at most* one fixed point.

^fActually, it can be shown by induction that

$$f_k^{(n)}(x) = p(x - a_k, b_k - x) e^{-\frac{1}{(x-a_k)^2(x-b_k)^2}},$$

where $p(x - a_k, b_k - x)$ is a polynomial in $\frac{1}{x - a_k}$ and $\frac{1}{b_k - x}$.

(b) If $f(t) = t + (1 + e^t)^{-1} = t$, then we have

$$\frac{1}{1 + e^t} = 0$$

which is impossible for a real t . Hence the function f has no fixed point in $(-\infty, \infty)$.

(c) Assume that $x_{n+1} \neq x_n$ for all positive integers n . Otherwise, if $x_{n_0+1} = x_{n_0}$ for some positive integer n_0 , then we have

$$x_n = x_{n_0}$$

for all $n \geq n_0 + 1$ [§] which means that x_{n_0} is a fixed point of f .

Suppose that there is a constant $A < 1$ such that

$$|f'(t)| \leq A$$

for all real t . Let $\{x_n\}$ be a sequence defined by

$$x_{n+1} = f(x_n)$$

and x_1 is an arbitrary real number. Since f is differentiable in $\mathbb{R}$, f is continuous on $[x_n, x_{n+1}]$ or $[x_{n+1}, x_n]$ and differentiable in (x_n, x_{n+1}) or (x_{n+1}, x_n) for all positive integers n . By Theorem 5.10, both cases imply that

$$\begin{aligned} |x_{n+2} - x_{n+1}| &= |f(x_{n+1}) - f(x_n)| \\ &= |x_{n+1} - x_n| |f'(x)| \\ &\leq A |x_{n+1} - x_n| \end{aligned} \quad (5.51)$$

for all positive integers n . By considering the inequality (5.51) recursively, we have

$$|x_{n+2} - x_{n+1}| \leq A^n |x_2 - x_1| \quad (5.52)$$

for all positive integers n . Since $0 < A < 1$, we have $A^n \rightarrow 0$ as $n \rightarrow \infty$. Therefore, if $m \geq n \geq 1$, then it follows from the inequality (5.52) that

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \cdots + |x_{n+1} - x_n| \\ &\leq (A^{m-2} + A^{m-3} + \cdots + A^{n-1}) |x_2 - x_1| \\ &\leq A^{n-1} (1 + A + A^2 + \cdots) |x_2 - x_1| \\ &\leq \frac{|x_2 - x_1|}{1 - A} A^{n-1}. \end{aligned} \quad (5.53)$$

Given $\epsilon > 0$. If we take N to be a positive integer such that

$$N > 1 + \left\lceil \frac{1}{\log A} \cdot \log \frac{\epsilon(1 - A)}{|x_2 - x_1|} \right\rceil$$

(note that $\log A < 0$), then we have

$$\frac{|x_2 - x_1|}{1 - A} A^{N-1} < \epsilon,$$

so this and the inequality (5.53) imply that for $m \geq n \geq N$, we have

$$|x_m - x_n| < \epsilon.$$

Thus $\{x_n\}$ is a Cauchy sequence so that $\lim_{n \rightarrow \infty} x_n = x$ exists. Since f is continuous on $\mathbb{R}$, we have $\lim_{n \rightarrow \infty} f(x_n) = f(x)$ and then

$$x = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} f(x_n) = f(x).$$

Hence x is a fixed point of f .

(d) The process described in part (c) can be “visualized” by the zig-zag path, where the points (x_n, x_{n+1}) are replaced by the points (p_{n-1}, p_n) (i.e., the red crosses on the blue curves).

[§]For example, we have $x_{n_0+2} = f(x_{n_0+1}) = f(x_{n_0}) = x_{n_0+1}$.

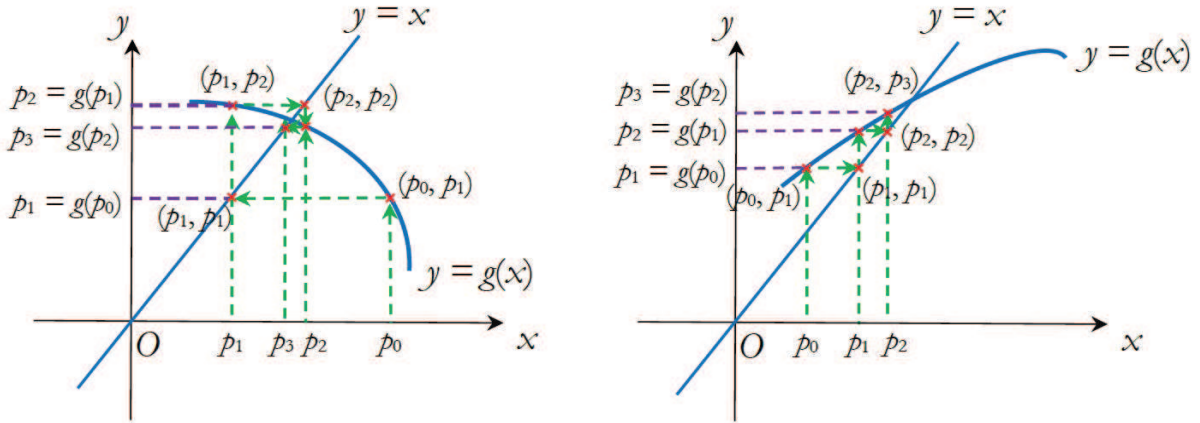


Figure 5.1: The zig-zag path of the process in (c).

Problem 5.23

Rudin Chapter 5 Exercise 23.

Proof. Since α, β and γ are fixed points of f , they are the solutions of the equation

$$x^3 - 3x + 1 = 0.$$

By using a scientific calculator, we can see easily that $\alpha = -1.87939\dots$, $\beta = 0.34730\dots$ and $\gamma = 1.53209\dots$. We define $g(x) = f(x) - x$. It is obvious that

$$g(x) = \frac{x^3 - 3x + 1}{3} = 0 \quad \text{if and only if} \quad f(x) = x.$$

In other words, fixed points of $f(x)$ are the *only* zeros of $g(x)$.^h Thus α, β and γ are the *only* zeros of $g(x)$.

- (a) Suppose that $x_1 < \alpha$. Since $g(-2) = -\frac{1}{3} < 0$, the continuity of g implies that $g(x) < 0$ on $(-\infty, \alpha)$. In particular, we have $g(x_1) < 0$. Now we apply induction to prove that $\{x_n\}$ is a strictly decreasing sequence. For $n = 1$, we have

$$x_2 - x_1 = f(x_1) - x_1 = g(x_1) < 0,$$

i.e., $x_2 < x_1$. The statement is true for $n = 1$. Assume that it is true for $n = k$ for some positive integer k , i.e.,

$$x_{k+1} < x_k.$$

For $n = k + 1$, we note that

$$g'(x) = x^2 - 1 > 0$$

on $(-\infty, -1)$, so Theorem 5.10 implies that g is strictly increasing on $(-\infty, -1)$. Therefore, our assumption gives the result that $g(x_{k+1}) < g(x_k)$ which is equivalent to

$$x_{k+2} - x_{k+1} < x_{k+1} - x_k < 0.$$

Thus it is also true for $n = k + 1$. By induction, it is true for all positive integers n .

^hThe following proof in this problem uses mainly the continuity property of the function g and induction. In fact, one may prove the problem by using mainly Theorem 5.11 and the differentiability and the convexity of the function f (Problems 4.23 and 5.14).

Next, we show that $\{x_n\}$ is unbounded. Assume that $\{x_n\}$ was bounded. By Theorem 3.14, we have $\{x_n\}$ is convergent. Let $\lim_{n \rightarrow \infty} x_n = \theta$. It is clear that $\theta < \alpha$. Furthermore, it follows from this and Theorem 4.2 that

$$\theta = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} f(x_n) = f(\theta). \quad (5.54)$$

Therefore, we arrive at a conclusion that θ is the fourth fixed point of f , a contradiction to our hypothesis. Hence, we have $x_n \rightarrow -\infty$ as $n \rightarrow \infty$.

(b) Suppose that $\alpha < x_1 < \gamma$. We note that $\alpha < \beta < \gamma$, so there are three cases:

– **Case (i):** $x_1 = \beta$. Then we have

$$x_2 = f(x_1) = \frac{x_1^3 + 1}{3} = \frac{3\beta}{3} = \beta.$$

Thus we have $x_n = \beta$ for all positive integers n and we are done in this case.

– **Case (ii):** $x_1 \in (\beta, \gamma)$. We claim that $x_n \in (\beta, \gamma)$ and $x_{n+1} < x_n$ for all positive integers n . We prove the claim by induction.

For the first assertion, it follows from $0 < \beta < x_1$ that $f(\beta) < f(x_1)$ which is equivalent to

$$\beta < x_2.$$

For the second assertion, since $g(\beta) = g(\gamma) = 0$ and $g(x) \neq 0$ on (β, γ) , we must have $g(x_1) \neq 0$. Since

$$g(1) = \frac{1^3 + 1}{3} - 1 = -\frac{1}{3},$$

the continuity of g guarantees that $g(x) < 0$ on (β, γ) . In particular, we have $g(x_1) < 0$ and then $f(x_1) < x_1$ which is equivalent to

$$x_2 < x_1.$$

Thus the statements are true for $n = 1$.

Assume that they are true for $n = k$ for some positive integer k , i.e., $x_k \in (\beta, \gamma)$ and $x_{k+1} < x_k$. For $n = k + 1$, it follows from $0 < \beta < x_k$ that $f(\beta) < f(x_k)$ so that

$$\beta < x_{k+1}.$$

Since $x_{k+1} \in (\beta, \gamma)$, we have $g(x_{k+1}) < 0$ so that $f(x_{k+1}) < x_{k+1}$ which is equivalent to

$$x_{k+2} < x_{k+1}.$$

Therefore, the statements are true for $n = k + 1$. Hence the claim follows from induction.

Since $\{x_n\}$ is bounded, Theorem 3.14 implies that $\{x_n\}$ converges. Let $\lim_{n \rightarrow \infty} x_n = \eta$. By using similar argument as obtaining the result (5.54), we have

$$\eta = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} f(x_n) = f(\eta).$$

Therefore, η is also a fixed point of f and $\eta \in [\beta, \gamma)$. Hence we have $\eta = \beta$ in this case.

– **Case (iii):** $x_1 \in (\alpha, \beta)$. We claim that $x_n \in (\alpha, \beta)$ and $x_n < x_{n+1}$ for all positive integers n . Since the proof of this part is very similar to that of **Case (ii)** above, we omit the details here.

Since $\{x_n\}$ is bounded, Theorem 3.14 implies that $\{x_n\}$ converges. Let $\lim_{n \rightarrow \infty} x_n = \tau$. By using similar argument as obtaining the result (5.54), we have

$$\tau = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} f(x_n) = f(\tau).$$

Therefore, τ is also a fixed point of f and $\tau \in (\alpha, \beta]$. Hence we have $\tau = \beta$ in this case.

- (c) Suppose that $\gamma < x_1$. Since $g(2) = 1 > 0$, the continuity of g implies that $g(x) > 0$ on (γ, ∞) . In particular, we have $g(x_1) > 0$. Now by applying similar argument as in part (a), we can show that $\{x_n\}$ is a strictly increasing sequence and it is unbounded. Hence we must have $x_n \rightarrow +\infty$ as $n \rightarrow \infty$.

This completes the proof of the problem. ■

Problem 5.24

Rudin Chapter 5 Exercise 24.

Proof. By Problem 3.16, we see that the convergence of the function f is much more rapid than that of the function g . Take the same starter $x_1 > \sqrt{\alpha}$ in the recursions of f and g . By the inequalities (5.51) to (5.53), the rate of the convergence depends on the bound of the recursion formula. It is easy to check that

$$\begin{aligned} f(x) - \sqrt{\alpha} &= \frac{1}{2} \left(x + \frac{\alpha}{x} \right) - \sqrt{\alpha} \\ &= \left(\frac{x}{2} - \frac{\sqrt{\alpha}}{2} \right) + \left(\frac{\alpha}{2x} - \frac{\sqrt{\alpha}}{2} \right) \\ &= \frac{1}{2} (x - \sqrt{\alpha}) + \frac{\sqrt{\alpha}}{2} \left(\frac{\sqrt{\alpha}}{x} - 1 \right) \\ f(x) - \sqrt{\alpha} &= \frac{1}{2} \left(1 - \frac{\sqrt{\alpha}}{x} \right) (x - \sqrt{\alpha}) \end{aligned} \quad (5.55)$$

and

$$g(x) - \sqrt{\alpha} = \frac{\alpha + x}{1 + x} - \sqrt{\alpha} = \frac{\alpha + x - \sqrt{\alpha} - x\sqrt{\alpha}}{1 + x} = \frac{1 - \sqrt{\alpha}}{1 + x} (x - \sqrt{\alpha}). \quad (5.56)$$

Let $\{x_n(f)\}$ and $\{x_n(g)\}$ be the two sequences generated by f and g respectively, where $x_1(f) = x_1(g)$. (Note that $x_n(f) \neq x_n(g)$ in general.) Then it follows from the formulas (5.55) and (5.56) that

$$|x_{n+1}(f) - \sqrt{\alpha}| = \frac{1}{2} \left| 1 - \frac{\sqrt{\alpha}}{x_n(f)} \right| \cdot |x_n(f) - \sqrt{\alpha}| \quad (5.57)$$

and

$$|x_{n+1}(g) - \sqrt{\alpha}| = \left| \frac{1 - \sqrt{\alpha}}{1 + x_n(g)} \right| \cdot |x_n(g) - \sqrt{\alpha}|. \quad (5.58)$$

- **Case (i):** $\alpha > 1$. By Problem 3.16, Lemma 3.1, Problem 3.17(a) and (b) and the fact that $x_1 > \sqrt{\alpha} > 1$, we consider the functions f and g defined on the interval $[1, x_1]$. Therefore, the expressions (5.57) and (5.58) imply that

$$|x_{n+1}(f) - \sqrt{\alpha}| = \frac{1}{2^n} \prod_{k=1}^n \left| 1 - \frac{\sqrt{\alpha}}{x_k(f)} \right| |x_1 - \sqrt{\alpha}| \quad (5.59)$$

and

$$|x_{n+1}(g) - \sqrt{\alpha}| = \prod_{k=1}^n \left| \frac{1 - \sqrt{\alpha}}{1 + x_k(g)} \right| |x_1 - \sqrt{\alpha}| \quad (5.60)$$

for all positive integers n . Since $\{x_n(f)\}$ decreases monotonically (see Problem 3.16(a)), we have from the expression (5.59) that

$$|x_{n+1}(f) - \sqrt{\alpha}| \leq \left(\frac{x_1 - \sqrt{\alpha}}{2x_1} \right)^n |x_1 - \sqrt{\alpha}|. \quad (5.61)$$

Similarly, since $x_n(g) \in [1, x_1]$, we have from the expression (5.60) that

$$\left(\frac{\sqrt{\alpha}-1}{1+x_1}\right)^n |x_1 - \sqrt{\alpha}| \leq |x_{n+1}(g) - \sqrt{\alpha}| \leq \left(\frac{\sqrt{\alpha}-1}{2}\right)^n |x_1 - \sqrt{\alpha}|. \quad (5.62)$$

We compare the magnitudes of the two constants

$$\frac{x_1 - \sqrt{\alpha}}{2x_1} \quad \text{and} \quad \frac{\sqrt{\alpha}-1}{1+x_1}.$$

In fact, if x_1 is chosen to be *close to* $\sqrt{\alpha}$,ⁱ then $x_1 - \sqrt{\alpha}$ is very small so that

$$\frac{x_1 - \sqrt{\alpha}}{2x_1} < \frac{\sqrt{\alpha}-1}{1+x_1} < 1.$$

Therefore, we follow from the inequalities (5.61) and (5.62) that

$$|x_{n+1}(f) - \sqrt{\alpha}| < |x_{n+1}(g) - \sqrt{\alpha}|.$$

This explains why the rate of convergence of $\{x_n(f)\}$ is *much more rapid* than that of $\{x_n(g)\}$ if n is large enough.

For example, take $\alpha = 10$, then we have

$$(\sqrt{\alpha}, \sqrt{\alpha}) = (3.162277660168379, 3.162277660168379).$$

Note that

$$x_1(f) = 5, \quad x_2(f) = \frac{7}{2}, \quad x_3(f) = \frac{89}{28}, \quad x_4(f) = \frac{15761}{4984}, \dots$$

and its zig-zag path is shown in Figure 5.2^j, where the **green**, **orange** and **purple** dots are

$$\left(5, \frac{7}{2}\right), \quad \left(\frac{7}{2}, \frac{89}{28}\right) \quad \text{and} \quad \left(\frac{89}{28}, \frac{15761}{4984}\right)$$

respectively.

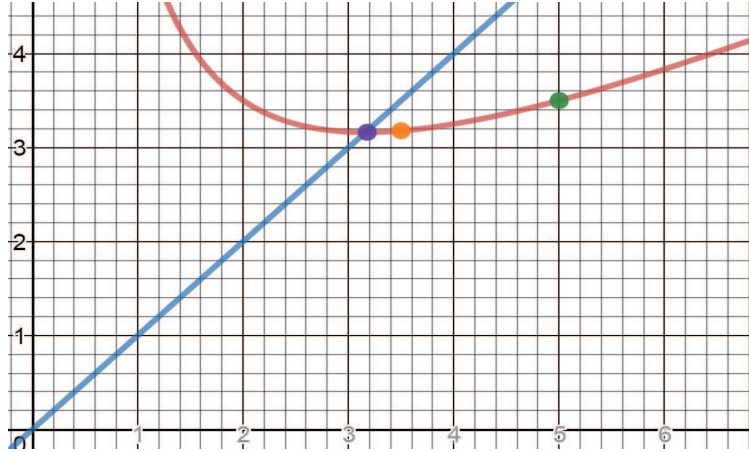


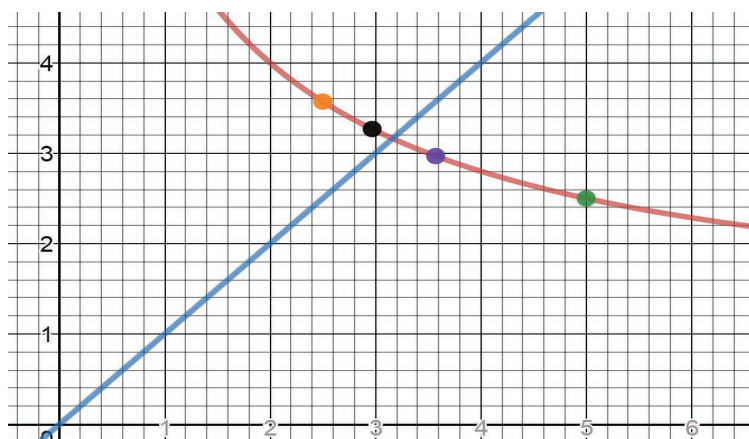
Figure 5.2: The zig-zag path induced by the function f in **Case (i)**.

Similarly, we have

$$x_1(g) = 5, \quad x_2(g) = \frac{15}{6}, \quad x_3(g) = \frac{75}{21}, \quad x_4(g) = \frac{286}{96}, \quad x_5(g) = \frac{1245}{381}, \dots$$

and its zig-zag path is shown in Figure 5.3, where the **green**, **orange**, **purple** and **black** dots are

$$\left(5, \frac{15}{6}\right), \quad \left(\frac{15}{6}, \frac{75}{21}\right), \quad \left(\frac{75}{21}, \frac{285}{96}\right) \quad \text{and} \quad \left(\frac{285}{96}, \frac{1245}{381}\right)$$

Figure 5.3: The zig-zag path induced by the function g in **Case (i)**.

respectively.

In conclusion, the pattern of the zig-zag path induced by the function f is approaching to the fixed point $(\sqrt{10}, \sqrt{10})$ from one-side, but the one induced by the function g goes near the fixed point “alternatively”.

- **Case (ii):** $0 < \alpha < 1$. Take $\alpha = \frac{1}{10}$ and $x_1 = \frac{1}{5}$. Then we have

$$(\sqrt{\alpha}, \sqrt{\alpha}) = (0.4472135954999579, 0.4472135954999579).$$

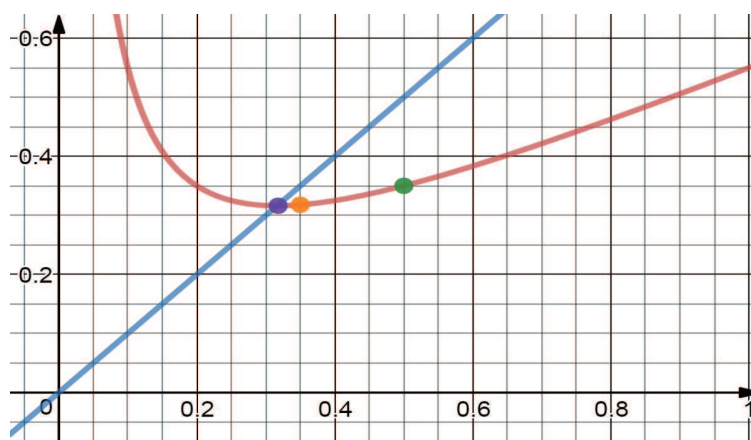
Therefore, we have

$$x_1(f) = \frac{1}{5}, \quad x_2(f) = \frac{7}{20}, \quad x_3(f) = \frac{89}{280} \quad \text{and} \quad x_4(f) = \frac{15761}{49840}$$

and its zig-zag path is shown in Figure 5.4, where the green, orange and purple dots are

$$\left(\frac{1}{5}, \frac{7}{20}\right), \quad \left(\frac{7}{20}, \frac{89}{280}\right) \quad \text{and} \quad \left(\frac{89}{280}, \frac{15761}{49840}\right)$$

respectively.

Figure 5.4: The zig-zag path induced by the function f in **Case (ii)**.

ⁱThis can be done by an initial estimation.

^jThe figures are drawn by “desmos”.

Similarly, we acquire

$$x_1(g) = \frac{1}{5}, \quad x_2(g) = \frac{1}{4}, \quad x_3(g) = \frac{7}{25}, \quad x_4(g) = \frac{19}{64}, \quad x_5(g) = \frac{127}{415}, \dots$$

and its zig-zag path is shown in Figure 5.3, where the green, orange, purple and black dots are

$$\left(\frac{1}{5}, \frac{1}{4}\right), \quad \left(\frac{1}{4}, \frac{7}{25}\right), \quad \left(\frac{7}{25}, \frac{19}{64}\right) \quad \text{and} \quad \left(\frac{19}{64}, \frac{127}{415}\right)$$

respectively.

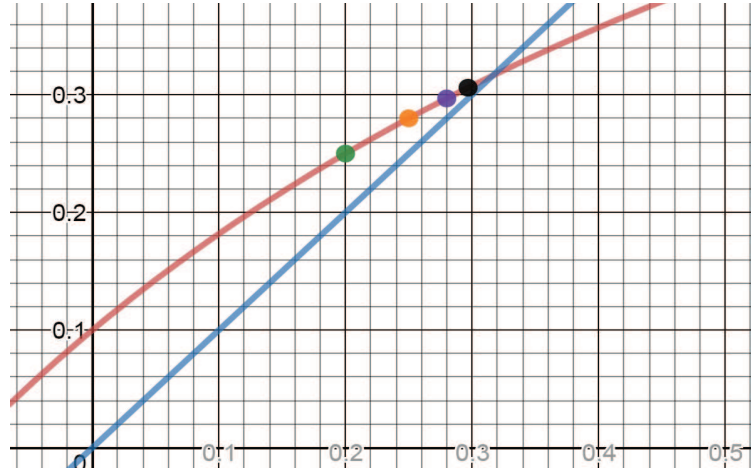


Figure 5.5: The zig-zag path induced by the function g in **Case (ii)**.

In this case, both the patterns of the zig-zag paths induced by the functions f and g are approaching to the fixed point $(\sqrt{\frac{1}{10}}, \sqrt{\frac{1}{10}})$ from one-side. This is main difference between **Case (i)** and **Case (ii)**.

We complete the proof of the problem. ■

Problem 5.25

Rudin Chapter 5 Exercise 25.

Proof.

(a) Now the given formula can be rewritten as

$$f'(x_n) = \frac{0 - f(x_n)}{x_{n+1} - x_n}. \quad (5.63)$$

Therefore, it means that the slope of the graph of f at x_n (the left-hand side of the formula (5.63)) equals to the slope of the straight line passing through

$$(x_n, f(x_n)) \quad \text{and} \quad (x_{n+1}, 0)$$

(the right-hand side of the formula (5.63)). See the following figure:

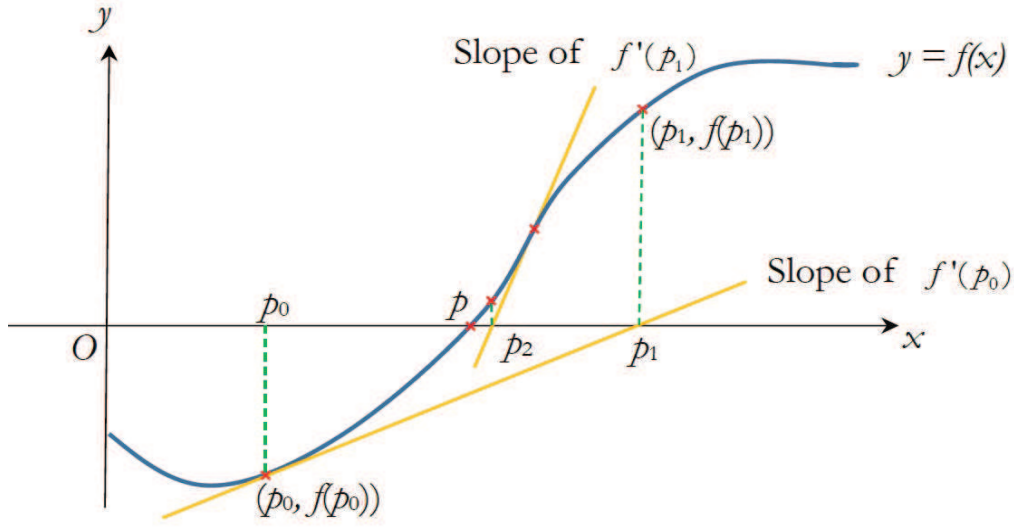


Figure 5.6: The geometrical interpretation of Newton's method.

- (b) We prove the result by induction. By the continuity of f and the fact that ξ is the *unique* point in (a, b) at which $f(\xi) = 0$, we have $f(x_1) > 0$. Otherwise, we have $f(t) = 0$ for some $t \in [x_1, b)$ contradicting to the uniqueness of ξ . Since $f'(x) > 0$ for all $x \in [a, b]$,

$$\frac{f(x_1)}{f'(x_1)} > 0$$

and so $x_2 < x_1$. This shows that the case for $n = 1$ is true.

Assume that it is also true for $n = k$ for some positive integer k , i.e., $x_{k+1} < x_k$. Note that $x_{k+1} < x_k$ is equivalent to $f(x_k) > 0$. For $n = k + 1$, we have

$$x_{k+2} = x_{k+1} - \frac{f(x_{k+1})}{f'(x_{k+1})}.$$

- (c) It is clear that the function f satisfies the conditions of Theorem 5.15 (Taylor's theorem) with $\alpha = \xi$ and $\beta = x_n$, so we have

$$\begin{aligned} f(x_n) &= f(\xi) + f'(x_n)(x_n - \xi) + \frac{f''(t)}{2}(x_n - \xi)^2 \\ &= f'(x_n)(x_n - \xi) + \frac{f''(t)}{2}(x_n - \xi)^2 \\ \frac{f'(x_n)}{f(x_n)} &= (x_n - \xi) + \frac{f''(t)}{2f'(x_n)}(x_n - \xi)^2 \end{aligned}$$

- (d) Since $f'(x) \geq \delta > 0$ and $0 \leq f''(x) \leq M$ for all $x \in [a, b]$, it follows from part (c) that

$$0 \leq x_{n+1} - \xi \leq \frac{M}{2\delta}(x_n - \xi)^2 = A(x_n - \xi)^2$$

which implies that

$$0 \leq x_{n+1} - \xi \leq \frac{1}{A}[A(x_1 - \xi)]^{2^n} \quad (5.64)$$

as required.

When we compare these inequalities (5.64) with Problems 3.16 and 3.18, we see that the recursion formula in Problem 3.18 which is

$$x_{n+1} = \frac{p-1}{p}x_n + \frac{\alpha}{p}x_n^{-p+1}$$

corresponds to applying Newton's method with the function $f(x) = x^p - \alpha$. Actually, we have $f'(x) = px^{p-1}$ so that

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^p - \alpha}{px_n^{p-1}} = \frac{p-1}{p}x_n + \frac{\alpha}{p}x_n^{-p+1}. \quad (5.65)$$

Put $p = 2$, the formula (5.65) reduces to Problem 3.16 as a special case and the error estimation

$$\epsilon_{n+1} < \beta \left(\frac{\epsilon_1}{\beta} \right)^{2^n}$$

is equivalent to the error estimation (5.64) deduced by Newton's method.

- (e) It is clear that $g(\xi) = \xi$ if and only if $f(\xi) = 0$. In other words, the fixed point ξ of g is exactly the zero of f . Hence, Newton's method of computing ξ as described in parts (a) to (c) amounts to finding the fixed point of the function g .

Since

$$g'(x) = \frac{f(x)f'(x)}{[f'(x)]^2},$$

we have

$$|g'(x)| = \left| \frac{f(x)f''(x)}{[f'(x)]^2} \right| \leq \frac{M}{\delta^2} |f(x)|$$

$$\lim_{x \rightarrow \xi} |g'(x)| \leq \lim_{x \rightarrow \xi} \frac{M}{\delta^2} |f(x)|$$

which implies that $\lim_{x \rightarrow \xi} g'(x) = 0$.

- (f) It is obvious that $f(x) = 0$ if and only if $x = 0$. Since $f'(x) = \frac{1}{3}x^{-\frac{2}{3}}$, $f'(x) \rightarrow 0$ as $x \rightarrow \infty$. This shows that the function f does not satisfy one of the conditions of Newton's method:

$$f'(x) \geq \delta > 0$$

for some positive constant δ and for all $x \in (-\infty, \infty)$. Therefore, Newton's method fails in this case. In fact, we have

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = -2x_n$$

so that $x_{n+1} = (-2)^n x_1$. Since $x_1 \in (0, \infty)$, the sequence $\{x_n\}$ is divergent.

This ends the proof of the problem. ■

Problem 5.26

Rudin Chapter 5 Exercise 26.

Proof. We follow the hint. Fix $x_0 \in [a, b]$, let

$$M_0(f) = \sup_{x \in [a, x_0]} |f(x)| \quad \text{and} \quad M_0(f') = \sup_{x \in [a, x_0]} |f'(x)|.$$

Since $|f'(x)| \leq A|f(x)|$ on $[a, b]$, we always have $M_0(f') \leq AM_0(f)$. Now for any $x \in [a, x_0]$, since f is differentiable in $[a, b]$, it is clear that it is continuous on $[a, x]$ and differentiable in (a, x) . By Theorem 5.10, we have

$$|f(x) - f(a)| = |(x - a)f'(t)|$$

for some $t \in (a, x)$. By our hypothesis, this implies that

$$|f(x)| \leq |f'(t)|(x_0 - a) \leq M_0(f')(x_0 - a) \leq A(x_0 - a)M_0(f). \quad (5.66)$$

Let x_0 be chosen so that $A(x_0 - a) < \frac{1}{2}$. Assume that $M_0(f) \neq 0$. Then $A(x_0 - a) < \frac{1}{2}$ and the inequality (5.66) imply that

$$|f(x)| < \frac{M_0(f)}{2}$$

for all $x \in [a, x_0]$ and this gives the contradiction that $M_0(f) \leq \frac{M_0(f)}{2}$. Hence we must have $M_0(f) = 0$ on $[a, x_0]$ which induces

$$f(x) = 0$$

on $[a, x_0]$.

Now we choose $x_0, x_1, \dots, x_n$ such that $a < x_0 < x_1 < \dots < x_n < b$ with

$$A(x_0 - a) < \frac{1}{2}, \quad A(x_k - x_{k-1}) < \frac{1}{2} \quad \text{and} \quad A(b - x_n) < \frac{1}{2},$$

where $k = 1, 2, \dots, n$. Next, we define

$$M_k(f) = \sup_{x \in [x_{k-1}, x_k]} |f(x)| \quad \text{and} \quad M_k(f') = \sup_{x \in [x_{k-1}, x_k]} |f'(x)|$$

for each $k = 1, 2, \dots, n$. Then the argument in the previous paragraph can be applied to each $M_k(f)$ to get the result that

$$f(x) = 0 \tag{5.67}$$

on $[x_{k-1}, x_k]$, where $k = 1, 2, \dots, n$. The remaining case for $[x_n, b]$ can be considered similarly and finally the equation (5.67) also holds on $[x_n, b]$.

Hence, by combining the above cases, we have $f(x) = 0$ on $[a, b]$, completing the proof of the problem. ■

5.4 Solutions of differential equations

Problem 5.27

Rudin Chapter 5 Exercise 27.

Proof. Assume that the initial-value problem had two solutions f_1 and f_2 . Let $f = f_1 - f_2$. Since f_1 and f_2 are differentiable in $[a, b]$, f is also differentiable in $[a, b]$ by Theorem 5.3(a). Furthermore, we have

$$f(a) = f_1(a) - f_2(a) = c - c = 0.$$

Since $\alpha \leq f_1(x) \leq \beta$ and $\alpha \leq f_2(x) \leq \beta$, we have $(x, f_1(x)), (x, f_2(x)) \in R$. Therefore, the hypothesis implies that

$$\begin{aligned} |\phi(x, f_2(x)) - \phi(x, f_1(x))| &\leq A|f_2(x) - f_1(x)| \\ |f_2'(x) - f_1'(x)| &\leq A|f_2(x) - f_1(x)| \\ |f'(x)| &\leq A|f(x)|. \end{aligned}$$

In other words, the function f satisfies the conditions of Problem 5.26 and it yields that $f \equiv 0$ on $[a, b]$, i.e., $f_1 \equiv f_2$ on $[a, b]$.

Now it is easy to check that the functions $f(x) = 0$ and $f(x) = \frac{x^2}{4}$ are two distinct solutions of the initial-value problem. To find all other solutions, we suppose first that the interval under consideration is $I = [0, \infty)$. Next, we let $f(x)$ be a (real) solution of the initial-value problem on I , $f(x) \not\equiv 0$ and $f(x) \not\equiv \frac{x^2}{4}$. Now we must have

$$f(x) \geq 0$$

on I . Otherwise, if $f(x_0) < 0$ for some $x_0 \in I$, then $(f(x_0))^{\frac{1}{2}}$ is a complex-value function in a neighborhood of x_0 which is a contradiction. Since $f(x) \not\equiv 0$, there exists $x_0 \in I$ such that $f(x_0) > 0$. Since f is continuous on I , we have $f(x) > 0$ in an interval (a, b) containing x_0 with

$$f(a) = 0,$$

where $0 \leq a < \infty$.^k

Define $F(x) = \sqrt{f(x)}$ on (a, b) . Then it is easy to see that $F'(x) = \frac{1}{2}$ so that $F(x) = \frac{1}{2}(x + C)$ and then $f(x) = \frac{1}{4}(x + C)^2$ for some constant C . Since $f(a) = 0$, we have $C = -a$ which gives

$$f(x) = \frac{1}{4}(x - a)^2$$

on (a, b) . The final task is to find the explicit form of $f(x)$ on $[0, a)$. By the continuity of f , the initial condition $f(0) = 0$ and $f(a+) = \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} \frac{1}{4}(x - a)^2 = 0$, we have

$$f(x) \equiv 0$$

on $[0, a)$. Hence, the other solutions must be in the form

$$f(x) = \begin{cases} 0, & \text{if } 0 \leq x < a; \\ \frac{1}{4}(x - a)^2, & \text{if } a \leq x < b. \end{cases}$$

As a final remark, since $f(x) \neq \frac{x^2}{4}$, it is impossible that $a = 0$ and $b = \infty$.^l This finishes the proof of the problem. ■

Problem 5.28

Rudin Chapter 5 Exercise 28.

Proof. A solution of the initial-value problem for systems of differential equations

$$\mathbf{y}' = \Phi(x, \mathbf{y}), \quad \mathbf{y}(a) = \mathbf{c} = (c_1, c_2, \dots, c_k) \quad (\alpha_i \leq c_i \leq \beta_i; i = 1, 2, \dots, k) \quad (5.68)$$

is, by definition, a differentiable function $\mathbf{f}$ on $[a, b]$ such that $\mathbf{f}(a) = \mathbf{c}$ and

$$\mathbf{f}' = \Phi(x, \mathbf{f}) \quad (a \leq x \leq b).$$

We claim that the initial-value problem (5.68) has at most one solution if there is a constant A such that

$$|\Phi(x, \mathbf{y}_2) - \Phi(x, \mathbf{y}_1)| \leq A|\mathbf{y}_2 - \mathbf{y}_1| \quad (5.69)$$

whenever $(x, \mathbf{y}_1), (x, \mathbf{y}_2) \in R$, where R is a $(k+1)$ -cell.

To prove the claim, let $\mathbf{y}_1$ and $\mathbf{y}_2$ be two solutions of the initial-value problem (5.68). Define the function $\mathbf{f}(x) = \mathbf{y}_2(x) - \mathbf{y}_1(x)$. It is clear that $\mathbf{f}$ is differentiable on $[a, b]$,

$$\mathbf{f}(a) = \mathbf{y}_2(a) - \mathbf{y}_1(a) = \mathbf{c} - \mathbf{c} = \mathbf{0}$$

and

$$|\mathbf{f}'(x)| = |\mathbf{y}_2'(x) - \mathbf{y}_1'(x)| = |\Phi(x, \mathbf{y}_2) - \Phi(x, \mathbf{y}_1)| \leq A|\mathbf{y}_2 - \mathbf{y}_1| = A|\mathbf{f}(x)|$$

on $[a, b]$. By using the vector-valued form of Problem 5.26, we have $\mathbf{f}(x) = \mathbf{0}$ for all $x \in [a, b]$. Hence, we have

$$\mathbf{y}_2(x) = \mathbf{y}_1(x)$$

for all $x \in [a, b]$ and the uniqueness of solutions of the initial-value problem (5.68) follows immediately. This completes the proof of the problem. ■

^kSuch an a must exist. Otherwise, if $f(x) > \epsilon > 0$ for some $\epsilon > 0$ and for all $x > 0$, then the continuity of f gives

$$0 = f(0) = \lim_{x \rightarrow 0} f(x) \geq \epsilon,$$

a contradiction.

^lWe shall remark that Problem 5.27 is actually part of the so-called Picard's existence theorem and the inequality in the problem is called the *Lipschitz condition*. See [5, §2.4, §2.8].

Problem 5.29*Rudin Chapter 5 Exercise 29.*

Proof. Let y be a solution of the differential equation with the initial conditions

$$\begin{cases} y^{(k)} + g_k(x)y^{(k-1)} + \cdots + g_2(x)y' + g_1(x)y = f(x), \\ y(a) = c_1, y'(a) = c_2, \dots, y^{(k-1)}(a) = c_k. \end{cases} \quad (5.70)$$

Suppose that $y_1 = y, y_2 = y'_1, y_3 = y'_2, \dots, y_k = y'_{k-1}, y'_k = y^{(k)}$. Then the differential equation (5.70) is equivalent to

$$y_k = f(x) - [g_1(x)y + g_2(x)y' + \cdots + g_k(x)y^{(k-1)}] = f(x) - \sum_{j=1}^k g_j(x)y_j.$$

Besides, the initial conditions $y^{(j-1)}(a) = c_j$ are equivalent to the conditions $y_j(a) = c_j$ respectively, where $j = 1, 2, \dots, k$. Therefore, what we have shown is that the differential equation (5.70) is equivalent to the initial-value problem

$$\begin{cases} y'_j = y_{j+1}, & y'_k = f(x) - \sum_{j=1}^k g_j(x)y_j, & (j = 1, 2, \dots, k-1); \\ y_j(a) = c_j & & (j = 1, 2, \dots, k). \end{cases} \quad (5.71)$$

We note that the initial-value problem (5.71) is a special case of the system of differential equations in Problem 5.28 with

$$\phi_1 = y_2, \quad \phi_2 = y_3, \dots, \phi_{k-1} = y_k \quad \text{and} \quad \phi_k = f(x) - \sum_{j=1}^k g_j(x)y_j.$$

Assume that u and v were two solutions of the differential equation (5.70). Then they induce two solutions

$$\mathbf{u} = (u_1, u_2, \dots, u_k) = (u, u', \dots, u^{(k-1)}) \quad \text{and} \quad \mathbf{v} = (v_1, v_2, \dots, v_k) = (v, v', \dots, v^{(k-1)})$$

of the initial-value problem (5.71) with

$$\Phi(x, \mathbf{y}) = (\phi_1, \phi_2, \dots, \phi_k) = \left(y_2, y_3, \dots, y_k, f(x) - \sum_{j=1}^k g_j(x)y_j \right).$$

To complete our problem, we have to show that the solutions $\mathbf{u}$ and $\mathbf{v}$ satisfy the inequality (5.69) for some positive constant A .

Since $g_1, g_2, \dots, g_k$ are continuous real functions on $[a, b]$, Theorem 4.16 ensures that maximums of $g_1, g_2, \dots, g_k$ can be found in $[a, b]$. Let $M_j = \max_{x \in [a, b]} g_j(x)$ and define

$$M = \max(M_1, M_2, \dots, M_k).$$

Therefore, it follows from the definition (5.71), Theorem 1.35 and Definition 1.36 that

$$\begin{aligned} |\Phi(x, \mathbf{u}) - \Phi(x, \mathbf{v})| &= |\mathbf{u}' - \mathbf{v}'| \\ &= |(u'_1, u'_2, \dots, u'_k) - (v'_1, v'_2, \dots, v'_k)| \\ &= \left| \left(u_2, u_3, \dots, u_k, f(x) - \sum_{j=1}^k g_j(x)u_j \right) - \left(v_2, v_3, \dots, v_k, f(x) - \sum_{j=1}^k g_j(x)v_j \right) \right| \\ &= \left| \left(u_2 - v_2, \dots, u_k - v_k, \sum_{j=1}^k g_j(x)(v_j - u_j) \right) \right| \end{aligned}$$

$$\begin{aligned}
&= \left\{ (u_2 - v_2)^2 + \cdots + (u_k - v_k)^2 + \left[\sum_{j=1}^k g_j(x)(v_j - u_j) \right]^2 \right\}^{\frac{1}{2}} \\
&\leq \left\{ (u_2 - v_2)^2 + \cdots + (u_k - v_k)^2 + M^2 \left[\sum_{j=1}^k (u_j - v_j) \right]^2 \right\}^{\frac{1}{2}} \\
&\leq \left\{ (u_2 - v_2)^2 + \cdots + (u_k - v_k)^2 + M^2 \times k \sum_{j=1}^k (u_j - v_j)^2 \right\}^{\frac{1}{2}} \\
&\leq (1 + kM^2)^{\frac{1}{2}} \times \left[(u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_k - v_k)^2 \right]^{\frac{1}{2}} \\
&\leq A|\mathbf{u} - \mathbf{v}|,
\end{aligned}$$

where $A = (1 + kM^2)^{\frac{1}{2}} > 0$. Hence, Problem 5.28 implies that $\mathbf{u} = \mathbf{v}$, as desired. This completes the proof of the problem. ■

CHAPTER 6

The Riemann-Stieltjes Integral

6.1 Problems on Riemann-Stieltjes integrals

Problem 6.1

Rudin Chapter 6 Exercise 1.

Proof. It is clear that f is bounded on $[a, b]$ and is discontinuous only at $x_0 \in [a, b]$. Since α is continuous at x_0 , it follows from Theorem 6.10 that $f \in \mathcal{R}(\alpha)$. Let $P = \{t_0, t_1, \dots, t_n\}$ be a partition of $[a, b]$ and $I_i = [t_{i-1}, t_i]$, where $i \in \{1, \dots, n\}$. Let, further, $x_0 \in I_i$ for some i . By Definition 6.2, we have

$$m_i = \inf_{x \in I_i} f(x) = 0 \quad (i = 1, 2, \dots, n).$$

Thus we have

$$L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta \alpha_i = 0$$

and then $\sup L(P, f, \alpha) = 0$ which gives

$$\int_a^b f \, d\alpha = 0.$$

Since $f \in \mathcal{R}(\alpha)$, we must have

$$\int_a^b f \, d\alpha = \int_a^b f \, d\alpha = 0$$

as desired. We complete the proof of the problem. ■

Problem 6.2

Rudin Chapter 6 Exercise 2.

Proof. Assume that $f(x_0) \neq 0$ for some $x_0 \in [a, b]$. Since $f \geq 0$ on $[a, b]$, we have $f(x_0) > 0$ and the continuity of f implies that

$$f(x) > \frac{f(x_0)}{2} > 0$$

on $(x_0 - \delta, x_0 + \delta) \subset [a, b]$ for some $\delta > 0$. Let $[c, d] \subset (x_0 - \delta, x_0 + \delta) \subset [a, b]$, where $c < d$. By Theorem 6.21 (Second Fundamental Theorem of Calculus), we know that

$$\int_a^b 0 \, dx = 0 \quad \text{and} \quad \int_a^b 1 \, dx = b - a.$$

By Theorem 6.12(b) and (c), we have

$$\int_a^b f \, dx = \int_a^c f \, dx + \int_c^d f \, dx + \int_d^b f \, dx \geq 0 + \int_c^d f \, dx + 0 \geq \frac{(d-c)f(x_0)}{2} > 0$$

which is a contradiction. Hence we have $f(x) = 0$ for all $x \in [a, b]$, completing the proof of the problem. ■

Problem 6.3

Rudin Chapter 6 Exercise 3.

Proof. Suppose that $P = \{x_0, x_1, \dots, x_n\}$ is a partition of $[-1, 1]$. By Theorem 6.4, we have for $j = 1, 2, 3$,

$$U(P^*, f, \beta_j) - L(P^*, f, \beta_j) \leq U(P, f, \beta_j) - L(P, f, \beta_j)$$

if P^* is a refinement of P . Thus, without loss of generality, we may assume that $0 \in P$. Let $x_k = 0$ for some positive integer k , where $0 < k < n$. Then we have

$$\Delta(\beta_1)_i = \beta_1(x_i) - \beta_1(x_{i-1}) = \begin{cases} \beta_1(x_i) - \beta_1(x_{i-1}) = 0, & \text{if } i = 1, 2, \dots, k; \\ 1, & \text{if } i = k+1; \\ \beta_1(x_i) - \beta_1(x_{i-1}) = 0, & \text{if } i = k+2, \dots, n, \end{cases} \quad (6.1)$$

$$\Delta(\beta_2)_i = \beta_2(x_i) - \beta_2(x_{i-1}) = \begin{cases} \beta_2(x_i) - \beta_2(x_{i-1}) = 0, & \text{if } i = 1, 2, \dots, k-1; \\ 1, & \text{if } i = k; \\ \beta_2(x_i) - \beta_2(x_{i-1}) = 0, & \text{if } i = k+1, \dots, n, \end{cases} \quad (6.2)$$

$$\Delta(\beta_3)_i = \beta_3(x_i) - \beta_3(x_{i-1}) = \begin{cases} \beta_3(x_i) - \beta_3(x_{i-1}) = 0, & \text{if } i = 1, 2, \dots, k-1; \\ \frac{1}{2}, & \text{if } i = k, k+1; \\ \beta_3(x_i) - \beta_3(x_{i-1}) = 0, & \text{if } i = k+2, \dots, n. \end{cases} \quad (6.3)$$

By the definitions (6.1), (6.2) and (6.3), we have

$$\begin{aligned} L(P, f, \beta_1) &= \sum_{i=1}^n m_i \Delta(\beta_1)_i = m_{k+1}, & U(P, f, \beta_1) &= \sum_{i=1}^n M_i \Delta(\beta_1)_i = M_{k+1}, \\ L(P, f, \beta_2) &= \sum_{i=1}^n m_i \Delta(\beta_2)_i = m_k, & U(P, f, \beta_2) &= \sum_{i=1}^n M_i \Delta(\beta_2)_i = M_k, \\ L(P, f, \beta_3) &= \sum_{i=1}^n m_i \Delta(\beta_3)_i = \frac{m_k + m_{k+1}}{2}, & U(P, f, \beta_3) &= \sum_{i=1}^n M_i \Delta(\beta_3)_i = \frac{M_k + M_{k+1}}{2} \end{aligned}$$

which imply that

$$U(P, f, \beta_1) - L(P, f, \beta_1) = M_{k+1} - m_{k+1}, \quad (6.4)$$

$$U(P, f, \beta_2) - L(P, f, \beta_2) = M_k - m_k, \quad (6.5)$$

$$U(P, f, \beta_3) - L(P, f, \beta_3) = \frac{(M_{k+1} - m_{k+1}) + (M_k - m_k)}{2}. \quad (6.6)$$

- (a) By Theorem 6.6 and the expression (6.4), $f \in \mathcal{R}(\beta_1)$ if and only if for every $\epsilon > 0$, there exists a partition P such that

$$M_{k+1} - m_{k+1} < \epsilon. \quad (6.7)$$

Suppose that such a partition P exists. If we take $\delta = x_{k+1} > 0$, then for all x with $x > 0$ and $x - 0 < \delta$, we have

$$|f(x) - f(0)| \leq M_{k+1} - m_{k+1} < \epsilon. \quad (6.8)$$

By Definition 4.25, this means that $f(0+) = f(0)$.

Conversely, suppose that $f(0+) = f(0)$. Then for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(0)| < \epsilon \quad (6.9)$$

for all x with $0 < x < \delta$. Let $P = \{-1, 0, \delta, 1\}$ be a partition of $[-1, 1]$. Thus the expression (6.4) and inequality (6.9) give

$$\begin{aligned} U(P, f, \beta_1) - L(P, f, \beta_1) &= \sup_{x \in [0, \delta]} f(x) - \inf_{x \in [0, \delta]} f(x) \\ &\leq \left| \sup_{x \in [0, \delta]} f(x) - f(0) \right| + \left| \inf_{x \in [0, \delta]} f(x) - f(0) \right| \\ &\leq \epsilon + \epsilon \\ &= 2\epsilon. \end{aligned}$$

Since ϵ is arbitrary, we know from Theorem 6.6 that $f \in \mathcal{R}(\beta_1)$. Hence we have shown that $f \in \mathcal{R}(\beta_1)$ if and only if $f(0+) = f(0)$.

If the inequality (6.7) holds for the partition P , then it follows from Theorem 6.7(c) that

$$\left| f(t_{k+1}) - \int f d\beta_1 \right| < \epsilon, \quad (6.10)$$

where $t_{k+1} \in [0, x_{k+1}]$. Hence we deduce from the inequalities (6.8) and (6.10) that, for all $t_{k+1} \in [0, x_{k+1}]$,

$$\left| f(0) - \int f d\beta_1 \right| \leq \left| f(0) - f(t_{k+1}) \right| + \left| f(t_{k+1}) - \int f d\beta_1 \right| < \epsilon + \epsilon = 2\epsilon.$$

Since ϵ is arbitrary, it means that

$$\int f d\beta_1 = f(0)$$

as desired.

- (b) We claim that $f \in \mathcal{R}(\beta_2)$ if and only if $f(0-) = f(0)$ and then

$$\int f d\beta_2 = f(0).$$

Since the proof of this is almost identical to that in part (a) with applying the expressions (6.2) and (6.5), we omit the details of its proof here.

- (c) By the analysis in part (a), we see that $M_{k+1} - m_{k+1} < \epsilon$ if and only if $f(0+) = f(0)$. Similarly, the analysis in part (b) implies that $M_k - m_k < \epsilon$ if and only if $f(0-) = f(0)$. By these, we obtain from the expression (6.6) that

$$U(P, f, \beta_3) - L(P, f, \beta_3) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

if and only if $f(0+) = f(0) = f(0-)$, i.e., f is continuous at 0 by Definition 4.25. Hence Theorem 6.6 shows that $f \in \mathcal{R}(\beta_3)$ if and only if f is continuous at 0.

By similar argument as in proving part (a), we can show that

$$\int f d\beta_3 = f(0).$$

- (d) If f is continuous at 0, then we must have $f(0) = f(0+) = f(0-)$ and the results of parts (a) - (c) imply that

$$\int f d\beta_1 = \int f d\beta_2 = \int f d\beta_3 = f(0).$$

This completes the proof of the problem. ■

Problem 6.4*Rudin Chapter 6 Exercise 4.*

Proof. Assume that $f \in \mathcal{R}$ on $[a, b]$ for some $a < b$. By Theorem 6.6, for every $\epsilon > 0$, there exists a partition $P = \{x_0, x_1, \dots, x_n\}$ such that

$$U(P, f) - L(P, f) < \epsilon. \quad (6.11)$$

By Definition 6.1, we have $M_i = 1$ and $m_i = 0$ for all $i = 1, 2, \dots, n$. Therefore, we have

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i = x_n - x_0 = b - a \quad \text{and} \quad L(P, f) = \sum_{i=1}^n m_i \Delta x_i = 0.$$

However, their difference is $b - a$ which contradicts the assumption (6.11). Hence we must have $f \notin \mathcal{R}$ on $[a, b]$ for any $a < b$, completing the proof of the problem. ■

Problem 6.5*Rudin Chapter 6 Exercise 5.*

Proof. Define the function

$$f(x) = \begin{cases} 1, & \text{if } x \in [a, b] \text{ and } x \text{ is rational;} \\ -1, & \text{if } x \in [a, b] \text{ and } x \text{ is irrational.} \end{cases}$$

Then it is bounded on $[a, b]$. Now $f^2(x) = 1$ for all $x \in [a, b]$ so that $f^2 \in \mathcal{R}$ on $[a, b]$. However, using similar argument as in the proof of Problem 6.4, we can show that $f \notin \mathcal{R}$ on $[a, b]$.

For the second assertion, we know from the hypothesis that there exist constants M and m such that

$$m \leq f(x) \leq M$$

for all $x \in [a, b]$. Define $\phi : [m^3, M^3] \rightarrow \mathbb{R}$ by

$$\phi(y) = y^{\frac{1}{3}}.$$

Then the function ϕ is continuous on $[m^3, M^3]$ and

$$\phi(f^3(x)) = f(x)$$

on $[a, b]$. By Theorem 6.11, we have $f \in \mathcal{R}(\alpha)$ on $[a, b]$. This completes the proof of the problem. ■

Problem 6.6*Rudin Chapter 6 Exercise 6.*

Proof. Recall the definition of the Cantor set P : Suppose that $E_0, E_1, E_2, \dots$ are the intervals defined in Sec. 2.44. Then we have

$$P = \bigcap_{n=1}^{\infty} E_n.$$

Table 6.1 below shows the number of intervals, the number of the end-points and the length of each interval for each E_n .

We apply Theorem 6.6 to show that $f \in \mathcal{R}$ on $[0, 1]$. In fact, our goal is to *construct* a suitable partition $P = \{x_0, x_1, \dots, x_n\}$ of $[0, 1]$ such that

$$U(P, f) - L(P, f) = \sum_{i=1}^n (M_i - m_i) \Delta x_i < \epsilon \quad (6.12)$$

Interval	Number of intervals	Number of end-points including 0 and 1	Length of each interval
E_0	2^0	2^1	$\frac{1}{3^0}$
E_1	2^1	2^2	$\frac{1}{3^1}$
E_2	2^2	2^3	$\frac{1}{3^2}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
E_n	2^n	2^{n+1}	$\frac{1}{3^n}$

Table 6.1: The number of intervals & end-points and the length of each interval for each E_n .

for every $\epsilon > 0$.

Since f is bounded on $[0, 1]$, there are constants M and m such that

$$m \leq f(x) \leq M$$

for all $x \in [0, 1]$. Suppose that n is a large positive integer such that

$$1 + \frac{\log \frac{\epsilon}{8(M-m)}}{\log \frac{2}{3}} < n. \quad (6.13)$$

Now for each E_n , we let

$$0 = a_1 < b_1 < a_2 < b_2 < \cdots < a_{2^n} < b_{2^n} = 1$$

be the end-points contained in E_n .^a By the information given in Table 6.1, we know two facts about the end-points:

- **Fact 1.** For $k = 1, 2, \dots, 2^n$ and $j = 1, 2, \dots, 2^n - 1$, we have

$$b_k - a_k = \frac{1}{3^n} \quad \text{and} \quad a_{j+1} - b_j \leq \frac{1}{3^{n-1}} \quad (6.14)$$

- **Fact 2.** If $x \in P$, then it follows from the definition of P and the footnote **a** that

$$x \in [a_k, b_k]$$

for some positive integer k . Furthermore, by the hypotheses, f is continuous at every point $x \in [0, 1] \setminus P$. However, it *does not* mean that f is discontinuous at every point of P . In fact, the only thing that we can say is that

“ f is *possibly* discontinuous at points of P .”

Now we can start to construct our partition based on **Fact 1**. Let $\delta > 0$ be a number such that $\delta < \frac{1}{2 \cdot 3^n}$. By the observations (6.14), we define a partition $P_n(\delta)$ by

$$P_n(\delta) = \{0 = a_1 < b_1 + \delta < a_2 - \delta < b_2 + \delta < \cdots < a_{2^n} - \delta < b_{2^n} = 1\}. \quad (6.15)$$

Since there are totally 2^{n+1} distinct points in $P_n(\delta)$, they make the interval $[0, 1]$ into $2^{n+1} - 1$ intervals.^b We consider

$$U(P_n(\delta), f) - L(P_n(\delta), f) = \sum_{i=1}^{2^{n+1}-1} (M_i - m_i) \Delta x_i. \quad (6.16)$$

Next, we estimate the magnitude of (6.16) by using **Fact 2**. In fact, we split the summation (6.16) into two parts: One part consists of only intervals on which f is continuous and the other part consisting of intervals that f *might* contain discontinuities. By the definition (6.15) of $P_n(\delta)$ and **Fact 2**, we know that

$$[a_1, b_1] \subset [a_1, b_1 + \delta], \quad [a_{2^n}, b_{2^n}] \subset [a_{2^n} - \delta, b_{2^n}] \quad \text{and} \quad [a_k, b_k] \subset [a_k - \delta, b_k + \delta]$$

^aWe note that each $[a_k, b_k]$ is an interval in E_n , where $k = 1, 2, \dots, n$.

^bYou should be careful that they are not the intervals in E_n !

for $k = 2, 3, \dots, 2^n - 1$. In other words, all *possible* discontinuities of f must fall into the intervals with *odd indices* of the summation (6.16). Hence we can rewrite the summation (6.16) as

$$\begin{aligned} U(P_n(\delta), f) - L(P_n(\delta), f) &= \underbrace{\sum_{s=1}^{2^n-1} (M_{2s} - m_{2s}) \Delta x_{2s}}_{\text{corresponding to the continuity of } f} + \underbrace{\sum_{r=1}^{2^n} (M_{2r-1} - m_{2r-1}) \Delta x_{2r-1}}_{\text{corresponding to possibly the discontinuity of } f} \\ &\leq (M - m) \sum_{s=1}^{2^n-1} \Delta x_{2s} + (M - m) \sum_{r=1}^{2^n} \Delta x_{2r-1}. \end{aligned} \quad (6.17)$$

In order to find the estimate of the inequality (6.17), we must find the bounds of the summations in (6.17). To this end, we obtain from the expressions (6.14) that

$$\Delta x_{2s} = x_{2s} - x_{2s-1} = (a_{s+1} - \delta) - (b_s + \delta) = a_{s+1} - b_s - 2\delta \leq \frac{1}{3^{n-1}} - 2\delta, \quad (6.18)$$

where $s = 1, 2, \dots, 2^n - 1$. Since δ is any number less than $\frac{1}{2 \cdot 3^n}$, we can further assume that

$$0 < \frac{3}{2} \left[\frac{1}{3^n} - \frac{\epsilon}{2(2^n - 1)(M - m)} \right] \leq \delta < \frac{1}{2 \cdot 3^n}. \quad (6.19)$$

Then we get from the expression (6.18) that

$$(M - m) \sum_{s=1}^{2^n-1} \Delta x_{2s} \leq 3(M - m) \sum_{s=1}^{2^n-1} \left(\frac{1}{3^n} - \frac{2\delta}{3} \right) \leq \sum_{s=1}^{2^n-1} \frac{3\epsilon}{2(2^n - 1)} = \frac{3\epsilon}{2}. \quad (6.20)$$

Similarly, we note from the expressions (6.14) that

$$\begin{aligned} \Delta x_1 &= x_1 - x_0 = b_1 - a_1 + \delta = \frac{1}{3^n} + \delta, \\ \Delta x_{2r-1} &= x_{2r-1} - x_{2r-2} = (b_r + \delta) - (a_r - \delta) = b_r - a_r + 2\delta = \frac{1}{3^n} + 2\delta, \\ \Delta x_{2^{n+1}-1} &= x_{2^{n+1}-1} - x_{2^{n+1}-2} = b_{2^n} - a_{2^n} + \delta = \frac{1}{3^n} + \delta, \end{aligned} \quad (6.21)$$

where $r = 2, 3, \dots, 2^n - 1$. Therefore, it follows from the inequality (6.13), the upper bound of δ and the expressions (6.21) that

$$(M - m) \sum_{r=1}^{2^n} \Delta x_{2r-1} < (M - m) \sum_{r=1}^{2^n} \left(\frac{1}{3^n} + 2\delta \right) \leq 4(M - m) \left(\frac{2}{3} \right)^{n-1} < \frac{\epsilon}{2}. \quad (6.22)$$

Hence it follows from the inequalities (6.20) and (6.22) that

$$U(P_n(\delta), f) - L(P_n(\delta), f) < 2\epsilon$$

holds for every n satisfying the inequality (6.13) and every δ satisfying the inequality (6.19). By Theorem 6.6, we have $f \in \mathcal{R}$ on $[0, 1]$. This completes the proof of the problem. ■

6.2 Definitions of improper integrals

Problem 6.7

Rudin Chapter 6 Exercise 7.

Proof. We use a red symbol on the left-hand side of the definition because we want to emphasize that it is a “new” definition of integral.

- (a) Since $f \in \mathcal{R}$ on $[0, 1]$ and $0 < c < 1$, Theorem 6.12(c) implies that $f \in \mathcal{R}$ on $[0, c]$ and on $[c, 1]$. In addition, $f \in \mathcal{R}$ on $[0, 1]$ also implies that f is bounded on $[0, 1]$ by Definition 6.1. Thus, we let

$$|f(x)| \leq M$$

on $[0, 1]$ for some $M > 0$. Therefore, we follow from Theorem 6.12(b) and (c) that

$$\left| \int_0^1 f \, dx - \int_c^1 f \, dx \right| = \left| \int_0^c f \, dx \right| \leq Mc$$

so that

$$\lim_{\substack{c \rightarrow 0 \\ c > 0}} \left| \int_0^1 f \, dx - \int_c^1 f \, dx \right| = \lim_{\substack{c \rightarrow 0 \\ c > 0}} Mc = 0.$$

Since $\int_0^1 f \, dx$ is a constant, we have

$$\int_0^1 f \, dx = \lim_{\substack{c \rightarrow 0 \\ c > 0}} \int_c^1 f \, dx = \int_0^1 f \, dx.$$

Hence this “new” definition of integral agrees with the old one.

- (b) Let $(0, 1] = \bigcup_{k=1}^{\infty} \left(\frac{1}{k+1}, \frac{1}{k} \right]$. Define $f(x) = (-1)^{k+1}(k+1)$ for all $x \in (\frac{1}{k+1}, \frac{1}{k}]$. Then we have

$$\int_{\frac{1}{k+1}}^{\frac{1}{k}} f(x) \, dx = (-1)^{k+1}(k+1) \left(\frac{1}{k} - \frac{1}{k+1} \right) = \frac{(-1)^{k+1}}{k}. \quad (6.23)$$

It is clear that $c \in (\frac{1}{k+1}, \frac{1}{k}]$ for one and only one positive integer k , so this and expression (6.23) imply that

$$\begin{aligned} \int_c^1 f(x) \, dx &= \int_{\frac{1}{2}}^1 f(x) \, dx + \int_{\frac{1}{3}}^{\frac{1}{2}} f(x) \, dx + \cdots + \int_{\frac{1}{k}}^{\frac{1}{k-1}} f(x) \, dx + \int_c^{\frac{1}{k}} f(x) \, dx \\ &= 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{(-1)^k}{k-1} + \int_c^{\frac{1}{k}} f(x) \, dx \\ \int_c^1 f(x) \, dx - \int_c^{\frac{1}{k}} f(x) \, dx &= 1 - \frac{1}{2} + \frac{1}{3} - \cdots + \frac{(-1)^k}{k-1}. \end{aligned} \quad (6.24)$$

Since $c \rightarrow 0$ if and only if $k \rightarrow \infty$, we have

$$\int_c^{\frac{1}{k}} f(x) \, dx \rightarrow 0$$

as $c \rightarrow 0$ and then we deduce from the expression (6.24) and Theorem 3.43 that

$$\int_0^1 f(x) \, dx = \lim_{\substack{c \rightarrow 0 \\ c > 0}} \int_c^1 f(x) \, dx = 1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2.$$

However, we have $|f(x)| = k+1 \geq 0$ for any positive integer k . Since $c \in (\frac{1}{k+1}, \frac{1}{k}]$, we follow from the expression (6.23) that

$$\sum_{j=1}^k \frac{1}{j} = \int_{\frac{1}{k+1}}^1 |f(x)| \, dx \geq \int_c^1 |f(x)| \, dx \geq \int_{\frac{1}{k}}^1 |f(x)| \, dx = \sum_{j=1}^{k-1} \frac{1}{j}.$$

By the fact that $c \rightarrow 0$ if and only if $k \rightarrow \infty$ and Theorem 3.28, we have

$$\lim_{\substack{c \rightarrow 0 \\ c > 0}} \int_c^1 |f(x)| \, dx$$

does not exist.

We end our analysis of the problem here. ■

Problem 6.8

Rudin Chapter 6 Exercise 8.

Proof. Let k be a positive integer such that $1 \leq k < n$. Since f decreases monotonically on $[1, \infty)$, we have

$$f(k+1) \leq f(x) \leq f(k) \quad (6.25)$$

for all $x \in [k, k+1]$. Recall that $f \in \mathcal{R}$ on $[1, b]$ for every $b > 1$, so we know from Theorem 6.12(c) and the inequalities (6.25) that $f \in \mathcal{R}$ on $[k, k+1]$ which certainly imply

$$f(k+1) \leq \int_k^{k+1} f(x) \, dx \leq f(k). \quad (6.26)$$

By the inequalities (6.26) and the hypothesis $f(x) \geq 0$,^c we have

$$\begin{aligned} \sum_{k=1}^{n-1} f(k+1) &\leq \sum_{k=1}^{n-1} \left(\int_k^{k+1} f(x) \, dx \right) \leq \sum_{k=1}^{n-1} f(k) \\ \sum_{k=2}^n f(k) &\leq \int_1^n f(x) \, dx \leq \sum_{k=1}^{n-1} f(k) \\ \sum_{k=2}^n f(k) &\leq \int_1^n f(x) \, dx \leq \sum_{k=1}^n f(k). \end{aligned} \quad (6.27)$$

Since $f(x) \geq 0$ on $[1, \infty)$, the sequences

$$\left\{ \int_1^n f(x) \, dx \right\} \quad \text{and} \quad \left\{ \sum_{k=1}^n f(k) \right\} \quad (6.28)$$

are increasing monotonically. If the series converges, then we deduce from the right-hand inequality in (6.27) that the sequence of integrals in (6.28) is bounded above. By Theorem 3.14, it converges. Conversely, if the sequence of integrals converges, then the left-hand inequality in (6.27) guarantees that the series in (6.28) is also bounded above. By Theorem 3.14 again, it converges. This completes the proof of the problem. ■

Problem 6.9

Rudin Chapter 6 Exercise 9.

Proof. The result is stated as follows:

Integration by Parts for Improper Integrals. Suppose that F and G are differentiable functions on $[a, \infty)$, $F' = f \in \mathcal{R}$ and $G' = g \in \mathcal{R}$ on $[a, \infty)$. If $\lim_{b \rightarrow \infty} F(b)G(b)$ and $\int_a^\infty f(x)G(x) \, dx$ exist, then

$$\int_a^\infty F(x)g(x) \, dx$$

also exists and we have

$$\int_a^\infty F(x)g(x) \, dx = \lim_{b \rightarrow \infty} F(b)G(b) - F(a)G(a) - \int_a^\infty f(x)G(x) \, dx. \quad (6.29)$$

^cThe hypothesis $f(x) \geq 0$ is used in the last set of inequalities.

Proof of Integration by Parts for Improper Integrals. Let a be fixed and $a < b$. By Theorem 6.22 (Integration by parts), we have

$$\int_a^b F(x)g(x) \, dx = F(b)G(b) - F(a)G(a) - \int_a^b f(x)G(x) \, dx. \quad (6.30)$$

Taking $b \rightarrow \infty$ to the right-hand side of the formula (6.30), since both

$$\lim_{b \rightarrow \infty} F(b)G(b) \quad \text{and} \quad \int_a^\infty f(x)G(x) \, dx$$

exist, they guarantee that the right-hand side of the formula (6.30) exists. Therefore, this shows that

$$\int_a^\infty F(x)g(x) \, dx$$

exists and then the formula (6.29) holds. ■

Now we return to the proof of the problem. Put $a = 0$, $F(x) = \sin x$ and $G(x) = \frac{1}{1+x}$ into the formula (6.29), we get

$$\begin{aligned} \int_0^\infty \frac{\sin x}{-(1+x)^2} \, dx &= \lim_{b \rightarrow \infty} \frac{\sin b}{1+b} - 0 - \int_0^\infty \frac{\cos x}{1+x} \, dx \\ \int_0^\infty \frac{\cos x}{1+x} \, dx &= \int_0^\infty \frac{\sin x}{(1+x)^2} \, dx. \end{aligned}$$

By Theorem 6.21 (Second Fundamental Theorem of Calculus), we have

$$\int_0^b \left| \frac{\sin x}{(1+x)^2} \right| \, dx \leq \int_0^b \frac{1}{(1+x)^2} \, dx = -\frac{1}{1+b} - \left(-\frac{1}{1+0} \right) = 1 - \frac{1}{1+b}$$

so that

$$\int_0^\infty \left| \frac{\sin x}{(1+x)^2} \right| \, dx = \lim_{b \rightarrow \infty} \int_0^b \left| \frac{\sin x}{(1+x)^2} \right| \, dx \leq 1.$$

In other words,

$$\int_0^\infty \frac{\sin x}{(1+x)^2} \, dx$$

converges absolutely.

To show that the integral

$$\int_0^\infty \frac{\cos x}{1+x} \, dx$$

does not converge absolutely, notice that

$$\begin{aligned} \int_0^{n\pi + \frac{\pi}{2}} \frac{|\cos x|}{1+x} \, dx &\geq \sum_{k=2}^n \int_{\frac{(k-1)\pi}{2}}^{\frac{(k+1)\pi}{2}} \frac{|\cos x|}{1+x} \, dx \\ &\geq \sum_{k=2}^n \frac{2}{(k+1)\pi + 2} \int_{\frac{(k-1)\pi}{2}}^{\frac{(k+1)\pi}{2}} |\cos x| \, dx \\ &\geq \sum_{k=2}^n \frac{4}{(k+1)\pi + 2} \\ &\geq \frac{4}{\pi} \sum_{k=2}^n \frac{1}{k+2} \end{aligned}$$

for every positive integer n . By Theorem 3.28, we know that

$$\sum_{k=2}^n \frac{1}{k+2}$$

diverges and hence this completes the proof of the problem. ■

6.3 Hölder's inequality

Problem 6.10*Rudin Chapter 6 Exercise 10.**Proof.*

- (a) The claim is certainly true if $u = 0$ or $v = 0$. Therefore, we assume that both u and v are positive. By Theorem 8.6(b) and (c), we have $(e^x)'' = e^x \geq 0$. Then Problem 5.14 implies that e^x is convex on $\mathbb{R}$ and so

$$uv = e^{\log(uv)} = e^{\frac{1}{p}\log(u^p) + \frac{1}{q}\log(v^q)} \leq \frac{1}{p}e^{\log u^p} + \frac{1}{q}e^{\log v^q} = \frac{u^p}{p} + \frac{v^q}{q}. \quad (6.31)$$

By Problem 4.23, we know that the equality of

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

holds if and only if $x = y$. Hence the equality of the inequality (6.31) holds if and only if

$$\log(u^p) = \log(v^q)$$

which is equivalent to $u^p = v^q$.^d

- (b) Let $u = f$ and $v = g$. By part (a), we have

$$fg \leq \frac{1}{p}f^p + \frac{1}{q}g^q. \quad (6.32)$$

Since $f, g \in \mathcal{R}(\alpha)$ on $[a, b]$, Theorem 6.13(a) shows that $f^p, g^q \in \mathcal{R}(\alpha)$ on $[a, b]$. Furthermore, there are constants m_f, m_g, M_f and M_g such that

$$m_f \leq |f(x)| \leq M_f \quad \text{and} \quad m_g \leq |g(x)| \leq M_g$$

on $[a, b]$. Since the functions $\phi_p(x) = x^p$ and $\phi_q(x) = x^q$ are obviously continuous on $[m_f, M_f]$ and $[m_g, M_g]$ respectively, Theorem 6.11 guarantees that

$$f^p, g^q \in \mathcal{R}(\alpha) \text{ on } [a, b].$$

Therefore, it follows from these, the inequality (6.32) and Theorem 6.12(a) that

$$\int_a^b fg \, d\alpha \leq \int_a^b \left(\frac{1}{p}f^p + \frac{1}{q}g^q \right) d\alpha = \frac{1}{p} \int_a^b f^p \, d\alpha + \frac{1}{q} \int_a^b g^q \, d\alpha = \frac{1}{p} + \frac{1}{q} = 1.$$

By part (a), we know that the equality holds if and only if $f^p = g^q$.

- (c) Suppose that f and g are complex-valued functions in $\mathcal{R}(\alpha)$ on $[a, b]$. Let

$$F(x) = \frac{|f(x)|}{\left\{ \int_a^b |f(x)|^p \, d\alpha \right\}^{\frac{1}{p}}} \quad \text{and} \quad G(x) = \frac{|g(x)|}{\left\{ \int_a^b |g(x)|^q \, d\alpha \right\}^{\frac{1}{q}}}.$$

Since $f, g \in \mathcal{R}(\alpha)$, we deduce from Theorem 6.13(b) that

$$|f(x)|, |g(x)| \in \mathcal{R}(\alpha) \text{ on } [a, b].$$

^dNote that the inequality proven in (a) is called Young's inequality.

Then Theorem 6.12(a) implies that $F, G \in \mathcal{R}(\alpha)$ on $[a, b]$. Now it is obvious that $F(x) \geq 0, G(x) \geq 0$ on $[a, b]$ and

$$\int_a^b F^p(x) d\alpha = \int_a^b G^q(x) d\alpha = 1.$$

Hence we obtain from part (b) and then Theorem 6.13(b) that

$$\begin{aligned} 1 &\geq \int_a^b F(x)G(x) d\alpha \\ \left\{ \int_a^b |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \left\{ \int_a^b |g(x)|^q d\alpha \right\}^{\frac{1}{q}} &\geq \int_a^b |f(x)||g(x)| d\alpha \\ \left\{ \int_a^b |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \left\{ \int_a^b |g(x)|^q d\alpha \right\}^{\frac{1}{q}} &\geq \left| \int_a^b f(x)g(x) d\alpha \right| \end{aligned}$$

which is our desired inequality. By part (b), the equality holds if and only if $F^p = G^q$ which is equivalent to

$$\frac{|f(x)|^p}{\int_a^b |f(x)|^p d\alpha} = \frac{|g(x)|^q}{\int_a^b |g(x)|^q d\alpha}.$$

- (d) – **Case (i): “Improper” integrals described in Problem 6.7.** Suppose that f and g are real functions on $(0, 1]$ and $f, g \in \mathcal{R}$ on $[c, 1]$ for every $c > 0$. Then Theorem 6.13(a) implies that $fg \in \mathcal{R}$ on $[c, 1]$. By part (c), we have

$$0 \leq \left| \int_c^1 f(x)g(x) d\alpha \right| \leq \left\{ \int_c^1 |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \left\{ \int_c^1 |g(x)|^q d\alpha \right\}^{\frac{1}{q}}. \quad (6.33)$$

If any integral on the right-hand side of the inequality (6.33) is divergent when $c \rightarrow 0$, then Hölder's inequality certainly holds in this case. If

$$\lim_{\substack{c \rightarrow 0 \\ c > 0}} \int_c^1 |f(x)|^p d\alpha = \int_0^1 |f(x)|^p d\alpha \quad \text{and} \quad \lim_{\substack{c \rightarrow 0 \\ c > 0}} \int_c^1 |g(x)|^q d\alpha = \int_0^1 |g(x)|^q d\alpha,$$

then we obtain from Theorem 4.2 that

$$\lim_{n \rightarrow \infty} \int_{c_n}^1 |f(x)|^p d\alpha = \int_0^1 |f(x)|^p d\alpha \quad \text{and} \quad \lim_{n \rightarrow \infty} \int_{c_n}^1 |g(x)|^q d\alpha = \int_0^1 |g(x)|^q d\alpha$$

for every sequence $\{c_n\}$ in $[c, 1]$ such that $c_n \neq c$ and $\lim_{n \rightarrow \infty} c_n = c$. By the inequality (6.33), we have

$$\begin{aligned} 0 \leq \lim_{n \rightarrow \infty} \left| \int_{c_n}^1 f(x)g(x) d\alpha \right| &\leq \lim_{n \rightarrow \infty} \left\{ \int_{c_n}^1 |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \left\{ \int_{c_n}^1 |g(x)|^q d\alpha \right\}^{\frac{1}{q}} \\ &= \lim_{n \rightarrow \infty} \left\{ \int_{c_n}^1 |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \times \lim_{n \rightarrow \infty} \left\{ \int_{c_n}^1 |g(x)|^q d\alpha \right\}^{\frac{1}{q}} \\ &= \left\{ \int_0^1 |f(x)|^p d\alpha \right\}^{\frac{1}{p}} \left\{ \int_0^1 |g(x)|^q d\alpha \right\}^{\frac{1}{q}} < \infty \end{aligned}$$

so that Theorem 4.2 ensures that

$$\left| \int_0^1 f(x)g(x) d\alpha \right|$$

exists. Hence Hölder's inequality also holds for “improper” integrals described in Problem 6.7.

- **Case (ii): “Improper” integrals described in Problem 6.8.** The argument for proving this part is very similar to that in **Case (i)**, so we omit the details here.

We complete the proof of the problem. ■

Problem 6.11

Rudin Chapter 6 Exercise 11.

Proof. Since $f, g, h \in \mathcal{R}(\alpha)$ on $[a, b]$, we follow from Theorem 6.12(a) that

$$f - h, f - g, g - h \in \mathcal{R}(\alpha)$$

on $[a, b]$. By Problem 6.10(c), when $p = q = 2$, we have

$$\left| \int_a^b |f - g| |g - h| d\alpha \right| \leq \left\{ \int_a^b |f - g|^2 d\alpha \right\}^{\frac{1}{2}} \times \left\{ \int_a^b |g - h|^2 d\alpha \right\}^{\frac{1}{2}}. \quad (6.34)$$

Hence, it follows from the inequality (6.34) and Theorem 1.33(e) that

$$\begin{aligned} \|f - h\|_2^2 &= \int_a^b |f - h|^2 d\alpha \\ &= \int_a^b |(f - g) + (g - h)|^2 d\alpha \\ &\leq \int_a^b (|f - g| + |g - h|)^2 d\alpha \\ &= \int_a^b |f - g|^2 d\alpha + 2 \int_a^b |f - g| |g - h| d\alpha + \int_a^b |g - h|^2 d\alpha \\ &\leq \|f - g\|_2^2 + 2\|f - g\|_2 \|g - h\|_2 + \|g - h\|_2^2 \\ &= (\|f - g\|_2 + \|g - h\|_2)^2 \end{aligned}$$

which implies that

$$\|f - h\|_2 \leq \|f - g\|_2 + \|g - h\|_2.$$

This completes the proof of the problem. ■

Problem 6.12

Rudin Chapter 6 Exercise 12.

Proof. We follow the given hint. Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$, define

$$g(t) = \frac{x_i - t}{\Delta x_i} f(x_{i-1}) + \frac{t - x_{i-1}}{\Delta x_i} f(x_i) \quad (6.35)$$

if $x_{i-1} \leq t \leq x_i$.^e As a function of t , it is clear that g is continuous on $[x_{i-1}, x_i]$. Since

$$g(x_i-) = \lim_{\substack{t \rightarrow x_i \\ t \in [x_{i-1}, x_i)}} g(t) = \lim_{\substack{t \rightarrow x_i \\ t \in [x_{i-1}, x_i)}} \left[\frac{x_i - t}{\Delta x_i} f(x_{i-1}) + \frac{t - x_{i-1}}{\Delta x_i} f(x_i) \right] = f(x_i)$$

and

$$g(x_i+) = \lim_{\substack{t \rightarrow x_i \\ t \in (x_i, x_{i+1}]}} g(t) = \lim_{\substack{t \rightarrow x_i \\ t \in (x_i, x_{i+1}]}} \left[\frac{x_{i+1} - t}{\Delta x_{i+1}} f(x_i) + \frac{t - x_i}{\Delta x_{i+1}} f(x_{i+1}) \right] = f(x_i),$$

^eIn fact, the geometric meaning of the function g is that if we consider the graph of $y = f(x)$, then $(t, g(t))$ is a point on the line segment joining the points $(x_{i-1}, f(x_{i-1}))$ and $(x_i, f(x_i))$.

we have

$$g(x_i+) = g(x_i-) = f(x_i)$$

for all $i = 1, 2, \dots, n-1$. Hence g is continuous on $[a, b]$.

To show the result $\|f - g\|_2 < \epsilon$, it is convenient to rewrite the function (6.35) as

$$g(t) = f(x_{i-1}) + \frac{f(x_i) - f(x_{i-1})}{\Delta x_i}(t - x_{i-1}).$$

If $f(x_i) \geq f(x_{i-1})$, then we have

$$\max_{t \in [x_{i-1}, x_i]} g(t) = f(x_i) \quad \text{and} \quad \min_{t \in [x_{i-1}, x_i]} g(t) = f(x_{i-1}). \quad (6.36)$$

If $f(x_i) < f(x_{i-1})$, then we have

$$\max_{t \in [x_{i-1}, x_i]} g(t) = f(x_{i-1}) \quad \text{and} \quad \min_{t \in [x_{i-1}, x_i]} g(t) = f(x_i). \quad (6.37)$$

Therefore, the observations (6.36) and (6.37) imply that

$$\min(f(x_{i-1}), f(x_i)) \leq g(t) \leq \max(f(x_{i-1}), f(x_i)) \quad (6.38)$$

on $[x_{i-1}, x_i]$. Since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, we have from Theorem 6.12(c) that $f \in \mathcal{R}(\alpha)$ on $[x_{i-1}, x_i]$ and then it is a bounded real function on $[x_{i-1}, x_i]$, i.e.,

$$m_i \leq f(x) \leq M_i \quad (6.39)$$

where $M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$ and $m_i = \inf_{x \in [x_{i-1}, x_i]} f(x)$. Thus we follow from the inequalities (6.38) and (6.39) that

$$m_i \leq g(x) \leq M_i \quad (6.40)$$

on $[x_{i-1}, x_i]$.^f Then we obtain from the inequalities (6.39) and (6.40) that

$$|f(x) - g(x)| \leq M_i - m_i \quad (6.41)$$

on $[x_{i-1}, x_i]$.

Since g is continuous on $[a, b]$, Theorem 6.8 implies that $g \in \mathcal{R}(\alpha)$ on $[a, b]$ and then Theorem 6.12(a) shows that $f - g \in \mathcal{R}(\alpha)$ on $[a, b]$. Hence we have from the inequalities (6.41) and the definition that

$$\|f - g\|_2^2 = \int_a^b |f - g|^2 d\alpha = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f - g|^2 d\alpha \leq \sum_{i=1}^n (M_i - m_i)^2 \Delta \alpha_i. \quad (6.42)$$

Again, since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, it is a bounded real function on $[a, b]$. Therefore, there exist constants M and m such that $m \leq f(x) \leq M$ for all $x \in [a, b]$. It is obvious that

$$M_i - m_i \leq M - m$$

for all $i = 1, 2, \dots, n$, so the inequality (6.42) implies that

$$\|f - g\|_2^2 \leq (M - m) \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i. \quad (6.43)$$

By Theorem 6.6, for every $\epsilon > 0$ there exists a partition $P = \{x_0, x_1, \dots, x_n\}$ such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \frac{\epsilon^2}{M - m}. \quad (6.44)$$

^fHere we change the independent variable of g from t to x .

By Definition 6.2, we have

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i.$$

Hence it follows from the inequalities (6.43) and (6.44) that

$$\|f - g\|_2^2 \leq (M - m)[U(P, f, \alpha) - L(P, f, \alpha)] < \epsilon^2$$

which implies that

$$\|f - g\|_2 < \epsilon.$$

This completes the proof of the problem. ■

6.4 Problems related to improper integrals

Problem 6.13

Rudin Chapter 6 Exercise 13.

Proof.

(a) Put $t^2 = u$. It follows from Theorem 6.21 (Integration by Parts) that

$$\begin{aligned} f(x) &= \int_x^{x+1} \sin(t^2) dt \\ &= \int_{x^2}^{(x+1)^2} \sin u \frac{du}{2\sqrt{u}} \\ &= - \int_{x^2}^{(x+1)^2} \frac{d(\cos u)}{2\sqrt{u}} \\ &= \frac{\cos(x^2)}{2x} - \frac{\cos(x+1)^2}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du. \end{aligned} \tag{6.45}$$

Hence, for $x > 0$, we get from the expression (6.45) and Theorem 6.13(b) that

$$\begin{aligned} |f(x)| &= \left| \frac{\cos(x^2)}{2x} - \frac{\cos(x+1)^2}{2(x+1)} - \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du \right| \\ &\leq \frac{1}{2x} + \frac{1}{2(x+1)} + \left| \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du \right| \\ &\leq \frac{1}{2x} + \frac{1}{2(x+1)} + \int_{x^2}^{(x+1)^2} \left| \frac{\cos u}{4u^{\frac{3}{2}}} \right| du \\ &\leq \frac{1}{2x} + \frac{1}{2(x+1)} + \int_{x^2}^{(x+1)^2} \frac{1}{4u^{\frac{3}{2}}} du \\ &= \frac{1}{2x} + \frac{1}{2(x+1)} - \frac{1}{2} \left(\frac{1}{x+1} - \frac{1}{x} \right) \\ &= \frac{1}{x}. \end{aligned} \tag{6.46}$$

Assume that $\cos x^2$ and $\cos(x+1)^2$ attained 1 at the same time. Then we have

$$x^2 = k\pi \quad \text{and} \quad (x+1)^2 = l\pi,$$

where $l > k > 0$. This leads $2x + 1 = (l - k)\pi$ and then $x = \frac{(l - k)\pi - 1}{2}$. Therefore, we have

$$\begin{aligned} x^2 &= \frac{[(l - k)\pi - 1]^2}{4} \\ 4k\pi &= (l - k)^2\pi^2 - 2(l - k)\pi + 1 \\ (l - k)^2\pi^2 - 2(l + k)\pi + 1 &= 0. \end{aligned} \quad (6.47)$$

However, the equation (6.47) means that π is an algebraic number which is a contradiction. Hence the inequality (6.46) is strict, i.e., $|f(x)| < \frac{1}{x}$ for $x > 0$.

(b) If we define

$$r(x) = \frac{\cos(x+1)^2}{x+1} - 2x \int_{x^2}^{(x+1)^2} \frac{\cos u}{4u^{\frac{3}{2}}} du, \quad (6.48)$$

then it follows from the expression (6.45) that

$$2xf(x) = \cos(x^2) - \cos[(x+1)^2] + r(x). \quad (6.49)$$

Hence we have from the expression (6.48) that

$$|r(x)| \leq \frac{1}{x+1} + 2x \int_{x^2}^{(x+1)^2} \frac{du}{4u^{\frac{3}{2}}} = \frac{1}{x+1} - x \left(\frac{1}{x+1} - \frac{1}{x} \right) = \frac{2}{x+1} < \frac{2}{x}$$

as required.

(c) Let's rewrite the expression (6.49) as

$$2xf(x) = 2 \sin \left(x^2 + x + \frac{1}{2} \right) \sin \left(x + \frac{1}{2} \right) + r(x). \quad (6.50)$$

By the result of part (b), we know that $r(x) \rightarrow 0$ as $x \rightarrow \infty$. Thus it follows from the expression (6.50) that

$$-1 \leq xf(x) \leq 1$$

for all $x > 0$. We claim that

$$\limsup_{x \rightarrow \infty} xf(x) = 1 \quad \text{and} \quad \liminf_{x \rightarrow \infty} xf(x) = -1.$$

To prove this claim, we note from the result of part (b) that $2xf(x)$ has the same magnitude as

$$\cos(x^2) - \cos[(x+1)^2] \quad (6.51)$$

as $x \rightarrow \infty$. Put $x_n = n\sqrt{2\pi}$ into the term (6.51) and then apply the periodicity of $\cos x$ to get

$$\cos(x_n^2) - \cos[(x_n+1)^2] = \cos(2n^2\pi) - \cos(2n^2\pi + n\sqrt{8\pi} + 1) = 1 - \cos(n\sqrt{8\pi} + 1).$$

It is clear that the number $\alpha = \sqrt{\frac{2}{\pi}}$ is irrational. By Lemma 4.6, we know that the set

$$S = \{k\alpha - h \mid k \in \mathbb{N}, h \in \mathbb{Z}\}$$

is dense in $\mathbb{R}$. In other words, we consider the number $\frac{1}{2} - \frac{1}{2\pi}$, then for every $\epsilon > 0$ there exists an integer N , sequences $\{h_m\} \subset \mathbb{Z}$ and $\{k_m\} \subset \mathbb{N}$ such that[§]

$$\left| k_m\alpha - h_m - \left(\frac{1}{2} - \frac{1}{2\pi} \right) \right| < \frac{\epsilon}{2\pi} \quad (6.52)$$

[§]It is obvious that

$$\lim_{m \rightarrow \infty} k_m = \infty.$$

for all $m \geq N$. It is clear that the inequality (6.52) is equivalent to

$$|k_m \sqrt{8\pi} + 1 - (2h_m \pi + \pi)| < \epsilon$$

for all $m \geq N$. Therefore, we deduce from this, the periodicity and the continuity of $\cos x$ that

$$\lim_{m \rightarrow \infty} \cos(k_m \sqrt{8\pi} + 1) = \lim_{m \rightarrow \infty} \cos(2h_m \pi + \pi) = \cos \pi = -1.$$

In other words, we have

$$\limsup_{x \rightarrow \infty} x f(x) = 1.$$

The case for

$$\liminf_{x \rightarrow \infty} x f(x) = -1$$

is similar, so we omit the details here.

(d) Let N be a positive integer. We consider

$$\begin{aligned} \int_0^N \sin t^2 dt &= \sum_{k=0}^{N-1} \int_k^{k+1} \sin t^2 dt \\ &= \sum_{k=0}^{N-1} f(k) \\ &= f(0) + \sum_{k=1}^{N-1} \left[\frac{\cos k^2}{2k} - \frac{\cos(k+1)^2}{2k} + \frac{r(k)}{2k} \right] \\ &= f(0) + \frac{1}{2} \sum_{k=1}^{N-1} \frac{r(k)}{k} + \frac{1}{2} \sum_{k=1}^{N-1} \left[\frac{\cos k^2}{k} - \frac{\cos(k+1)^2}{k} \right]. \end{aligned} \quad (6.53)$$

Since

$$\frac{\cos(k+1)^2}{k+1} - \frac{\cos(k+1)^2}{k} = -\frac{\cos(k+1)^2}{k(k+1)},$$

we follow from the expression (6.53) that

$$\int_0^N \sin t^2 dt = f(0) + \frac{1}{2} \sum_{k=1}^{N-1} \frac{r(k)}{k} + \frac{1}{2} \left(\cos 1 - \frac{\cos N^2}{N-1} \right) - \frac{1}{2} \sum_{k=2}^{N-1} \frac{\cos k^2}{k(k-1)}. \quad (6.54)$$

Recall from part (b) that $|r(k)| < \frac{c}{k}$ for a constant c , so we have

$$\left| \sum_{k=1}^{N-1} \frac{r(k)}{k} \right| \leq \sum_{k=1}^{N-1} \left| \frac{r(k)}{k} \right| < \sum_{k=1}^{N-1} \frac{c}{k^2}.$$

Thus Theorems 3.25 and 3.28 imply that

$$\sum_{k=1}^{\infty} \frac{r(k)}{k}$$

converges. Similarly, since $|\cos k^2| \leq 1$, by applying Theorems 3.25 and 3.28 again, we can show that

$$\sum_{k=2}^{\infty} \frac{\cos k^2}{k(k-1)}$$

converges. By Theorem 3.20(a), it is easy to see that

$$\lim_{N \rightarrow \infty} \frac{\cos N^2}{N-1} = 0.$$

Hence every term in the right-hand side of the expression (6.54) converges as $N \rightarrow \infty$ and then

$$\int_0^\infty \sin t^2 dt = \lim_{N \rightarrow \infty} \int_0^N \sin t^2 dt$$

converges.

This ends the proof of the problem. ■

Problem 6.14

Rudin Chapter 6 Exercise 14.

Proof. Put $e^t = u$. It follows from Theorem 6.21 (Integration by Parts) that

$$\begin{aligned} f(x) &= \int_x^{x+1} \sin(e^t) dt \\ &= \int_{e^x}^{e^{x+1}} \frac{\sin u}{u} du \\ &= - \int_{e^x}^{e^{x+1}} \frac{d(\cos u)}{u} \\ &= - \frac{\cos(e^{x+1})}{e^{x+1}} + \frac{\cos(e^x)}{e^x} - \int_{e^x}^{e^{x+1}} \frac{\cos u}{u^2} du. \end{aligned} \quad (6.55)$$

Hence, we have

$$|f(x)| \leq \frac{1}{e^{x+1}} + \frac{1}{e^x} + \int_{e^x}^{e^{x+1}} \frac{du}{u^2} = \frac{1}{e^{x+1}} + \frac{1}{e^x} - \left(\frac{1}{e^{x+1}} - \frac{1}{e^x} \right) = \frac{2}{e^x}. \quad (6.56)$$

Since $\cos u < 1$ for some $u \in [e^x, e^{x+1}]$, the inequality (6.56) is strict, i.e., $e^x |f(x)| < 2$. If we define

$$r(x) = -e^x \int_{e^x}^{e^{x+1}} \frac{\cos u}{u^2} du, \quad (6.57)$$

then the expression (6.55) becomes

$$e^x f(x) = \cos(e^x) - e^{-1} \cos(e^{x+1}) + r(x).$$

Therefore, it follows from the definition (6.57) that

$$|r(x)| = \left| -e^x \int_{e^x}^{e^{x+1}} \frac{\cos u}{u^2} du \right| \leq e^x \int_{e^x}^{e^{x+1}} \frac{du}{u^2} = e^x \left(\frac{1}{e^x} - \frac{1}{e^{x+1}} \right) = \frac{e-1}{e} < 2e^{-1}.$$

This finishes the proof of the problem. ■

6.5 Applications and a generalization of integration by parts

Problem 6.15

Rudin Chapter 6 Exercise 15.

Proof. Since f is a continuously differentiable function on $[a, b]$, Theorem 6.8 and Theorem 6.13(a) imply that

$$f, f^2, f', xff' \in \mathcal{R}$$

on $[a, b]$. Let $F(x) = xf(x)$ and $G(x) = f(x)$. Then we have

$$F'(x) = xf'(x) + f(x) \quad \text{and} \quad G'(x) = f'(x)$$

which are also Riemann-integrable on $[a, b]$. Therefore, Theorem 6.22 (Integration by Parts) implies that

$$\begin{aligned} \int_a^b F(x)G'(x) \, dx &= F(b)G(b) - F(a)G(a) - \int_a^b F'(x)G(x) \, dx \\ \int_a^b xf(x)f'(x) \, dx &= - \int_a^b [xf'(x) + f(x)]f(x) \, dx \\ &= - \int_a^b xf(x)f'(x) \, dx - \int_a^b f^2(x) \, dx \\ \int_a^b xf(x)f'(x) \, dx &= -\frac{1}{2}. \end{aligned}$$

Apply Problem 6.10(c) to the functions $f'(x)$ and $xf(x)$ with $p = q = 2$, we have

$$\begin{aligned} \left\{ \int_a^b [f'(x)]^2 \, dx \right\}^{\frac{1}{2}} \times \left\{ \int_a^b [xf(x)]^2 \, dx \right\}^{\frac{1}{2}} &\geq \left| \int_a^b xf(x)f'(x) \, dx \right| \\ \int_a^b [f'(x)]^2 \, dx \cdot \int_a^b x^2 f^2(x) \, dx &\geq \frac{1}{4}. \end{aligned} \quad (6.58)$$

Next we show that the inequality (6.58) is strict. By Problem 6.10(c), the equality holds if and only if

$$\frac{[f'(x)]^2}{A} = \frac{x^2 f(x)^2}{B}, \quad (6.59)$$

where

$$A = \int_a^b [f'(x)]^2 \, dx \quad \text{and} \quad B = \int_a^b x^2 f(x)^2 \, dx.$$

From the inequality (6.58), we have $AB = \frac{1}{4}$ so that the expression (6.59) becomes either

$$\frac{f'(x)}{f(x)} = 2Ax \quad \text{or} \quad \frac{f'(x)}{f(x)} = -2Ax$$

which yield either

$$f(x) = e^{Ax^2} \quad \text{or} \quad f(x) = e^{-Ax^2}$$

respectively. However, both cases contradict the hypothesis that $f(a) = f(b) = 0$. Hence the inequality (6.58) must be strict. This completes the proof of the problem. ■

Problem 6.16

Rudin Chapter 6 Exercise 16.

Proof.

(a) As suggested by the hint, we consider

$$f(N) = s \int_1^N \frac{[x]}{x^{s+1}} \, dx - \sum_{n=1}^N \frac{1}{n^s}.$$

Since $[x]$ is the greatest integer less than or equal to x , we have $[x] = n$ on $[n, n+1)$ so that

$$f(N) = s \int_1^N \frac{[x]}{x^{s+1}} \, dx - \sum_{n=1}^N \frac{1}{n^s}$$

$$\begin{aligned}
&= s \sum_{n=1}^{N-1} \int_n^{n+1} \frac{[x]}{x^{s+1}} dx - \sum_{n=1}^N \frac{1}{n^s} \\
&= s \sum_{n=1}^{N-1} \int_n^{n+1} \frac{n}{x^{s+1}} dx - \sum_{n=1}^N \frac{1}{n^s} \\
&= - \sum_{n=1}^{N-1} \left[\frac{n}{(n+1)^s} - \frac{n}{n^s} \right] - \sum_{n=1}^N \frac{1}{n^s} \\
&= 1 + \sum_{n=2}^{N-1} \frac{n}{n^s} - \sum_{n=2}^N \frac{n-1}{n^s} - \sum_{n=1}^N \frac{1}{n^s} \\
&= 1 + \sum_{n=2}^{N-1} \frac{n}{n^s} - \sum_{n=2}^N \frac{n}{n^s} + \sum_{n=2}^N \frac{1}{n^s} - \sum_{n=1}^N \frac{1}{n^s} \\
&= -\frac{1}{N^{s-1}}.
\end{aligned} \tag{6.60}$$

Since $s > 1$, we get from Theorem 3.20(a) that

$$\lim_{N \rightarrow \infty} \frac{1}{N^{s-1}} = 0.$$

Hence it yields from the expression (6.60) that $\lim_{N \rightarrow \infty} f(N) = 0$ and then

$$\zeta(s) = s \int_1^{\infty} \frac{[x]}{x^{s+1}} dx$$

as required.

(b) By Problem 6.8 and Theorem 6.21 (Second Fundamental Theorem of Calculus), we have

$$s \int_1^{\infty} \frac{dx}{x^s} = \frac{s}{s-1}.$$

Therefore, the result of part (a) implies that we can split the integral of the right-hand side into two integrals. Hence we have

$$\begin{aligned}
\frac{s}{s-1} - s \int_1^{\infty} \frac{x - [x]}{x^{s+1}} dx &= \frac{s}{s-1} - s \lim_{N \rightarrow \infty} \int_1^N \frac{x - [x]}{x^{s+1}} dx \\
&= \frac{s}{s-1} - s \lim_{N \rightarrow \infty} \left(\int_1^N \frac{dx}{x^s} - \int_1^N \frac{[x]}{x^{s+1}} dx \right) \\
&= \frac{s}{s-1} - s \lim_{N \rightarrow \infty} \int_1^N \frac{dx}{x^s} + s \lim_{N \rightarrow \infty} \int_1^N \frac{[x]}{x^{s+1}} dx \\
&= s \int_1^{\infty} \frac{[x]}{x^{s+1}} dx \\
&= \zeta(s).
\end{aligned}$$

For the second assertion, we note that $0 \leq x - [x] < 1$ for all $x > 1$, so we have

$$0 \leq \int_1^{\infty} \frac{x - [x]}{x^{s+1}} dx = \lim_{N \rightarrow \infty} \int_1^N \frac{x - [x]}{x^{s+1}} dx < \lim_{N \rightarrow \infty} \int_1^N \frac{1}{x^{s+1}} dx = \lim_{N \rightarrow \infty} -\frac{1}{s} \left(\frac{1}{N} - 1 \right) = \frac{1}{s}$$

for all $s > 0$. Hence the integral in part (b) converges for all $s > 0$. This ends the proof of the problem. ■

Problem 6.17

Rudin Chapter 6 Exercise 17.

Proof. We follow the hint. Take g to be real, without loss of generality. Given $P = \{x_0, x_1, \dots, x_n\}$ which will be determined later. Since g is a real continuous function on $[x_{i-1}, x_i]$, Theorem 5.10 (Mean Value Theorem) implies that

$$G(x_i) - G(x_{i-1}) = (x_i - x_{i-1})G'(t_i) = g(t_i)\Delta x_i$$

for some $t_i \in (x_{i-1}, x_i)$. Therefore, we have

$$\begin{aligned} \sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i &= \sum_{i=1}^n \alpha(x_i)[G(x_i) - G(x_{i-1})] \\ &= \sum_{i=1}^n G(x_i)\alpha(x_i) - \sum_{i=1}^n G(x_{i-1})\alpha(x_i) + \sum_{i=1}^n G(x_{i-1})\alpha(x_{i-1}) \\ &\quad - \sum_{i=1}^n G(x_{i-1})\alpha(x_{i-1}) \\ &= \sum_{i=1}^n G(x_i)\alpha(x_i) - \sum_{i=1}^n G(x_{i-1})\alpha(x_{i-1}) - \sum_{i=1}^n G(x_{i-1})\Delta\alpha_i \\ &= [G(x_n)\alpha(x_n) - G(x_0)\alpha(x_0)] + \sum_{i=1}^{n-1} G(x_i)\alpha(x_i) - \sum_{i=1}^{n-1} G(x_i)\alpha(x_i) - \sum_{i=1}^n G(x_{i-1})\Delta\alpha_i \\ &= G(b)\alpha(b) - G(a)\alpha(a) - \sum_{i=1}^n G(x_{i-1})\Delta\alpha_i. \end{aligned} \tag{6.61}$$

Since $g(x) = G'(x)$ on $[a, b]$, we have G is continuous on $[a, b]$ and then Theorem 6.8 implies that $G \in \mathcal{R}(\alpha)$ on $[a, b]$. By Theorem 6.6, for every $\epsilon > 0$, there exists a partition P_1 such that

$$U(P_1, G, \alpha) - L(P_1, G, \alpha) < \epsilon$$

and so we obtain from Theorem 6.7(c) that

$$\left| \sum_{i=1}^n G(t_i)\Delta\alpha_i - \int_a^b G d\alpha \right| < \epsilon, \tag{6.62}$$

where $t_i \in [x_{i-1}, x_i]$. Since t_i is arbitrary, we may take $t_i = x_{i-1}$ in the inequality (6.62). Hence the summation on the right-hand side of the expression (6.61) can be made arbitrarily close to

$$\int_a^b G d\alpha.$$

Now it remains to show that

$$\sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i$$

can also be made arbitrarily close to

$$\int_a^b \alpha(x)g(x) dx.$$

To this end, since α increases monotonically on $[a, b]$, Theorem 6.9 implies that $\alpha \in \mathcal{R}$ on $[a, b]$ and then it follows from Theorem 6.13(a) that $\alpha g \in \mathcal{R}$ on $[a, b]$. By arguing as above, we know that for every $\epsilon > 0$, there exists a partition P_2 such that

$$U(P_2, \alpha g) - L(P_2, \alpha g) < \epsilon$$

and so we obtain from Theorem 6.7(c) that

$$\left| \sum_{i=1}^n \alpha(t_i)g(t_i)\Delta x_i - \int_a^b \alpha(x)g(x) dx \right| < \frac{\epsilon}{2}, \tag{6.63}$$

where $t_i \in [x_{i-1}, x_i]$. Since t_i is arbitrary, we may take $t_i = x_i$ in the inequality (6.63) so that

$$\left| \sum_{i=1}^n \alpha(x_i)g(x_i)\Delta x_i - \int_a^b \alpha(x)g(x) dx \right| < \frac{\epsilon}{2}. \quad (6.64)$$

It is clear that

$$\begin{aligned} \left| \sum_{i=1}^n \alpha(x_i)g(x_i)\Delta x_i - \sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i \right| &= \left| \sum_{i=1}^n \alpha(x_i)[g(x_i) - g(t_i)]\Delta x_i \right| \\ &\leq |\max(\alpha(a), \alpha(b))| \times \left| \sum_{i=1}^n [g(x_i) - g(t_i)]\Delta x_i \right|, \end{aligned} \quad (6.65)$$

where $t_i \in [x_{i-1}, x_i]$. Since g is continuous on $[x_{i-1}, x_i]$, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|g(x_i) - g(t)| < \frac{\epsilon}{2(b-a)|\max(\alpha(a), \alpha(b))|}$$

for every $t \in [x_{i-1}, x_i]$ such that $|x_i - t| < \delta$. Make t_i be such t , so the inequality (6.65) reduces to

$$\begin{aligned} \left| \sum_{i=1}^n \alpha(x_i)g(x_i)\Delta x_i - \sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i \right| &\leq |\max(\alpha(a), \alpha(b))| \sum_{i=1}^n |g(x_i) - g(t_i)|\Delta x_i \\ &< \frac{\epsilon}{2(b-a)} \sum_{i=1}^n \Delta x_i \\ &= \frac{\epsilon}{2}. \end{aligned} \quad (6.66)$$

Therefore, it follows from the inequalities (6.64) and (6.66) that

$$\begin{aligned} \left| \sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i - \int_a^b \alpha(x)g(x) dx \right| &\leq \left| \sum_{i=1}^n \alpha(x_i)g(x_i)\Delta x_i - \sum_{i=1}^n \alpha(x_i)g(t_i)\Delta x_i \right| \\ &\quad + \left| \sum_{i=1}^n \alpha(x_i)g(x_i)\Delta x_i - \int_a^b \alpha(x)g(x) dx \right| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned} \quad (6.67)$$

In other words, the summation on the left-hand side of the expression (6.61) can be made arbitrarily close to

$$\int_a^b \alpha(x)g(x) dx.$$

Hence, if we take P to be the common refinement of P_1 and P_2 (i.e., $P = P_1 \cup P_2$), then the inequalities (6.62) and (6.67) will hold *simultaneously* and they imply the desired formula that

$$\int_a^b \alpha(x)g(x) dx = G(b)\alpha(b) - G(a)\alpha(a) - \int_a^b G d\alpha.$$

This completes the proof of the problem. ^h ■

6.6 Problems on rectifiable curves

Problem 6.18

Rudin Chapter 6 Exercise 18.

^hWe remark that we can't apply Theorem 6.17 directly because α may not be differentiable.

Proof. Suppose that

$$F(t) = t, \quad G(t) = 2t \quad \text{and} \quad H(t) = t \sin \frac{1}{t} - \frac{1}{2},$$

where $t \in [0, 2\pi]$. It is clear that they are continuous functions on $[0, 2\pi]$. By Theorem 8.7(a), the function e^{it} is periodic with the period 2π . Since the lengths of $F([0, 2\pi])$ and $G([0, 2\pi])$ are 2π and 4π respectively, we have

$$\gamma_1([0, 2\pi]) = \gamma_2([0, 2\pi]).$$

For the function $H(t)$, since we have

$$H(0) = -\frac{1}{2} < 0 \quad \text{and} \quad H\left(\frac{3}{\pi}\right) = \frac{3}{\pi} \sin \frac{\pi}{3} - \frac{1}{2} = \frac{3\sqrt{3}}{2\pi} - \frac{1}{2} > \frac{1}{2},$$

it follows from Theorem 4.23 (Intermediate Value Theorem) that $H([0, \frac{3}{\pi}])$ is an interval of length at least 1. Since $H([0, \frac{3}{\pi}]) \subseteq H([0, 2\pi])$, $2\pi H([0, 2\pi])$ is an interval of length at least 2π . This implies that

$$\gamma_1([0, 2\pi]) = \gamma_2([0, 2\pi]) = \gamma_3([0, 2\pi]),$$

i.e., the three curves have the same range.

By Theorems 5.5 and 8.6(b), the functions $\gamma_1(t)$, $\gamma_2(t)$ and $\gamma_3(t)$ are differentiable on $[0, 2\pi]$ and we have

$$\gamma_1'(t) = ie^{it}, \quad \gamma_2'(t) = 2ie^{2it} \quad \text{and} \quad \gamma_3'(t) = 2\pi i \left(\sin \frac{1}{t} - \frac{1}{t} \cos \frac{1}{t} \right) e^{2\pi i t \sin \frac{1}{t}}. \quad (6.68)$$

By the first two equations in (6.68), we know that $\gamma_1'(t)$ and $\gamma_2'(t)$ are continuous on $[0, 2\pi]$. By Theorem 6.27, we conclude that $\gamma_1(t)$ and $\gamma_2(t)$ are rectifiable and then

$$\Lambda(\gamma_1) = \int_0^{2\pi} |\gamma_1'(t)| dt = \int_0^{2\pi} dt = 2\pi \quad \text{and} \quad \Lambda(\gamma_2) = \int_0^{2\pi} |\gamma_2'(t)| dt = \int_0^{2\pi} 2 dt = 4\pi.$$

By the third equation in (6.68), we have

$$\Lambda(\gamma_3) = \int_0^{2\pi} |\gamma_3'(t)| dt = 2\pi \int_0^{2\pi} \left| \sin \frac{1}{t} - \frac{1}{t} \cos \frac{1}{t} \right| dt. \quad (6.69)$$

By the triangle inequality that $|a| - |b| \leq |a - b|$ and the fact that $|\sin \frac{1}{t}| \leq 1$, we have

$$\left| \sin \frac{1}{t} - \frac{1}{t} \cos \frac{1}{t} \right| \geq \left| \frac{1}{t} \cos \frac{1}{t} \right| - \left| \sin \frac{1}{t} \right| \geq \left| \frac{1}{t} \cos \frac{1}{t} \right| - 1$$

so that the integral (6.69) reduces to

$$\Lambda(\gamma_3) \geq 2\pi \int_0^{2\pi} \left(\left| \frac{1}{t} \cos \frac{1}{t} \right| - 1 \right) dt = 2\pi \int_0^{2\pi} \left| \frac{1}{t} \cos \frac{1}{t} \right| dt - 4\pi^2. \quad (6.70)$$

Now we know from Problem 6.7 that we must evaluate the limit

$$\lim_{c \rightarrow 0} \int_c^{2\pi} \left| \frac{1}{t} \cos \frac{1}{t} \right| dt. \quad (6.71)$$

By applying the substitution $x = \frac{1}{t}$ to the integral in (6.71), we have

$$\int_c^{2\pi} \left| \frac{1}{t} \cos \frac{1}{t} \right| dt = - \int_{\frac{1}{c}}^{\frac{1}{2\pi}} \left| \frac{\cos x}{x} \right| dx = \int_{\frac{1}{2\pi}}^{\frac{1}{c}} \left| \frac{\cos x}{x} \right| dx \geq \int_{2\pi}^{\frac{1}{c}} \left| \frac{\cos x}{x} \right| dx.$$

Since c is assumed to be very small, let N be the *largest* positive integer such that $2N\pi \leq \frac{1}{c}$. Then we have

$$\int_{2\pi}^{\frac{1}{c}} \left| \frac{\cos x}{x} \right| dx \geq \int_{2\pi}^{2N\pi} \left| \frac{\cos x}{x} \right| dx$$

$$\begin{aligned}
&= \sum_{k=1}^{N-1} \left(\int_{2k\pi}^{2k\pi + \frac{\pi}{2}} \frac{\cos x}{x} dx - \int_{2k\pi + \frac{\pi}{2}}^{2k\pi + \frac{3\pi}{2}} \frac{\cos x}{x} dx + \int_{2k\pi + \frac{3\pi}{2}}^{2(k+1)\pi} \frac{\cos x}{x} dx \right) \\
&\geq \sum_{k=1}^{N-1} \int_{2k\pi}^{2k\pi + \frac{\pi}{2}} \frac{\cos x}{x} dx \\
&\geq \sum_{k=1}^{N-1} \frac{1}{2k\pi + \frac{\pi}{2}} \int_{2k\pi}^{2k\pi + \frac{\pi}{2}} \cos x dx \\
&= \sum_{k=1}^{N-1} \frac{1}{2k\pi + \frac{\pi}{2}}.
\end{aligned}$$

If $c \rightarrow 0$, then $N \rightarrow \infty$ and we conclude from Theorem 3.28 that the series

$$\sum_{k=1}^{N-1} \frac{1}{2k\pi + \frac{\pi}{2}}$$

and then the integral (6.71) diverges. Hence γ_3 is not rectifiable. This completes the proof of the problem. ■

Problem 6.19

Rudin Chapter 6 Exercise 19.

Proof. We have $\gamma_1 : [a, b] \rightarrow \mathbb{R}^k$. Since $\gamma_2 = \gamma_1 \circ \phi$, we get from Theorem 4.7 that γ_2 is continuous and thus a curve. By Theorem 4.17, the inverse mapping $\phi^{-1} : [a, b] \rightarrow [c, d]$ is also bijective and continuous. Here we need a lemma for proving one of our results:

Lemma 6.1

Suppose that $f : [a, b] \rightarrow [c, d]$ is a continuous and one-to-one function. Then f is monotonic.

Proof of Lemma 6.1. Assume that f was not monotonic. Then we have $x, y, z \in [a, b]$ with $x < y < z$ such that either

$$f(x) \leq f(y) \quad \text{and} \quad f(y) \geq f(z)$$

or

$$f(x) \geq f(y) \quad \text{and} \quad f(y) \leq f(z).$$

Suppose that $f(x) \leq f(y)$ and $f(y) \geq f(z)$. In fact, it is impossible that $f(x) = f(y)$ or $f(y) = f(z)$ because f is one-to-one. Thus we have $f(x) < f(y)$ and $f(y) > f(z)$. It is clear that we can find a number c such that

$$f(x) < c < f(y) \quad \text{and} \quad f(y) > c > f(z).$$

For such number c , Theorem 4.23 (Intermediate Value Theorem) implies that there exist $p \in (x, y)$ and $q \in (y, z)$ such that $f(p) = f(q) = c$. However, this contradicts the hypothesis that f is one-to-one. The case for $f(y) \geq f(z)$ or $f(x) \geq f(y)$ and $f(y) \leq f(z)$ can be proven similarly, so we omit the details here. Hence we have the desired result that f is monotonic. This completes the proof of the lemma. ■

Since the mapping $\phi : [c, d] \rightarrow [a, b]$ is bijective, continuous and $\phi(c) = a$, we conclude that $\phi(d) = b$. Now we prove the remaining results one by one.

- **γ_1 is an arc if and only if γ_2 is an arc.** Suppose that γ_1 is one-to-one. If $\gamma_2(s) = \gamma_2(t)$, then

$$\gamma_1(\phi(s)) = \gamma_1(\phi(t)).$$

Since γ_1 is one-to-one, we have $\phi(s) = \phi(t)$. Since ϕ is one-to-one, we have $s = t$. Therefore, γ_2 is one-to-one.

Suppose that γ_2 is one-to-one. Let $\gamma_1(u) = \gamma_1(v)$, where $u, v \in [a, b]$. By the above observation, there exist $s, t \in [c, d]$ such that

$$\phi(s) = u \quad \text{and} \quad \phi(t) = v.$$

Since $\gamma_1(u) = \gamma_1(v)$, we have $\gamma_1(\phi(s)) = \gamma_1(\phi(t))$ and then $\gamma_2(s) = \gamma_2(t)$. Since γ_2 is one-to-one, we have $s = t$ and then $u = v$. Therefore, γ_1 is one-to-one.

- **γ_1 is a closed curve if and only if γ_2 is a closed curve.** We note that

$$\gamma_1(t) = \gamma_2(\phi^{-1}(t))$$

for all $t \in [a, b]$. By Lemma 6.1, ϕ is monotonic. Since $\phi(c) = a$, it is monotonically increasing and also $\phi(d) = b$. Thus we have $\gamma_1(a) = \gamma_1(b)$ if and only if $\gamma_2(\phi^{-1}(a)) = \gamma_2(\phi^{-1}(b))$ if and only if $\gamma_2(c) = \gamma_2(d)$. Hence γ_1 is a closed curve if and only if γ_2 is a closed curve.

- **γ_1 is a rectifiable curve if and only if γ_2 is a rectifiable curve.** The result can be easily seen from the formula (6.72) below.ⁱ

Before proving that they have the same length, we have an observation first. If $P = \{x_0, x_1, \dots, x_n\}$ is a partition of $[a, b]$, then

$$P(\phi^{-1}) = \{\phi^{-1}(x_0), \phi^{-1}(x_1), \dots, \phi^{-1}(x_n)\}$$

is a partition of $[c, d]$. Similarly, if $P^* = \{x_0, x_1, \dots, x_n\}$ is a partition of $[c, d]$, then

$$P^*(\phi) = \{\phi(x_0), \phi(x_1), \dots, \phi(x_n)\}$$

is a partition of $[a, b]$. In other words, we see that there is a bijective mapping between partitions of $[a, b]$ and partitions of $[c, d]$.

Now let $P^* = \{x_0, x_1, \dots, x_n\}$ be a partition of $[c, d]$. By Definition 6.26 and the definition of γ_2 , we have

$$\Lambda(P, \gamma_2) = \sum_{i=1}^n |\gamma_2(x_i) - \gamma_2(x_{i-1})| = \sum_{i=1}^n |\gamma_1(\phi(x_i)) - \gamma_1(\phi(x_{i-1}))| = \Lambda(P^*(\phi), \gamma_1). \quad (6.72)$$

Since there is a bijective mapping between partitions of $[a, b]$ and partitions of $[c, d]$, we conclude from (6.72) that

$$\Lambda(\gamma_2) = \sup \Lambda(P, \gamma_2) = \sup \Lambda(P^*(\phi), \gamma_1) = \Lambda(\gamma_1),$$

where the leftmost and rightmost supremums are taken over all partitions of $[a, b]$ and $[c, d]$ respectively. Hence they have the same length and this completes the proof of the problem. ■

ⁱNote that we can't apply Theorem 6.27 directly because we are not sure whether γ_1 is differentiable or not.

CHAPTER 7

Sequences and Series of Functions

7.1 Problems on uniform convergence of sequences of functions

Problem 7.1

Rudin Chapter 7 Exercise 1.

Proof. Suppose that $\{f_n\}$ is a uniformly convergent sequence of bounded functions. Let $\epsilon = 1$. Since $\{f_n\}$ is uniformly convergent, there exists an integer N_1 such that $m \geq n \geq N_1$ implies

$$|f_m - f_n| \leq 1.$$

Let the limit function be f . Then there exists an integer N_2 such that $n \geq N_2$ implies

$$|f_n - f| \leq 1.$$

Let $N = \max(N_1, N_2)$. Then, for $m \geq n \geq N$, the triangle inequality shows that

$$|f_n| \leq |f_n - f_m| + |f_m - f| + |f - f_{N+1}| + |f_{N+1}| = 3 + |f_{N+1}|. \quad (7.1)$$

Since f_{N+1} is bounded, we have $|f_{N+1}(x)| \leq M$ for all $x \in \mathbb{R}$. Therefore, we follow from the inequality (7.1) that

$$|f_n| \leq 3 + M$$

for all $n \geq N$ and $x \in \mathbb{R}$. Since $f_1, f_2, \dots, f_N$ are bounded functions, we have $|f_k(x)| \leq M_k$ for all $x \in \mathbb{R}$ and $k = 1, 2, \dots, N$. Hence we have

$$|f_n(x)| \leq \max(M_1, M_2, \dots, M_N, M)$$

for all $x \in \mathbb{R}$ and all $n \geq 1$, completing the proof of the problem. ■

Problem 7.2

Rudin Chapter 7 Exercise 2.

Proof. For every $\epsilon > 0$, there exist integers N_1 and N_2 such that

$$|f_m - f_n| < \frac{\epsilon}{2}$$

if $m \geq n \geq N_1$ and

$$|g_s - g_t| < \frac{\epsilon}{2}$$

if $s \geq t \geq N_2$. Let $N = \max(N_1, N_2)$. Then, for $u, v \geq N$, we have

$$|f_u + g_u - f_v - g_v| \leq |f_u - f_v| + |g_u - g_v| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence $\{f_n + g_n\}$ converges uniformly on E .

Since $\{f_n\}$ and $\{g_n\}$ are uniformly convergent sequences of bounded functions, we conclude from Problem 7.1 that $\{f_n\}$ and $\{g_n\}$ are uniformly bounded. Let A and B be non-negative numbers such that

$$|f_n(x)| \leq A \quad \text{and} \quad |g_n(x)| \leq B$$

for all $n = 1, 2, \dots$ and for all $x \in E$. Given that $\epsilon > 0$. Then there exist integers N_3 and N_4 such that $m \geq n \geq N_3$ implies

$$|f_m - f_n| < \frac{\epsilon}{2B}$$

and $s \geq t \geq N_4$ implies

$$|g_s - g_t| < \frac{\epsilon}{2A}.$$

Let $N^* = \max(N_3, N_4)$. Then, for $u, v \geq N^*$, we have

$$\begin{aligned} |f_u g_u - f_v g_v| &\leq |f_u g_u - f_u g_v| + |f_u g_v - f_v g_v| \\ &\leq |f_u| \times |g_u - g_v| + |g_v| \times |f_u - f_v| \\ &< A \cdot \frac{\epsilon}{2A} + B \cdot \frac{\epsilon}{2B} \\ &= \epsilon. \end{aligned}$$

Hence $\{f_n g_n\}$ converges uniformly on E . This completes the proof of the problem. ■

Problem 7.3

Rudin Chapter 7 Exercise 3.

Proof. Suppose that $f_n(x) = x$ and $g_n(x) = \frac{1}{n}$ are functions defined on $E = \mathbb{N}$, where $n = 1, 2, \dots$. Let $f(x) = x$ and $g(x) = 0$. For every $\epsilon > 0$, we have

$$|f_n(x) - f(x)| = |x - x| = 0 \leq \epsilon$$

for all $n \geq 1$ and $x \in \mathbb{N}$. By Definition 7.7, $\{f_n\}$ converges uniformly to f on $\mathbb{N}$. Similarly, for every $\epsilon > 0$, let N be a positive integer such that $N \geq \frac{1}{\epsilon}$. Then we have

$$|g_n(x) - g(x)| = \left| \frac{1}{n} - 0 \right| \leq \frac{1}{N} \leq \epsilon$$

for all $n \geq N$ and $x \in \mathbb{N}$. Thus $\{g_n\}$ also converges uniformly to g on $\mathbb{N}$.

Now we have

$$f_n(x)g_n(x) = \frac{x}{n}$$

for all $n = 1, 2, \dots$. For each $x \in \mathbb{N}$, it is obvious that

$$f_n(x)g_n(x) \rightarrow 0$$

as $n \rightarrow \infty$. Therefore, the limit function of $\{f_n g_n\}$ must be $F(x) = 0$ for all $x \in \mathbb{N}$. Assume that the convergence was uniform. Then for $0 < \epsilon < 1$, there exists an integer N such that

$$|f_n(x)g_n(x) - 0| = \frac{x}{n} \leq \epsilon \tag{7.2}$$

for all $n \geq N$ and $x \in \mathbb{N}$. In particular, if we take $n = x = N$ in the inequality (7.2), then we have

$$|f_N(N)g_N(N) - 0| = \left| \frac{N}{N} - 0 \right| = 1 \leq \epsilon < 1$$

which is a contradiction. Hence the sequence $\{f_n g_n\}$ does not converge uniformly on $\mathbb{N}$. This completes the proof of the problem. ■

Problem 7.4*Rudin Chapter 7 Exercise 4.**Proof.* Suppose that

$$g_n(x) = \frac{1}{1+n^2x} \quad \text{and} \quad f_n(x) = \sum_{k=1}^n g_k(x).$$

We answer the questions one by one:

- **For what values of x does the series converge absolutely?**

– **Case (i):** $x > 0$. Then we always have $|1+n^2x| = 1+n^2x \geq n^2x$ which leads to

$$|g_n(x)| = \frac{1}{|1+n^2x|} \leq \frac{1}{n^2x}. \quad (7.3)$$

Since $\frac{1}{x} \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by Theorem 3.28,

$$\sum_{n=1}^{\infty} \frac{1}{|1+n^2x|}$$

converges by Theorem 3.25(a).

– **Case (ii):** $x = 0$. Then we have

$$f_n(0) = \sum_{k=1}^n \frac{1}{1+k^2 \cdot 0} = n. \quad (7.4)$$

Thus the series diverges in this case.

– **Case (iii):** $x < 0$. Since $g_n(x)$ is undefined at $x = -\frac{1}{n^2}$, the function $f(x)$ is undefined on the set

$$S = \left\{ -1, -\frac{1}{2^2}, -\frac{1}{3^2}, \dots \right\}. \quad (7.5)$$

So let $x \notin S$. By this, we get $x \in (-\infty, -a] \setminus S$ for some positive a . Thus we have

$$\frac{1}{an^2-1} \geq \left| \frac{1}{1+n^2x} \right| = |g_n(x)|.$$

By Theorems 3.25(a) and 3.28 again, we have $\sum_{n=1}^{\infty} \frac{1}{an^2-1}$ and then

$$\sum_{n=1}^{\infty} \left| \frac{1}{1+n^2x} \right|$$

converges.

In conclusion, the series converges absolutely only if $x > 0$ or $x < 0$ and $x \notin S$ and diverges in other cases.

- **On what intervals does it converge uniformly?** Let $I = [a, b]$ be an interval by Definition 2.17, where $a < b$. Thus it is obvious that the uniform convergence of $\{f_n\}$ on $[a, b]$ depends *solely* on the values of a and b .

- **Case (i):** $a = \alpha > 0$. Then we have $I = [\alpha, b]$ and it follows from the inequality (7.3) that

$$|g_n(x)| = \frac{1}{|1 + n^2x|} \leq \frac{1}{\alpha n^2}$$

holds on I . Since $\sum_{n=1}^{\infty} \frac{1}{\alpha n^2}$ converges by Theorem 3.28, we conclude from Theorem 7.10 (Weierstrass M -Test) that the series converges uniformly on $I = [\alpha, b]$.

- **Case (ii):** $a = 0$. Then we have

$$|g_n(x)| = \left| \frac{1}{1 + n^2x} \right| \leq 1$$

for all $x \in I$. Assume that the series converged uniformly on I . Then Problem 7.1 implies that the sequence $\{f_n\}$ is uniformly bounded on I . However, since I is an interval, there exists a positive integer N such that $\frac{1}{N^2} \in I$. Thus we have $\frac{1}{n^2} \in I$ for all $n \geq N$ and then

$$f_n\left(\frac{1}{n^2}\right) = \sum_{k=1}^n \frac{1}{1 + \frac{k^2}{n^2}} \geq \sum_{k=1}^n \frac{1}{2} = \frac{n}{2} \rightarrow \infty$$

as $n \rightarrow \infty$. This contradicts Definition 7.19.

- **Case (iii):** $b = 0$. Then we see from the expression (7.4) that the series diverges on $[a, 0]$.
- **Case (iv):** $b = \beta < 0$. Then there are two subcases:

- * **I does not contain any element of the set (7.5).** Then we have $\beta < -1$. Since $x < \beta < 0$, we have $1 + n^2x \leq 1 + \beta n^2 < 0$ for $n \geq 1$. Therefore, we have

$$|g_n(x)| = \left| \frac{1}{1 + n^2x} \right| \leq \left| \frac{1}{1 + \beta n^2} \right|$$

on $[a, \beta]$ for $n = 1, 3, \dots$. Let m and n be positive integers such that $m \geq n \geq 2$. Then we have

$$|f_m(x) - f_n(x)| = \left| \sum_{k=n+1}^m \frac{1}{1 + k^2x} \right| \leq \sum_{k=n+1}^m \left| \frac{1}{1 + k^2x} \right| \leq \sum_{k=n+1}^m \left| \frac{1}{1 + \beta k^2} \right| \quad (7.6)$$

for all $x \in [a, \beta]$. By Theorem 3.28, the series $\sum_{k=1}^{\infty} \left| \frac{1}{1 + \beta k^2} \right|$ converges and Theorem 3.22 implies that for every $\epsilon > 0$, there exists an integer N such that

$$\sum_{k=n+1}^m \left| \frac{1}{1 + \beta k^2} \right| \leq \epsilon \quad (7.7)$$

for $m \geq n \geq N$. Hence we establish from the inequalities (7.6) and (7.7) that

$$|f_m(x) - f_n(x)| \leq \epsilon$$

for all $m \geq n \geq N$ and $x \in [a, \beta]$. By Theorem 7.8, the sequence of functions $\{f_n\}$ converges uniformly on $[a, \beta]$.

- * **I contains an element of the set (7.5).** Then we have $I = [a, \beta]$, where $-1 \leq \beta < 0$. This means that $-\frac{1}{n^2} \in I$ for some $n \geq 1$ and so the function $f_n(x)$ is undefined at $x = -\frac{1}{n^2}$ for such $n \geq 1$. Hence the sequence of functions $\{f_n\}$ does not converge uniformly on $[a, \beta]$.

In conclusion, $\{f_n\}$ converges uniformly to f only on $[\alpha, b]$ or $[a, \beta]$, where $\alpha > 0$ and $\beta < -1$.

- **On what intervals does it fail to converge uniformly?** By the above analysis, we see that it fails to converge uniformly on the intervals $[0, b]$ and $[a, \beta]$, where $-1 \leq \beta \leq 0$.

- **Is f continuous wherever the series converges?** Using the same notations as above, we know that $\{f_n\}$ converges uniformly to f on the intervals $[\alpha, b]$ or $[a, \beta]$, where $\alpha > 0$ and $\beta < -1$. Since each f_n is continuous on $[\alpha, b]$ or $[a, \beta]$, then Theorem 7.12 implies that the function f is continuous on $[\alpha, b]$ or $[a, \beta]$.
- **Is f bounded?** We can see from the above argument that f is not bounded on any set containing an element of the set S .

This completes the proof of the problem. ■

Problem 7.5

Rudin Chapter 7 Exercise 5.

Proof. We claim that for each $x \in \mathbb{R}$, $f_n(x) \rightarrow 0$ as $n \rightarrow \infty$. In other words, $\{f_n\}$ converges to 0 *pointwise* on $\mathbb{R}$. In fact, if $x \leq 0$, then we have $f_n(x) = 0$ for all positive integers n and thus we have

$$|f_n(x) - 0| = |0 - 0| \leq \epsilon$$

for all $n \geq 1$. If $x > 0$, then we let N be a positive integer such that $N > \frac{1}{x}$. For all $n \geq N$, since $x > \frac{1}{n}$, we have

$$|f_n(x) - 0| = |0 - 0| \leq \epsilon.$$

This proves our claim.

However, it does not converge to 0 uniformly on $\mathbb{R}$. In fact, for every positive integer n , we take $x = \frac{1}{n+\frac{1}{2}} \in [\frac{1}{n+1}, \frac{1}{n}]$ and so

$$f_n(x) = f_n\left(\frac{1}{n+\frac{1}{2}}\right) = \sin^2\left(n + \frac{1}{2}\right)\pi = 1.$$

Now this implies that

$$|f_n(x) - 0| = |1 - 0| = 1.$$

Hence $\{f_n\}$ does not converge to 0 uniformly on $\mathbb{R}$ by Definition 7.7.

If $x \leq 0$, then it is clear from the definition that $f_n(x) = 0$ for all $n \geq 1$. We note that

$$f_1(x) = \begin{cases} 0, & \left(x < \frac{1}{2}\right); \\ \sin^2 \frac{\pi}{x}, & \left(\frac{1}{2} \leq x \leq 1\right); \\ 0, & (1 < x). \end{cases}$$

Thus if $x \geq 1$, then we also have $f_n(x) = 0$ for all $n \geq 1$. Therefore, if $x \leq 0$ or $x \geq 1$, we have

$$\sum f_n = 0.$$

If $0 < x < 1$, then Theorem 1.20(a) (Archimedean property) guarantees that there is a positive integer $N \geq 2$ such that $x \geq \frac{1}{N}$. In fact, we may assume without loss of generality that this N is the *smallest* integer with this property. This implies that we must have $x \leq \frac{1}{N-1}$. Thus we have

$$\frac{1}{N} \leq x \leq \frac{1}{N-1}$$

and then

$$\sum_{n=1}^{\infty} f_n(x) = f_{N-1}(x) = \sin^2 \frac{\pi}{x}.$$

Now we have

$$\sum_{n=1}^{\infty} f_n(x) = \begin{cases} 0, & \text{if } x \leq 0 \text{ or } x \geq 1; \\ \sin^2 \frac{\pi}{x}, & \text{if } 0 < x < 1. \end{cases}$$

Since $\sin^2 \frac{\pi}{x}$ is clearly non-negative on $(0, 1)$, the series must converge absolutely. Assume that $\sum f_n$ converged uniformly to some function f . In other words, for every $\epsilon > 0$, there is an integer N such that

$$\left| \sum_{k=1}^n f_k(x) - f(x) \right| \leq \frac{\epsilon}{2}$$

for all $n \geq N$ and $x \in \mathbb{R}$. By the fact that

$$f_n = \sum_{k=1}^n f_k - \sum_{k=1}^{n-1} f_k,$$

we have

$$\begin{aligned} |f_n(x) - 0| &= \left| \sum_{k=1}^n f_k(x) - \sum_{k=1}^{n-1} f_k(x) - f(x) + f(x) \right| \\ &\leq \left| \sum_{k=1}^n f_k(x) - f(x) \right| + \left| \sum_{k=1}^{n-1} f_k(x) - f(x) \right| \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon \end{aligned}$$

for all $n \geq N + 1$ and $x \in \mathbb{R}$. Hence this means that $\{f_n\}$ converges to 0 uniformly on $\mathbb{R}$ which is a contradiction. In conclusion, we have the result that absolute convergence on $\mathbb{R}$ does not imply uniform convergence. This ends the proof of the problem. ■

Problem 7.6

Rudin Chapter 7 Exercise 6.

Proof. For every bounded interval $[a, b]$, where $-\infty < a < b < \infty$, we have $|x^2| \leq M = \max(a^2, b^2)$. Suppose that

$$f_m(x) = \sum_{k=1}^m (-1)^k \frac{x^2 + k}{k^2},$$

where $m = 1, 2, \dots$. Then we have

$$f_m(x) = x^2 \sum_{k=1}^m \frac{(-1)^k}{k^2} + \sum_{k=1}^m \frac{(-1)^k}{k}.$$

Given $\epsilon > 0$. By Theorem 3.43, we know that $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ converges. Then Theorem 3.22 implies that there is an integer N_1 such that

$$\left| \sum_{k=n+1}^m \frac{(-1)^k}{k} \right| \leq \frac{\epsilon}{2} \quad (7.8)$$

if $m \geq n \geq N_1$. By Theorem 3.28, we have $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ converges. Similarly, there is an integer N_2 such that

$$\left| \sum_{k=n+1}^m \frac{(-1)^k}{k^2} \right| \leq \frac{\epsilon}{2M} \quad (7.9)$$

if $m \geq n \geq N_2$. Let $N = \max(N_1, N_2)$. Then it yields from the inequalities (7.8) and (7.9) that

$$|f_m(x) - f_n(x)| = \left| x^2 \sum_{k=1}^m \frac{(-1)^k}{k^2} + \sum_{k=1}^m \frac{(-1)^k}{k} - x^2 \sum_{k=1}^n \frac{(-1)^k}{k^2} - \sum_{k=1}^n \frac{(-1)^k}{k} \right|$$

$$\begin{aligned}
&\leq M \left| \sum_{k=n+1}^m \frac{(-1)^k}{k^2} \right| + \left| \sum_{k=n+1}^m \frac{(-1)^k}{k} \right| \\
&\leq M \cdot \frac{\epsilon}{2M} + \frac{\epsilon}{2} \\
&= \epsilon
\end{aligned}$$

for all $m \geq n \geq N$ and $x \in [a, b]$. By Theorem 7.8, the series converges uniformly on every bounded interval.

However, since $\frac{x^2+n}{n^2} \geq \frac{1}{n} > 0$ for all positive integers n and the series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges by Theorem 3.28, the series

$$\sum_{n=1}^{\infty} (-1)^n \frac{x^2+n}{n^2}$$

does not converge absolutely for any x by Theorem 3.25(b). We complete the proof of the problem. ■

Problem 7.7

Rudin Chapter 7 Exercise 7.

Proof. For each $x \in \mathbb{R}$, we have

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \frac{x}{1+nx^2} = 0.$$

Thus the limit function of the sequence of functions $\{f_n\}$ must be $f(x) = 0$. Given that $\epsilon > 0$. Then there is a positive integer N such that $N \geq \frac{1}{4\epsilon^2}$ and so we have

$$|f_n(x) - 0| = \left| \frac{x}{1+nx^2} \right| = \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{n}|x|}{1+nx^2} \quad (7.10)$$

for all $n \geq N$ and $x \in \mathbb{R}$. By applying A.M. $\geq$ G.M. to 1 and nx^2 , we have

$$\frac{1+nx^2}{2} \geq \sqrt{n}|x|.$$

Therefore we follow from this and the expression (7.10) that

$$|f_n(x) - 0| \leq \frac{1}{2\sqrt{n}} \leq \frac{1}{2\sqrt{N}} \leq \epsilon$$

for all $n \geq N$ and $x \in \mathbb{R}$. By Definition 7.7, we have the desired result that $\{f_n\}$ converges uniformly to a function $f = 0$ on $\mathbb{R}$.

Since $f(x) = 0$ on $\mathbb{R}$, we have $f'(x) = 0$ for every $x \in \mathbb{R}$. Now f_n is clearly differentiable and we have

$$f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2}$$

which gives

$$\lim_{n \rightarrow \infty} f'_n(x) = \frac{1-nx^2}{(1+nx^2)^2} = \begin{cases} 0, & \text{if } x \neq 0; \\ 1, & \text{if } x = 0. \end{cases}$$

Hence we have

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$$

is correct if $x \neq 0$, but false if $x = 0$, finishing the proof of the problem. ■

Problem 7.8

Rudin Chapter 7 Exercise 8.

Proof. Suppose that

$$g_n(x) = c_n I(x - x_n) \quad \text{and} \quad f_n(x) = \sum_{k=1}^n g_k(x),$$

where $x \in [a, b]$. By the definition of $I(x)$, we know that

$$|g_n(x)| = |c_n I(x - x_n)| \leq |c_n|.$$

Since $\sum |c_n|$ converges, we conclude from Theorem 7.10 that $f_n = \sum g_n$ converges uniformly to f on $[a, b]$.

Let $x \neq x_n$. If each f_n is continuous at x , then Theorem 7.12 implies that f is also continuous at x . We check Definition 4.5 for

$$f_n(x) = \sum_{k=1}^n c_k I(x - x_k).$$

Since $x \neq x_k$ for $k = 1, 2, \dots, n$, we let $\delta = \min(|x - x_1|, |x - x_2|, \dots, |x - x_n|) > 0$. Let $x - x_k > 0$ for some k so that $x - x_k \geq \delta > 0$. If $x - \delta < t < x + \delta$, then we have

$$t - x_k > x - \delta - x_k = x - x_k - \delta \geq \delta - \delta = 0.$$

This implies that

$$I(x - x_k) = I(t - x_k) = 1.$$

Similarly, let $x - x_k < 0$ for some k so that $x - x_k \leq -\delta < 0$. If $x - \delta < t < x + \delta$, then we have

$$t - x_k < x + \delta - x_k = x - x_k + \delta \leq \delta - \delta = 0.$$

In this case, we have

$$I(x - x_k) = I(t - x_k) = 0.$$

By these, if $|x - t| < \delta$, then we have

$$|f_n(x) - f_n(t)| = \left| \sum_{k=1}^n c_k [I(x - x_k) - I(t - x_k)] \right| = 0 < \epsilon.$$

By Definition 4.5, each f_n is continuous at every $x \neq x_n$. Hence our desired result follows from Theorem 7.12. This completes the proof of the problem. ■

Problem 7.9

Rudin Chapter 7 Exercise 9.

Proof. By Theorem 7.12, we have f is continuous on E . Let $x \in E$ and $\{x_n\}$ be a sequence in E such that $x_n \neq x$ and $\lim_{n \rightarrow \infty} x_n = x$. Given $\epsilon > 0$. Then there is an integer N_1 such that for all $n \geq N_1$, we have

$$|f(x_n) - f(x)| < \frac{\epsilon}{2}. \quad (7.11)$$

Since $\{f_n\}$ converges uniformly to f on E , there is an integer N_2 such that

$$|f_n(x) - f(x)| \leq \frac{\epsilon}{2} \quad (7.12)$$

for all $n \geq N_2$ and $x \in E$. In particular, we put $x = x_n$ into the inequality (7.12) so that

$$|f_n(x_n) - f(x_n)| \leq \frac{\epsilon}{2} \quad (7.13)$$

for all $n \geq N_2$. Let $N = \max(N_1, N_2)$. Then for $n \geq N$, we follow from the inequalities (7.11) and (7.13) that

$$|f_n(x_n) - f(x)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(x)| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence we have the required result that

$$\lim_{n \rightarrow \infty} f_n(x_n) = f(x)$$

for every sequence of points $x_n \in E$ such that $x_n \rightarrow x$, and $x \in E$.

The converse is false. For example, let $f_n(x) = \frac{1}{nx}$ and $E = (0, 1)$. It is clear that each f_n is a continuous function on E . Fix $x \in E$. Then we have

$$\lim_{n \rightarrow \infty} f_n(x) = 0.$$

In other words, $\{f_n\}$ converges pointwise to $f = 0$ on E .

Given that $\epsilon > 0$. Let $\{x_n\}$ be a sequence of E such that $x_n \rightarrow x$. Since $0 < x < 1$ and x is fixed, there is an integer N_1 such that $|x - x_n| < \frac{x}{2}$ for all $n \geq N_1$ and this inequality is equivalent to

$$0 < \frac{x}{2} < x_n < \frac{3x}{2}$$

for all $n \geq N_1$. Besides, we also have $0 < x_n < 1$ for all positive integers n . If we take N to be an integer such that $N > \max(N_1, \frac{2}{\epsilon x})$, then for all $n \geq N$, we have

$$|f_n(x_n) - 0| = \frac{1}{nx_n} - 0 < \frac{2}{nx} \leq \frac{2}{Nx} < \epsilon.$$

This means that

$$\lim_{n \rightarrow \infty} f_n(x_n) = 0$$

holds for every sequence of points $x_n \in E$ such that $x_n \rightarrow x$ and $x \in E$. However, $\{f_n\}$ does not converge uniformly to $f = 0$ on E . In fact, if $m > n \geq 1$, then we have

$$|f_n(x) - f_m(x)| = \left| \frac{1}{nx} - \frac{1}{mx} \right| = \frac{(m-n)}{mnx} \geq \frac{1}{mnx} \geq 1 > \epsilon$$

as $x \rightarrow 0$. Hence the desired result follows from Theorem 7.8 and this completes the proof of the problem. ■

Problem 7.10

Rudin Chapter 7 Exercise 10.

Proof. Suppose that

$$g_n(x) = \frac{(nx)}{n^2} \quad \text{and} \quad f_n(x) = \sum_{k=1}^n g_k(x),$$

where n is a positive integer. By the definition of (x) , we know that $g_n(x) \geq 0$ for every positive integer n and $x \in \mathbb{R}$. Furthermore, the fact $|(nx)| \leq 1$ for every positive integer n implies that

$$|g_n(x)| = \left| \frac{(nx)}{n^2} \right| \leq \frac{1}{n^2}. \quad (7.14)$$

Thus we follow from Theorems 3.28 and 7.10 that the series $\sum g_n$ converges uniformly to f on $\mathbb{R}$ which is equivalent to saying that $f_n \rightarrow f$ uniformly on $\mathbb{R}$.

By Figure 4.3, we know that the function $g(x) = (x)$ is continuous on $\mathbb{R} \setminus \mathbb{Z}$ and discontinuous on $\mathbb{Z}$. Therefore, each function $g_n(x)$ is discontinuous *only* at every point $x \in \mathbb{R}$ such that $nx \in \mathbb{Z}$. Since $nx \in \mathbb{Z}$ if and only if $x \in \mathbb{Q}$, we have $g_n(x)$ is discontinuous only on $\mathbb{Q}$. Then this means that the function $f_n(x)$ is discontinuous on $\mathbb{Q}$ and is continuous on $\mathbb{R} \setminus \mathbb{Q}$ (the set of irrational numbers). Since $f_n \rightarrow f$ uniformly on $\mathbb{R}$, we have $f_n \rightarrow f$ uniformly on $\mathbb{R} \setminus \mathbb{Q}$ and then Theorem 7.12 implies that f is continuous on $\mathbb{R} \setminus \mathbb{Q}$. Now f may be discontinuous on $\mathbb{Q}$.

Let $x = \frac{p}{q} \in \mathbb{Q}$, where $p \in \mathbb{Z}$, $q \in \mathbb{N}$, p and q are relative prime (the only positive integer which divides them is 1). We check Definition 4.25 at $x = \frac{p}{q}$. In other words, we want to show that

$$f(x+) \neq f(x-).$$

To this end, we notice that if n is an integer, then

$$\lim_{\substack{t \rightarrow n \\ t > n}} [t] = n \quad \text{and} \quad \lim_{\substack{t \rightarrow n \\ t < n}} [t] = n - 1. \quad (7.15)$$

Therefore, it follows from the limits (7.15) and the definition of g_n that

$$g_q(x+) = \lim_{\substack{t \rightarrow x \\ t > x}} g_q(t) = \lim_{\substack{t \rightarrow x \\ t > x}} \frac{(qt)}{q^2} = \lim_{\substack{r \rightarrow p \\ r > p}} \frac{(r)}{q^2} = \lim_{\substack{r \rightarrow p \\ r > p}} \frac{r - [r]}{q^2} = 0$$

and

$$g_q(x-) = \lim_{\substack{t \rightarrow x \\ t < x}} g_q(t) = \lim_{\substack{t \rightarrow x \\ t < x}} \frac{(qt)}{q^2} = \lim_{\substack{r \rightarrow p \\ r < p}} \frac{(r)}{q^2} = \lim_{\substack{r \rightarrow p \\ r < p}} \frac{r - [r]}{q^2} = \frac{1}{q^2}.$$

Thus we always have $g_q(x-) - g_q(x+) > 0$ for every positive integer q . Suppose that N_1 is an integer such that

$$\sum_{n=N_1}^{\infty} \frac{1}{n^2} < \frac{1}{2q^2}. \quad (7.16)$$

Let $N = \max(N_1, q)$. On the one hand, we have

$$\begin{aligned} f_N(x-) - f_N(x+) &= \lim_{\substack{t \rightarrow x \\ t < x}} f_N(t) - \lim_{\substack{t \rightarrow x \\ t > x}} f_N(t) \\ &= \lim_{\substack{t \rightarrow x \\ t < x}} \sum_{k=1}^N g_k(t) - \lim_{\substack{t \rightarrow x \\ t > x}} \sum_{k=1}^N g_k(t) \\ &= \sum_{k=1}^N \left[\lim_{\substack{t \rightarrow x \\ t < x}} g_k(t) - \lim_{\substack{t \rightarrow x \\ t > x}} g_k(t) \right] \\ &= \sum_{k=1}^N [g_k(t-) - g_k(t+)] \\ &= \sum_{k=1}^N \frac{1}{k^2} \\ &> \frac{1}{q^2}. \end{aligned} \quad (7.17)$$

On the other hand, we follow from the inequalities (7.14) and (7.16) that

$$0 \leq \sum_{k=N+1}^{\infty} g_k(t) \leq \sum_{k=N_1}^{\infty} g_k(t) \leq \sum_{k=N_1}^{\infty} \frac{1}{k^2} < \frac{1}{2q^2}. \quad (7.18)$$

Since $|t - \frac{p}{q}| < \delta$ if and only if $\frac{p}{q} - \delta < t < \frac{p}{q} + \delta$, we conclude from the inequalities (7.17) and (7.18) that

$$\begin{aligned} f(x-) - f(x+) &= \lim_{s \rightarrow \infty} f\left(x - \frac{1}{s}\right) - \lim_{s \rightarrow \infty} f\left(x + \frac{1}{s}\right) \\ &= \lim_{s \rightarrow \infty} \left[\sum_{k=1}^{\infty} g_k\left(x - \frac{1}{s}\right) - \sum_{k=1}^{\infty} g_k\left(x + \frac{1}{s}\right) \right] \\ &= \lim_{s \rightarrow \infty} \left[\sum_{k=1}^N g_k\left(x - \frac{1}{s}\right) - \sum_{k=1}^N g_k\left(x + \frac{1}{s}\right) \right] \end{aligned}$$

$$\begin{aligned}
& + \lim_{s \rightarrow \infty} \left[\sum_{k=N+1}^{\infty} g_k \left(x - \frac{1}{s} \right) - \sum_{k=N+1}^{\infty} g_k \left(x + \frac{1}{s} \right) \right] \\
& > \lim_{s \rightarrow \infty} \left[\sum_{k=1}^N g_k \left(x - \frac{1}{s} \right) - \sum_{k=1}^N g_k \left(x + \frac{1}{s} \right) \right] - \lim_{s \rightarrow \infty} \sum_{k=N+1}^{\infty} g_k \left(x + \frac{1}{s} \right) \\
& = [f_N(x-) - f_N(x+)] - \lim_{s \rightarrow \infty} \sum_{k=N+1}^{\infty} g_k \left(x + \frac{1}{s} \right) \\
& > \frac{1}{q^2} - \frac{1}{2q^2} \\
& = \frac{1}{2q^2} \\
& > 0.
\end{aligned}$$

This means that f is discontinuous at $x = \frac{p}{q}$ and then on $\mathbb{Q}$. Since $\mathbb{Q}$ is a countable dense subset of $\mathbb{R}$, the second assertion follows immediately.

For the third assertion, let $[a, b]$ be a bounded interval. Recall that $(nx) = nx - [nx]$ so that we may rewrite $g_n(x)$ as

$$g_n(x) = \frac{x}{n} - \frac{[nx]}{n^2}.$$

For every positive integer n , we have $h_n(x) = [nx]$ is clearly monotonic on $[a, b]$ (see Figure 4.3 again). By Theorem 6.9, we have $h_n \in \mathcal{R}$ on $[a, b]$. It is obvious that $\frac{x}{n} \in \mathcal{R}$ on $[a, b]$, so we follow from Theorem 6.12(a) that $g_n \in \mathcal{R}$ on $[a, b]$ and then $f_n \in \mathcal{R}$ on $[a, b]$. Recall the fact that f_n converges uniformly to f on $\mathbb{R}$, so we conclude from Theorem 7.16 that

$$f \in \mathcal{R}$$

on $[a, b]$ which is our desired result. This completes the proof of the problem. ■

Problem 7.11

Rudin Chapter 7 Exercise 11.

Proof. We follow the idea of the proof of Theorem 3.42. Suppose that

$$h_n(x) = \sum_{k=1}^n f_k(x).$$

By the hypothesis (a), $\{h_n\}$ is uniformly bounded on E . By Definition 7.19, there is a positive number M such that

$$|h_n(x)| < M$$

for all $x \in E$ and $n = 1, 2, \dots$. Given $\epsilon > 0$. By the hypothesis (b), there is an integer N such that

$$|g_n(x)| = |g_n(x) - 0| \leq \frac{\epsilon}{2M} \quad (7.19)$$

for all $n \geq N$ and all $x \in E$. If $m \geq n \geq N$ and $x \in E$, then it follows from Theorem 3.41 and the inequality (7.19) that

$$\begin{aligned}
\left| \sum_{k=n}^m f_k g_k \right| &= \left| h_m g_m - h_{n-1} g_n + \sum_{k=n}^{m-1} h_k (g_k - g_{k+1}) \right| \\
&\leq M \left| \sum_{k=n}^{m-1} (g_k - g_{k+1}) + g_n + g_m \right| \\
&= 2M g_n
\end{aligned}$$

$$\begin{aligned} &\leq 2Mg_N \\ &\leq \epsilon. \end{aligned}$$

By Theorem 7.8, the series $\sum f_n g_n$ converges uniformly on E . This finishes the proof of the problem. $\blacksquare$

Problem 7.12

Rudin Chapter 7 Exercise 12.

Proof. Given that $\epsilon > 0$. Our goal is to prove that there is an integer N such that

$$\left| \int_0^\infty f_n(x) dx - \int_0^\infty f(x) dx \right| < \epsilon \quad (7.20)$$

for all $n \geq N$. To this end, we divide the proof into a few steps.

- **Step 1: Convergence of the integrals in (7.20).** We first show that both integrals

$$\int_0^\infty f_n(x) dx \quad \text{and} \quad \int_0^\infty f(x) dx \quad (7.21)$$

are *well-defined* in the sense of Problems 6.7 and 6.8. Since $|f_n| \leq g$ on $[t, T]$ for every positive integer n , we have from Theorem 6.13(b) that

$$\left| \int_t^T f_n(x) dx \right| \leq \int_t^T |f_n(x)| dx \leq \int_t^T g(x) dx \quad (7.22)$$

for every positive integer n . Similarly, for every $x \in [t, T]$ we have

$$|f(x)| = \left| \lim_{n \rightarrow \infty} f_n(x) \right| \leq \lim_{n \rightarrow \infty} |f_n(x)| \leq g(x) \quad (7.23)$$

so that

$$\left| \int_t^T f(x) dx \right| \leq \int_t^T |f(x)| dx \leq \int_t^T g(x) dx. \quad (7.24)$$

Then the convergence of the two integrals (7.21) follow immediately from the fact that

$$\int_0^\infty g(x) dx < \infty$$

and by letting $t \rightarrow 0$ and $T \rightarrow \infty$ one by one in the inequalities (7.22) and (7.24). In particular, we have the following formulas:

Lemma 7.1

Let $0 < t < T < \infty$. Then we have

$$\left| \int_0^t f(x) dx \right| \leq \int_0^t g(x) dx, \quad \left| \int_T^\infty f(x) dx \right| \leq \int_T^\infty g(x) dx \quad (7.25)$$

and

$$\left| \int_0^t f_n(x) dx \right| \leq \int_0^t g(x) dx, \quad \left| \int_T^\infty f_n(x) dx \right| \leq \int_T^\infty g(x) dx \quad (7.26)$$

where $n = 1, 2, 3, \dots$

- **Step 2: Establish a “temporary” upper bound of (7.20) by other integrals.** For any $0 < t < T < \infty$ and every positive integer n , the triangle inequality shows that the left-hand side of the inequality (7.20) can be reduced as

$$\begin{aligned} \left| \int_0^\infty f_n(x) dx - \int_0^\infty f(x) dx \right| &\leq \left| \int_0^\infty f_n(x) dx - \int_t^T f_n(x) dx \right| + \left| \int_t^T [f_n(x) - f(x)] dx \right| \\ &\quad + \left| \int_t^T f(x) dx - \int_0^\infty f(x) dx \right|. \end{aligned} \quad (7.27)$$

Thus we show the inequality (7.20) is valid by investigating the absolute value of each term in the right-hand side of the inequality (7.27). Before this, we need a lemma:

Lemma 7.2

Theorem 6.12(c) is also valid for improper integrals in the sense of Problems 6.7 and 6.8.

Proof of Lemma 7.2. Let $b > c > a > 0$. Then it follows from Theorem 6.12(c) that

$$\int_0^b f d\alpha = \lim_{a \rightarrow 0} \int_a^b f d\alpha = \lim_{a \rightarrow 0} \left[\int_a^c f d\alpha + \int_c^b f d\alpha \right] = \int_0^c f d\alpha + \int_c^b f d\alpha$$

and

$$\int_a^\infty f d\alpha = \lim_{b \rightarrow \infty} \int_a^b f d\alpha = \lim_{b \rightarrow \infty} \left[\int_a^c f d\alpha + \int_c^b f d\alpha \right] = \int_a^c f d\alpha + \int_c^\infty f d\alpha.$$

■

- **Step 3: Estimate the absolute value of each term in (7.27).** By Lemmas 7.1 and 7.2 as well as the fact that $|f_n| \leq g$ for every positive integer n , we have

$$\begin{aligned} \left| \int_0^\infty f_n(x) dx - \int_t^T f_n(x) dx \right| &\leq \left| \int_0^t f_n(x) dx + \int_T^\infty f_n(x) dx \right| \\ &\leq \int_0^t g(x) dx + \int_T^\infty g(x) dx. \end{aligned} \quad (7.28)$$

Let $0 < t \leq a < b \leq T < \infty$. Since $g \in \mathcal{R}$ on $[t, T]$, it follows from Theorem 6.20 (First Fundamental Theorem of Calculus) that the function $G : [a, b] \rightarrow \mathbb{R}$ defined by

$$G(u) = \int_t^u g(x) dx$$

is continuous on $[t, T]$. Furthermore, since $0 \leq |f_n(x)| \leq g(x)$ for all $x \in [a, b]$, we deduce from Theorem 6.12(b) that $G(x) \geq 0$ for all $x \in [a, b]$ and G is monotonically increasing on $[a, b]$. By Problem 6.7, we know that

$$\int_0^a g(x) dx = \lim_{t \rightarrow 0} \int_t^a g(x) dx$$

which means that for every $\epsilon > 0$, there exists a $\delta_1 > 0$ such that

$$\int_0^t g(x) dx = \int_0^a g(x) dx - \int_t^a g(x) dx < \frac{\epsilon}{6} \quad (7.29)$$

for all t with $0 < t < \delta_1$. Similarly, we see from Problem 6.8 that there exists a $M_1 > 0$ such that

$$\int_T^\infty g(x) dx = \int_t^\infty g(x) dx - \int_t^T g(x) dx < \frac{\epsilon}{6} \quad (7.30)$$

for all $T > M_1$. Therefore, we apply the inequalities (7.29) and (7.30) to the inequality (7.28) to establish

$$\left| \int_0^\infty f_n(x) dx - \int_t^T f_n(x) dx \right| < \frac{\epsilon}{6} + \frac{\epsilon}{6} = \frac{\epsilon}{3} \quad (7.31)$$

for $0 < t < \delta_1$ and $T > M_1$. By a similar argument, there exist $\delta_2 > 0$ and $M_2 > 0$ such that

$$\left| \int_0^\infty f(x) dx - \int_t^T f(x) dx \right| < \frac{\epsilon}{6} + \frac{\epsilon}{6} = \frac{\epsilon}{3} \quad (7.32)$$

for $0 < t < \delta_2$ and $T > M_2$. Now it remains to evaluate the middle integral in (7.27). Since $f_n \rightarrow f$ uniformly on $[t, T]$, there is an integer N such that $n \geq N$ implies^a

$$|f_n(x) - f(x)| \leq \frac{\epsilon}{3(T-t)}$$

for all $x \in [t, T]$ so that

$$\left| \int_t^T [f_n(x) - f(x)] dx \right| \leq \int_t^T |f_n(x) - f(x)| dx \leq \frac{\epsilon}{3}. \quad (7.33)$$

- **Step 4: Obtain the desired inequality (7.20).** Let $\delta = \min(\delta_1, \delta_2)$ and $M = \max(M_1, M_2)$. We pick a t and a T satisfying $0 < t < \delta$ and $T > M$ so that both the inequalities (7.31) and (7.32) hold. After selecting the values of t and T , we can choose an integer N such that the inequality (7.33) holds. Hence, our desired inequality (7.20) follows from the inequality (7.27) by combining the three inequalities (7.31), (7.32) and (7.33).

This completes the proof of the problem. ■

Problem 7.13

Rudin Chapter 7 Exercise 13.

Proof.

(a) We follow the given hint.

- (i) By the hypothesis, we have $|f_n(x)| \leq 1$ for all n and all $x \in \mathbb{R}$. In particular, it is also true that $|f_n(x)| \leq 1$ for all n and all $x \in \mathbb{Q}$. By Definition 7.19, $\{f_n\}$ is uniformly bounded (thus pointwise bounded) on $\mathbb{Q}$. Recall that $\mathbb{Q}$ is countable, so Theorem 7.23 implies that $\{f_n\}$ contains a subsequence $\{f_{n_i}\}$ such that $\{f_{n_i}(r)\}$ converges for every $r \in \mathbb{Q}$.

- (ii) Define $f : \mathbb{R} \rightarrow \mathbb{R}$ to be

$$f(x) = \sup_{\substack{r \leq x \\ r \in \mathbb{Q}}} f(r), \quad (7.34)$$

where $f(r) = \lim_{i \rightarrow \infty} f_{n_i}(r)$. By (i), the value $f(r)$ is well-defined. Thus the supremum (7.34) is well-defined because $\mathbb{R}$ has the least-upper-bound property (see Definition 1.10 and Theorem 1.19).

- (iii) Given that $\epsilon > 0$. Since f is continuous at x , there exists a $\delta > 0$ such that

$$|f(p) - f(x)| < \frac{\epsilon}{2} \quad (7.35)$$

for all $|p - x| < \delta$. By Theorem 1.20(b), we may take p to be rational. Now the inequality (7.35) can be rewritten as

$$f(x) - \frac{\epsilon}{2} < f(p) < f(x) + \frac{\epsilon}{2} \quad (7.36)$$

^aObviously, the integer N depends on t and T .

for all $p \in (x - \delta, x + \delta)$. If we choose $r, s \in \mathbb{Q}$ and $x - \delta < r < x < s < x + \delta$, then since f is a monotonically increasing function on $\mathbb{R}$, we obtain from the inequality (7.36) that

$$f(x) - \frac{\epsilon}{2} < f(r) \leq f(x) \leq f(s) < f(x) + \frac{\epsilon}{2}. \quad (7.37)$$

Besides, since every f_{n_i} is also a monotonically increasing function on $\mathbb{R}$, we have

$$0 \leq f_{n_i}(r) \leq f_{n_i}(x) \leq f_{n_i}(s). \quad (7.38)$$

Since $f(p) = \lim_{i \rightarrow \infty} f_{n_i}(p)$, there exists an integer N_p such that $i \geq N_p$ implies

$$|f_{n_i}(p) - f(p)| < \frac{\epsilon}{2}. \quad (7.39)$$

If we take $N = \max(N_r, N_s)$, then it follows from the inequality (7.38) and then the inequality (7.39) that for all $i \geq N$, we have

$$f(r) - \frac{\epsilon}{2} < f_{n_i}(r) \leq f_{n_i}(x) \leq f_{n_i}(s) < f(s) + \frac{\epsilon}{2}. \quad (7.40)$$

Combining the inequalities (7.37) and (7.40), we conclude that

$$f(x) - \epsilon < f(r) - \frac{\epsilon}{2} < f_{n_i}(r) \leq f_{n_i}(x) \leq f_{n_i}(s) < f(s) + \frac{\epsilon}{2} < f(x) + \epsilon$$

for all $i \geq N$. In other words, we have

$$|f_{n_i}(x) - f(x)| < \epsilon$$

for all $i \geq N$ and this means that $f_{n_i}(x) \rightarrow f(x)$ at every x at which f is continuous.

- (iv) Since every f_{n_i} is a monotonically increasing function on $\mathbb{R}$, the definition (7.34) says that f is also a monotonically increasing function on $\mathbb{R}$. By Theorem 4.30, the set of points of $\mathbb{R}$ at which f is discontinuous is at most countable. Let this set be $E = \{p_1, p_2, \dots\} \subseteq \mathbb{R}$. By Theorem 7.23 again, since $\{f_{n_i}\}$ is still a uniformly bounded (thus pointwise bounded) sequence on the countable set E , $\{f_{n_i}\}$ has a subsequence $\{f_{n_{i_k}}\}$ such that $\{f_{n_{i_k}}\}$ converges for every $x \in E$.

By the above steps, we define a function $f : \mathbb{R} \rightarrow \mathbb{R}$ and obtain a sequence $\{n_{i_k}\}$ such that

$$f(x) = \lim_{k \rightarrow \infty} f_{n_{i_k}}(x)$$

for every $x \in \mathbb{R}$.

- (b) Suppose that K is a compact subset of $\mathbb{R}$, each $f_{n_k} : K \rightarrow \mathbb{R}$ is monotonically increasing and $f : K \rightarrow \mathbb{R}$ is continuous. We want to show that for every $\epsilon > 0$, there exists an integer N such that $n_k \geq N$ implies

$$|f_{n_k}(x) - f(x)| \leq \epsilon$$

for all $x \in K$.

Since f is continuous on K , we know from Theorem 4.14 that f is uniformly continuous on K , i.e., for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \frac{\epsilon}{4} \quad (7.41)$$

for all $x, y \in K$ for which $|x - y| < \delta$. It is clear that $\{(x - \delta, x + \delta) \mid x \in K\}$ is an open cover of K . Since K is compact, we have

$$K \subseteq \bigcup_{i=1}^m (x_i - \delta, x_i + \delta)$$

for some $x_1, \dots, x_m \in K$. Since K is bounded, we may assume further that $x_{i+1} - x_i < \delta$ for $1 \leq i \leq m - 1$.

To simplify our notations, we replace $f_{n_{i_k}}$ by f_{n_k} in the following discussion. Let $x \in K$. Then we must have $x \in [x_i, x_{i+1}]$ for some $1 \leq i \leq m-1$. We fix this i and consider the following inequality

$$|f_{n_k}(x) - f(x)| \leq |f_{n_k}(x) - f_{n_k}(x_i)| + |f_{n_k}(x_i) - f(x_i)| + |f(x_i) - f(x)|. \quad (7.42)$$

Now we are going to find an estimate of each modulus in the right-hand side of the inequality (7.42).

- Since $\{f_{n_k}\}$ converges to f pointwise on K , for each x_i , there exists an integer N_i such that $n_k \geq N_i$ implies that

$$|f_{n_k}(x_i) - f(x_i)| \leq \frac{\epsilon}{5}. \quad (7.43)$$

Take $N = \max(N_1, N_2, \dots, N_m)$. Then the inequality (7.43) certainly holds for all $n_k \geq N$ and all $1 \leq i \leq m$.

- Next, since $x_{i+1} - x_i < \delta$ and $x \in [x_i, x_{i+1}]$, we have $|x_i - x| < \delta$ and thus the inequality (7.41) implies that

$$|f(x_i) - f(x)| < \frac{\epsilon}{5}. \quad (7.44)$$

- By the hypothesis, each f_{n_k} is a monotonically increasing function on K and $x \in [x_i, x_{i+1}]$ for some $1 \leq i \leq m-1$, we have

$$f_{n_k}(x_i) \leq f_{n_k}(x) \leq f_{n_k}(x_{i+1})$$

which implies

$$0 \leq f_{n_k}(x) - f_{n_k}(x_i) \leq f_{n_k}(x_{i+1}) - f_{n_k}(x_i). \quad (7.45)$$

Now we apply the triangle inequality to (7.45) and then the inequalities (7.43) and (7.44) come into play, we get

$$\begin{aligned} |f_{n_k}(x) - f_{n_k}(x_i)| &\leq |f_{n_k}(x_{i+1}) - f_{n_k}(x_i)| \\ &\leq |f_{n_k}(x_{i+1}) - f(x_{i+1})| + |f(x_{i+1}) - f(x_i)| + |f_{n_k}(x_i) - f(x_i)| \\ &< \frac{\epsilon}{5} + \frac{\epsilon}{5} + \frac{\epsilon}{5} \\ &= \frac{3\epsilon}{5}. \end{aligned} \quad (7.46)$$

Hence, by substituting the results (7.43), (7.44) and (7.46) into the inequality (7.42), we derive that

$$|f_{n_k}(x) - f(x)| \leq \frac{3\epsilon}{5} + \frac{\epsilon}{5} + \frac{\epsilon}{5} = \epsilon$$

for all $n_k \geq N$ and $x \in K$. This is our expected inequality and so $f_{n_k} \rightarrow f$ uniformly on compact sets.

Hence, this completes the proof of the problem. ■

Problem 7.14

Rudin Chapter 7 Exercise 14.

Proof. By the property $0 \leq f(t) \leq 1$ for every $t \in \mathbb{R}$, we have

$$\begin{aligned} 0 \leq x(t) &= \sum_{n=1}^{\infty} 2^{-n} f(3^{2n-1}t) \leq \frac{1}{2} + \frac{1}{2^2} + \cdots = 1, \\ 0 \leq y(t) &= \sum_{n=1}^{\infty} 2^{-n} f(3^{2n}t) \leq \frac{1}{2} + \frac{1}{2^2} + \cdots = 1 \end{aligned}$$

for every $t \in \mathbb{R}$. Furthermore, since

$$|2^{-n} f(3^{2n-1}t)| \leq 2^{-n} \quad \text{and} \quad |2^{-n} f(3^{2n}t)| \leq 2^{-n}$$

for every $t \in \mathbb{R}$, Theorems 7.10 and 3.26 imply that $x(t)$ and $y(t)$ are well-defined by the uniform convergence of both series. Since f is continuous on $\mathbb{R}$, we have $\{2^{-n}f(3^{2n-1}t)\}$ and $\{2^{-n}f(3^{2n}t)\}$ are sequences of continuous functions on $\mathbb{R}$. By Theorem 7.12, the functions $x(t)$ and $y(t)$ are continuous on $\mathbb{R}$. Hence $\Phi(t)$ is continuous on $\mathbb{R}$.

We follow the given hint. Let

$$t_0 = \sum_{i=1}^{\infty} 3^{-i-1}(2a_i) = \frac{2a_1}{3^2} + \frac{2a_2}{3^3} + \cdots. \quad (7.47)$$

By Problem 3.19, we see that the set of all points t_0 defined in (7.47) is exactly the Cantor set. If $a_1 = 0$, then we have

$$0 \leq 3t_0 \leq \frac{2}{3^2} + \frac{2}{3^3} + \cdots = \frac{1}{3}$$

so that $f(3t_0) = 0$ in this case. If $a_1 = 1$, then we have

$$\frac{2}{3} \leq 3t_0 \leq \frac{2}{3^2} + \frac{2}{3^3} + \cdots = 1$$

so that $f(3t_0) = 1$. Thus they mean that

$$f(3t_0) = a_1. \quad (7.48)$$

For $k = 2, 3, \dots$, the definition (7.47) gives

$$3^k t_0 = 2(a_1 + a_2 + \cdots + a_{k-1}) + \frac{2a_k}{3} + \frac{2a_{k+1}}{3^2} + \cdots.$$

Therefore, a similar argument shows that

$$f(3^k t_0) = f\left(2(a_1 + a_2 + \cdots + a_{k-1}) + \frac{2a_k}{3} + \frac{2a_{k+1}}{3^2} + \cdots\right) = f\left(\frac{2a_k}{3} + \frac{2a_{k+1}}{3^2} + \cdots\right) = a_k. \quad (7.49)$$

Hence we follow from the expressions (7.48) and (7.49) that

$$x(t_0) = x_0 \quad \text{and} \quad y(t_0) = y_0.$$

This completes the proof of the problem. ■

7.2 Problems on equicontinuous families of functions

Problem 7.15

Rudin Chapter 7 Exercise 15.

Proof. Let $\epsilon > 0$. Since $\{f_n\}$ is equicontinuous on $[0, 1]$, there exists a $\delta > 0$ such that

$$|f_n(t) - f_n(y)| < \epsilon \quad (7.50)$$

for all $|t - y| < \delta$ and f_n , where $t, y \in [0, 1]$. In particular, we may choose $y = 0$ so that the inequality (7.50) becomes

$$|f_n(t) - f_n(0)| < \epsilon \quad (7.51)$$

for all $0 \leq t < \delta$, $t \in [0, 1]$ and f_n . By definition, the inequality (7.51) is equivalent to

$$|f(nt) - f(0)| < \epsilon$$

for all $t \in [0, 1]$ with $0 \leq t < \delta$ and for all positive integers n . Let $x \in \mathbb{R}$. By Theorem 1.20(a) (Archimedean property), there exists a positive integer n such that $n\delta > x$ and this implies that we have $nt = x$ for some $t \in [0, 1]$ with $0 \leq t < \delta$.^b Hence we obtain from the inequality (7.51) that

$$|f(x) - f(0)| < \epsilon$$

for all $x \in \mathbb{R}$. Since ϵ is arbitrary, we have $f(x) = f(0)$ on $\mathbb{R}$, i.e., f is a constant function. ■

^bIn fact, we have $t = \frac{x}{n} < \delta$.

Problem 7.16*Rudin Chapter 7 Exercise 16.*

Proof. Given that $\epsilon > 0$. Suppose that X is the metric space containing K . Since $\{f_n\}$ is equicontinuous on K , there is a $\delta > 0$ such that

$$|f_n(x) - f_n(y)| < \frac{\epsilon}{3} \quad (7.52)$$

whenever $d_X(x, y) < \delta$, $x \in K$, $y \in K$ and every positive integer n . For any $x \in K$, let

$$K_x = \{y \in K \mid d_X(x, y) < \delta\}.$$

Then K_x is an open subset of X and $K = \bigcup_{x \in K} K_x$. Since K is compact, we have

$$K \subseteq K_{x_1} \cup K_{x_2} \cup \cdots \cup K_{x_s}$$

for some positive integer s . Since f_n converges pointwise on K , $\{f_n(x)\}$ is a Cauchy sequence for every $x \in K$. Therefore, for each x_i , there is an integer N_i such that $m, n \geq N_i$ implies

$$|f_m(x_i) - f_n(x_i)| < \frac{\epsilon}{3}, \quad (7.53)$$

where $i = 1, 2, \dots, s$. Put $N = \max(N_1, N_2, \dots, N_s)$. Thus if $m, n \geq N$, then we obtain from (7.53) that

$$|f_m(x_i) - f_n(x_i)| < \frac{\epsilon}{3}, \quad (7.54)$$

where $i = 1, 2, \dots, s$.

Let $x \in K$. We know that $x \in K_{x_i}$ for some $1 \leq i \leq s$. Then we have $d_X(x, x_i) < \delta$ and it follows from the inequality (7.52) that

$$|f_n(x) - f_n(x_i)| < \frac{\epsilon}{3} \quad (7.55)$$

for every positive integer n . Now if $m, n \geq N$, then we see from the inequalities (7.54) and (7.55) that

$$\begin{aligned} |f_m(x) - f_n(x)| &\leq |f_m(x) - f_m(x_i)| + |f_m(x_i) - f_n(x_i)| + |f_n(x_i) - f_n(x)| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon \end{aligned}$$

for all $x \in K$. Hence we obtain from Theorem 7.8 (Cauchy Criterion for Uniform Convergence) that $\{f_n\}$ converges uniformly on K . This completes the proof of the problem. ■

Problem 7.17*Rudin Chapter 7 Exercise 17.*

Proof. Let X and Y be metric spaces and $f_n : E \subseteq X \rightarrow Y$ for $n = 1, 2, 3, \dots$. We state the definitions of uniform convergence and equicontinuity for the sequence of mappings $\{f_n\}$.

Definition 1. We say that $\{f_n\}$ converges uniformly on $E \subseteq X$ to a mapping $f : X \rightarrow Y$ if for every $\epsilon > 0$, there is an integer N such that $n \geq N$ implies that

$$d_Y(f_n(x), f(x)) < \epsilon$$

for all $x \in E$.

Definition 2. A family $\mathcal{F}$ of mappings $f : E \subseteq X \rightarrow Y$ is said to be equicontinuous on E if for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$d_Y(f(x), f(y)) < \epsilon$$

whenever $d_X(x, y) < \delta$, $x \in E, y \in E$ and $f \in \mathcal{F}$.

- **Generalized Theorems 7.9 and 7.12.** Given that $\epsilon > 0$. Define

$$M_n = \sup_{x \in E} d_Y(f_n(x), f(x)).$$

By Definition 1, we see that $d_Y(f_n(x), f(x)) < \epsilon$ if and only if $M_n \rightarrow 0$ as $n \rightarrow \infty$. This proves the generalized Theorem 7.9.

Suppose that $p \in E$. Since there is nothing to prove if p is an *isolated point* of E , we may assume that p is a limit point of E . By the triangle inequality, we have

$$d_Y(f(x), f(p)) \leq d_Y(f(x), f_n(x)) + d_Y(f_n(x), f_n(p)) + d_Y(f_n(p), f(p)) \quad (7.56)$$

for every $x \in E$, where the positive integer n will be determined very soon. Since $f_n \rightarrow f$ uniformly on E , there is an integer N such that $n \geq N$ implies

$$d_Y(f_n(x), f(x)) \leq \frac{\epsilon}{3} \quad (7.57)$$

for all $x \in E$. In particular, we also have

$$d_Y(f_n(p), f(p)) \leq \frac{\epsilon}{3}. \quad (7.58)$$

Now we fix $n = N$ in the inequalities (7.56), (7.57) and (7.58), i.e.,

$$\begin{aligned} d_Y(f(x), f(p)) &\leq d_Y(f(x), f_N(x)) + d_Y(f_N(x), f_N(p)) + d_Y(f_N(p), f(p)) \\ &\leq \frac{2\epsilon}{3} + d_Y(f_N(x), f_N(p)). \end{aligned} \quad (7.59)$$

It remains to find an estimate of $d_Y(f_N(x), f_N(p))$. In fact, since f_N is continuous on E , there is a $\delta > 0$ such that

$$d_Y(f_N(x), f_N(p)) < \frac{\epsilon}{3} \quad (7.60)$$

for all $x \in E$ with $0 < d_X(x, p) < \delta$. Hence, for these $x \in E$ with $0 < d_X(x, p) < \delta$, we may substitute the inequality (7.60) into the inequality (7.59) to get

$$d_Y(f(x), f(p)) < \frac{2\epsilon}{3} + \frac{\epsilon}{3} = \epsilon.$$

Hence f is continuous at p . This proves the generalized Theorem 7.12.^c

- **Generalized Theorems 7.8 and 7.11.** Suppose that the metric space Y is complete and $\epsilon > 0$. It is easy to see the the part of proof on [21, p. 147] remains valid when the absolute values are replaced by the metric d_Y . Conversely, suppose that N is an integer such that $m, n \geq N$ implies that

$$d_Y(f_n(x), f_m(x)) \leq \epsilon \quad (7.61)$$

for every $x \in E$. Since Y complete, it follows from Definition 3.12 that the sequence $\{f_n(x)\}$ converges for every $x \in E$, to a limit which we may call $f(x)$. Thus the sequence $\{f_n\}$ converges on E , to f . Next, we have to prove that the convergence is *uniform*. To this end, fix $n(\geq N)$ temporary and let $m \rightarrow \infty$ in the inequality (7.61). Since $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$, this gives

$$d_Y(f_n(x), f(x)) \leq \epsilon$$

^cWe note that Rudin used Theorem 7.11 to prove Theorem 7.12, but we haven't applied any generalized Theorem 7.11 to prove our generalized Theorem 7.12 here because Theorem 7.11 is not valid for *any* metric space.

for every $n \geq N$ and every $x \in E$. This proves the generalized Theorem 7.8.

One of the core parts of the proof of Theorem 7.11 (see [21, p. 149]) is that the sequence $\{A_n\}$, where

$$A_n = \lim_{t \rightarrow x} f_n(t),$$

is Cauchy in the complete metric space $\mathbb{R}$ and therefore converges to a real number A . By this, one can prove the inequalities (19) to (22) in [21, p. 149] and then finally the result of Theorem 7.11. Since our Y is supposed to be *complete*, the Cauchy sequence $\{A_n\}$ still converges to $A \in Y$. Hence all the inequalities (19) to (22) in [21, p. 149] remain valid when the absolute values are replaced by the metric d_Y and this shows the generalized Theorem 7.12 is true.

- **Generalized Theorems 7.10, 7.16, 7.17, 7.24 and 7.25.** Let k be a positive integer,

$$\mathbf{f}_n = (f_{n1}, f_{n2}, \dots, f_{nk}) : E \subseteq X \rightarrow \mathbb{R}^k$$

be vector-valued functions for every positive integer n . Here each $f_{ni} : E \subseteq X \rightarrow \mathbb{R}$ is a real-valued function, where $1 \leq i \leq k$. Suppose that

$$|\mathbf{f}_n(x)| \leq M_n$$

for all $x \in E$ and all positive integers n . If $\sum M_n$ converges, then for m and n sufficiently large enough, we have

$$\left| \sum_{j=n}^m \mathbf{f}_j(x) \right| \leq \sum_{j=n}^m |\mathbf{f}_j(x)| \leq \sum_{j=n}^m M_j \leq \epsilon$$

for all $x \in E$. Therefore, the sequence $\{\sum \mathbf{f}_n\}$ satisfies the condition (7.61) on E . Hence it follows from the generalized Theorem 7.8 that $\sum \mathbf{f}_n$ converges uniformly on E . This is the generalized Theorem 7.10.

We put $\epsilon_n = \sup_{a \leq x \leq b} |\mathbf{f}_n(x) - \mathbf{f}(x)|$. By Definition 1.36, we know that

$$|f_{ni}(x) - f_i(x)| \leq |\mathbf{f}_n(x) - \mathbf{f}(x)| \leq \epsilon_n$$

for all $i = 1, 2, \dots, k$ and for all $x \in [a, b]$. Then, for each $i = 1, 2, \dots, k$, we have

$$f_{ni} - \epsilon_n \leq f_i \leq f_{ni} + \epsilon_n.$$

Therefore, by applying the argument as in the proof of Theorem 7.16 to each function f_{ni} , we can show that $f_i \in \mathcal{R}(\alpha)$ on $[a, b]$ and

$$\int_a^b f_i d\alpha = \lim_{n \rightarrow \infty} \int_a^b f_{ni} d\alpha. \quad (7.62)$$

Hence we deduce from Definition 6.23 and the expression (7.62) that $\mathbf{f} \in \mathcal{R}(\alpha)$ on $[a, b]$ and

$$\int_a^b \mathbf{f} d\alpha = \lim_{n \rightarrow \infty} \int_a^b \mathbf{f}_n d\alpha$$

as required. This proves the generalized Theorem 7.16.

Given that $\epsilon > 0$. Since $\{\mathbf{f}_n(x_0)\}$ converges for some $x_0 \in [a, b]$ and $\{\mathbf{f}'_n\}$ converges uniformly on $[a, b]$, we choose an integer N such that $m \geq N$ and $n \geq N$ imply that

$$|\mathbf{f}_n(x_0) - \mathbf{f}_m(x_0)| < \frac{\epsilon}{2} \quad (7.63)$$

and

$$|\mathbf{f}'_n(t) - \mathbf{f}'_m(t)| < \frac{\epsilon}{2(b-a)} \quad (7.64)$$

for all $t \in [a, b]$.

If we apply Theorem 5.19 to the vector-valued function $\mathbf{f}_n - \mathbf{f}_m$, then the inequality (7.64) becomes

$$|\mathbf{f}_n(x) - \mathbf{f}_m(x) - \mathbf{f}_n(t) + \mathbf{f}_m(t)| \leq |x - t| \sup_{\substack{y \in (x,t) \text{ or} \\ y \in (t,x)}} |\mathbf{f}'_n(y) - \mathbf{f}'_m(y)| \leq |x - t| \cdot \frac{\epsilon}{2(b-a)} \leq \frac{\epsilon}{2} \quad (7.65)$$

for any $x, t \in [a, b]$ and $n, m \geq N$. By the inequalities (7.63) and (7.65), we have

$$\begin{aligned} |\mathbf{f}_n(x) - \mathbf{f}_m(x)| &\leq |\mathbf{f}_n(x) - \mathbf{f}_m(x) - \mathbf{f}_n(x_0) + \mathbf{f}_m(x_0)| + |\mathbf{f}_n(x_0) - \mathbf{f}_m(x_0)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon \end{aligned}$$

for any $x \in [a, b]$, $n \geq N$ and $m \geq N$. This means that $\{\mathbf{f}_n\}$ converges uniformly on $[a, b]$. Let

$$\mathbf{f}(x) = \lim_{n \rightarrow \infty} \mathbf{f}_n(x),$$

where $x \in [a, b]$. We now fix a point x in $[a, b]$ and define

$$\Phi_n(t) = \frac{\mathbf{f}_n(t) - \mathbf{f}_n(x)}{t - x} \quad \text{and} \quad \Phi(t) = \frac{\mathbf{f}(t) - \mathbf{f}(x)}{t - x} \quad (7.66)$$

for $t \in [a, b]$ but $t \neq x$. Then we have

$$\lim_{t \rightarrow x} \Phi_n(t) = \mathbf{f}'_n(x) \quad (7.67)$$

for $n = 1, 2, \dots$. Now the second inequality in (7.65) shows that

$$|\Phi_n(t) - \Phi_m(t)| \leq \frac{\epsilon}{2(b-a)}$$

if $n \geq N$ and $m \geq N$ so that the sequence $\{\Phi_n\}$ converges uniformly for $t \neq x$. Since $\{\mathbf{f}_n\}$ converges (uniformly) to $\mathbf{f}$ on $[a, b]$, we conclude from the definition (7.66) that

$$\lim_{n \rightarrow \infty} \Phi_n(t) = \Phi(t) \quad (7.68)$$

uniformly for $t \in [a, b]$ but $t \neq x$.

If we apply Theorem 7.11 to $\{\Phi_n\}$, then the limits (7.67) and (7.68) imply that

$$\lim_{t \rightarrow x} \Phi(t) = \lim_{n \rightarrow \infty} \mathbf{f}'_n(x)$$

which is our desired result

$$\mathbf{f}'(x) = \lim_{n \rightarrow \infty} \mathbf{f}'_n(x).$$

This completes the proof of the generalized Theorem 7.17.

We follow the proof of Theorem 7.24 to prove the generalized Theorem 7.24. Let K be a compact metric space and $\mathbf{f}_n \in \mathcal{C}(K)$ for $n = 1, 2, 3, \dots$. Suppose that $\{\mathbf{f}_n\}$ converges uniformly on K . Then there exists an integer N such that

$$\|\mathbf{f}_n - \mathbf{f}_N\| < \epsilon \quad (7.69)$$

for $n > N$. Since $\mathbf{f}_n$ is continuous on the compact set K , Theorem 4.19 implies that $\mathbf{f}_n$ is uniformly continuous on K . Thus there is a $\delta > 0$ such that

$$|\mathbf{f}_i(x) - \mathbf{f}_i(y)| < \epsilon \quad (7.70)$$

if $1 \leq i \leq N$ and $x, y \in K$ with $d(x, y) < \delta$. If $n > N$ and $x, y \in K$ with $d(x, y) < \delta$, we obtain from the inequality (7.69) that

$$|\mathbf{f}_n(x) - \mathbf{f}_n(y)| \leq |\mathbf{f}_n(x) - \mathbf{f}_N(x)| + |\mathbf{f}_N(x) - \mathbf{f}_N(y)| + |\mathbf{f}_N(y) - \mathbf{f}_n(y)| < 3\epsilon.$$

Combining this with the inequality (7.70), we see that the sequence $\{\mathbf{f}_n\}$ satisfies Definition 2. Hence it is equicontinuous on K . This shows the generalized Theorem 7.24.

Finally, we prove the generalized Theorem 7.25 by following the proof of Theorem 7.25:

- (a) Let $\epsilon > 0$ be given and choose a $\delta > 0$ such that

$$|\mathbf{f}_n(x) - \mathbf{f}_n(y)| < \epsilon \quad (7.71)$$

for $n = 1, 2, \dots$ and $x, y \in K$ with $d(x, y) < \delta$.

Since K is compact, there are finitely many points $p_1, p_2, \dots, p_r$ in K such that to every $x \in K$ corresponds at least one p_i with $d(x, p_i) < \delta$. Since $\{\mathbf{f}_n\}$ is pointwise bounded on K , there exists $M_i < \infty$ such that

$$|\mathbf{f}_n(p_i)| < M_i$$

for $n = 1, 2, \dots$. If $M = \max(M_1, M_2, \dots, M_r)$, then we have

$$|\mathbf{f}_n(x)| = |\mathbf{f}_n(x) - \mathbf{f}_n(p_i) + \mathbf{f}_n(p_i)| \leq |\mathbf{f}_n(x) - \mathbf{f}_n(p_i)| + |\mathbf{f}_n(p_i)| < M_i + \epsilon \leq M + \epsilon$$

for all $x \in K$. This shows that $\{\mathbf{f}_n\}$ is uniformly bounded on K .

- (b) By Problem 2.25, let E be a countable dense subset of K . Then Theorem 7.23 shows that $\{\mathbf{f}_n\}$ has a subsequence $\{\mathbf{f}_{n_i}\}$ such that $\{\mathbf{f}_{n_i}(x)\}$ converges for every $x \in E$.^d

Put $\mathbf{g}_i = \mathbf{f}_{n_i}$. By the assumptions in the proof of part (a), we let

$$V(x, \delta) = \{y \in K \mid d(x, y) < \delta\}.$$

Since E is dense in K and K is compact, there are finitely many points $x_1, \dots, x_m$ in E such that

$$K \subseteq V(x_1, \delta) \cup V(x_2, \delta) \cup \dots \cup V(x_m, \delta). \quad (7.72)$$

Since $\{\mathbf{g}_i(x)\}$ converges for every $x \in E$, there is an integer N such that

$$|\mathbf{g}_i(x_s) - \mathbf{g}_j(x_s)| < \epsilon \quad (7.73)$$

whenever $i \geq N, j \geq N$ and $1 \leq s \leq m$. If $x \in K$, then the relation (7.72) implies that $x \in V(x_s, \delta)$ for some s so that the inequality (7.71) gives

$$|\mathbf{g}_i(x) - \mathbf{g}_i(x_s)| < \epsilon \quad (7.74)$$

for every positive integer i . Now if $i \geq N$ and $j \geq N$, then it follows from the inequalities (7.73) and (7.74) that

$$|\mathbf{g}_i(x) - \mathbf{g}_j(x)| \leq |\mathbf{g}_i(x) - \mathbf{g}_i(x_s)| + |\mathbf{g}_i(x_s) - \mathbf{g}_j(x_s)| + |\mathbf{g}_j(x_s) - \mathbf{g}_j(x)| < 3\epsilon$$

for all $x \in K$. Hence we obtain from the generalized Theorem 7.8 that $\{\mathbf{g}_i\}$ converges uniformly on K .

This completes the proof of the generalized Theorem 7.25.^e

Hence we finish the proof of the problem. ■

Problem 7.18

Rudin Chapter 7 Exercise 18.

Proof. Since $\{f_n\}$ is uniformly bounded on $[a, b]$, we have $|f_n(x)| < M$ for all $x \in [a, b]$ and $n = 1, 2, \dots$, where M is a positive number. Since $f_n \in \mathcal{R}$ on $[a, b]$, each F_n is well-defined on $[a, b]$. By Theorem 6.12(d), we have

$$|F_n(x)| = \left| \int_a^x f_n(t) dt \right| \leq M(b-a).$$

^dHere we assume without proof that Theorem 7.23 is also valid for vector-valued functions.

^eThis version of Theorem 7.25 will be applied to prove results in Problem 7.26.

This means that $\{F_n\}$ is uniformly bounded on $[a, b]$. By Theorem 6.20 (First Fundamental Theorem of Calculus), we know that F_n is continuous on $[a, b]$ so that $F_n \in \mathcal{C}([a, b])$ for every positive integer n .

Given that $\epsilon > 0$. Let $\delta = \frac{\epsilon}{M} > 0$. Thus if $x, y \in [a, b]$ and $|x - y| < \delta$, then we have

$$|F_n(x) - F_n(y)| = \left| \int_a^x f_n(t) dt - \int_a^y f_n(t) dt \right| \leq \left| \int_y^x f_n(t) dt \right| \leq M|y - x| < \epsilon$$

which means that $\{F_n\}$ is equicontinuous on $[a, b]$ by Definition 7.22. Hence it contains a subsequence $\{F_{n_k}\}$ converges uniformly on $[a, b]$ by Theorem 7.25. This completes the proof of the problem. ■

Problem 7.19

Rudin Chapter 7 Exercise 19.

Proof. We prove the necessity and the sufficiency of the problem as follows:

- **Necessity.** Suppose that S is compact.

- **Uniformly closed.** By Theorem 2.34, S is closed in $\mathcal{C}(K)$, so S is uniformly closed in $\mathcal{C}(K)$ by Definition 7.14.
- **Pointwise bounded.** Let $x \in K$. Define the set

$$V_n = \{f \in \mathcal{C}(K) \mid \|f\| < n\},$$

where $n = 1, 2, 3, \dots$. By Definition 2.18, each V_n is a neighborhood of the zero function 0. Then Theorem 2.19 implies that the V_n is open in $\mathcal{C}(K)$. Since

$$S \subseteq \mathcal{C}(K) \subseteq \bigcup_{n=1}^{\infty} V_n,$$

the collection $\{V_n\}$ forms an open cover of S . Since S is compact, we have

$$S \subseteq V_{n_1} \cup V_{n_2} \cup \dots \cup V_{n_k} \quad (7.75)$$

for some positive integers $n_1, n_2, \dots, n_k$. Without loss of generality, we may assume that $n_1 < n_2 < \dots < n_k$ so that the relation (7.75) reduces to

$$S \subseteq V_{n_k}.$$

In other words, we have $\|f\| < n_k$ for all $f \in S$. By Definition 7.19, S is uniformly bounded and thus pointwise bounded.

- **Equicontinuous.** We follow the hint to prove the last assertion. Assume that S was not equicontinuous on K . Then there is a $\epsilon > 0$ such that for every $\delta > 0$, we have

$$|f(x) - f(y)| \geq \epsilon$$

for some $f \in S$ and some $x, y \in K$ with $d(x, y) < \delta$. Now we take $\delta = \frac{1}{n}$, where $n = 1, 2, \dots$ in the above consideration so that a sequence of functions $\{f_n\} \subseteq S$ plus sequences of points $\{x_n\}$ and $\{y_n\}$ in K with $d(x_n, y_n) < \frac{1}{n}$ are constructed. In other words, we have

$$|f_n(x_n) - f_n(y_n)| \geq \epsilon, \quad (7.76)$$

where $x_n, y_n \in K$ and $d(x_n, y_n) < \frac{1}{n}$ for $n = 1, 2, \dots$

To complete this part, we need a lemma:

Lemma 7.3

The above sequence of functions $\{f_n\}$ does not contain any equicontinuous subsequence on K .

Proof of Lemma 7.3. Otherwise, the family of functions

$$\mathcal{F} = \{f_{n_k}\} \subseteq \{f_n\}$$

was equicontinuous on K . By Definition 7.22, there exists a $\delta' > 0$ such that

$$|f_{n_k}(x) - f_{n_k}(y)| < \epsilon \quad (7.77)$$

whenever $x, y \in K$ with $d(x, y) < \delta'$ and all $f_{n_k} \in \mathcal{F}$. However, if we choose n_k large enough so that $\frac{1}{n_k} < \delta'$, then the points x_{n_k}, y_{n_k} satisfy

$$d(x_{n_k}, y_{n_k}) < \delta'$$

and thus we obtain from the inequality (7.76) that

$$|f_{n_k}(x_{n_k}) - f_{n_k}(y_{n_k})| \geq \epsilon$$

which contradicts the inequality (7.77). ■

If $\{f_n\} \subseteq S$ contains a uniformly convergent subsequence on K , then Theorem 7.24 implies that such subsequence must be equicontinuous on K which obviously contradicts our Lemma 7.3. Therefore, no subsequence of $\{f_n\}$ converges in $\mathcal{C}(K)$ by the rephrased Theorem 7.9 on [21, p. 151]. However, this contradicts to the fact that S is compact.^f Hence we conclude that S is equicontinuous on K .

- **Sufficiency.** Now we suppose that the subset S is uniformly closed, pointwise bounded and equicontinuous on K . If S is a finite set, then there is nothing to prove by Definition 2.32. Thus we may assume without loss of generality that S is infinite. Let $\{f_n\} \subseteq S$. Then $\{f_n\}$ must be pointwise bounded and equicontinuous on K . By Theorem 7.25, we know that $\{f_n\}$ contains a uniformly convergent subsequence $\{f_{n_k}\}$ on K . By the rephrased Theorem 7.9 on [21, p. 151] again, such subsequence $\{f_{n_k}\}$ must converge to a function f in $\mathcal{C}(K)$. Since S is closed in $\mathcal{C}(K)$, we have $f \in S$. By Problem 2.26, S is compact.

This completes the proof of the problem. ■

7.3 Applications of the (Stone-)Weierstrass theorem

Problem 7.20

Rudin Chapter 7 Exercise 20.

Proof. Let $P(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n$, where $a_0, a_1, \dots, a_n \in \mathbb{R}$. Now it follows from the hypotheses and Theorem 6.12(a) that

$$\int_0^1 f(x)P(x) dx = 0. \quad (7.78)$$

Since f is continuous on $[0, 1]$, Theorem 7.26 implies that there exists a sequence of polynomial P_n converging uniformly to f on $[0, 1]$. In particular, we have from the integral equation (7.78) that

$$\int_0^1 f(x)P_n(x) dx = 0 \quad (7.79)$$

for all $n = 1, 2, \dots$. Since $f, P_n \in \mathcal{R}$ on $[0, 1]$ for all $n = 1, 2, \dots$, we have

$$fP_n \in \mathcal{R}$$

^fIn fact, since S is compact, we deduce from Theorem 3.6(a) that $\{f_n\}$ has a subsequence converging in S and then in $\mathcal{C}(K)$.

on $[0, 1]$ from Theorem 6.13(a). Since $P_n \rightarrow f$ uniformly on $[0, 1]$, $fP_n \rightarrow f^2$ uniformly on $[0, 1]$. Therefore, we conclude from Theorem 7.16 and the integral equation (7.79) that

$$\int_0^1 f^2(x) dx = \lim_{n \rightarrow \infty} \int_0^1 f(x)P_n(x) dx = 0.$$

Since $f^2 \geq 0$ and f^2 is continuous on $[0, 1]$, Problem 6.2 implies that $f^2(x) = 0$ on $[0, 1]$. Hence $f(x) = 0$ for all $x \in [0, 1]$, finishing the proof of the problem. ■

Problem 7.21

Rudin Chapter 7 Exercise 21.

Proof. We have $K = \{z \in \mathbb{C} \mid |z| = 1\}$. It is clear that $f(z) = z \in \mathcal{A}$ because we have

$$f(e^{i\theta}) = e^{i\theta} = 0 \cdot e^0 + e^{i\theta}.$$

Now the algebra $\mathcal{A}$ separates points on K by this f . Furthermore, we see that $f(z) \neq 0$ for all $z \in K$ so that $\mathcal{A}$ vanishes at no point of K . We notice that for every $f \in \mathcal{A}$, we have

$$\int_0^{2\pi} f(e^{i\theta})e^{i\theta} d\theta = \int_0^{2\pi} \left(\sum_{n=0}^N c_n e^{in\theta} \right) e^{i\theta} d\theta = \sum_{n=0}^N c_n \int_0^{2\pi} e^{i(n+1)\theta} d\theta = 0. \quad (7.80)$$

By Definition 7.28, the closure $\mathcal{B}$ of $\mathcal{A}$ is the set of all functions which are limits f of *uniformly convergent sequences* $\{f_n\}$ in $\mathcal{A}$. Define

$$g_n(\theta) = f_n(e^{i\theta}) \quad \text{and} \quad g(\theta) = f(e^{i\theta}).$$

Since $f_n \rightarrow f$ uniformly on K , we have $g_n \rightarrow g$ uniformly on $[0, 2\pi]$. By Theorem 7.16, we can conclude that the integral equation (7.80) is also true for every f in the closure of $\mathcal{A}$. It is obvious that the function $F(z) = \frac{1}{z}$ is a continuous function on K .

However, $F \notin \mathcal{B}$ because

$$\int_0^{2\pi} F(e^{i\theta})e^{i\theta} d\theta = \int_0^{2\pi} e^{-i\theta}e^{i\theta} d\theta = 2\pi. \quad (7.81)$$

This completes the proof of the problem. ■

Problem 7.22

Rudin Chapter 7 Exercise 22.

Proof. Suppose that $\epsilon > 0$. Since $f \in \mathcal{R}(\alpha)$ on $[a, b]$, Problem 6.12 ensures that there exists a continuous function g on $[a, b]$ such that

$$\|f - g\|_2 < \frac{\epsilon}{2}. \quad (7.82)$$

Since g is continuous on $[a, b]$, we follow from Theorem 7.26 (Weierstrass's Theorem) that there exists a sequence of polynomials P_n converging uniformly to g on $[a, b]$. By Definition 7.7, there is an integer N such that $n \geq N$ implies that

$$|P_n(x) - g(x)| \leq \frac{\epsilon^2}{4[\alpha(b) - \alpha(a)]}$$

for all $x \in [a, b]$. By this and Theorem 6.12(d), we have

$$\|P_n - g\|_2 = \left\{ \int_a^b |P_n(x) - g(x)|^2 dx \right\}^{\frac{1}{2}}$$

$$\begin{aligned}
&\leq \left\{ \frac{\epsilon^2}{4[\alpha(b) - \alpha(a)]} [\alpha(b) - \alpha(a)] \right\}^{\frac{1}{2}} \\
&= \frac{\epsilon}{2}
\end{aligned} \tag{7.83}$$

for all $n \geq N$. Hence it follows from the inequalities (7.82), (7.83) and Problem 6.11 that

$$\|f - P_n\|_2 \leq \|f - g\|_2 + \|g - P_n\|_2 < \epsilon$$

for all $n \geq N$ which is exactly what we want. We complete the proof of the problem. ■

Problem 7.23

Rudin Chapter 7 Exercise 23.

Proof. We follow the given hint. For $n = 0$, we know that

$$P_1(x) = P_0(x) + \frac{x^2 - P_0^2(x)}{2} = \frac{x^2}{2}.$$

Since $|x| \leq 1$, we have $0 \leq P_0(x) \leq P_1(x) \leq |x|$. This means that the statement is true for $n = 0$. Assume that the statement is also true for $n = k$ for some non-negative integer k , i.e.,

$$0 \leq P_k(x) \leq P_{k+1}(x) \leq |x|$$

if $|x| \leq 1$. Now for $n = k + 1$, we note from the assumption that $|x| - P_{k+1}(x) \geq 0$ if $|x| \leq 1$. Furthermore, if $|x| \leq 1$, then the assumption also implies that

$$0 \leq \frac{|x| + P_{k+1}(x)}{2} \leq \frac{|x| + |x|}{2} = |x| \leq 1,$$

so we have

$$1 - \frac{|x| + P_{k+1}(x)}{2} \geq 0.$$

Therefore, we obtain from these and the given identity that

$$|x| - P_{k+2}(x) = [|x| - P_{k+1}(x)] \left[1 - \frac{|x| + P_{k+1}(x)}{2} \right] \geq 0$$

if $|x| \leq 1$. By definition and the assumption, we have

$$P_{k+2}(x) - P_{k+1}(x) = \frac{x^2 - P_{k+1}^2(x)}{2} = \frac{[|x| - P_{k+1}(x)][|x| + P_{k+1}(x)]}{2} \geq 0$$

if $|x| \leq 1$. Therefore, the statement is true for $n = k + 1$ if it is true for $n = k$. Hence it follows from induction that it is true for all non-negative integers n .

By using the identity repeatedly, we have

$$\begin{aligned}
|x| - P_n(x) &= [|x| - P_{n-1}(x)] \left[1 - \frac{|x| + P_{n-1}(x)}{2} \right] \\
|x| - P_{n-1}(x) &= [|x| - P_{n-2}(x)] \left[1 - \frac{|x| + P_{n-2}(x)}{2} \right] \\
&\vdots \\
|x| - P_1(x) &= [|x| - P_0(x)] \left[1 - \frac{|x| + P_0(x)}{2} \right].
\end{aligned} \tag{7.84}$$

We remark that if $|x| \leq 1$, then the inequalities $0 \leq P_n(x) \leq P_{n+1}(x) \leq |x|$ certainly imply that

$$1 - \frac{|x| + P_n(x)}{2} \leq 1 - \frac{|x|}{2} \quad (7.85)$$

for all non-negative integer n . Therefore, if $|x| \leq 1$ and n is a non-negative integer, then the relations (7.84) and the inequalities (7.85) give

$$\begin{aligned} 0 \leq |x| - P_n(x) &= \left[1 - \frac{|x| + P_{n-1}(x)}{2} \right] \times \left[1 - \frac{|x| + P_{n-2}(x)}{2} \right] \times \cdots \times \left(1 - \frac{|x|}{2} \right) |x| \\ &\leq |x| \left(1 - \frac{|x|}{2} \right)^n \\ &< \frac{2}{n+1}. \end{aligned}$$

Given that $\epsilon > 0$. There exists an integer $N \geq \frac{2}{\epsilon} - 1$ such that for $n \geq N$, we have

$$||x| - P_n(x)| < \frac{2}{n+1} \leq \frac{2}{N+1} \leq \epsilon$$

for all $x \in [-1, 1]$. By Definition 7.7, we have

$$\lim_{n \rightarrow \infty} P_n(x) = |x|,$$

uniformly on $[-1, 1]$, completing our proof of the problem. ■

7.4 Isometric mappings and initial-value problems

Problem 7.24

Rudin Chapter 7 Exercise 24.

Proof. If $x \in X$, then it deduces from Definition 2.15(c) that $f_p(x) = d(x, p) - d(x, a) \leq d(a, p)$ which gives the desired result that $|f_p(x)| \leq d(a, p)$. This proves our first assertion.

For the second assertion, since a and p are fixed, we note from the first assertion that f_p is bounded on X . Next, let $q \in X$ and $\epsilon > 0$. We take $\delta = \frac{\epsilon}{2}$. Thus if $x \in X$ and $d(x, q) < \delta$, then the triangle inequality implies that

$$\begin{aligned} |f_p(x) - f_p(q)| &\leq |d(x, p) - d(q, p)| + |d(q, a) - d(x, a)| \\ &\leq d(x, q) + d(q, x) \\ &= 2d(x, q) \\ &< 2\delta \\ &= \epsilon. \end{aligned}$$

By Definition 4.5, f_p is continuous at q and thus on X . By Definition 7.14, we have $f_p \in \mathcal{C}(X)$.

For all $p, q, x \in X$, we have $|f_p(x) - f_q(x)| = |d(x, p) - d(x, q)| \leq d(p, q)$ and the equality holds if $x = p$ or $x = q$. Therefore, we have

$$\|f_p - f_q\| = \sup_{x \in X} |f_p(x) - f_q(x)| = d(p, q).$$

This proves the third assertion.

If we define $\Phi : X \rightarrow \mathcal{C}(X)$ by $\Phi(p) = f_p$, then it follows from the third assertion that Φ is an isometry of X onto $\Phi(X)$. Suppose that $Y = \overline{\Phi(X)}$. By Theorem 2.27(a), Y is closed in $\mathcal{C}(X)$. By the paragraph following Definition 3.12, we know that every closed subset E of a complete metric space X is also complete. Thus Y must be complete which completes the proof of the fourth assertion. We end the proof of the problem. ■

Problem 7.25*Rudin Chapter 7 Exercise 25.**Proof.* We follow the given hint.

(a) We prove the assertions as follows:

- It is clear that $0 = x_0 < x_1 < \cdots < x_n = 1$. If $t = x_i$ for some $0 \leq i \leq n$, then we have $\Delta_n(x_i) = 0$ which means that

$$f'_n(x_i) = \phi(x_i, f_n(x_i)).$$

Since $|\phi(x, y)| \leq M$ for all $x \in [0, 1]$ and $y \in \mathbb{R}$, we have $|f'_n(x_i)| \leq M$ for all $i = 0, 1, 2, \dots, n$. If $t \neq x_i$ for all $i = 0, 1, 2, \dots, n$, then $t \in (x_i, x_{i+1})$ for some $i = 0, 1, \dots, n-1$. In this case, we have

$$|f'_n(t)| = |\phi(x_i, f_n(x_i))| \leq M.$$

In conclusion, we have

$$|f'_n(t)| \leq M$$

for all $t \in [0, 1]$.

- Next the second assertion $|\Delta_n(t)| \leq 2M$, where $t \in [0, 1]$, follows immediately from the first assertion and the hypothesis that $|\phi(x, y)| \leq M$ for all $x \in [0, 1]$ and $y \in \mathbb{R}$.
- Since ϕ is a continuous bounded real function on $[0, 1] \times \mathbb{R}$ and

$$f'_n(t) = \phi(x_i, f_n(x_i))$$

if $x_i < t < x_{i+1}$, the function f'_n is bounded and continuous on $[0, 1] \setminus \{x_0, x_1, \dots, x_n\}$. By the definition of Δ_n , we know that Δ_n is bounded on $[0, 1]$ and continuous on $[0, 1] \setminus \{x_0, x_1, \dots, x_n\}$. By Theorem 6.10, we have $\Delta_n \in \mathcal{R}$ on $[0, 1]$.

- Since $\phi(t, f_n(t)) + \Delta_n(t) = f'_n(t)$ on $[0, 1] \setminus \{x_0, x_1, \dots, x_n\}$, we have for every $x \in [0, 1]$,

$$|f_n(x)| = \left| c + \int_0^x [\phi(t, f_n(t)) + \Delta_n(t)] dt \right| \leq |c| + \int_0^x |f'_n(t)| dt \leq |c| + Mx \leq |c| + M.$$

- (b) Given that $\epsilon > 0$. Let $\delta = \frac{\epsilon}{M} > 0$. Now for all $x, y \in [0, 1]$ with $|x - y| < \delta$, since $|f'_n| \leq M$ on $[0, 1]$, we obtain from Theorem 6.12(c) that

$$|f_n(x) - f_n(y)| = \left| \int_0^x f'_n(t) dt - \int_0^y f'_n(t) dt \right| = \left| \int_y^x f'_n(t) dt \right| \leq M|x - y| < M\delta = \epsilon,$$

where $n = 1, 2, \dots$ [§] By Definition 7.22, $\{f_n\}$ is equicontinuous on $[0, 1]$.

- (c) By part (a), $\{f_n\}$ is uniformly bounded (and thus pointwise bounded) on $[0, 1]$. Since $\{f_n\}$ is equicontinuous on $[0, 1]$, Theorem 7.25(b) implies that it contains a uniformly convergent subsequence on $[0, 1]$.

- (d) By Definition 2.17, the rectangle

$$K = \{(x, y) \mid 0 \leq x \leq 1, |y| \leq M_1\}$$

is a 2-cell. Then it follows from Theorem 2.40 that K is compact. Since ϕ is continuous on K , we deduce from Theorem 4.19 that ϕ is uniformly continuous on K . Thus, given $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|\phi(x, y) - \phi(x', y')| < \epsilon \tag{7.86}$$

[§]Here we assume that $x > y$. If $x < y$, then we have the integral $\int_x^y f'_n(t) dt$ instead of $\int_y^x f'_n(t) dt$.

for all $(x, y), (x', y') \in K$ with $\sqrt{(x - x')^2 + (y - y')^2} < \delta$. In particular, we take $x = x' = t \in [0, 1]$ in the inequality (7.86) so that

$$|\phi(t, y) - \phi(t, y')| < \epsilon \quad (7.87)$$

for all $|y| \leq M_1$ and $|y'| \leq M_1$ with $|y - y'| < \delta$. By part (c) with the fixed δ , there exists an integer N such that $n_k \geq N$ implies that

$$|f_{n_k}(t) - f(t)| < \delta \quad (7.88)$$

for all $t \in [0, 1]$. If we put $y = f_{n_k}(t)$ and $y' = f(t)$, then the inequality (7.88) and part (a) show that y and y' satisfy the hypotheses $|y| \leq M_1, |y'| \leq M_1$ and $|y - y'| < \delta$ as required in (7.87). Therefore, for $n_k \geq N$, we have

$$|\phi(t, f_{n_k}(t)) - \phi(t, f(t))| < \epsilon$$

for all $t \in [0, 1]$. Hence, by Definition 7.7, the sequence $\{\phi(t, f_{n_k}(t))\}$ converges uniformly to $\phi(t, f(t))$ on $[0, 1]$.

(e) Our goal is to show that for every $\epsilon > 0$, there exists an integer N such that $n \geq N$ implies that

$$|\Delta_n(t)| \leq \epsilon \quad (7.89)$$

for all $t \in [0, 1]$.

Since $f'_n(t) = \phi(x_i, f_n(x_i))$ on (x_i, x_{i+1}) and $\Delta_n(t) = f'_n(t) - \phi(t, f_n(t))$, we have

$$\Delta_n(t) = \phi(x_i, f_n(x_i)) - \phi(t, f_n(t))$$

for all $t \in (x_i, x_{i+1})$, where $i = 0, 1, \dots, n-1$.

For every $t \in [0, 1]$, we know that $t \in [x_i, x_{i+1}]$ for some $0 \leq i \leq n-1$ so that

$$|t - x_i| \leq \frac{1}{n}.^{\text{h}}$$

Recall that ϕ is uniformly continuous on K . In particular, if we put $x = x_i, x' = t, y = f_n(x_i)$ and $y' = f_n(t)$ into the inequality (7.86), then *there is* a $\delta > 0$ such that

$$|\phi(x_i, f_n(x_i)) - \phi(t, f_n(t))| < \frac{\epsilon}{2} \quad (7.90)$$

for all $t \in [0, 1]$ with

$$\sqrt{(x_i - t)^2 + [f_n(x_i) - f_n(t)]^2} < \delta. \quad (7.91)$$

Now it is easy to see that the inequality (7.89) follows immediately from the inequality (7.90) if we can find an integer N such that the inequality (7.91) holds for all $n \geq N$.ⁱ

By part (b), since $\{f_n\}$ is equicontinuous on $[0, 1]$, for the *fixed* $\delta > 0$, there is a $\eta > 0$ such that

$$|f_n(x) - f_n(t)| < \frac{\delta}{\sqrt{2}} \quad (7.92)$$

for all $x, t \in [0, 1]$ with $|x - t| < \eta$ and $n = 1, 2, \dots$. We take N to be an integer such that $N > \max(\frac{1}{\eta}, \frac{\sqrt{2}}{\delta})$. Then for all $n \geq N$, we get

$$|t - x_i| \leq \frac{1}{n} \leq \frac{1}{N} < \eta \quad \text{and} \quad |t - x_i| < \frac{\delta}{\sqrt{2}}. \quad (7.93)$$

Therefore, for $n \geq N$, we follow from the inequalities (7.92) and (7.93) that

$$\sqrt{(x_i - t)^2 + [f_n(x_i) - f_n(t)]^2} < \sqrt{\frac{\delta^2}{2} + \frac{\delta^2}{2}} = \delta$$

which is exactly (7.91). Hence we obtain our desired inequality (7.89) and then

$$\Delta_n(t) \rightarrow 0$$

uniformly on $[0, 1]$.

^hThis means that $x_i \rightarrow t$ as $n \rightarrow \infty$.

ⁱThis corresponds a partition such that every t must be a point in some interval with length less than or equal to $\frac{1}{N}$.

(f) Combining parts (d) and (e), we have

$$\phi(t, f_{n_k}(t)) + \Delta_{n_k}(t) \rightarrow \phi(t, f(t))$$

uniformly on $[0, 1]$. Hence we establish from Theorem 7.16 that

$$\begin{aligned} f(x) &= \lim_{k \rightarrow \infty} f_{n_k}(x) \\ &= \lim_{k \rightarrow \infty} \left(c + \int_0^x [\phi(t, f_{n_k}(t)) + \Delta_{n_k}(t)] dt \right) \\ &= c + \int_0^x \lim_{k \rightarrow \infty} [\phi(t, f_{n_k}(t)) + \Delta_{n_k}(t)] dt \\ &= c + \int_0^x \phi(t, f(t)) dt, \end{aligned} \tag{7.94}$$

where $x \in [0, 1]$. It is clear that $f(0) = c$. Since $\phi(t, f(t))$ is continuous on $[0, x]$, Theorem 6.20 (First Fundamental Theorem of Calculus) implies that the integral on the right-hand side in the expression (7.94) is differentiable on $[0, x]$ and

$$f'(x) = \frac{d}{dx} \left(\int_0^x \phi(t, f(t)) dt \right) = \phi(x, f(x)).$$

Hence the function f is a solution of the given problem.

This completes the proof of the problem. ■

Problem 7.26

Rudin Chapter 7 Exercise 26.

Proof. Basically, we follow the setting of the proof of Problem 7.25. Fix n . For $i = 0, 1, \dots, n$, put $x_i = \frac{i}{n}$. Let $\mathbf{f}_n$ be a continuous mapping on $[0, 1]$ such that $\mathbf{f}_n(0) = \mathbf{c}$,

$$\mathbf{f}'_n(t) = \Phi(x_i, \mathbf{f}_n(x_i)) \quad \text{if } x_i < t < x_{i+1},$$

and put

$$\Delta_n(t) = \mathbf{f}'_n(t) - \Phi(t, \mathbf{f}_n(t)),$$

except at the points x_i , where $\Delta_n(t) = \mathbf{0}$. Then

$$\mathbf{f}_n(x) = \mathbf{c} + \int_0^x [\Phi(t, \mathbf{f}_n(t)) + \Delta_n(t)] dt.$$

Choose $M < \infty$ so that $|\Phi| \leq M$.

(a) Then by using similar argument as in the proof of Problem 7.25(a), we have

$$|\mathbf{f}'_n| \leq M, |\Delta_n| \leq M \quad \text{and} \quad |f_n| \leq |\mathbf{c}| + M = M_1$$

on $[0, 1]$ for all positive integers n . For $\Delta_n \in \mathcal{R}$ on $[0, 1]$, we note that Δ_n is continuous on $[0, 1] \setminus \{x_0, x_1, \dots, x_n\}$. If we write

$$\Delta_n = (\Delta_{n1}, \dots, \Delta_{nk}),$$

then it follows from Theorem 4.10(a) that each $\Delta_{nj} : [0, 1] \rightarrow \mathbb{R}$ is continuous on $[0, 1] \setminus \{x_0, x_1, \dots, x_n\}$ so that $\Delta_{nj} \in \mathcal{R}$ on $[0, 1]$ by Theorem 6.10, where $j = 1, 2, \dots, k$. Thus we have $\Delta_n \in \mathcal{R}$ on $[0, 1]$ by Definition 6.23.

(b) By applying similar argument as in the proof of Problem 7.25(b), we have $\{\mathbf{f}_n\}$ is equicontinuous on $[0, 1]$.

- (c) By the vector-valued version of Theorem 7.25 (see Problem 7.17), we have the result that a subsequence $\{\mathbf{f}_{n_k}\}$ converges to some $\mathbf{f}$ uniformly on $[0, 1]$.
- (d) Similarly, the argument in the proof of Problem 7.25(d) can be repeated to show that

$$\Phi(t, \mathbf{f}_{n_j}(t)) \rightarrow \Phi(t, \mathbf{f}(t))$$

uniformly on $[0, 1]$.

- (e) It can be shown, by an argument similar to the proof of Problem 7.25(e), that

$$\Delta_n(t) \rightarrow 0$$

uniformly on $[0, 1]$. As a remark, we note that the inequality (7.91) is replaced by

$$\sqrt{(x_i - t)^2 + |\mathbf{f}_n(x_i) - \mathbf{f}_n(t)|^2} < \delta.$$

- (f) Suppose that

$$\begin{aligned}\Phi(t, \mathbf{f}_{n_j}(t)) &= (\phi_1(t, \mathbf{f}_{n_j}(t)), \dots, \phi_k(t, \mathbf{f}_{n_j}(t))), \\ \Phi(t, \mathbf{f}(t)) &= (\phi_1(t, \mathbf{f}(t)), \dots, \phi_k(t, \mathbf{f}(t))), \\ \Delta_{n_j}(t) &= (\Delta_{n_j1}(t), \dots, \Delta_{n_jk}(t)).\end{aligned}$$

By using part (d), part (e) and the proof of Theorem 4.10 on [21, p. 88], we see that

$$\Phi(t, \mathbf{f}_{n_j}(t)) + \Delta_{n_j}(t) \rightarrow \Phi(t, \mathbf{f}(t))$$

uniformly on $[0, 1]$ if and only if

$$\phi_i(t, \mathbf{f}_{n_j}(t)) + \Delta_{n_ji}(t) \rightarrow \phi_i(t, \mathbf{f}(t)) \quad (7.95)$$

uniformly on $[0, 1]$, where $i = 1, 2, \dots, k$. Applying Theorem 7.16 to each equation in (7.95), we get

$$f_i(x) = c_i + \int_0^x \phi_i(t, \mathbf{f}(t)) dt,$$

where $\mathbf{f} = (f_1, \dots, f_k)$, $\mathbf{c} = (c_1, \dots, c_k)$, $c_i \in [0, 1]$ and $i = 1, 2, \dots, k$. Hence these give

$$\mathbf{f}(x) = \mathbf{c} + \int_0^x \Phi(t, \mathbf{f}(t)) dt$$

and it is easily shown that it is a solution of the given initial-value problem.

Hence, we have completed the proof of the problem. ■

Some Special Functions

8.1 Problems related to special functions

Problem 8.1

Rudin Chapter 8 Exercise 1.

Proof. We complete the proof by proving two claims:

- **Claim 1:** If $x \neq 0$, then we have

$$f^{(n)}(x) = \frac{p(x)}{q(x)} e^{-\frac{1}{x^2}}$$

for every positive integer n , where $p(x)$ and $q(x)$ are polynomials with $q(x) \neq 0$.^a We prove the claim by induction. For $x \neq 0$, since

$$f'(x) = \frac{2}{x^3} e^{-\frac{1}{x^2}},$$

the statement is true for $n = 1$. Assume that the statement is true for $n = k$, where k is a positive integer, i.e.,

$$f^{(k)}(x) = \frac{p(x)}{q(x)} e^{-\frac{1}{x^2}}, \quad (8.1)$$

where $x \neq 0$, $p(x)$ and $q(x)$ are polynomials with $q(x) \neq 0$. Now, for $x \neq 0$, it follows from the assumption (8.1), Theorem 5.3(b) and (c) that

$$\begin{aligned} f^{(k+1)}(x) &= \left[\frac{2}{x^3} \cdot \frac{p(x)}{q(x)} + \frac{d}{dx} \left(\frac{p(x)}{q(x)} \right) \right] e^{-\frac{1}{x^2}} \\ &= \left[\frac{2}{x^3} \cdot \frac{p(x)}{q(x)} + \frac{q(x)p'(x) - q'(x)p(x)}{q^2(x)} \right] e^{-\frac{1}{x^2}} \\ &= \frac{2p(x)q(x) + x^3[q(x)p'(x) - q'(x)p(x)]}{x^3q^2(x)} e^{-\frac{1}{x^2}}. \end{aligned} \quad (8.2)$$

Since $q(x) \neq 0$, $q^2(x) \neq 0$ too. Thus it is also true for $n = k + 1$ if it is true for $n = k$. By induction, the statement is true for all positive integers n .

- **Claim 2:** Suppose that $p(x)$ and $q(x)$ are polynomials with $q(x) \neq 0$. Then we have

$$\lim_{x \rightarrow 0} \frac{p(x)}{q(x)} e^{-\frac{1}{x^2}} = 0. \quad (8.3)$$

^aThe degrees of $p(x)$ and $q(x)$ are not important in the proof.

Let $p(x) = p_0x^s + p_1x^{s-1} + \cdots + p_s$ and $q(x) = q_0x^t + q_1x^{t-1} + \cdots + q_t$, where $p_0, p_1, \dots, p_s, q_0, q_1, \dots, q_t$ are all real and $p_0q_0 \neq 0$. If $x = \frac{1}{y}$, then the left-hand side of the limit (8.3) can be rewritten as

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{p(x)}{q(x)} e^{-\frac{1}{x^2}} &= \lim_{y \rightarrow +\infty} \frac{p(\frac{1}{y})}{q(\frac{1}{y})} e^{-y^2} \\ &= \lim_{y \rightarrow +\infty} \frac{\frac{p_0}{y^s} + \frac{p_1}{y^{s-1}} + \cdots + p_s}{\frac{q_0}{y^t} + \frac{q_1}{y^{t-1}} + \cdots + q_t} e^{-y^2} \\ &= \lim_{y \rightarrow +\infty} \frac{p_0 + p_1y + \cdots + p_sy^s}{q_0 + q_1y + \cdots + q_ty^t} y^{t-s} e^{-y^2}. \end{aligned} \quad (8.4)$$

Now it is clear that

$$\lim_{y \rightarrow +\infty} \frac{1}{q_0 + q_1y + \cdots + q_ty^t} = 0. \quad (8.5)$$

Furthermore, we know from Theorem 8.6(c) that $e^{-y} > e^{-y^2} > 0$ for large positive y so that

$$0 < y^i e^{-y^2} < y^i e^{-y}$$

for large positive y , where $i = 0, 1, \dots, s$. Therefore, we deduce from Theorem 8.6(f) that

$$\lim_{y \rightarrow +\infty} p_i y^i e^{-y^2} = 0, \quad (8.6)$$

where $i = 0, 1, \dots, s$. Hence we obtain from the limits (8.5) and (8.6) that the right-hand side of the limit (8.4) is 0 and this is exactly our desired result (8.3).

For every positive integer n , it follows from the limit (8.3) that

$$f^{(n)}(0) = \lim_{x \rightarrow 0} \frac{f^{(n-1)}(x) - f^{(n-1)}(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{p(x)}{xq(x)} e^{-\frac{1}{x^2}} = 0.$$

Hence this shows that f has derivatives of all orders at $x = 0$ and $f^{(n)}(0) = 0$ for $n = 1, 2, 3, \dots$. This completes the proof of the problem. ■

Problem 8.2

Rudin Chapter 8 Exercise 2.

Proof. We note that

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{i=1}^{\infty} \left(a_{ii} + \sum_{j=1}^{i-1} 2^{j-i} \right) = \sum_{i=1}^{\infty} \left(-1 + \frac{1}{2} + \cdots + \frac{1}{2^{i-1}} \right) = - \sum_{i=1}^{\infty} \frac{1}{2^{i-1}} = -2$$

and

$$\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \left(a_{jj} + \sum_{i=j+1}^{\infty} 2^{j-i} \right) = \sum_{j=1}^{\infty} (-1 + 1) = 0.$$

This completes the proof of the problem. ■

Problem 8.3

Rudin Chapter 8 Exercise 3.

Proof. Since $a_{ij} \geq 0$ for all i and j , we have $|a_{ij}| = a_{ij}$. For each $i = 1, 2, \dots$, let $\sum_{j=1}^{\infty} a_{ij} = b_i$. There are two cases for consideration.

Case (i): $\sum b_i$ converges. Then it follows from Theorem 8.3 that we have

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}. \quad (8.7)$$

We note that the double series in the expression (8.7) are finite.

Case (ii): $\sum b_i$ diverges. Let

$$s_n = \sum_{i=1}^n b_i.$$

We claim that $\lim_{n \rightarrow \infty} s_n = +\infty$. Otherwise, $\{s_n\}$ is bounded. Since $a_{ij} \geq 0$, we have $b_i \geq 0$ and thus the definition implies that $\{s_n\}$ is a monotonic increasing sequence. By Theorem 3.14 (Monotone Convergence Theorem), the sequence $\{s_n\}$ converges. By definition, $\sum b_i$ also converges which is a contradiction. Therefore, we have

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = +\infty \quad (8.8)$$

in this case.

Next, for each $j = 1, 2, \dots$, we define $\sum_{i=1}^n a_{ij} = c_j$. If $\sum c_j$ converges, then we may apply Theorem 8.3 to obtain the expression (8.7) and both double series are finite. This certainly contradicts the result (8.8) and thus we must have $\sum c_j$ diverges. By an argument in the previous paragraph, we can show that

$$\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij} = +\infty.$$

Hence the expression (8.7) also holds in this subcase.

This finishes the proof of the problem. ■

Problem 8.4

Rudin Chapter 8 Exercise 4.

Proof.

(a) We follow from Theorem 5.13 (L'Hospital's Rule) that, for $b > 0$,

$$\lim_{x \rightarrow 0} \frac{b^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^{x \log b} - 1}{x} = \lim_{x \rightarrow 0} \frac{\log b e^{x \log b}}{1} = \log b.$$

(b) By Theorem 5.13 (L'Hospital's Rule) and [21, Eqn. (38), p. 180], we have

$$\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1} = 1.$$

(c) Let $y = (1+x)^{\frac{1}{x}}$. Then we have

$$y = e^{\frac{\log(1+x)}{x}}.$$

By part (b), we have

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\frac{\log(1+x)}{x}} = \exp\left(\lim_{x \rightarrow 0} \frac{\log(1+x)}{x}\right) = e.$$

(d) Notice that

$$\left(1 + \frac{x}{n}\right)^n = \left[\left(1 + \frac{x}{n}\right)^{\frac{1}{n}}\right]^n.$$

Since $n \rightarrow \infty$ if and only if $\frac{x}{n} \rightarrow 0$, it follows from part (c) that

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = \lim_{\frac{x}{n} \rightarrow 0} \left[\left(1 + \frac{x}{n}\right)^{\frac{1}{n}}\right]^n = \left[\lim_{\frac{x}{n} \rightarrow 0} \left(1 + \frac{x}{n}\right)^{\frac{1}{n}}\right]^x = e^x.$$

We end the proof of the problem. ■

Problem 8.5

Rudin Chapter 8 Exercise 5.

Proof.

(a) If $y = (1+x)^{\frac{1}{x}}$, then we have

$$\log y = \frac{\log(1+x)}{x}.$$

We deduce from [21, Eqn. (38), p.180] and Theorem 5.5 (Chain Rule) that

$$\frac{y'}{y} = \frac{x - (1+x)\log(1+x)}{x^2(1+x)}.$$

By Theorem 5.13 (L'Hospital's Rule) twice, we have

$$\lim_{x \rightarrow 0} \frac{y'}{y} = \lim_{x \rightarrow 0} \frac{x - (1+x)\log(1+x)}{x^2(1+x)} = \lim_{x \rightarrow 0} \frac{-\log(1+x)}{2x + 3x^2} = \lim_{x \rightarrow 0} \frac{-\frac{1}{1+x}}{2 + 6x} = -\frac{1}{2}$$

so that

$$\lim_{x \rightarrow 0} y' = -\frac{1}{2} \lim_{x \rightarrow 0} y = -\frac{e}{2}.$$

Hence Theorem 5.13 (L'Hospital's Rule) implies that

$$\lim_{x \rightarrow 0} \frac{e - (1+x)^{\frac{1}{x}}}{x} = \lim_{x \rightarrow 0} -y' = \frac{e}{2}.$$

(b) Since $n^{\frac{1}{n}} = e^{\frac{\log n}{n}}$, we have

$$\frac{n}{\log n} [n^{\frac{1}{n}} - 1] = \frac{e^{\frac{\log n}{n}} - 1}{\frac{\log n}{n}}. \quad (8.9)$$

Since $n \rightarrow \infty$ if and only if $\frac{\log n}{n} \rightarrow 0$, it follows from the identity (8.9) that

$$\lim_{n \rightarrow \infty} \frac{n}{\log n} [n^{\frac{1}{n}} - 1] = \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x - e^0}{x - 0} = e^0 = 1,$$

where $x = \frac{\log n}{n}$.

(c) By applying Theorem 5.13 (L'Hospital's Rule) three times, we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - x}{x(1 - \cos x)} &= \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{1 - \cos x + x \sin x} \\ &= \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{\sin x + \sin x + x \cos x} \\ &= \lim_{x \rightarrow 0} \frac{2 \sec^3 x \tan x}{2 \tan x + x} \end{aligned}$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{2(3 \sec^3 x \tan^2 x + \sec^5 x)}{2 \sec^2 x + 1} \\
&= \frac{2}{3}.
\end{aligned}$$

(d) By Theorem 5.13 (L'Hospital's Rule), we have

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{\tan x - x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sec^2 x - 1} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{1 - \cos^2 x} \cdot \cos^2 x = \lim_{x \rightarrow 0} \frac{\cos^2 x}{1 + \cos x} = \frac{1}{2}.$$

We complete the proof of the problem. ■

Problem 8.6

Rudin Chapter 8 Exercise 6.

Proof.

(a) Put $x = y = 0$ into the given equation, we have $f^2(0) = f(0)$ which means that

$$f(0) = 0 \quad \text{or} \quad f(0) = 1.$$

Since f is not zero, we have $f(0) = 1$. Since f is differentiable on $\mathbb{R}$, we have^b

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h} = f(x) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h - 0} = cf(x),$$

where $c = f'(0)$. Let $F(x) = e^{-cx}f(x)$. Then we have

$$F'(x) = e^{-cx}f'(x) - ce^{-cx}f(x) = ce^{-cx}f(x) - ce^{-cx}f(x) = 0$$

for every $x \in \mathbb{R}$. By Theorem 5.11(b), we have $F(x) = C$ for some constant C . Since we have $F(0) = f(0) = 1$, $C = 1$ and thus

$$f(x) = e^{cx}$$

as required.

(b) Suppose that we can prove that

$$f(r) = [f(1)]^r, \tag{8.10}$$

where $r \in \mathbb{Q}$.

For any $x \in \mathbb{R}$, since $\mathbb{Q}$ is dense in $\mathbb{R}$ (Theorem 1.20(b)), there exists a sequence $\{r_n\} \subseteq \mathbb{Q}$ such that $r_n \rightarrow x$ as $n \rightarrow \infty$. Since f is continuous on $\mathbb{R}$, this and Theorem 4.2 imply that

$$f(x) = \lim_{n \rightarrow \infty} f(r_n) = \lim_{n \rightarrow \infty} [f(1)]^{r_n} = [f(1)]^x. \tag{8.11}$$

By the hypothesis, we have

$$f(x) = f\left(\frac{x}{2} + \frac{x}{2}\right) = f\left(\frac{x}{2}\right)f\left(\frac{x}{2}\right) = f^2\left(\frac{x}{2}\right) > 0$$

for all real x . In particular, we have $f(1) > 0$ so that $e^{\log f(1)} = f(1)$ by [21, Eqn. (36), p.180]. If we let $c = \log f(1)$, then we deduce from the expression (8.11) that

$$f(x) = e^{cx}.$$

Thus we have transformed our problem into the question of the validity of the formula (8.10).

To this end, we divide the proof into several steps:

^bThis is an equivalent definition of the derivative of f at x (by letting $h = t - x$ in [21, Eqn. (2), p. 103]).

- **Step 1:** $f(n) = [f(1)]^n$ for all $n \in \mathbb{N}$. The result follows from induction and the hypothesis $f(x)f(y) = f(x+y)$, so we skip its proof here.
- **Step 2:** $f(n) = [f(1)]^n$ for all $n \in \mathbb{Z}$. Since $f(-1)f(1) = f(1-1) = f(0) = 1$, we have

$$f(-1) = [f(1)]^{-1}.$$

Assume that $f(-k) = [f(1)]^{-k}$ for some positive integer k . Since

$$f(-k-1)f(1) = f(-k-1+1) = f(-k) = [f(1)]^{-k},$$

we have $f(-(k+1)) = [f(1)]^{-(k+1)}$. By induction, we have the claim.

- **Step 3: The formula (8.10) holds.** Suppose that $r = \frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. It is clear that

$$f\left(\frac{p}{q}\right) = f\left(\underbrace{\frac{1}{q} + \cdots + \frac{1}{q}}_{p \text{ terms}}\right) = f\left(\underbrace{\frac{1}{q} + \cdots + \frac{1}{q}}_{(p-1) \text{ terms}}\right) \times f\left(\frac{1}{q}\right) = \left[f\left(\frac{1}{q}\right)\right]^p. \quad (8.12)$$

Furthermore, since

$$\left[f\left(\frac{1}{q}\right)\right]^q = f\left(\frac{1}{q}\right) \times f\left(\frac{1}{q}\right) \times \cdots \times f\left(\frac{1}{q}\right) = f\left(\frac{1}{q} + \frac{1}{q} + \cdots + \frac{1}{q}\right) = f(1),$$

we have

$$f\left(\frac{1}{q}\right) = [f(1)]^{\frac{1}{q}}. \quad (8.13)$$

Combining the expressions (8.12) and (8.13), we get

$$f\left(\frac{p}{q}\right) = [f(1)]^{\frac{p}{q}}$$

which is exactly formula (8.10).

Hence we have completed the proof of our problem. ■

Problem 8.7

Rudin Chapter 8 Exercise 7.

Proof. We consider the function $f(x) = \frac{\sin x}{x}$ defined on $(0, \frac{\pi}{2})$. Then we have

$$f'(x) = \frac{x \cos x - \sin x}{x^2}.$$

Let $g(x) = x \cos x - \sin x$ be a function defined on $(0, \frac{\pi}{2})$. Since

$$g'(x) = -x \sin x + \cos x - \cos x = -x \sin x < 0 \quad (8.14)$$

on $(0, \frac{\pi}{2})$, g is strictly decreasing on $(0, \frac{\pi}{2})$ by Problem 5.2. Since g is continuous on $[0, \frac{\pi}{2}]$, it is also strictly decreasing on $[0, \frac{\pi}{2}]$.^c Since $g(0) = 0$, we always have $g(x) < 0$ on $(0, \frac{\pi}{2})$ and thus $f'(x) < 0$ on $(0, \frac{\pi}{2})$. Therefore, f is strictly decreasing on $(0, \frac{\pi}{2})$. By the continuity of f , it yields that f is also strictly decreasing on $(0, \frac{\pi}{2}]$. Hence we have $f(x) > f(\frac{\pi}{2})$ which gives

$$\frac{\sin x}{x} > \frac{\sin \frac{\pi}{2}}{\frac{\pi}{2}} = \frac{2}{\pi} \quad (8.15)$$

^cIf $g(0) = g(p)$ for some $p \in (0, \frac{\pi}{2})$, then Theorem 5.10 (Mean Value Theorem) implies that $g'(x) = 0$ for some $x \in (0, p) \subseteq (0, \frac{\pi}{2})$ which contradicts the inequality (8.14).

for every $x \in (0, \frac{\pi}{2})$.

For the second inequality, we consider another function $h(x) = \sin x - x$ on $(0, \frac{\pi}{2})$. Now we have

$$h'(x) = \cos x - 1 < 0$$

on $(0, \frac{\pi}{2})$ so that h is strictly decreasing on $(0, \frac{\pi}{2})$ by Problem 5.2. By the continuity of h , we know that h is also strictly decreasing on $[0, \frac{\pi}{2}]$ and this implies that $h(x) < h(0)$ which is equivalent to

$$\frac{\sin x}{x} < 1 \quad (8.16)$$

for every $x \in (0, \frac{\pi}{2})$. Hence our desired result follows immediately by combining the inequalities (8.15) and (8.16), completing the proof of the problem. ■

Problem 8.8

Rudin Chapter 8 Exercise 8.

Proof. The inequality holds obviously if $n = 0$. Assume that

$$|\sin k| \leq k |\sin x|$$

for some non-negative integer k . If $n = k + 1$, then, by basic properties of trigonometric functions, we have

$$\begin{aligned} |\sin(k+1)x| &= |\sin kx \cos x + \cos kx \sin x| \\ &\leq |\sin kx| |\cos x| + |\cos kx| |\sin x| \\ &\leq |\sin kx| + |\sin x| \\ &\leq k |\sin x| + |\sin x| \\ &= (k+1) |\sin x|. \end{aligned}$$

Hence the inequality also holds in this case. By induction, the inequality holds for all non-negative integers n . This completes the proof of the problem. ■

Problem 8.9

Rudin Chapter 8 Exercise 9.

Proof.

(a) Let $\{p_N\} = \{s_N - \log N\}$, where $N \geq 1$. Then we follow from [21, Eqn. (39), p.180] that

$$p_{N+1} - p_N = s_{N+1} - \log(N+1) - s_N + \log N = \frac{1}{N+1} - \int_N^{N+1} \frac{dx}{x}. \quad (8.17)$$

On $[N, N+1]$, we have $\frac{1}{N+1} \leq \frac{1}{x}$ so that the expression (8.17) yields

$$p_{N+1} - p_N \leq \frac{1}{N+1} - \int_N^{N+1} \frac{dx}{N+1} = \frac{1}{N+1} - \frac{1}{N+1} = 0.$$

Thus the sequence $\{p_N\}$ is monotonic decreasing. Furthermore, we have

$$s_N = \frac{1}{N} + \sum_{k=1}^{N-1} \int_k^{k+1} \frac{dx}{k} \geq \frac{1}{N} + \sum_{k=1}^{N-1} \int_k^{k+1} \frac{dx}{x} = \frac{1}{N} + \int_1^N \frac{dx}{x} = \frac{1}{N} + \log N \quad (8.18)$$

so that

$$p_N = s_N - \log N \geq \frac{1}{N} > 0.$$

In other words, $\{p_N\}$ is bounded below. By Theorem 3.14 (Monotone Convergence Theorem), we have

$$\lim_{N \rightarrow \infty} p_N = \lim_{N \rightarrow \infty} (s_N - \log N)$$

converges.

(b) When $N = 10^m$, the inequality (8.18) gives^d

$$s_N = s_{10^m} > \log 10^m = \frac{\log_{10} 10^m}{\log_{10} e} = \frac{m}{\log_{10} e}.$$

Therefore, if $m \geq 44$, then we have

$$\frac{m}{\log_{10} e} \geq \frac{44}{0.434294481} \approx 101.3137441 > 100.$$

Hence the value of m is at least 44 to make sure that $s_N > 100$.

This ends the analysis of the problem. ■

Problem 8.10

Rudin Chapter 8 Exercise 10.

Proof. We follow the hint. Let $p_1, \dots, p_k$ be those primes that divide at least one integer $\leq N$. Then $p_1, \dots, p_k$ are all primes less than or equal to N and *each* integer $\leq N$ must have the form

$$p_1^{m_1} p_2^{m_2} \cdots p_k^{m_k}$$

for some non-negative integers $m_1, m_2, \dots, m_k$. This fact allows us to have the inequality

$$\sum_{n=1}^N \frac{1}{n} \leq \prod_{j=1}^k \left(1 + \frac{1}{p_j} + \frac{1}{p_j^2} + \cdots\right). \quad (8.19)$$

Since $p_j \geq 2$, it follows from Theorem 3.26 that

$$1 + \frac{1}{p_j} + \frac{1}{p_j^2} + \cdots = \frac{1}{1 - \frac{1}{p_j}}.$$

By [21, Eqn. (25), p.178], we know that

$$e^{2x} \geq 1 + 2x$$

for all $x \geq 0$ so that

$$(1 - x)e^{2x} \geq (1 - x)(1 + 2x) = 1 + x(1 - 2x) \geq 1 \quad (8.20)$$

on $[0, \frac{1}{2}]$. Therefore, combining inequalities (8.19) and (8.20), we have

$$\sum_{n=1}^N \frac{1}{n} \leq \prod_{j=1}^k \left(1 - \frac{1}{p_j}\right)^{-1} \leq \prod_{j=1}^k e^{\frac{2}{p_j}} = \exp\left(2 \sum_{j=1}^k \frac{1}{p_j}\right). \quad (8.21)$$

Since $N \rightarrow \infty$ if and only if $k \rightarrow \infty$, we deduce immediately from the inequality (8.21), Theorem 3.25(b) (Comparison Test) and Theorem 3.28 that

$$\sum_{\text{all primes } p} \frac{1}{p}$$

diverges. This completes the proof of the problem. ■

^dWe have applied the change of base formula for logarithmic in the second equality.

Problem 8.11*Rudin Chapter 8 Exercise 11.*

Proof. We prove the result by considering two steps:

- **Step 1: The improper integral in the question is well-defined.** Since $f(x) \rightarrow 1$ as $x \rightarrow \infty$, for every $\epsilon > 0$, there exists an integer $N > 0$ such that

$$0 < 1 - \frac{\epsilon}{4} < f(x) < 1 + \frac{\epsilon}{4} \quad (8.22)$$

for all $x > N$. By the inequalities (8.22), we have

$$\begin{aligned} 0 &< \int_N^n \left(1 - \frac{\epsilon}{4}\right) e^{-tx} dx < \int_N^n e^{-tx} f(x) dx < \int_N^n \left(1 + \frac{\epsilon}{4}\right) e^{-tx} dx \\ 0 &< \left(1 - \frac{\epsilon}{4}\right) \left(\frac{e^{-nt}}{-t} + \frac{e^{-Nt}}{t}\right) < \int_N^n e^{-tx} f(x) dx < \left(1 + \frac{\epsilon}{4}\right) \left(\frac{e^{-nt}}{-t} + \frac{e^{-Nt}}{t}\right). \end{aligned} \quad (8.23)$$

By the most left-hand inequality in (8.23) the sequence

$$\left\{ \int_N^n e^{-tx} f(x) dx \right\}$$

is increasing monotonically. Fix $t > 0$. By the right-hand side inequality in (8.23), we know that

$$\int_N^n e^{-tx} f(x) dx < \left(1 + \frac{\epsilon}{4}\right) \frac{e^{-Nt}}{t}$$

so that the sequence is bounded above. By Theorem 3.14 (Monotone Convergence Theorem), the limit

$$\lim_{n \rightarrow \infty} \int_N^n e^{-tx} f(x) dx$$

exists. Since $f(x), e^{-tx} \in \mathcal{R}$ on $[0, N]$, Theorem 6.13(a) implies that $e^{-tx} f(x) \in \mathcal{R}$ on $[0, N]$. Hence the limit

$$\begin{aligned} \lim_{n \rightarrow \infty} \int_0^n e^{-tx} f(x) dx &= \lim_{n \rightarrow \infty} \left(\int_0^N e^{-tx} f(x) dx + \int_N^n e^{-tx} f(x) dx \right) \\ &= \int_0^N e^{-tx} f(x) dx + \lim_{n \rightarrow \infty} \int_N^n e^{-tx} f(x) dx \end{aligned}$$

certainly exists.

- **Step 2: The limit holds.** Next, we are going to show the desired limit holds. When $n \rightarrow \infty$, the inequalities (8.23) become

$$\left(1 - \frac{\epsilon}{4}\right) e^{-Nt} < t \int_N^\infty e^{-tx} f(x) dx < \left(1 + \frac{\epsilon}{4}\right) e^{-Nt} \quad (8.24)$$

for all $t > 0$. Since $f \in \mathcal{R}$ on $[0, N]$, f is clearly bounded on $[0, N]$. In other words, there is a positive number M such that

$$-M \leq f(x) \leq M$$

for all $x \in [0, N]$. By this, we have

$$-Mt \int_0^N e^{-tx} dx \leq t \int_0^N e^{-tx} f(x) dx \leq Mt \int_0^N e^{-tx} dx$$

$$-M(1 - e^{-Nt}) \leq t \int_0^N e^{-tx} f(x) dx \leq M(1 - e^{-Nt}). \quad (8.25)$$

We note from Theorem 8.6(c) that e^x is a strictly increasing function of x and $e^x \geq 1$ for all $x \geq 0$, so we have $e^{-Nt} \leq 1$ for all $t \geq 0$ and there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$-\frac{\epsilon}{4M} < 1 - e^{-Nt} < \frac{\epsilon}{4M} \quad (8.26)$$

for $0 < t < \delta_1$ and

$$-\frac{\epsilon}{4} < 1 - e^{-Nt} < \frac{\epsilon}{4} \quad (8.27)$$

for $0 < t < \delta_2$ respectively. By inequalities (8.26), the inequalities (8.25) imply that

$$-\frac{\epsilon}{4} < t \int_0^N e^{-tx} f(x) dx < \frac{\epsilon}{4} \quad (8.28)$$

for $0 < t < \delta_1$. Similarly, we apply the inequalities (8.27) to the inequalities (8.24) to get

$$\left(1 - \frac{\epsilon}{4}\right)^2 < t \int_N^\infty e^{-tx} f(x) dx < \left(1 + \frac{\epsilon}{4}\right)^2 \quad (8.29)$$

for all $0 < t < \delta_2$. Rewrite the inequalities (8.29) as

$$1 - \frac{\epsilon}{2} + \frac{\epsilon^2}{16} < t \int_N^\infty e^{-tx} f(x) dx < 1 + \frac{\epsilon}{2} + \frac{\epsilon^2}{16} \quad (8.30)$$

for all $0 < t < \delta_2$.

Therefore, if we take $\delta = \min(\delta_1, \delta_2)$, then both the inequalities (8.26) and (8.27) hold for $0 < t < \delta$ which imply that we can add the inequalities (8.28) and (8.30) together to get

$$\begin{aligned} 1 - \frac{3\epsilon}{4} + \frac{\epsilon^2}{16} &< t \int_0^\infty e^{-tx} f(x) dx < 1 + \frac{3\epsilon}{4} + \frac{\epsilon^2}{16} \\ -\epsilon &< -\frac{3\epsilon}{4} + \frac{\epsilon^2}{16} < t \int_0^\infty e^{-tx} f(x) dx - 1 < \frac{3\epsilon}{4} + \frac{\epsilon^2}{16} < \epsilon \end{aligned} \quad (8.31)$$

for $0 < t < \delta$. Hence we deduce from the inequality (8.31) that

$$\lim_{t \rightarrow 0} t \int_0^\infty e^{-tx} f(x) dx = 1 \quad (t > 0).$$

This completes the proof of the problem. ■

Problem 8.12

Rudin Chapter 8 Exercise 12.

Proof.

(a) It is clear that $f \in \mathcal{R}$ on $[-\pi, \pi]$. By Definition 8.9 and [21, Eqn. (46), p. 182], we have

$$c_m = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-imx} dx = \frac{1}{2\pi} \int_{-\delta}^{\delta} e^{-imx} dx = \frac{i}{2m\pi} (e^{-im\delta} - e^{im\delta}) = \frac{\sin m\delta}{m\pi} \quad (8.32)$$

for all non-zero integers m . If $m = 0$, then we have

$$c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \int_{-\delta}^{\delta} f(x) dx = \frac{\delta}{\pi}. \quad (8.33)$$

(b) Since

$$|f(0+t) - f(0)| = |f(t) - f(0)| = 1 - 1 \leq |t|$$

for all $t \in (-\delta, \delta)$, it follows from Theorem 8.14 that

$$f(0) = \lim_{N \rightarrow \infty} s_N(f; 0). \quad (8.34)$$

By the formulas (8.32) and (8.33), we have

$$s_N(f; 0) = \sum_{-N}^N c_n = c_0 + \sum_{-N}^1 c_n + \sum_1^N c_n = \frac{\delta}{\pi} + \sum_{n=1}^N \frac{2 \sin n\delta}{n\pi}.$$

Therefore, the limit (8.34) implies that

$$\begin{aligned} 1 = f(0) &= \frac{\delta}{\pi} + \sum_{n=1}^{\infty} \frac{2 \sin n\delta}{n\pi} \\ \sum_{n=1}^{\infty} \frac{2 \sin n\delta}{n\pi} &= \frac{\pi - \delta}{\pi} \\ \sum_{n=1}^{\infty} \frac{\sin n\delta}{n} &= \frac{\pi - \delta}{2} \end{aligned}$$

which is our desired formula.

(c) Since $f \in \mathcal{R}$ on $[-\pi, \pi]$ with period 2π , we deduce from Theorem 8.16 (Parseval's Theorem), the formulas (8.32) and (8.33) that

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x)|^2 dx &= \frac{\delta^2}{\pi^2} + \sum_{n=1}^{\infty} \frac{2 \sin^2(n\delta)}{n^2 \pi^2} \\ \frac{\delta}{\pi} &= \frac{\delta^2}{\pi^2} + \sum_{n=1}^{\infty} \frac{2 \sin^2(n\delta)}{n^2 \pi^2} \\ \frac{\pi\delta - \delta^2}{\pi^2} &= \sum_{n=1}^{\infty} \frac{2 \sin^2(n\delta)}{n^2 \pi^2} \\ \sum_{n=1}^{\infty} \frac{\sin^2(n\delta)}{n^2 \delta} &= \frac{\pi - \delta}{2} \end{aligned} \quad (8.35)$$

as required.

(d) Given that $\epsilon > 0$. Let $b > 0$ be a fixed large number such that

$$\frac{1}{b} < \frac{\epsilon}{3}.$$

We consider $\delta_n = \frac{b}{n}$ and the partition $\{x_0, x_1, \dots, x_n\}$ of $[0, b]^e$ such that $x_j = j\delta_n$ for $0 \leq j \leq n$. Then we have

$$\Delta x_i = x_i - x_{i-1} = i\delta_n - (i-1)\delta_n = \delta_n$$

for $1 \leq i \leq n$. We want to show that there exists an integer N such that

$$\left| \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2 \delta_n} - \int_0^{\infty} \frac{\sin^2 x}{x^2} dx \right| < \epsilon$$

for all $n \geq N$.

^eThus we must have $b < n$.

To this end, suppose that

$$f(x) = \frac{\sin^2 x}{x^2}, \quad M_i = \sup_{x_{i-1} \leq x \leq x_i} f(x) \quad \text{and} \quad m_i = \inf_{x_{i-1} \leq x \leq x_i} f(x),$$

where $i = 1, 2, \dots, n$. Then we have

$$\begin{aligned} L(P, f) &\leq \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n^2} \cdot \delta_n \leq U(P, f) \\ L(P, f) &\leq \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n} \leq U(P, f) \end{aligned} \quad (8.36)$$

Since f is bounded on $[0, b]$ ^f, Theorem 6.10 implies that $f \in \mathcal{R}$ on $[0, b]$. By the inequalities (8.36), we have

$$\int_0^b \frac{\sin^2 x}{x^2} dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n} \quad (8.37)$$

which means that there exists an integer N such that

$$\left| \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n} - \int_0^b \frac{\sin^2 x}{x^2} dx \right| < \frac{\epsilon}{3} \quad (8.38)$$

for all $n \geq N$.

Furthermore, since $\frac{1}{b} < \frac{\epsilon}{3}$, we have

$$\left| \sum_{i=n+1}^{\infty} \frac{\sin^2(i\delta_n)}{i^2\delta_n} \right| \leq \sum_{i=n+1}^{\infty} \left| \frac{\sin^2(i\delta_n)}{i^2\delta_n} \right| \leq \frac{1}{\delta_n} \sum_{i=n+1}^{\infty} \frac{1}{i^2} \leq \frac{1}{\delta_n} \int_n^{\infty} \frac{1}{t^2} dt = \frac{1}{n\delta_n} = \frac{1}{b} < \frac{\epsilon}{3} \quad (8.39)$$

and

$$\left| \int_0^{\infty} \frac{\sin^2 x}{x^2} dx - \int_0^b \frac{\sin^2 x}{x^2} dx \right| = \left| \int_b^{\infty} \frac{\sin^2 x}{x^2} dx \right| \leq \int_b^{\infty} \frac{1}{x^2} dx = \frac{1}{b} < \frac{\epsilon}{3}. \quad (8.40)$$

Thus, for $n \geq N$, combining the inequalities (8.38), (8.39) and (8.40), we have

$$\begin{aligned} \left| \sum_{i=1}^{\infty} \frac{\sin^2(i\delta_n)}{i^2\delta_n} - \int_0^{\infty} \frac{\sin^2 x}{x^2} dx \right| &\leq \left| \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n} - \int_0^b \frac{\sin^2 x}{x^2} dx \right| \\ &\quad + \left| \int_0^b \frac{\sin^2 x}{x^2} dx - \int_b^{\infty} \frac{\sin^2 x}{x^2} dx + \sum_{i=n+1}^{\infty} \frac{\sin^2(i\delta_n)}{i^2\delta_n} \right| \\ &\leq \left| \sum_{i=1}^n \frac{\sin^2(i\delta_n)}{i^2\delta_n} - \int_0^b \frac{\sin^2 x}{x^2} dx \right| + \left| \sum_{i=n+1}^{\infty} \frac{\sin^2(i\delta_n)}{i^2\delta_n} \right| \\ &\quad + \left| \int_0^b \frac{\sin^2 x}{x^2} dx - \int_b^{\infty} \frac{\sin^2 x}{x^2} dx \right| \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon \end{aligned}$$

which is our required inequality. By part (b), we have

$$\left| \frac{\pi - \delta_n}{2} - \int_0^{\infty} \frac{\sin^2 x}{x^2} dx \right| < \epsilon$$

for all $n \geq N$.

^fNote the classical result that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

(e) Put $\delta = \frac{\pi}{2}$ in the formula (8.35), we have

$$\sum_{n=1}^{\infty} \frac{\sin^2(\frac{n\pi}{2})}{n^2} = \frac{\pi^2}{8}. \quad (8.41)$$

Since $\sin^2(\frac{n\pi}{2}) = 1$ if n is odd and $\sin^2(\frac{n\pi}{2}) = 0$ if n is even, we deduce from the formula (8.41) that

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}.$$

We complete the proof of the problem. ■

Problem 8.13

Rudin Chapter 8 Exercise 13.

Proof. Since the f is *not* a function with period 2π , we can't apply Parseval's Theorem directly and we have to find a function based on f which satisfies the hypotheses of Parseval's Theorem. Define the function $F : \mathbb{R} \rightarrow \mathbb{R}$ by

$$F(x) = f(x - 2n\pi) = x - 2n\pi \quad (2n\pi \leq x < (2n+2)\pi),$$

where n is an integer. Now if $2n\pi \leq x < (2n+2)\pi$, then

$$2(n+1)\pi \leq x + 2\pi < [2(n+1) + 2]\pi.$$

This implies that

$$F(x + 2\pi) = f(x + 2\pi - 2(n+1)\pi) = x + 2\pi - 2(n+1)\pi = x - 2n\pi = F(x).$$

Thus F is a function with period 2π . Since F is bounded on $[-\pi, \pi]$ and is discontinuous only at 0, we follow from Theorem 6.10 that $F \in \mathcal{R}$ on $[-\pi, \pi]$. By Definition 8.9 and Theorem 6.22 (Integration by Parts), we have

$$\begin{aligned} c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^0 F(x) e^{-inx} dx + \int_0^{\pi} F(x) e^{-inx} dx \right] \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^0 (x + 2\pi) e^{-inx} dx + \int_0^{\pi} x e^{-inx} dx \right] \\ &= \frac{1}{2\pi} \left[2\pi \int_{-\pi}^0 e^{-inx} dx + \int_{-\pi}^{\pi} x e^{-inx} dx \right] \\ &= \frac{1}{2\pi} \left[2\pi \frac{e^{-inx}}{-in} \Big|_{-\pi}^0 + \frac{x e^{-inx}}{-in} \Big|_{-\pi}^{\pi} + \frac{1}{in} \int_{-\pi}^{\pi} e^{-inx} dx \right] \\ &= \frac{1}{2\pi} \left[-\frac{2\pi}{in} - \frac{2\pi(-1)^n}{-in} + \frac{2\pi(-1)^n}{-in} + 0 \right] \\ &= -\frac{1}{in}, \end{aligned}$$

where n is a non-zero integer. If $n = 0$, then we have

$$c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} F(x) dx = \frac{1}{2\pi} \left[\int_{-\pi}^0 (x + 2\pi) dx + \int_0^{\pi} x dx \right] = \frac{1}{2\pi} \times 2\pi^2 = \pi.$$

Hence it follows from Theorem 8.16 (Parseval's Theorem) that

$$\begin{aligned}\frac{1}{2\pi} \int_{-\pi}^{\pi} |F(x)|^2 dx &= \pi^2 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ \frac{1}{2\pi} \times \frac{(2\pi)^3}{3} &= \pi^2 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ \frac{4\pi^2}{3} &= \pi^2 + 2 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ \sum_{n=1}^{\infty} \frac{1}{n^2} &= \frac{\pi^2}{6}.\end{aligned}$$

This completes the proof of the problem. ■

Problem 8.14

Rudin Chapter 8 Exercise 14.

Proof. Since $f(-\pi) = f(\pi) = 0$, we can extend the function f to be a continuous function on $\mathbb{R}$:

$$f(x) = (\pi - |x - (2n+2)\pi|)^2,$$

where $x \in [(2n+1)\pi, (2n+3)\pi]$ and $n \in \mathbb{Z}$, see Figure 8.1 for the illustration.

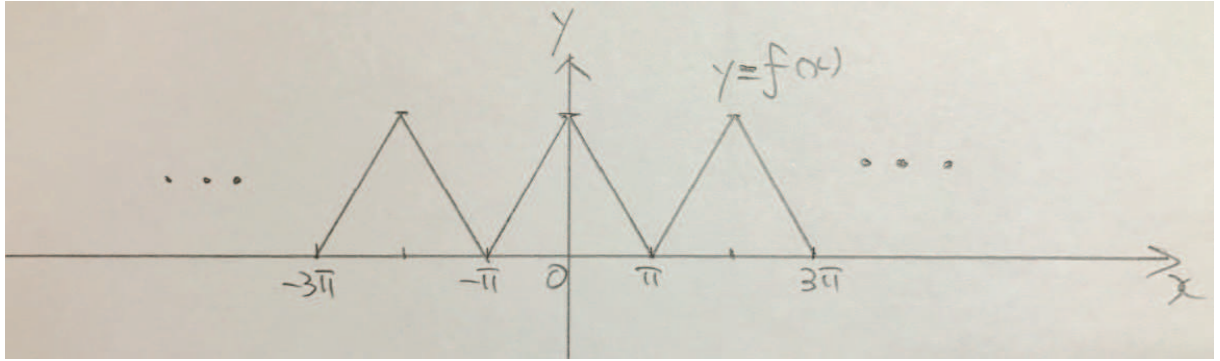


Figure 8.1: The graph of the continuous function $y = f(x) = (\pi - |x|)^2$ on $[-\pi, \pi]$.

Since f is integrable on $[-\pi, \pi]$, we apply Theorem 6.22 (Integration by Parts) to get the Fourier coefficients of f :

$$\begin{aligned}c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (\pi - |x|)^2 e^{-inx} dx \\ &= \frac{1}{2\pi} \left[\int_{-\pi}^0 (\pi + x)^2 e^{-inx} dx + \int_0^{\pi} (\pi - x)^2 e^{-inx} dx \right] \\ &= \begin{cases} \frac{2}{n^2}, & \text{if } n \neq 0; \\ \frac{\pi^2}{3}, & \text{if } n = 0. \end{cases}\end{aligned}\tag{8.42}$$

For every $x \in [-\pi, \pi]$ and $t \in (-\pi, \pi)$, we have

$$|f(x+t) - f(x)| = |(\pi - |x+t|)^2 - (\pi - |x|)^2|$$

$$\begin{aligned}
&= |\pi^2 - 2\pi|x + t| + |x + t|^2 - \pi^2 + 2\pi|x| - |x|^2| \\
&= |2\pi|x| - 2\pi|x + t| + |x + t|^2 - |x|^2| \\
&\leq |2\pi|x| - 2\pi(|x| - |t|) + (|x| + |t|)^2 - |x|^2| \\
&\leq |2\pi|t| + 2|x||t| + |t|^2| \\
&\leq 2\pi|t| + 2\pi|t| + \pi|t| \\
&= 5\pi|t|.
\end{aligned}$$

By Theorem 8.14 and the definitions (8.42), we have

$$\begin{aligned}
f(x) &= \lim_{N \rightarrow \infty} s_N(f; x) \\
&= \sum_{-\infty}^{-1} c_n e^{inx} + c_0 + \sum_1^{\infty} c_n e^{inx} \\
&= \sum_{-\infty}^{-1} \frac{2}{n^2} e^{inx} + \frac{\pi^2}{3} + \sum_1^{\infty} \frac{2}{n^2} e^{inx} \\
&= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{2}{n^2} (e^{inx} + e^{-inx}) \\
&= \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4}{n^2} \cos nx. \tag{8.43}
\end{aligned}$$

To prove the two identities in the problem, we first put $x = 0$ into the expression (8.43), since $f(0) = \pi^2$, we have

$$\begin{aligned}
\pi^2 &= \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2} \\
\sum_{n=1}^{\infty} \frac{1}{n^2} &= \frac{\pi^2}{6}.
\end{aligned}$$

Next, Theorem 8.16 (Parseval's Theorem) implies that

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\pi}^{\pi} (\pi - |x|)^4 dx &= \sum_{-\infty}^{-1} |c_n|^2 + |c_0|^2 + \sum_1^{\infty} |c_n|^2 \\
\frac{\pi^4}{5} &= \sum_{-\infty}^{-1} \frac{4}{n^4} + \frac{\pi^4}{9} + \sum_1^{\infty} \frac{4}{n^4} \\
\frac{\pi^4}{5} - \frac{\pi^4}{9} &= \sum_{n=1}^{\infty} \frac{8}{n^4} \\
\sum_{n=1}^{\infty} \frac{1}{n^4} &= \frac{\pi^4}{90}.
\end{aligned}$$

This completes the proof of the problem. ■

Problem 8.15

Rudin Chapter 8 Exercise 15.

Proof. By the definition of the Dirichlet kernel, we have

$$D_n(x) = \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}}$$

so that

$$K_N(x) = \frac{1}{N+1} \sum_{n=0}^N D_n(x) = \frac{1}{N+1} \sum_{n=0}^N \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}}.$$

We claim that

$$\sum_{n=0}^N \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}} = \frac{1 - \cos(N+1)x}{1 - \cos x} \quad (8.44)$$

for every non-negative integer N . This can be proven by induction. The case for $N = 0$ is easy. Assume that for some non-negative integer k , we have

$$\sum_{n=0}^k \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}} = \frac{1 - \cos(k+1)x}{1 - \cos x}.$$

If $N = k + 1$, then we have

$$\begin{aligned} \sum_{n=0}^{k+1} \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}} &= \sum_{n=0}^k \frac{\sin(n + \frac{1}{2})x}{\sin \frac{x}{2}} + \frac{\sin(k + \frac{3}{2})x}{\sin \frac{x}{2}} \\ &= \frac{1 - \cos(k+1)x}{1 - \cos x} + \frac{\sin(k + \frac{3}{2})x}{\sin \frac{x}{2}} \\ &= \frac{[1 - \cos(k+1)x] \sin \frac{x}{2} + (1 - \cos x) \sin(k + \frac{3}{2})x}{\sin \frac{x}{2} (1 - \cos x)} \\ &= \frac{\sin \frac{x}{2} - \frac{1}{2} [\sin(k + \frac{3}{2})x - \sin(k + \frac{1}{2})x] + \sin(k + \frac{3}{2})x - \frac{1}{2} [\sin(k + \frac{5}{2})x + \sin(k + \frac{1}{2})x]}{\sin \frac{x}{2} (1 - \cos x)} \\ &= \frac{\sin \frac{x}{2} + \frac{1}{2} [\sin(k + \frac{3}{2})x - \sin(k + \frac{5}{2})x]}{\sin \frac{x}{2} (1 - \cos x)} \\ &= \frac{\sin \frac{x}{2} - \cos(k+2)x \sin \frac{x}{2}}{\sin \frac{x}{2} (1 - \cos x)} \\ &= \frac{1 - \cos(k+2)x}{1 - \cos x}. \end{aligned}$$

Thus the statement is true for $N = k + 1$ if it is true for $N = k$. By induction, the formula (8.44) holds for every non-negative integer N . Hence we have

$$K_N(x) = \frac{1}{N+1} \cdot \frac{1 - \cos(N+1)x}{1 - \cos x}. \quad (8.45)$$

(a) Since $-1 \leq \cos x < 1$ for every $x \in \mathbb{R} \setminus \{0, \pm 2\pi, \pm 4\pi, \dots\}$, we have

$$K_N(x) \geq 0$$

for all $x \in \mathbb{R} \setminus \{0, \pm 2\pi, \pm 4\pi, \dots\}$. If $x \in \{0, \pm 2\pi, \pm 4\pi, \dots\}$, then $\cos x = 1$ and thus Theorem 5.13 (L'Hospital's Rule) implies that

$$K_N(x) = \lim_{t \rightarrow x} \frac{1}{N+1} \cdot \frac{1 - \cos(N+1)t}{1 - \cos t} = \lim_{t \rightarrow x} \frac{\sin(N+1)t}{\sin t} = \lim_{t \rightarrow x} \frac{(N+1) \cos(N+1)t}{\cos t} = N+1.$$

Hence we have

$$K_N(x) \geq 0$$

for every $x \in \mathbb{R}$.

(b) By definition, we have

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} K_N(x) dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\frac{1}{N+1} \sum_{n=0}^N D_n(x) \right] dx$$

$$\begin{aligned}
&= \frac{1}{2(N+1)\pi} \sum_{n=0}^N \int_{-\pi}^{\pi} D_n(x) dx \\
&= \frac{1}{2(N+1)\pi} \left[\int_{-\pi}^{\pi} D_0(x) dx + \sum_{n=1}^N \int_{-\pi}^{\pi} D_n(x) dx \right] \\
&= \frac{1}{2(N+1)\pi} \left[2\pi + \sum_{n=1}^N \int_{-\pi}^{\pi} \left(\sum_{k=-n}^n e^{ikx} \right) dx \right] \\
&= \frac{1}{2(N+1)\pi} \left[2\pi + \sum_{n=1}^N \left(\int_{-\pi}^{\pi} dx + \sum_{\substack{k=-n \\ k \neq 0}}^n \int_{-\pi}^{\pi} e^{ikx} dx \right) \right] \\
&= \frac{1}{2(N+1)\pi} \left[2\pi + \sum_{n=1}^N \left(\int_{-\pi}^{\pi} dx + \sum_{\substack{k=-n \\ k \neq 0}}^n \int_{-\pi}^{\pi} e^{ikx} dx \right) \right] \\
&= \frac{1}{2(N+1)\pi} \left[2\pi + \sum_{n=1}^N \left(2\pi + \sum_{\substack{k=-n \\ k \neq 0}}^n 2 \sin kx \right) \right] \\
&= \frac{1}{2(N+1)\pi} \left(2\pi + \sum_{n=1}^N 2\pi \right) \\
&= 1.
\end{aligned}$$

(c) Let $\delta > 0$. If $\delta \leq |x| \leq \pi$, then we have $\cos x \leq \cos \delta < 1$ which implies that

$$1 - \cos x \geq 1 - \cos \delta > 0. \quad (8.46)$$

By the equivalent definition of (8.45), the inequality (8.46) and part (a), we have

$$0 \leq K_N(x) = |K_N(x)| = \left| \frac{1}{N+1} \cdot \frac{1 - \cos(N+1)x}{1 - \cos x} \right| \leq \frac{1}{N+1} \cdot \frac{1}{1 - \cos \delta}.$$

By [21, Eqn. (78), p. 189], we have

$$s_N(f; x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) D_N(t) dt$$

so that

$$\begin{aligned}
\sigma_N(f; x) &= \frac{1}{N+1} \sum_{n=0}^N s_n(f; x) \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) \frac{1}{N+1} \sum_{n=0}^N D_n(t) dt \\
&= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) K_N(t) dt.
\end{aligned} \quad (8.47)$$

Proof of Fejér's theorem. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function with period 2π . We proceed as in Theorem 7.26. Given that $\epsilon > 0$, by the continuity of the function f , we choose a $\delta > 0$ such that $|y - x| < \delta$ implies

$$|f(y) - f(x)| < \frac{\epsilon}{2}. \quad (8.48)$$

Let $M = \sup_{x \in [-\pi, \pi]} |f(x)|$. By properties (a), (b) and (c), we see that for $x \in [-\pi, \pi]$,

$$|\sigma_n(f; x) - f(x)| = \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x-t) K_n(t) dt - f(x) \cdot \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(t) dt \right|$$

$$\begin{aligned}
&= \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(x-t) - f(x)] K_n(t) dt \right| \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(x-t) - f(x)| K_n(t) dt \\
&\leq \frac{1}{2\pi} \int_{-\pi}^{-\delta} |f(x-t) - f(x)| K_n(t) dt + \frac{1}{2\pi} \int_{-\delta}^{\delta} |f(x-t) - f(x)| K_n(t) dt \\
&\quad + \frac{1}{2\pi} \int_{\delta}^{\pi} |f(x-t) - f(x)| K_n(t) dt \\
&< 2M \cdot \frac{1}{2\pi} \int_{-\pi}^{-\delta} K_n(t) dt + \frac{\epsilon}{2} \cdot \frac{1}{2\pi} \int_{-\delta}^{\delta} K_n(t) dt + 2M \cdot \frac{1}{2\pi} \int_{\delta}^{\pi} K_n(t) dt \\
&\leq \frac{4M}{n+1} \cdot \frac{2}{1-\cos\delta} + \frac{\epsilon}{2} \\
&< \epsilon
\end{aligned}$$

for all large enough n , which proves Fejér's theorem. ■

This completes the proof of the problem. ■

Problem 8.16

Rudin Chapter 8 Exercise 16.

Proof. We follow the argument as in the proof of Fejér's theorem in Problem 8.15. Suppose that x is a real number such that both $f(x+)$ and $f(x-)$ exist. Given that $\epsilon > 0$. We have

$$\left| \sigma_N(f; x) - \frac{f(x+) + f(x-)}{2} \right| = \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt \right|. \quad (8.49)$$

Now we may split the integral on the right-hand side of (8.49) into two parts:

$$\begin{aligned}
&\frac{1}{2\pi} \int_{-\pi}^{\pi} \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt \\
&= \frac{1}{2\pi} \int_{-\pi}^0 \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt \\
&\quad + \frac{1}{2\pi} \int_0^{\pi} \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt. \quad (8.50)
\end{aligned}$$

Let $t = -y$. If $t = -\pi$, then $y = \pi$. If $t = 0$, then $y = 0$. Therefore, the first integral on the right-hand side in the expression (8.50) can be rewritten as[§]

$$\begin{aligned}
\frac{1}{2\pi} \int_{-\pi}^0 \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt &= \frac{1}{2\pi} \int_{\pi}^0 \left[f(x+y) - \frac{f(x+) + f(x-)}{2} \right] K_N(-y)(-dy) \\
&= \frac{1}{2\pi} \int_0^{\pi} \left[f(x+t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt. \quad (8.51)
\end{aligned}$$

Substitute the expression (8.51) back into the expression (8.50) and apply property (a) in Problem 8.15, we get

$$\begin{aligned}
\left| \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[f(x-t) - \frac{f(x+) + f(x-)}{2} \right] K_N(t) dt \right| &= \left| \frac{1}{2\pi} \int_0^{\pi} F(x, t) K_N(t) dt \right| \\
&\leq \frac{1}{2\pi} \int_0^{\pi} |F(x, t)| K_N(t) dt, \quad (8.52)
\end{aligned}$$

[§]Note that $K_N(x)$ is an even function in x .

where $F(x, t) = f(x+t) + f(x-t) - f(x+) - f(x-)$.

By definitions of $f(x+)$ and $f(x-)$, there exists a $\delta > 0$ such that $0 < t < \delta$ implies

$$|f(x-t) - f(x-)| < \frac{\epsilon}{2} \quad \text{and} \quad |f(x+t) - f(x+)| < \frac{\epsilon}{2}$$

which show that

$$|F(x, t)| \leq |f(x-t) - f(x-)| + |f(x+t) - f(x+)| < \epsilon. \quad (8.53)$$

Now we deduce from property (b) of Problem 8.15 and the inequality (8.53) that

$$\frac{1}{2\pi} \int_0^\delta |F(x, t)| K_N(t) dt < \frac{\epsilon}{2\pi} \int_0^\delta K_N(t) dt \leq \frac{\epsilon}{2\pi} \cdot \pi = \frac{\epsilon}{2}. \quad (8.54)$$

Since $f \in \mathcal{R}$ on $[a, b]$, f is bounded on $[a, b]$. Therefore, we may define $M = \sup_{x \in [-\pi, \pi]} |f(x)|$. Then it follows from property (c) of Problem 8.15 that

$$\begin{aligned} \frac{1}{2\pi} \int_\delta^\pi |F(x, t)| K_N(t) dt &\leq \frac{1}{2\pi} \cdot 4M \int_\delta^\pi K_N(t) dt \\ &\leq \frac{2M}{\pi} \cdot \frac{1}{N+1} \cdot \frac{2}{1-\cos\delta} \int_\delta^\pi dt \\ &< \frac{1}{N+1} \cdot \frac{4M}{1-\cos\delta} \\ &< \frac{\epsilon}{2} \end{aligned} \quad (8.55)$$

for large enough N . Hence after putting the inequalities (8.54) and (8.55) into the inequality (8.52), we deduce from the expression (8.49) that for large enough N , we have

$$\begin{aligned} \left| \sigma_N(f; x) - \frac{f(x+) + f(x-)}{2} \right| &\leq \frac{1}{2\pi} \int_0^\pi |F(x, t)| K_N(t) dt \\ &= \frac{1}{2\pi} \int_0^\delta |F(x, t)| K_N(t) dt + \frac{1}{2\pi} \int_\delta^\pi |F(x, t)| K_N(t) dt \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

In other words, we have the desired result that

$$\lim_{N \rightarrow \infty} \sigma_N(f; x) = \frac{1}{2} [f(x+) + f(x-)].$$

This completes the proof of the problem. ■

Problem 8.17

Rudin Chapter 8 Exercise 17.

Proof.

- (a) Here we must assume that $f \in \mathcal{R}$ on $[-\pi, \pi]$ so that the coefficients c_n are well-defined. By definition, we have

$$nc_n = \frac{n}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

Since $(-\frac{1}{in} e^{-inx})' = e^{-inx}$, we deduce from Problem 6.17 and then Theorem 6.13(b) that

$$|nc_n| = \left| \frac{n}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \right|$$

$$\begin{aligned}
&= \left| -\frac{1}{2\pi i} \left[e^{-in\pi} f(\pi) - e^{in\pi} f(-\pi) - \int_{-\pi}^{\pi} e^{-inx} df \right] \right| \\
&\leq \frac{1}{2\pi} \left[f(\pi) + f(-\pi) + \left| \int_{-\pi}^{\pi} e^{-inx} df \right| \right] \\
&\leq \frac{1}{2\pi} \left[f(\pi) + f(-\pi) + \int_{-\pi}^{\pi} |e^{-inx}| df \right] \\
&\leq \frac{f(\pi)}{\pi}
\end{aligned}$$

for every positive integer n . Hence $\{nc_n\}$ is a bounded sequence.

(b) For every positive integer N , let $a_N = s_N(f; x) - s_{N-1}(f; x)$. Then we have

$$a_N = \sum_{n=-N}^N c_n e^{inx} - \sum_{n=-(N-1)}^{N-1} c_n e^{inx} = c_{-N} e^{-iNx} + c_N e^{iNx}$$

so that

$$|Na_N| \leq |Nc_{-N}| + |Nc_N|.$$

By part (a), $\{Nc_N\}$ is a bounded sequence which implies that there is $M < \infty$ such that $|Na_N| \leq M$ for all positive integers N . Since f is monotonic on $[-\pi, \pi)$, Theorem 4.29 guarantees that $f(x+)$ and $f(x-)$ exist for every $x \in [-\pi, \pi)$. By Problem 8.16, we have

$$\lim_{N \rightarrow \infty} \sigma_N(f; x) = \frac{1}{2}[f(x+) + f(x-)].$$

Therefore, Problem 3.14(e) shows that

$$\lim_{N \rightarrow \infty} s_N(f; x) = \frac{1}{2}[f(x+) + f(x-)]$$

for every $x \in [-\pi, \pi)$.

(c) Since $f \in \mathcal{R}$ on $[-\pi, \pi]$, it is a bounded function on $[-\pi, \pi]$ so that $f(\alpha)$ and $f(\beta)$ are finite. Define a real function $g : [-\pi, \pi] \rightarrow \mathbb{R}$ as follows:

$$g(x) = \begin{cases} f(\alpha), & \text{if } x \in [-\pi, \alpha]; \\ f(x), & \text{if } x \in (\alpha, \beta); \\ f(\beta), & \text{if } x \in [\beta, \pi]. \end{cases} \quad (8.56)$$

Next, we define $h : [-\pi, \pi] \rightarrow \mathbb{R}$ by

$$h(x) = f(x) - g(x).$$

Since $h(x) = 0$ for all $x \in (\alpha, \beta)$, the corollary of Theorem 8.14 (Localization Theorem) implies that

$$\lim_{N \rightarrow \infty} s_N(h; x) = 0 \quad (8.57)$$

for every $x \in (\alpha, \beta)$. Since $s_N(h; x) = s_N(f; x) - s_N(g; x)$ for every $x \in (\alpha, \beta)$, the limit (8.57) can be rewritten as

$$\lim_{N \rightarrow \infty} s_N(f; x) = \lim_{N \rightarrow \infty} s_N(g; x) \quad (8.58)$$

on (α, β) .

If g is monotonic on $[-\pi, \pi]$, then Theorem 6.9 implies that $g \in \mathcal{R}$ on $[-\pi, \pi]$.^h Since it is bounded and monotonic on $[-\pi, \pi]$, part (b) implies that

$$\lim_{N \rightarrow \infty} s_N(g; x) = \frac{1}{2}[g(x+) + g(x-)] \quad (8.59)$$

^hTherefore we can apply the pointwise version of Fejér's theorem (Problem 8.16) to g .

for every $x \in [-\pi, \pi]$. Thus we substitute the expression (8.59) into the limit relation (8.58) to get

$$\lim_{N \rightarrow \infty} s_N(f; x) = \frac{1}{2}[g(x+) + g(x-)] = \frac{1}{2}[f(x+) + f(x-)] \quad (8.60)$$

for every $x \in (\alpha, \beta)$.

If g is not monotonic on $[-\pi, \pi]$, then we may modify the definition (8.56) as

$$g(x) = \begin{cases} \frac{f(\alpha)}{M}, & \text{if } x \in [-\pi, \alpha]; \\ f(x), & \text{if } x \in (\alpha, \beta); \\ Mf(\beta), & \text{if } x \in [\beta, \pi], \end{cases}$$

where M is a positive constant such that

$$\frac{f(\alpha)}{M} < f(x) < Mf(\beta)$$

for all $x \in (\alpha, \beta)$. In this case, this modified function g will be monotonic on $[-\pi, \pi]$ and the above argument can be repeated to show that the expression (8.60) still holds on (α, β) .

This completes the proof of the problem. ■

Problem 8.18

Rudin Chapter 8 Exercise 18.

Proof. Since $\tan x \rightarrow +\infty$ as $x \rightarrow \frac{\pi}{2}$ and $x \in (0, \frac{\pi}{2})$, we have $f(x) < 0$ and $g(x) < 0$ as $x \rightarrow \frac{\pi}{2}$ and $x \in (0, \frac{\pi}{2})$. Thus it is reasonable to conjecture that

$$f(x) < 0 \quad \text{and} \quad g(x) < 0$$

on $(0, \frac{\pi}{2})$. By applying [21, Eqn. (49), p. 182] repeatedly, it is clear that

$$\begin{aligned} f'(x) &= 3x^2 - (\tan^2 x + 2 \sin^2 x) \\ f''(x) &= 6x - 2(\tan x + \tan^3 x + \sin 2x) \\ f'''(x) &= 6 - 2(\sec^2 x + 3 \tan^2 x \sec^2 x + 2 \cos 2x) \\ &= 6 - 2(3 \tan^4 x + 4 \tan^2 x + 1 + 2 \cos 2x) \\ f^{(4)}(x) &= -2(12 \tan^3 x \sec^2 x + 8 \tan x \sec^2 x - 4 \sin 2x) \\ &= -8(3 \tan^3 x + 5 \tan^3 x + 2 \tan x - \sin 2x) \\ &= -8(3 \tan^3 x + 5 \tan^3 x + 2 \tan x \sin^2 x) \\ &= -8 \tan x (3 \tan^4 x + 5 \tan^2 x + 2 \sin^2 x). \end{aligned} \quad (8.61)$$

Furthermore, it is easy to check that

$$f(0) = f'(0) = f''(0) = f'''(0) = f^{(4)}(0) = 0. \quad (8.62)$$

Since $\tan x$ and $\sin x$ are positive on $(0, \frac{\pi}{2})$, the expression (8.61) yields that

$$f^{(4)}(x) < 0$$

for all $x \in (0, \frac{\pi}{2})$. By Problem 5.2, we have $f'''(x)$ is strictly decreasing in $(0, \frac{\pi}{2})$. Thus we follow from the values (8.62) that $f'''(x) < 0$ for all $x \in (0, \frac{\pi}{2})$. By repeating this kind of argument, we can show that

$$f''(x) < 0, \quad f'(x) < 0$$

and then

$$f(x) < 0$$

for all $x \in (0, \frac{\pi}{2})$, as desired.

Similarly, we have

$$\begin{aligned} g'(x) &= 4x - (\sin 2x + \tan x + x + x \tan^2 x) \\ g''(x) &= 4 - (2 \cos 2x + 2 + 2 \tan^2 x + 2x \tan x + 2x \tan^3 x) \\ g'''(x) &= -(-4 \sin 2x + 6 \tan x + 6 \tan^3 x + 2x + 8x \tan^2 x + 6x \tan^4 x) \\ g^{(4)}(x) &= -(-8 \cos 2x + 6 \sec^2 x + 18 \tan^2 x \sec^2 x + 2 \\ &\quad + 8 \tan^2 x + 16x \tan x \sec^2 x + 6 \tan^4 x + 24x \tan^3 x \sec^2 x) \\ &= -(-8 \cos 2x + 8 + 32 \tan^2 x + 24 \tan^4 x + 16x \tan x \\ &\quad + 40x \tan^3 x + 24x \tan^5 x) \end{aligned} \quad (8.63)$$

$$\begin{aligned} g^{(5)}(x) &= -(16 \sin 2x + 16x + 80 \tan x + 200 \tan^3 x + 120 \tan^5 x \\ &\quad + 136x \tan^2 x + 240x \tan^4 x + 120x \tan^6 x). \end{aligned} \quad (8.64)$$

In addition, we also have

$$g(0) = g'(0) = g''(0) = g'''(0) = g^{(4)}(0) = g^{(5)}(0) = 0. \quad (8.65)$$

Since $\tan x$ and $\sin 2x$ are clearly positive on $(0, \frac{\pi}{2})$, the expression (8.63) yields that

$$g^{(5)}(x) < 0$$

for all $x \in (0, \frac{\pi}{2})$. Thus, by using similar argument as above and the values (8.65), we obtain that

$$g^{(4)}(x) < 0, \quad g'''(x) < 0, \quad g''(x) < 0, \quad g'(x) < 0$$

and then

$$g(x) < 0$$

for all $x \in (0, \frac{\pi}{2})$, completing the proof of the problem. ■

Problem 8.19

Rudin Chapter 8 Exercise 19.

Proof. Suppose that $f(x) = e^{ikx}$. If $k = 0$, then we have $f(x) = 1$ which means that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x + n\alpha) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N 1 = 1 \quad \text{and} \quad \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} dt = 1.$$

Therefore, we have the desired result in this special case. Next, suppose that $k \neq 0$. Since $\frac{\alpha}{\pi}$ is irrational, $k\alpha$ cannot be a multiple of π . In other words, we have $e^{ikn\alpha} \neq 1$ for every positive integer n . By this, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x + n\alpha) &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{ik(x+n\alpha)} \\ &= e^{ikx} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N e^{ikn\alpha} \\ &= e^{ikx} \lim_{N \rightarrow \infty} \frac{1}{N} \cdot e^{ik\alpha} \cdot \frac{1 - e^{ikN\alpha}}{1 - e^{ik\alpha}} \\ &= 0 \end{aligned}$$

and

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{ikt} dt = \frac{1}{2\pi} \left(\frac{e^{ik\pi} - e^{-ik\pi}}{ik} \right) = \frac{\sin k\pi}{k\pi} = 0.$$

Therefore, we also have the desired result in this case.

For the general case, since f is continuous with period 2π , we obtain from Theorem 8.15 (Stone–Weierstrass theorem) that for every $\epsilon > 0$, there is a trigonometric polynomial P such that

$$|P(x) - f(x)| < \frac{\epsilon}{3} \quad (8.66)$$

for all real x . Therefore, we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N f(x + n\alpha) &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \{[f(x + n\alpha) - P(x + n\alpha)] + P(x + n\alpha)\} \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N [f(x + n\alpha) - P(x + n\alpha)] + \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N P(x + n\alpha) \end{aligned} \quad (8.67)$$

and

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \{[f(t) - P(t)] + P(t)\} dt \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(t) - P(t)] dt + \frac{1}{2\pi} \int_{-\pi}^{\pi} P(t) dt. \end{aligned} \quad (8.68)$$

By Definition 8.9, we have

$$P(x) = \sum_{n=-N}^N c_n e^{inx},$$

where c_n ($-N \leq n \leq N$) are some constants. Thus the analysis for the special case $f(x) = e^{ikx}$ implies that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N P(x + n\alpha) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P(t) dt$$

for every x . In other words, given $\epsilon > 0$, there exists an integer N such that

$$\left| \frac{1}{N} \sum_{n=1}^N P(x + n\alpha) - \frac{1}{2\pi} \int_{-\pi}^{\pi} P(t) dt \right| < \frac{\epsilon}{3} \quad (8.69)$$

for all $n \geq N$.

Therefore, we deduce from the expressions (8.67), (8.68) and the inequality (8.69) with the integer N defined in the inequality (8.69) that for all $n \geq N$, we have

$$\begin{aligned} &\left| \frac{1}{N} \sum_{n=1}^N f(x + n\alpha) - \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt \right| \\ &= \left| \frac{1}{N} \sum_{n=1}^N [f(x + n\alpha) - P(x + n\alpha)] + \frac{1}{N} \sum_{n=1}^N P(x + n\alpha) - \frac{1}{2\pi} \int_{-\pi}^{\pi} P(t) dt \right. \\ &\quad \left. - \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(t) - P(t)] dt \right| \\ &\leq \left| \frac{1}{N} \sum_{n=1}^N [f(x + n\alpha) - P(x + n\alpha)] \right| + \left| \frac{1}{N} \sum_{n=1}^N P(x + n\alpha) - \frac{1}{2\pi} \int_{-\pi}^{\pi} P(t) dt \right| \\ &\quad + \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} [f(t) - P(t)] dt \right| \end{aligned}$$

$$< \frac{1}{N} \sum_{n=1}^N |f(x + n\alpha) - P(x + n\alpha)| + \frac{\epsilon}{3} + \frac{1}{2\pi} \int_{-\pi}^{\pi} |f(t) - P(t)| dt. \quad (8.70)$$

Finally, we apply the inequality (8.66) to the inequality (8.70), so we establish

$$\left| \frac{1}{N} \sum_{n=1}^N f(x + n\alpha) - \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) dt \right| < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon.$$

Hence we have the desired result and this completes the proof of the problem. ■

Problem 8.20

Rudin Chapter 8 Exercise 20.

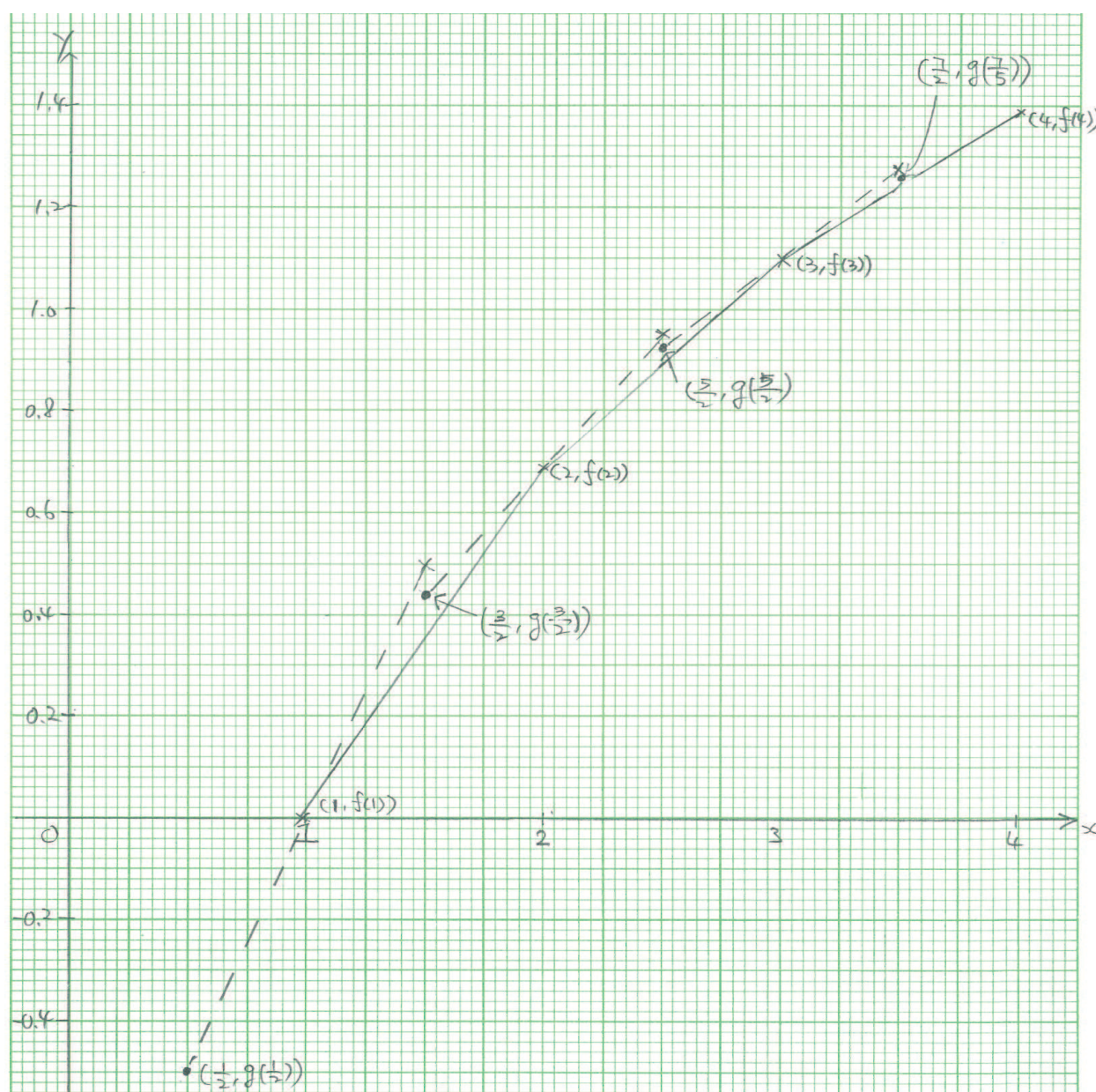
Proof. For examples, if $m = 1, 2, 3, 4$, then we have

$$\begin{aligned} f(x) &= \begin{cases} (x-1) \log 2, & \text{if } 1 \leq x \leq 2; \\ (3-x) \log 2 + (x-2) \log 3, & \text{if } 2 \leq x \leq 3; \\ (4-x) \log 3 + (x-3) \log 4, & \text{if } 3 \leq x \leq 4; \\ (5-x) \log 4 + (x-4) \log 5, & \text{if } 4 \leq x \leq 5 \end{cases} \\ &= \begin{cases} (x-1) \log 2, & \text{if } 1 \leq x \leq 2; \\ \left(\log \frac{3}{2}\right)x + \log \frac{2^3}{3^2}, & \text{if } 2 \leq x \leq 3; \\ \left(\log \frac{4}{3}\right)x + \log \frac{3^4}{4^3}, & \text{if } 3 \leq x \leq 4; \\ \left(\log \frac{5}{4}\right)x + \log \frac{4^5}{5^4}, & \text{if } 4 \leq x \leq 5 \end{cases} \end{aligned}$$

and

$$g(x) = \begin{cases} x-1, & \text{if } \frac{1}{2} \leq x < \frac{3}{2}; \\ \frac{x}{2} - 1 + \log 2, & \text{if } \frac{3}{2} \leq x < \frac{5}{2}; \\ \frac{x}{3} - 1 + \log 3, & \text{if } \frac{5}{2} \leq x < \frac{7}{2}; \\ \frac{x}{4} - 1 + \log 4, & \text{if } \frac{7}{2} \leq x < \frac{9}{2}. \end{cases}$$

Therefore, their graphs are given by

Figure 8.2: The graphs of the two functions f and g .

Now for a positive integer n , we have

$$\begin{aligned}
 \int_1^n f(x) dx &= \sum_{m=1}^{n-1} \int_m^{m+1} f(x) dx \\
 &= \sum_{m=1}^{n-1} \int_m^{m+1} [(m+1-x) \log m + (x-m) \log(m+1)] dx \\
 &= \sum_{m=1}^{n-1} \frac{1}{2} [\log m + \log(m+1)] \\
 &= \frac{1}{2} \log(n-1)! + \frac{1}{2} \log(n!) \\
 &= \log(n!) - \frac{1}{2} \log n.
 \end{aligned} \tag{8.71}$$

Furthermore, we have

$$\begin{aligned}
 \int_1^n g(x) dx &= \int_1^{\frac{3}{2}} g(x) dx + \int_{n-\frac{1}{2}}^n g(x) dx + \sum_{m=2}^{n-1} \int_{m-\frac{1}{2}}^{m+\frac{1}{2}} g(x) dx \\
 &= \frac{1}{8} - \frac{1}{8n} + \frac{1}{2} \log n + \sum_{m=2}^{n-1} \int_{m-\frac{1}{2}}^{m+\frac{1}{2}} \left(\frac{x}{m} - 1 + \log m \right) dx \\
 &= \frac{1}{8} - \frac{1}{8n} + \frac{1}{2} \log n + \sum_{m=2}^{n-1} \log m \\
 &= \frac{1}{8} - \frac{1}{8n} + \frac{1}{2} \log n + \log(n-1)! \\
 &= \log(n!) - \frac{1}{2} \log n + \frac{1}{8} - \frac{1}{8n}.
 \end{aligned} \tag{8.72}$$

Combining the expressions (8.71) and (8.72), we have for $x \geq 1$,

$$\int_1^n f(x) dx = \log(n!) - \frac{1}{2} \log n > -\frac{1}{8} + \int_1^n g(x) dx. \tag{8.73}$$

By Theorem 6.22, we have

$$\int_1^n \log x dx = n \log n - 1 \cdot \log 1 - \int_1^n x d(\log x) = n \log n - \int_1^n dx = n \log n - n + 1. \tag{8.74}$$

Since $f(x) \leq \log x \leq g(x)$ if $x \geq 1$, we follow from the inequality (8.73) and the expression (8.74) that

$$\begin{aligned}
 \int_1^n f(x) dx &\leq \int_1^n \log x dx \leq \int_1^n g(x) dx \\
 \log(n!) - \frac{1}{2} \log n &\leq n \log n - n + 1 < \log(n!) - \frac{1}{2} \log n + \frac{1}{8}
 \end{aligned}$$

which deduce the inequalities

$$\frac{7}{8} < \log(n!) - \left(n + \frac{1}{2}\right) \log n + n < 1 \tag{8.75}$$

for $n = 2, 3, 4, \dots$. Hence, by taking exponential to each part of the inequalities (8.75), we have eventually the desired formula

$$e^{\frac{7}{8}} < \frac{n!}{\left(\frac{n}{e}\right)^n \sqrt{n}} < e.$$

This completes the proof of the problem. ■

Problem 8.21

Rudin Chapter 8 Exercise 21.

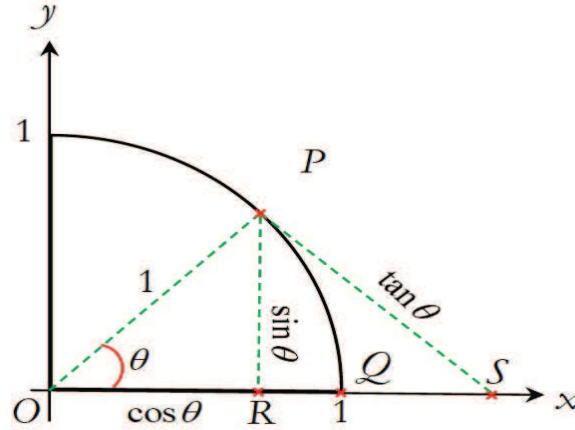
Proof. Since $|D_n(t)|$ is an even function on $[-\pi, \pi]$, Theorem 6.12(c) and Theorem 6.19 give

$$L_n = \frac{1}{2\pi} \left[\int_{-\pi}^0 |D_n(t)| dt + \int_0^{\pi} |D_n(t)| dt \right] = \frac{1}{\pi} \int_0^{\pi} |D_n(t)| dt = \frac{1}{\pi} \int_0^{\pi} \left| \frac{\sin(n + \frac{1}{2})t}{\sin \frac{t}{2}} \right| dt. \tag{8.76}$$

By Theorem 6.19 again with $t = 2x$, the expression (8.76) implies that

$$L_n = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \left| \frac{\sin(2n+1)x}{\sin x} \right| dx. \tag{8.77}$$

We know that $0 < \sin x \leq x$ on $(0, \frac{\pi}{2}]$, see Figure 8.3 for a geometric proof of this.

Figure 8.3: A geometric proof of $0 < \sin x \leq x$ on $(0, \frac{\pi}{2}]$.

Thus we can deduce from the formula (8.77) that

$$L_n \geq \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \left| \frac{\sin(2n+1)x}{x} \right| dx. \quad (8.78)$$

Apply Theorem 6.19 with $y = (2n+1)x$ to the inequality (8.78), we yield

$$L_n \geq \frac{2}{\pi} \int_0^{(2n+1)\frac{\pi}{2}} \left| \frac{\sin y}{y} \right| dy \geq \frac{2}{\pi} \int_0^{n\pi} \left| \frac{\sin y}{y} \right| dy = \frac{2}{\pi} \sum_{k=0}^{n-1} \int_{k\pi}^{(k+1)\pi} \left| \frac{\sin y}{y} \right| dy. \quad (8.79)$$

It is clear that

$$\frac{1}{y} \geq \frac{1}{(k+1)\pi}$$

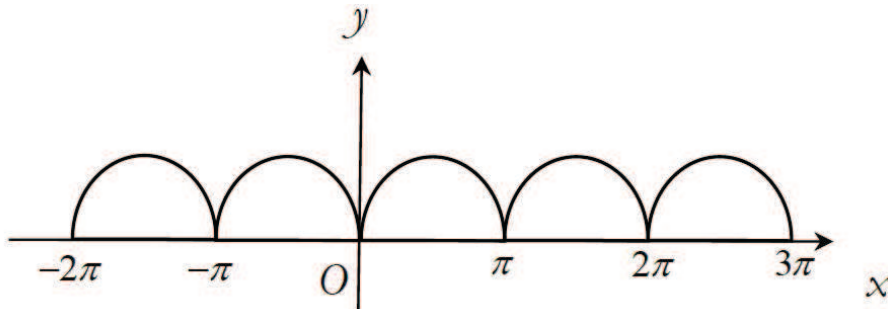
on $[k\pi, (k+1)\pi]$, where $k = 0, 1, \dots, n-1$. Therefore, the inequality (8.79) implies that

$$L_n \geq \frac{2}{\pi^2} \sum_{k=0}^{n-1} \frac{1}{k+1} \int_{k\pi}^{(k+1)\pi} |\sin y| dy. \quad (8.80)$$

For each $k \in \{0, 1, 2, \dots, n-1\}$, we acquire

$$\int_{k\pi}^{(k+1)\pi} |\sin x| dx = \int_0^{\pi} \sin x dx, \quad (8.81)$$

see Figure 8.4 for this geometric meaning.

Figure 8.4: The graph of $y = |\sin x|$.

By substituting the expression (8.81) into the inequality (8.80) and applying Problem 8.9(a), we obtain

$$L_n \geq \frac{2}{\pi^2} \sum_{k=0}^{n-1} \frac{1}{k+1} \int_0^\pi \sin x \, dx = \frac{4}{\pi^2} \sum_{k=1}^n \frac{1}{k} \geq \frac{4}{\pi^2} \log n.$$

This finishes the analysis of the problem. ■

Problem 8.22

Rudin Chapter 8 Exercise 22.

Proof. We follow the given hint. Let

$$f(x) = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n. \quad (8.82)$$

Since

$$\limsup_{n \rightarrow \infty} \left| \frac{\alpha(\alpha-1) \cdots (\alpha-n)}{(n+1)!} \times \frac{n!}{\alpha(\alpha-1) \cdots (\alpha-n+1)} x \right| = \lim_{n \rightarrow \infty} \left| \frac{\alpha-n}{n+1} \right| |x| = |x|,$$

we follow from Theorem 3.34(a) that the series converges if $|x| < 1$. By Theorem 8.1, the function defined by the series (8.82) is differentiable in $(-1, 1)$ and

$$f'(x) = \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^{n-1}. \quad (8.83)$$

Thus we follow from the definition (8.82) and the derivative (8.83) that

$$\begin{aligned} (1+x)f'(x) &= (1+x) \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^{n-1} \\ &= \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^{n-1} + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^n \\ &= \alpha + \sum_{n=2}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^{n-1} + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^n \\ &= \alpha + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n)}{n!} x^n + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} x^n \\ &= \alpha + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{(n-1)!} \left(\frac{\alpha-n}{n} + 1 \right) x^n \\ &= \alpha + \alpha \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!} x^n \\ &= \alpha f(x). \end{aligned}$$

By [21, Eqn. (38), p. 180] and Theorem 5.5, we have

$$\begin{aligned} \frac{d}{dx}(\log f(x)) &= \frac{f'(x)}{f(x)} \\ \frac{d}{dx}(\log f(x)) &= \frac{\alpha}{1+x} \\ \log f(x) - \log f(0) &= \int_0^x \frac{\alpha}{1+t} dt \\ \log f(x) &= \alpha \log(1+x) \end{aligned}$$

$$f(x) = (1+x)^\alpha. \quad (8.84)$$

Hence, by the definition (8.82) and the expression (8.84), we have the desired result that

$$(1+x)^\alpha = 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!} x^n. \quad (8.85)$$

By replacing x and α by $-x$ and $-\alpha$ respectively in the expression (8.85), we have

$$\begin{aligned} (1-x)^{-\alpha} &= 1 + \sum_{n=1}^{\infty} \frac{(-\alpha)(-\alpha-1)\cdots(-\alpha-n+1)}{n!} (-x)^n \\ &= 1 + \sum_{n=1}^{\infty} \frac{\alpha(\alpha+1)\cdots(\alpha+n-1)}{n!} x^n. \end{aligned} \quad (8.86)$$

In particular, we suppose that $-1 < x < 1$ and $\alpha > 0$ in the equation (8.86). By Theorem 8.18(a), we know that

$$\begin{aligned} \frac{\Gamma(n+\alpha)}{\Gamma(\alpha)} &= \frac{\Gamma((n+\alpha-1)+1)}{\Gamma(\alpha)} \\ &= \frac{(\alpha+n-1)\Gamma(\alpha+n-1)}{\Gamma(\alpha)} \\ &= \cdots \\ &= \frac{(\alpha+n-1)\cdots(\alpha+1)\alpha\Gamma(\alpha)}{\Gamma(\alpha)} \\ &= (\alpha+n-1)\cdots(\alpha+1)\alpha. \end{aligned} \quad (8.87)$$

By substituting the identity (8.87) into the expression (8.86), we get

$$(1-x)^{-\alpha} = 1 + \sum_{n=1}^{\infty} \frac{\Gamma(n+\alpha)}{n!\Gamma(\alpha)} x^n = \sum_{n=0}^{\infty} \frac{\Gamma(n+\alpha)}{n!\Gamma(\alpha)} x^n,$$

where $-1 < x < 1$ and $\alpha > 0$. This completes the proof of the problem. ■

8.2 Index of a curve

Problem 8.23

Rudin Chapter 8 Exercise 23.

Proof. We follow the given hint. Since γ is continuously differentiable and $\gamma(t) \neq 0$ on $[a, b]$, the function

$$\frac{\gamma'}{\gamma}$$

is well-defined and continuous on $[a, b]$. Thus $\frac{\gamma'}{\gamma} \in \mathcal{R}$ on $[a, b]$ by Theorem 6.8 and we can define $\varphi : [a, b] \rightarrow \mathbb{C}$ to be the function given by

$$\varphi(x) = \int_a^x \frac{\gamma'(t)}{\gamma(t)} dt. \quad (8.88)$$

By Theorem 6.20 (First Fundamental Theorem of Calculus), the function φ is differentiable on $[a, b]$ and

$$\varphi'(t) = \frac{\gamma'(t)}{\gamma(t)}. \quad (8.89)$$

Furthermore, we have $\varphi(a) = 0$. Let $f(t) = \gamma(t)e^{-\varphi(t)}$, where $t \in [a, b]$. It is clear that f is differentiable on $[a, b]$ and we deduce from the expression (8.89) that

$$f'(t) = \gamma'(t)e^{-\varphi(t)} + \gamma(t)[- \varphi'(t)]e^{-\varphi(t)} = \gamma'(t)e^{-\varphi(t)} - \gamma'(t)e^{-\varphi(t)} = 0$$

for all $t \in (a, b)$. By Theorem 5.11(b), f is a constant on (a, b) . Since f is continuous on $[a, b]$, it must be a constant on $[a, b]$. Since $\gamma(a) = \gamma(b)$, we have

$$e^{\varphi(b)} = e^{\varphi(a)} = e^0 = 1. \quad (8.90)$$

By [21, Eqn. (52), p. 183], Theorem 8.7(a) and (c), the expression (8.90) implies that

$$\varphi(b) = 2n\pi i$$

for some integer n . We note that $\varphi(b) = 2\pi i \text{Ind}(\gamma)$, so we have

$$2\pi i \text{Ind}(\gamma) = 2n\pi i$$

which is equivalent to

$$\text{Ind}(\gamma) = n$$

for some integer n . This proves the first assertion.

For the second assertion, if $\gamma(t) = e^{int}$, $a = 0$ and $b = 2\pi$, then we have

$$\text{Ind}(\gamma) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{ine^{int}}{e^{int}} dt = \frac{n}{2\pi} \int_0^{2\pi} dt = n.$$

Now the number $\text{Ind}(\gamma)$ is called the winding number of γ around 0 because it counts the total number of times that the curve γ travels *counterclockwise* around the origin 0. This number certainly depends on the *orientation of the curve*, so it is negative if the curve travels around the point clockwise. See Figure 8.5 about the winding number of γ around an arbitrary point p .

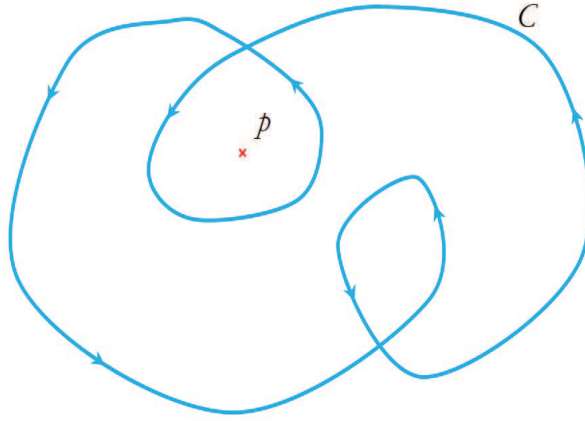


Figure 8.5: The winding number of γ around an arbitrary point p .

We end the analysis of the problem here. ■

Problem 8.24

Rudin Chapter 8 Exercise 24.

Proof. We follow the given hint. Let $0 \leq c < \infty$. It is clear that $\gamma + c : [a, b] \rightarrow \mathbb{C}$ is still a continuously differentiable closed curve. Since γ does not intersect the negative real axis, we must have

$$\gamma(t) + c \neq 0$$

for every $t \in [a, b]$. Thus $\gamma + c$ satisfies all the hypotheses of Problem 8.23 and it is meaningful to talk about $\text{Ind}(\gamma + c)$.

We define $f : [0, \infty) \rightarrow \mathbb{Z}$ by

$$f(c) = \text{Ind}(\gamma + c) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t) + c} dt,$$

where $c \in [0, \infty)$. We claim that f is a continuous function of c . To this end, we check the definition of continuity (Definition 4.5). Given that $\epsilon > 0$ and $c \in [0, \infty)$. It is easy to see that

$$\begin{aligned} |f(x) - f(c)| &= \left| \frac{1}{2\pi i} \int_a^b \left[\frac{\gamma'(t)}{\gamma(t) + x} - \frac{\gamma'(t)}{\gamma(t) + c} \right] dt \right| \\ &\leq \frac{1}{2\pi} \int_a^b \left| \frac{\gamma'(t)}{\gamma(t) + x} - \frac{\gamma'(t)}{\gamma(t) + c} \right| dt \\ &\leq \frac{1}{2\pi} \int_a^b \left| \frac{\gamma'(t)(c - x)}{(\gamma(t) + x)(\gamma(t) + c)} \right| dt. \end{aligned} \quad (8.91)$$

Since γ and γ' are continuous on $[a, b]$, we know from Theorem 4.16 (Extreme Value Theorem) that there exist real numbers m, m', M and M' such that

$$0 < m \leq |\gamma(t)| \leq M \quad \text{and} \quad m' \leq |\gamma'(t)| \leq M' \quad (8.92)$$

for all $t \in [a, b]$.ⁱ

If $M' = 0$, then $\gamma'(t) = 0$ for all $t \in [a, b]$ which imply that $\gamma = A$ for some constant A on (a, b) . Since γ is continuous on $[a, b]$, we have $\gamma = A$ on $[a, b]$. However, γ cannot be a closed curve in this case, a contradiction. Thus we must have

$$M' > 0.$$

Furthermore, we note that $c > 0$ and $x \geq 0$ imply that

$$m + c > m \quad \text{and} \quad m + x \geq m,$$

so we deduce easily from the inequalities (8.91) and (8.92) that

$$|f(x) - f(c)| \leq \frac{1}{2\pi} \int_a^b \frac{M'}{(m + c)(m + x)} |c - x| dt < \frac{M'(b - a)}{2\pi m^2} |c - x|. \quad (8.93)$$

If we take $\delta = \frac{2\pi m^2}{M'(b - a)} \epsilon > 0$, then we obtain from the inequality (8.93) that

$$|f(x) - f(c)| < \epsilon$$

for all $x \in [0, \infty)$ with $|x - c| < \delta$. Hence we prove our claim that f is continuous on $[0, \infty)$.

Next, the definition of f and the inequalities (8.92) also imply that

$$|f(c)| \leq \frac{1}{2\pi} \int_a^b \left| \frac{\gamma'(t)}{\gamma(t) + c} \right| dt \leq \frac{M'(b - a)}{2\pi(m + c)}.$$

Thus we have

$$\lim_{c \rightarrow \infty} |f(c)| = 0.$$

Since the range of f is $\mathbb{Z}$, we have $f(c) = 0$ for all $c \in [0, \infty)$. In particular, we take $c = 0$ to get

$$f(0) = \text{Ind}(\gamma) = 0,$$

as required. This completes the proof of the problem. ■

ⁱNote that it is possible that $m' = 0$, but $m > 0$ because $\gamma(t) \neq 0$ on $[a, b]$.

Problem 8.25*Rudin Chapter 8 Exercise 25.*

Proof. We follow the given hint. Let $\gamma = \frac{\gamma_1}{\gamma_2}$. Since $\gamma_1 : [a, b] \rightarrow \mathbb{C}$ and $\gamma_2 : [a, b] \rightarrow \mathbb{C}$ are continuously differentiable closed curves and $\gamma_1(t)\gamma_2(t) \neq 0$ for every $t \in [a, b]$, the function $\gamma : [a, b] \rightarrow \mathbb{C}$ is also a continuously differentiable closed curve and $\gamma(t) \neq 0$ for every $t \in [a, b]$.

Now the inequality $|\gamma_1(t) - \gamma_2(t)| < |\gamma_1(t)|$ shows that

$$|1 - \gamma(t)| < 1$$

on $[a, b]$, so we have $0 < \gamma(t) < 2$ for all $t \in [a, b]$. Next, by Problem 8.24, we have $\text{Ind}(\gamma) = 0$. In addition, a direct computation shows that

$$\frac{\gamma'}{\gamma} = \frac{\gamma_2'}{\gamma_2} - \frac{\gamma_1'}{\gamma_1}$$

which gives

$$\text{Ind}(\gamma) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t)} dt = \frac{1}{2\pi i} \int_a^b \frac{\gamma_2'(t)}{\gamma_2(t)} dt - \frac{1}{2\pi i} \int_a^b \frac{\gamma_1'(t)}{\gamma_1(t)} dt = \text{Ind}(\gamma_2) - \text{Ind}(\gamma_1).$$

Hence we have

$$\text{Ind}(\gamma_1) = \text{Ind}(\gamma_2)$$

as required, completing the proof of the problem. ■

Problem 8.26*Rudin Chapter 8 Exercise 26.*

Proof. For all $t \in [0, 2\pi]$, we obtain from the triangle inequality that

$$\delta < |\gamma(t)| \leq |\gamma(t) - P_1(t)| + |P_1(t)| < \frac{\delta}{4} + |P_1(t)|$$

so that

$$|P_1(t)| > \frac{3\delta}{4} > \frac{\delta}{2} > 0$$

on $[0, 2\pi]$. By this and the triangle inequality, we have

$$|P_1(t) - P_2(t)| = |P_1(t) - \gamma(t) + \gamma(t) - P_2(t)| \leq |P_1(t) - \gamma(t)| + |P_2(t) - \gamma(t)| < \frac{\delta}{2} < |P_1(t)|$$

for all $t \in [0, 2\pi]$. Therefore, it follows from Problem 8.25 that

$$\text{Ind}(P_1) = \text{Ind}(P_2).$$

Define this common value to be $\text{Ind}(\gamma)$. In other words, the winding number of the closed curve γ can be defined in terms of that of any trigonometric polynomials $P(t)$ satisfying the inequality

$$|P(t) - \gamma(t)| < \frac{\delta}{4}$$

on $[0, 2\pi]$.

- **Extension of Problem 8.24.** Suppose that the range of γ does not intersect the negative real axis. Then there exists a $\eta > 0$ such that $|\gamma(t) - x| > \eta$ for all $t \in [0, 2\pi]$ and $x \leq 0$. Put $\kappa = \min(\delta, \eta)$. Then both

$$|\gamma(t)| > \kappa \quad \text{and} \quad |\gamma(t) - x| > \kappa$$

are valid on $[0, 2\pi]$ and on $x \leq 0$. If $P(t)$ is a trigonometric polynomial such that

$$|P(t) - \gamma(t)| < \frac{\kappa}{4},$$

then the triangle inequality implies that

$$|\gamma(t) - x| \leq |\gamma(t) - P(t)| + |P(t) - x|$$

and then it gives

$$|P(t) - x| \geq |\gamma(t) - x| - |\gamma(t) - P(t)| > \kappa - \frac{\kappa}{4} = \frac{3\kappa}{4} > 0$$

for all $t \in [0, 2\pi]$. Therefore, the range of $P(t)$ does not intersect the negative real axis. By Problem 8.24, we have $\text{Ind}(P) = 0$. By definition, we have

$$\text{Ind}(\gamma) = 0$$

which extends the result of Problem 8.24 to any closed curve (not necessarily differentiable) in $\mathbb{C}$ with domain $[0, 2\pi]$ and $\gamma(t) \neq 0$ for every $t \in [0, 2\pi]$.

- **Extension of Problem 8.25.** Suppose that γ_1 and γ_2 are two closed curves in $\mathbb{C}$ with domain $[0, 2\pi]$, and $\gamma_1(t)\gamma_2(t) \neq 0$ on $[0, 2\pi]$. Then there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that

$$|\gamma_1(t)| > \delta_1 \quad \text{and} \quad |\gamma_2(t)| > \delta_2$$

on $[0, 2\pi]$. By Theorem 8.15 (Stone–Weierstrass theorem), there exists trigonometric polynomials P_1 and P_2 such that

$$|\gamma_1(t) - P_1(t)| < \frac{\delta_1}{4} \quad \text{and} \quad |\gamma_2(t) - P_2(t)| < \frac{\delta_2}{4}$$

on $[0, 2\pi]$. Furthermore, we suppose that

$$|\gamma_1(t) - \gamma_2(t)| < |\gamma_1(t)|$$

for every $t \in [0, 2\pi]$. Thus there exists a $\delta_3 > 0$ such that

$$|\gamma_1(t)| - |\gamma_1(t) - \gamma_2(t)| > \delta_3$$

for all $t \in [0, 2\pi]$. Let $\delta = \min(\delta_1, \delta_2, \delta_3)$. Then we have

$$|\gamma_1(t) - P_1(t)| < \frac{\delta}{4}, \quad |\gamma_2(t) - P_2(t)| < \frac{\delta}{4} \quad \text{and} \quad |\gamma_1(t)| - |\gamma_1(t) - \gamma_2(t)| > \delta$$

for all $t \in [0, 2\pi]$. By these and the triangle inequality, we have

$$\begin{aligned} |P_1(t) - P_2(t)| &\leq |P_1(t) - \gamma_1(t)| + |\gamma_1(t) - \gamma_2(t)| + |\gamma_2(t) - P_2(t)| \\ &< \frac{\delta}{4} + \frac{\delta}{4} + |\gamma_1(t)| - \delta \\ &= |\gamma_1(t)| - \frac{\delta}{2} \\ &\leq |\gamma_1(t) - P_1(t)| + |P_1(t)| - \frac{\delta}{2} \\ &< |P_1(t)| + \frac{\delta}{4} - \frac{\delta}{2} \\ &< |P_1(t)| \end{aligned}$$

for every $t \in [0, 2\pi]$. By Problem 8.25, we have $\text{Ind}(P_1) = \text{Ind}(P_2)$ which implies that

$$\text{Ind}(\gamma_1) = \text{Ind}(\gamma_2).$$

This definitely extends the result of Problem 8.25 to any closed curves γ_1 and γ_2 (not necessarily differentiable) in $\mathbb{C}$ with domain $[0, 2\pi]$ and $\gamma_1(t)\gamma_2(t) \neq 0$ for every $t \in [0, 2\pi]$.

This completes the proof of the problem. ■

Problem 8.27

Rudin Chapter 8 Exercise 27.

Proof. Assume that $f(z) \neq 0$ for all $z \in \mathbb{C}$. Define $\gamma_r(t) = f(re^{it})$ for $0 \leq r < \infty$ and $0 \leq t \leq 2\pi$.

- (a) Since $\gamma_0(t) = f(0) \neq 0$ for all $t \in [0, 2\pi]$, we have $\gamma'_0(t) = 0$ and thus

$$\text{Ind}(\gamma_0) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'_0(t)}{\gamma_0(t)} dt = 0.$$

- (b) Since

$$\lim_{|z| \rightarrow \infty} z^{-n} f(z) = c,$$

for every $\epsilon > 0$ with $\epsilon < |c|$ there exists a $R > 0$ such that

$$|z^{-n} f(z) - c| < \epsilon$$

for all $|z| \geq R$ which is equivalent to

$$|\gamma_r(t) - cr^n e^{int}| = |f(re^{it}) - cr^n e^{int}| < \epsilon |r^n e^{int}| < |cr^n e^{int}| \quad (8.94)$$

for all $r \geq R$ and $0 \leq t \leq 2\pi$. Now we apply Problem 8.25 to the inequality (8.94) and then the definition, we get

$$\text{Ind}(\gamma_r(t)) = \text{Ind}(cr^n e^{int}) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{cinr^n e^{int}}{cr^n e^{int}} dt = n.$$

Therefore, we have $\text{Ind}(\gamma_r(t)) = n$ for all sufficiently large r .

- (c) Let $p, r \in [0, \infty)$. Define $d = |p - r| \geq 0$ and $I(p, d) = [\min(0, p - d), p + d]$. Now we want to show that for every $\epsilon > 0$, there exists a $\delta > 0$ such that $r \in I(p, d)$ and $|p - r| < \delta$ imply that^j

$$|\text{Ind}(\gamma_p) - \text{Ind}(\gamma_r)| < \epsilon. \quad (8.95)$$

Next, we define the set

$$K(r, d) = \{ae^{it} \mid a \in I(r, d), 0 \leq t \leq 2\pi\}.$$

Since $f : \mathbb{C} \rightarrow \mathbb{C}$ is a continuous function and the set $K(p, 0)$ is compact, Theorem 4.16 (Extreme Value Theorem) ensures that the value

$$m = \min_{t \in [0, 2\pi]} |f(pe^{it})|$$

is finite.

Given that $\epsilon > 0$ with $m > \epsilon$. By Theorem 4.19, f is uniformly continuous on $K(p, d)$. Thus there exists a $\delta > 0$ such that for all $z_1, z_2 \in K(p, d)$ with $|z_1 - z_2| < \delta$ implies

$$|f(z_1) - f(z_2)| < \epsilon. \quad (8.96)$$

In particular, we may assume that $z_1 = pe^{it}$ and $z_2 = re^{it}$, where p is fixed and r varies. Thus the inequality (8.96) implies that

$$|f(pe^{it}) - f(re^{it})| < \epsilon < m \leq |f(pe^{it})| \quad (8.97)$$

^jThe introduction of the set $I(p, d)$ makes sure that if $r \in I(p, d)$, then $r \geq 0$.

for all $r \in I(p, d)$ with $|p - r| < \delta$ and $0 \leq t \leq 2\pi$. By definition, the inequality (8.97) can be rewritten as

$$|\gamma_p(t) - \gamma_r(t)| < |\gamma_p(t)|$$

for all $r \in I(p, d)$ with $|p - r| < \delta$ and $0 \leq t \leq 2\pi$. By Problem 8.25 or Problem 8.26, we have

$$\text{Ind}(\gamma_p) = \text{Ind}(\gamma_r) \quad (8.98)$$

for all $r \in I(p, d)$ with $|p - r| < \delta$. In other words, the identity (8.98) means that the inequality (8.94) holds, i.e., the function $\text{Ind}(\gamma_p)$ is continuous at $p \in [0, \infty)$.

Recall from Problem 8.23 that $\text{Ind}(\gamma_r) \in \mathbb{Z}$. Since $[0, \infty)$ is connected by Theorem 2.47, we can see from Theorem 4.22 and part (c) that

$$\text{Ind}(\gamma_r)([0, \infty))$$

is also connected. By part (b), we know that $\text{Ind}(\gamma_r) = n$ for all sufficiently large r and thus we must have $\text{Ind}(\gamma_r) = n$ for every $r \in [0, \infty)$ and for some $n \in \mathbb{Z}$. In particular, we have $\text{Ind}(\gamma_0) = n > 0$ which contradicts the result of (a). Hence we have $f(z) = 0$ for *at least* one complex number z . This completes the proof of the problem. ■

Problem 8.28

Rudin Chapter 8 Exercise 28.

Proof. For $0 \leq r \leq 1$, $0 \leq t \leq 2\pi$, we put

$$\gamma_r(t) = g(re^{it}) \quad \text{and} \quad \psi(t) = e^{-it}\gamma_1(t). \quad (8.99)$$

Assume that $g(z) \neq -z$ for every $z \in T$. Then we have

$$\psi(t) \neq -1 \quad (8.100)$$

for every $t \in [0, 2\pi]$. Since $|g(z)| = 1$ for every $z \in \overline{D}$, we have

$$|\psi(t)| = |e^{-it}\gamma_1(t)| = |e^{-it}g(e^{it})| = 1 \neq 0 \quad (8.101)$$

for every $t \in [0, 2\pi]$, i.e., ψ maps $[0, 2\pi]$ into the unit circle T . By these two facts (8.100) and (8.101), we have the result that the range of ψ *does not* intersect the negative real axis. Furthermore, we have

$$\psi(0) = \gamma_1(0) = g(1) \quad \text{and} \quad \psi(2\pi) = e^{-2\pi i}\gamma_1(2\pi) = g(e^{2\pi i}) = g(1)$$

so that it is a closed curve. In conclusion, the curve ψ satisfies the hypotheses of Problem 8.24 or Problem 8.26. Therefore, we must have

$$\text{Ind}(\psi) = 0. \quad (8.102)$$

By a similar argument as in Problem 8.27(c), we know that $\text{Ind}(\gamma_r)$ is a continuous function of r , on $[0, 1]$. Since $[0, 1]$ is connected, Theorem 4.22 implies that $\text{Ind}(\gamma_r)([0, 1])$ is also connected. Recall that $\text{Ind}(\gamma_r) \in \mathbb{Z}$, so it must be a *fixed* integer for all $r \in [0, 1]$. To derive a contradiction from this result, we compute the values of $\text{Ind}(\gamma_0)$ and $\text{Ind}(\gamma_1)$.

We note that $\gamma_0(t) = g(0) \neq 0$ which is a non-zero constant, the definition of the winding number of a closed curve (see Problem 8.23) gives

$$\text{Ind}(\gamma_0) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{0}{g(0)} dt = 0.$$

To find $\text{Ind}(\gamma_1)$, we need a lemma first.

Lemma 8.1

Let $\alpha, \beta : [a, b] \rightarrow \mathbb{C}$ be closed curves, $\alpha(t) \neq 0$ and $\beta(t) \neq 0$ for every $t \in [a, b]$. Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be defined by $\gamma = \alpha \times \beta$. Then we have $\text{Ind}(\gamma) = \text{Ind}(\alpha) + \text{Ind}(\beta)$.

Proof of Lemma 8.1. We suppose that α and β are continuously differentiable. Then it is easy to check that γ is also a continuously differentiable closed curve and $\gamma(t) \neq 0$ for every $t \in [a, b]$. By definition and the fact that $\gamma' = \alpha'\beta + \beta'\alpha$, we have

$$\text{Ind}(\gamma) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'}{\gamma} dt = \frac{1}{2\pi i} \int_a^b \frac{\alpha'}{\alpha} dt + \frac{1}{2\pi i} \int_a^b \frac{\beta'}{\beta} dt = \text{Ind}(\alpha) + \text{Ind}(\beta).$$

Next, suppose that α and β are *not* differentiable so that γ may not be differentiable. However, the numbers $\text{Ind}(\alpha)$, $\text{Ind}(\beta)$ and $\text{Ind}(\gamma)$ are still well-defined by Problem 8.26. Let

$$M_1 = \max_{t \in [0, 2\pi]} |\alpha(t)| \quad \text{and} \quad M_2 = \max_{t \in [0, 2\pi]} |\beta(t)|.$$

Since α, β and γ are non-zero on $[0, 2\pi]$, there exists a small $\delta > 0$ such that

$$|\alpha(t)| > \delta, \quad |\beta(t)| > \delta \quad \text{and} \quad |\gamma(t)| > \delta$$

for every $t \in [0, 2\pi]$.

- **Case (i):** $M_1 + M_2 \geq 1$. By Theorem 8.15 (Stone–Weierstrass theorem), there are trigonometric polynomials P_1 and P_2 such that

$$|P_1(t) - \alpha(t)| < \frac{\delta}{8(M_1 + M_2)} < \frac{\delta}{4} \quad \text{and} \quad |P_2(t) - \beta(t)| < \frac{\delta}{8(M_1 + M_2)} < \frac{\delta}{4} \quad (8.103)$$

for every $t \in [0, 2\pi]$. By Problem 8.26, we have $\text{Ind}(\alpha) = \text{Ind}(P_1)$ and $\text{Ind}(\beta) = \text{Ind}(P_2)$, so we derive from the triangle inequality and the inequalities (8.103) that

$$\begin{aligned} |P_1(t)P_2(t) - \gamma(t)| &= |P_1(t)P_2(t) - P_2(t)\alpha(t) + P_2(t)\alpha(t) - \alpha(t)\beta(t)| \\ &\leq |P_2(t)||P_1(t) - \alpha(t)| + |\alpha(t)||P_2(t) - \beta(t)| \\ &< (|P_2(t)| + |\alpha(t)|) \frac{\delta}{8(M_1 + M_2)} \\ &\leq (|P_2(t) - \beta(t)| + |\beta(t)| + |\alpha(t)|) \frac{\delta}{8(M_1 + M_2)} \\ &< \frac{\frac{\delta}{4} + M_1 + M_2}{2(M_1 + M_2)} \cdot \frac{\delta}{4} \\ &\leq \frac{\delta}{4} \end{aligned}$$

for every $t \in [0, 2\pi]$. In this case, we have the formula

$$\text{Ind}(\gamma) = \text{Ind}(P_1P_2) = \text{Ind}(P_1) + \text{Ind}(P_2) = \text{Ind}(\alpha) + \text{Ind}(\beta). \quad (8.104)$$

- **Case (ii):** $0 < M_1 + M_2 < 1$. Instead of the inequalities (8.103), we have

$$|P_1(t) - \alpha(t)| < \frac{\delta}{8} < 1 \quad \text{and} \quad |P_2(t) - \beta(t)| < \frac{\delta}{8} < 1 \quad (8.105)$$

for every $t \in [0, 2\pi]$. Similarly, we can derive from the triangle inequality and the inequalities (8.105) that

$$|P_1(t)P_2(t) - \gamma(t)| < (|P_2(t)| + |\alpha(t)|) \frac{\delta}{8} \leq (|P_2(t) - \beta(t)| + |\beta(t)| + |\alpha(t)|) \frac{\delta}{8} < \frac{\delta}{4}$$

for every $t \in [0, 2\pi]$. Thus we also have the formula (8.104).

This completes the proof of Lemma 8.1. ■

Now we may go back to the proof of Problem 8.28. Apply Lemma 8.1 to (8.102), we obtain

$$0 = \text{Ind}(\psi) = \text{Ind}(e^{-it}\gamma_1) = \text{Ind}(e^{-it}) + \text{Ind}(\gamma_1) = -1 + \text{Ind}(\gamma_1)$$

which means that $\text{Ind}(\gamma_1) = 1$. As a result, the consequences

$$\text{Ind}(\gamma_0) = 0 \quad \text{and} \quad \text{Ind}(\gamma_1) = 1$$

certainly contradict to the fact that $\text{Ind}(\gamma_r)$ is a fixed integer for every $r \in [0, 1]$. Hence $g(z) = -z$ for at least one $z \in T$ and this ends the proof of the problem. ■

Problem 8.29

Rudin Chapter 8 Exercise 29.

Proof. Let $T = \{z \in \mathbb{C} \mid |z| = 1\}$ and $f : \overline{D} \rightarrow \overline{D}$, where $\overline{D} = \{z \in \mathbb{C} \mid |z| \leq 1\}$. Assume that $f(z) \neq z$ for every $z \in \overline{D}$. Then we have

$$|z - f(z)| \neq 0$$

for every $z \in \overline{D}$. Now we associate to each $z \in \overline{D}$ the point $g(z) \in T$ which lies on the ray that starts at $f(z)$ and passes through the point z . See Figure 8.6 for an illustration.

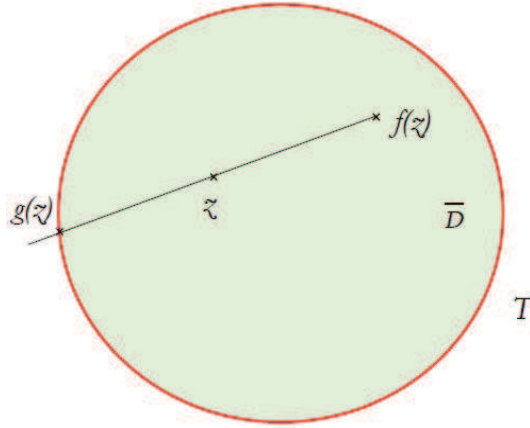


Figure 8.6: The geometry of the points z , $f(z)$ and $g(z)$.

In fact, we have

$$g(z) = \frac{z - f(z)}{|z - f(z)|}. \quad (8.106)$$

By the definition (8.106), we have $g(\overline{D}) \subseteq T$ and we follow from Theorem 4.4 that g is continuous on $\overline{D}$. Thus g is a continuous mapping of $\overline{D}$ into the unit circle T . By Problem 8.28,

$$g(z_0) = -z_0 \quad (8.107)$$

for some $z_0 \in T$. However, the construction of g implies that

$$g(z) = z \quad (8.108)$$

if $z \in T$. Therefore, we deduce a contradiction from the expressions (8.107) and (8.108) that

$$z_0 = 0 \notin T.$$

Hence we have $f(z) = z$ for some $z \in \overline{D}$. This completes the proof of the problem. ■

8.3 Stirling's formula

Problem 8.30*Rudin Chapter 8 Exercise 30.*

Proof. Let $c \in \mathbb{R}$ and $y = x + c$. By Theorem 8.18(a), we have

$$\lim_{x \rightarrow \infty} \frac{\Gamma(x+c)}{x^c \Gamma(x)} = \lim_{y \rightarrow \infty} \frac{\Gamma(y)}{(y-c)^c \Gamma(y-c)} = \lim_{y \rightarrow \infty} \frac{\Gamma(y+1)}{y} \cdot \frac{1}{(y-c)^{c-1} \Gamma(y-c+1)}.$$

It is easy to see that

$$\frac{\Gamma(y+1)}{y} \cdot \frac{1}{(y-c)^{c-1} \Gamma(y-c+1)} = \frac{\Gamma(y+1)}{\left(\frac{y}{e}\right)^y \sqrt{2\pi y}} \cdot \frac{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}}{y(y-c)^{c-1} \Gamma(y-c+1)} \cdot \frac{\left(\frac{y}{e}\right)^y \sqrt{2\pi y}}{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}}.$$

Since

$$\lim_{y \rightarrow \infty} \frac{\left(\frac{y}{e}\right)^y \sqrt{2\pi y}}{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}} = 1,$$

Stirling's formula implies that

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\Gamma(x+c)}{x^c \Gamma(x)} &= \lim_{y \rightarrow \infty} \frac{\Gamma(y+1)}{y} \cdot \frac{1}{(y-c)^{c-1} \Gamma(y-c+1)} \\ &= \lim_{y \rightarrow \infty} \frac{\Gamma(y+1)}{\left(\frac{y}{e}\right)^y \sqrt{2\pi y}} \times \lim_{y \rightarrow \infty} \frac{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}}{y(y-c)^{c-1} \Gamma(y-c+1)} \\ &\quad \times \lim_{y \rightarrow \infty} \frac{\left(\frac{y}{e}\right)^y \sqrt{2\pi y}}{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}} \\ &= \lim_{y \rightarrow \infty} \frac{\left(\frac{y-c}{e}\right)^{y-c} \sqrt{2\pi(y-c)}}{y(y-c)^{c-1} \Gamma(y-c+1)} \\ &= 1. \end{aligned}$$

This completes the proof of the problem. ■

Problem 8.31*Rudin Chapter 8 Exercise 31.*

Proof. Since $(1-x^2)^n$ is an even function in x , we have

$$\int_{-1}^1 (1-x^2)^n dx = 2 \int_0^1 (1-x^2)^n dx. \quad (8.109)$$

By the change of variable $t = x^2$, the integral on the right-hand side of (8.109) becomes

$$\int_0^1 t^{-\frac{1}{2}} (1-t)^n dt. \quad (8.110)$$

Apply Theorem 8.20 with $x = \frac{1}{2}$ and $y = n+1$ to the integral (8.110), we have

$$\int_0^1 t^{-\frac{1}{2}} (1-t)^n dt = \frac{\Gamma(\frac{1}{2}) \Gamma(n+1)}{\Gamma(n+\frac{3}{2})}. \quad (8.111)$$

Substituting the identity (8.111) into the integral equation (8.109) and applying [21, Eqn. (99), p. 194] and Problem 8.30, we have

$$\begin{aligned}\int_{-1}^1 (1-x^2)^n dx &= \frac{\sqrt{\pi}\Gamma(n+1)}{\Gamma(n+\frac{3}{2})} \\ \lim_{n \rightarrow \infty} \sqrt{n} \int_{-1}^1 (1-x^2)^n dx &= \sqrt{\pi} \lim_{n \rightarrow \infty} \frac{n^{\frac{1}{2}}\Gamma(n+1)}{\Gamma(n+\frac{3}{2})} \\ &= \sqrt{\pi}.\end{aligned}$$

This completes the proof of the problem. ■

CHAPTER 9

Functions of Several Variables

9.1 Linear transformations

Problem 9.1

Rudin Chapter 9 Exercise 1.

Proof. Let $\text{span}(S)$ be the span of S . Suppose that $\mathbf{x}, \mathbf{y} \in \text{span}(S)$ and c is a scalar. By Definition 9.1(b), we have

$$\mathbf{x} = c_1 \mathbf{s}_1 + c_2 \mathbf{s}_2 + \cdots + c_p \mathbf{s}_p \quad \text{and} \quad \mathbf{y} = d_1 \mathbf{s}'_1 + d_2 \mathbf{s}'_2 + \cdots + d_q \mathbf{s}'_q,$$

where $c_1, c_2, \dots, c_p, d_1, d_2, \dots, d_q$ are scalars and $\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_p, \mathbf{s}'_1, \mathbf{s}'_2, \dots, \mathbf{s}'_q \in \text{span}(S)$. Then we have

$$\mathbf{x} + \mathbf{y} = c_1 \mathbf{s}_1 + c_2 \mathbf{s}_2 + \cdots + c_p \mathbf{s}_p + d_1 \mathbf{s}'_1 + d_2 \mathbf{s}'_2 + \cdots + d_q \mathbf{s}'_q \in \text{span}(S)$$

and

$$c\mathbf{x} = (cc_1)\mathbf{s}_1 + (cc_2)\mathbf{s}_2 + \cdots + (cc_p)\mathbf{s}_p \in \text{span}(S).$$

By Definition 9.1(a), $\text{span}(S)$ is a vector space. This completes the proof of the problem. ■

Problem 9.2

Rudin Chapter 9 Exercise 2.

Proof. Suppose that $A \in L(X, Y)$ and $B \in L(Y, Z)$. Let $\mathbf{x}_1, \mathbf{x}_2 \in X$ and c be a scalar. Then we have

$$(BA)(\mathbf{x}_1 + \mathbf{x}_2) = (BA)\mathbf{x}_1 + (BA)\mathbf{x}_2 = B(A\mathbf{x}_1) + B(A\mathbf{x}_2) = (BA)\mathbf{x}_1 + (BA)\mathbf{x}_2$$

and

$$(BA)(c\mathbf{x}) = B(A(c\mathbf{x})) = B(cA\mathbf{x}) = cB(A\mathbf{x}) = c(BA)\mathbf{x}.$$

By Definition 9.4, we obtain $BA \in L(X, Z)$.

For $\mathbf{y}, \mathbf{y}_1, \mathbf{y}_2 \in Y$, there exist unique $\mathbf{x}, \mathbf{x}_1, \mathbf{x}_2 \in X$ such that $A\mathbf{x}_1 = \mathbf{y}_1, A\mathbf{x}_2 = \mathbf{y}_2$ and $A\mathbf{x} = \mathbf{y}$. By Definition 9.4, we have

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2, \quad A(c\mathbf{x}) = cA\mathbf{x}, \quad A^{-1}\mathbf{y}_1 = \mathbf{x}_1, \quad A^{-1}\mathbf{y}_2 = \mathbf{x}_2, \quad A^{-1}\mathbf{y} = \mathbf{x}. \quad (9.1)$$

Thus we deduce from the expressions (9.1) that

$$A^{-1}(\mathbf{y}_1 + \mathbf{y}_2) = A^{-1}(A\mathbf{x}_1 + A\mathbf{x}_2) = A^{-1}(A(\mathbf{x}_1 + \mathbf{x}_2)) = \mathbf{x}_1 + \mathbf{x}_2 = A^{-1}\mathbf{y}_1 + A^{-1}\mathbf{y}_2$$

and

$$A^{-1}(c\mathbf{y}) = A^{-1}(cA\mathbf{x}) = A^{-1}(A(c\mathbf{x})) = c\mathbf{x} = cA^{-1}\mathbf{y}.$$

By Definition 9.4, A^{-1} is linear. If $A^{-1}\mathbf{y}_1 = A^{-1}\mathbf{y}_2$, then we have

$$A(A^{-1}\mathbf{y}_1) = A(A^{-1}\mathbf{y}_2)$$

so that $\mathbf{y}_1 = \mathbf{y}_2$. Therefore, A^{-1} is one-to-one. For every $\mathbf{x} \in X$, there exists a $\mathbf{y} \in X$ such that $A\mathbf{x} = \mathbf{y}$ which means

$$A^{-1}\mathbf{y} = \mathbf{x}$$

by the expressions (9.1). Thus A^{-1} is onto. Hence A^{-1} is invertible. This completes the proof of the problem. ■

Problem 9.3

Rudin Chapter 9 Exercise 3.

Proof. If $A\mathbf{x} = A\mathbf{y}$, then we have

$$A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$$

which implies that $\mathbf{x} - \mathbf{y} = \mathbf{0}$, i.e., $\mathbf{x} = \mathbf{y}$. Hence A is 1-1, completing the proof of the problem. ■

Problem 9.4

Rudin Chapter 9 Exercise 4.

Proof. Let X and Y be vector spaces and $A \in L(X, Y)$. Furthermore, let $\mathcal{N}(A) = \{\mathbf{x} \in X \mid A\mathbf{x} = \mathbf{0}\}$ be the null space of A . If $\mathbf{x}, \mathbf{y} \in \mathcal{N}(A)$ and c is a scalar, then we follow from Definition 9.4 that

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0} \quad \text{and} \quad A(c\mathbf{x}) = cA\mathbf{x} = \mathbf{0}.$$

Hence $\mathcal{N}(A)$ is a vector space in X .

Let $\mathcal{R}(A) = \{A\mathbf{x} \in Y \mid \mathbf{x} \in X\}$ be the range of A . If $\mathbf{x}, \mathbf{x}_1, \mathbf{x}_2 \in X$ and c is a scalar, then we have

$$A\mathbf{x} = \mathbf{y} \in Y, \quad A\mathbf{x}_1 = \mathbf{y}_1 \in Y \quad \text{and} \quad A\mathbf{x}_2 = \mathbf{y}_2 \in Y.$$

Thus it follows from Definitions 9.1 and 9.4 that

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \mathbf{y}_1 + \mathbf{y}_2 \in Y \quad \text{and} \quad A(c\mathbf{x}) = cA\mathbf{x} = c\mathbf{y} \in Y.$$

Hence $\mathcal{R}(A)$ is also a vector space in Y . We finish the proof of the problem. ■

Problem 9.5

Rudin Chapter 9 Exercise 5.

Proof. Suppose that $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$ and $\mathbf{x} = \sum c_j \mathbf{e}_j$, where $c_j \in \mathbb{R}$ for all $j = 1, 2, \dots, n$. By Definition 9.4, we have

$$A\mathbf{x} = A\left(\sum c_j \mathbf{e}_j\right) = \sum c_j A\mathbf{e}_j = \mathbf{x} \cdot \mathbf{y}, \tag{9.2}$$

where $\mathbf{y} = \sum (A\mathbf{e}_j) \mathbf{e}_j$. This proves the existence of the vector $\mathbf{y}$. In fact, the uniqueness of such vector follows immediately from the form

$$\mathbf{y} = \sum (A\mathbf{e}_j) \mathbf{e}_j$$

because for every $A \in L(\mathbb{R}^n, \mathbb{R})$, the real numbers $A\mathbf{e}_j$ ($1 \leq j \leq n$) are uniquely determined.

By Theorem 1.37(d) (Schwarz inequality) and the expression (9.2), we have

$$|A\mathbf{x}| = |\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{y}||\mathbf{x}|$$

for all $\mathbf{x} \in \mathbb{R}^n$. Then Definition 9.6(c) implies that

$$\|A\| \leq |\mathbf{y}|. \quad (9.3)$$

If $\mathbf{x} = \frac{\mathbf{y}}{|\mathbf{y}|}$, then we have $|\mathbf{x}| = 1$ so that

$$A\mathbf{x} = \mathbf{x} \cdot \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{y}}{|\mathbf{y}|} = |\mathbf{y}|.$$

By Definition 9.6(c), we have

$$\|A\| \geq |A\mathbf{x}| = |\mathbf{y}|. \quad (9.4)$$

Hence our desired result follows immediately from the inequalities (9.3) and (9.4). This completes the proof of the problem. ■

9.2 Differentiable mappings

Problem 9.6

Rudin Chapter 9 Exercise 6.

Proof. By Definition 9.16 and Theorem 5.13 (L'Hospital's Rule), for all $(x, y) \neq (0, 0)$, we have

$$\begin{aligned} (D_1 f)(x, y) &= \lim_{t \rightarrow 0} \frac{f((x, y) + t(1, 0)) - f(x, y)}{t} \\ &= \lim_{t \rightarrow 0} \frac{\frac{(x+t)y}{(x+t)^2 + y^2} - \frac{xy}{x^2 + y^2}}{t} \\ &= \lim_{t \rightarrow 0} \frac{[(x+t)^2 + y^2]y - (x+t)y \cdot 2(x+t)}{[(x+t)^2 + y^2]^2} \\ &= \frac{y^3 - 2x^2y}{(x^2 + y^2)^2} \end{aligned}$$

and

$$\begin{aligned} (D_2 f)(x, y) &= \lim_{t \rightarrow 0} \frac{f((x, y) + t(0, 1)) - f(x, y)}{t} \\ &= \lim_{t \rightarrow 0} \frac{\frac{x(y+t)}{x^2 + (y+t)^2} - \frac{xy}{x^2 + y^2}}{t} \\ &= \lim_{t \rightarrow 0} \frac{[x^2 + (y+t)^2]x - x(y+t) \cdot 2(y+t)}{[x^2 + (y+t)^2]^2} \\ &= \frac{x^3 - 2y^2x}{(x^2 + y^2)^2}. \end{aligned}$$

For $(x, y) = (0, 0)$, we note that

$$(D_1 f)(0, 0) = \lim_{t \rightarrow 0} \frac{f((0, 0) + t(1, 0)) - f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{0}{t} = 0$$

and

$$(D_2 f)(0, 0) = \lim_{t \rightarrow 0} \frac{f((0, 0) + t(0, 1)) - f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{0}{t} = 0.$$

Therefore, $(D_1f)(x, y)$ and $(D_2f)(x, y)$ exist at every point of $\mathbb{R}^2$. However, if $x = y \neq 0$, then

$$|f(x, y) - f(0, 0)| = |f(x, x) - (0, 0)| = \frac{1}{2},$$

so f is not continuous at $(0, 0)$. This completes the proof of the problem. ■

Problem 9.7

Rudin Chapter 9 Exercise 7.

Proof. We have $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Fix $\mathbf{x} \in E$ and $\epsilon > 0$. Since the partial derivatives are bounded in E , there exists a positive number M such that

$$|(D_jf)(\mathbf{y})| \leq M \quad (9.5)$$

for all $\mathbf{y} \in E$ and $1 \leq j \leq n$. Since E is open, there is an open ball $S \subseteq E$, with center at $\mathbf{x}$ and radius $\frac{\epsilon}{nM}$. Suppose that $\mathbf{y} \in S$. Then we have

$$\mathbf{y} - \mathbf{x} = \mathbf{h} = \sum h_j \mathbf{e}_j,$$

where $|\mathbf{h}| < \frac{\epsilon}{nM}$. Put $\mathbf{v}_0 = \mathbf{0}$ and $\mathbf{v}_k = h_1 \mathbf{e}_1 + \cdots + h_k \mathbf{e}_k$, where $1 \leq k \leq n$. Then we have

$$f(\mathbf{y}) - f(\mathbf{x}) = f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = \sum_{j=1}^n [f(\mathbf{x} + \mathbf{v}_j) - f(\mathbf{x} + \mathbf{v}_{j-1})]. \quad (9.6)$$

Since $|\mathbf{v}_k| < \frac{\epsilon}{nM}$ for $1 \leq k \leq n$ and S is convex by Definition 2.17, the segments with end points

$$\mathbf{x} + \mathbf{v}_{j-1} \quad \text{and} \quad \mathbf{x} + \mathbf{v}_j$$

must lie in S . Since $\mathbf{v}_j = \mathbf{v}_{j-1} + h_j \mathbf{e}_j$, Theorem 5.10 (Mean Value Theorem) implies that

$$f(\mathbf{x} + \mathbf{v}_j) - f(\mathbf{x} + \mathbf{v}_{j-1}) = h_j (D_jf)(\mathbf{x} + \mathbf{v}_{j-1} + \theta_j h_j \mathbf{e}_j) \quad (9.7)$$

for some $\theta_j \in (0, 1)$. By the bounds (9.5), the expression (9.7) gives

$$|f(\mathbf{x} + \mathbf{v}_j) - f(\mathbf{x} + \mathbf{v}_{j-1})| \leq M |h_j|.$$

Therefore, the expression (9.6) reduces to

$$|f(\mathbf{y}) - f(\mathbf{x})| \leq M \sum_{j=1}^n |h_j| \leq nM |\mathbf{h}| < \epsilon.$$

By Definition 4.5, f is continuous at $\mathbf{x}$. Since $\mathbf{x}$ is arbitrary, f is continuous on E and this completes the proof of the problem. ■

Problem 9.8

Rudin Chapter 9 Exercise 8.

Proof. We have $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Since E is open and f is differentiable at $\mathbf{x}$, it follows from Theorem 9.17 that

$$f'(\mathbf{x}) \mathbf{e}_j = (D_jf)(\mathbf{x})$$

for $1 \leq j \leq n$ so that its corresponding matrix is

$$[f'(\mathbf{x})] = \begin{pmatrix} (D_1f)(\mathbf{x}) & (D_2f)(\mathbf{x}) & \cdots & (D_nf)(\mathbf{x}) \end{pmatrix} \quad (9.8)$$

Since f has a local maximum at $\mathbf{x} \in E$, there exists a $\delta > 0$ such that

$$f(\mathbf{y}) \leq f(\mathbf{x})$$

for all $\mathbf{y} \in N_\delta(\mathbf{x})$. Note that, for each $1 \leq j \leq n$, if $\mathbf{y} = \mathbf{x} + t\mathbf{e}_j$ with $|t| < \delta$, then we have $\mathbf{y} \in N_\delta(\mathbf{x})$.

Suppose that $F_j : (-\delta, \delta) \rightarrow \mathbb{R}$ is defined by

$$F_j(t) = f(\mathbf{x} + t\mathbf{e}_j),$$

where $1 \leq j \leq n$. By the previous analysis, we see that $\mathbf{x} + t\mathbf{e}_j \in N_\delta(\mathbf{x})$ so that each F_j is well-defined. By the hypotheses, the real-valued function F_j is differentiable on $(-\delta, \delta)$ and has a local maximum at 0. By Theorem 5.8, we see that

$$F'_j(0) = 0$$

for $1 \leq j \leq n$. By Definition 5.1 and [21, Eqn. (25), p. 215], we have

$$(D_j f)(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{e}_j) - f(\mathbf{x})}{t} = \lim_{t \rightarrow 0} \frac{F_j(t) - F_j(0)}{t} = F'_j(0) = 0$$

for all $1 \leq j \leq n$. Hence we obtain from the matrix representation (9.8) that

$$[f'(\mathbf{x})] = \begin{pmatrix} 0 & 0 & \cdots & 0 \end{pmatrix}$$

which is equivalent to $f'(\mathbf{x}) = \mathbf{0}$. This completes the proof of the problem. ■

Problem 9.9

Rudin Chapter 9 Exercise 9.

Proof. We have $\mathbf{f} : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$. We borrow a result from general topology, see [18, §23, p. 148].

Lemma 9.1

A space E is connected if and only if the only subsets of E that are both open and closed in E are the empty set and E itself.

Pick and fix $\mathbf{x} \in E$. Consider the set

$$A = \{\mathbf{y} \in E \mid f(\mathbf{y}) = f(\mathbf{x})\}.$$

Now for every $\mathbf{y} \in A$, since E is open, there is an open ball $G_{\mathbf{y}}$ such that $\mathbf{y} \in G_{\mathbf{y}} \subseteq E$. It is clear from the hypothesis that $\mathbf{f}'(\mathbf{x}) = \mathbf{0}$ for every $\mathbf{x} \in G_{\mathbf{y}}$. Since $G_{\mathbf{y}}$ is convex and open, we follow from the corollary of Theorem 9.19 that

$$\mathbf{f}(\mathbf{z}) = \mathbf{f}(\mathbf{x})$$

for all $\mathbf{z} \in G_{\mathbf{y}}$, i.e., $G_{\mathbf{y}} \subseteq A$. By Definition 2.18(f), A is open in E .

Let $\mathbf{u} \in A^c$. Then we have $\mathbf{f}(\mathbf{u}) \neq \mathbf{f}(\mathbf{x})$. Again, the openness of E implies that

$$\mathbf{u} \in G_{\mathbf{u}} \subseteq E$$

for some open ball $G_{\mathbf{u}}$. By the corollary of Theorem 9.19, we have

$$\mathbf{f}(\mathbf{v}) = \mathbf{f}(\mathbf{u}) \neq \mathbf{f}(\mathbf{x})$$

for all $\mathbf{v} \in G_{\mathbf{u}}$. Therefore, $G_{\mathbf{u}} \subseteq A^c$ and then A^c is open. By Theorem 2.23, A is closed in E . Hence it follows from Lemma 9.1 that $A = E$ which means $\mathbf{f}$ is constant in E , finishing the proof of the problem. ■

Problem 9.10*Rudin Chapter 9 Exercise 10.*

Proof. We have $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Let $\mathbf{x}, \mathbf{x}' \in E$ and they differ *only* at the first coordinate, i.e., $\mathbf{x} = (a, x_2, \dots, x_n)$ and $\mathbf{x}' = (b, x_2, \dots, x_n)$. Let, further, $b > a$ and consider the function $g : [a, b] \rightarrow \mathbb{R}$ defined by

$$g(t) = f(t, x_2, \dots, x_n). \quad (9.9)$$

We first show that g is well-defined. To this end, note that

$$(1 - \lambda)\mathbf{x} + \lambda\mathbf{x}' \in E \quad (9.10)$$

for all $\lambda \in [0, 1]$ because E is convex. Thus this implies that

$$((1 - \lambda)a + \lambda b, x_2, \dots, x_n) \in E$$

for all $\lambda \in [0, 1]$ which means that $(t, x_2, \dots, x_n) \in E$ for $t \in [a, b]$ and then the function g given by (9.9) is well-defined. Since $D_1 f$ exists on E , the function g is differentiable on $[a, b]$. Apply Theorem 5.5 (Mean Value Theorem) to g , we have

$$g(b) - g(a) = (b - a)g'(\theta) \quad (9.11)$$

for some $\theta \in (a, b)$. By the hypothesis, we have

$$g'(\theta) = (D_1 f)(\theta, x_2, \dots, x_n) = 0$$

so that we follow from the expression (9.11) that $g(b) = g(a)$ which is equivalent to $f(\mathbf{x}) = f(\mathbf{x}')$, i.e., $f(\mathbf{x})$ depends only on $x_2, \dots, x_n$.

It is easy to see that the relation (9.10) still holds by a weaker condition that E is convex in the first coordinate. Let $W_1 = \{(x, y) \in \mathbb{R}^2 \mid x > 0, y > 0\}$, $W_2 = \{(x, y) \in \mathbb{R}^2 \mid x < 0, y > 0\}$ and $E = W_1 \cup W_2$.^a Define the real-valued function $f : E \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} y, & \text{if } (x, y) \in W_1; \\ -y, & \text{if } (x, y) \in W_2. \end{cases}$$

Since W_1 and W_2 are open, E is open. Obviously, E is not convex. Furthermore, we have

$$(D_1 f)(x, y) = 0$$

for every $(x, y) \in E$. However, we have $f(1, 1) = 1$ and $f(-1, 1) = -1$ so that f depends on the first coordinate. This completes the proof of the problem. ■

Problem 9.11*Rudin Chapter 9 Exercise 11.*

Proof. By [21, Eqn. (34), p. 217], we have for $\mathbf{x} \in \mathbb{R}^n$,

$$\nabla(fg)(\mathbf{x}) = \sum_{i=1}^n (D_i(fg))(\mathbf{x})\mathbf{e}_i. \quad (9.12)$$

Since $D_i(fg)(\mathbf{x}) = f(D_i g)(\mathbf{x}) + g(D_i f)(\mathbf{x})$, the expression (9.12) reduces to

$$\nabla(fg)(\mathbf{x}) = \sum_{i=1}^n [f(\mathbf{x})(D_i g)(\mathbf{x}) + g(\mathbf{x})(D_i f)(\mathbf{x})]\mathbf{e}_i$$

^aIn other words, E is the union of the first and second quadrants which is shaped like a horseshoe.

$$\begin{aligned}
&= f(\mathbf{x}) \sum_{i=1}^n (D_i g)(\mathbf{x}) \mathbf{e}_i + g(\mathbf{x}) \sum_{i=1}^n (D_i f)(\mathbf{x}) \mathbf{e}_i \\
&= f(\mathbf{x}) \nabla(g)(\mathbf{x}) + g(\mathbf{x}) \nabla(f)(\mathbf{x}).
\end{aligned}$$

Thus we have

$$\nabla(fg) = f \nabla g + g \nabla f. \quad (9.13)$$

Suppose that $f \neq 0$ in $\mathbb{R}^n$. Since $1 = f \cdot \frac{1}{f}$ and $\nabla(1) = 0$, we put $g = \frac{1}{f}$ in the identity (9.13) to obtain

$$\begin{aligned}
0 &= f \nabla\left(\frac{1}{f}\right) + \frac{1}{f} \nabla(f) \\
\nabla\left(\frac{1}{f}\right) &= f^{-2} \nabla(f).
\end{aligned}$$

Hence, we complete the proof of the problem. ■

9.3 Local maxima and minima

Problem 9.12

Rudin Chapter 9 Exercise 12.

Proof. The range K of $\mathbf{f}$ is in fact the (ring) torus generated by rotating the circle given by

$$(x - b)^2 + z^2 = a^2$$

about the z -axis, see Figure 9.1 for an example produced by WolframAlpha^b with $a = 1$ and $b = 2$.

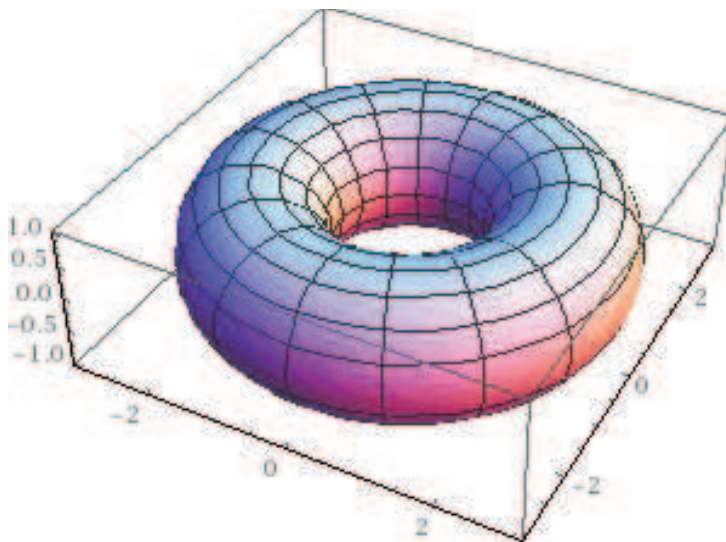


Figure 9.1: An example of the range K of $\mathbf{f}$.

(a) By definition, we have

$$(\nabla f_1)(\mathbf{x}) = (D_1 f_1)(\mathbf{x}) \mathbf{e}_1 + (D_2 f_1)(\mathbf{x}) \mathbf{e}_2 = (-a \sin s \cos t, -(b + a \cos s) \sin t),$$

where $\mathbf{x} = (s, t)$. Thus $(\nabla f_1)(\mathbf{x}) = \mathbf{0}$ if and only if

$$\sin s \cos t = 0 \quad \text{and} \quad (b + a \cos s) \sin t = 0. \quad (9.14)$$

^bSee <https://www.wolframalpha.com/>.

Since $b > a > 0$, we have $(b + a \cos s) > 0$ for any s , so the second equation in (9.14) implies that $t = 0$ or π . In both cases, the first equation in (9.14) shows that $s = 0$ or $s = \pi$. Hence there are *exactly* four points $\mathbf{p} \in K$ such that $(\nabla f_1)(\mathbf{f}^{-1}(\mathbf{p})) = \mathbf{0}$. In fact, they are the images of the four points $(0, 0)$, $(0, \pi)$, $(\pi, 0)$ and (π, π) which are

$$\mathbf{f}(0, 0) = (a + b, 0, 0), \quad \mathbf{f}(0, \pi) = (-a - b, 0, 0), \quad \mathbf{f}(\pi, 0) = (b - a, 0, 0)$$

and

$$\mathbf{f}(\pi, \pi) = (-(b - a), 0, 0).$$

(b) Again, by definition, we have

$$(\nabla f_3)(\mathbf{x}) = (D_1 f_3)(\mathbf{x})\mathbf{e}_1 + (D_2 f_3)(\mathbf{x})\mathbf{e}_2 = (a \cos s, 0),$$

where $\mathbf{x} = (s, t)$. Thus $(\nabla f_3)(\mathbf{x}) = \mathbf{0}$ if and only if $\cos s = 0$ if and only if $s = \frac{\pi}{2}$ or $s = \frac{3\pi}{2}$. Therefore, the required set is given by

$$\begin{array}{ll} f_1(\frac{\pi}{2}, t) = b \cos t, & f_1(\frac{3\pi}{2}, t) = b \cos t, \\ f_2(\frac{\pi}{2}, t) = b \sin t, & \text{or } f_2(\frac{3\pi}{2}, t) = b \sin t, \\ f_3(\frac{\pi}{2}, t) = a & f_3(\frac{3\pi}{2}, t) = -a. \end{array} \quad (9.15)$$

Geometrically, the locus of the left-hand side in (9.15) is the circle center at $z = a$ with radius b and the locus of the right-hand side in (9.15) is the circle center at $z = -a$ with radius b . The graph of the loci can be seen in Figure 9.2.

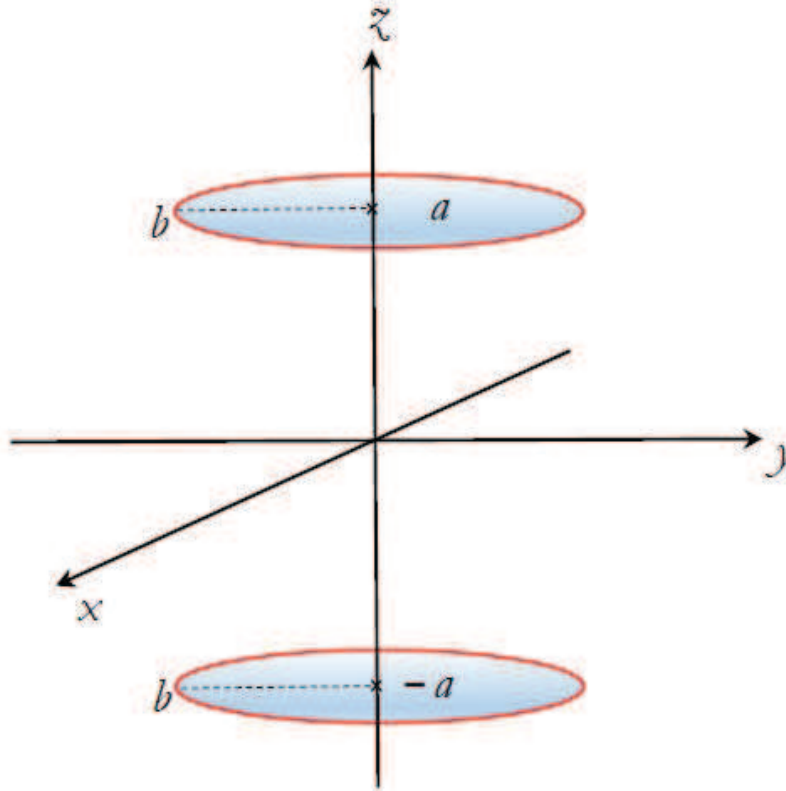


Figure 9.2: The set of $\mathbf{q} \in K$ such that $(\nabla f_3)(\mathbf{f}^{-1}(\mathbf{q})) = \mathbf{0}$.

(c) For any $(s, t) \in [0, 2\pi] \times [0, 2\pi]$, we have

$$-(a + b) \leq f_1(s, t) \leq a + b.$$

Since

$$f_1(0,0) = a + b, \quad f_1(0,\pi) = -(a + b), \quad f_1(\pi,0) = b - a \quad \text{and} \quad f_1(\pi,\pi) = -(b - a),$$

the points $(0,0)$ and $(0,\pi)$ correspond to the local maximum $(a + b, 0, 0)$ and the local minimum $(-(a + b), 0, 0)$ of f_1 respectively. Finally, it is easy to see that any of the remaining two points is neither a local maximum or a local minimum.

For $f_3(s, t)$, we know that

$$-a \leq f_3(s, t) \leq a$$

for all $(s, t) \in [0, 2\pi] \times [0, 2\pi]$. Since

$$f_3\left(\frac{3\pi}{2}, t\right) = -a \quad \text{and} \quad f_3\left(\frac{\pi}{2}, t\right) = a$$

for every $t \in [0, 2\pi]$, the points a and $-a$ are obviously the local maximum and the local minimum of f_3 respectively.

(d) We have $\mathbf{g} : \mathbb{R} \rightarrow \mathbb{R}^3$. By definition, we have

$$\mathbf{g}(t) = \mathbf{f}(t, \lambda t) = ((b + a \cos t) \cos \lambda t, (b + a \cos t) \sin \lambda t, a \sin t) \quad (9.16)$$

which implies that

$$\mathbf{g}'(t) = (-\lambda(b + a \cos t) \sin \lambda t - a \sin t \cos \lambda t, \lambda(b + a \cos t) \cos \lambda t - a \sin t \sin \lambda t, a \cos t).$$

Therefore, we have

$$\begin{aligned} |\mathbf{g}'(t)|^2 &= \mathbf{g}'(t) \cdot \mathbf{g}'(t) \\ &= [-\lambda(b + a \cos t) \sin \lambda t - a \sin t \cos \lambda t]^2 \\ &\quad + [\lambda(b + a \cos t) \cos \lambda t - a \sin t \sin \lambda t]^2 + a^2 \cos^2 t \\ &= \lambda^2(b + a \cos t)^2 + a^2 \sin^2 t + a^2 \cos^2 t \\ &= a^2 + \lambda^2(b + a \cos t)^2. \end{aligned}$$

This proves the second assertion of this part.

For the first assertion, we note from the definition (9.16) that if $\mathbf{g}(u) = \mathbf{g}(v)$ for $u, v \in \mathbb{R}$, then we have $\sin u = \sin v$ which means that

$$u = v + 2k\pi$$

for some $k \in \mathbb{Z}$. Since $b + a \cos t \geq b - a > 0$ for every $t \in [0, 2\pi]$, we have

$$\begin{aligned} (b + a \cos u) \cos \lambda u &= (b + a \cos v) \cos \lambda v \\ (b + a \cos v) \cos \lambda(v + 2k\pi) &= (b + a \cos v) \cos \lambda v \\ \cos \lambda(v + 2k\pi) &= \cos \lambda v \\ \lambda(v + 2k\pi) &= \lambda v + 2m\pi \\ k\lambda &= m \end{aligned}$$

for some $m \in \mathbb{Z}$. If $k \neq 0$, then $\lambda = \frac{m}{k} \in \mathbb{Q}$, a contradiction. Thus $k = 0$ and $m = 0$, so we have $u = v$, i.e., $\mathbf{g}$ is 1-1.

Finally, we show that $\mathbf{g}(\mathbb{R})$ is dense in K and we divide its proof into several steps:

- **Step 1: Rephrasing the problem.** Pick $\mathbf{x} \in K$. For every $\epsilon > 0$, there exists a $t \in \mathbb{R}$ such that

$$|\mathbf{g}(t) - \mathbf{x}| < \epsilon.$$

To this end, suppose that $\mathbf{x} = ((b + a \cos p) \cos q, (b + a \cos p) \sin q, a \sin p)$, where $p, q \in \mathbb{R}$. By the definition (9.16), we have

$$\begin{aligned} |\mathbf{g}(t) - \mathbf{x}|^2 &= [(b + a \cos t) \cos \lambda t - (b + a \cos p) \cos q]^2 \\ &\quad + [(b + a \cos t) \sin \lambda t - (b + a \cos p) \sin q]^2 \\ &\quad + (a \sin t - a \sin p)^2. \end{aligned} \quad (9.17)$$

- **Step 2: Simplification of the expression (9.17).** Now the expression in the right-hand side of (9.17) is too complicated for computation, so we need to simplify it. Since $\sin t$ and $\cos t$ are periodic functions, if we take

$$t = p + 2n\pi$$

for some integer n , then the expression (9.17) reduces to

$$\begin{aligned} |\mathbf{g}(p + 2n\pi) - \mathbf{x}|^2 &= [(b + a \cos p) \cos \lambda(p + 2n\pi) - (b + a \cos p) \cos q]^2 \\ &\quad + [(b + a \cos p) \sin \lambda(p + 2n\pi) - (b + a \cos p) \sin q]^2 \\ &\leq (b + a)^2 \left\{ [\cos(\lambda(p + 2n\pi) - 2m\pi) - \cos q]^2 \right. \\ &\quad \left. + [\sin(\lambda(p + 2n\pi) - 2m\pi) - \sin q]^2 \right\} \end{aligned} \quad (9.18)$$

for some $m \in \mathbb{Z}$. (The reason why the term $2m\pi$ is inserted in the inequality (9.18) will be clear very soon in **Step 3** and **Step 4** below.)

- **Step 3: Transformation of the problem.** The expression in the right-hand side of the inequality (9.18) is quite simple now, but we can do much better. In fact, by [21, Eqn. (49), p. 182] and Definition 5.1, we have

$$|\cos t - \cos x| \leq |t - x| \quad \text{and} \quad |\sin t - \sin x| \leq |t - x|$$

as $t \rightarrow x$. Thus we can further reduce the inequality (9.18) to

$$|\mathbf{g}(p + 2n\pi) - \mathbf{x}|^2 \leq 2(b + a)^2 |\lambda(p + 2n\pi) - 2m\pi - q|^2. \quad (9.19)$$

If we can show that for every $\epsilon > 0$, there exist $m, n \in \mathbb{Z}$ such that

$$|2n\lambda\pi - 2m\pi + \lambda p - q| < \frac{\epsilon}{\sqrt{2}(b + a)},^c \quad (9.20)$$

then the inequalities (9.19) and (9.20) imply that

$$|\mathbf{g}(p + 2n\pi) - \mathbf{x}| < \epsilon \quad (9.21)$$

for some integer n .

Let's summarize what we have done so far. We have shown that our original problem

$$|\mathbf{g}(t) - \mathbf{x}| < \epsilon$$

follows immediately from the validity of the inequality (9.20) for a sequence of integers, so **Step 4** comes into play.

- **Step 4: Application of the Kronecker's Approximation Theorem (Lemma 4.6).** To show that (9.20) holds, we need a previous result in Chapter 4. That is the Kronecker's Approximation Theorem (Lemma 4.6): *Let $0 < \theta < 1$. Given $\epsilon > 0$, there exists $n \in \mathbb{N}$ such that*

$$|n\lambda - h - \theta| < \frac{\epsilon}{2\sqrt{2}\pi(b + a)}, \quad (9.22)$$

where a, b are constants with $0 < a < b$ and $h = [n\lambda] \in \mathbb{Z}$. We notice that the θ in the inequality (9.22) can be taken to be any real number. In particular, we substitute $\theta = \frac{q - \lambda p}{2\pi}$ in the inequality (9.22) to get

$$|2n\lambda\pi - 2h\pi + (\lambda p - q)| < \frac{\epsilon}{\sqrt{2}(b + a)}$$

which is exactly the inequality (9.20) with $m = [n\lambda]$.

^cThis inequality may not hold without the existence of the term $2m\pi$. For example, if $p = q = \frac{\pi}{2}$ and $\lambda = \sqrt{2}$, then we have $|2\sqrt{2}n\pi + \frac{\pi}{2}(\sqrt{2} - 1)|$ which cannot be arbitrary small.

Hence the set $\mathbf{g}(\mathbb{R})$ is dense in K .

This completes the proof of the problem.^d ■

Problem 9.13

Rudin Chapter 9 Exercise 13.

Proof. Suppose that $\mathbf{f} : \mathbb{R} \rightarrow \mathbb{R}^3$ is given by

$$\mathbf{f}(t) = (f_1(t), f_2(t), f_3(t)).$$

Now the differentiability of $\mathbf{f}$ implies the differentiability of f_1, f_2 and f_3 by Remark 5.16. Since $\mathbf{f}(t) \cdot \mathbf{f}(t) = 1$, we have $f_1(t)f_1(t) + f_2(t)f_2(t) + f_3(t)f_3(t) = 1$ and thus

$$\begin{aligned} \frac{d}{dt}[\mathbf{f}(t) \cdot \mathbf{f}(t)] &= 0 \\ \frac{d}{dt}[f_1(t)f_1(t) + f_2(t)f_2(t) + f_3(t)f_3(t)] &= 0 \\ 2f_1'(t)f_1(t) + 2f_2'(t)f_2(t) + 2f_3'(t)f_3(t) &= 0 \\ (f_1'(t), f_2'(t), f_3'(t)) \cdot (f_1(t), f_2(t), f_3(t)) &= 0 \\ \mathbf{f}'(t) \cdot \mathbf{f}(t) &= 0 \end{aligned}$$

as required, completing the proof of the problem. ■

Problem 9.14

Rudin Chapter 9 Exercise 14.

Proof.

(a) By simple computation, we have

$$(D_1 f)(x, y) = \frac{x^2(x^2 + 3y^2)}{(x^2 + y^2)^2} \quad \text{and} \quad (D_2 f)(x, y) = \frac{-2x^3 y}{(x^2 + y^2)^2},$$

where $(x, y) \neq (0, 0)$. Thus we have

$$0 \leq |(D_1 f)(x, y)| \leq \frac{x^2(3x^2 + 3y^2)}{(x^2 + y^2)^2} = \frac{3x^2}{x^2 + y^2} \leq 3$$

and the A.M. $\geq$ G.M. implies that

$$0 \leq |(D_2 f)(x, y)| = \frac{x^2(2|x||y|)}{(x^2 + y^2)^2} = \frac{x^2(2\sqrt{x^2 y^2})}{(x^2 + y^2)^2} \leq \frac{x^2(x^2 + y^2)}{(x^2 + y^2)^2} = \frac{x^2}{x^2 + y^2} = 1.$$

If $(x, y) = (0, 0)$, then we have

$$(D_1 f)(0, 0) = \lim_{t \rightarrow 0} \frac{f(t, 0) - f(0, 0)}{t} = 1 \quad \text{and} \quad (D_2 f)(0, 0) = \lim_{t \rightarrow 0} \frac{f(0, t) - f(0, 0)}{t} = 0. \quad (9.23)$$

Hence both $D_1 f$ and $D_2 f$ are bounded in $\mathbb{R}^2$.

^dThis problem can also be found in [6, Exercises 11.5, pp. 265, 266].

(b) Let $\mathbf{u} = u_1 \mathbf{e}_1 + u_2 \mathbf{e}_2$, where $u_1^2 + u_2^2 = 1$. By [21, Eqn. (39), p. 217], we know that

$$\begin{aligned} (D_{\mathbf{u}}f)(0, 0) &= \lim_{t \rightarrow 0} \frac{f((0, 0) + t(u_1, u_2)) - f(0, 0)}{t} \\ &= \lim_{t \rightarrow 0} \frac{f(tu_1, tu_2)}{t} \\ &= \lim_{t \rightarrow 0} \frac{t^3 u_1^3}{t(t^2 u_1^2 + t^2 u_2^2)} \\ &= u_1^3. \end{aligned}$$

Since $\mathbf{u}$ is a unit vector, $|u_1^3| \leq |\mathbf{u}|^3 \leq 1$ and the result follows.

(c) We have $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ with $\gamma(0) = (0, 0)$ and $|\gamma'(0)| > 0$. Let $\gamma(t) = (\gamma_1(t), \gamma_2(t))$. Then we have

$$g(t) = f(\gamma(t)) = \frac{\gamma_1^3(t)}{\gamma_1^2(t) + \gamma_2^2(t)}. \quad (9.24)$$

Since γ is differentiable in $\mathbb{R}$, γ_1 and γ_2 are differentiable in $\mathbb{R}$. Thus it is easy to see from the right-hand side in the expression (9.24) that g is differentiable at every point t with $(\gamma_1(t), \gamma_2(t)) \neq (0, 0)$. Now the remaining case is the differentiability of g at the point a where

$$\gamma_1(a) = \gamma_2(a) = 0.$$

In this case, we have

$$g(a) = f(\gamma(a)) = f(0, 0) = 0$$

which gives

$$\frac{g(t) - g(a)}{t - a} = \frac{\gamma_1^3(t)}{(t - a)[\gamma_1^2(t) + \gamma_2^2(t)]} = \frac{\left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a}\right]^3}{\left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a}\right]^2 + \left[\frac{\gamma_2(t) - \gamma_2(a)}{t - a}\right]^2}, \quad (9.25)$$

where $t \neq a$. Here we must impose an addition assumption that

$$\gamma_1(a) \neq 0 \quad \text{or} \quad \gamma_2(a) \neq 0.$$

Thus it follows from the expression (9.25) that

$$g'(a) = \frac{[\gamma_1'(a)]^3}{[\gamma_1'(a)]^2 + [\gamma_2'(a)]^2}. \quad (9.26)$$

This proves our first assertion.

For the second assertion, suppose that γ is continuously differentiable on $\mathbb{R}$, i.e., γ' is continuous on $\mathbb{R}$. Then both γ_1' and γ_2' are continuous on $\mathbb{R}$. By the expression (9.24) again, we know that g' is continuous at every point t with $(\gamma_1(t), \gamma_2(t)) \neq (0, 0)$. It remains to check the continuity of g' at every point a such that

$$\gamma_1(a) = \gamma_2(a) = 0.$$

To this end, we establish from the expression (9.24) that

$$g'(t) = \frac{\gamma_1^4(t)\gamma_1'(t) + 3\gamma_1^2(t)\gamma_2^2(t)\gamma_1'(t) - 2\gamma_1^3(t)\gamma_2(t)\gamma_2'(t)}{[\gamma_1^2(t) + \gamma_2^2(t)]^2}.$$

Then we follow from the equation (9.26) that

$$\begin{aligned} \lim_{t \rightarrow a} g'(t) &= \lim_{t \rightarrow a} \frac{\gamma_1^4(t)\gamma_1'(t) + 3\gamma_1^2(t)\gamma_2^2(t)\gamma_1'(t) - 2\gamma_1^3(t)\gamma_2(t)\gamma_2'(t)}{[\gamma_1^2(t) + \gamma_2^2(t)]^2} \\ &= \lim_{t \rightarrow a} \left\{ \left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a}\right]^4 \gamma_1'(t) + 3 \left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a}\right]^2 \left[\frac{\gamma_2(t) - \gamma_2(a)}{t - a}\right]^2 \gamma_1'(t) \right. \end{aligned}$$

$$\begin{aligned}
& -2 \left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a} \right]^3 \left[\frac{\gamma_2(t) - \gamma_2(a)}{t - a} \right] \gamma_2'(t) \Big\} \times \left\{ \left[\frac{\gamma_1(t) - \gamma_1(a)}{t - a} \right]^2 + \left[\frac{\gamma_2(t) - \gamma_2(a)}{t - a} \right]^2 \right\}^{-2} \\
&= \frac{[\gamma_1'(a)]^5 + 3[\gamma_1'(a)]^3[\gamma_2'(a)]^2 - 2[\gamma_1'(a)]^3[\gamma_2'(a)]^2}{\{[\gamma_1'(a)]^2 + [\gamma_2'(a)]^2\}^2} \\
&= \frac{[\gamma_1'(a)]^3}{[\gamma_1'(a)]^2 + [\gamma_2'(a)]^2} \\
&= g'(a).
\end{aligned}$$

Hence g' is continuous at a and our desired result follows.^e

- (d) Assume that f was differentiable at $(0, 0)$. Let $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$. By [21, Eqn. (40), p. 218] and the limits (9.23), we acquire

$$(D_{\mathbf{u}}f)(0, 0) = (D_1f)(0, 0)u_1 + (D_2f)(0, 0)u_2 = u_1$$

which contradicts the result of part (b).

Hence, we end the analysis of the problem. ■

Problem 9.15

Rudin Chapter 9 Exercise 15.

Proof.

- (a) If $x = 0$ or $y = 0$, then the inequality clearly holds. Suppose that $x \neq 0$ and $y \neq 0$. Apply the A.M. $\geq$ G.M. to the positive numbers x^4 and y^2 , we get the desired result.
- (b) By direct computation, we have

$$\begin{aligned}
g_\theta(t) &= f(t \cos \theta, t \sin \theta) = t^2 - 2t^3 \cos^2 \theta \sin \theta - \frac{4t^4 \cos^6 \theta \sin^2 \theta}{(t^2 \cos^4 \theta + \sin^2 \theta)^2}, \\
g'_\theta(t) &= 2t - 6t^2 \cos^2 \theta \sin \theta - \frac{16t^3 \cos^6 \theta \sin^4 \theta}{(t^2 \cos^4 \theta + \sin^2 \theta)^3}, \\
g''_\theta(t) &= 2 - 12t \cos^2 \theta \sin \theta - \frac{48t^2 \cos^6 \theta \sin^4 \theta (\sin^2 \theta - t^2 \cos^4 \theta)}{(t^2 \cos^4 \theta + \sin^2 \theta)^4}.
\end{aligned}$$

Therefore, we have

$$g_\theta(0) = 0, \quad g'_\theta(0) = 0 \quad \text{and} \quad g''_\theta(0) = 2.$$

Hence, for each $\theta \in [0, 2\pi]$, the function g_θ has a strict local minimum at $t = 0$.

- (c) By direct computation, it is clear that

$$f(x, x^2) = -x^4 < 0 = f(0, 0)$$

so that the point $(0, 0)$ is not a local minimum for f .

This completes the proof of the problem. ■

^eWe remark that the condition $|\gamma'(0)| > 0$ has not been used in our argument, so the author wonders that this may happen to be a typo and the correct condition *should be* $|\gamma'(t)| > 0$ for all $t \in \mathbb{R}$. However, the author can't find a counterexample to show that the condition $|\gamma'(0)| > 0$ for all $t \in \mathbb{R}$ is necessary for the first assertion. Thus, can someone find a differentiable curve γ in $\mathbb{R}^2$ with $\gamma(0) = 0$, $|\gamma'(0)| > 0$ and $\gamma(1) = 0$ (take $a = 1$ for example), *but g is differentiable at 1?*

9.4 The inverse function theorem and the implicit function theorem

Problem 9.16*Rudin Chapter 9 Exercise 16.*

Proof. We remark that this exercise was also been discussed in Hardy's book [10, p. 236]. In this book, Hardy used the function $\phi(x) = \alpha x + x^2 \sin \frac{1}{x}$ instead of the function $f(t)$ Rudin used here.

By [21, Example 5.6(b), p. 106] and the fact that

$$\lim_{t \rightarrow 0} t \sin \frac{1}{t},$$

we have

$$f(0) = 0 \quad \text{and} \quad f'(0) = \lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t} = \lim_{t \rightarrow 0} \left(1 + 2t \sin \frac{1}{t}\right) = 1.$$

Furthermore, for all $t \neq 0$, we have

$$f'(t) = 1 + 4t \sin \frac{1}{t} - 2 \cos \frac{1}{t}. \quad (9.27)$$

Therefore, we have $|f'(t)| \leq 7$ for all $t \in (-1, 1)$. Now we put $t = \frac{1}{2k\pi}$ into the derivative (9.27), where $k \in \mathbb{N}$ and $k \rightarrow \infty$, so

$$f'\left(\frac{1}{2k\pi}\right) = 1 + \frac{4}{2k\pi} \sin(2k\pi) - 2 \cos(2k\pi) = -1 \quad (9.28)$$

and then

$$\lim_{k \rightarrow \infty} f'\left(\frac{1}{2k\pi}\right) = -1 \neq f'(0),$$

i.e., f' is not continuous at 0.

Assume that f was one-to-one in $(-\delta, \delta)$ for some $\delta > 0$. Since f is continuous on $(-\delta, \delta)$, it follows from Lemma 6.1 that f is monotonic on any $[a, b] \subseteq (-\delta, \delta)$, where $a < b$.

- **f is monotonically increasing in $(-\delta, \delta)$.** Suppose that t is an arbitrary point in $(-\delta, \delta)$. Then we have

$$f(x) - f(t) \begin{cases} \geq 0, & \text{if } x - t > 0; \\ \leq 0, & \text{if } x - t < 0. \end{cases}$$

Both cases imply that

$$\frac{f(x) - f(t)}{x - t} \geq 0 \quad (9.29)$$

for all $x, t \in (-\delta, \delta)$ with $x \neq t$. Therefore, we must have

$$f'(t) \geq 0$$

on $(-\delta, \delta)$. If the positive integer k is chosen such that $\frac{1}{2k\pi} \in (0, \delta)$, then the inequality (9.29) is contradictory to the result (9.28).

- **f is monotonically decreasing in $(-\delta, \delta)$.** Instead of the inequality (9.29), we have

$$\frac{f(x) - f(t)}{x - t} \leq 0 \quad (9.30)$$

for all $x, t \in (-\delta, \delta)$ with $x \neq t$. In this case, we have

$$f'(t) \leq 0$$

on $(-\delta, \delta)$. Now if we take the positive integer k to be large enough so that $\frac{1}{(2k+1)\pi} \in (0, \delta)$, then we derive from the expression (9.27) that

$$f'\left(\frac{1}{(2k+1)\pi}\right) = 1 + \frac{4}{(2k+1)\pi} \sin(2k+1)\pi - 2 \cos(2k+1)\pi = 1 + 2 = 3 \quad (9.31)$$

which gives a contradiction.

Hence f is not one-to-one in any neighborhood of 0 and this completes the proof of the problem. ■

Problem 9.17

Rudin Chapter 9 Exercise 17.

Proof.

(a) Since $f_1^2(x, y) + f_2^2(x, y) = e^{2x}(\cos^2 y + \sin^2 y) = e^{2x}$, the range of $\mathbf{f}$ is $\mathbb{R}^2 \setminus \{(0, 0)\}$.

(b) Now we have

$$D_1 f_1 = e^x \cos y, \quad D_2 f_1 = -e^x \sin y, \quad D_1 f_2 = e^x \sin y \quad \text{and} \quad D_2 f_2 = e^x \cos y$$

so that

$$[\mathbf{f}'(x, y)] = \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}, \quad (9.32)$$

where $(x, y) \in \mathbb{R}^2$. Hence we have

$$J_{\mathbf{f}}(x, y) = \det[\mathbf{f}'(x, y)] = \begin{vmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{vmatrix} = e^{2x} \neq 0$$

for every $(x, y) \in \mathbb{R}^2$.

By Theorem 9.36, the linear operator $\mathbf{f}'(x, y)$ is invertible for every $(x, y) \in \mathbb{R}^2$ and then we deduce from Theorem 9.24 (The Inverse Function Theorem) that there exists a neighborhood of (x, y) such that $\mathbf{f}$ is one-to-one. However, we note that

$$\mathbf{f}(x, y + 2\pi) = (e^x \sin(y + 2\pi), e^x \cos(y + 2\pi)) = (e^x \sin y, e^x \cos y) = \mathbf{f}(x, y),$$

so $\mathbf{f}$ is not one-to-one on $\mathbb{R}^2$.

(c) We have $\mathbf{b} = \mathbf{f}(0, \frac{\pi}{3}) = (\cos \frac{\pi}{3}, \sin \frac{\pi}{3}) = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Now we want the formula

$$\mathbf{g}(e^x \cos y, e^x \sin y) = (x, y)$$

holds in a neighborhood of $\mathbf{a}$. It is easy to see that if $p = e^x \cos y$ and $q = e^x \sin y$, then we have

$$x = \log \sqrt{p^2 + q^2} \quad \text{and} \quad y = \tan^{-1} \frac{q}{p},$$

where $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Thus the mapping $\mathbf{g}$ is given by

$$\mathbf{g}(p, q) = \left(\log \sqrt{p^2 + q^2}, \tan^{-1} \frac{q}{p} \right).$$

On the one hand, we have from the matrix (9.32) that

$$\begin{aligned} [\mathbf{f}'(\mathbf{g}(p, q))] &= \left[\mathbf{f}' \left(\log \sqrt{p^2 + q^2}, \tan^{-1} \frac{q}{p} \right) \right] \\ &= \begin{pmatrix} \sqrt{p^2 + q^2} \times \frac{p}{\sqrt{p^2 + q^2}} & -\sqrt{p^2 + q^2} \times \frac{q}{\sqrt{p^2 + q^2}} \\ \sqrt{p^2 + q^2} \times \frac{q}{\sqrt{p^2 + q^2}} & \sqrt{p^2 + q^2} \times \frac{p}{\sqrt{p^2 + q^2}} \end{pmatrix} \\ &= \begin{pmatrix} p & -q \\ q & p \end{pmatrix} \end{aligned}$$

which implies that

$$[\mathbf{f}'(\mathbf{g}(p, q))]^{-1} = \frac{1}{p^2 + q^2} \begin{pmatrix} p & q \\ -q & p \end{pmatrix}. \quad (9.33)$$

On the other hand, we have

$$D_1g_1 = \frac{p}{p^2 + q^2}, \quad D_2g_1 = \frac{q}{p^2 + q^2}, \quad D_1g_2 = -\frac{q}{p^2 + q^2} \quad \text{and} \quad D_2g_2 = \frac{p}{p^2 + q^2}$$

so that

$$[\mathbf{g}'(p, q)] = \begin{pmatrix} \frac{p}{p^2+q^2} & \frac{q}{p^2+q^2} \\ -\frac{q}{p^2+q^2} & \frac{p}{p^2+q^2} \end{pmatrix} = \frac{1}{p^2 + q^2} \begin{pmatrix} p & q \\ -q & p \end{pmatrix} \quad (9.34)$$

which is exactly the matrix (9.33).

Finally, it is easy to check from the matrices (9.32) and (9.34) that

$$[\mathbf{f}'(\mathbf{a})] = \left[\mathbf{f}'\left(0, \frac{\pi}{3}\right) \right] = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} \quad \text{and} \quad [\mathbf{g}'(\mathbf{b})] = \left[\mathbf{g}'\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \right] = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}.$$

(d) For the vertical line $x = c$ for some constant c , we have

$$\mathbf{f}(c, y) = (e^c \cos y, e^c \sin y).$$

Let $X = e^c \cos y$ and $Y = e^c \sin y$. Then we have

$$X^2 + Y^2 = e^{2c}$$

which is the circle centered at the origin with radius e^c in the XY -plane.

For the horizontal line $y = c$ for some constant c , we have

$$\mathbf{f}(x, c) = (e^x \cos c, e^x \sin c).$$

Let $X = e^x \cos c$ and $Y = e^x \sin c$. If $c \neq \frac{n\pi}{2}$, where $n \in \mathbb{Z}$, then the locus of the image is given by

$$Y = (\tan c)X$$

which is the straight line but *not* passing through the origin in the XY -plane. If c is a multiple of $\frac{\pi}{2}$, then there are two cases:

– **Case (i):** $c = \frac{(2m+1)\pi}{2}$ for some $m \in \mathbb{Z}$. Then we have

$$X = 0 \quad \text{and} \quad Y = (-1)^m e^x$$

which is the positive or negative Y -axis depending on the value of m .

– **Case (ii):** $c = m\pi$ for some $m \in \mathbb{Z}$. In this case, we have

$$X = (-1)^m e^x \quad \text{and} \quad Y = 0$$

which is the positive or negative X -axis depending on the value of m .

We finish the proof of the problem. ■

Problem 9.18

Rudin Chapter 9 Exercise 18.

Proof. Let $\mathbf{f} = (f_1, f_2)$ be the mapping of $\mathbb{R}^2$ into $\mathbb{R}^2$ given by

$$f_1(x, y) = x^2 - y^2 \quad \text{and} \quad f_2(x, y) = 2xy.$$

(a) Given that $u = x^2 - y^2$ and $v = 2xy$. If $u = v = 0$, then we have $x = y = 0$. In this case, we have

$$\mathbf{f}(0, 0) = (0, 0).$$

Suppose that $(u, v) \neq (0, 0)$. There are two cases for consideration.

- **Case (i):** $v \neq 0$. Then both x and y are non-zero. Let $x = \frac{v}{2y}$. By putting this into the first equation, we have

$$u = \frac{v^2}{4y^2} - x^2$$

which implies that

$$4y^4 + 4uy^2 - v^2 = 0.$$

By the quadratic formula, we have

$$y^2 = \frac{-u \pm \sqrt{u^2 + v^2}}{2}. \quad (9.35)$$

Next, by taking the positive value on the right-hand side in the solutions (9.35), we have

$$y = \sqrt{\frac{u + \sqrt{u^2 + v^2}}{2}} \quad (9.36)$$

and then

$$x = \frac{v}{2} \times \frac{1}{\sqrt{\frac{u + \sqrt{u^2 + v^2}}{2}}} \times \frac{\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}}{\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}} = \frac{v}{2} \times \frac{\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}}{\sqrt{\frac{u^2 + v^2 - u^2}{4}}} = \operatorname{sgn}(v) \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}. \quad (9.37)$$

Therefore, if x and y are given by the equations (9.36) and (9.37), then we must have

$$\mathbf{f}(x, y) = (u, v).$$

- **Case (ii):** $v = 0$. Then we have

$$\begin{cases} f(\sqrt{u}, 0) = (u, 0), & \text{if } u > 0; \\ f(0, \sqrt{-u}) = (u, 0), & \text{if } u < 0. \end{cases}$$

Hence we have shown that $\mathbf{f}(\mathbb{R}^2) = \mathbb{R}^2$.

- (b) Since

$$D_1 f_1 = 2x, \quad D_2 f_1 = -2y, \quad D_1 f_2 = 2y \quad \text{and} \quad D_2 f_2 = 2x, \quad (9.38)$$

we have

$$J_{\mathbf{f}}(x, y) = \det[\mathbf{f}'(x, y)] = \begin{vmatrix} 2x & -2y \\ 2y & 2x \end{vmatrix} = 4(x^2 + y^2).$$

Thus $J_{\mathbf{f}}(x, y) \neq 0$ if and only if $(x, y) \neq (0, 0)$, and then it follows from Theorem 9.24 (Inverse Function Theorem) that every point of $\mathbb{R}^2 \setminus \{(0, 0)\}$ has a neighborhood in which $\mathbf{f}$ is one-to-one. However, it is clear that $\mathbf{f}$ is not one-to-one because

$$\mathbf{f}(1, -1) = (0, 2) = \mathbf{f}(-1, 1).$$

- (c) Suppose that $\mathbf{a} = (2, 1)$. Then we have $\mathbf{b} = \mathbf{f}(2, 1) = (3, 4)$. Let $\mathbf{g}$ be the continuous inverse of $\mathbf{f}$, defined in a neighborhood of $\mathbf{b}$, such that $\mathbf{g}(\mathbf{b}) = \mathbf{a}$. By the analysis of part (a), we know the explicit formula for $\mathbf{g}$:

$$\mathbf{g}(u, v) = \left(\sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}}, \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} \right).$$

On the one hand, we note from the partial derivatives (9.38) that

$$[\mathbf{f}'(x, y)] = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}$$

so that

$$[\mathbf{f}'(\mathbf{g}(u, v))] = 2 \begin{pmatrix} \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} & -\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} \\ \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} & \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} \end{pmatrix}$$

and then

$$[\mathbf{f}'(\mathbf{g}(u, v))]^{-1} = \frac{1}{2\sqrt{u^2 + v^2}} \begin{pmatrix} \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} & \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} \\ -\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} & \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} \end{pmatrix}. \quad (9.39)$$

On the other hand, we have

$$\begin{aligned} D_1 g_1 &= \frac{1}{4} \sqrt{\frac{2}{\sqrt{u^2 + v^2} + u}} \left(\frac{u}{\sqrt{u^2 + v^2}} + 1 \right), & D_2 g_1 &= \frac{1}{4} \sqrt{\frac{2}{\sqrt{u^2 + v^2} + u}} \times \frac{v}{\sqrt{u^2 + v^2}}, \\ D_1 g_2 &= \frac{1}{4} \sqrt{\frac{2}{\sqrt{u^2 + v^2} - u}} \left(\frac{u}{\sqrt{u^2 + v^2}} - 1 \right), & D_2 g_2 &= \frac{1}{4} \sqrt{\frac{2}{\sqrt{u^2 + v^2} - u}} \times \frac{v}{\sqrt{u^2 + v^2}}. \end{aligned}$$

After simplification, these expressions become

$$\begin{aligned} D_1 g_1 &= \frac{1}{2\sqrt{u^2 + v^2}} \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}}, & D_2 g_1 &= \frac{1}{2\sqrt{u^2 + v^2}} \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}, \\ D_1 g_2 &= -\frac{1}{2\sqrt{u^2 + v^2}} \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}}, & D_2 g_2 &= \frac{1}{2\sqrt{u^2 + v^2}} \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}}. \end{aligned}$$

Therefore, we have

$$[\mathbf{g}'(u, v)] = \frac{1}{2\sqrt{u^2 + v^2}} \begin{pmatrix} \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} & \sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} \\ -\sqrt{\frac{\sqrt{u^2 + v^2} - u}{2}} & \sqrt{\frac{\sqrt{u^2 + v^2} + u}{2}} \end{pmatrix}$$

which is exactly the matrix (9.39).

Finally, it is easy to see that

$$[\mathbf{f}'(\mathbf{a})] = [\mathbf{f}'(2, 1)] = \begin{pmatrix} 4 & -2 \\ 2 & 4 \end{pmatrix} \quad \text{and} \quad [\mathbf{g}'(\mathbf{b})] = [\mathbf{g}'(3, 4)] = \frac{1}{10} \begin{pmatrix} \sqrt{\frac{13}{2}} & \sqrt{\frac{7}{2}} \\ -\sqrt{\frac{7}{2}} & \sqrt{\frac{13}{2}} \end{pmatrix}.$$

This completes the proof of the problem. ■

Problem 9.19

Rudin Chapter 9 Exercise 19.

Proof. The system of equations is equivalent to the mapping $\mathbf{f} : \mathbb{R}^{3+1} \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{f}(x, y, z, u) = (3x + y - z + u^2, x - y + 2z + u, 2x + 2y - 3z + 2u). \quad (9.40)$$

If $\mathbf{a} = (x, y, z) = (0, 0, 0)$ and $\mathbf{b} = u = 0$, then we have $\mathbf{f}(\mathbf{a}, \mathbf{b}) = (0, 0, 0)$. Therefore, its corresponding matrix at $(0, 0, 0, 0)$ is given by

$$[\mathbf{f}'(0, 0, 0, 0)] = \begin{pmatrix} 3 & 1 & -1 & 0 \\ 1 & -1 & 2 & 1 \\ 2 & 2 & -3 & 2 \end{pmatrix}. \quad (9.41)$$

The determinants of the submatrices

$$\begin{pmatrix} 3 & 1 & 0 \\ 1 & -1 & 1 \\ 2 & 2 & 2 \end{pmatrix}, \quad \begin{pmatrix} 3 & -1 & 0 \\ 1 & 2 & 1 \\ 2 & -3 & 2 \end{pmatrix}, \quad \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 2 & -3 & 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 3 & 1 & -1 \\ 1 & -1 & 2 \\ 2 & 2 & -3 \end{pmatrix}$$

are

$$-12, \quad 21, \quad 3 \quad \text{and} \quad 0$$

respectively. Hence, it follows from Theorem 9.28 (Implicit Function Theorem) that the system of equations can be solved for x, y, u in terms of z ; for x, z, u in terms of y ; for y, z, u in terms of x ; but *not* for x, y, z in terms of u . This completes the proof of the problem. ■

Problem 9.20

Rudin Chapter 9 Exercise 20.

Proof. Let us restate the implicit function theorem in the case $n = m = 1$ first. Suppose that $E \subseteq \mathbb{R}^2$ is an open set, $f : E \rightarrow \mathbb{R}$ is a $\mathcal{C}'$ -mapping and (a, b) is a point in $\mathbb{R}^2$ such that $f(a, b) = 0$. Suppose, further, that $\partial_x f(a, b) \neq 0$.^f Then there exists an open set $U \subseteq \mathbb{R}^2$ and an interval $I \subseteq \mathbb{R}$ with $(a, b) \in U$ and $b \in I$, having the following property: For every $y \in I$ corresponds a unique $x \in \mathbb{R}$ such that

$$(x, y) \in U \quad \text{and} \quad f(x, y) = 0.$$

If this x is defined to be $g(y)$, where $g : I \subseteq \mathbb{R} \rightarrow \mathbb{R}$, then the function g is $\mathcal{C}'$, $g(b) = a$, $f(g(y), y) = 0$ for $y \in I$ and

$$g'(b) = -\frac{\partial_y f(a, b)}{\partial_x f(a, b)}. \quad (9.42)$$

We can interpret the implicit function theorem in two approaches:

- **Approach 1:** By the explanation of Apostol [1, pp. 373, 374], the expression $f(x, y) = 0$ does not necessarily represent a function. Then one may ask when the relation can be *solved explicitly* for x in terms of y . The implicit function theorem solves this problem *locally*. Geometrically, it means that given a point (a, b) such that $f(a, b) = 0$, if $\partial_x f(a, b) \neq 0$, then there will be an *interval* I of b such that the relation $f(x, y) = 0$ is in fact a function in I . In other words, we can find a continuously differentiable function $g : I \rightarrow \mathbb{R}$ *implicitly* such that

$$f(g(y), y) = 0,$$

i.e., x can be *solved explicitly* in terms of y in this neighborhood.

- **Approach 2:** Another way to look at the theorem is that the *level curve*

$$S = \{(x, y) \in \mathbb{R}^2 \mid f(x, y) = 0\} \quad (9.43)$$

is *locally* a graph of a function. Here the word “locally” means that for every $(a, b) \in S$, there exist an interval I of b and an open set $U \subseteq \mathbb{R}^2$ of (a, b) such that $U \cap S$ is the graph of a continuously differentiable function $x = g(y)$, i.e.,

$$U \cap S = \{(x, y) = (g(y), y) \mid y \in I\}.$$

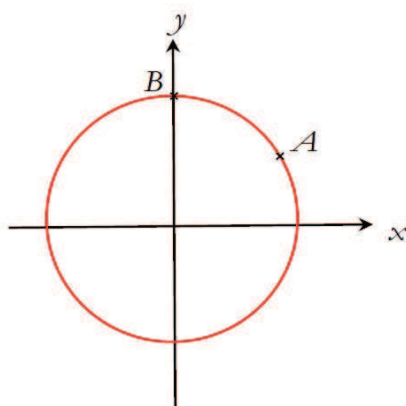


Figure 9.3: Geometric meaning of the implicit function theorem.

Furthermore, the slope of the tangent to the curve at the point (a, b) is given by the derivative (9.42). Since $\partial_x f(a, b) \neq 0$, we see that the tangent is *not* vertical.

For instance, if $f(x, y) = x^2 + y^2 - 1$, then the level curve S defined by (9.43) is the unit circle in $\mathbb{R}^2$, see Figure 9.3. Around the point $A(x_1, y_1)$, x can be expressed in terms of y , i.e., $x = g(y) = \sqrt{1 - y^2}$. Furthermore, the slope of the tangent at A is given by

$$g'(x_1) = -\frac{2y_1}{2x_1} = -\frac{y_1}{x_1}.$$

However, there is no such function around the point $B(0, 1)$ because $\partial_x f(0, 1) = 0$ so that g' is not well-defined at B .

We complete the analysis of the problem. ■

Problem 9.21

Rudin Chapter 9 Exercise 21.

Proof.

- (a) By [21, Eqn. (34), p. 217], we have $\nabla f(x, y) = (6x^2 - 6x)\mathbf{e}_1 + (6y^2 + 6y)\mathbf{e}_2$. Therefore, $\nabla f(x, y) = \mathbf{0}$ if and only if $x^2 - x = 0$ and $y^2 + y = 0$ if and only if $x = 0, 1$ and $y = 0, -1$. Hence the four points are $(0, 0)$, $(0, -1)$, $(1, 0)$ and $(1, -1)$. To find the local extreme of the function f , we need a result from calculus of several variables, see [1, Theorem 13.11, p. 379].

Lemma 9.2

Let f be a real-valued function with continuous second-order partial derivatives at a stationary point $(a, b) \in \mathbb{R}^2$. Let $A = \partial_{xx}f(a, b)$, $B = \partial_{xy}f(a, b)$, $C = \partial_{yy}f(a, b)$ and

$$\Delta = \det \begin{pmatrix} A & B \\ B & C \end{pmatrix} = AC - B^2.$$

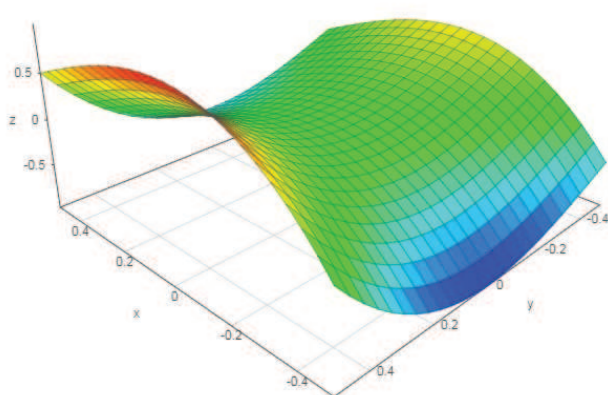
- (a) If $\Delta > 0$ and $A > 0$, then f has a locally minimum at (a, b) .
- (b) If $\Delta > 0$ and $A < 0$, then f has a locally maximum at (a, b) .
- (c) If $\Delta < 0$, then f has a saddle point at (a, b) .

^fHere the notation $\partial_x f$ means $\frac{\partial f}{\partial x}$.

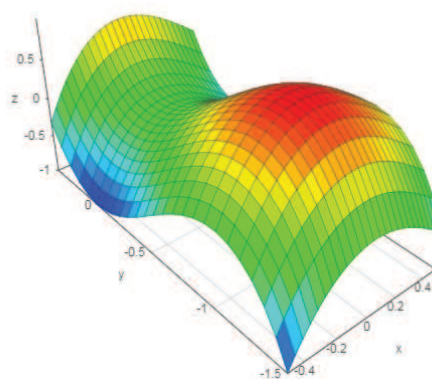
Now we have $\partial_{xx}f = 12x - 6$, $\partial_{yy}f = 12y + 6$ and $\partial_{xy}f = 0$.

- At $(0, 0)$: $A = -6, B = 0$ and $C = 6$. Since $\Delta = -36 < 0$, $(0, 0)$ is a saddle point by Lemma 9.2(c).
- At $(0, -1)$: $A = -6, B = 0$ and $C = -6$. Since $\Delta = 36 > 0$ and $A < 0$, $(0, -1)$ is a local maximum by Lemma 9.2(b).
- At $(1, 0)$: $A = 6, B = 0$ and $C = 6$. Since $\Delta = 36 > 0$ and $A > 0$, $(1, 0)$ is a local minimum by Lemma 25(a).
- At $(1, -1)$: $A = 6, B = 0$ and $C = -6$. Since $\Delta = -36 < 0$, $(1, -1)$ is a saddle point by Lemma 9.2(c).

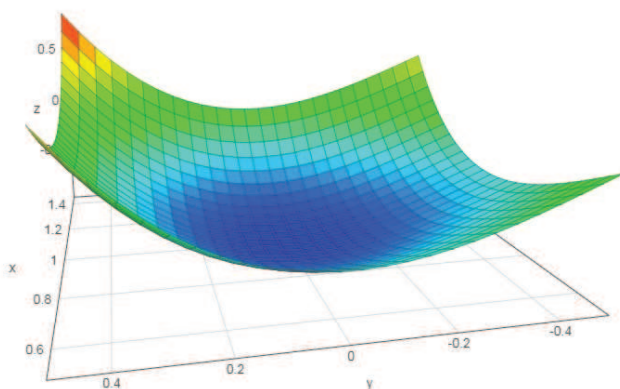
The behaviours of the four points are shown in Figures 9.4(a) to (d) below.^g



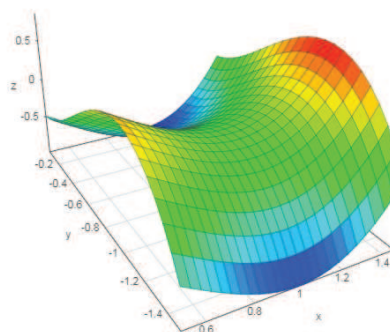
(a) The saddle point $(0, 0)$.



(b) The local maximum point $(0, -1)$.



(c) The local minimum point $(1, 0)$.



(d) The saddle point $(1, -1)$.

Figure 9.4: The graphs around the four points.

(b) By computation, we have

$$f(x, y) = (x + y)(2x^2 - 2xy + 2y^2 - 3x + 3y).$$

By definition, we have

$$\begin{aligned} S &= \{(x, y) \in \mathbb{R}^2 \mid f(x, y) = 0\} \\ &= \{(x, y) \in \mathbb{R}^2 \mid (x + y)(2x^2 - 2xy + 2y^2 - 3x + 3y) = 0\}. \end{aligned}$$

^gThe graphs in Figures 9.4 and 9.5 are produced by using the free online software “3D Surface Plotter”, see <https://academo.org/demos/3d-surface-plotter/>.

Therefore, $f(x, y) = 0$ if and only if

$$x + y = 0 \quad \text{or} \quad 2x^2 - 2xy + 2y^2 - 3x + 3y = 0. \quad (9.44)$$

In other words, points of S are exactly the zeros of one of the equations in (9.44).

- **Case (i): x as a function of y .** For every y , we have from the leftmost equation in (9.44) always gives the solution

$$x = -y. \quad (9.45)$$

Recall the fact that

$$D_1 f(x, y) = 6x(x - 1),$$

so Theorem 9.28 (Implicit Function Theorem) implies that for every $x \in \mathbb{R} \setminus \{0, 1\}$, x can be expressed as a function of y locally. When $x = 0$, it follows from the rightmost equation in (9.44) that

$$y = 0 \quad \text{or} \quad y = -\frac{3}{2};$$

when $x = 1$, we obtain

$$y = -1 \quad \text{or} \quad y = \frac{1}{2}.$$

Therefore, at points

$$(0, 0), \quad \left(0, -\frac{3}{2}\right), \quad (1, -1) \quad \text{and} \quad \left(1, \frac{1}{2}\right), \quad (9.46)$$

x *might not possibly* be expressed as a function of y .

To check whether x can be solved in terms of y around these four points, we rewrite the rightmost equation in (9.44) as

$$2x^2 - (3 + 2y)x + (2y^2 + 3y) = 0$$

so that

$$x = \frac{(3 + 2y) \pm \sqrt{(3 + 2y)^2 - 8(2y^2 + 3y)}}{4}. \quad (9.47)$$

By the forms of solutions (9.45) and (9.47), we have the following table:

Expressions of x	$y \rightarrow 0$	$y \rightarrow -\frac{3}{2}$	$y \rightarrow -1$	$y \rightarrow \frac{1}{2}$
$-y$	0	$\frac{3}{2}$	1	$-\frac{1}{2}$
$\frac{(3 + 2y) + \sqrt{3(1 - 2y)(3 + 2y)}}{4}$	$\frac{3}{2}$	0	1	1
$\frac{(3 + 2y) - \sqrt{3(1 - 2y)(3 + 2y)}}{4}$	0	0	$-\frac{1}{2}$	1

Table 9.1: Expressions of x around four points.

Hence we can conclude from Table 9.1 that x *cannot* be expressed uniquely as a function of y around these four points. For instance, as $y \rightarrow 0$, both

$$x = -y \quad \text{and} \quad x = \frac{(3 + 2y) - \sqrt{3(1 - 2y)(3 + 2y)}}{4}$$

tend to 0. This means that x has two *different* expressions around the point $(0, 0)$.

- **Case (ii): y as a function of x .** Similarly, for every x , we have

$$y = -x \quad (9.48)$$

and for $-\frac{1}{2} \leq x \leq \frac{3}{2}$, we have

$$y = \frac{(2x - 3) \pm \sqrt{3(3 - 2x)(1 + 2x)}}{4}. \quad (9.49)$$

Since

$$D_2 f(x, y) = 6y(y + 1),$$

we know from Theorem 9.8 (Implicit Function Theorem) that for every $y \in \mathbb{R} \setminus \{-1, 0\}$, y can be expressed as a function of x locally and the only points of uncertainty are

$$\left(-\frac{1}{2}, -1\right), (1, -1), (0, 0) \text{ and } \left(\frac{3}{2}, 0\right). \quad (9.50)$$

By the forms of solutions (9.48) and (9.49), we have the following table:

Expressions of y	$x \rightarrow -\frac{1}{2}$	$x \rightarrow 1$	$x \rightarrow 0$	$x \rightarrow \frac{3}{2}$
$-x$	$\frac{1}{2}$	-1	-1	$-\frac{3}{2}$
$\frac{(2x-3) + \sqrt{3(3-2x)(1+2x)}}{4}$	-1	$\frac{1}{2}$	-1	-1
$\frac{(2x-3) - \sqrt{3(3-2x)(1+2x)}}{4}$	-1	-1	$-\frac{3}{2}$	-1

Table 9.2: Expressions of y around four points.

Hence, by similar analysis as in **Case (i)**, we can conclude from Table 9.2 that y *cannot* be expressed uniquely as a function of x around the four points.

Finally, we see easily from the sets of points (9.46) and (9.50) that there is no neighborhoods around the points $(0, 0)$ and $(1, -1)$ such that $f(x, y) = 0$ cannot be solved for y in terms of x or for x in terms of y .

This completes the proof of the problem. ■

Problem 9.22

Rudin Chapter 9 Exercise 22.

Proof. We follow the flow of the proof of Problem 9.21.

(a) By [21, Eqn. (34), p.217], we have

$$\nabla f(x, y) = 6(x^2 + y^2 - x)\mathbf{e}_1 + 6y(2x + 1)\mathbf{e}_2.$$

Therefore, $\nabla f(x, y) = \mathbf{0}$ if and only if

$$x^2 + y^2 - x = 0 \quad \text{and} \quad y(2x + 1) = 0$$

if and only if^h

$$x^2 - x = 0 \quad \text{and} \quad y = 0$$

if and only if

$$x = 0, 1 \quad \text{and} \quad y = 0.$$

Hence $\nabla f(x, y) = \mathbf{0}$ only at the two points $(0, 0)$ and $(1, 0)$. Since

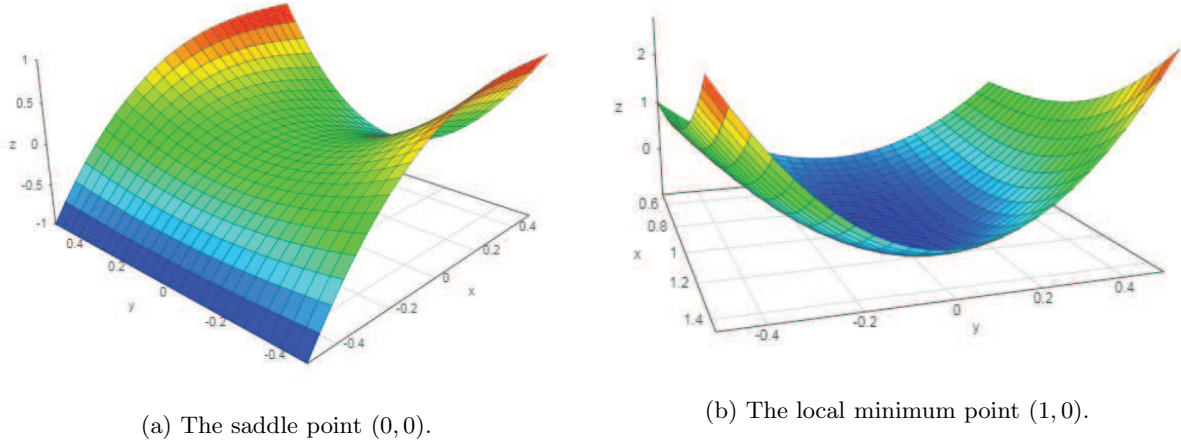
$$\partial_{xx} f = 12x - 6, \quad \partial_{yy} f = 12x + 6 \quad \text{and} \quad \partial_{xy} f = 12y.$$

– At $(0, 0)$: $A = -6$, $B = 0$ and $C = 6$. Since $\Delta = -36 < 0$, $(0, 0)$ is a saddle point by Lemma 9.2(c).

– At $(1, 0)$: $A = 6$, $B = 0$ and $C = 18$. Since $\Delta = 108 > 0$ and $A > 0$, $(1, 0)$ is a local minimum by Lemma 9.2(a).

The behaviours of the four points are shown in Figures 9.5(a) and (b) below.

^hIt is impossible that $x = -\frac{1}{2}$ because $y^2 + \frac{3}{4} = 0$ gives non-real y .

Figure 9.5: The graphs around $(0, 0)$ and $(1, 0)$.

(b) By definition, we have

$$S = \{(x, y) \in \mathbb{R}^2 \mid f(x, y) = 2x^3 - 3x^2 + 6y^2x + 3y^2 = 0\}.$$

- **Case (i): x as a function of y .** We note from Theorem 9.28 (Implicit Function Theorem) that if $D_1f(x, y) \neq 0$, then x in terms of y . Therefore, points of S that have no neighborhoods in which the equation $f(x, y) = 0$ cannot be solved for x in terms of y must satisfy

$$D_1f(x, y) = 0. \quad (9.51)$$

By the result of part (a), we know that $D_1f(x, y) = 0$ is equivalent to $y^2 = x - x^2$. Put this into the equation $f(x, y) = 0$ to reduce it to

$$3x - 4x^3 = 0$$

which gives

$$x = 0 \quad \text{or} \quad x = \pm \frac{\sqrt{3}}{2}.$$

If $x = 0$, then $y = 0$. If $x = \frac{\sqrt{3}}{2}$, then we have

$$y = \pm \frac{\sqrt{2\sqrt{3}-3}}{2}.$$

Now the point $x = -\frac{\sqrt{3}}{2}$ is rejected because $y^2 = \frac{-2\sqrt{3}-3}{4} < 0$. In conclusion, x *might possibly* be expressed as a function of y around the points

$$(0, 0) \quad \text{and} \quad \left(\frac{\sqrt{3}}{2}, \pm \frac{\sqrt{2\sqrt{3}-3}}{2}\right). \quad (9.52)$$

It is well-known that the discriminant Δ of the cubic equation

$$2x^3 - 3x^2 + 6y^2x + 3y^2 = 0 \quad (9.53)$$

is given byⁱ

$$\Delta = -108y^2(16y^4 + 24y^2 - 3) = -108y^2 \left[16 \left(y^2 + \frac{3}{4} \right)^2 - 12 \right] < 0,$$

so the equation (9.53) has one real root and a pair of complex conjugate roots for every real y . Hence x cannot be expressed uniquely as a function of y at the points (9.52).

ⁱSee https://en.wikipedia.org/wiki/Cubic_function.

- **Case (ii): y as a function of x .** Similarly, we consider $D_2f(x, y) = 0$ which is equivalent to $6y(2x + 1) = 0$. If $x = -\frac{1}{2}$, then we have

$$f\left(-\frac{1}{2}, y\right) = -1 \neq 0,$$

so $(-\frac{1}{2}, y) \notin S$ for every real y . If $y = 0$, then $f(x, 0) = 0$ if and only if $x = 0$ or $x = \frac{3}{2}$. As a result, y *might possibly* be expressed as a function of x around the points

$$(0, 0) \quad \text{and} \quad \left(\frac{3}{2}, 0\right). \quad (9.54)$$

For fixed x , we have

$$y^2 = \frac{x^2(3 - 2x)}{3(2x + 1)}$$

so that $-\frac{1}{2} < x \leq \frac{3}{2}$ for real solutions y . Hence, this shows that at the points (9.54), y cannot be solved in terms of x .

We complete the analysis of the problem. ■

Problem 9.23

Rudin Chapter 9 Exercise 23.

Proof. It is obvious that $f(0, 1, -1) = 0$. By definition, we have

$$D_1f = 2xy_1 + e^x, \quad D_2f = x^2 \quad \text{and} \quad D_3f = 1$$

and then

$$D_1f(0, 1, -1) = 1 \neq 0, \quad D_2f(0, 1, -1) = 0 \quad \text{and} \quad D_3f(0, 1, -1) = 1. \quad (9.55)$$

Therefore, we deduce from Theorem 9.28 (Implicit Function Theorem) that there exists a differentiable function g in a neighborhood of $(1, -1)$ in $\mathbb{R}^2$ such that $g(1, -1) = 0$ and

$$f(g(y_1, y_2), y_1, y_2) = 0.$$

To find $(D_1g)(1, -1)$ and $(D_2g)(1, -1)$, we derive from the note following [21, Eqn. (65), p.226] that

$$(D_1f)(0, 1, -1)(D_1g)(1, -1) = -(D_2f)(0, 1, -1)$$

and

$$(D_1f)(0, 1, -1)(D_2g)(1, -1) = -(D_3f)(0, 1, -1).$$

Hence we obtain from the values (9.55) that

$$(D_1g)(1, -1) = 0 \quad \text{and} \quad (D_2g)(1, -1) = -1.$$

This completes the proof of the problem. ■

9.5 The rank of a linear transformation

Problem 9.24

Rudin Chapter 9 Exercise 24.

Proof. We have the mapping $\mathbf{f} : \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow \mathbb{R}^2$. Thus for every $(x, y) \neq (0, 0)$, since

$$D_1 f_1 = \frac{4xy^2}{(x^2 + y^2)^2}, \quad D_2 f_1 = \frac{-4x^2 y}{(x^2 + y^2)^2}, \quad D_1 f_2 = \frac{y(y^2 - x^2)}{(x^2 + y^2)^2} \quad \text{and} \quad D_2 f_2 = \frac{x(x^2 - y^2)}{(x^2 + y^2)^2},$$

we have

$$[\mathbf{f}'(x, y)] = \begin{pmatrix} \frac{4xy^2}{(x^2 + y^2)^2} & \frac{-4x^2 y}{(x^2 + y^2)^2} \\ \frac{y(y^2 - x^2)}{(x^2 + y^2)^2} & \frac{x(x^2 - y^2)}{(x^2 + y^2)^2} \end{pmatrix}. \quad (9.56)$$

Now it is easy to see that

$$\det[\mathbf{f}'(x, y)] = \frac{1}{(x^2 + y^2)^4} [(4xy^2)x(x^2 - y^2) + (4x^2 y)y(y^2 - x^2)] = 0$$

for any $(x, y) \neq (0, 0)$. By Theorem 9.36, the matrix (9.56) is not invertible. Then we deduce from [15, Theorem 8, p. 112] that the column vectors of $[\mathbf{f}'(x, y)]$ are dependent. Since $\mathcal{R}([\mathbf{f}'(x, y)])$ is a vector space in $\mathbb{R}^2$, we have

$$\text{rank}([\mathbf{f}'(x, y)]) = \dim \mathcal{R}([\mathbf{f}'(x, y)]) \leq 2.$$

Since $\mathcal{R}([\mathbf{f}'(x, y)])$ is spanned by the column vectors of $[\mathbf{f}'(x, y)]$ (see [21, p. 210]), we follow from Theorem 9.3(a) that $\text{rank}([\mathbf{f}'(x, y)]) = 2$ is impossible. Therefore, we have either

$$\text{rank}([\mathbf{f}'(x, y)]) = 0 \quad \text{and} \quad \text{rank}([\mathbf{f}'(x, y)]) = 1.$$

Suppose that $\text{rank}([\mathbf{f}'(x, y)]) = 0$. Then Definition 9.30 says that

$$[\mathbf{f}'(x, y)](u, v) = (0, 0)$$

for all $(u, v) \in \mathbb{R}^2 \setminus \{(0, 0)\}$. If $x \neq -y$, then we choose $u = 1$ and $v = -1$ so that the matrix (9.56) implies that

$$[\mathbf{f}'(x, y)](1, -1) = \begin{pmatrix} \frac{4xy^2}{(x^2 + y^2)^2} & \frac{-4x^2 y}{(x^2 + y^2)^2} \\ \frac{y(y^2 - x^2)}{(x^2 + y^2)^2} & \frac{x(x^2 - y^2)}{(x^2 + y^2)^2} \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{(x^2 + y^2)^2} \begin{pmatrix} 4xy(x + y) \\ (x - y)(x^2 - y^2) \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If $x = -y$, then we choose $u = v = 1$ so that the matrix (9.56) gives

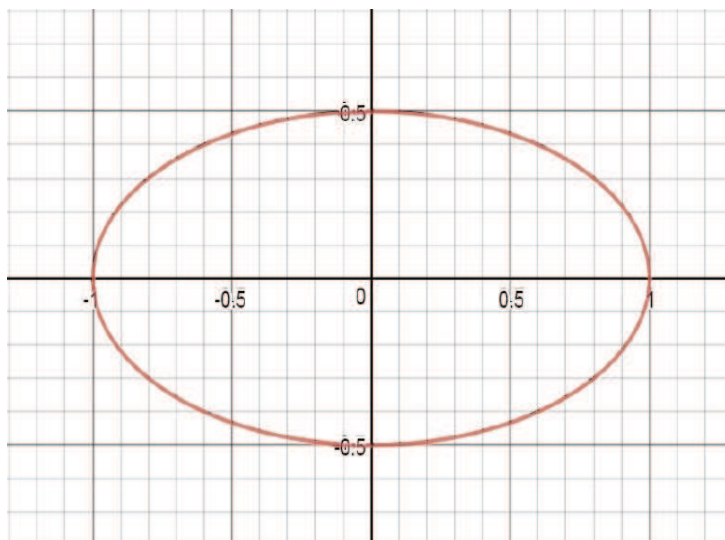
$$[\mathbf{f}'(x, -x)](1, -1) = \frac{1}{4x^4} \begin{pmatrix} 4x^3 & 4x^3 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{2}{x} \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Hence it is impossible to have $\text{rank}([\mathbf{f}'(x, y)]) = 0$ and so $\text{rank}([\mathbf{f}'(x, y)]) = 1$. This answers the first assertion.

For the second assertion, let $X = f_1(x, y)$ and $Y = f_2(x, y)$. Since

$$X^2 + 4Y^2 = f_1^2(x, y) + 4f_2^2(x, y) = \frac{x^4 - 2x^2 y^2 + y^4}{(x^2 + y^2)^2} + \frac{4x^2 y^2}{(x^2 + y^2)^2} = \frac{x^4 + 2x^2 y^2 + y^4}{(x^2 + y^2)^2} = 1,$$

the range of $\mathbf{f}$ is a *subset* of the ellipse with radii 1 and $\frac{1}{2}$ on the X and Y axes respectively, see Figure 9.6.

Figure 9.6: The graph of the ellipse $X^2 + 4Y^2 = 1$.

Now we show that the range of $\mathbf{f}$ is exactly the ellipse $X^2 + 4Y^2 = 1$. Suppose that (X, Y) is a point on the ellipse such that

$$X = \frac{x^2 - y^2}{x^2 + y^2} \quad \text{and} \quad Y = \frac{xy}{x^2 + y^2} \quad (9.57)$$

for some x and y with $(x, y) \neq (0, 0)$. Fix $x = 1$. Then we have $X = \frac{1-y^2}{1+y^2}$ which implies that

$$y = \pm \sqrt{\frac{1-X}{1+X}}, \quad (9.58)$$

where $X \neq -1$. By the expressions (9.58) and the fact that $X^2 + 4Y^2 = 1$, when $X \neq -1$, we have

$$\begin{aligned} \mathbf{f}\left(1, \pm \sqrt{\frac{1-X}{1+X}}\right) &= \left(f_1\left(1, \pm \sqrt{\frac{1-X}{1+X}}\right), f_2\left(1, \pm \sqrt{\frac{1-X}{1+X}}\right)\right) \\ &= \left(\frac{1 - \frac{1-X}{1+X}}{1 + \frac{1-X}{1+X}}, \frac{\pm \sqrt{\frac{1-X}{1+X}}}{1 + \frac{1-X}{1+X}}\right) \\ &= \left(X, \pm \frac{1}{2} \sqrt{1-X^2}\right) \\ &= (X, Y). \end{aligned}$$

If $X = -1$, then we deduce from the definition (9.57) that $x = 0$ and thus $Y = 0$. In this case, we have

$$\mathbf{f}(0, 2) = (f_1(0, 2), f_2(0, 2)) = \left(\frac{0^2 - 2^2}{0^2 + 2^2}, \frac{0 \cdot 2}{0^2 + 2^2}\right) = (-1, 0).$$

Hence the range of $\mathbf{f}$ is exactly the graph of the ellipse $X^2 + 4Y^2 = 1$. This shows the second assertion and thus completes the proof of the problem. ■

Problem 9.25

Rudin Chapter 9 Exercise 25.

Proof.

- (a) Recall the definitions from the proof of Theorem 9.32 that $Y_1 = \mathcal{R}(A)$, $\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$ is a basis of Y_1 , $\mathbf{z}_i \in \mathbb{R}^n$ is defined by

$$A\mathbf{z}_i = \mathbf{y}_i \quad (9.59)$$

for $1 \leq i \leq r$, and a linear mapping $S : Y_1 \rightarrow \mathbb{R}^n$ is given by

$$S(c_1\mathbf{y}_1 + \dots + c_r\mathbf{y}_r) = c_1\mathbf{z}_1 + \dots + c_r\mathbf{z}_r \quad (9.60)$$

for all scalars $c_1, \dots, c_r$.

We note from Theorem 9.3(c) that $\mathbb{R}^n$ has a basis containing $\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$. Let such a basis be $\{\mathbf{y}_1, \dots, \mathbf{y}_r, \mathbf{x}_{r+1}, \dots, \mathbf{x}_n\}$. Then we have

$$\mathbf{x} = c_1\mathbf{y}_1 + \dots + c_r\mathbf{y}_r + c_{r+1}\mathbf{x}_{r+1} + \dots + c_n\mathbf{x}_n$$

for some scalars $c_1, \dots, c_n$. By Definition 9.30, we have $A\mathbf{x} \in Y_1 = \mathcal{R}(A)$ so that

$$A\mathbf{x} = c_1\mathbf{y}_1 + \dots + c_r\mathbf{y}_r. \quad (9.61)$$

By using the expressions (9.59) and (9.61), we have

$$AS A\mathbf{x} = AS(c_1\mathbf{y}_1 + \dots + c_r\mathbf{y}_r) = A(c_1\mathbf{z}_1 + \dots + c_r\mathbf{z}_r) = c_1\mathbf{y}_1 + \dots + c_r\mathbf{y}_r = A\mathbf{x}. \quad (9.62)$$

Thus it deduces from the expression (9.62) that

$$SAS A\mathbf{x} = SA\mathbf{x}$$

for every $\mathbf{x} \in \mathbb{R}^n$. Hence SA is a projection in $\mathbb{R}^n$.

For the second assertion, since SA is a projection in $\mathbb{R}^n$, we follow from the property [21, p. 228] that every $\mathbf{x} \in \mathbb{R}^n$ has a unique representation of the form

$$\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2, \quad (9.63)$$

where $\mathbf{x}_1 \in \mathcal{R}(SA)$ and $\mathbf{x}_2 \in \mathcal{N}(SA)$.

To finish the proof of this part, we have to prove two steps:

- **Step 1:** $\mathcal{R}(SA) = \mathcal{R}(S)$. On the one hand, if $\mathbf{z} \in \mathcal{R}(S)$, then we have $\mathbf{z} = S\mathbf{y}$ for some $\mathbf{y} \in Y_1 = \mathcal{R}(A)$. Since $\mathbf{y} = A\mathbf{w}$ for some $\mathbf{x} \in \mathbb{R}^n$, we have

$$\mathbf{z} = S\mathbf{y} = SA\mathbf{x}.$$

Thus $\mathbf{z} \in \mathcal{R}(SA)$, i.e., $\mathcal{R}(S) \subseteq \mathcal{R}(SA)$. On the other hand, if $\mathbf{z} \in \mathcal{R}(SA)$, then $\mathbf{z} = SA\mathbf{y}$ for some $\mathbf{y} \in \mathbb{R}^n$. By definition, $A\mathbf{y} \in Y_1$ so that $\mathbf{z} \in \mathcal{R}(S)$, i.e., $\mathcal{R}(SA) \subseteq \mathcal{R}(S)$ and then

$$\mathcal{R}(SA) = \mathcal{R}(S).$$

- **Step 2:** $\mathcal{N}(SA) = \mathcal{N}(A)$. On the one hand, if $\mathbf{x} \in \mathcal{N}(A)$, then $A\mathbf{x} = \mathbf{0}$ which shows clearly that

$$SA\mathbf{x} = S\mathbf{0} = \mathbf{0}.$$

In other words, $\mathbf{x} \in \mathcal{N}(SA)$, i.e., $\mathcal{N}(A) \subseteq \mathcal{N}(SA)$. On the other hand, if $\mathbf{x} \in \mathcal{N}(SA)$, then we have $SA\mathbf{x} = \mathbf{0}$ and it follows from the expression (9.62) that

$$A\mathbf{x} = AS A\mathbf{x} = A\mathbf{0} = \mathbf{0}.$$

Thus, $\mathbf{x} \in \mathcal{N}(A)$, i.e., $\mathcal{N}(SA) \subseteq \mathcal{N}(A)$ and then

$$\mathcal{N}(SA) = \mathcal{N}(A).$$

Hence the vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ in the representation (9.63) are elements of $\mathcal{R}(S)$ and $\mathcal{N}(A)$ respectively.

(b) We divide the proof into two steps:

- **Step 1:** $\dim \mathcal{R}(S) = \dim \mathcal{R}(A)$. Let $\mathbf{y}, \mathbf{y}' \in Y_1$ be such that $S(\mathbf{y}) = S(\mathbf{y}')$. Then we have

$$S(c_1\mathbf{y}_1 + \cdots + c_r\mathbf{y}_r) = S(c'_1\mathbf{y}_1 + \cdots + c'_r\mathbf{y}_r) \quad (9.64)$$

for some scalars $c_1, \dots, c_r, c'_1, \dots, c'_r$. By definition, we have from the expression (9.64) that

$$c_1\mathbf{z}_1 + \cdots + c_r\mathbf{z}_r = c'_1\mathbf{z}_1 + \cdots + c'_r\mathbf{z}_r. \quad (9.65)$$

Apply A to both sides of the expression (9.65), we have

$$c_1\mathbf{y}_1 + \cdots + c_r\mathbf{y}_r = c'_1\mathbf{y}_1 + \cdots + c'_r\mathbf{y}_r.$$

Since $\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$ is a basis of Y_1 , we have

$$c_1 = c'_1, \dots, c_r = c'_r$$

so that S is an one-to-one linear mapping. Therefore, the mapping $S : Y_1 \rightarrow S(Y_1)$ is actually an *isomorphism*^j and we have

$$\dim \mathcal{R}(S) = \dim S(Y_1) = \dim Y_1 = \dim \mathcal{R}(A) = r. \quad (9.66)$$

- **Step 2:** $\dim \mathcal{N}(A) = n - r$. We know from part (a) that $\{\mathbf{y}_1, \dots, \mathbf{y}_r, \mathbf{x}_{r+1}, \dots, \mathbf{x}_n\}$ is a basis of $\mathbb{R}^n$. For every $\mathbf{x} \in \mathbb{R}^n$, we have

$$\mathbf{x} = c_1\mathbf{y}_1 + \cdots + c_r\mathbf{y}_r + c_{r+1}\mathbf{x}_{r+1} + \cdots + c_n\mathbf{x}_n$$

for some scalars $c_1, \dots, c_r$. In particular, if $\mathbf{x} \in \mathcal{N}(A)$, then we have $A\mathbf{x} = \mathbf{0}$ and by the expression (9.61), we have

$$c_1\mathbf{y}_1 + \cdots + c_r\mathbf{y}_r = A\mathbf{x} = \mathbf{0}.$$

Since $\{\mathbf{y}_1, \dots, \mathbf{y}_r\}$ is a basis of Y_1 , it gives

$$c_1 = c_2 = \cdots = c_r = 0.$$

Thus every $\mathbf{x} \in \mathcal{N}(A)$ is a linear combination of $\mathbf{x}_{r+1}, \dots, \mathbf{x}_n$, i.e., $\mathcal{N}(A)$ is spanned by $\{\mathbf{x}_{r+1}, \dots, \mathbf{x}_n\}$.

Next, it is clear that the set $\{\mathbf{x}_{r+1}, \dots, \mathbf{x}_n\}$ is linearly independent, so it is a basis of $\mathcal{N}(A)$ and then

$$\dim \mathcal{N}(A) = n - r, \quad (9.67)$$

as desired.

Combining the numbers (9.66) and (9.67), we have

$$\dim \mathcal{N}(A) + \dim \mathcal{R}(A) = n - r + r = n.$$

This completes the proof of the problem. ■

9.6 Derivatives of higher order

Problem 9.26

Rudin Chapter 9 Exercise 26.

^jA linear mapping $f : V \rightarrow W$ between two vector spaces is called an isomorphism if it is one-to-one and onto. See, for example, [15, p. 155].

Proof. By Theorem 7.18, there exists a real continuous function on $\mathbb{R}$ which is nowhere differentiable. Let this function be g . Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = g(x).$$

Then $D_1f(x, y)$ does not exist for every $(x, y) \in \mathbb{R}^2$, but $D_{12}f(x, y) = 0$. We have completed the proof of the problem. ■

Problem 9.27

Rudin Chapter 9 Exercise 27.

Proof. We have

$$f(x, y) = \begin{cases} 0, & \text{if } (x, y) = (0, 0); \\ \frac{xy(x^2 - y^2)}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0). \end{cases} \quad (9.68)$$

(a) If $(x, y) \neq (0, 0)$, then we have

$$D_1f(x, y) = \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \quad \text{and} \quad D_2f(x, y) = \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}. \quad (9.69)$$

Thus it is obvious that f, D_1f and D_2f are continuous at every point $(x, y) \neq (0, 0)$.

Next we have to check their continuity at the point $(0, 0)$. By the A.M. $\geq$ G.M., we have

$$|f(x, y) - f(0, 0)| = \left| \frac{xy(x^2 - y^2)}{x^2 + y^2} \right| \leq \frac{|x^2 - y^2|}{2}.$$

We observe that

$$\frac{|x^2 - y^2|}{2} \rightarrow 0$$

as $(x, y) \rightarrow (0, 0)$, so this means that f is continuous at $(0, 0)$.

By the definition (9.68) and the A.M. $\geq$ G.M., we have

$$D_1f(0, 0) = \lim_{t \rightarrow 0} \frac{f(t, 0) - f(0, 0)}{t} = 0 \quad \text{and} \quad D_2f(0, 0) = \lim_{t \rightarrow 0} \frac{f(0, t) - f(0, 0)}{t} = 0.$$

Therefore, we have from the expressions (9.69) that

$$|D_1f(x, y) - D_1f(0, 0)| = \left| \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \right| \leq |y| \left| \frac{(x^2 + y^2)^2 - 2y^4}{(x^2 + y^2)^2} \right| \leq |y| \rightarrow 0$$

and

$$|D_2f(x, y) - D_2f(0, 0)| = \left| \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \right| \leq |x| \left| \frac{(x^2 + y^2)^2 - 2x^4}{(x^2 + y^2)^2} \right| \leq |x| \rightarrow 0$$

as $(x, y) \rightarrow (0, 0)$. Thus D_1f and D_2f are continuous at $(0, 0)$.

(b) We have

$$D_1f(x, y) = \begin{cases} 0, & \text{if } (x, y) = (0, 0); \\ \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, & \text{if } (x, y) \neq (0, 0). \end{cases} \quad (9.70)$$

and

$$D_2f(x, y) = \begin{cases} 0, & \text{if } (x, y) = (0, 0); \\ \frac{x(x^4 - 4x^2y^2 - y^4)}{(x^2 + y^2)^2}, & \text{if } (x, y) \neq (0, 0). \end{cases} \quad (9.71)$$

For $(x, y) \neq (0, 0)$, we see that

$$D_{12}f(x, y) = D_{21}f(x, y) = \frac{x^6 + 9x^4y^2 - 9x^2y^4 - y^6}{(x^2 + y^2)^3} = \frac{(x^2 - y^2)(x^4 + 10x^2y^2 + y^4)}{(x^2 + y^2)^3}. \quad (9.72)$$

For $(x, y) = (0, 0)$, we have

$$D_{12}f(0, 0) = \lim_{t \rightarrow 0} \frac{D_2f(t, 0) - D_2f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{t - 0}{t} = 1, \quad (9.73)$$

$$D_{21}f(0, 0) = \lim_{t \rightarrow 0} \frac{D_1f(0, t) - D_1f(0, 0)}{t} = \lim_{t \rightarrow 0} \frac{-t - 0}{t} = -1. \quad (9.74)$$

Therefore, both $D_{12}f$ and $D_{21}f$ exist at every point of $\mathbb{R}^2$. In addition, we deduce from the expressions (9.72) that they are continuous at every point except possibly the origin $(0, 0)$. However, if $x = y = t$, then we have

$$D_{12}f(t, t) = D_{21}f(t, t) = 0$$

so that

$$|D_{12}f(t, t) - D_{12}f(0, 0)| = 1 \quad \text{and} \quad |D_{21}f(t, t) - D_{21}f(0, 0)| = 1,$$

i.e., they are discontinuous at $(0, 0)$.

(c) The results have already been shown in the limits (9.73) and (9.74).

We end the proof of the problem. ■

Problem 9.28

Rudin Chapter 9 Exercise 28.

Proof. For $t < 0$, we have

$$\varphi(x, t) = -\varphi(x, |t|) = \begin{cases} -x, & (0 \leq x \leq \sqrt{|t|}); \\ x - 2\sqrt{|t|}, & (\sqrt{|t|} \leq x \leq 2\sqrt{|t|}); \\ 0, & (\text{otherwise}). \end{cases} \quad (9.75)$$

Now we can divide the xt -plane into six regions, see Figure 9.7:

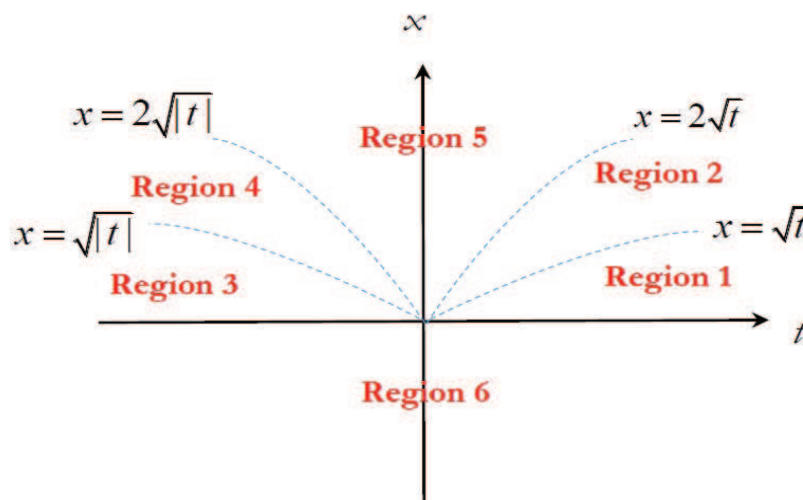


Figure 9.7: The definition of the function $\varphi(x, t)$.

In fact, we have

$$\varphi(x, t) = \begin{cases} x, & \text{Region 1;} \\ -x + 2\sqrt{t} & \text{Region 2;} \\ -x, & \text{Region 3;} \\ x - 2\sqrt{|t|}, & \text{Region 4;} \\ 0, & \text{Regions 5 and 6.} \end{cases} \quad (9.76)$$

Therefore, it is easy to see from the definition (9.76) that $\varphi(x, t)$ is continuous on each region, so we only need to check its continuity on the curves $x = \sqrt{t}$, $x = 2\sqrt{t}$ when $t \geq 0$ and $x = \sqrt{|t|}$, $x = 2\sqrt{|t|}$ when $t < 0$.^k

- **Case (i): On the curve $x = \sqrt{t}$.** Since

$$\varphi(\sqrt{t}+, t) = \lim_{\substack{x \rightarrow \sqrt{t} \\ \sqrt{t} < x \leq 2\sqrt{t}}} \varphi(x, t) = \lim_{x \rightarrow \sqrt{t}} (-x + 2\sqrt{t}) = \sqrt{t}$$

and

$$\varphi(\sqrt{t}-, t) = \lim_{\substack{x \rightarrow \sqrt{t} \\ 0 \leq x < \sqrt{t}}} \varphi(x, t) = \lim_{x \rightarrow \sqrt{t}} x = \sqrt{t},$$

we have

$$\varphi(\sqrt{t}+, t) = \varphi(\sqrt{t}-, t)$$

and so $\varphi(x, t)$ is continuous on the curve $x = \sqrt{t}$.

- **Case (i): On the curve $x = 2\sqrt{t}$.** Since

$$\varphi(2\sqrt{t}+, t) = \lim_{\substack{x \rightarrow 2\sqrt{t} \\ 2\sqrt{t} < x}} \varphi(x, t) = \lim_{x \rightarrow 2\sqrt{t}} 0 = 0$$

and

$$\varphi(2\sqrt{t}-, t) = \lim_{\substack{x \rightarrow 2\sqrt{t} \\ \sqrt{t} \leq x < 2\sqrt{t}}} \varphi(x, t) = \lim_{x \rightarrow 2\sqrt{t}} (-x + 2\sqrt{t}) = 0,$$

we have

$$\varphi(2\sqrt{t}+, t) = \varphi(2\sqrt{t}-, t)$$

and thus $\varphi(x, t)$ is continuous on the curve $x = 2\sqrt{t}$.

- **Case (i): On the curves $x = \sqrt{|t|}$ and $x = 2\sqrt{|t|}$.** Similar argument can be applied to the curves $x = \sqrt{|t|}$ and $x = 2\sqrt{|t|}$ when $t < 0$, so $\varphi(x, t)$ is also continuous on them.

In conclusion, φ is continuous on $\mathbb{R}^2$. This proves the first assertion.

For the second assertion, we have three cases as follows:

- **Case (i): $x = 0$.** Then we observe that

$$(D_2\varphi)(0, 0+) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{\varphi(0, h) - \varphi(0, 0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{\varphi(0, h) - 0}{h} = 0$$

and

$$(D_2\varphi)(0, 0-) = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{\varphi(0, h) - \varphi(0, 0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{-\varphi(0, |h|) - 0}{h} = 0.$$

Thus we have $(D_2\varphi)(0, 0) = 0$.

^kThey are the boundaries between Regions 1 and 2, Regions 2 and 5, Regions 3 and 4, as well as Regions 4 and 5 respectively.

- **Case (ii):** $x > 0$. If $h > 0$ and h is so small such that $0 < 2\sqrt{h} \leq x$, then we have

$$\varphi(x, h) = 0$$

which implies that

$$(D_2\varphi)(x, 0+) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{\varphi(x, h) - \varphi(x, 0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{\varphi(x, h) - 0}{h} = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{0}{h} = 0.$$

If $h < 0$ and $|h|$ is so small such that $0 < 2\sqrt{|h|} \leq x$, then we have

$$\varphi(x, h) = -\varphi(x, |h|) = 0$$

from the definition (9.75). Therefore, this implies that

$$(D_2\varphi)(x, 0-) = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{\varphi(x, h) - \varphi(x, 0)}{h} = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{-\varphi(x, |h|) - 0}{h} = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{0}{h} = 0.$$

Thus we have $(D_2\varphi)(x, 0) = 0$ if $x > 0$.

- **Case (iii):** $x < 0$. If $h > 0$, then we have

$$\varphi(x, h) = 0$$

by definition. Thus we have

$$(D_2\varphi)(x, 0+) = \lim_{\substack{h \rightarrow 0 \\ h > 0}} \frac{\varphi(x, h) - \varphi(x, 0)}{h} = 0.$$

If $h < 0$, then we also have

$$\varphi(x, h) = -\varphi(x, |h|) = 0$$

from the definition (9.75) which implies that

$$(D_2\varphi)(x, 0-) = \lim_{\substack{h \rightarrow 0 \\ h < 0}} \frac{\varphi(x, h) - \varphi(x, 0)}{h} = 0.$$

Thus we have $(D_2\varphi)(x, 0) = 0$ if $x < 0$.

For the third assertion, if $0 \leq t < \frac{1}{4}$, then we know from the definition of $\varphi(x, t)$ that

$$\begin{aligned} f(t) &= \int_{-1}^1 \varphi(x, t) \, dx \\ &= \int_{-1}^0 \varphi(x, t) \, dx + \int_0^1 \varphi(x, t) \, dx \\ &= 0 + \int_0^{\sqrt{t}} \varphi(x, t) \, dx + \int_{\sqrt{t}}^{2\sqrt{t}} \varphi(x, t) \, dx + \int_{2\sqrt{t}}^1 \varphi(x, t) \, dx \\ &= \int_0^{\sqrt{t}} x \, dx + \int_{\sqrt{t}}^{2\sqrt{t}} (-x + 2\sqrt{t}) \, dx \\ &= \frac{t}{2} + \left[-\frac{x^2}{2} + 2\sqrt{t}x \right]_{\sqrt{t}}^{2\sqrt{t}} \\ &= \frac{t}{2} + 2t - \frac{3t}{2} \\ &= t. \end{aligned} \tag{9.77}$$

Similarly, if $-\frac{1}{4} < t < 0$, then we have

$$f(t) = \int_{-1}^1 \varphi(x, t) \, dx = - \int_{-1}^1 \varphi(x, |t|) \, dx = -f(|t|) = -f(-t) = t. \tag{9.78}$$

Hence we follow from the expressions (9.77) and (9.78) that $f(t) = t$ if $|t| < \frac{1}{4}$. Since $(D_2\varphi)(x, 0) = 0$ for all x , we obtain that

$$\int_{-1}^1 (D_2\varphi)(x, 0) dx = 0.$$

However, since $f'(t) = 1$, we have $f'(0) = 1$. In other words,

$$f'(0) \neq \int_{-1}^1 (D_2\varphi)(x, 0) dx.$$

This completes the proof of the problem. ■

Problem 9.29

Rudin Chapter 9 Exercise 29.

Proof. This result is sometimes called Schwarz's theorem, Clairaut's theorem or Young's theorem.

We have $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Since $f \in \mathcal{C}^{(k)}(E)$, the k th-order derivative

$$D_{i_1 i_2 \dots i_k} f = D_{i_1} D_{i_2} \cdots D_{i_k} f$$

is well-defined and continuous on E , where $i_1, i_2, \dots, i_k = 1, 2, \dots, n$. Here we provide two solutions and both use induction.

- **Method 1:** We prove the assertion by induction. If $k = 2$, then the corollary of Theorem 9.41 shows that $D_{21}f = D_{12}f$.¹ Thus the statement is true for $k = 2$.

Assume that the statement is also true for $k = m \geq 2$. That is, if $f \in \mathcal{C}^{(m)}(E)$, then we have

$$D_{i_1 i_2 \dots i_m} f = D_{i_1} D_{i_2} \cdots D_{i_m} f$$

is unchanged if the subscripts $i_1, \dots, i_m$ are permuted, i.e.,

$$D_{\varphi(i_1)} D_{\varphi(i_2)} \cdots D_{\varphi(i_m)} f = D_{\varphi(i_1)\varphi(i_2)\dots\varphi(i_m)} f = D_{i_1 i_2 \dots i_m} f = D_{i_1} D_{i_2} \cdots D_{i_m} f \quad (9.79)$$

if φ is a *permutation*^m on $\{i_1, \dots, i_m\}$ and $i_1, i_2, \dots, i_m = 1, 2, \dots, n$.

Let $k = m + 1$ and $A = \{i_1, i_2, \dots, i_{m+1}\}$, where $m \geq 2$ and $i_1, i_2, \dots, i_{m+1} = 1, 2, \dots, n$. Suppose that $f \in \mathcal{C}^{(m+1)}(E)$ and $\varphi : A \rightarrow A$ is a permutation. By the hypothesis, we know that

$$D_1 f, \dots, D_n f \in \mathcal{C}^{(m)}(E).$$

- **Case (i):** $\varphi(i_{m+1}) = i_{m+1}$. Then it is clear that φ is also a permutation on $\{i_1, i_2, \dots, i_m\}$. Thus we obtain from the expression (9.79) and Theorem 9.41 that

$$\begin{aligned} D_{\varphi(i_1)\varphi(i_2)\dots\varphi(i_{m+1})} f &= [D_{\varphi(i_1)} D_{\varphi(i_2)} \cdots D_{\varphi(i_m)}] D_{\varphi(i_{m+1})} f \\ &= (D_{i_1} D_{i_2} \cdots D_{i_m}) D_{i_{m+1}} f \\ &= D_{i_1 i_2 \dots i_m i_{m+1}} f. \end{aligned}$$

- **Case (ii):** φ **interchanges the indices i_m and i_{m+1}** . In this case, φ is also a permutation on the restricted set $\{i_1, i_2, \dots, i_{m-1}\}$ and we follow from the induction hypothesis and Theorem 9.41 that

$$\begin{aligned} D_{\varphi(i_1)\varphi(i_2)\dots\varphi(i_{m+1})} f &= [D_{\varphi(i_1)} D_{\varphi(i_2)} \cdots D_{i_{m-1}}] [D_{\varphi(i_m)} D_{\varphi(i_{m+1})} f] \\ &= [D_{\varphi(i_1)} D_{\varphi(i_2)} \cdots D_{\varphi(i_{m-1})}] (D_{i_{m+1}} D_{i_m} f) \\ &= [D_{\varphi(i_1)} D_{\varphi(i_2)} \cdots D_{\varphi(i_{m-1})}] (D_{i_m} D_{i_{m+1}} f) \\ &= D_{i_1} D_{i_2} \cdots D_{i_{m-1}} D_{i_m} D_{i_{m+1}} f \\ &= D_{i_1 i_2 \dots i_m i_{m+1}} f. \end{aligned}$$

¹As mentioned by Rudin [21, p. 235], the results of Theorem 9.41 and its corollary are also true when $\mathbb{R}^2$ is replaced by $\mathbb{R}^n$.

^mA bijective mapping of a set S to itself is called a permutation of the set.

- **Case (iii): φ is a permutation such that $\varphi(i_m) = i_{m+1}$.** Let $\alpha, \beta : A \rightarrow A$ be permutations such that

$$\alpha(i_m) = i_{m+1}, \quad \alpha(i_{m+1}) = i_m \quad \text{and} \quad \alpha(i_s) = i_s \quad \text{for } s \neq m, m+1$$

and

$$\beta(i_{m+1}) = i_{m+1}, \quad \beta(i_m) = \varphi(i_{m+1}) \quad \text{and} \quad \beta(i_s) = \varphi(i_s) \quad \text{for } s \neq m, m+1.$$

We claim that $\varphi = \beta \circ \alpha$ on A , i.e.,

$$\varphi(i_s) = \beta(\alpha(i_s)) \tag{9.80}$$

for all $s = 1, 2, \dots, m+1$. To prove this claim, we consider three subcases:

- * **Subcase (i): $s = m$.** In this case, we have $\beta(\alpha(i_m)) = \beta(i_{m+1}) = i_{m+1} = \varphi(i_m)$.
- * **Subcase (ii): $s = m+1$.** We acquire $\beta(\alpha(i_{m+1})) = \beta(i_m) = \varphi(i_{m+1})$.
- * **Subcase (iii): $s \neq m, m+1$.** This implies that $\beta(\alpha(i_s)) = \beta(i_s) = \varphi(i_s)$.

Therefore, the formula (9.80) holds for all $s = 1, 2, \dots, m+1$ and our claim is proven. Since α and β are permutations as described in **Case (ii)** and **Case (i)** respectively, we have

$$D_{\alpha(i_1)\alpha(i_2)\dots\alpha(i_{m+1})}f = D_{i_1i_2\dots i_{m+1}}f, \tag{9.81}$$

$$D_{\beta(\alpha(i_1))\beta(\alpha(i_2))\dots\beta(\alpha(i_{m+1}))}f = D_{\alpha(i_1)\alpha(i_2)\dots\alpha(i_{m+1})}f. \tag{9.82}$$

After combining the two expressions (9.81) and (9.82), we apply formula (9.80) to deduce that

$$D_{\varphi(i_1)\varphi(i_2)\dots\varphi(i_{m+1})}f = D_{i_1i_2\dots i_{m+1}}f.$$

- **Case (iv): φ is a permutation such that $\varphi(i_r) = i_{m+1}$, where $r \neq m, m+1$.** Let $\rho, \sigma, \tau : A \rightarrow A$ be permutations such that

$$\rho(i_r) = i_m, \quad \rho(i_m) = i_r, \quad \rho(i_{m+1}) = i_{m+1} \quad \text{and} \quad \rho(i_s) = i_s \quad \text{for } s \neq r, m, m+1,$$

and

$$\sigma(i_m) = i_{m+1}, \quad \sigma(i_{m+1}) = i_m \quad \text{and} \quad \sigma(i_s) = i_s \quad \text{for } s \neq m, m+1$$

and

$$\tau(i_{m+1}) = i_{m+1}, \quad \tau(i_r) = \varphi(i_m), \quad \tau(i_m) = \varphi(i_{m+1}) \quad \text{and} \quad \tau(i_s) = \varphi(i_s) \quad \text{for } s \neq r, m, m+1.$$

We claim that $\varphi = \tau \circ \sigma \circ \rho$ on A , i.e.,

$$\varphi(i_s) = \tau(\sigma(\rho(i_s))) \tag{9.83}$$

for all $s = 1, 2, \dots, m+1$. To prove this claim, we consider four subcases:

- * **Subcase (i): $s = r$.** Then we have $\tau(\sigma(\rho(i_r))) = \tau(\sigma(i_m)) = \tau(i_{m+1}) = i_{m+1} = \varphi(i_r)$.
 - * **Subcase (ii): $s = m$.** In this case, we obtain that
- $$\tau(\sigma(\rho(i_m))) = \tau(\sigma(i_r)) = \tau(i_r) = \varphi(i_m).$$
- * **Subcase (iii): $s = m+1$.** We get $\tau(\sigma(\rho(i_{m+1}))) = \tau(\sigma(i_{m+1})) = \tau(i_m) = \varphi(i_{m+1})$.
 - * **Subcase (iv): $s \neq r, m, m+1$.** By definition of ρ , we have $\rho(i_s) = \varphi(i_s) \neq i_m, i_{m+1}$. This implies that

$$\tau(\sigma(\rho(i_s))) = \tau(\sigma(i_s)) = \tau(i_s) = \varphi(i_s).$$

Therefore, the formula (9.83) holds for all $s = 1, 2, \dots, m+1$ and our claim is proven. Since σ is a permutation as described in **Case (ii)** and ρ and τ are permutations as described in **Case (i)**, we have

$$D_{\rho(i_1)\rho(i_2)\dots\rho(i_{m+1})}f = D_{i_1i_2\dots i_{m+1}}f, \tag{9.84}$$

$$D_{\sigma(\rho(i_1))\sigma(\rho(i_2))\dots\sigma(\rho(i_{m+1}))}f = D_{\rho(i_1)\rho(i_2)\dots\rho(i_{m+1})}f, \tag{9.85}$$

$$D_{\tau(\sigma(\rho(i_1)))\tau(\sigma(\rho(i_2)))\dots\tau(\sigma(\rho(i_{m+1})))}f = D_{\sigma(\rho(i_1))\sigma(\rho(i_2))\dots\sigma(\rho(i_{m+1}))}f. \tag{9.86}$$

After combining the expressions (9.84), (9.85) and (9.86), then we use the formula (9.83) to acquire

$$D_{\varphi(i_1)\varphi(i_2)\dots\varphi(i_{m+1})}f = D_{i_1i_2\dots i_{m+1}}f.$$

Hence it follows from induction that the k th-order derivative

$$D_{i_1 i_2 \dots i_k} f = D_{i_1} D_{i_2} \cdots D_{i_k} f$$

is unchanged if the subscripts $i_1, \dots, i_k$ are permuted.ⁿ

- **Method 2:** Besides the above argument, the author discovered that there is a shorter and nicer solution provided by Christian Blatter on web^o. Here we repeat his argument with a slight modification.

Denote by W_k the set of *words* $\alpha = \alpha_1 \alpha_2 \cdots \alpha_k$ of length $k \geq 1$, where $\alpha_1, \alpha_2, \dots, \alpha_k = 1, 2, \dots, n$. See, for example, [12, §39, p. 341]. For every $\alpha \in W_k$, denote by $\hat{\alpha} : \{1, 2, \dots, n\} \rightarrow \mathbb{N} \cup \{0\}$ the function giving the multiplicity with which each letter j occurs in α , i.e.,

$$\hat{\alpha}(j) = \#\{r \in \{1, 2, \dots, k\} \mid \alpha_r = j\}.$$

We say that two words $\alpha, \beta \in W_k$ with $\hat{\alpha} = \hat{\beta}$ are *equivalent*.

For $\alpha \in W_k$, we denote by $D^\alpha f$ the k th partial derivative of f with respect to $x_{\alpha_k}, x_{\alpha_{k-1}}, \dots, x_{\alpha_1}$ in this order, i.e.,

$$D^\alpha f = D_{\alpha_1} D_{\alpha_2} \cdots D_{\alpha_k} f.$$

Now we have to prove that if $\alpha, \beta \in W_k$ and $\alpha \sim \beta$, then we must have

$$D^\alpha f = D^\beta f.$$

The case for $k = 2$ is obviously true, so we may assume that $k \geq 3$ and that the statement is true for $m < k$. Take two equivalent words $\alpha, \beta \in W_k$. Then we have

$$\alpha = \alpha' \alpha_k \quad \text{and} \quad \beta = \beta' \beta_k,$$

where $\alpha', \beta' \in W_{k-1}$. If $\alpha_k = \beta_k$, then we have $\alpha' \sim \beta'$ and the induction hypothesis implies that

$$D^\alpha f = \frac{\partial}{\partial x_{\alpha_k}} D^{\alpha'} f = \frac{\partial}{\partial x_{\alpha_k}} D^{\beta'} f = D^\beta f.$$

If $\alpha_k \neq \beta_k$, then β_k must occur in α' and there is a word $\alpha'' \in W_{k-2}$ such that

$$\alpha' \sim \alpha'' \beta_k.$$

Similarly, there is a word $\beta'' \in W_{k-2}$ such that

$$\beta' \sim \beta'' \alpha_k.$$

Since $\alpha \sim \beta$, we deduce from this and the corollary of Theorem 9.41 that

$$\alpha'' \sim \beta''.$$

Therefore, it follows from our induction hypothesis that

$$\begin{aligned} D^\alpha f &= \frac{\partial}{\partial x_{\alpha_k}} D^{\alpha'} f \\ &= \frac{\partial^2}{\partial x_{\alpha_k} \partial x_{\beta_k}} D^{\alpha''} f \\ &= \frac{\partial^2}{\partial x_{\alpha_k} \partial x_{\beta_k}} D^{\beta''} f \end{aligned}$$

ⁿHere we haven't applied the hint given by Rudin: repeated application of Theorem 9.41. If someone wants to use this hint, he/she needs some results from abstract algebra first: By definition, a *transposition* on $\{1, 2, \dots, n\}$ is a permutation that leaves all elements but two fixed and it maps those elements onto the other, see [23, p. 13]. Then every permutation on $\{1, 2, \dots, n\}$ is a product of adjacent transpositions $(1, 2), (2, 3), \dots, (n-1, n)$, see [23, Exercise 38(b), p. 79]. Thus, if φ is an adjacent transposition, then a repeated application of the corollary of Theorem 9.41 implies our result (9.79).

^oSee <https://math.stackexchange.com/questions/2234894/proof-of-the-generalized-clairauts-theorem>.

$$\begin{aligned}
&= \frac{\partial^2}{\partial x_{\beta_k} \partial x_{\alpha_k}} D^{\beta''} f \\
&= \frac{\partial}{\partial x_{\beta_k}} D^{\beta'} f \\
&= D^{\beta} f.
\end{aligned}$$

By induction, we have the desired result.

This completes the proof of the problem. ■

Problem 9.30

Rudin Chapter 9 Exercise 30.

Proof. Now we have $\mathbf{p} : [0, 1] \rightarrow E \subseteq \mathbb{R}^n$ and $h : S \subseteq \mathbb{R} \rightarrow \mathbb{R}$, where S is a set containing the closed interval $[0, 1]$.

(a) Since $\mathbf{p}$ is differentiable on $[0, 1]$ and $f \in \mathcal{C}^{(k)}(E)$, we deduce from Theorem 9.15 (Chain Rule) that

$$h'(t) = [f'(\mathbf{p}(t))] [\mathbf{p}'(t)], \quad (9.87)$$

where $[f'(\mathbf{p}(t))]$ and $[\mathbf{p}'(t)]$ are the matrices of the linear transformations $f'(\mathbf{p}(t))$ and $\mathbf{p}'(t)$ respectively. By the result of Theorem 9.17, we have

$$[f'(\mathbf{p}(t))] = \begin{pmatrix} D_1 f(\mathbf{p}(t)) & D_2 f(\mathbf{p}(t)) & \cdots & D_n f(\mathbf{p}(t)) \end{pmatrix} \quad \text{and} \quad [\mathbf{p}'(t)] = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix},$$

where $\mathbf{x} = (x_1, x_2, \dots, x_n)$. By these, we conclude from the expression (9.87) that

$$h'(t) = D_1 f(\mathbf{p}(t))x_1 + D_2 f(\mathbf{p}(t))x_2 + \cdots + D_n f(\mathbf{p}(t))x_n. \quad (9.88)$$

Since $D_j f(\mathbf{p}(t)) \in \mathcal{C}^{(m-1)}(E)$ for each $j = 1, 2, \dots, n$, we apply Theorem 9.15 (Chain Rule) to each $D_j f(\mathbf{p}(t))$ to acquire that

$$\begin{aligned}
(D_j f(\mathbf{p}(t)))' &= [(D_j f)'(\mathbf{p}(t))] [\mathbf{p}'(t)] \\
&= \begin{pmatrix} D_1 D_j f(\mathbf{p}(t)) & D_2 D_j f(\mathbf{p}(t)) & \cdots & D_n D_j f(\mathbf{p}(t)) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\
&= D_1 D_j f(\mathbf{p}(t))x_1 + D_2 D_j f(\mathbf{p}(t))x_2 + \cdots + D_n D_j f(\mathbf{p}(t))x_n
\end{aligned}$$

for $j = 1, 2, \dots, n$. Therefore, we follow from the expression (9.88) that

$$\begin{aligned}
h''(t) &= [D_1 D_1 f(\mathbf{p}(t))x_1x_1 + D_2 D_1 f(\mathbf{p}(t))x_2x_1 + \cdots + D_n D_1 f(\mathbf{p}(t))x_nx_1] \\
&\quad + [D_1 D_2 f(\mathbf{p}(t))x_1x_2 + D_2 D_2 f(\mathbf{p}(t))x_2x_2 + \cdots + D_n D_2 f(\mathbf{p}(t))x_nx_2] \\
&\quad + \cdots + [D_1 D_n f(\mathbf{p}(t))x_1x_n + D_2 D_n f(\mathbf{p}(t))x_2x_n + \cdots + D_n D_n f(\mathbf{p}(t))x_nx_n] \\
&= \sum_{i,j=1}^n (D_{ij} f)(\mathbf{p}(t))x_i x_j.
\end{aligned} \quad (9.89)$$

Hence, if $k \geq 3$, then by applying Theorem 9.15 (Chain Rule) $(k-2)$ -times to the expression (9.89), we obtain our desired result.

(b) Since $h(1) = f(\mathbf{p}(1)) = f(\mathbf{a} + \mathbf{x})$, it follows from Theorem 5.15 (Taylor's Theorem) that

$$f(\mathbf{a} + \mathbf{x}) = \sum_{k=0}^{m-1} \frac{h^{(k)}(0)}{k!} + \frac{h^{(m)}(t)}{m!} \quad (9.90)$$

for some $t \in (0, 1)$. By the result of part (a), we have

$$h^{(k)}(0) = \sum (D_{i_1 i_2 \dots i_k} f)(\mathbf{p}(0)) x_{i_1} \cdots x_{i_k} = \sum (D_{i_1 i_2 \dots i_k} f)(\mathbf{a}) x_{i_1} \cdots x_{i_k},$$

where the sum extends over all ordered k -tuples $(i_1, \dots, i_k)$ in which each i_j is one of the integers $1, \dots, n$. Thus the expression (9.90) can be rewritten as^P

$$f(\mathbf{a} + \mathbf{x}) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum (D_{i_1 i_2 \dots i_k} f)(\mathbf{a}) x_{i_1} \cdots x_{i_k} + \frac{h^{(m)}(t)}{m!} \quad (9.91)$$

for some $t \in (0, 1)$. Let $r(\mathbf{x}) = \frac{h^{(m)}(t)}{m!}$. Then we have

$$\frac{r(\mathbf{x})}{|\mathbf{x}|^{m-1}} = \frac{1}{|\mathbf{x}|^{m-1}} \sum (D_{i_1 i_2 \dots i_m} f)(\mathbf{p}(t)) x_{i_1} \cdots x_{i_m} \quad (9.92)$$

for some $t \in (0, 1)$. The sum extends over all ordered m -tuples $(i_1, \dots, i_m)$ in which each i_j is one of the integers $1, \dots, n$. By the definition given in Problem 9.29, we see that each $D_{i_1 i_2 \dots i_m} f \in \mathcal{C}(E)$, i.e., $D_{i_1 i_2 \dots i_m} f$ is a real-valued, continuous and bounded function on E . Since $\mathbf{p}$ is continuous on $[0, 1]$, the set $\mathbf{p}([0, 1])$ is compact by Theorem 4.14. By Theorem 4.16 (Extreme Value Theorem), there exists a positive integer M such that

$$|(D_{i_1 i_2 \dots i_m} f)(\mathbf{p}(t))| \leq M$$

for all m -tuples $(i_1, \dots, i_m)$ and $t \in [0, 1]$. Therefore, the expression (9.92) gives

$$\begin{aligned} \left| \frac{r(\mathbf{x})}{|\mathbf{x}|^{m-1}} \right| &\leq \frac{1}{|\mathbf{x}|^{m-1}} \left| \sum (D_{i_1 i_2 \dots i_m} f)(\mathbf{p}(t)) x_{i_1} \cdots x_{i_m} \right| \\ &\leq \frac{1}{|\mathbf{x}|^{m-1}} M \left| \sum x_{i_1} \cdots x_{i_m} \right| \\ &\leq M |\mathbf{x}| \end{aligned}$$

which implies that

$$\lim_{\mathbf{x} \rightarrow \mathbf{0}} \frac{r(\mathbf{x})}{|\mathbf{x}|^{m-1}} = 0.$$

(c) Fix the positive integer n . For each nonnegative k , we have the k letters $i_1, i_2, \dots, i_k$ which are elements of the set of numbers $\{1, 2, \dots, n\}$. It is clear that there are $k!$ ways to permute $i_1, i_2, \dots, i_k$. However, some letters may repeat. Let $s_1, s_2, \dots, s_n$ be the multiplicities of the numbers $1, 2, \dots, n$ in the set $\{i_1, i_2, \dots, i_k\}$ respectively. Then they are nonnegative and

$$s_1 + s_2 + \cdots + s_n = k$$

so that the number^Q

$$\frac{k!}{s_1! s_2! \cdots s_n!} \quad (9.93)$$

represents the number of *distinct* ways to permute the numbers $1, 2, \dots, n$.

Apply this formula (9.93) and Problem 9.29 to the inner summation (9.91), we have

$$f(\mathbf{a} + \mathbf{x}) = \sum_{k=0}^{m-1} \frac{1}{k!} \sum (D_{i_1 i_2 \dots i_k} f)(\mathbf{a}) x_{i_1} \cdots x_{i_k} + r(\mathbf{x})$$

^PAs mentioned in the problem, we may simply define $(D_{i_1 i_2 \dots i_k} f)(\mathbf{a}) = f(\mathbf{a})$ and $x_{i_1} \cdots x_{i_k} = 1$ when $k = 0$.

^QThis number is the multinomial coefficient in the Multinomial Theorem.

$$\begin{aligned}
&= \sum_{k=0}^{m-1} \frac{1}{k!} \sum \frac{k!}{s_1! s_2! \cdots s_n!} (D_1^{s_1} D_2^{s_2} \cdots D_n^{s_n} f)(\mathbf{a}) x_1^{s_1} \cdots x_n^{s_n} + r(\mathbf{x}) \\
&= \sum_{s_1 + \cdots + s_n \leq m-1} \frac{(D_1^{s_1} D_2^{s_2} \cdots D_n^{s_n} f)(\mathbf{a})}{s_1! s_2! \cdots s_n!} x_1^{s_1} \cdots x_n^{s_n} + r(\mathbf{x}).
\end{aligned}$$

Hence this completes the proof of the problem. ■

Problem 9.31

Rudin Chapter 9 Exercise 31.

Proof. We have $f : E \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, where E is a neighborhood of $\mathbf{a}$. By definition, we have

$$\nabla f(\mathbf{a}) = (D_1 f)(\mathbf{a}) \mathbf{e}_1 + (D_2 f)(\mathbf{a}) \mathbf{e}_2 = \mathbf{0}$$

which is equivalent to the fact that^r

$$(D_1 f)(\mathbf{a}) = (D_2 f)(\mathbf{a}) = 0.$$

By these and the result of Problem 9.30, the Taylor polynomial of f at $\mathbf{a}$ is given by

$$\begin{aligned}
f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) &= \sum_{1 \leq s_1 + s_2 \leq 2} \frac{(D_1^{s_1} D_2^{s_2} f)(\mathbf{a})}{s_1! s_2!} x_1^{s_1} x_2^{s_2} \\
&= (D_1 f)(\mathbf{a}) x_1 + (D_2 f)(\mathbf{a}) x_2 \\
&\quad + \frac{1}{2} [2(D_1 D_2 f)(\mathbf{a}) x_1 x_2 + (D_1^2 f)(\mathbf{a}) x_1^2 + (D_2^2 f)(\mathbf{a}) x_2^2] + r(\mathbf{x}) \\
&= \frac{1}{2} [2(D_1 D_2 f)(\mathbf{a}) x_1 x_2 + (D_1^2 f)(\mathbf{a}) x_1^2 + (D_2^2 f)(\mathbf{a}) x_2^2] + r(\mathbf{x}), \tag{9.94}
\end{aligned}$$

where $\mathbf{x} = (x_1, x_2)$ is close to $\mathbf{0} = (0, 0)$ and $r(\mathbf{x})$ is the remainder such that

$$\frac{r(\mathbf{x})}{|\mathbf{x}|^2} \rightarrow 0$$

as $\mathbf{x} \rightarrow \mathbf{0}$. Therefore, the sign of $f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a})$ is determined by that of the brackets in the right-hand side of the expression (9.94).

By the hypothesis, we know that not all $(D_1 D_2 f)(\mathbf{a})$, $(D_1^2 f)(\mathbf{a})$ and $(D_2^2 f)(\mathbf{a})$ are zero. Let

$$H_{\mathbf{a}} = \begin{pmatrix} (D_1^2 f)(\mathbf{a}) & (D_{12} f)(\mathbf{a}) \\ (D_{21} f)(\mathbf{a}) & (D_2^2 f)(\mathbf{a}) \end{pmatrix}$$

be the Hessian matrix at the point $\mathbf{a}$. Now we can rewrite the expression (9.94) as

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) = Q(\mathbf{x}) + r(\mathbf{x}), \tag{9.95}$$

where $Q(\mathbf{x})$ is defined by^s

$$Q(\mathbf{x}) = \frac{1}{2} [(D_1^2 f)(\mathbf{a}) x_1^2 + 2(D_1 D_2 f)(\mathbf{a}) x_1 x_2 + (D_2^2 f)(\mathbf{a}) x_2^2]. \tag{9.96}$$

Suppose that $(D_1^2 f)(\mathbf{a}) \neq 0$. Then we can rewrite the expression (9.96) as

$$Q(\mathbf{x}) = \frac{1}{2(D_1^2 f)(\mathbf{a})} \left\{ [(D_1^2 f)(\mathbf{a}) x_1 + (D_1 D_2 f)(\mathbf{a}) x_2]^2 + (\det H_{\mathbf{a}}) x_2^2 \right\}. \tag{9.97}$$

^rBy definition, the point $\mathbf{a}$ is a critical point of f .

^sNote that $(D_{12} f)(\mathbf{a}) = (D_{21} f)(\mathbf{a})$ because of the corollary of Theorem 9.41.

- **Case (i):** $\det H_{\mathbf{a}} > 0$. In this case, the brackets in the right-hand side of the expression (9.97) is the sum of two squares. In this case, we have $Q(\mathbf{x}) > 0$ if $(D_1^2 f)(\mathbf{a}) > 0$ and $Q(\mathbf{x}) < 0$ if $(D_1^2 f)(\mathbf{a}) < 0$. Hence it follows from this and the expression (9.95) that

- f has a local minimum at $\mathbf{a}$ if $(D_1^2 f)(\mathbf{a}) > 0$ and $\det H_{\mathbf{a}} > 0$,
- f has a local maximum at $\mathbf{a}$ if $(D_1^2 f)(\mathbf{a}) < 0$ and $\det H_{\mathbf{a}} > 0$.

- **Case (ii):** $\det H_{\mathbf{a}} < 0$. By the form (9.97), $Q(\mathbf{x})$ can be expressed in the form

$$Q(\mathbf{x}) = \gamma(\alpha x_1 + \beta x_2)(\alpha x_1 - \beta x_2), \quad (9.98)$$

where α, β, γ are some constants and $\gamma \neq 0$. Now it is clear from the form (9.98) that $Q(\mathbf{x}) = 0$ if and only if

$$\alpha x_1 + \beta x_2 = 0 \quad \text{or} \quad \alpha x_1 - \beta x_2 = 0.$$

Besides, the lines $\alpha x_1 + \beta x_2 = 0$ and $\alpha x_1 - \beta x_2 = 0$ divides the $x_1 x_2$ -plane into four regions. See Figure 9.8 for details.^t

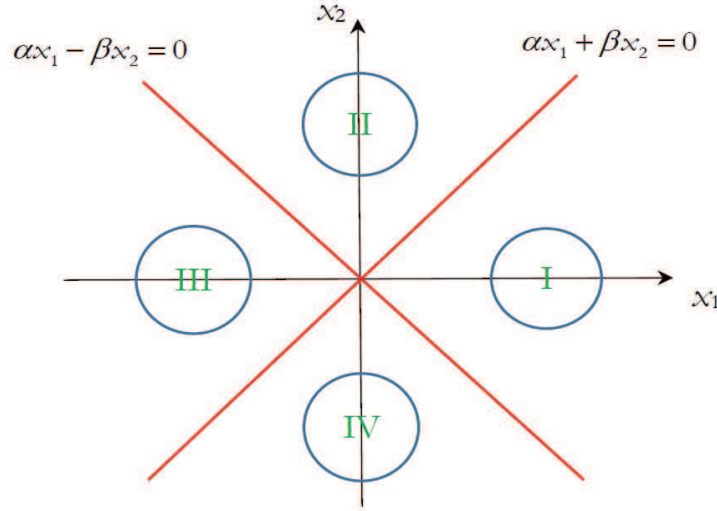


Figure 9.8: The four regions divided by the two lines $\alpha x_1 + \beta x_2 = 0$ and $\alpha x_1 - \beta x_2 = 0$.

By direct computation, we have

$$Q(1, 0) = Q(-1, 0) = \alpha^2 \gamma \quad \text{and} \quad Q(0, 1) = Q(0, -1) = -\beta^2 \gamma.$$

When $\gamma > 0$, then $Q(\mathbf{x}) > 0$ in Regions I and III and $Q(\mathbf{x}) < 0$ in Regions II and IV; when $\gamma < 0$, then $Q(\mathbf{x}) < 0$ in Regions I and III and $Q(\mathbf{x}) > 0$ in Regions II and IV. Hence f has a saddle point at $\mathbf{a}$.

- **Case (iii):** $\det H_{\mathbf{a}} = 0$. Then the point $\mathbf{a}$ may be a local maximum, a local minimum or a saddle point.

This finishes the discussion of the first assertion.

We start to prove the second assertion. We consider the function $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, where E is a neighborhood of $\mathbf{a}$. Suppose that

$$\nabla f(\mathbf{a}) = \sum_{i=1}^n (D_i f)(\mathbf{a}) \mathbf{e}_i = \mathbf{0}$$

^tHere we suppose that the slope of the line $\alpha x_1 + \beta x_2 = 0$ is positive. The other case can be done similarly.

and not all second-order derivatives of f are 0 at $\mathbf{a}$. By these and the results from Problem 9.30, we have

$$\begin{aligned} f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) &= \sum_{1 \leq s_1 + \dots + s_n \leq 2} \frac{(D_1^{s_1} \dots D_n^{s_n} f)(\mathbf{a})}{s_1! \dots s_n!} x_1^{s_1} \dots x_n^{s_n} + r(\mathbf{x}) \\ &= \frac{1}{2} \sum_{i,j=1}^n (D_i D_j f)(\mathbf{a}) x_i x_j + r(\mathbf{x}). \end{aligned} \quad (9.99)$$

Similar as in the case of $\mathbb{R}^2$, we define

$$H_{\mathbf{a}} = [(D_{ij} f)(\mathbf{a})] \quad (9.100)$$

to be the Hessian matrix at the point $\mathbf{a}$, where $(D_{ij} f)(\mathbf{a})$ is the entry appears in the i th row and j th column of the matrix (9.100). By this, the expression (9.99) can be written in the form

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) = \frac{1}{2} \mathbf{x}^T H_{\mathbf{a}} \mathbf{x} + r(\mathbf{x}), \quad (9.101)$$

where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

and $\mathbf{x}^T$ is the transpose of the vector $\mathbf{x}$. Recall that a symmetric matrix A (i.e., $A^T = A$) is called *positive definite* if the quadratic form

$$\mathbf{x}^T A \mathbf{x} > 0$$

for all $\mathbf{x} \neq 0$. To finish our proof, we need to find bounds of the number $\mathbf{x}^T A \mathbf{x}$ when A is symmetric. For this purpose, some basic results about *eigenvalues* of a matrix are needed, see [15, Theorem 5, §7.2, Chap. 7, p. 405] or [3, Exercise 7, §5, Chap. 7, p. 266]:

Lemma 9.3

Let A be an $n \times n$ symmetric matrix. Then a quadratic form $\mathbf{x}^T A \mathbf{x}$ is

- (a) positive definite if and only if the eigenvalues of A are all positive,
- (b) negative definite if and only if the eigenvalues of A are all negative, or
- (c) indefinite if and only if A has both positive and negative eigenvalues.

Lemma 9.3 provides a quick way to check whether a symmetric matrix A is positive definite or negative definite. Another simple way to show that a matrix is positive definite is by using its submatrices: A real symmetric $n \times n$ matrix A is positive definite if and only if

$$\det A_i > 0$$

for each $i = 1, 2, \dots, n$, where A_i is the upper left $i \times i$ submatrix of A . We need one more lemma, see [3, Proposition 5.7, §5, Chap. 7, p. 255]:

Lemma 9.4 Spectral Theorem (real case)

Let T be a symmetric operator on a real vector space V with a positive definite bilinear form. Then there is an orthonormal basis of V consisting of eigenvectors of T .

Let $\lambda_1, \lambda_2, \dots, \lambda_n$ be eigenvalues of an $n \times n$ symmetric and positive definite matrix A with real entries and $\mathbf{x}_1, \dots, \mathbf{x}_n$ be their corresponding eigenvectors. By Lemma 9.3(a), $\lambda_1, \lambda_2, \dots, \lambda_n$ are real and positive.

By Lemma 9.4, without loss of generality, we may assume that the set $\{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ is an orthonormal basis of $\mathbb{R}^n$. Given non-zero $\mathbf{x} \in \mathbb{R}^n$, we have

$$\mathbf{x} = c_1 \mathbf{x}_1 + \dots + c_n \mathbf{x}_n$$

for some scalar $c_1, \dots, c_n$. It follows that

$$\mathbf{x}^T A \mathbf{x} = \left(\sum_{i=1}^n c_i \mathbf{x}_i \right)^T A \left(\sum_{i=1}^n c_i \mathbf{x}_i \right) = \left(\sum_{i=1}^n c_i \mathbf{x}_i \right)^T \left(\sum_{i=1}^n c_i \lambda_i \mathbf{x}_i \right) = \sum_{i=1}^n \lambda_i c_i^2 \geq \lambda |\mathbf{x}|^2, \quad (9.102)$$

where $\lambda = \min_{1 \leq i \leq n} (\lambda_i) > 0$. Now it is time to determine whether f has a local maximum, or a local minimum, or a saddle point, at $\mathbf{a}$ as follows:

- **Case (i): $H_{\mathbf{a}}$ is positive definite.** By the corollary of Theorem 9.41, the matrix $H_{\mathbf{a}}$ is symmetric. Put $A = H_{\mathbf{a}}$ into the inequality (9.102) so that the expression (9.101) becomes

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) \geq \frac{1}{2} \lambda |\mathbf{x}|^2 + r(\mathbf{x}), \quad (9.103)$$

where $\mathbf{x}$ is non-zero and so close to $\mathbf{0}$. Since $\frac{r(\mathbf{x})}{|\mathbf{x}|^2} \rightarrow 0$ as $\mathbf{x} \rightarrow \mathbf{0}$, for $\frac{\lambda}{2} > 0$ there exists a $\delta > 0$ such that

$$\left| \frac{r(\mathbf{x})}{|\mathbf{x}|^2} \right| < \frac{\lambda}{2} \quad (9.104)$$

for $0 < |\mathbf{x}| < \delta$. By the equivalent form

$$-\frac{\lambda}{2} |\mathbf{x}|^2 < r(\mathbf{x}) < \frac{\lambda}{2} |\mathbf{x}|^2$$

of the inequality (9.104), the inequality (9.103) can be reduced to

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) > 0$$

for $0 < |\mathbf{x}| < \delta$. This means that f has a local minimum at the point $\mathbf{a}$.

- **Case (ii): $H_{\mathbf{a}}$ is negative definite.** In this case, Lemma 9.3(b) says that $\lambda_1, \dots, \lambda_n$ are all negative. Instead of the inequalities (9.102) and (9.103), we have

$$\mathbf{x}^T H_{\mathbf{a}} \mathbf{x} \leq \rho |\mathbf{x}|^2$$

and

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) \leq \frac{\rho}{2} |\mathbf{x}|^2 + r(\mathbf{x}), \quad (9.105)$$

where $\mathbf{x}$ is non-zero and so close to $\mathbf{0}$, and $\rho = \max_{1 \leq i \leq n} (\lambda_i) < 0$. Again, since $\frac{r(\mathbf{x})}{|\mathbf{x}|^2} \rightarrow 0$ as $\mathbf{x} \rightarrow \mathbf{0}$, for $\frac{|\rho|}{2} > 0$ there exists a $\delta > 0$ such that

$$\left| \frac{r(\mathbf{x})}{|\mathbf{x}|^2} \right| < \frac{|\rho|}{2} \quad (9.106)$$

for $0 < |\mathbf{x}| < \delta$. By the equivalent form

$$-\frac{|\rho|}{2} |\mathbf{x}|^2 < r(\mathbf{x}) < \frac{|\rho|}{2} |\mathbf{x}|^2$$

of the inequality (9.106), we see that the inequality (9.105) induces

$$f(\mathbf{a} + \mathbf{x}) - f(\mathbf{a}) \leq \frac{\rho}{2} |\mathbf{x}|^2 + r(\mathbf{x}) < \frac{\rho + |\rho|}{2} |\mathbf{x}|^2 = 0$$

for $0 < |\mathbf{x}| < \delta$. In other words, f has a local maximum at the point $\mathbf{a}$.

- **Case (iii): $H_{\mathbf{a}}$ is indefinite.** In this case, the point $\mathbf{a}$ may be a local maximum, a local minimum or a saddle point in this case.

This proves our second assertion, completing the proof of the problem. ■

CHAPTER 10

Integration of Differential Forms

10.1 Integration over sets in $\mathbb{R}^k$ and primitive mappings

Problem 10.1

Rudin Chapter 10 Exercise 1.

Proof. Let H be a compact convex set in $\mathbb{R}^k$, $\text{supp}(f) \subseteq H$ and $H^\circ \neq \emptyset$, where H° denotes the *interior* of H (see Problem 2.9). If $f \in \mathcal{C}(H)$, we extend f to a function on I^k containing H by setting $f(\mathbf{x}) = 0$ for all $\mathbf{x} \in I^k \setminus H$, and define

$$\int_H f = \int_{I^k} f, \quad (10.1)$$

where

$$I^k = \{(x_1, \dots, x_k) \mid 0 \leq x_i \leq 1, i = 1, 2, \dots, k\}.$$

As suggested by the hint, we are going to show that f can be approximated by functions F that are continuous on $\mathbb{R}^k$ and $\text{supp}(F) \subseteq H$. Before that, we need to show the existence of the *integral* on the right-hand side in (10.1).

Denote ∂H to be the boundary of the compact convex set H . Since $H^\circ \neq \emptyset$, we define the function $\rho : I^k \rightarrow \mathbb{R}$ by

$$\begin{aligned} \rho(\mathbf{x}) &= \begin{cases} \rho_{\partial H}(\mathbf{x}), & \text{if } \mathbf{x} \in H; \\ 0, & \text{if } \mathbf{x} \in I^k \setminus H, \end{cases} \\ &= \begin{cases} \inf\{|\mathbf{x} - \mathbf{y}| \mid \mathbf{y} \in \partial H\}, & \text{if } \mathbf{x} \in H; \\ 0, & \text{if } \mathbf{x} \in I^k \setminus H. \end{cases} \end{aligned} \quad (10.2)$$

By Problem 4.20(a),

$$\rho_{\partial H}(\mathbf{x}) = 0 \quad \text{if and only if} \quad \mathbf{x} \in \overline{\partial H} = \partial H.$$

By this and Problem 4.20(b), ρ is (uniformly) continuous on I^k . Suppose $0 < \delta < 1$, put

$$\varphi(t) = \begin{cases} 1, & \text{if } \delta \leq t < +\infty; \\ \frac{t}{\delta}, & \text{if } 0 < t < \delta; \\ 0, & \text{if } t = 0. \end{cases}$$

Now it is clear that φ is continuous $[0, +\infty)$. Define $F : I^k \rightarrow \mathbb{R}$ by $F(\mathbf{x}) = \varphi(\rho(\mathbf{x}))f(\mathbf{x})$. Since $f \in \mathcal{C}(H)$, F is obviously continuous on H . Since $f(\mathbf{x}) = 0$ on $I^k \setminus H$, F is continuous on $I^k \setminus H$ too. However, it is *not* clear whether F is continuous on ∂H or not. In fact, the answer to this question is affirmative: Let $\mathbf{p} \in \partial H$ and $\{\mathbf{x}_n\}$ be a sequence in $I^k \setminus \partial H$ converging to $\mathbf{p}$. By the continuity of the functions φ and ρ , we observe that

$$\lim_{n \rightarrow \infty} \varphi(\rho(\mathbf{x}_n)) = \varphi(\rho(\mathbf{p})) = 0.$$

Since H is compact, we have $\mathbf{p} \in H$ so that $f(\mathbf{p})$ is well-defined. Furthermore, Theorem 4.15 implies that f is bounded by a positive constant M on H . Therefore, we have

$$0 \leq \lim_{n \rightarrow \infty} |F(\mathbf{x}_n)| = \lim_{n \rightarrow \infty} |\varphi(\rho(\mathbf{x}_n))f(\mathbf{x}_n)| \leq M \lim_{n \rightarrow \infty} |\varphi(\rho(\mathbf{x}_n))| = M|\varphi(\rho(\mathbf{p}))| = 0$$

which means that

$$\lim_{n \rightarrow \infty} F(\mathbf{x}_n) = 0 = \varphi(\rho(\mathbf{p}))f(\mathbf{p}) = F(\mathbf{p}).$$

Hence F is also continuous on ∂H and then $F \in \mathcal{C}(I^k)$.^a

Lemma 10.1

Put $\mathbf{y} = (x_1, \dots, x_{k-1}) \in I^{k-1}$. Let $S = \{x_k \in [0, 1] \mid F(\mathbf{y}, x_k) \neq f(\mathbf{y}, x_k)\}$. Then the set S is either empty or is a line segment whose length does not exceed δ .

Proof of Lemma 10.1. Suppose that $S \neq \emptyset$. Then we have $F(\mathbf{y}, x_k) \neq f(\mathbf{y}, x_k)$ for some $x_k \in [0, 1]$. By definition, this means that

$$0 \leq \varphi(\rho(\mathbf{y}, x_k)) < 1 \quad \text{and} \quad f(\mathbf{y}, x_k) \neq 0.$$

In addition, these conditions are equivalent to

$$0 \leq \rho(\mathbf{y}, x_k) < \delta \quad \text{and} \quad f(\mathbf{y}, x_k) \neq 0. \quad (10.3)$$

By the second condition in (10.3), we must have $(\mathbf{y}, x_k) \notin I^k \setminus H$. Thus, $(\mathbf{y}, x_k) \in H$ so that the first condition in (10.3) reduces to

$$0 < \rho(\mathbf{y}, x_k) = \rho_{\partial H}(\mathbf{y}, x_k) < \delta.$$

Since H is convex, the line segment joining the point $(\mathbf{y}, x_k)$ and any point on ∂H is still inside H . See Figure 10.1 for details. Therefore, S must be a line segment whose length is less than δ . This completes the proof of the lemma. ■

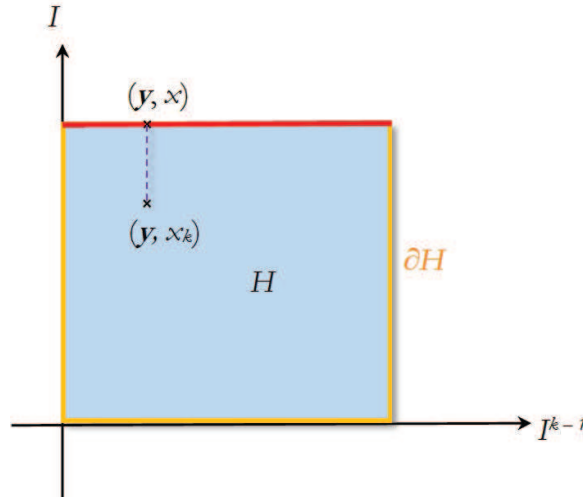


Figure 10.1: The compact convex set H and its boundary ∂H .

^aIt should be noted that since f may *not* be continuous on I^k , we can't conclude that

$$\lim_{n \rightarrow \infty} f(\mathbf{x}_n) = f(\mathbf{p}).$$

Let return to the proof of the problem. Since $0 \leq \varphi \leq 1$, it follows that

$$|F_{k-1}(\mathbf{y}) - f_{k-1}(\mathbf{y})| \leq \int_0^1 |F_k(\mathbf{y}, x_k) - f_k(\mathbf{y}, x_k)| dx_k \leq \delta \|f\|, \quad (10.4)$$

where $\|f\| = \max_{\mathbf{x} \in I^k} |f(\mathbf{x})|$. As $\delta \rightarrow 0$, the inequality (10.4) exhibits f_{k-1} as a uniform limit (with respect to δ) of a sequence of continuous functions $\{F_{k-1}\}$ on I^{k-1} . This proves that $f_{k-1} \in \mathcal{C}(I^{k-1})$ and also the existence of the **integral** on the right-hand side in (10.1).

Furthermore, if $F = F_k$ and $f = f_k$, then we can rewrite the inequality (10.4) as

$$\left| \int_{I^k} F - \int_{I^k} f \right| \leq \delta \|f\|$$

which is true no matter what the order of the k single integrations is. As we have shown above, $F \in \mathcal{C}(I^k)$, so

$$L(F) = L'(F)$$

by Theorem 10.2. Hence the same is true for f , i.e., $\int f$ is unaffected by any change of the order of integration. This completes the proof of the problem. ■

Problem 10.2

Rudin Chapter 10 Exercise 2.

Proof. By Definition 10.3, we have

$$\text{supp}(f) = \overline{\{(x, y) \in \mathbb{R}^2 \mid f(x, y) \neq 0\}} \subseteq \mathbb{R}^2.$$

Let $S_i = \text{supp}(\varphi_i) \subseteq (2^{-i}, 2^{1-i}) \subset (0, 1)$ for each $i \in \mathbb{N}$ and

$$S = \bigcup_{i=1}^{\infty} S_i.$$

We need a lemma:

Lemma 10.2

We have $f(x, y) \neq 0$ if and only if $x \in S_i$ and $y \in S_i \cup S_{i-1}$ for some $i \in \mathbb{N}$.

Proof of Lemma 10.2. By the hypothesis, we note that $S_i \cap S_j = \emptyset$ for all $i, j \in \mathbb{N}$ and $i \neq j$. In other words, the supports $\{S_i\}$ are *mutually disjoint*. Suppose that $x \in S_i$ and $y \in S_i \cup S_{i-1}$ for some $i \in \mathbb{N}$. Then we have $\varphi_i(x) \neq 0, \varphi_j(x) = 0$ for all $j \neq i$ and either $\varphi_i(y) \neq 0$ or $\varphi_{i-1}(y) \neq 0$ so that

$$\begin{aligned} f(x, y) &= [\varphi_{i-1}(x) - \varphi_i(x)]\varphi_{i-1}(y) + [\varphi_i(x) - \varphi_{i+1}(x)]\varphi_i(y) \\ &= -\varphi_i(x)\varphi_{i-1}(y) + \varphi_i(x)\varphi_i(y) \\ &= [\varphi_i(y) - \varphi_{i-1}(y)]\varphi_i(x) \\ &\neq 0. \end{aligned}$$

Next, suppose that $x \notin S_i$ or $y \notin S_i$ for all $i \in \mathbb{N}$. Then we have

$$\varphi_i(x) = 0 \quad \text{or} \quad \varphi_i(y) = 0$$

for all $i \in \mathbb{N}$. In this case, the definition of f gives $f(x, y) = 0$. This completes the proof of the lemma. ■

By Lemma 10.2, if $f(x, y) \neq 0$, then $(x, y) \in S_i \times (S_i \cup S_{i-1}) \subset (0, 1) \times (0, 1)$ for some $i \in \mathbb{N}$. This means that

$$S = \{(x, y) \in \mathbb{R}^2 \mid f(x, y) \neq 0\} \subset (0, 1) \times (0, 1). \quad (10.5)$$

Since $S \subset (0, 1) \times (0, 1) \subset [0, 1] \times [0, 1]$ and $[0, 1] \times [0, 1]$ is closed in $\mathbb{R}^2$, Theorem 2.27(c) implies that $\overline{S} \subseteq [0, 1] \times [0, 1]$. Since $\overline{S} = \text{supp}(f)$, we have

$$\text{supp}(f) \subseteq [0, 1] \times [0, 1]$$

which means $\text{supp}(f)$ is a bounded set in $\mathbb{R}^2$. By Theorem 2.27(a), $\text{supp}(f)$ is a closed set in $\mathbb{R}^2$. Hence we conclude from Theorem 2.41 (Heine-Borel Theorem) that $\text{supp}(f)$ is compact in $\mathbb{R}^2$. This proves our first assertion.

For the second assertion, we consider two cases:

- **Case (i): f is continuous at $(x, y) \neq (0, 0)$.** Let $(x, y) \neq (0, 0)$. If both x and y are nonzero, then it is clear that

$$x \in S_i \subseteq (2^{-i}, 2^{1-i}) \quad \text{and} \quad y \in S_j \subseteq (2^{-j}, 2^{1-j})$$

for some $i, j \in \mathbb{N}$. Denote

$$\eta = \min(|x - 2^{-i}|, |x - 2^{1-i}|, |y - 2^{-j}|, |y - 2^{1-j}|).$$

Then we have

$$N_\eta((x, y)) \subseteq (2^{-i}, 2^{1-i}) \times (2^{-j}, 2^{1-j}).$$

Furthermore, for every $(p, q) \in N_\eta((x, y))$, we see that $p \notin S_r$ for all $r \neq i$ and $q \notin S_t$ for all $t \neq j$. These mean that $\varphi_r(p) = 0$ for all $r \neq i$ and $\varphi_t(q) = 0$ for all $t \neq j$, and they imply that

$$f(p, q) = \begin{cases} 0, & \text{if } i < j; \\ \varphi_i(p)\varphi_i(q), & \text{if } i = j; \\ -\varphi_i(p)\varphi_{i-1}(q), & \text{if } i = j + 1; \\ 0, & \text{if } i > j + 1 \end{cases} \quad (10.6)$$

for all $(p, q) \in N_\eta((x, y))$.^b By the definition (10.6), we always have

$$|f(x, y) - f(p, q)| = \begin{cases} 0, & \text{if } i < j; \\ |\varphi_i(x)\varphi_i(y) - \varphi_i(p)\varphi_i(q)|, & \text{if } i = j; \\ |\varphi_i(x)\varphi_{i-1}(y) - \varphi_i(p)\varphi_{i-1}(q)|, & \text{if } i = j + 1; \\ 0, & \text{if } i > j + 1 \end{cases} \quad (10.7)$$

for all $(x, y), (p, q) \in N_\eta((x, y))$. Since each φ_i is continuous on $\mathbb{R}$, it is also continuous on $(2^{-i}, 2^{1-i})$. Hence, given $\epsilon > 0$, it is easily seen from the expressions in (10.7) that we can always find $0 < \delta < \eta$ small enough so that

$$|f(x, y) - f(p, q)| < \epsilon$$

if $|(x, y) - (p, q)| < \delta$. In other words, f is continuous at every (x, y) whenever x and y are nonzero. The remaining cases $x \neq 0$ and $y = 0$ or $x = 0$ and $y \neq 0$ can be done similarly, so we don't repeat the details here and we simply conclude that f is continuous at every point $(x, y) \neq (0, 0)$.

- **Case (ii): f is discontinuous at $(0, 0)$.** Next we show that f is discontinuous at $(0, 0)$. Consider the sequence of points $\{(2^{-k}, 2^{-k})\}$. It is obvious that

$$(2^{-k}, 2^{-k}) \rightarrow (0, 0)$$

as $k \rightarrow \infty$. For each $k \in \mathbb{N}$, since $2^k \notin (2^{-i}, 2^{1-i})$ for all $i \in \mathbb{N}$, we have $2^k \notin \text{supp}(\varphi_i)$ for all $i \in \mathbb{N}$. In other words, we have $\varphi_i(2^k) = 0$ for all $i \in \mathbb{N}$ and then

$$f(2^{-k}, 2^{-k}) = 0 \quad (10.8)$$

^bBy the hypothesis, we just require that $S_k = \text{supp}(\varphi_k)$ is in $(2^{-k}, 2^{1-k})$ for each $k \in \mathbb{N}$, not the whole interval. Thus it may happen that $p \in (2^{-i}, 2^{1-i})$ but not in S_i or $q \in (2^{-j}, 2^{1-j})$ but not in S_j . As a result, it is still possible that $f(p, q) = 0$ in the second or third case in the definition (10.6). However, this does not give any trouble to our argument.

for all $k \in \mathbb{N}$. Since $\int_{\mathbb{R}} \varphi_k = 1$ for every $k \in \mathbb{N}$, we have

$$1 = \int_{\mathbb{R}} \varphi_k = \int_{2^{-k}}^{2^{1-k}} \varphi_k(x) dx \leq \int_{2^{-k}}^{2^{1-k}} \max_{x \in S_k} \varphi_k(x) dx = \max_{x \in S_k} \varphi_k(x) \int_{2^{-k}}^{2^{1-k}} dx = 2^{-k} \times \max_{x \in S_k} \varphi_k(x).$$

Thus we have

$$\max_{x \in S_k} \varphi_k(x) \geq 2^k$$

for every $k \in \mathbb{N}$. By Theorem 2.27(a), S_k is closed. Since $S_k \subset (0, 1)$ for all $k \in \mathbb{N}$, each S_k is bounded and Theorem 2.41 (Heine-Borel Theorem) implies that each S_k is compact. Since $\varphi_k : S_k \rightarrow \mathbb{R}$ is continuous, Theorem 4.16 (Extreme Value Theorem) ensures that φ_k attains its maximum value 2^k at some point in the interval S_k . Let one of these points be $p_k \in S_k$, i.e., $\varphi_k(p_k) = 2^k$. Consider another sequence of points $\{(p_k, p_k)\}$. By the construction, it is clear that $p_k \in (2^{-k}, 2^{1-k})$ and so

$$(p_k, p_k) \rightarrow (0, 0)$$

as $k \rightarrow \infty$. By the definition of f , we have

$$f(p_k, p_k) = [\varphi_k(p_k) - \varphi_{k+1}(p_k)]\varphi_k(k) = [\varphi_k(k)] = 2^{k+1}$$

giving

$$f(p_k, p_k) \rightarrow \infty \quad (10.9)$$

as $k \rightarrow \infty$. By comparing the two results (10.8) and (10.9), we conclude that f is discontinuous at $(0, 0)$. This shows the second assertion.^c

To prove the equations in the third assertion, we basically follow the consideration of Definition 10.1. If $y \notin S_i$ for all $i \in \mathbb{N}$, then we have $\varphi_i(y) = 0$ for all $i \in \mathbb{N}$ so that $f(x, y) = 0$ which gives

$$\int f(x, y) dx = 0$$

and thus

$$\int_{\mathbb{R}} dy \int_{\mathbb{R}} f(x, y) dx = \int_0^1 dy \int_{\mathbb{R}} f(x, y) dx = \sum_{i=1}^{\infty} \int_{2^{-i}}^{2^{1-i}} dy \int_{\mathbb{R}} f(x, y) dx. \quad (10.10)$$

On each S_i , we know that $f(x, y) = [\varphi_i(x) - \varphi_{i+1}(x)]\varphi_i(y)$ and thus

$$\begin{aligned} \int_{2^{-i}}^{2^{1-i}} dy \int_{\mathbb{R}} f(x, y) dx &= \int_{2^{-i}}^{2^{1-i}} \varphi_i(y) \left(\int_{\mathbb{R}} \varphi_i(x) dx - \int_{\mathbb{R}} \varphi_{i+1}(x) dx \right) dy \\ &= \int_{2^{-i}}^{2^{1-i}} \varphi_i(y) (1 - 1) dy \\ &= 0. \end{aligned} \quad (10.11)$$

Combining the expressions (10.10) and (10.11), we get

$$\int_{\mathbb{R}} dy \int_{\mathbb{R}} f(x, y) dx = 0.$$

Similarly, if $x \notin S_i$ for all $i \in \mathbb{N}$, then we have $\varphi_i(x) = 0$ for all $i \in \mathbb{N}$ so that $f(x, y) = 0$ which gives

$$\int f(x, y) dy = 0$$

and thus

$$\int_{\mathbb{R}} dx \int_{\mathbb{R}} f(x, y) dy = \int_0^1 dx \int_{\mathbb{R}} f(x, y) dy = \sum_{i=1}^{\infty} \int_{2^{-i}}^{2^{1-i}} dx \int_{\mathbb{R}} f(x, y) dy. \quad (10.12)$$

^cThe limit (10.9) also shows that f is unbounded in every neighborhood of $(0, 0)$.

On each S_i , we have

$$f(x, y) = \begin{cases} \varphi_i(x)[\varphi_i(y) - \varphi_{i-1}(y)], & \text{if } i \neq 1; \\ \varphi_1(x)\varphi_1(y), & \text{if } i = 1. \end{cases}$$

Thus we obtain

$$\begin{aligned} \int_{\mathbb{R}} f(x, y) dy &= \begin{cases} \varphi_i(x) \int_{\mathbb{R}} [\varphi_i(y) - \varphi_{i-1}(y)] dy, & \text{if } i \neq 1; \\ \varphi_1(x) \int_{\mathbb{R}} \varphi_1(y) dy, & \text{if } i = 1, \end{cases} \\ &= \begin{cases} 0, & \text{if } i \neq 1; \\ \varphi_1(x), & \text{if } i = 1. \end{cases} \end{aligned} \quad (10.13)$$

Combining the expressions (10.12) and (10.13), we obtain

$$\begin{aligned} \int_{\mathbb{R}} dx \int_{\mathbb{R}} f(x, y) dy &= \sum_{i=1}^{\infty} \int_{2^{-i}}^{2^{1-i}} dx \int_{\mathbb{R}} f(x, y) dy \\ &= \int_{\frac{1}{2}}^1 dx \int_{\mathbb{R}} f(x, y) dy + \sum_{i=2}^{\infty} \int_{2^{-i}}^{2^{1-i}} dx \int_{\mathbb{R}} f(x, y) dy \\ &= \int_{\frac{1}{2}}^1 \varphi_1(x) dx + \sum_{i=2}^{\infty} \int_{2^{-i}}^{2^{1-i}} 0 dx \\ &= 1. \end{aligned}$$

This completes the proof of the problem. ■

Problem 10.3

Rudin Chapter 10 Exercise 3.

Proof.

- (a) We modify the proof of Theorem 10.7 to obtain the result. In fact, put $\mathbf{F}_1 = \mathbf{F}$. Since $\mathbf{F}'_1(\mathbf{0}) = \mathbf{I}$, we also have $\mathbf{F}'(\mathbf{0}) = \mathbf{I}$. By the assumption, we know that $\mathbf{F}_1 \in \mathcal{C}'(V_1)$ for some neighborhood $V_1 \subseteq E$ of $\mathbf{0}$. Then we still have [21, Eqn. (19), p. 249] in the case $m = 1$:

$$\mathbf{F}'_1(\mathbf{0})\mathbf{e}_1 = \sum_{i=1}^n (D_1 F_i)(\mathbf{0})\mathbf{e}_i, \quad (10.14)$$

where $F_1, \dots, F_n$ are real $\mathcal{C}'$ -functions in V_1 . Recall from the proof of Theorem 10.7 that there is a k such that $1 \leq k \leq n$, $(D_1 F_k)(\mathbf{0}) \neq 0$ and B_1 is defined to be the flip that interchanges 1 and this k . Next, we apply the fact $\mathbf{F}'_1(\mathbf{0}) = \mathbf{I}$ to the formula (10.14) to obtain

$$\mathbf{e}_1 = (D_1 F_1)(\mathbf{0})\mathbf{e}_1 + (D_1 F_2)(\mathbf{0})\mathbf{e}_2 + \cdots + (D_1 F_n)(\mathbf{0})\mathbf{e}_n.$$

Thus, we have $(D_1 F_1)(\mathbf{0}) = 1$ and $(D_1 F_i)(\mathbf{0}) = 0$ for all $i = 2, 3, \dots, n$. This means that $k = m = 1$, so $B_1 = \mathbf{I}$. Since $(D_1 F_1)(\mathbf{0}) = 1$, we obtain from the expression

$$\mathbf{G}_1(\mathbf{x}) = \mathbf{x} + [F_1(\mathbf{x}) - x_1]\mathbf{e}_1$$

that $\mathbf{G}_1 \in \mathcal{C}'(V_1)$, $\mathbf{G}_1$ is primitive and

$$\mathbf{G}'_1(\mathbf{0}) = \mathbf{I}.$$

Therefore, we have

$$\mathbf{F}'_1(\mathbf{0}) = \mathbf{G}'_1(\mathbf{0}) = B_1 = \mathbf{I}.$$

Assume that $\mathbf{F}'_i(\mathbf{0}) = \mathbf{G}'_i(\mathbf{0}) = B_i = \mathbf{I}$, $\mathbf{G}_i \in \mathcal{C}'(V_i)$ and $\mathbf{G}_i$ is primitive for all $1 \leq i \leq m$. By [21, Eqn. (21), p. 250], we still have

$$\mathbf{F}_{m+1}(\mathbf{y}) = B_m \mathbf{F}_m \circ \mathbf{G}_m^{-1}(\mathbf{y}) = \mathbf{F}_m \circ \mathbf{G}_m^{-1}(\mathbf{y}) \quad (\mathbf{y} \in V_{m+1}). \quad (10.15)$$

Apply Theorem 9.15 (Chain Rule) and the induction hypothesis to the expression (10.15) to get

$$\mathbf{F}'_{m+1}(\mathbf{0}) = \mathbf{F}'_m(\mathbf{G}_m^{-1}(\mathbf{0}))(\mathbf{G}_m^{-1}(\mathbf{0}))' = \mathbf{F}'_m(\mathbf{0})(\mathbf{G}_m^{-1}(\mathbf{0}))' = (\mathbf{G}_m^{-1}(\mathbf{0}))'. \quad (10.16)$$

By using [21, Eqn. (52), p. 223] in the proof of Theorem 9.24 (Inverse Function Theorem) and then the induction hypothesis, we acquire that

$$(\mathbf{G}_m^{-1}(\mathbf{0}))' = (\mathbf{G}'_m(\mathbf{G}_m^{-1}(\mathbf{0})))^{-1} = (\mathbf{G}'_m(\mathbf{0}))^{-1} = \mathbf{I}^{-1} = \mathbf{I}.$$

Thus the expression (10.16) implies that $\mathbf{F}'_{m+1}(\mathbf{0}) = \mathbf{I}$. By Theorem 9.17, we have

$$\mathbf{F}'_{m+1}(\mathbf{0})\mathbf{e}_{m+1} = \sum_{i=m+1}^n (D_{m+1}F_i)(\mathbf{0})\mathbf{e}_i. \quad (10.17)$$

Recall from the proof of Theorem 10.7 that there is a k such that $m+1 \leq k \leq n$, $(D_{m+1}F_k)(\mathbf{0}) \neq 0$ and B_{m+1} is defined to be the flip that interchanges $m+1$ and this k . Since $\mathbf{F}'_{m+1}(\mathbf{0}) = \mathbf{I}$, we deduce from the formula (10.17) that

$$\mathbf{e}_{m+1} = (D_{m+1}F_{m+1})(\mathbf{0})\mathbf{e}_{m+1} + \cdots + (D_{m+1}F_n)(\mathbf{0})\mathbf{e}_n$$

so that

$$(D_{m+1}F_{m+1})(\mathbf{0}) = 1 \quad \text{and} \quad (D_{m+1}F_i)(\mathbf{0}) = 0$$

for $i = 2, \dots, n$. As a result, we have $k = m+1$ and then $B_{m+1} = \mathbf{I}$. Similarly, since $(D_{m+1}F_{m+1})(\mathbf{0}) = 1$, we obtain from the expression

$$\mathbf{G}_{m+1}(\mathbf{x}) = \mathbf{x} + [\mathbf{F}_{m+1}(\mathbf{x}) - x_{m+1}]\mathbf{e}_{m+1}$$

that

$$\mathbf{G}'_{m+1}(\mathbf{0}) = \mathbf{I}.$$

To sum up, what we have shown is that $\mathbf{F}'_i(\mathbf{0}) = \mathbf{G}'_i(\mathbf{0}) = B_i = \mathbf{I}$, $\mathbf{G}_i \in \mathcal{C}'(V_i)$ and $\mathbf{G}_i$ is primitive for all $1 \leq i \leq n$. The expression (10.15), with $\mathbf{y} = \mathbf{G}_m(\mathbf{x})$, is equivalent to

$$\mathbf{F}_m(\mathbf{x}) = \mathbf{F}_{m+1}(\mathbf{G}_m(\mathbf{x})) \quad (\mathbf{x} \in U_m). \quad (10.18)$$

By applying this with $m = 1, 2, \dots, n-1$, we establish that

$$\mathbf{F}_1 = \mathbf{F}_2 \circ \mathbf{G}_1 = \mathbf{F}_3 \circ \mathbf{G}_2 \circ \mathbf{G}_1 = \cdots = \mathbf{F}_n \circ \mathbf{G}_{n-1} \circ \cdots \circ \mathbf{G}_1$$

in some neighborhood of $\mathbf{0}$. By [21, Eqn. (18), p. 249], we have

$$\mathbf{F}_n(\mathbf{x}) = P_{n-1}\mathbf{x} + \alpha_n(\mathbf{x})\mathbf{e}_n, \quad (10.19)$$

so $\mathbf{F}_n$ is also primitive by Definition 10.5. Hence we just rename $\mathbf{F}_n$ by $\mathbf{G}_n$ in the formula (10.19) so as to obtain our desired result.

(b) Let $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the mapping defined by

$$\mathbf{F}(x, y) = (y, x).$$

Assume that $\mathbf{F} = \mathbf{G}_2 \circ \mathbf{G}_1$, where each $\mathbf{G}_i : V_i \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a primitive $\mathcal{C}'$ -mapping in some neighborhood of $\mathbf{0}$, $\mathbf{G}_i(\mathbf{0}) = \mathbf{0}$ and $\mathbf{G}'_i(\mathbf{0})$ is invertible. By Theorem 9.15 (Chain Rule), we have

$$\mathbf{F}'(\mathbf{0}) = \mathbf{G}'_2(\mathbf{G}_1(\mathbf{0}))\mathbf{G}'_1(\mathbf{0}) = \mathbf{G}'_2(\mathbf{0})\mathbf{G}'_1(\mathbf{0}). \quad (10.20)$$

Now we have

$$\mathbf{F}'(\mathbf{0}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (10.21)$$

Since $\mathbf{G}_1$ is a primitive, we have either

$$\mathbf{G}_1(x, y) = (x, g_1(x, y)) \quad \text{or} \quad \mathbf{G}_1(x, y) = (g_1(x, y), y),$$

where $g_1 : V_1 \rightarrow \mathbb{R}$ is a real function and it is differentiable at $\mathbf{0}$. By direct computation, we have either

$$\mathbf{G}'_1(\mathbf{0}) = \begin{pmatrix} 1 & 0 \\ a & b \end{pmatrix} \quad \text{and} \quad \mathbf{G}'_1(\mathbf{0}) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix},$$

where a and b are real numbers. Similarly, we have either

$$\mathbf{G}'_2(\mathbf{0}) = \begin{pmatrix} 1 & 0 \\ c & d \end{pmatrix} \quad \text{and} \quad \mathbf{G}'_2(\mathbf{0}) = \begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix},$$

where c and d are real numbers. Thus the right-hand side of the expression (10.20) is in one of the following forms:

$$\begin{pmatrix} 1 & 0 \\ c+ad & bd \end{pmatrix}, \quad \begin{pmatrix} c+ad & bd \\ 0 & b \end{pmatrix}, \quad \begin{pmatrix} a & b \\ ac & bc+d \end{pmatrix}, \quad \begin{pmatrix} ac & bc+d \\ 0 & 1 \end{pmatrix},$$

but none of them is the matrix (10.21). Hence, the mapping $\mathbf{F}$ is not the composition of any two primitive mappings.

We complete the proof of the problem. ■

Problem 10.4

Rudin Chapter 10 Exercise 4.

Proof. By direct computation, we have

$$\begin{aligned} (\mathbf{G}_2 \circ \mathbf{G}_1)(x, y) &= \mathbf{G}_2(\mathbf{G}_1(x, y)) \\ &= \mathbf{G}_2(e^x \cos y - 1, y) \\ &= (e^x \cos y - 1, (1 + e^x \cos y - 1) \tan y) \\ &= (e^x \cos y - 1, e^x \sin y) \\ &= \mathbf{F}(x, y) \end{aligned}$$

for every $(x, y) \in \mathbb{R}^2$. Suppose that $g_1, g_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ are given by

$$g_1(x, y) = e^x \cos y - 1 \quad \text{and} \quad g_2(x, y) = (1 + x) \tan y.$$

Then we have

$$\mathbf{G}_1(x, y) = (0, y) + (e^x \cos y - 1, 0) = y\mathbf{e}_2 + g_1(x, y)\mathbf{e}_1$$

and

$$\mathbf{G}_2(x, y) = (x, 0) + (0, (1 + x) \tan y) = x\mathbf{e}_1 + g_2(x, y)\mathbf{e}_2.$$

By Definition 10.5, $\mathbf{G}_1$ and $\mathbf{G}_2$ are primitive. By Definition 9.38, we have

$$\begin{aligned} J_{\mathbf{G}_1}(x, y) &= \begin{pmatrix} e^x \cos y & -e^x \sin y \\ 0 & 1 \end{pmatrix}, \\ J_{\mathbf{G}_2}(x, y) &= \begin{pmatrix} 1 & 0 \\ \tan y & (1 + x) \sec^2 y \end{pmatrix}, \\ J_{\mathbf{F}}(x, y) &= \begin{pmatrix} e^x \cos y & -e^x \sin y \\ e^x \sin y & e^x \cos y \end{pmatrix}. \end{aligned}$$

Thus we have

$$J_{\mathbf{G}_1}(0,0) = J_{\mathbf{G}_2}(0,0) = J_{\mathbf{F}}(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Let $D = \{(u,v) \in \mathbb{R}^2 \mid u^2 + v^2 \leq 1\}$ be the unit disk. Then we have $e^{2u} - v^2 \geq 0$ for all $(u,v) \in D$, so

$$\mathbf{H}_1(u,v) = (\sqrt{e^{2u} - v^2} - 1, v)$$

is well-defined. Then we have

$$\begin{aligned} (\mathbf{H}_1 \circ \mathbf{H}_2)(x,y) &= \mathbf{H}_1(\mathbf{H}_2(x,y)) \\ &= \mathbf{H}_1(x, e^x \sin y) \\ &= (\sqrt{e^{2x} - (e^x \sin y)^2} - 1, e^x \sin y) \\ &= (e^x \cos y - 1, e^x \sin y) \\ &= \mathbf{F}(x,y) \end{aligned}$$

in the unit disk D , completing the proof of the problem. ■

10.2 Generalizations of partitions of unity

Problem 10.5

Rudin Chapter 10 Exercise 5.

Proof. Here is the analogue of Theorem 10.8: Suppose K is a compact subset of a metric space X with metric d , and $\{V_\alpha\}$ is an open cover of K . Then there exists functions $\psi_1, \dots, \psi_s \in \mathcal{C}(X)$ such that

- (a) $0 \leq \psi_i \leq 1$ for $1 \leq i \leq s$;
- (b) each ψ_i has its support in some V_α , and
- (c) $\psi_1(x) + \dots + \psi_s(x) = 1$ for every $x \in K$.

When we read Rudin's proof carefully, we see that the key step of the construction there are relations [21, Eqns. (26) & (27), p.251]. More precisely, the relation [21, Eqn. (26), p.251] tells us that

$$W(\mathbf{x}_i) \setminus B(\mathbf{x}_i) \neq \emptyset \quad \text{and} \quad \overline{B(\mathbf{x}_i)} \cap W(\mathbf{x}_i)^c = \emptyset. \quad (10.22)$$

How do these relations (10.22) motivate ideas of the construction of functions in our question? We observe a few points first. Functions of the type in Problem 4.22 are constructed on disjoint nonempty *closed* sets A and B . Since

$$\overline{B(\mathbf{x}_i)} \quad \text{and} \quad W(\mathbf{x}_i)^c$$

are disjoint nonempty closed subsets of $\mathbb{R}^n$, they satisfy this condition. By constructing each $\varphi_i \in \mathcal{C}(\mathbb{R}^n)$, the relation [21, Eqn. (27), p.251] plays the role of “gluing” them together to give functions satisfying the unity requirement. Thus these give us a “direction” of proving our result. In fact, we first construct sets having properties similar to relations [21, Eqns. (26) & (27), p.251]. Since K is compact, $\{V_\alpha\}$ has a finite subcover and then there is a positive integer s such that

$$K \subseteq V_1 \cup V_2 \cup \dots \cup V_s.$$

We need some results from topology:

Lemma 10.3

If X is a metric space, then it is Hausdorff.

Proof of Lemma 10.3. See [18, p. 98] for the definition of a Hausdroff space. Let $x, y \in X$ with $x \neq y$. Let $\delta = d(x, y)$, $U = N_{\frac{\delta}{2}}(x)$ and $V = N_{\frac{\delta}{2}}(y)$. Then it is easy to check that

$$U \cap V = \emptyset,$$

proving the lemma. ■

Since K is a subset of X , it is also Hausdroff by [18, Theorem 17.11, p. 100]. Furthermore, it follows from Lemma 10.3 and [18, Theorem 32.3, p. 202] that K is *normal*.^d Now, by using **Step 1** of the proof of Theorem 36.1 in [18, p. 225], we can show that it is possible to find open coverings $\{U_1, \dots, U_s\}$ and $\{W_1, \dots, W_s\}$ of K such that

$$\overline{U_i} \subset W_i \subset \overline{W_i} \subset V_i \quad (10.23)$$

for each $i = 1, \dots, s$. Therefore, the sets U_i and W_i are what we need, see Figure 10.2 for the sets U_i, W_i and V_i .

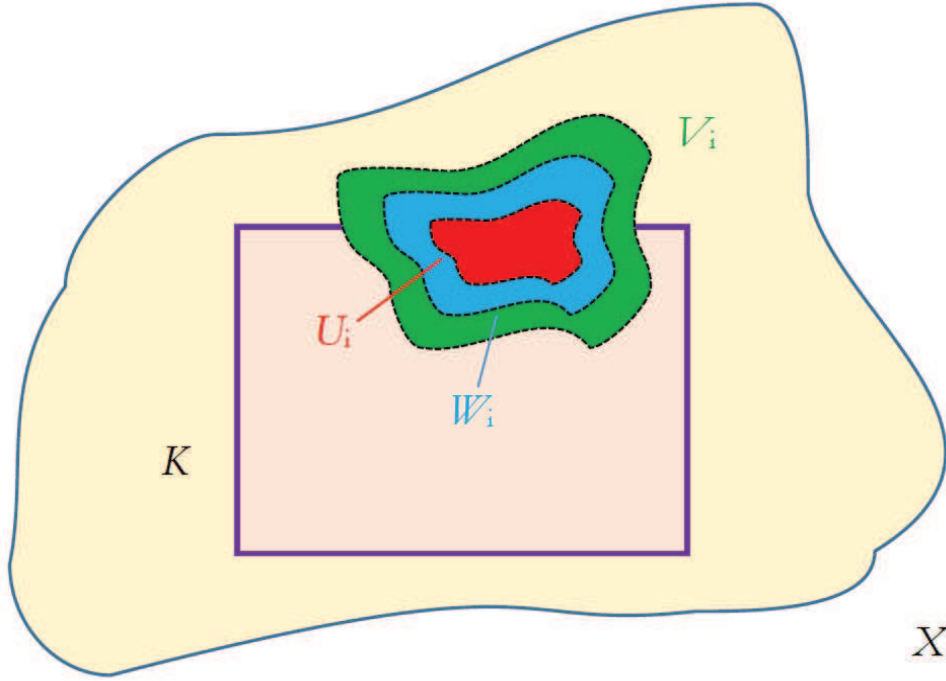


Figure 10.2: The figures of the sets U_i, W_i and V_i .

Next, we apply Problem 4.20 to construct the functions satisfying the requirements of the partitions of unity. To do this, let

$$A_i = \overline{U_i} \quad \text{and} \quad B_i = W_i^c,$$

where $i = 1, \dots, s$. Then the relations (10.23) show that A_i and B_i are disjoint nonempty closed sets. By Problem 4.20, we consider the continuous function $\varphi_i : X \rightarrow [0, 1]$ defined by

$$\varphi_i(x) = \frac{\rho_{B_i}(x)}{\rho_{A_i}(x) + \rho_{B_i}(x)}.$$

Then we have $\varphi_i(x) = 0$ precisely on B_i , $\varphi_i(x) = 1$ precisely on A_i and $0 \leq \varphi_i(x) \leq 1$ for all $x \in X$. In addition, since $\varphi_i^{-1}((0, 1]) = B_i^c = W_i$, Definition 10.3 and relations (10.23) imply that

$$\text{supp}(\varphi_i) = \overline{\varphi_i^{-1}((0, 1])} = \overline{W_i} \subset V_i.$$

^dSee [18, p. 195] for the definition of a normal space.

Let $W = W_1 \cup W_2 \cup \cdots \cup W_s$. Define $\varphi : W \rightarrow \mathbb{R}$ by

$$\varphi(x) = \sum_{i=1}^s \varphi_i(x).$$

If $x \in W$, then $x \in W_i$ for some i which means that $\varphi_i(x) > 0$. As a result, we have

$$\varphi(x) > 0 \quad (10.24)$$

for all $x \in W$. Let $Y = W^c$. By Theorem 2.24(a), W is open in X and then Theorem 2.23 ensures that Y is closed in X . Since K and Y are disjoint nonempty closed subsets of X , Problem 4.20 again implies that the function $f : X \rightarrow [0, 1]$ defined by

$$f(x) = \frac{\rho_Y(x)}{\rho_K(x) + \rho_Y(x)}$$

is a continuous function on X , $0 \leq f(x) \leq 1$ for all $x \in X$, $f(x) = 0$ precisely on Y and $f(x) = 1$ precisely on K .

Finally, for each $i = 1, \dots, s$, we define the function $\psi_i : X \rightarrow [0, 1]$ by

$$\psi_i(x) = \begin{cases} \frac{\varphi_i(x)f(x)}{\varphi(x)}, & \text{if } x \in W; \\ 0, & \text{if } x \in X \setminus W. \end{cases} \quad (10.25)$$

We claim that the functions $\psi_1, \dots, \psi_s$ satisfy conditions (a) to (c):^e

- **Proof of condition (a).** To this end, we follow from the facts $0 \leq \varphi_i(x) \leq 1$, $0 \leq f(x) \leq 1$ for all $x \in X$, the inequality (10.24) and the definition (10.25) that

$$0 \leq \psi_i(x) \leq 1$$

on X . This proves condition (a).

- **Proof of condition (b).** The definition (10.25) and the inequality (10.24) imply that $\psi(x_0) \neq 0$ if and only if $\frac{\varphi_i(x_0)f(x_0)}{\varphi(x_0)} \neq 0$ if and only if $\varphi_i(x_0)f(x_0) \neq 0$ if and only if $\varphi_i(x_0) \neq 0$ and $f(x_0) \neq 0$. By the definition of φ_i , we have

$$\varphi_i(x_0) \neq 0 \quad \text{if and only if} \quad x_0 \in B_i^c = W_i. \quad (10.26)$$

Similarly, the definition of f implies that $f(x_0) \neq 0$ if and only if $x_0 \notin Y$. Since $Y = W^c$, we have

$$f(x_0) \neq 0 \quad \text{if and only if} \quad x_0 \in W. \quad (10.27)$$

In conclusion, statements (10.26) and (10.27) tell us that

$$\psi_i(x) \neq 0 \quad \text{if and only if} \quad x \in W_i.$$

Therefore, we deduce from this and the relations (10.23) that

$$\text{supp}(\psi_i) = \overline{\{x \in X \mid \psi_i(x) \neq 0\}} = \overline{W_i} \subset V_i.$$

This is exactly condition (b).

- **Proof of condition (c).** Since $\{W_1, \dots, W_s\}$ is an open cover of K , if $x \in K$, then $x \in W$ and it follows from the definition (10.25) that

$$\sum_{i=1}^s \psi_i(x) = \frac{f(x)}{\varphi(x)} \sum_{i=1}^s \varphi_i(x) = \frac{1}{\varphi(x)} \cdot \varphi(x) = 1.$$

This shows condition (c).

^eSuch functions are called *bump functions* in X .

To finish our proof, we have to show that each function ψ_i is continuous on X . It is trivial to see that ψ_i is continuous on W and on $X \setminus W$ by its definition (10.25). Thus what is left is to show that it is continuous on the *boundary* of W .

Recall that the boundary of a set A , denoted by ∂A , is defined to be

$$\overline{A} \setminus A^\circ.$$

Let $a \in \partial W$. Since W is open, $a \in X \setminus W^f$ and then $\psi_i(a) = 0$. Let $\{a_n\}$ be a sequence^g in W such that

$$\lim_{n \rightarrow \infty} a_n = a.$$

Since f is continuous on X and $f(x) = 0$ on $Y = W^c$, we have

$$\lim_{n \rightarrow \infty} f(a_n) = f(a) = 0.$$

By the inequality (10.24), we know that $\frac{\varphi_i(x)}{\varphi(x)}$ is bounded by 1 on W . Then we obtain from the remark preceding Theorem 3.20 that

$$\lim_{n \rightarrow \infty} \psi_i(a_n) = 0 = \psi_i(a).$$

Hence, by Theorem 4.2 and Definition 4.5, ψ_i is also continuous on ∂W and then on X , i.e., $\psi_i \in \mathcal{C}(X)$ for $i = 1, 2, \dots, s$. We end the proof of the problem. ■

Problem 10.6

Rudin Chapter 10 Exercise 6.

Proof. As a remark, we note that a function f is called *smooth* if it has derivatives of all orders in its domain.

Firstly, we recall the function in Problem 8.1:

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & (x \neq 0); \\ 0, & (x = 0). \end{cases}$$

This is a function satisfying the conditions that $f(x) > 0$ for all $x \in \mathbb{R}$, $f \in \mathcal{C}^\infty(\mathbb{R})$ and $f^{(m)}(0) = 0$ for all $m = 1, 2, \dots$ ^h Secondly, we define the function $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$\begin{aligned} g(x) &= \begin{cases} f(\sqrt{x}), & \text{if } x > 0; \\ 0, & \text{if } x \leq 0, \end{cases} \\ &= \begin{cases} e^{-\frac{1}{x}}, & \text{if } x > 0; \\ 0, & \text{if } x \leq 0. \end{cases} \end{aligned}$$

This function is nonnegative on $\mathbb{R}$. Furthermore, it can be shown in the same way as the proof of Problem 8.1 that this function g has derivatives of all orders for all real x (i.e., $g \in \mathcal{C}^\infty(\mathbb{R})$) and $g^{(m)}(0) = 0$ for all $m = 1, 2, \dots$). Thirdly, we define the function $h: \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$h(\mathbf{x}) = \frac{g(2 - |\mathbf{x}|)}{g(2 - |\mathbf{x}|) + g(|\mathbf{x}| - 1)}. \quad (10.28)$$

If $|\mathbf{x}| < 2$, then we have $g(2 - |\mathbf{x}|) > 0$. If $|\mathbf{x}| \geq 2$, then we have $g(2 - |\mathbf{x}|) > 0$ but $g(|\mathbf{x}| - 1) > 0$. Thus we always have

$$g(2 - |\mathbf{x}|) + g(|\mathbf{x}| - 1) > 0$$

^fSee [18, Exercise 19(a), p. 102]

^gSuch a sequence exists because of the equivalent definition of the boundary of a set: $a \in \partial A$ if and only if $N_\delta(a) \cap A \neq \emptyset$ and $N_\delta(a) \cap A^c \neq \emptyset$ for every $\delta > 0$.

^hThe notation $\mathcal{C}^\infty(\mathbb{R}^n)$ denotes the set of all infinitely differentiable functions on $\mathbb{R}^n$ for some positive integer n .

on $\mathbb{R}^n$. Apart from this, since $g(|\mathbf{x}| - 1) = 0$ for $|\mathbf{x}| \leq 1$, we have $h(\mathbf{x}) = 1$ for $|\mathbf{x}| \leq 1$. Similarly, we have $h(\mathbf{x}) = 0$ for $|\mathbf{x}| \geq 2$ and $0 \leq h(\mathbf{x}) \leq 1$ for $1 \leq |\mathbf{x}| \leq 2$. By repeated applications of Theorems 9.15 (Chain Rule), 9.21 and the fact that g has derivatives of all orders, we conclude that $h \in \mathcal{C}^\infty(\mathbb{R}^n)$.

It is time to construct our desired function based on the function (10.28). We follow the proof of Theorem 10.8. Without loss of generality, we may assume further that

$$\overline{B_{r_i}(\mathbf{x}_i)} \subset W_{2r_i}(\mathbf{x}_i) \subset \overline{W_{2r_i}(\mathbf{x}_i)} \subset V_{\alpha(\mathbf{x}_i)} \quad \text{and} \quad K \subseteq B_{r_1}(\mathbf{x}_1) \cup \cdots \cup B_{r_s}(\mathbf{x}_s), \quad (10.29)$$

where $\mathbf{x}_i \in K$, $B_{r_i}(\mathbf{x}_i)$ and $W_{2r_i}(\mathbf{x}_i)$ are open balls, centered at $\mathbf{x}_i$, with rational radii r_i and $2r_i$ respectively, $1 \leq i \leq s$. Now we define $\varphi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$\varphi_i(\mathbf{x}) = h\left(\frac{\mathbf{x}}{r_i}\right) \quad \text{and} \quad \varphi(\mathbf{x}) = \sum_{i=1}^s \varphi_i(\mathbf{x}) \quad (10.30)$$

respectively. By the construction of the function h , we know that $\varphi_i(\mathbf{x}) = 1$ if $\mathbf{x} \in B_{r_i}(\mathbf{x}_i)$, $\varphi_i(\mathbf{x}) = 0$ if $\mathbf{x} \in (W_{2r_i}(\mathbf{x}_i))^c$ and $0 \leq \varphi_i(\mathbf{x}) \leq 1$ on $\mathbb{R}^n$. Besides, if $\mathbf{x} \in K$, then $\mathbf{x} \in B_{r_i}(\mathbf{x}_i)$ for some i so that $\varphi_i(\mathbf{x}) > 0$. Thus we always have

$$\varphi(\mathbf{x}) > 0$$

on K and we can further define the function $\psi_i : \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$\psi_i(\mathbf{x}) = \frac{\varphi_i(\mathbf{x})}{\varphi(\mathbf{x})}. \quad (10.31)$$

We check that the functions $\psi_1, \dots, \psi_s$ satisfy conditions (a) to (c):

- **Proof of condition (a).** It follows from the definitions (10.30) and (10.31) trivially.
- **Proof of condition (b).** Since $\varphi_i(\mathbf{x}) = 0$ on $(W_{2r_i}(\mathbf{x}_i))^c$, we have

$$\{\mathbf{x} \in \mathbb{R}^n \mid \varphi_i(\mathbf{x}) \neq 0\} \subset W_{2r_i}(\mathbf{x}_i).$$

Therefore, we follow from the left-most relation in (10.29) that

$$\text{supp}(\varphi_i) \subset W_{2r_i}(\mathbf{x}_i) \subset V_{\alpha(\mathbf{x}_i)}.$$

This verifies condition (b).

- **Proof of condition (c).** For every $\mathbf{x} \in K$, we have

$$\sum_{i=1}^s \psi_i(\mathbf{x}) = \frac{1}{\varphi(\mathbf{x})} \sum_{i=1}^s \varphi_i(\mathbf{x}) = 1$$

so that condition (c) is satisfied.

This completes the proof of the problem. ■

10.3 Applications of Theorem 10.9 (Change of Variables Theorem)

Problem 10.7

Rudin Chapter 10 Exercise 7.

Proof.

(a) Recall that

$$Q^k = \{\mathbf{x} = (x_1, \dots, x_k) \in \mathbb{R}^k \mid x_1 + \dots + x_k \leq 1 \text{ and } x_1, \dots, x_k \geq 0\}. \quad (10.32)$$

Let $S = \{\mathbf{0}, \mathbf{e}_1, \dots, \mathbf{e}_k\}$. Then the smallest convex subset of $\mathbb{R}^k$ containing S is exactly the *convex hull* of S . If we denote the convex hull of S by $\text{Conv}(S)$, then we have

$$\text{Conv}(S) = \{c_0\mathbf{0} + c_1\mathbf{e}_1 + \dots + c_k\mathbf{e}_k \mid c_0 + c_1 + \dots + c_k = 1 \text{ and } c_0, c_1, \dots, c_k \geq 0\}. \quad (10.33)$$

We have to prove that $Q^k = \text{Conv}(S)$. To this end, we suppose that $\mathbf{x} \in \text{Conv}(S)$. Then we have

$$\mathbf{x} = c_0\mathbf{0} + c_1\mathbf{e}_1 + \dots + c_k\mathbf{e}_k,$$

where $c_0 + c_1 + \dots + c_k = 1$ and $c_0, c_1, \dots, c_k \geq 0$. If $x_1 = c_1, x_2 = c_2, \dots, x_k = c_k$, then since $c_0\mathbf{0} = \mathbf{0}$, we have

$$\mathbf{x} = c_1\mathbf{e}_1 + \dots + c_k\mathbf{e}_k = (c_1, \dots, c_k) = (x_1, \dots, x_k),$$

where $x_1 + \dots + x_k \leq 1$ and $x_1, \dots, x_k \geq 0$. By definition (10.32), we have $\mathbf{x} \in Q^k$. Conversely, we suppose that $\mathbf{x} \in Q^k$, so

$$\mathbf{x} = (x_1, \dots, x_k) = x_1\mathbf{e}_1 + \dots + x_k\mathbf{e}_k,$$

where $x_1 + \dots + x_k \leq 1$ and $x_1, \dots, x_k \geq 0$. Let $c_0 = 1 - (x_1 + \dots + x_k), c_1 = x_1, \dots, c_k = x_k$. We can see from these that

$$c_0 + c_1 + \dots + c_k = 1, \quad c_0, c_1, \dots, c_k \geq 0$$

and

$$\mathbf{x} = x_1\mathbf{e}_1 + \dots + x_k\mathbf{e}_k = c_0\mathbf{0} + c_1\mathbf{e}_1 + \dots + c_k\mathbf{e}_k.$$

By definition (10.33), we have $\mathbf{x} \in \text{Conv}(S)$. Hence, we establish

$$Q^k = \text{Conv}(S).$$

(b) Let $\mathbf{f} : X \rightarrow Y$, where X is a convex set. Without loss of generality, we may assume that $Y = \mathbf{f}(X)$. Our purpose is to show that Y is convex. Suppose that $\mathbf{u}, \mathbf{v} \in Y$. By Definition 10.26, we know that

$$\mathbf{f}(\mathbf{x}) = \mathbf{f}(\mathbf{0}) + A\mathbf{x}$$

for some $A \in L(X, Y)$. Thus there are $\mathbf{a}, \mathbf{b} \in X$ such that

$$\mathbf{u} = \mathbf{f}(\mathbf{a}) = \mathbf{f}(\mathbf{0}) + A\mathbf{a} \quad \text{and} \quad \mathbf{v} = \mathbf{f}(\mathbf{b}) = \mathbf{f}(\mathbf{0}) + A\mathbf{b}.$$

Let $0 < \lambda < 1$. Since X is convex, we have $\lambda\mathbf{a} + (1 - \lambda)\mathbf{b} \in X$ and then

$$\lambda\mathbf{u} + (1 - \lambda)\mathbf{v} = \mathbf{f}(\mathbf{0}) + A[\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}] = \mathbf{f}(\lambda\mathbf{a} + (1 - \lambda)\mathbf{b}) \in Y.$$

In other words, Y is also convex which is our desired result.

This finishes the proof of the problem. ■

Problem 10.8

Rudin Chapter 10 Exercise 8.

Proof. Before proving the result, it is believed that Rudin missed the few words “and (1, 1) to (4, 5)” at the end of the second sentence in the question.

By Definition 10.26, since $T(0, 0) = (1, 1)$, the affine map $T : I^2 \rightarrow H$ has the form

$$T(\mathbf{x}) = T(\mathbf{0}) + A\mathbf{x} = (1, 1) + A\mathbf{x},$$

where A is a 2×2 matrix and $I^2 = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Then we have

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax + by + 1 \\ cx + dy + 1 \end{pmatrix}. \quad (10.34)$$

In particular, since $T(1, 0) = (3, 2)$ and $T(0, 1) = (2, 4)$, we obtain from the expression (10.34) that

$$\begin{pmatrix} a+1 \\ c+1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} b+1 \\ d+1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

which imply that $a = 2, b = c = 1$ and $b = 3$. Therefore, the affine map T is given by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x + y + 1 \\ x + 3y + 1 \end{pmatrix}. \quad (10.35)$$

See 10.3 for the mapping T .

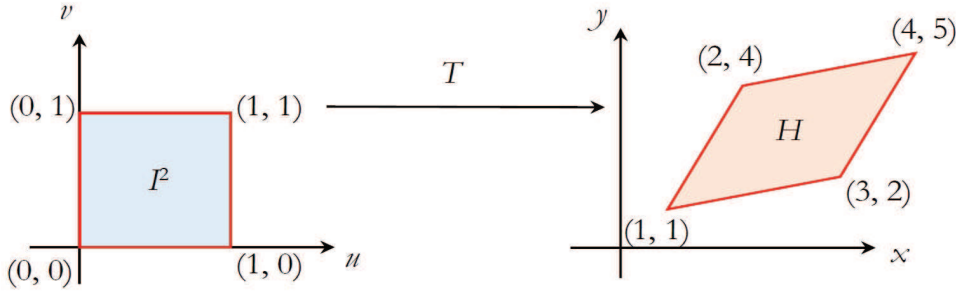


Figure 10.3: The mapping $T : I^2 \rightarrow H$.

By the representation (10.35) and Definition 9.38, we have

$$J_T(\mathbf{x}) = \det[T'(\mathbf{x})] = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5 \quad (10.36)$$

for every $\mathbf{x} \in I^2$.

Let I° and H° be the interiors of the square I^2 and the parallelogram H respectively. To evaluate the integral α , one may think that Theorem 10.9 should be applied to the function $f : H \rightarrow \mathbb{R}$ defined by

$$f(x, y) = e^{x-y}.$$

Although the map T satisfies the hypotheses of the theorem (i.e., $T \in \mathcal{C}(I^\circ)$ by Theorem 9.21 and the fact that $T \in \mathcal{C}(I^2)$; $T : I^\circ \rightarrow H^\circ$ is bijective by the fact that A is invertible; and $J_T(\mathbf{x}) \neq 0$ on I°), the function f fails in this case: $f(x, y) \neq 0$ for every $(x, y) \in H^\circ$ so that

$$\text{supp}(f) = \overline{H^\circ} = H$$

which is compact in $\mathbb{R}^2$, but

$$\text{supp}(f) \not\subseteq T(I^\circ) = H^\circ.$$

Therefore, we have to seek a variation of Theorem 10.9 which is applicable to this situation. In fact, the version considered by Fitzpatrick ([8, Theorem 19.9, p. 506]) serves this purpose.ⁱ

Lemma 10.4

Suppose that U is open in $\mathbb{R}^n$ and the mapping $T : U \rightarrow \mathbb{R}^n$ is a *smooth change of variables*. Let D be an open Jordan domain such that $K = D \cup \partial D \subseteq U$. Then $T(K)$ is a Jordan domain with the property that for any continuous function $f : T(K) \rightarrow \mathbb{R}$, the following formula holds:

$$\int_{T(K)} f(\mathbf{y}) \, d\mathbf{y} = \int_K f(T(\mathbf{x})) |J_T(\mathbf{x})| \, d\mathbf{x}.$$

ⁱ T is called *smooth* if it is one-to-one and $J_T(\mathbf{x}) \neq 0$ on U . Besides, a bounded subset of $\mathbb{R}^n$ is said to be a Jordan domain if its boundary has Jordan content 0, see [8, p. 492].

To apply Lemma 10.4, we need to extend the domain of T from the unit square I^2 to the open 2-cell $R^2 = (0, 2) \times (0, 2)$. Now it is easy to check that T is also one-to-one and $J_T(\mathbf{x}) \neq 0$ on R^2 . By [8, Exercise 1, p. 496], the unit square I^2 is a Jordan domain. By definition, it means that ∂I^2 has Jordan content 0. Since $\partial I^\circ = \partial I^2$, the set I° is an open Jordan domain. Furthermore, we have

$$K = I^\circ \cup \partial I^\circ = I^2 \subset R^2$$

and then $T(K) = T(I^2) = H$. Since f is obviously continuous on H , we follow from Lemma 10.4 that

$$\alpha = \int_H e^{x-y} dx dy = \int_H f(\mathbf{y}) d\mathbf{y} = \int_{T^{-1}(H)} f(T(\mathbf{x})) |J_T(\mathbf{x})| d\mathbf{x} = \int_{I^2} f(T(\mathbf{x})) |J_T(\mathbf{x})| d\mathbf{x}. \quad (10.37)$$

By the formula (10.35), we have

$$f(T(\mathbf{x})) = f(2u + v + 1, x + 3v + 1) = e^{u-2v}$$

and then it deduces from the formula (10.37) that

$$\alpha = 5 \int_0^1 \int_0^1 e^{u-2v} du dv. \quad (10.38)$$

Since $e^{u-2v} \in \mathcal{C}(I^2)$, Theorem 10.2^j says that the *order of the integration* on the right-hand side of (10.38) does not matter. Hence we obtain from the integral (10.38) that

$$\alpha = 5 \int_0^1 \int_0^1 e^{u-2v} du dv = 5 \left(\int_0^1 e^u du \right) \left(\int_0^1 e^{-2v} dv \right) = \frac{5}{2} (e - 1)(e^{-2} + 1),$$

completing the proof of the problem. ■

Problem 10.9

Rudin Chapter 10 Exercise 9.

Proof. Rudin said that the interval is from $(0, 0)$ to $(0, a)$, but the end point is a typo, see [1, Example 1, p. 418]. Let $A = \{(r, \theta) \mid 0 \leq r \leq a, 0 \leq \theta \leq 2\pi\}$ and $D = \{(x, y) \mid x^2 + y^2 = a^2\}$ be the rectangle in the $r\theta$ -plane and the closed disc with center $(0, 0)$ and radius a respectively. See Figure 10.4 below:

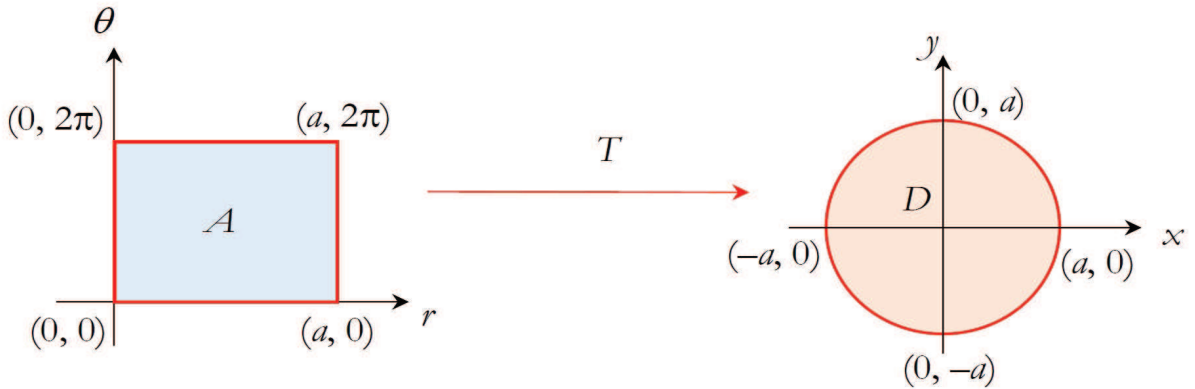


Figure 10.4: The mapping $T : A \rightarrow D$.

^jIt is a kind of Fubini's Theorem: If $f(x, y) = g(x)h(y)$ and $I^2 = [a, b] \times [c, d]$, where a, b, c and d are some constants, then we have $\int_{I^2} f(x, y) dx dy = \left(\int_a^b g(x) dx \right) \left(\int_c^d h(y) dy \right)$. See, for instance, [24, Theorem 3.10, p. 58].

Let $T : A \rightarrow D$ be defined by

$$T(r, \theta) = (r \cos \theta, r \sin \theta).$$

Since $(r \cos \theta)^2 + (r \sin \theta)^2 = r^2$, T maps A into D . Given any $(x, y) \in D$. If $x \neq 0$, then let $\theta = \tan^{-1} \frac{y}{x}$ (thus $\theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$) and $r = \sqrt{x^2 + y^2}$ which imply that

$$T(r, \theta) = (x, y).$$

If $x = 0$ and $y = r$, then

$$T\left(r, \frac{\pi}{2}\right) = (0, r).$$

If $x = 0$ and $y = -r$, then

$$T\left(r, \frac{3\pi}{2}\right) = (0, -r).$$

Therefore, the mapping T is onto.

Next, suppose that $A^\circ = \{(r, \theta) \mid 0 < r < a, 0 < \theta < 2\pi\}$ which is open by Problem 2.9(a). We consider the mapping

$$T : A^\circ \rightarrow D_0, \quad (10.39)$$

where $D_0 = \{(x, y) \mid x^2 + y^2 < a^2\} \setminus \{(r, 0) \mid 0 \leq r \leq a\}$.^k See Figure 10.5 for the illustration.

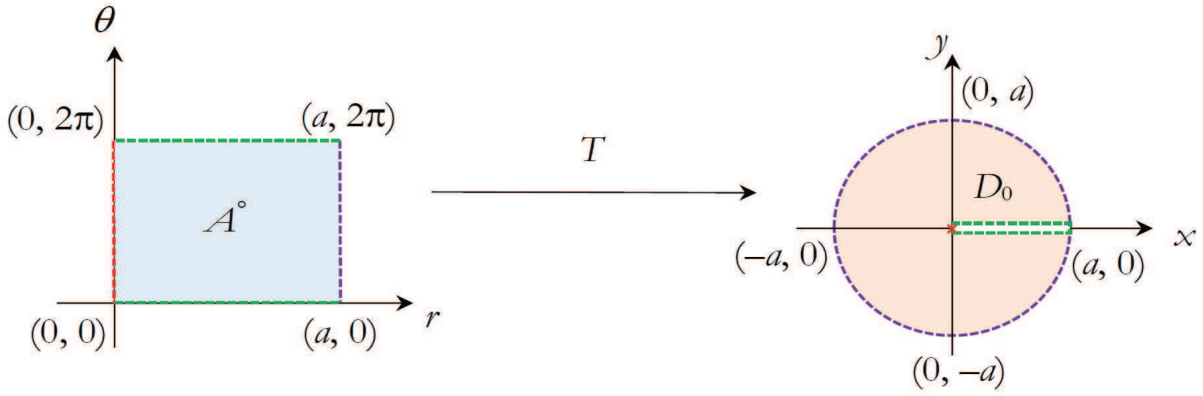


Figure 10.5: The mapping $T : A^\circ \rightarrow D_0$.

We check the hypotheses of Theorem 10.9 for the map (10.39):

- Let $(r, \theta), (r', \theta') \in A^\circ$. If $T(r, \theta) = T(r', \theta')$, then we have $(r \cos \theta, r \sin \theta) = (r' \cos \theta', r' \sin \theta')$ which gives

$$r \cos \theta = r' \cos \theta' \quad \text{and} \quad r \sin \theta = r' \sin \theta'.$$

Thus $\tan \theta = \tan \theta'$ and either $\theta' = \theta$ or $\theta' = \pi + \theta$. However, the relation $\theta' = \pi + \theta$ implies that

$$r \cos \theta = r' \cos(\pi + \theta) = -r' \cos \theta \quad \text{and} \quad r \sin \theta = r' \sin(\pi + \theta) = -r' \sin \theta. \quad (10.40)$$

Since $r > 0$ and $r' > 0$, there is no θ such that the two equations in (10.40) hold *simultaneously*. Therefore, we have $\theta' = \theta$ and then $r = r'$. In other words, the mapping $T : A^\circ \rightarrow D_0$ is one-to-one.

- For the Jacobian of T , since

$$[T'(r, \theta)] = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix},$$

we have

$$J_T(r, \theta) = \det[T'(r, \theta)] = r \neq 0$$

for every $(r, \theta) \in A^\circ$.

^kWe notice that $T(0, \theta) = (0, 0)$ for every $0 \leq \theta \leq 2\pi$; $T(r, 0) = T(r, 2\pi) = (r, 0)$ for every $0 \leq r \leq a$; and $T(a, \theta) = (a \cos \theta, a \sin \theta)$ for every $0 \leq \theta \leq 2\pi$ which is the boundary of D .

Suppose that $f \in \mathcal{C}(D)$. Then f is a continuous and bounded function on the compact set D . By Theorem 4.15, $f(D)$ is bounded too. Since

$$\text{supp}(f) = \overline{\{(x, y) \in D \mid f(x, y) \neq 0\}} \subseteq f(D),$$

the set $\text{supp}(f)$ is also bounded. Since $\text{supp}(f)$ is closed, Theorem 2.41 (Heine-Borel Theorem) shows that it is compact. In addition, since $D_0 \subset D$, we know from Definition 4.5 that f is also continuous on D_0 . Let f_{D_0} be the restriction of f to D_0 (see Problem 4.7 for the definition). Then

$$\text{supp}(f_{D_0}) = \overline{\{(x, y) \in D_0 \mid f(x, y) \neq 0\}}$$

is a closed subset of the compact set $\text{supp}(f)$, so $\text{supp}(f_{D_0})$ is also compact by Theorem 2.35. Furthermore, it is clear that $\text{supp}(f_{D_0})$ lies in $D_0 = T(A^\circ)$. By Theorem 10.9 with $\mathbf{y} = (x, y)$ and $\mathbf{x} = (r, \theta)$, we must have

$$\int_{D_0} f(x, y) \, dx \, dy = \int_{D_0} f(\mathbf{y}) \, d\mathbf{y} = \int_{A^\circ} f(T(\mathbf{x})) |J_T(\mathbf{x})| \, d\mathbf{x}. \quad (10.41)$$

Since $f \in \mathcal{C}(D)$ and $T, J_T \in \mathcal{C}(A)$, we have

$$(f \circ T) \times J_T \in \mathcal{C}(A)$$

and then Theorem 10.2 implies that the mapping $(f \circ T) \times J_T : A \rightarrow \mathbb{R}$ is integrable. By [17, Theorem 13.6, p. 110], the restriction

$$(f \circ T) \times J_T : A^\circ \rightarrow \mathbb{R}$$

is also integrable and

$$\int_{A^\circ} f(T(\mathbf{x})) |J_T(\mathbf{x})| \, d\mathbf{x} = \int_A f(T(\mathbf{x})) |J_T(\mathbf{x})| \, d\mathbf{x}. \quad (10.42)$$

Combining formulas (10.41) and (10.42), we obtain

$$\int_{D_0} f(x, y) \, dx \, dy = \int_0^a \int_0^{2\pi} f(T(r, \theta)) r \, dr \, d\theta. \quad (10.43)$$

To summary, what we have shown in the previous paragraph is that if $f \in \mathcal{C}(D)$, then the equality (10.43) holds on the interior D_0 of D . To remove this restriction, we proceed as in Example 10.4. In other words, we want to show something like [21, Eqn. (8), p. 248]. Since $f \in \mathcal{C}(D)$, we extend f to a function on I^2 by setting $f(x, y) = 0$ on $I^2 \setminus D$, and define

$$\int_D f(x, y) \, dx \, dy = \int_{I^2} f(x, y) \, dx \, dy. \quad (10.44)$$

Here I^2 is the 2-cell defined by

$$-1 \leq x, y \leq 1.$$

Since f may be discontinuous on I^2 , the existence of the **integral** on the right-hand side of (10.44) needs proof. To do this, suppose $0 < \delta < 1$ and let¹

$$\varphi_\delta(t) = \begin{cases} 1, & \text{if } 0 \leq t \leq 1 - \delta; \\ \frac{1-t}{\delta}, & \text{if } 1 - \delta < t \leq 1; \\ 0, & \text{if } t > 1. \end{cases} \quad (10.45)$$

It is easy to see that the function (10.45) is continuous on $[0, \infty)$. Define $F_\delta : I^2 \rightarrow \mathbb{R}$ by

$$F_\delta(x, y) = \varphi_\delta(x^2 + y^2) f(x, y). \quad (10.46)$$

Here we have a result about this F_δ :

Lemma 10.5

The function $F_\delta : I^2 \rightarrow \mathbb{R}$ defined by (10.46) is continuous and bounded on I^2 .

¹The functions φ and F defined in [21, Eqn. (5) & (6)] depend on δ , but Rudin didn't mention this point clearly in the text.

Proof of Lemma 10.5. If $(x, y) \in D^\circ$, then we have $F_\delta(x, y) = f(x, y)$ which is clearly continuous on D° . Similarly, if $(x, y) \in I^2 \setminus D$, then we have $F_\delta(x, y) = 0$ so that it is also continuous on $D \setminus I^2$. Suppose that $(x, y) \in \partial D$. Then we have

$$\lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 < 1}} F_\delta(x, y) = \lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 < 1}} \varphi_\delta(x^2 + y^2) f(x, y) = \lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 < 1}} \frac{1 - (x^2 + y^2)}{\delta} f(x, y) = 0$$

and

$$\lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 > 1}} F_\delta(x, y) = \lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 > 1}} \varphi_\delta(x^2 + y^2) f(x, y) = \lim_{\substack{x^2+y^2 \rightarrow 1 \\ x^2+y^2 > 1}} 0 \times f(x, y) = 0.$$

These mean that F_δ is continuous on ∂D . Since D is compact, we conclude from Theorem 4.15 that $F_\delta \in \mathcal{C}(I^2)$. ■

Let's return to the proof of the problem. For each $x_0 \in [-1, 1]$, let

$$S_{x_0} = \{y \in [-1, 1] \mid F_\delta(x_0, y) \neq f(x_0, y)\}.$$

If $S_{x_0} \neq \emptyset$, then $F_\delta(x_0, y) \neq f(x_0, y)$ for some $y \in [-1, 1]$ which amounts to $\varphi(x_0^2 + y^2) \neq 1$ and $f(x_0, y) \neq 0$ and then it is equivalent to

$$1 - \delta \leq x_0^2 + y^2 \leq 1 \quad \text{and} \quad f(x_0, y) \neq 0.$$

In this case, the set S_{x_0} is a segment of length does not exceed $2(1 - \sqrt{1 - \delta})$. By definition (10.45), we have $0 \leq \varphi \leq 1$ which shows that

$$|F_{1\delta}(x) - f_1(x)| \leq 2(1 - \sqrt{1 - \delta}) \cdot \max_{(x,y) \in I^2} |f(x, y)| \quad (10.47)$$

for all $x \in [-1, 1]$, where

$$F_{1\delta}(x) = \int_{-1}^1 F_\delta(x, y) dy \quad \text{and} \quad f_1(x) = \int_{-1}^1 f(x, y) dy.$$

As $\delta \rightarrow 0$, the inequality (10.47) implies that f_1 is a uniform limit of a sequence of continuous functions $\{F_{1\delta}\}$ on $[-1, 1]$. By Theorem 7.12, we have $f_1 \in \mathcal{C}([-1, 1])$. This proves the existence of the integral on the right-hand side of (10.44). Furthermore, we follow from the inequality (10.47) that

$$\left| \int_{I^2} F_\delta(x, y) dx dy - \int_{I^2} f(x, y) dx dy \right| \leq 2^2(1 - \sqrt{1 - \delta}) \cdot \max_{(x,y) \in I^2} |f(x, y)|.$$

Since $F_\delta \in \mathcal{C}(I^2)$, the order of integration of

$$\int_{I^2} F_\delta$$

is irrelevant and then the same is true for

$$\int_{I^2} f.$$

Hence, by the definition (10.44), f is integrable over D .

To get our final conclusion, we need a lemma (see [17, Theorem 13.6, p. 110]):

Lemma 10.6

Let S be a bounded set in $\mathbb{R}^n$; let $f : S \rightarrow \mathbb{R}$ be a bounded continuous function. If f is integrable over S , then f is integrable over S° , and

$$\int_S f = \int_{S^\circ} f.$$

Now the closed disk D is bounded and $f : D \rightarrow \mathbb{R}$ is a bounded continuous function. By the above analysis, f is integrable over D , so Lemma 10.6 shows that

$$\int_D f(x, y) \, dx \, dy = \int_{D^\circ} f(x, y) \, dx \, dy. \quad (10.48)$$

Hence, we establish from the equalities (10.43) and (10.48) that

$$\int_D f(x, y) \, dx \, dy = \int_0^a \int_0^{2\pi} f(T(r, \theta)) r \, dr \, d\theta. \quad (10.49)$$

This completes the proof of the problem. ■

Problem 10.10

Rudin Chapter 10 Exercise 10.

Proof. Let's make clear what is the meaning of “ f decreases sufficiently rapidly as $|x| + |y| \rightarrow \infty$ ”. In fact, it means that

$$\lim_{|x|+|y| \rightarrow \infty} |(x^2 + y^2)f(x, y)| = 0 \quad (10.50)$$

or equivalently

$$|f(x, y)| \leq \frac{A}{(x^2 + y^2)^{1+c}} \quad (10.51)$$

for some positive constants A and c and for all large $|x| + |y|$. If we put $x = r \cos \theta$ and $y = r \sin \theta$, then the inequality (10.51) becomes

$$|f(r \cos \theta, r \sin \theta)| \leq \frac{A}{r^{2+2c}} \quad (10.52)$$

for all large enough r . Define $\hat{f} : [0, b] \rightarrow \mathbb{R}$ by

$$\hat{f}(r) = r \int_0^{2\pi} f(T(r, \theta)) \, d\theta = r \int_0^{2\pi} f(r \cos \theta, r \sin \theta) \, d\theta$$

so that

$$\int_0^b \int_0^{2\pi} f(T(r, \theta)) r \, dr \, d\theta = \int_0^b \hat{f}(r) \, dr,$$

where $b > 0$. Now we show that the improper integral

$$\int_0^\infty \hat{f}(r) \, dr$$

exists. To this end, we need a comparison result of improper integrals.

Lemma 10.7

Suppose that $F : (a, \infty) \rightarrow \mathbb{R}$ is increasing. If there exists a constant M such that $F(x) \leq M$ for all $x \in (a, \infty)$, then $\lim_{x \rightarrow \infty} F(x)$ exists and

$$\lim_{x \rightarrow \infty} F(x) \leq M.$$

Proof of Lemma 10.7. Suppose that

$$E = F((a, \infty)) = \{y = F(x) \mid x > a\} \subseteq \mathbb{R}.$$

By the hypothesis, we know that E is bounded by M . By Theorem 1.19, $\mathbb{R}$ is an ordered field with the least-upper-bound property. By Definition 1.10, E has a least upper bound in $\mathbb{R}$. Suppose that $S = \sup E$. Given $\epsilon > 0$, then the number $S - \epsilon$ is *not* an upper bound of E , so we have

$$S - \epsilon < F(x_0) \quad (10.53)$$

for some $x_0 > a$. Since F is increasing, the inequality (10.53) shows that

$$S - \epsilon < F(x_0) \leq F(x) \leq S$$

for all $x > x_0$. In other words, we have

$$|F(x) - S| < \epsilon$$

for all $x > x_0$. By Definition 4.33, we have

$$\lim_{x \rightarrow \infty} F(x) = S.$$

Since $S = \sup E$, it is trivial that $S \leq M$ which is our desired result. ■

Lemma 10.8

Let $f(x)$ and $g(x)$ be continuous functions and a be a constant. Suppose that

$$0 \leq g(x) \leq f(x)$$

for $x \geq a$. If $\int_a^\infty f(x) \, dx$ exists, then the limit $\int_a^\infty g(x) \, dx$ also exists and

$$\int_a^\infty g(x) \, dx \leq \int_a^\infty f(x) \, dx.$$

Proof of Lemma 10.8. Suppose that

$$F(y) = \int_a^y f(x) \, dx \quad \text{and} \quad G(y) = \int_a^y g(x) \, dx,$$

where $y \geq a$. Since $f(x) \geq 0$ and $g(x) \geq 0$ for all $x \geq a$, the functions F and G are increasing. Since $0 \leq g(x) \leq f(x)$ for all $x \geq a$, we have

$$G(y) \leq F(y) \quad (10.54)$$

for all $y \geq a$. By the hypothesis, the number

$$M = \int_a^\infty f(x) \, dx$$

is finite. Since we always have $F(y) \leq M$ for all $y \geq a$, we deduce from the inequality (10.54) that

$$G(y) \leq M$$

for all $y \geq a$. Hence the desired results follow immediately from Lemma 10.8. ■

Lemma 10.9

The improper integral

$$\int_0^\infty |\widehat{f}(r)| \, dr$$

exists.

Proof of Lemma 10.9. Let N be a *fixed* positive integer such that the inequality (10.52) holds for all $r \geq N$. If $b > N$, then we get from Theorems 6.12(c) and 6.13(b) that

$$\begin{aligned} \int_0^b |\widehat{f}(r)| \, dr &= \int_0^N |\widehat{f}(r)| \, dr + \int_N^b |\widehat{f}(r)| \, dr \\ &\leq \int_0^N |\widehat{f}(r)| \, dr + \int_N^b \left| r \int_0^{2\pi} f(r \cos \theta, r \sin \theta) \, d\theta \right| \, dr \\ &\leq \int_0^N |\widehat{f}(r)| \, dr + \int_N^b \frac{2\pi A}{r^{1+2c}} \, dr \end{aligned} \quad (10.55)$$

Since $\widehat{f}$ is continuous on the compact interval $[0, N]$ and N is *fixed*, Theorem 4.14 implies that $|\widehat{f}(r)| \leq m$ on $[0, N]$ for a positive constant m . Thus the inequality (10.55) becomes

$$0 \leq \int_0^b |\widehat{f}(r)| \, dr \leq mN + \int_N^b \frac{2\pi A}{r^{1+2\delta}} \, dr.$$

Since

$$\int_N^b \frac{2\pi A}{r^{1+2\delta}} \, dr = \frac{\pi A}{\delta} \left(\frac{1}{r^{2N}} - \frac{1}{r^{2b}} \right)$$

and

$$\lim_{b \rightarrow \infty} \frac{\pi A}{\delta} \left(\frac{1}{r^{2N}} - \frac{1}{r^{2b}} \right) = \frac{\pi A}{\delta r^{2N}},$$

Lemma 10.8 implies that the limit

$$\int_0^\infty |\widehat{f}(r)| \, dr$$

exists. This completes the proof of this lemma. ■

Since $0 \leq |\widehat{f}(r)| - \widehat{f}(r) \leq 2|\widehat{f}(r)|$ for every $0 \leq r \leq b$, it follows from Lemmas 10.8 and 10.9 that

$$\int_0^\infty \widehat{f}(r) \, dr$$

exists. Next, since D is the closed disk centered at the origin with radius a , D becomes the whole plane $\mathbb{R}^2$ as $a \rightarrow \infty$. Hence, we deduce from the equality (10.49) that

$$\int_{\mathbb{R}^2} f(x, y) \, dx \, dy = \int_0^\infty \int_0^{2\pi} f(T(r, \theta)) r \, dr \, d\theta.$$

Now the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = \exp(-x^2 - y^2)$$

is clearly a continuous and bounded function on the closed disk D with center at $(0, 0)$ and radius a so that $f \in \mathcal{C}(D)$. Furthermore, f is a function satisfying the condition (10.50). Therefore, on the one hand, we have

$$\int_{\mathbb{R}^2} \exp(-x^2 - y^2) \, dx \, dy = \int_0^\infty \int_0^{2\pi} f(T(r, \theta)) r \, dr \, d\theta$$

$$\begin{aligned}
&= \int_0^\infty \int_0^{2\pi} e^{-r^2} r \, dr \, d\theta \\
&= \frac{1}{2} \int_0^{2\pi} \left[\int_0^\infty e^{-r^2} d(r^2) \right] d\theta \\
&= \frac{1}{2} \int_0^{2\pi} \left(-e^{-r^2} \Big|_0^\infty \right) d\theta \\
&= \pi.
\end{aligned} \tag{10.56}$$

On the other hand, we have

$$\int_{\mathbb{R}^2} \exp(-x^2 - y^2) \, dx \, dy = \left(\int_{-\infty}^\infty e^{-x^2} \, dx \right) \times \left(\int_{-\infty}^\infty e^{-y^2} \, dy \right) = \left(\int_{-\infty}^\infty e^{-x^2} \, dx \right)^2. \tag{10.57}$$

Combining the expressions (10.56) and (10.57), we have

$$\int_{-\infty}^\infty e^{-x^2} \, dx = \sqrt{\pi}$$

which is exactly formula (101) of Chap. 8 (see [21, p. 194]). This completes the proof of our problem. ■

Problem 10.11

Rudin Chapter 10 Exercise 11.

Proof. Let $S = \{(s, t) \mid 0 < s < \infty, 0 < t < 1\}$ and $Q = \{(u, v) \mid u > 0, v > 0\}$ be the strip in the (s, t) -plane and the positive quadrant in the (u, v) -plane respectively. Define $T : S \rightarrow \mathbb{R}$ by

$$T(s, t) = (s - st, st). \tag{10.58}$$

Since $s > 0$ and $0 < t < 1$, $s - st = s(1 - t) > 0$ and $st > 0$. Thus we have $T(S) \subseteq Q$. If $T(s, t) = T(s', t')$, then the definition (10.58) gives $s - st = s' - s't'$ and $st = s't'$ which show that $s = s'$ and $t = t'$ immediately. Thus T is one-to-one. Besides, given $u > 0$ and $v > 0$, we define $s = u + v$ and $t = \frac{v}{u+v}$. Then we have $(s, t) \in S$ and

$$T(s, t) = (s - st, st) = \left(u + v - (u + v) \frac{v}{u + v}, (u + v) \frac{v}{u + v} \right) = (u, v).$$

Therefore, we have $T(S) = Q$, i.e., the map $T : S \rightarrow Q$ is onto. See Figure 10.6 for the mapping T .

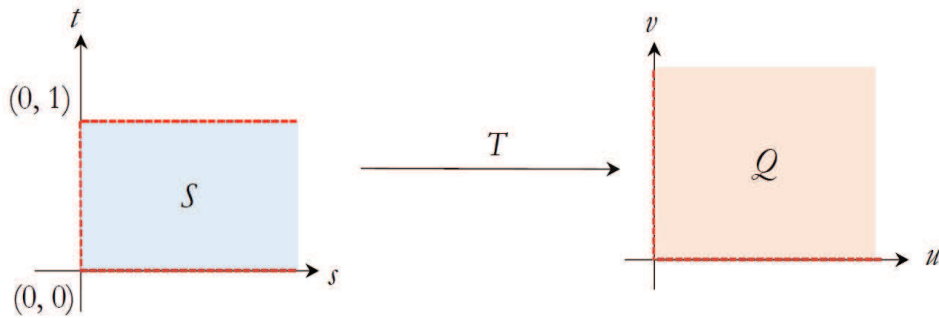


Figure 10.6: The mapping $T : S \rightarrow Q$.

It is obvious from the definition (10.58) that

$$J_T(s, t) = \det \begin{pmatrix} \frac{\partial u}{\partial s} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial s} & \frac{\partial v}{\partial t} \end{pmatrix} = \det \begin{pmatrix} 1 - t & -s \\ t & s \end{pmatrix} = s(1 - t) + st = s.$$

This proves our second assertion.

For $x > 0, y > 0$, we consider the function $f : Q \rightarrow \mathbb{R}$ and the integral given by

$$f(u, v) = \frac{u^{x-1}}{e^u} \cdot \frac{v^{y-1}}{e^v} \quad \text{and} \quad \alpha = \int_Q f(u, v) du dv = \int_Q u^{x-1} e^{-u} v^{y-1} e^{-v} du dv$$

respectively. Obviously, S is an open set and T is an one-to-one (in fact, bijective) $\mathcal{C}'$ -mapping. Although the function $f(u, v)$ is clearly continuous on Q , its support is

$$\text{supp}(f) = \overline{\{(u, v) \in Q \mid f(u, v) \neq 0\}} = Q \cup \{(u, 0) \mid u \geq 0\} \cup \{(0, v) \mid v \geq 0\}$$

which consists of the positive quadrant plus the nonnegative u and v axes. Thus the set $\text{supp}(f)$ is unbounded (and hence not compact in $\mathbb{R}^2$ by Theorem 2.41 (Heine-Borel Theorem)) and does not lie in $T(S) = Q$. In other words, the hypotheses of Theorem 10.9 do not satisfy in this case.

To overcome these two problems, we try to consider subsets of S and Q and then we apply Lemma 10.4 to such subsets. To this end, for every small enough $\epsilon > 0$, we consider the set

$$S_\epsilon = \{(s, t) \mid \epsilon < s < \epsilon^{-2}, \epsilon < t < 1 - \epsilon\} \subset S.$$

Let $A_\epsilon = T(\epsilon, \epsilon)$, $B_\epsilon = T(\epsilon, 1 - \epsilon)$, $C_\epsilon = T(\epsilon^{-2}, \epsilon)$ and $D_\epsilon = T(\epsilon^{-2}, 1 - \epsilon)$. Then direct computation gives

$$A_\epsilon = (\epsilon(1 - \epsilon), \epsilon^2), \quad B_\epsilon = (\epsilon^2, \epsilon(1 - \epsilon)), \quad C_\epsilon = (\epsilon^{-2}(1 - \epsilon), \epsilon^{-1}) \quad \text{and} \quad D_\epsilon = (\epsilon^{-1}, \epsilon^{-2}(1 - \epsilon)).$$

Now let Q_ϵ be the interior of the *convex hull* of the points $A_\epsilon, B_\epsilon, C_\epsilon$ and D_ϵ in the (u, v) -plane. That is, if

$$H_\epsilon = \{\lambda_1 A_\epsilon + \lambda_2 B_\epsilon + \lambda_3 C_\epsilon + \lambda_4 D_\epsilon \mid \lambda_i \geq 0 \text{ for } i = 1, 2, 3, 4 \text{ and } \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 1\},$$

then we have

$$Q_\epsilon = H_\epsilon^\circ \subset Q. \quad (10.59)$$

For examples, if $\epsilon = 0.1$, then the four points are

$$A_{0.1} = (0.09, 0.01), \quad B_{0.1} = (0.01, 0.09), \quad C_{0.1} = (90, 10) \quad \text{and} \quad D_{0.1} = (10, 90);$$

if $\epsilon = 0.2$, then they are

$$A_{0.2} = (0.16, 0.04), \quad B_{0.2} = (0.04, 0.16), \quad C_{0.2} = (20, 5) \quad \text{and} \quad D_{0.2} = (5, 20).$$

See Figure 10.7 for the open sets $Q_{0.1}, Q_{0.2}$ and Q .

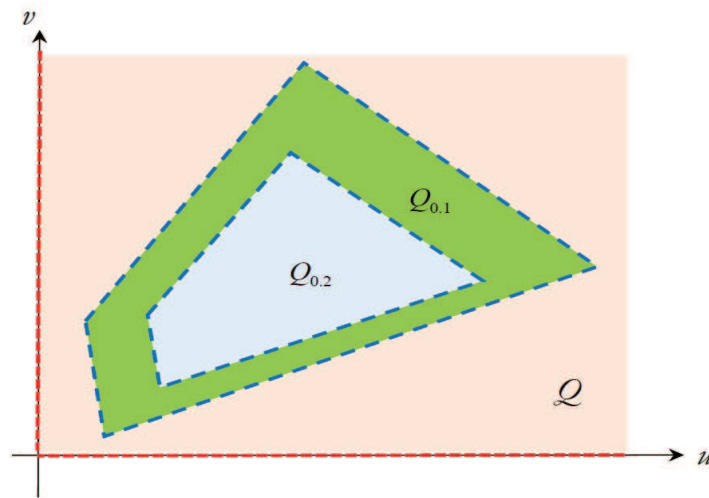


Figure 10.7: The open sets $Q_{0.1}, Q_{0.2}$ and Q .

By definition, we know that both S_ϵ and Q_ϵ are open and bounded in $\mathbb{R}^2$. Next, we have

$$S_\epsilon \subset S_\delta \quad \text{and} \quad Q_\epsilon \subset Q_\delta$$

if $0 < \delta < \epsilon$. As $\epsilon \rightarrow 0$, we know from the representations of S_ϵ and Q_ϵ that

$$S_\epsilon \rightarrow S \quad \text{and} \quad Q_\epsilon \rightarrow Q.$$

Furthermore, by the construction of the sets S_ϵ and Q_ϵ , it is easily seen that the restriction $T : S_\epsilon \rightarrow Q_\epsilon$ is a bijective $\mathcal{C}'$ -mapping. Since S_ϵ is a rectangle in $\mathbb{R}^2$, it follows from [8, Exercise 1, p. 496] that it is an open Jordan domain. Since $K_\epsilon = S_\epsilon \cup \partial S_\epsilon = \overline{S_\epsilon} \subset S$, we know from the definition (10.59) of Q_ϵ that

$$T(K_\epsilon) = \overline{Q_\epsilon} = H_\epsilon.$$

Now our function f is obviously continuous on H_ϵ , so we may apply Theorem 10.9 to the restriction $f : H_\epsilon \rightarrow \mathbb{R}$ to obtain

$$\int_{H_\epsilon} f(u, v) \, du \, dv = \int_{\overline{S_\epsilon}} f(T(s, t)) |J_T(s, t)| \, ds \, dt. \quad (10.60)$$

On the one hand, we know that

$$\int_{H_\epsilon} f(u, v) \, du \, dv = \left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} u^{x-1} e^{-u} \, du \right) \left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} v^{y-1} e^{-v} \, dv \right) \quad (10.61)$$

and on the other hand, we have

$$\int_{\overline{S_\epsilon}} f(T(s, t)) |J_T(s, t)| \, ds \, dt = \int_{\epsilon}^{1-\epsilon} \int_{\epsilon}^{\epsilon^{-1}} s^{x+y-1} t^{y-1} (1-t)^{x-1} e^{-s} \, ds \, dt. \quad (10.62)$$

By substituting the integrals (10.61) and (10.62) into the integral relation (10.60), we have

$$\begin{aligned} \left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} u^{x-1} e^{-u} \, du \right) \left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} v^{y-1} e^{-v} \, dv \right) &= \left[\int_{\epsilon}^{1-\epsilon} t^{y-1} (1-t)^{x-1} \, dt \right] \\ &\quad \times \left(\int_{\epsilon}^{\epsilon^{-1}} s^{x+y-1} e^{-s} \, ds \right) \end{aligned}$$

and since

$$\int_{\epsilon}^{\epsilon^{-1}} s^{x+y-1} e^{-s} \, ds > 0$$

for every $\epsilon > 0$, we must have

$$\int_{\epsilon}^{1-\epsilon} t^{y-1} (1-t)^{x-1} \, dt = \frac{\left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} u^{x-1} e^{-u} \, du \right) \left(\int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} v^{y-1} e^{-v} \, dv \right)}{\int_{\epsilon}^{\epsilon^{-1}} s^{x+y-1} e^{-s} \, ds} \quad (10.63)$$

for every $\epsilon > 0$. Recall from Definition 8.17 that the integral

$$\int_0^\infty t^{x-1} e^{-t} \, dt$$

converges for all $x > 0$, so we can conclude that

$$\lim_{\epsilon \rightarrow 0} \int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} u^{x-1} e^{-u} \, du = \int_0^\infty u^{x-1} e^{-u} \, du = \Gamma(x),$$

$$\lim_{\epsilon \rightarrow 0} \int_{\epsilon^2}^{\epsilon^{-2}(1-\epsilon)} v^{y-1} e^{-v} dv = \int_0^\infty v^{x-1} e^{-v} dv = \Gamma(y),$$

$$\lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{\epsilon^{-1}} s^{x+y-1} e^{-s} ds = \int_0^\infty s^{x+y-1} e^{-s} ds = \Gamma(x+y).$$

Since $\Gamma(x)$ is nonzero on $(0, \infty)$, they imply that the integral on the left-hand side of the relation (10.63) also converges and we have

$$\int_0^1 t^{y-1} (1-t)^{x-1} dt = \lim_{\epsilon \rightarrow 0} \int_{\epsilon}^{1-\epsilon} t^{y-1} (1-t)^{x-1} dt = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}$$

which is exactly formula (96) of Chap. 8. This completes the proof of the problem. ■

As a remark to Problem 10.11, Rudin mentioned that Theorem 10.9 has to be extended to cover the case of improper integrals. The proof we present here does not use this approach. If the reader wants to apply such extension to prove the problem, we recommend the following version of Theorem 10.9 which can be found in [25, Theorem 1, p. 156]:

Lemma 10.10

Suppose that $T : U \rightarrow V$ is a diffeomorphism of the open set $U \subseteq \mathbb{R}^n$ onto the open set $V \subseteq \mathbb{R}^n$ and $f : U \rightarrow \mathbb{R}$ is integrable on all measurable compact subsets of U . If the improper integral

$$\int_V f(\mathbf{y}) d\mathbf{y}$$

converges, then the improper integral

$$\int_U f(T(\mathbf{x})) |J_T(\mathbf{x})| d\mathbf{x}$$

converges and has the same value.

Problem 10.12

Rudin Chapter 10 Exercise 12.

Proof. See Figure 10.8 for the example $T : I^3 \rightarrow Q^3$.

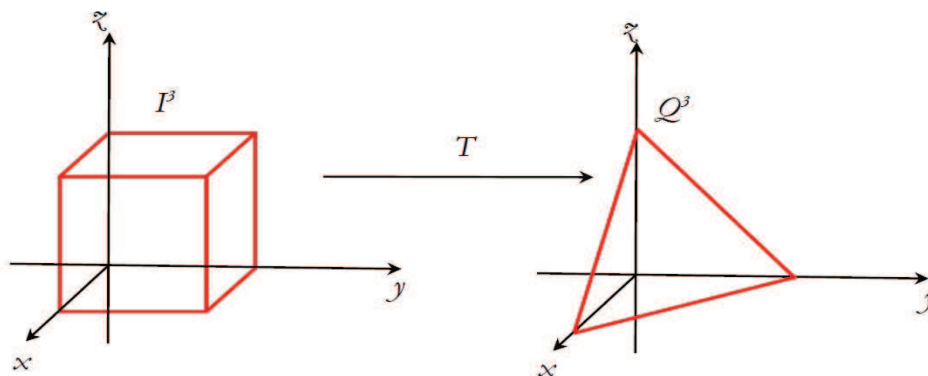


Figure 10.8: The mapping $T : I^3 \rightarrow Q^3$.

We divide the proof into several steps:

- **Proof of the formula.** The formula can be shown by induction. The case $k = 1$ is trivial. Assume that the formula is true for $k = n$, i.e.,

$$\sum_{i=1}^n x_i = 1 - \prod_{i=1}^n (1 - u_i). \quad (10.64)$$

If $k = n + 1$, then we follow from the formula (10.64) and the definition that

$$\begin{aligned} \sum_{i=1}^{n+1} x_i &= x_{n+1} + \sum_{i=1}^n x_i \\ &= x_{n+1} + 1 - \prod_{i=1}^n (1 - u_i) \\ &= 1 + (1 - u_1) \cdots (1 - u_n) u_{n+1} - \prod_{i=1}^n (1 - u_i) \\ &= 1 + u_{n+1} \prod_{i=1}^n (1 - u_i) - \prod_{i=1}^n (1 - u_i) \\ &= 1 + \prod_{i=1}^n (1 - u_i) (u_{n+1} - 1) \\ &= 1 - \prod_{i=1}^{n+1} (1 - u_i). \end{aligned}$$

Thus the formula is also true in the case $k = n + 1$. Hence we follow from induction that the formula is true for all positive integers k .

- **The surjectivity of the map $T : I^k \rightarrow Q^k$.** Since $0 \leq u_i \leq 1$ for all $i = 1, 2, \dots, k$, we have $x_i \geq 0$ for all $i = 1, 2, \dots, k$. By the formula (10.64), we have

$$\sum_{i=1}^k x_i \leq 1,$$

so T maps I^k into Q^k . Given $(x_1, \dots, x_k) \in Q^k$. If $\sum_{i=1}^m x_i \neq 1$ for all $m = 1, 2, \dots, k - 1$, then the points

$$\begin{aligned} u_1 &= x_1, \\ u_2 &= \frac{x_2}{1 - x_1}, \\ u_3 &= \frac{x_3}{(1 - x_1)(1 - x_2)} = \frac{x_3}{1 - x_1 - x_2}, \\ &\dots\dots\dots, \\ u_k &= \frac{x_k}{(1 - x_1) \cdots (1 - x_{k-1})} = \frac{x_k}{1 - x_1 - \cdots - x_{k-1}} \end{aligned} \quad (10.65)$$

imply that

$$T(u_1, \dots, u_k) = (x_1, \dots, x_k).$$

If m is the *least* positive integer in the set $\{1, 2, \dots, k - 1\}$ such that

$$\sum_{i=1}^m x_i = 1,$$

then we have

$$\sum_{i=1}^{m-1} x_i < 1 \quad \text{and} \quad x_{m+1} = x_{m+2} = \cdots = x_k = 0.$$

By these and the formula (10.64), the point $(u_1, \dots, u_k)$ defined by

$$u_1 = x_1, \quad u_2 = \frac{x_2}{1 - x_1}, \dots, u_m = x_m \div \left(1 - \sum_{i=1}^{m-1} x_i\right), \quad u_i = x_i,$$

where $i = m + 1, \dots, k$, implies that

$$T(u_1, \dots, u_m, u_{m+1}, \dots, u_k) = (x_1, x_2, \dots, x_m, 0, \dots, 0).$$

If we have

$$\sum_{i=1}^k x_i = 1 \quad \text{and} \quad \sum_{i=1}^{k-1} x_i < 1,$$

then the point $(u_1, \dots, u_k)$ defined by

$$u_1 = x_1, u_2 = \frac{x_2}{1 - x_1}, \dots, u_k = x_k \div \left(1 - \sum_{i=1}^{k-1} x_i\right)$$

implies that

$$T(u_1, \dots, u_k) = (x_1, x_2, \dots, x_k).$$

Thus it means that the mapping T is onto.

- **The injectivity of the mapping $T : I \rightarrow Q$.** Let I and Q be the interiors of I^k and Q^k respectively, i.e.,

$$I = \{(u_1, \dots, u_k) \in \mathbb{R}^k \mid 0 < u_1, \dots, u_k < 1\} \quad (10.66)$$

and

$$Q = \left\{ (x_1, \dots, x_k) \in \mathbb{R}^k \mid x_1, \dots, x_k > 0, \sum_{i=1}^k x_i < 1 \right\}. \quad (10.67)$$

We first show that the mapping T is one-to-one in I . To this end, let $\mathbf{u} = (u_1, \dots, u_k)$ and $\mathbf{v} = (v_1, \dots, v_k)$ be points in I such that

$$T(\mathbf{u}) = \mathbf{x} = (x_1, \dots, x_k) \quad \text{and} \quad T(\mathbf{v}) = \mathbf{y} = (y_1, \dots, y_k).$$

If $T(\mathbf{u}) = T(\mathbf{v})$, then we have $x_1 = y_1, x_2 = y_2, \dots, x_k = y_k$. Note that $0 < u_i < 1$ and $0 < v_i < 1$ for $1 \leq i \leq k$. By definition, the relation $x_1 = y_1$ implies trivially that $u_1 = v_1$. Similarly, the inequality $0 < u_1 = v_1 < 1$ and the relation $x_2 = y_2$ show that $u_2 = v_2$. By using the inequalities $0 < u_i = v_i < 1$ and the relations $x_i = y_i$ for $1 \leq i \leq k-1$, it deduces immediately from the definition that $x_k = y_k$. Hence T is one-to-one in I .

Next, we show that $T(I) = Q$. Let $\mathbf{x} \in Q$ so that $\mathbf{x} = (x_1, \dots, x_k)$, where $x_1, \dots, x_k > 0$ and $\sum_{i=1}^k x_k < 1$. If we define $u_1, \dots, u_k$ by the equalities (10.65), then we obtain immediately that

$$T(u_1, \dots, u_k) = (x_1, \dots, x_k).$$

Thus $T(I) = Q$, as required.

In addition, it is clearly that the inverse $S : Q \rightarrow I$ is well-defined by the formulas (10.65).

- **Computation of the $J_T(\mathbf{u})$ and $J_S(\mathbf{x})$.** To find $J_T(\mathbf{u})$, we note that

$$T(u_1, \dots, u_k) = (u_1, (1 - u_1)u_2, \dots, (1 - u_1) \cdots (1 - u_{k-1})u_k). \quad (10.68)$$

Apply Theorem 9.17 to (10.68), we get

$$[T'(\mathbf{u})] = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ -u_2 & 1-u_1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\prod_{i=2}^{k-1}(1-u_i)u_k & -\prod_{\substack{i=1 \\ i \neq 2}}^{k-1}(1-u_i)u_k & -\prod_{\substack{i=1 \\ i \neq 3}}^{k-1}(1-u_i)u_k & \cdots & \prod_{i=1}^{k-1}(1-u_i) \end{pmatrix}$$

which is a lower triangular matrix. It is well-known that the determinant of a triangular matrix is the product of the entries on the diagonal (see [15, Theorem 2, 167]), so we have

$$\begin{aligned} J_T(\mathbf{u}) &= \det[T'(\mathbf{u})] \\ &= 1 \times (1-u_1) \times [(1-u_1)(1-u_2)] \times \cdots \times \prod_{i=1}^{k-1}(1-u_i) \\ &= (1-u_1)^{k-1}(1-u_2)^{k-2} \cdots (1-u_{k-1}), \end{aligned} \quad (10.69)$$

where $\mathbf{u} \in I$. To find $J_S(\mathbf{x})$, we notice that

$$S(x_1, \dots, x_k) = \left(x_1, \frac{x_2}{1-x_1}, \dots, \frac{x_k}{1-x_1-\cdots-x_{k-1}} \right). \quad (10.70)$$

Similarly, we apply Theorem 9.17 to the expression (10.70), we get

$$[S'(\mathbf{x})] = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ \frac{x_2}{(1-x_1)^2} & \frac{1}{1-x_1} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{x_k}{\left(1-\sum_{i=1}^{k-1}x_i\right)^2} & \frac{x_k}{\left(1-\sum_{i=1}^{k-1}x_i\right)^2} & \frac{x_k}{\left(1-\sum_{i=1}^{k-1}x_i\right)^2} & \cdots & \frac{1}{\left(1-\sum_{i=1}^{k-1}x_i\right)^2} \end{pmatrix}$$

and therefore,

$$\begin{aligned} J_S(\mathbf{x}) &= \det[S'(\mathbf{x})] \\ &= 1 \times \frac{1}{1-x_1} \times \cdots \times \frac{1}{1-x_1-\cdots-x_{k-1}} \\ &= [(1-x_1)(1-x_1-x_2) \cdots (1-x_1-x_2-\cdots-x_{k-1})]^{-1}, \end{aligned}$$

where $\mathbf{x} \in Q$.

We end the proof of the problem. ■

Problem 10.13

Rudin Chapter 10 Exercise 13.

Proof. Let $T : I^k \rightarrow Q^k$ be the mapping as defined in Problem 10.12. Then I is an open set, $T : I \rightarrow Q$ is bijective and we deduce from the definition (10.66) and the formula (10.69) that

$$J_T(\mathbf{u}) \neq 0$$

for all $\mathbf{u} \in I$. Define the function $f : Q \rightarrow \mathbb{R}$ by

$$f(\mathbf{x}) = x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k}$$

which is clearly continuous on Q . By definition (10.67), we know that

$$\text{supp}(f) = \overline{\{\mathbf{x} \in Q \mid f(\mathbf{x}) \neq 0\}} = \overline{Q} = Q^k$$

which is compact by Theorem 2.41 (Heine-Borel Theorem).

Hence, it follows from Theorem 10.9 and the formula (10.69) that

$$\begin{aligned} \int_Q x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k} d\mathbf{x} &= \int_I f(T(\mathbf{u})) |J_T(\mathbf{u})| d\mathbf{u} \\ &= \int_I f(T(\mathbf{u})) (1-u_1)^{k-1} (1-u_2)^{k-2} \cdots (1-u_{k-1}) du_1 du_2 \cdots du_k. \end{aligned} \quad (10.71)$$

By the definition of $\mathbf{x}$, the integral (10.71) reduces to

$$\begin{aligned} \int_Q x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k} d\mathbf{x} &= \int_I f(T(\mathbf{u})) (1-u_1)^{k-1} (1-u_2)^{k-2} \cdots (1-u_{k-1}) du_1 du_2 \cdots du_k \\ &= \int_I u_1^{r_1} [(1-u_1)u_2]^{r_2} [(1-u_1)(1-u_2)u_3]^{r_3} \cdots [(1-u_1) \cdots (1-u_{k-1})u_k]^{r_k} \\ &\quad \times |(1-u_1)^{k-1} (1-u_2)^{k-2} \cdots (1-u_{k-1})| du_1 du_2 \cdots du_k \\ &= \int_I u_1^{r_1} \cdots u_k^{r_k} \left\{ (1-u_1)^{r_2+\cdots+r_k} (1-u_2)^{r_3+\cdots+r_k} \cdots (1-u_{k-1})^{r_k} \right\} \\ &\quad (1-u_1)^{k-1} (1-u_2)^{k-2} \cdots (1-u_{k-1}) du_1 du_2 \cdots du_k \\ &= \left[\int_0^1 u_1^{r_1} (1-u_1)^{k+r_2+\cdots+r_k-1} du_1 \right] \times \left[\int_0^1 u_2^{r_2} (1-u_2)^{k+r_3+\cdots+r_k-2} du_2 \right] \\ &\quad \times \cdots \times \left[\int_0^1 u_{k-1}^{r_{k-1}} (1-u_{k-1})^{1+r_k} du_{k-1} \right] \times \left(\int_0^1 u_k^{r_k} du_k \right). \end{aligned} \quad (10.72)$$

Finally, we apply Theorem 8.20 to each integral on the right-hand side of (10.72) and then Theorem 8.18(b), cancellations happen and it further reduces to

$$\begin{aligned} \int_Q x_1^{r_1} x_2^{r_2} \cdots x_k^{r_k} d\mathbf{x} &= \frac{\Gamma(r_1+1) \Gamma(k+r_2+\cdots+r_k)}{\Gamma(k+r_1+\cdots+r_k+1)} \times \frac{\Gamma(r_2+1) \Gamma(k-1+r_3+\cdots+r_k)}{\Gamma(k+r_2+\cdots+r_k)} \\ &\quad \times \cdots \times \frac{\Gamma(r_{k-1}+1) \Gamma(2+r_k)}{\Gamma(3+r_{k-1}+r_k)} \times \frac{\Gamma(r_k+1) \Gamma(1)}{\Gamma(2+r_k)} \\ &= \frac{\Gamma(r_1+1) \Gamma(r_2+1) \cdots \Gamma(r_k+1)}{\Gamma(k+r_1+\cdots+r_k+1)} \\ &= \frac{r_1! r_2! \cdots r_k!}{(k+r_1+\cdots+r_k)!}. \end{aligned} \quad (10.73)$$

In particular, if we take $r_1 = \cdots = r_k = 0$ into the integral (10.73), then we get

$$\int_{Q^k} d\mathbf{x} = \frac{1}{k!}$$

which gives the volume of the k -simplex Q^k . This completes the proof of the problem. ■

10.4 Properties of k -forms and k -simplexes

Problem 10.14

Rudin Chapter 10 Exercise 14.

Proof. Recall from Definition 9.33 that if $\{j_1, \dots, j_k\}$ is an ordered k -tuple of *distinct* integers, then we have

$$s(j_1, \dots, j_k) = \prod_{q=2}^k \prod_{p=1}^{q-1} \operatorname{sgn}(j_q - j_p). \quad (10.74)$$

Let $k = 2$. Then we have

$$s(j_1, j_2) = \operatorname{sgn}(j_2 - j_1) = \begin{cases} 1, & \text{if } j_2 > j_1; \\ -1, & \text{if } j_2 < j_1. \end{cases}$$

If $j_2 > j_1$, then $dx_{j_1} \wedge dx_{j_2}$ is an increasing 2-index so that $\varepsilon(j_1, j_2) = 1$. Similarly, if $j_2 < j_1$, then $dx_{j_1} \wedge dx_{j_2} = -dx_{j_2} \wedge dx_{j_1}$ so that $\varepsilon(j_1, j_2) = -1$ by [21, Eqn. (42), p. 256]. Therefore, we have

$$\varepsilon(j_1, j_2) = s(j_1, j_2).$$

Assume that we have

$$\varepsilon(j_1, \dots, j_m) = s(j_1, \dots, j_m), \quad (10.75)$$

where $2 \leq m < k$. Let $k = m + 1$. By definition (10.74) and the equality (10.75), we have

$$\begin{aligned} s(j_1, \dots, j_m, j_{m+1}) &= \prod_{q=2}^{m+1} \prod_{p=1}^{q-1} \operatorname{sgn}(j_q - j_p) \\ &= \prod_{p=1}^m \operatorname{sgn}(j_{m+1} - j_p) \times \prod_{q=2}^m \prod_{p=1}^{q-1} \operatorname{sgn}(j_q - j_p) \\ &= \varepsilon(j_1, \dots, j_m) \times \prod_{p=1}^m \operatorname{sgn}(j_{m+1} - j_p). \end{aligned} \quad (10.76)$$

To make the $(m+1)$ -tuple $\{j_1, \dots, j_m, j_{m+1}\}$ of distinct integers into an increasing $(m+1)$ -index, we can first make the m -tuple $\{j_1, \dots, j_m\}$ into an increasing m -index, namely $\{j_{r_1}, \dots, j_{r_m}\}$ where

$$j_{r_1} < \dots < j_{r_m},$$

and then add the integer j_{m+1} to that increasing m -index to produce the increasing $(m+1)$ -index. Suppose that

$$j_{r_1} < \dots < j_{r_s} < \textcolor{red}{j_{m+1}} < j_{r_{s+1}} < \dots < j_{r_m}. \quad (10.77)$$

Then we have

$$\varepsilon(j_1, \dots, j_m, j_{m+1}) = \varepsilon(j_1, \dots, j_m) \times (-1)^{m-s-1}. \quad (10.78)$$

By the inequalities (10.77), we have

$$\prod_{p=1}^m \operatorname{sgn}(j_{m+1} - j_p) = \prod_{p=1}^m \operatorname{sgn}(j_{m+1} - j_{r_p}) = (-1)^{m-s-1}. \quad (10.79)$$

Hence we follow from the equalities (10.76), (10.78) and (10.79) that

$$s(j_1, \dots, j_m, j_{m+1}) = \varepsilon(j_1, \dots, j_m, j_{m+1})$$

which implies that the statement is true for $k = m + 1$. By induction, formula (46) is true for all integers k . This completes the proof of the problem. ■

Problem 10.15

Rudin Chapter 10 Exercise 15.

Proof. Suppose that ω and λ are represented in the standard presentation:

$$\omega = \sum_I a_I(\mathbf{x}) dx_I \quad \text{and} \quad \lambda = \sum_J b_J(\mathbf{x}) dx_J, \quad (10.80)$$

where the summations in (10.80) extend over all increasing k -indices I and m -indices J respectively. By Definition 10.17, we have the $(k+m)$ -form in an open set $E \subseteq \mathbb{R}^n$

$$\omega \wedge \lambda = \sum_{I,J} a_I(\mathbf{x}) b_J(\mathbf{x}) dx_I \wedge dx_J \quad \text{and} \quad \lambda \wedge \omega = \sum_{I,J} b_J(\mathbf{x}) a_I(\mathbf{x}) dx_J \wedge dx_I,$$

where I and J range *independently* over their

$$a_I(\mathbf{x}) b_J(\mathbf{x}) = b_J(\mathbf{x}) a_I(\mathbf{x})$$

for every $\mathbf{x} \in E$. Therefore, our result follows immediately if we can show that

$$dx_I \wedge dx_J = (-1)^{km} dx_J \wedge dx_I \quad (10.81)$$

for each increasing k -indices I and increasing m -indices J .

Suppose that I and J have an element in common, then we know from [21, Eqn. (43), p. 256] that

$$dx_I \wedge dx_J = dx_J \wedge dx_I = 0$$

so that the formula (10.81) holds. Next, we suppose that $\{i_1, \dots, i_k\}$ and $\{j_1, \dots, j_m\}$ are increasing k -indices and m -indices respectively with *no* element in common. By repeated application of the *anti-commutative relation* [21, Eqn. (42), p. 256], we have

$$\begin{aligned} dx_I \wedge dx_J &= (dx_{i_1} \wedge \cdots \wedge dx_{i_k}) \wedge (dx_{j_1} \wedge \cdots \wedge dx_{j_m}) \\ &= (dx_{i_1} \wedge \cdots \wedge dx_{i_{k-1}} \wedge (-1) dx_{j_1}) \wedge (dx_{i_k} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_m}) \\ &= (dx_{i_1} \wedge \cdots \wedge dx_{i_{k-2}} \wedge (-1)^2 dx_{j_1} \wedge dx_{i_{k-1}}) \wedge (dx_{i_k} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_m}) \\ &= ((-1)^k dx_{j_1} \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_{k-1}}) \wedge (dx_{i_k} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_m}) \\ &= ((-1)^{2k} dx_{j_1} \wedge dx_{j_2} \wedge dx_{i_1} \wedge \cdots \wedge dx_{i_{k-2}}) \\ &\quad \wedge (dx_{i_{k-1}} \wedge dx_{i_k} \wedge dx_{j_3} \wedge \cdots \wedge dx_{j_m}) \\ &\quad \vdots \\ &= (-1)^{km} (dx_{j_1} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_m}) \wedge (dx_{i_1} \wedge dx_{i_2} \wedge \cdots \wedge dx_{i_k}) \\ &= (-1)^{km} dx_J \wedge dx_I \end{aligned}$$

which is exactly the equality (10.81). This completes the proof of the problem. ■

Problem 10.16

Rudin Chapter 10 Exercise 16.

Proof. Let $k = 2$. Then we get from Definition 10.29 that

$$\partial\sigma = [\mathbf{p}_1, \mathbf{p}_2] - [\mathbf{p}_0, \mathbf{p}_2] + [\mathbf{p}_0, \mathbf{p}_1]$$

which gives

$$\partial^2\sigma = \partial[\mathbf{p}_1, \mathbf{p}_2] - \partial[\mathbf{p}_0, \mathbf{p}_2] + \partial[\mathbf{p}_0, \mathbf{p}_1] = [\mathbf{p}_2] - [\mathbf{p}_1] - ([\mathbf{p}_2] - [\mathbf{p}_0]) + [\mathbf{p}_1] - [\mathbf{p}_0] = 0.$$

Let $k = 3$. Similarly, we obtain from Definition 10.29 that

$$\partial\sigma = [\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3] - [\mathbf{p}_0, \mathbf{p}_2, \mathbf{p}_3] + [\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_3] - [\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2]$$

which implies that

$$\begin{aligned}\partial^2\sigma &= \partial[\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3] - \partial[\mathbf{p}_0, \mathbf{p}_2, \mathbf{p}_3] + \partial[\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_3] - [\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2] \\ &= [\mathbf{p}_2, \mathbf{p}_3] - [\mathbf{p}_1, \mathbf{p}_3] + [\mathbf{p}_1, \mathbf{p}_2] - ([\mathbf{p}_2, \mathbf{p}_3] - [\mathbf{p}_0, \mathbf{p}_3] + [\mathbf{p}_0, \mathbf{p}_2]) \\ &\quad + [\mathbf{p}_1, \mathbf{p}_3] - [\mathbf{p}_0, \mathbf{p}_3] + [\mathbf{p}_0, \mathbf{p}_1] - ([\mathbf{p}_1, \mathbf{p}_2] - [\mathbf{p}_0, \mathbf{p}_2]) + [\mathbf{p}_0, \mathbf{p}_1] \\ &= 0.\end{aligned}$$

For the general case, let σ_i and σ_{ij} be the $(k-1)$ -simplex and $(k-2)$ -simplex obtained by deleting $\mathbf{p}_i$ and $\mathbf{p}_i$ plus $\mathbf{p}_j$ from σ respectively, where $i < j$. That is

$$\sigma_i = [\mathbf{p}_0, \dots, \mathbf{p}_{i-1}, \mathbf{p}_{i+1}, \dots, \mathbf{p}_k] \quad \text{and} \quad \sigma_{ij} = [\mathbf{p}_0, \dots, \mathbf{p}_{i-1}, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k]$$

where $i, j = 0, \dots, k$ and $i < j$. Now each σ_{ij} occurs exactly *twice* in $\partial^2\sigma$, one from deleting the $\mathbf{p}_j$ first and then the $\mathbf{p}_i$ next, and the other one from deleting the $\mathbf{p}_i$ first and then the $\mathbf{p}_j$ next. We claim that the resulting $(k-2)$ -simplex have opposite sign. To this end, we notice that the positions of $\mathbf{p}_i$ and $\mathbf{p}_j$ in the oriented affine k -simplex σ first:

$$\sigma = [\underbrace{\mathbf{p}_0, \dots, \mathbf{p}_{i-1}}_{i \text{ terms before } \mathbf{p}_i}, \mathbf{p}_i, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \underbrace{\mathbf{p}_j, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k}_{j \text{ terms before } \mathbf{p}_j}]. \quad (10.82)$$

- **Case (i): Delete the $\mathbf{p}_j$ from the expression (10.82).** We have

$$\sigma_j = [\underbrace{\mathbf{p}_0, \dots, \mathbf{p}_{i-1}}_{i \text{ terms before } \mathbf{p}_i}, \mathbf{p}_i, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k] \quad (10.83)$$

and this contributes a factor $(-1)^j$. We observe from the expression (10.83) that there are i terms before the $\mathbf{p}_i$, so when we delete the $\mathbf{p}_i$ from the expression (10.83), we have

$$\sigma_{ij} = [\mathbf{p}_0, \dots, \mathbf{p}_{i-1}, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k]$$

and this contributes another factor $(-1)^i$. Thus we obtain

$$(-1)^{i+j}\sigma_{ij}$$

in this way.

- **Case (ii): Delete the $\mathbf{p}_i$ from (10.82).** We get

$$\sigma_i = [\mathbf{p}_0, \dots, \mathbf{p}_{i-1}, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \underbrace{\mathbf{p}_j, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k}_{(j-1) \text{ terms before } \mathbf{p}_j}] \quad (10.84)$$

and this contributes a factor $(-1)^i$. Now we notice from the form (10.84) that there are $(j-1)$ terms before the $\mathbf{p}_j$, so if we delete the $\mathbf{p}_j$ from the expression (10.84), then we have

$$\sigma_{ij} = [\mathbf{p}_0, \dots, \mathbf{p}_{i-1}, \mathbf{p}_{i+1}, \dots, \mathbf{p}_{j-1}, \mathbf{p}_{j+1}, \dots, \mathbf{p}_k]$$

and this contributes another factor $(-1)^{j-1}$. Thus we obtain

$$(-1)^{i+j-1}\sigma_{ij}$$

in this way.

This proves our claim and then completes the proof of our problem. ■

Problem 10.17

Rudin Chapter 10 Exercise 17.

As a remark to Definition 10.28, we recall from Rudin's explanation [21, p. 268] that the notation “+” used in the chain $J^2 = \tau_1 + \tau_2$ does not mean the addition of mappings. In fact, if we denote $\Omega_k(E)$ to be the collection of all k -forms in E and $\Phi : D \subset \mathbb{R}^k \rightarrow E \subseteq \mathbb{R}^n$ is a k -surface in E , then we define $\Phi : \Omega_k(E) \rightarrow \mathbb{R}$ to be^m

$$\Phi(\omega) = \int_{\Phi} \omega.$$

Hence the “+” in the expression $J^2 = \tau_1 + \tau_2$ means that

$$J^2(\omega) = \tau_1(\omega) + \tau_2(\omega) = \int_{\tau_1} \omega + \int_{\tau_2} \omega \quad (10.85)$$

for every 2-form ω . To avoid any ambiguity, we write $\tau_1 +_c \tau_2$ to replace the original affine chain $\tau_1 + \tau_2$ with the meaning shown in the integrals (10.85).

Proof. We first find the explicit representations of τ_1 and τ_2 . Note that τ_1 is characterized by

$$\tau_1(\mathbf{0}) = \mathbf{0}, \quad \tau_1(\mathbf{e}_1) = \mathbf{e}_1 \quad \text{and} \quad \tau_1(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2.$$

By [21, Eqn. (80), p. 267], we know that $\tau_2 = [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_1, \mathbf{e}_2]$ and then it is characterized by

$$\tau_2(\mathbf{0}) = \mathbf{0}, \quad \tau_2(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_2 \quad \text{and} \quad \tau_2(\mathbf{e}_2) = \mathbf{e}_2.$$

By [21, Eqn. (78), p. 266], we acquire the mappings $\tau_1 : Q^2 \rightarrow \mathbb{R}^2$ and $\tau_2 : Q^2 \rightarrow \mathbb{R}^2$ by

$$\tau_1(\mathbf{u}) = A\mathbf{u} \quad \text{and} \quad \tau_2(\mathbf{u}) = B\mathbf{u} \quad (10.86)$$

for all $\mathbf{u} \in Q^2$, where

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad (10.87)$$

Next, we find $\tau_1(Q^2)$ and $\tau_2(Q^2)$. By Definition 10.26,

$$Q^2 = \{a\mathbf{e}_1 + b\mathbf{e}_2 \mid a, b \geq 0, a + b \leq 1\}.$$

Therefore, by this and the mappings (10.86) with the matrices (10.87), we deduce that

$$\tau_1(a\mathbf{e}_1 + b\mathbf{e}_2) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a+b \\ b \end{pmatrix} = (a+b)\mathbf{e}_1 + b\mathbf{e}_2 \quad (10.88)$$

which imply that

$$\tau_1(Q^2) = \{(a+b)\mathbf{e}_1 + b\mathbf{e}_2 \mid a, b \geq 0, a + b \leq 1\} \quad (10.89)$$

and it is the “lower right” half of the unit square I^2 , see Figure 10.9:

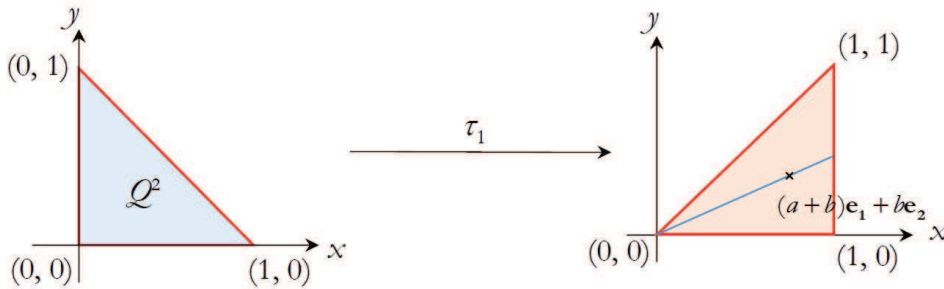


Figure 10.9: The mapping $\tau_1 : Q^2 \rightarrow I^2$.

^mWe use different colors for Φ so as to make clear that it has “different” meanings in “different” situations: For Φ , it means a k -surface and for Φ , it means a function.

Similarly, we have

$$\tau_2(a\mathbf{e}_1 + b\mathbf{e}_2) = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ a+b \end{pmatrix} = a\mathbf{e}_1 + (a+b)\mathbf{e}_2 \quad (10.90)$$

which imply that

$$\tau_2(Q^2) = \{a\mathbf{e}_1 + (a+b)\mathbf{e}_2 \mid a, b \geq 0, a+b \leq 1\} \quad (10.91)$$

and it is the “upper left” half of the unit square I^2 , see Figure 10.10:

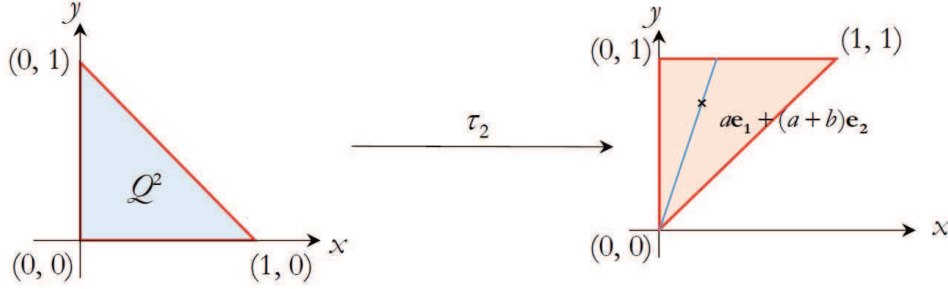


Figure 10.10: The mapping $\tau_2 : Q^2 \rightarrow I^2$.

Thus we follow from the ranges (10.89) and (10.91) that

$$I^2 = \tau_1(Q^2) \cup \tau_2(Q^2)$$

and the interiors of $\tau_1(Q^2)$ and $\tau_2(Q^2)$ are disjoint. By the matrices (10.87), since

$$\det A = \det B = 1 > 0,$$

τ_1 and τ_2 are obviously one-to-one mappings of class $\mathcal{C}''$. Furthermore, we know from Definition 10.31 that J^2 has the positively oriented boundary. Hence it is reasonable to say that J^2 the positively oriented square in $\mathbb{R}^2$. See Figure 10.11 for the orientation of J^2 , where the red and green arrows connecting the points $(0,0)$ and $(1,1)$ have opposite orientation so that they *cancel* each other.

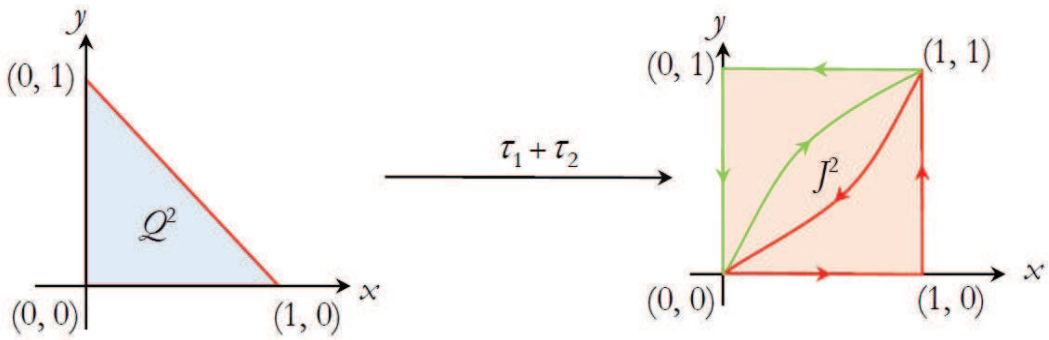


Figure 10.11: The mapping $\tau_2 : Q^2 \rightarrow I^2$.

Finally, we compute ∂J^2 and $\partial(\tau_1 - \tau_2)$. By routine computation and [21, Eqn. (80), p. 267], we have

$$\begin{aligned} \partial J^2 &= \partial\tau_1 + \partial\tau_2 \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_1] - ([\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_2]) \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_1] - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_2] \\ &= [\mathbf{0}, \mathbf{e}_1] + [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2] + [\mathbf{e}_2, \mathbf{0}] \end{aligned}$$

which is the sum of 4 oriented affine 1-simplexes.ⁿ Similarly, we have

$$\begin{aligned}\partial(\tau_1 - \tau_2) &= \partial\tau_1 - \partial\tau_2 \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_1] \\ &\quad + ([\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_2]) \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_1] + [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_2] - 2[\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2].\end{aligned}$$

Therefore, we have

$$\partial J^2 \neq \partial(\tau_1 - \tau_2)$$

and this completes the proof of the problem. ■

Problem 10.18

Rudin Chapter 10 Exercise 18.

Proof. Recall from Definition 10.26 that

$$Q^3 = \{a\mathbf{e}_1 + b\mathbf{e}_2 + c\mathbf{e}_3 \mid a, b, c \geq 0, a + b + c \leq 1\}$$

and σ_1 is characterized by

$$\sigma_1(\mathbf{0}) = \mathbf{0}, \quad \sigma_1(\mathbf{e}_1) = \mathbf{e}_1, \quad \sigma_1(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2 \quad \text{and} \quad \sigma_1(\mathbf{e}_3) = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3.$$

By [21, Eqn. (78), p. 266], we have the mapping $\sigma_1 : Q^3 \rightarrow \mathbb{R}^3$ defined by

$$\sigma_1(\mathbf{u}) = A_1 \mathbf{u}$$

for all $\mathbf{u} \in Q^3$, where

$$A_1 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}. \quad (10.92)$$

Since $\det A_1 = 1 > 0$, σ_1 is positively oriented. The five permutations of $(1, 2, 3)$ other than $(1, 2, 3)$ are

$$(1, 3, 2), \quad (2, 1, 3), \quad (2, 3, 1), \quad (3, 1, 2) \quad \text{and} \quad (3, 2, 1).$$

By Problem 10.14, we have

$$s(1, 3, 2) = -1, \quad s(2, 1, 3) = -1, \quad s(2, 3, 1) = 1, \quad s(3, 1, 2) = 1 \quad \text{and} \quad s(3, 2, 1) = -1.$$

Therefore, we have

$$\begin{aligned}\sigma_2 &= -[\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] = [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3], \\ \sigma_3 &= -[\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] = [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2], \\ \sigma_4 &= [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3], \\ \sigma_5 &= [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3], \\ \sigma_6 &= -[\mathbf{0}, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] = [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3]\end{aligned}$$

so that

$$A_2 = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad A_5 = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

and

$$A_6 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

ⁿActually, the ∂J^2 is exactly the positively oriented boundary ∂I^2 of the unit square I^2 given in [21, p. 271].

Since

$$\det A_2 = \det A_3 = \det A_4 = \det A_5 = \det A_6 = 1 > 0,$$

$\sigma_2, \dots, \sigma_6$ are positively oriented by Definition 10.26.

Put $J^3 = \sigma_1 + \dots + \sigma_6$. By Definition 10.29, we have

$$\begin{aligned} \partial J^3 &= \sum_{m=1}^6 \partial \sigma_m \\ &= ([\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2]) \\ &\quad - ([\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3]) \\ &\quad - ([\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2]) \\ &\quad + ([\mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3]) \\ &\quad + ([\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3]) \\ &\quad - ([\mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3]) \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] \\ &\quad - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3] \\ &\quad - [\mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3] \end{aligned} \quad (10.93)$$

which consists exactly 12 oriented affine 2-simplexes.

By the matrix representation (10.92), $A_1 \mathbf{u} = \mathbf{x}$ if and only if

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

if and only if

$$a + b + c = x_1, \quad b + c = x_2 \quad \text{and} \quad c = x_3.$$

Thus $\mathbf{x} \in \sigma_1(Q^3)$ if and only if $0 \leq x_3 \leq x_2 \leq x_1 \leq 1$. Similarly, we have

$$\begin{aligned} A_2 \mathbf{u} = \mathbf{x} &\quad \text{if and only if} \quad 0 \leq x_2 \leq x_3 \leq x_1 \leq 1, \\ A_3 \mathbf{u} = \mathbf{x} &\quad \text{if and only if} \quad 0 \leq x_3 \leq x_1 \leq x_2 \leq 1, \\ A_4 \mathbf{u} = \mathbf{x} &\quad \text{if and only if} \quad 0 \leq x_1 \leq x_3 \leq x_2 \leq 1, \\ A_5 \mathbf{u} = \mathbf{x} &\quad \text{if and only if} \quad 0 \leq x_2 \leq x_1 \leq x_3 \leq 1, \\ A_6 \mathbf{u} = \mathbf{x} &\quad \text{if and only if} \quad 0 \leq x_1 \leq x_2 \leq x_3 \leq 1. \end{aligned}$$

These mean that

$$\begin{aligned} \mathbf{x} \in \sigma_2(Q^3) &\quad \text{if and only if} \quad 0 \leq x_2 \leq x_3 \leq x_1 \leq 1, \\ \mathbf{x} \in \sigma_3(Q^3) &\quad \text{if and only if} \quad 0 \leq x_3 \leq x_1 \leq x_2 \leq 1, \\ \mathbf{x} \in \sigma_4(Q^3) &\quad \text{if and only if} \quad 0 \leq x_1 \leq x_3 \leq x_2 \leq 1, \\ \mathbf{x} \in \sigma_5(Q^3) &\quad \text{if and only if} \quad 0 \leq x_2 \leq x_1 \leq x_3 \leq 1, \\ \mathbf{x} \in \sigma_6(Q^3) &\quad \text{if and only if} \quad 0 \leq x_1 \leq x_2 \leq x_3 \leq 1. \end{aligned}$$

In other words, we have

$$\begin{aligned} \sigma_1(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_3 \leq x_2 \leq x_1 \leq 1\}, \\ \sigma_2(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_2 \leq x_3 \leq x_1 \leq 1\}, \\ \sigma_3(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_3 \leq x_1 \leq x_2 \leq 1\}, \\ \sigma_4(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_1 \leq x_3 \leq x_2 \leq 1\}, \\ \sigma_5(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_2 \leq x_1 \leq x_3 \leq 1\}, \\ \sigma_6(Q^3) &= \{(x_1, x_2, x_3) \mid 0 \leq x_1 \leq x_2 \leq x_3 \leq 1\}. \end{aligned} \quad (10.94)$$

Hence it follows from the ranges (10.94) that they have disjoint interiors and

$$I^3 = \sigma_1(Q^3) \cup \sigma_2(Q^3) \cup \cdots \cup \sigma_6(Q^3).$$

This completes the proof of the problem. ■

Problem 10.19

Rudin Chapter 10 Exercise 19.

Proof. By rewriting the expression (10.93), we get the following new expression for ∂J^3 :

$$\begin{aligned} \partial J^3 = & \left\{ [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] \right. \\ & \left. + [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3] \right\} + \left\{ [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3] - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \right. \\ & \left. + [\mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \right\} + \left\{ [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \right. \\ & \left. - [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] \right\}. \end{aligned} \quad (10.95)$$

For $r = 0, 1, i = 1, 2, 3$, $B_{ri} : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ are $\mathcal{C}''$ -mappings so that each β_{ri} is an oriented 3-simplex of class $\mathcal{C}''$. Since $J^2 = \tau_1 + \tau_2$, we deduce from [21, Eqn. (88, p. 270)] that

$$\beta_{ri} = B_{ri}(\tau_1 + \tau_2) = B_{ri}(\tau_1) + B_{ri}(\tau_2). \quad (10.96)$$

Let $b_{ri}^1 = B_{ri}(\tau_1)$ and $b_{ri}^2 = B_{ri}(\tau_2)$, where $r = 0, 1, i = 1, 2, 3$. Then the expression (10.96) can be rewritten as

$$\beta_{ri} = b_{ri}^1 + b_{ri}^2. \quad (10.97)$$

Now we must find the *explicit forms* of the oriented affine 2-simplexes b_{ri}^1 and b_{ri}^2 in the definition (10.97) in order to compute each $(-1)^i(\beta_{0i} - \beta_{1i})$. To this end, we recall the representations (10.88) and (10.90) first:

$$\tau_1(a\mathbf{e}_1 + b\mathbf{e}_2) = (a + b)\mathbf{e}_1 + b\mathbf{e}_2 = ((a + b), b) \quad \text{and} \quad \tau_2(a\mathbf{e}_1 + b\mathbf{e}_2) = a\mathbf{e}_1 + (a + b)\mathbf{e}_2 = (a, a + b).$$

- **Computation of $-(\beta_{01} - \beta_{11})$.** For b_{01}^1 and b_{01}^2 , since

$$b_{01}^1(a\mathbf{e}_1 + b\mathbf{e}_2) = (0, a + b, b) \quad \text{and} \quad b_{01}^2(a\mathbf{e}_1 + b\mathbf{e}_2) = (0, a, a + b),$$

we have

$$\begin{aligned} b_{01}^1(\mathbf{0}) &= \mathbf{0}, & b_{01}^1(\mathbf{e}_1) &= \mathbf{e}_2 & \text{and} & & b_{01}^1(\mathbf{e}_2) &= \mathbf{e}_2 + \mathbf{e}_3, \\ b_{01}^2(\mathbf{0}) &= \mathbf{0}, & b_{01}^2(\mathbf{e}_1) &= \mathbf{e}_2 + \mathbf{e}_3 & \text{and} & & b_{01}^2(\mathbf{e}_2) &= \mathbf{e}_3. \end{aligned}$$

By [21, Eqn. (77), p. 266], we have

$$b_{01}^1 = [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] \quad \text{and} \quad b_{01}^2 = [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_3],$$

so the definition (10.97) gives

$$\beta_{01} = [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_3]. \quad (10.98)$$

Similarly, since

$$b_{11}^1(a\mathbf{e}_1 + b\mathbf{e}_2) = (1, a + b, b) \quad \text{and} \quad b_{11}^2(a\mathbf{e}_1 + b\mathbf{e}_2) = (1, a, a + b),$$

we have

$$b_{11}^1(\mathbf{0}) = \mathbf{e}_1, \quad b_{11}^1(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_2 \quad \text{and} \quad b_{11}^1(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3,$$

$$b_{11}^2(\mathbf{0}) = \mathbf{e}_1, \quad b_{11}^2(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3 \quad \text{and} \quad b_{11}^2(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_3.$$

By [21, Eqn. (77), p. 266] again, we have

$$b_{11}^1 = [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \quad \text{and} \quad b_{12}^2 = [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3],$$

so the definition (10.97) gives

$$\beta_{11} = [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3]. \quad (10.99)$$

Therefore, we deduce from the two expressions (10.98) and (10.99) that

$$\begin{aligned} (-1)^1(\beta_{01} - \beta_{11}) &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3] \\ &\quad - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_3] \\ &= [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad - [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3] \end{aligned}$$

which is exactly the first brackets in the expression (10.95).

- **Computation of $\beta_{02} - \beta_{12}$.** For b_{02}^1 and b_{02}^2 , since

$$b_{02}^1(a\mathbf{e}_1 + b\mathbf{e}_2) = (a + b, 0, b) \quad \text{and} \quad b_{02}^2(a\mathbf{e}_1 + b\mathbf{e}_2) = (a, 0, a + b),$$

we have

$$\begin{aligned} b_{02}^1(\mathbf{0}) &= \mathbf{0}, \quad b_{02}^1(\mathbf{e}_1) = \mathbf{e}_1 \quad \text{and} \quad b_{02}^1(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_3, \\ b_{02}^2(\mathbf{0}) &= \mathbf{0}, \quad b_{02}^2(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_3 \quad \text{and} \quad b_{02}^2(\mathbf{e}_2) = \mathbf{e}_3. \end{aligned}$$

Since

$$b_{02}^1 = [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] \quad \text{and} \quad b_{02}^2 = [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_3],$$

we have

$$\beta_{02} = [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_3]. \quad (10.100)$$

Now we have

$$b_{12}^1(a\mathbf{e}_1 + b\mathbf{e}_2) = (a + b, 1, b) \quad \text{and} \quad b_{12}^2(a\mathbf{e}_1 + b\mathbf{e}_2) = (a, 1, a + b)$$

which imply that

$$b_{12}^1 = [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \quad \text{and} \quad b_{12}^2 = [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3].$$

Thus we have

$$\beta_{12} = [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3]. \quad (10.101)$$

Combining the expressions (10.100) and (10.101), we obtain

$$\begin{aligned} \beta_{02} - \beta_{12} &= [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] + [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_3] - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3] \\ &= [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_3] - [\mathbf{0}, \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3] - [\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad + [\mathbf{e}_2, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \end{aligned}$$

which is exactly the second brackets in the expression (10.95).

- **Computation of $-(\beta_{03} - \beta_{13})$.** The computation of $(-1)^3(\beta_{03} - \beta_{13})$ can be done similarly as above. We have

$$b_{03}^1(a\mathbf{e}_1 + b\mathbf{e}_2) = (a + b, b, 0) \quad \text{and} \quad b_{03}^2(a\mathbf{e}_1 + b\mathbf{e}_2) = (a, a + b, 0)$$

which imply that

$$b_{03}^1(\mathbf{0}) = \mathbf{0}, \quad b_{03}^1(\mathbf{e}_1) = \mathbf{e}_1 \quad \text{and} \quad b_{03}^1(\mathbf{e}_2) = \mathbf{e}_1 + \mathbf{e}_2,$$

$$b_{03}^2(\mathbf{0}) = \mathbf{0}, \quad b_{03}^2(\mathbf{e}_1) = \mathbf{e}_1 + \mathbf{e}_2 \quad \text{and} \quad b_{03}^2(\mathbf{e}_2) = \mathbf{e}_2.$$

These mean that

$$b_{03}^1 = [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] \quad \text{and} \quad b_{03}^2 = [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2],$$

so we have

$$\beta_{03} = [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2]. \quad (10.102)$$

Similarly, we have

$$\beta_{13} = [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] + [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3]. \quad (10.103)$$

Therefore we follow from the expressions (10.102) and (10.103) that

$$\begin{aligned} -(\beta_{03} - \beta_{13}) &= -[\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] - [\mathbf{0}, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2] + [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad + [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3] \\ &= [\mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] - [\mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_3, \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3] \\ &\quad - [\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2] + [\mathbf{0}, \mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_2] \end{aligned}$$

which is the third brackets in the expression (10.95).

Hence, by the above computations, we get our desired result that

$$\partial J^3 = \sum_{i=1}^3 (-1)^i (\beta_{0i} - \beta_{1i}).$$

This ends the proof of the problem. ■

10.5 Problems on closed forms and exact forms

Problem 10.20

Rudin Chapter 10 Exercise 20.

Proof. Suppose that E is an open set in $\mathbb{R}^n$, $f \in \mathcal{C}'(E)$, ω is a k -form of class $\mathcal{C}'$ in E and Φ is a $(k+1)$ -chain of class $\mathcal{C}''$ in E . By Theorem 10.20(a), we have

$$d(f\omega) = (df) \wedge \omega + (-1)^0 f d\omega = (df) \wedge \omega + f d\omega. \quad (10.104)$$

Apply Theorem 10.33 (Stokes' Theorem) to the left-hand side in the expression (10.104), we get

$$\int_{\partial\Phi} f\omega = \int_{\Phi} d(f\omega) = \int_{\Phi} (df) \wedge \omega + \int_{\Phi} f d\omega$$

which implies that the desired equality

$$\int_{\Phi} f d\omega = \int_{\partial\Phi} f\omega - \int_{\Phi} (df) \wedge \omega.$$

Let $n = 1$ and $k = 0$ in the above consideration. Let, further, E be an open set in $\mathbb{R}$ containing the interval $[a, b]$, where a and b are real numbers with $a < b$. Now we consider the oriented affine 1-simplex $\Phi : [0, 1] \rightarrow \mathbb{R}$, where

$$\Phi(0) = a \quad \text{and} \quad \Phi(1) = b.$$

By Definition 10.29, we have

$$\partial\Phi = [b] - [a]$$

which is an oriented 0-simplex. If $\omega = g$ is a 0-form of class $\mathcal{C}'$ in E , then fg is also a 0-form of class $\mathcal{C}'$ in E . Thus, by the equation just preceding Theorem 10.27 (see [21, p. 267]), we acquire

$$\int_{\partial\Phi} f\omega = \int_{+b} fg + \int_{-a} fg = f(b)g(b) - f(a)g(a). \quad (10.105)$$

On the other hand, it is clear that $f dg$ and $(df)g$ are 1-forms by Definition 10.18. In addition, we obtain from [21, Eqn. (59)] that

$$dg = g'(x) dx,$$

where $x = \Phi(t)$, thus we see from [21, Eqn. (35), p. 254] that

$$\begin{aligned} \int_{\Phi} f dg &= \int_{\Phi} \underbrace{f(x)g'(x)}_{\substack{\text{This is the } a(x) \text{ in} \\ [21, \text{Eqn. (34), p. 254}]} } dx \\ &= \int_{[0,1]} f(\Phi(t))g'(\Phi(t))\Phi'(t) dt \\ &= \int_0^1 f(\Phi(t))g'(\Phi(t))\Phi'(t) dt. \end{aligned}$$

If Φ is supposed to be strictly increasing on $[0, 1]$, then [21, Eqn. (39), p. 133] implies that

$$\int_0^1 \underbrace{f(\Phi(t))g'(\Phi(t))}_{\substack{\text{This is the } f(\varphi(y)) \text{ in} \\ [21, \text{Eqn. (39), p. 133}]} } \Phi'(t) dt = \int_a^b f(u)g'(u) du. \quad (10.106)$$

Similarly, we have $df = f'(x) dx$, where $x = \Phi(t)$. Then it deduces from [21, Eqn. (35), p. 254] that

$$\int_{\Phi} (df) \wedge \omega = \int_{\Phi} g(df) = \int_{\Phi} g(x)f'(x) dx = \int_{[0,1]} g(\Phi(t))f'(\Phi(t))\Phi'(t) dt = \int_0^1 g(\Phi(t))f'(\Phi(t))\Phi'(t) dt.$$

By [21, Eqn. (39), p. 133] again, we obtain

$$\int_0^1 g(\Phi(t))f'(\Phi(t))\Phi'(t) dt = \int_a^b g(u)f'(u) du. \quad (10.107)$$

Combining the equalities (10.105) to (10.107), we establish that

$$\int_a^b f(u)g'(u) du = f(b)g(b) - f(a)g(a) - \int_a^b f(u)g'(u) du$$

which is exactly Theorem 6.22 (Integration of Parts). This completes the proof of the problem. ■

Problem 10.21

Rudin Chapter 10 Exercise 21.

Proof.

- (a) Since $x = r \cos t$ and $y = r \sin t$, the direct computation of the formula [21, Eqn. (113), p. 277] is given as follows:

$$\int_{\gamma} \eta = \int_0^{2\pi} \frac{r^2 \cos^2 t dt + r^2 \sin^2 t dt}{r^2} = \int_0^{2\pi} dt = 2\pi.$$

However, it follows from Theorem 10.20 and [21, Eqn. (59), p. 260] that

$$d\eta = d\left(\frac{x}{x^2 + y^2}\right) \wedge dy + \left(\frac{x}{x^2 + y^2}\right) \wedge d^2y - d\left(\frac{y}{x^2 + y^2}\right) \wedge dx - \left(\frac{y}{x^2 + y^2}\right) \wedge d^2x$$

$$\begin{aligned}
&= \left[\frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} dx + \frac{-x(2y)}{(x^2 + y^2)^2} dy \right] \wedge dy \\
&\quad - \left[\frac{-y(2x)}{(x^2 + y^2)^2} dx + \frac{(x^2 + y^2) - y(2y)}{(x^2 + y^2)^2} dy \right] \wedge dx.
\end{aligned} \tag{10.108}$$

By the anticommutative relation ($dy \wedge dx = -dx \wedge dy$) and [21, Eqn. (43), p. 256], we deduce from the expression (10.108) that

$$\begin{aligned}
d\eta &= \left[\frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2} dx \wedge dy - \frac{(x^2 + y^2) - y(2y)}{(x^2 + y^2)^2} dx \wedge dy \right] \\
&\quad + \left[\frac{-x(2y)}{(x^2 + y^2)^2} dy \wedge dy - \frac{-y(2x)}{(x^2 + y^2)^2} dx \wedge dx \right] \\
&= \frac{1}{(x^2 + y^2)^2} (-2xy dy \wedge dy + 2xy dx \wedge dx) \\
&= 0.
\end{aligned}$$

- (b) Let $D = \{(t, u) \mid 0 \leq t \leq 2\pi, 0 \leq u \leq 1\}$ and $\Phi : D \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ be given as in the hint. Since $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ and $\Gamma : [0, 2\pi] \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ are $\mathcal{C}''$ -mappings, Φ is a 2-surface in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ by Definition 10.10. The geometric interpretation of the mapping Φ is given in Figure 10.12, where a **purple arrow** is an interval $[\gamma(t), \Gamma(t)]$ for some $t \in [0, 2\pi]$ which does not contain the origin $\mathbf{0}$.^o

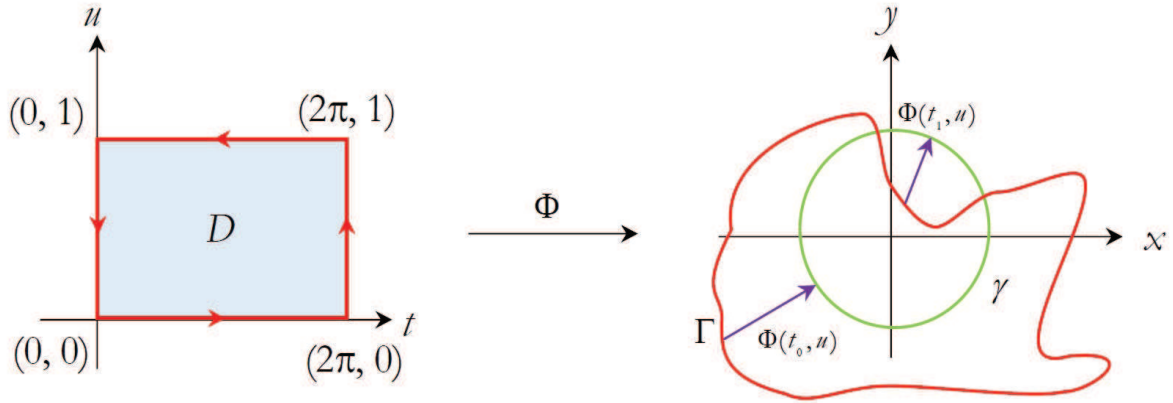


Figure 10.12: The mapping $\Phi : D \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$.

By a similar analysis as in Example 10.32, we know that

$$\partial\Phi = \Phi(\partial D) = \sigma_1 + \sigma_2 + \sigma_3 + \sigma_4, \tag{10.109}$$

where

$$\begin{aligned}
\sigma_1(t) &= \Phi(t, 0) = \Gamma(t), \\
\sigma_2(u) &= \Phi(2\pi, u) = (1 - u)\Gamma(2\pi) + u\gamma(2\pi), \\
\sigma_3(t) &= \Phi(2\pi - t, 1) = \gamma(2\pi - t), \\
\sigma_4(u) &= \Phi(0, 1 - u) = u\Gamma(0) + (1 - u)\gamma(0).
\end{aligned}$$

Since $\gamma(0) = \gamma(2\pi)$ and $\Gamma(0) = \Gamma(2\pi)$, we follow from [21, Eqn. (77) & (80), pp. 266, 267] in Definition 10.26 that

$$\sigma_2 = [\Gamma(0), \gamma(0)] = -[\gamma(0), \Gamma(0)] = -\sigma_4.$$

^oIn fact, the Φ is a *homotopy* between γ and Γ , see [18, p. 323].

Similarly, by direct application of [21, Eqn. (35), p. 254], we obtain

$$\int_{\sigma_3} \omega = - \int_{\gamma} \omega$$

for every 1-form ω . In other words, we get from the relation (10.109) that

$$\partial\Phi = \Gamma - \gamma \quad (10.110)$$

By Theorem 10.33 (Stokes' Theorem) and part (a), we get

$$\int_{\partial\Phi} \eta = \int_{\Phi} d\eta = \int_{\Phi} 0 = 0. \quad (10.111)$$

Hence it follows from the expressions (10.110) and (10.111) that

$$\begin{aligned} \int_{\sigma_1} \eta + \int_{\sigma_2} \eta + \int_{\sigma_3} \eta + \int_{\sigma_4} \eta &= 0 \\ \int_{\Gamma} \eta - \int_{\sigma_4} \eta - \int_{\gamma} \eta + \int_{\sigma_4} \eta &= 0 \\ \int_{\Gamma} \eta &= \int_{\gamma} \eta. \end{aligned} \quad (10.112)$$

Now, by the result of part (a), we have the desired result from (10.112)

$$\int_{\Gamma} \eta = 2\pi.$$

(c) Suppose that

$$\Gamma(t) = (a \cos t, b \sin t),$$

where $a > 0, b > 0$ are fixed. Now Γ is a $\mathcal{C}''$ -curve in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ with parameter interval $[0, 2\pi]$ and $\Gamma(0) = \Gamma(2\pi)$. Let $[\gamma(t), \Gamma(t)]$ be the interval joining the points $\gamma(t)$ and $\Gamma(t)$ for each $t \in [0, 2\pi]$. Since $[\gamma(t), \Gamma(t)]$ does not contain the origin $\mathbf{0}$, we deduce from part (b) that

$$\begin{aligned} \int_{\Gamma} \eta &= 2\pi \\ \int_{[0, 2\pi]} \frac{x dy - y dx}{x^2 + y^2} &= 2\pi \\ \int_0^{2\pi} \frac{ab \cos^2 t dt + ab \sin^2 t dt}{a^2 \cos^2 t + b^2 \sin^2 t} &= 2\pi \\ \int_0^{2\pi} \frac{ab}{a^2 \cos^2 t + b^2 \sin^2 t} dt &= 2\pi. \end{aligned}$$

(d) Recall from [21, Remark 10.35(a), p. 275] that a 1-form

$$\omega = \sum_{i=1}^n f_i(\mathbf{x}) dx_i$$

is exact in an open set $E \subseteq \mathbb{R}^n$ if and only if there is a function (0-form) $g \in \mathcal{C}'(E)$ such that

$$(D_i g)(\mathbf{x}) = f_i(\mathbf{x}) \quad (\mathbf{x} \in E, 1 \leq i \leq n). \quad (10.113)$$

Consider $n = 2$ and E to be any convex open set in which $x \neq 0$.^P By the definition of η , we have

$$f_1(x, y) = \frac{-y}{x^2 + y^2} \quad \text{and} \quad f_2(x, y) = \frac{x}{x^2 + y^2}.$$

^PIn other words, E does not intersect with the y -axis and this means that E lies either entirely in the left or right half plane.

Let $g : E \rightarrow \mathbb{R}$ be defined by

$$g(x, y) = \arctan \frac{y}{x}.$$

Since $\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$, we can easily see that

$$\frac{\partial}{\partial x} \left(\arctan \frac{y}{x} \right) = \frac{-y}{x^2} \cdot \frac{1}{1 + \frac{y^2}{x^2}} = -\frac{y}{x^2 + y^2} \quad \text{and} \quad \frac{\partial}{\partial y} \left(\arctan \frac{y}{x} \right) = \frac{x}{x^2 + y^2}.$$

Now the function g satisfies the conditions (10.113), so the definition gives

$$\eta = dg = d\left(\arctan \frac{x}{y}\right)$$

in E .

Similarly, we suppose that F is any convex open set in which $y \neq 0$.[¶] A direct computation shows that

$$\frac{\partial}{\partial x} \left(-\arctan \frac{x}{y} \right) = -\frac{y}{x^2 + y^2} \quad \text{and} \quad \frac{\partial}{\partial y} \left(-\arctan \frac{x}{y} \right) = \frac{x}{x^2 + y^2},$$

thus we have

$$\begin{aligned} d\left(-\arctan \frac{x}{y}\right) &= \frac{\partial}{\partial x} \left(-\arctan \frac{x}{y} \right) dx + \frac{\partial}{\partial y} \left(-\arctan \frac{x}{y} \right) dy \\ &= -\frac{y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy \\ &= \frac{x dy - y dx}{x^2 + y^2} \\ &= \eta. \end{aligned}$$

By the result of Example 10.36, η is not exact in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, but the analysis verifies that we can say that η is exact locally in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, so it is reasonable to denote

$$\eta = d\theta$$

for some 0-form θ .

(e) We write

$$[0, 2\pi] = I_1 \cup I_2 \cup \cdots \cup I_6,$$

where $I_i = \left[\frac{(i-1)\pi}{3}, \frac{i\pi}{3}\right]$ and $i = 1, 2, \dots, 6$. Now

$$\gamma\left([0, \frac{\pi}{3}]\right), \quad \gamma\left([\frac{2\pi}{3}, \pi]\right), \quad \gamma\left([\pi, \frac{4\pi}{3}]\right) \quad \text{and} \quad \gamma\left([\frac{5\pi}{3}, 2\pi]\right)$$

lie in E , while

$$\gamma\left([\frac{\pi}{3}, \frac{2\pi}{3}]\right) \quad \text{and} \quad \gamma\left([\frac{4\pi}{3}, \frac{5\pi}{3}]\right)$$

lie in F . Therefore, it follows from (d) that

$$\begin{aligned} \int_{\gamma} \eta &= \sum_{i=1}^6 \int_{\gamma(I_i)} \eta \\ &= \left(\int_{\gamma(I_1)} + \int_{\gamma(I_3)} + \int_{\gamma(I_4)} + \int_{\gamma(I_6)} \right) d\left(\arctan \frac{y}{x}\right) \\ &\quad + \left(\int_{\gamma(I_2)} + \int_{\gamma(I_5)} \right) d\left(-\arctan \frac{x}{y}\right). \end{aligned} \tag{10.114}$$

[¶]Now F does not intersect with the x -axis and so F lies either entirely in the upper or lower half plane.

We apply Theorem 10.33 (Stokes' Theorem) to the integrals on the right hand side in the expression (10.114) and then [21, Eqn. (62), p. 261], we obtain

$$\begin{aligned}
 \int_{\gamma} \eta &= \tan^{-1} \tan t \Big|_0^{\frac{\pi}{3}} + \tan^{-1} \tan t \Big|_{\frac{2\pi}{3}}^{\pi} + \tan^{-1} \tan t \Big|_{\pi}^{\frac{4\pi}{3}} + \tan^{-1} \tan t \Big|_{\frac{5\pi}{2}}^{2\pi} \\
 &\quad - \tan^{-1} \cot t \Big|_{\frac{\pi}{3}}^{\frac{2\pi}{3}} - \tan^{-1} \cot t \Big|_{\frac{4\pi}{3}}^{\frac{5\pi}{3}} \\
 &= \frac{\pi}{3} \times 4 - \tan^{-1} \tan \left(\frac{\pi}{2} - t \right) \Big|_{\frac{\pi}{3}}^{\frac{2\pi}{3}} - \tan^{-1} \left(\frac{\pi}{2} - t \right) \Big|_{\frac{4\pi}{3}}^{\frac{5\pi}{3}} \\
 &= \frac{4\pi}{3} - \left(\frac{\pi}{2} - t \right) \Big|_{\frac{\pi}{3}}^{\frac{2\pi}{3}} - \left(\frac{\pi}{2} - t \right) \Big|_{\frac{4\pi}{3}}^{\frac{5\pi}{3}} \\
 &= \frac{4\pi}{3} + \frac{2\pi}{3} \\
 &= 2\pi.
 \end{aligned}$$

This means that part (d) implies part (b).

(f) Given that $\Gamma(t) = (\Gamma_1(t), \Gamma_2(t))$, where $t \in [0, 2\pi]$. We also write

$$\Gamma = \Gamma_1 + i\Gamma_2$$

in the complex plane $\mathbb{C}$. Since Γ is assumed to be a closed $\mathcal{C}'$ -curve in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$, it is a continuously differentiable closed curve in $\mathbb{C}$ and $\Gamma(t) \neq 0$ for every $t \in [0, 2\pi]$. Thus, by the definition of the index of Γ in Problem 8.23, we have

$$\begin{aligned}
 \text{Ind}(\Gamma) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{\Gamma'_1 + i\Gamma'_2}{\Gamma_1 + i\Gamma_2} dt \\
 &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{\Gamma'_1 + i\Gamma'_2}{\Gamma_1 + i\Gamma_2} \cdot \frac{\Gamma_1 - i\Gamma_2}{\Gamma_1 - i\Gamma_2} dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} \frac{\Gamma_1\Gamma'_2 - \Gamma'_1\Gamma_2}{\Gamma_1^2 + \Gamma_2^2} dt + \frac{1}{2\pi i} \int_0^{2\pi} \frac{\Gamma_1\Gamma'_1 + \Gamma_2\Gamma'_2}{\Gamma_1^2 + \Gamma_2^2} dt. \tag{10.115}
 \end{aligned}$$

Since $\eta = \frac{x dy - y dx}{x^2 + y^2}$, Definition 10.11 implies that

$$\frac{1}{2\pi} \int_{\Gamma} \eta = \frac{1}{2\pi} \int_0^{2\pi} \frac{\Gamma_1\Gamma'_2 - \Gamma_2\Gamma'_1}{\Gamma_1^2 + \Gamma_2^2} dt \tag{10.116}$$

which is exactly the real part of the complex number (10.115). Furthermore, since $\Gamma(2\pi) = \Gamma(0)$, we have

$$\int_0^{2\pi} \frac{\Gamma_1\Gamma'_1 + \Gamma_2\Gamma'_2}{\Gamma_1^2 + \Gamma_2^2} dt = \int_0^{2\pi} d(\ln(\Gamma_1^2 + \Gamma_2^2)) = \ln[\Gamma_1^2(x) + \Gamma_2^2(x)] \Big|_0^{2\pi} = 0. \tag{10.117}$$

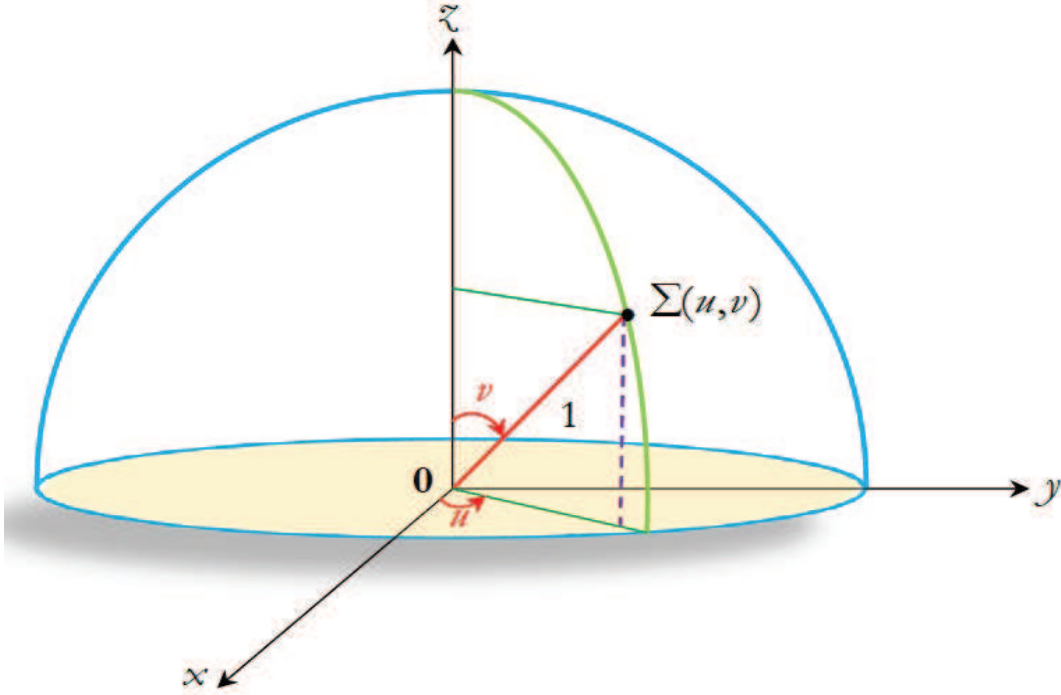
Hence we reach the desired result by comparing the expressions (10.115) to (10.117). This completes the proof of the problem. ■

Problem 10.22

Rudin Chapter 10 Exercise 22.

Proof. It should be noted that the equations of x, y and z expressed in terms of u and v are actually the spherical coordinates of the point (x, y, z) on the unit sphere.

Let's "see" the spherical coordinates for the point $\Sigma(u, v)$ in Figure 10.13.

Figure 10.13: The spherical coordinates for the point $\Sigma(u, v)$.

(a) By Definition 10.18, we see that

$$d\zeta = d\left(\frac{x}{r^3}\right) dy \wedge dz + d\left(\frac{y}{r^3}\right) dz \wedge dx + d\left(\frac{z}{r^3}\right) dx \wedge dy. \quad (10.118)$$

Since

$$\begin{aligned} d\left(\frac{x}{r^3}\right) &= \frac{\partial}{\partial x}\left(\frac{x}{r^3}\right) dx + \frac{\partial}{\partial y}\left(\frac{x}{r^3}\right) dy + \frac{\partial}{\partial z}\left(\frac{x}{r^3}\right) dz \\ &= \frac{r^2 - 3x^2}{r^6} dx - \frac{3xy}{r^5} dy - \frac{3xz}{r^5} dz, \\ d\left(\frac{y}{r^3}\right) &= \frac{\partial}{\partial x}\left(\frac{y}{r^3}\right) dx + \frac{\partial}{\partial y}\left(\frac{y}{r^3}\right) dy + \frac{\partial}{\partial z}\left(\frac{y}{r^3}\right) dz \\ &= -\frac{3xy}{r^5} dx + \frac{r^2 - 3y^2}{r^6} dy - \frac{3yz}{r^5} dz, \\ d\left(\frac{z}{r^3}\right) &= \frac{\partial}{\partial x}\left(\frac{z}{r^3}\right) dx + \frac{\partial}{\partial y}\left(\frac{z}{r^3}\right) dy + \frac{\partial}{\partial z}\left(\frac{z}{r^3}\right) dz \\ &= -\frac{3xz}{r^5} dx - \frac{3yz}{r^5} dy + \frac{r^2 - 3z^2}{r^6} dz, \end{aligned}$$

we apply these, the anticommutative relation and the facts $dx \wedge dx = dy \wedge dy = dz \wedge dz = 0$ to the expression (10.118) to obtain

$$d\zeta = \frac{3r^2 - 3(x^2 + y^2 + z^2)}{r^6} \cdot dx \wedge dy \wedge dz = \frac{3r^2 - 3r^2}{r^6} \cdot dx \wedge dy \wedge dz = 0$$

in $\mathbb{R}^3 \setminus \{0\}$.

(b) We remark that the first place where the concept of the area of a 2-surface in $\mathbb{R}^3$ occurs is Sec. 10.46, not Sec. 10.43., so it is believed that this is a typo.

Suppose that $E \subseteq D$ is a compact set and $S = \Sigma_E : E \rightarrow \mathbb{R}^3$. Then S is also a 2-surface in $\mathbb{R}^3$, of class $\mathcal{C}''$. Notice that

$$\begin{aligned}\frac{\partial(y, z)}{\partial(u, v)} &= \begin{vmatrix} \cos u \sin v & \sin u \cos v \\ -\sin u & 0 \end{vmatrix} = \sin^2 u \cos v, \\ \frac{\partial(z, x)}{\partial(u, v)} &= \begin{vmatrix} -\sin u & 0 \\ \cos u \cos v & -\sin u \sin v \end{vmatrix} = \sin^2 u \sin v, \\ \frac{\partial(x, y)}{\partial(u, v)} &= \begin{vmatrix} \cos u \cos v & -\sin u \sin v \\ \cos u \sin v & \sin u \cos v \end{vmatrix} = \sin u \cos u.\end{aligned}\tag{10.119}$$

Therefore, it follows from the Jacobians (10.119) and Definition 10.11 that

$$\begin{aligned}\int_S \zeta &= \int_S \sin u \cos v \cdot \frac{\partial(y, z)}{\partial(u, v)} du \wedge dv + \int_S \sin u \sin v \cdot \frac{\partial(z, x)}{\partial(u, v)} du \wedge dv \\ &\quad + \int_S \cos u \cdot \frac{\partial(x, y)}{\partial(u, v)} du \wedge dv \\ &= \int_E \sin u \cos v \cdot \frac{\partial(y, z)}{\partial(u, v)} du dv + \int_E \sin u \sin v \cdot \frac{\partial(z, x)}{\partial(u, v)} du dv + \int_E \cos u \cdot \frac{\partial(x, y)}{\partial(u, v)} du dv \\ &= \int_E \sin^3 u \cos^2 v du dv + \int_E \sin^3 u \sin^2 v du dv + \int_E \sin u \cos^2 u du dv \\ &= \int_E \sin u du dv.\end{aligned}\tag{10.120}$$

By Definition 10.46, since

$$\mathbf{N}(u, v) = (\sin^2 u \cos v)\mathbf{e}_1 + (\sin^2 u \sin v)\mathbf{e}_2 + (\sin u \cos u)\mathbf{e}_3,$$

we have

$$A(S) = \int_S |\mathbf{N}(u, v)| du dv = \int_E \sin u du dv.\tag{10.121}$$

Hence our result follows immediately from the expressions (10.120) and (10.121).

(c) By a bit algebra, we know that

$$\begin{aligned}\frac{\partial(y, z)}{\partial(t, s)} &= \begin{vmatrix} g'(t)h_2(s) & g(t)h'_2(s) \\ g'(t)h_3(s) & g(t)h'_3(s) \end{vmatrix} = g'(t)g(t)[h_2(s)h'_3(s) - h'_2(s)h_3(s)], \\ \frac{\partial(z, x)}{\partial(t, s)} &= \begin{vmatrix} g'(t)h_3(s) & g(t)h'_3(s) \\ g'(t)h_1(s) & g(t)h'_1(s) \end{vmatrix} = g'(t)g(t)[h_3(s)h'_1(s) - h_1(s)h'_3(s)], \\ \frac{\partial(x, y)}{\partial(t, s)} &= \begin{vmatrix} g'(t)h_1(s) & g(t)h'_1(s) \\ g'(t)h_2(s) & g(t)h'_2(s) \end{vmatrix} = g'(t)g(t)[h_1(s)h'_2(s) - h'_1(s)h_2(s)].\end{aligned}\tag{10.122}$$

Therefore, it follows from the Jacobians (10.122) and Definition 10.11 that

$$\begin{aligned}\int_{\Phi} \zeta &= \int_{I^2} \frac{1}{g^3(t)[h_1^2(s) + h_2^2(s) + h_3^2(s)]^{\frac{3}{2}}} \times \left\{ g^2(t)g'(t)h_1(s)[h_2(s)h'_3(s) - h'_2(s)h_3(s)] \right. \\ &\quad + g^2(t)g'(t)h_2(s)[h_3(s)h'_1(s) - h_1(s)h'_3(s)] \\ &\quad \left. + g^2(t)g'(t)h_3(s)[h_1(s)h'_2(s) - h'_1(s)h_2(s)] \right\} dt ds \\ &= \int_{I^2} \frac{g'(t)}{g(t)[h_1^2(s) + h_2^2(s) + h_3^2(s)]^{\frac{3}{2}}} \times \left\{ h_1(s)[h_2(s)h'_3(s) - h'_2(s)h_3(s)] \right. \\ &\quad \left. + h_2(s)[h_3(s)h'_1(s) - h_1(s)h'_3(s)] + h_3(s)[h_1(s)h'_2(s) - h'_1(s)h_2(s)] \right\} dt ds\end{aligned}$$

$$\begin{aligned}
&= \int_{I^2} \frac{g'(t)}{g(t)[h_1^2(s) + h_2^2(s) + h_3^2(s)]^{\frac{3}{2}}} \times [h_1(s)h_2(s)h_3'(s) - h_1(s)h_2'(s)h_3(s) \\
&\quad + h_1'(s)h_2(s)h_3(s) - h_1(s)h_2(s)h_3'(s) + h_1(s)h_2'(s)h_3(s) - h_1'(s)h_2(s)h_3(s)] dt ds \\
&= 0.
\end{aligned}$$

(d) We follow the given hint. Since E is a closed rectangle, we have

$$E = [a, b] \times [c, d]$$

for some constants a, b, c and d with $0 < a < b < \pi$ and $0 < c < d < 2\pi$, see Figure 10.14 below:

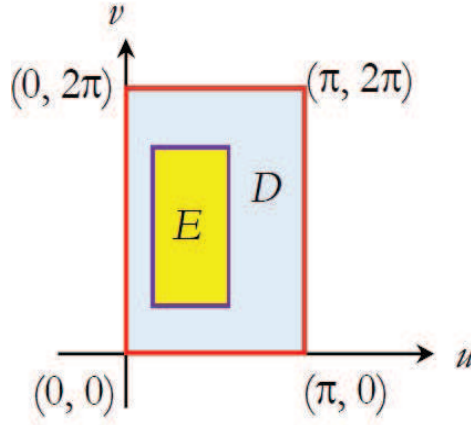


Figure 10.14: The rectangles D and E .

Consider the 3-surface $\Psi : [0, 1] \times E \rightarrow \mathbb{R}^3 \setminus \{\mathbf{0}\}$ given by

$$\Psi(t, u, v) = [1 - t + tf(u, v)]\Sigma(u, v),$$

where $(u, v) \in E, 0 \leq t \leq 1$. For fixed v , define

$$E_u = \{u \in [0, \pi] \mid (u, v) \in E\} = [a, b]$$

and the mapping $\Phi : [0, 1] \times [a, b] \rightarrow \mathbb{R}^3 \setminus \{\mathbf{0}\}$ given by

$$\Phi(t, u) = \Psi(t, u, v).$$

Now $[0, 1] \times [a, b]$ is a 2-cell, thus it is compact by Theorem 2.40 and Φ is a 2-surface of class $\mathcal{C}''$ with parameter domain $[0, 1] \times [a, b]$. Since v is fixed, we have $\Phi(t, u) = (x, y, z)$, where

$$x = g(t, u) \cos v \sin u, \quad y = g(t, u) \sin v \sin u, \quad z = g(t, u) \cos u$$

and

$$g(t, u) = [1 - t + tf(u, v)].$$

Since $0 \leq t \leq 1$ and $f(u, v) > 0$ on D , $g(t, u) > 0$ on D .

By a similar argument as in part (c), instead of the Jacobians (10.122), we have^r

$$\begin{aligned}
\frac{\partial(y, z)}{\partial(t, s)} &= \begin{vmatrix} g_t \sin v \sin u & \sin v(g \cos u + g_u \sin u) \\ g_t \cos u & g_u \cos u - g \sin u \end{vmatrix} = -gg_t \sin v, \\
\frac{\partial(z, x)}{\partial(t, s)} &= \begin{vmatrix} g_t \cos u & g_u \cos u - g \sin u \\ g_t \cos v \sin u & \cos v(g \cos u + g_u \sin u) \end{vmatrix} = gg_t \cos v, \\
\frac{\partial(x, y)}{\partial(t, s)} &= \begin{vmatrix} g_t \cos v \sin u & \cos v(g \cos u + g_u \sin u) \\ g_t \sin v \sin u & \sin v(g \cos u + g_u \sin u) \end{vmatrix} = 0.
\end{aligned} \tag{10.123}$$

^rHere we denote $g_t = \frac{\partial g(t, u)}{\partial t}$ and $g_u = \frac{\partial g(t, u)}{\partial u}$.

Thus, by the Jacobians (10.123), we have

$$\begin{aligned} \int_{\Phi} \zeta &= \int_{[0,1] \times [a,b]} g^2 g_t \sin v \cos v \sin u \, du \, dv - \int_{[0,1] \times [a,b]} g^2 g_t \sin v \cos v \sin u \, du \, dv \\ &= 0. \end{aligned}$$

We notice that the same thing holds when u is fixed.

By the definition of Ψ , we get

$$\begin{aligned} \partial\Psi &= \Psi(0, u, v) - \Psi(1, u, v) + \Psi(t, a, v) - \Psi(t, b, v) + \Psi(t, u, c) - \Psi(t, u, d) \\ &= S(u, v) - \Omega(u, v) + \Phi_a(t, v) - \Phi_b(t, v) + \Phi_c(t, u) - \Phi_d(t, u), \end{aligned} \quad (10.124)$$

where Φ_c and Φ_d are the mappings defined by

$$(t, u) \mapsto \Psi(t, u, c) \quad \text{and} \quad (t, u) \mapsto \Psi(t, u, d)$$

respectively, while Φ_a and Φ_b are the mappings defined by

$$(t, v) \mapsto \Psi(t, a, v) \quad \text{and} \quad (t, v) \mapsto \Psi(t, b, v)$$

respectively. By the above analysis, we know that

$$\int_{\Phi_a} \eta = \int_{\Phi_b} \eta = \int_{\Phi_c} \eta = \int_{\Phi_d} \eta = 0. \quad (10.125)$$

Since $f \in \mathcal{C}''(D)$, Ψ is a 3-chain of class $\mathcal{C}''$ in $\mathbb{R}^3 \setminus \{\mathbf{0}\}$. In addition, since ζ is a 2-form of class $\mathcal{C}'$ in $\mathbb{R}^3 \setminus \{\mathbf{0}\}$, Theorem 10.33 (Stokes' Theorem) implies that

$$\int_{\Psi} d\zeta = \int_{\partial\Psi} \zeta. \quad (10.126)$$

By part (a), $d\zeta = 0$ so that the integral (10.126) shows that

$$\int_{\partial\Psi} \zeta = 0. \quad (10.127)$$

Hence we follow from the relations (10.124), the integrals (10.125) and the expression (10.127) that

$$\begin{aligned} 0 &= \int_{\partial\Psi} \zeta \\ &= \int_S \zeta - \int_{\Omega} \zeta + \int_{\Phi_a} \zeta - \int_{\Phi_b} \zeta + \int_{\Phi_c} \zeta - \int_{\Phi_d} \zeta \\ &= \int_S \zeta - \int_{\Omega} \zeta \end{aligned}$$

and it is equivalent to

$$\int_{\Omega} \zeta = \int_S \zeta = A(S).$$

For a better illustration of the set $S = \Sigma_E$, by the analysis in Example 10.32, we have

$$\partial S = \partial(\Sigma_E) = \Sigma(\partial E) = \gamma_1 + \gamma_2 + \gamma_3 + \gamma_4,$$

where

$$\begin{aligned} \gamma_1(u) &= \Sigma(u, c) \\ &= (\sin u \cos c, \sin u \sin c, \cos u), \\ \gamma_2(v) &= \Sigma(b, v) \end{aligned}$$

$$\begin{aligned}
&= (\sin b \cos v, \sin b \sin v, \cos b), \\
\gamma_3(u) &= \Sigma(b + a - u, d) \\
&= (\sin(b + a - u) \cos d, \sin(b + a - u) \sin d, \cos(b + a - u)), \\
\gamma_4(v) &= \Sigma(a, c + d - v) \\
&= (\sin a \cos(c + d - v), \sin a \sin(c + d - v), \cos a),
\end{aligned}$$

with $a \leq u \leq b$ and $c \leq v \leq d$. In particular, we consider the example that $E = [\frac{\pi}{4}, \frac{\pi}{2}] \times [\frac{\pi}{2}, \pi]$. Then we have

$$\begin{aligned}
\gamma_1(u) &= (0, \sin u, \cos u), \\
\gamma_2(v) &= (\cos v, \sin v, 0), \\
\gamma_3(u) &= \left(-\sin\left(\frac{3\pi}{4} - u\right), 0, \cos\left(\frac{3\pi}{4} - u\right)\right), \\
\gamma_4(v) &= \left(\frac{\sqrt{2}}{2} \cos\left(\frac{3\pi}{2} - v\right), \frac{\sqrt{2}}{2} \sin\left(\frac{3\pi}{2} - v\right), \frac{\sqrt{2}}{2}\right),
\end{aligned}$$

where $\frac{\pi}{4} \leq u \leq \frac{\pi}{2}$ and $\frac{\pi}{2} \leq v \leq \pi$. The corresponding 2-surface S and its boundary ∂S is shown in Figure 10.15 below:

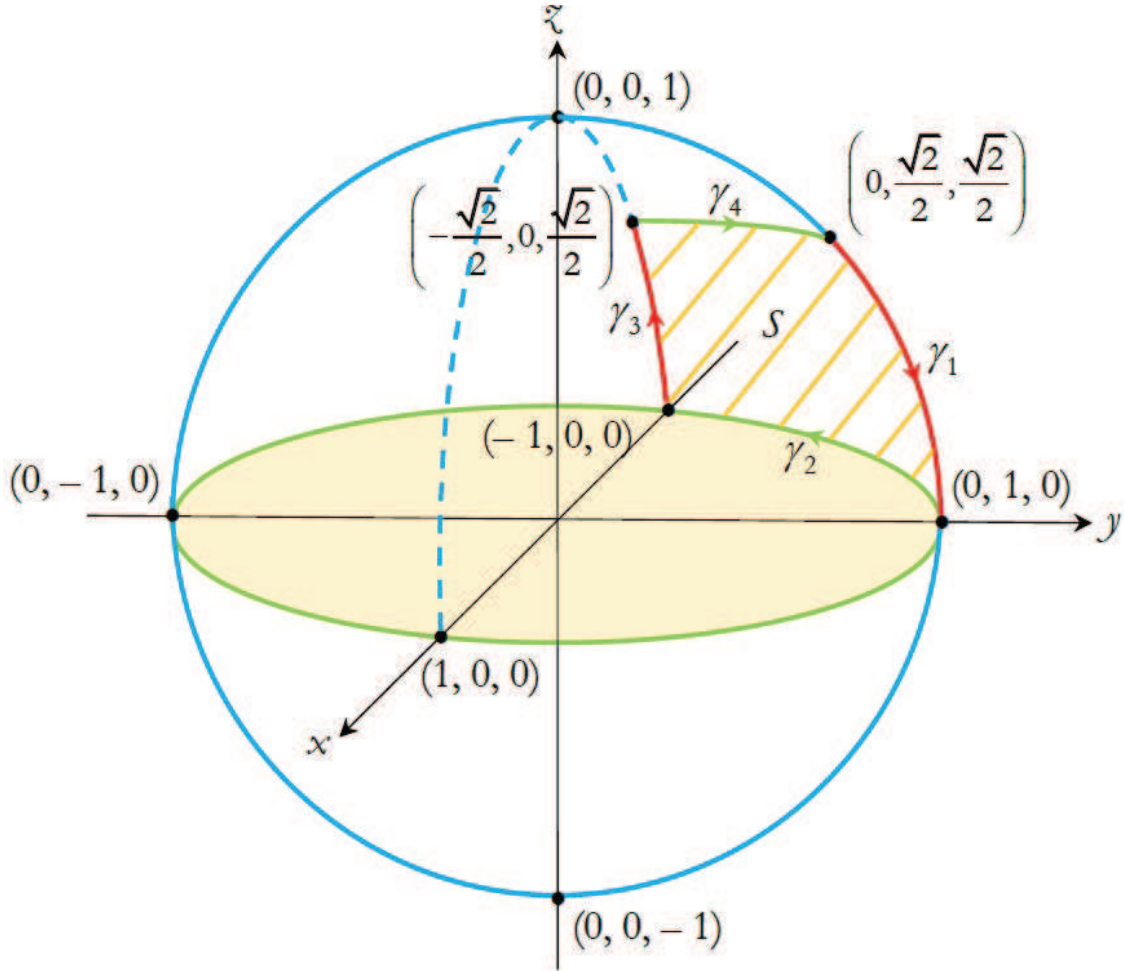


Figure 10.15: An example of the 2-surface S and its boundary ∂S .

(e) Let

$$V = \{(x, y, z) \mid x^2 + y^2 > 0, z \in \mathbb{R}\} = \mathbb{R}^3 \setminus \{(0, 0, z) \mid z \in \mathbb{R}\}$$

which is an open set in $\mathbb{R}^3$. In other words, V is the 3-dimensional space with deleted z -axis. Now

$$-\frac{z}{r} \quad \text{and} \quad \eta$$

are 0- and 1-forms respectively, we follow from Theorem 10.20(a), Definition 10.18 and Problem 10.21(a) that

$$\begin{aligned} d\lambda &= -\left[d\left(\frac{z}{r}\right)\right] \wedge \eta + (-1)^0 \frac{z}{r} \wedge d\eta \\ &= \left(\frac{xz}{r^3} dx + \frac{yz}{r^3} dy - \frac{r^2 - z^2}{r^3} dz\right) \wedge \left(\frac{x dy - y dx}{x^2 + y^2}\right) \\ &= \frac{z}{r^3} dx \wedge dy - \frac{x^2 + y^2}{r^3} dz \wedge \left(\frac{x dy - y dx}{x^2 + y^2}\right) \\ &= \frac{z}{r^3} dx \wedge dy - \frac{x}{r^3} dz \wedge dy + \frac{y}{r^3} dz \wedge dx \\ &= \frac{x dy \wedge dz + y dz \wedge dx + z dx \wedge dy}{r^3} \\ &= \zeta. \end{aligned}$$

Hence ζ is exact in the open set V .

(f) Note that

$$\Omega(u, v) = f(u, v)\Sigma(u, v) = (f(u, v) \sin u \cos v, f(u, v) \sin u \sin v, f(u, v) \cos u),$$

where $(u, v) \in E$. By the hypothesis $E = [a, b] \times [c, d]$ made in part (b), if we further assume that $0 < a < b < \pi$, then it is easy to see that

$$[f(u, v) \sin u \cos v]^2 + [f(u, v) \sin u \sin v]^2 = f^2(u, v) \sin^2 u > 0$$

which means

$$\Omega \subseteq V. \quad (10.128)$$

Recall that $f \in \mathcal{C}''(D)$ and Σ is a 2-surface of class $\mathcal{C}''$ in $\mathbb{R}^3 \setminus \{\mathbf{0}\}$, we have Ω is a 2-surface of class $\mathcal{C}''$ in $\mathbb{R}^3 \setminus \{\mathbf{0}\}$ too. Since λ is clearly a 1-form of class $\mathcal{C}'$ in V ,

$$\zeta = d\lambda$$

in V by part (d) and the subset relation (10.128), the exactness of ζ still hold in Ω . Hence it follows from Theorem 10.33 (Stokes' Theorem) that

$$\int_{\Omega} \zeta = \int_{\Omega} d\lambda = \int_{\partial\Omega} \lambda = \int_{\partial\Omega} \frac{z}{r} \eta. \quad (10.129)$$

Similarly, the definition of S implies that $S \subseteq V$. Since S is a 2-surface of class $\mathcal{C}''$ in $\mathbb{R}^3 \setminus \{\mathbf{0}\}$, Theorem 10.33 (Stokes' Theorem) again implies that

$$\int_S \zeta = \int_S d\lambda = \int_{\partial S} \lambda = \int_{\partial S} \frac{z}{r} \eta. \quad (10.130)$$

Now we want to show that the two right-most integrals in the relations (10.129) and (10.130) are equal. Such a proof is presented in two steps below.

– **Step 1: Analysis of $\frac{z}{r}$ on $\partial\Omega$ and ∂S .** We notice that if $(x, y, z) \in \Sigma_E(u, v)$, then

$$x = \sin u \cos v, \quad y = \sin u \sin v, \quad z = \cos u \quad (10.131)$$

so that

$$\frac{z}{r} = \frac{\cos u}{1} = \cos u.$$

Similarly, if $(x, y, z) \in \Omega(u, v)$, then

$$x = f(u, v) \sin u \cos v, \quad y = f(u, v) \sin u \sin v, \quad z = f(u, v) \cos u,$$

where $f(u, v) > 0$ so that

$$\frac{z}{r} = \frac{f(u, v) \cos u}{f(u, v)} = \cos u.$$

In other words, $\frac{z}{r}$ is the same at $\Sigma_E(u, v)$ as at $\Omega(u, v)$. By part (b), we know that

$$S = \Sigma_E$$

and thus $\frac{z}{r}$ is the same at ∂S as at $\partial \Omega$.

- **Step 2: Analysis of ζ on $\partial \Omega$ and ∂S .** By Figure 10.14, we know that E does not intersect $u = 0$, therefore we have

$$\eta = d\left(\arctan \frac{y}{x}\right)$$

by Problem 10.21(d).^s On ∂S , we have

$$\arctan \frac{y}{x} = \arctan \left(\frac{\sin u \sin v}{\sin u \cos v} \right) = \arctan(\tan v) = v.$$

Similarly, on $\partial \Omega$, we have

$$\arctan \frac{y}{x} = \arctan \left(\frac{f(u, v) \sin u \sin v}{f(u, v) \sin u \cos v} \right) = \arctan(\tan v) = v.$$

Thus the 1-form η is the same at ∂S as at $\partial \Omega$ too.

Hence we deduce from the above analysis and part (b) that

$$\int_{\Omega} \zeta = \int_{\partial \Omega} \lambda = \int_{\partial \Omega} \frac{z}{r} \eta = \int_{\partial S} \frac{z}{r} \eta = \int_{\partial S} \lambda = \int_S \zeta = A(S)$$

which is our expected result.

- (g) The answer is affirmative. Let L be a straight line through the origin. Recall that the spherical coordinates for a unit vector $\mathbf{x} = (x, y, z)$ are given by the formulas (10.131). It is well-known^t that the matrices

$$\begin{aligned} \mathbf{R}_{xy}(u) &= \begin{pmatrix} \cos u & -\sin u & 0 \\ \sin u & \cos u & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ \mathbf{R}_{zx}(v) &= \begin{pmatrix} \cos v & 0 & \sin v \\ 0 & 1 & 0 \\ -\sin v & 0 & \cos v \end{pmatrix}, \\ \mathbf{R}_{yz}(w) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos w & -\sin w \\ 0 & \sin w & \cos w \end{pmatrix} \end{aligned}$$

represent rotations by angle u, v and w counterclockwise in the (x, y) -plane, the (z, x) -plane and the (y, z) -plane respectively.

^sNote that E does not intersect $v = 0$, so the formula $\eta = d(-\arctan \frac{x}{y})$ is also applicable to get the same result.

^tSee, for instance, [16, pp. 328 - 332].

Lemma 10.11

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the transformation with matrix $\mathbf{M}$ defined by

$$\mathbf{M} = \mathbf{R}_{zx}(-u)\mathbf{R}_{xy}(-v).$$

Then T transforms the straight line L onto the z -axis.

Proof of Lemma 10.11. We have

$$\begin{aligned} \mathbf{M} &= \mathbf{R}_{zx}(-u)\mathbf{R}_{xy}(-v) \\ &= \begin{pmatrix} \cos u & 0 & -\sin u \\ 0 & 1 & 0 \\ \sin u & 0 & \cos u \end{pmatrix} \begin{pmatrix} \cos v & \sin v & 0 \\ -\sin v & \cos v & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \cos u \cos v & \cos u \sin v & -\sin u \\ -\sin v & \cos v & 0 \\ \sin u \cos v & \sin u \sin v & \cos u \end{pmatrix}, \end{aligned} \quad (10.132)$$

where $0 \leq u \leq \pi$ and $0 \leq v \leq 2\pi$. Then direct computation shows that

$$\begin{aligned} T(\mathbf{x}) &= \mathbf{M}\mathbf{x} \\ &= \begin{pmatrix} \cos u \cos v & \cos u \sin v & -\sin u \\ -\sin v & \cos v & 0 \\ \sin u \cos v & \sin u \sin v & \cos u \end{pmatrix} \begin{pmatrix} \sin u \cos v \\ \sin u \sin v \\ \cos u \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \end{aligned}$$

In other words, the mapping T transforms L onto the z -axis, completing the proof of the lemma. ■

Let $E = \mathbb{R}^3 \setminus L$. Since T is bijective^u, we consider the mapping

$$T_E = T : E \rightarrow V$$

which is also bijective. Since λ is of class $\mathcal{C}'$ in V and the formula (10.132) of $\mathbf{M}$ implies that T and then T_E is of class $\mathcal{C}''$, we deduce from Theorem 10.22(c) and part (e) that

$$d(\lambda_{T_E}) = (d\lambda)_{T_E} = \zeta_{T_E}.$$

Since ζ is a 2-form in V , ζ_{T_E} is a 2-form in E . We claim that

$$\zeta = \zeta_{T_E}.$$

To this end, we get from the formula (10.132) that

$$\begin{aligned} T(\mathbf{x}) &= \begin{pmatrix} \cos u \cos v & \cos u \sin v & -\sin u \\ -\sin v & \cos v & 0 \\ \sin u \cos v & \sin u \sin v & \cos u \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \\ &= \begin{pmatrix} (\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z \\ (-\sin v)x + (\cos v)y \\ (\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z \end{pmatrix}. \end{aligned}$$

Following the notations used in Definition 10.21, we have

$$t_1(\mathbf{x}) = (\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z,$$

^uThis can be easily checked by the matrix form (10.132).

$$\begin{aligned} t_2(\mathbf{x}) &= (-\sin v)x + (\cos v)y, \\ t_3(\mathbf{x}) &= (\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z \end{aligned}$$

which imply that

$$\begin{aligned} dt_1 &= (\cos u \cos v) dx + (\cos u \sin v) dy - \sin u dz, \\ dt_2 &= -\sin u dx + \cos v dy, \\ dt_3 &= (\sin u \cos v) dx + (\sin u \sin v) dy + \cos u dz. \end{aligned} \tag{10.133}$$

Thus we obtain from [21, Eqn. (67), p.262] that

$$\begin{aligned} \zeta_{T_E} &= [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z] dt_2 \wedge dt_3 \\ &\quad + [(-\sin u)x + (\cos v)y] dt_3 \wedge dt_1 \\ &\quad + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z] dt_1 \wedge dt_2. \end{aligned} \tag{10.134}$$

We need to compute

$$dt_2 \wedge dt_3, \quad dt_3 \wedge dt_1 \quad \text{and} \quad dt_1 \wedge dt_2.$$

To do this, we know from the formulas (10.133) that

$$\begin{aligned} dt_2 \wedge dt_3 &= [-\sin v dx + \cos v dy] \wedge [(\sin u \cos v) dx + (\sin u \sin v) dy + \cos u dz] \\ &= -(\sin u \sin^2 v + \sin u \cos^2 v) dx \wedge dy - (\sin v \cos u) dx \wedge dz \\ &\quad + (\cos u \cos v) dy \wedge dz \\ &= (-\sin u) dx \wedge dy - (\sin v \cos u) dx \wedge dz + (\cos u \cos v) dy \wedge dz, \end{aligned} \tag{10.135}$$

$$\begin{aligned} dt_3 \wedge dt_1 &= [(\sin u \cos v) dx + (\sin u \sin v) dy + \cos u dz] \\ &\quad \wedge [(\cos u \cos v) dx + (\cos u \sin v) dy - \sin u dz] \\ &= (\sin u \cos u \sin^2 v) dx \wedge dy + (\sin u^2 \cos v) dz \wedge dx \\ &\quad - (\sin u \cos u \sin^2 v) dx \wedge dy - (\sin^2 u \sin v) dy \wedge dz \\ &\quad + (\cos^2 u \cos v) dz \wedge dx - (\cos^2 u \sin v) dy \wedge dz \\ &= (\cos v) dz \wedge dx - (\sin v) dy \wedge dz \end{aligned} \tag{10.136}$$

and

$$\begin{aligned} dt_1 \wedge dt_2 &= [(\cos u \cos v) dx + (\cos u \sin v) dy - \sin u dz] \wedge [-\sin v dx + \cos v dy] \\ &= (\cos u \cos^2 v + \cos u \sin^2 v) dx \wedge dy + (\sin u \sin v) dz \wedge dx \\ &\quad + (\sin u \cos v) dy \wedge dz \\ &= (\cos u) dx \wedge dy + (\sin u \sin v) dz \wedge dx + (\sin u \cos v) dy \wedge dz. \end{aligned} \tag{10.137}$$

Now we show that r is *invariant* under the rotation $\mathbf{R}_{zx}(-u)\mathbf{R}_{xy}(-v)$:

Lemma 10.12

If we denote $r_{T_E} = (t_1^2 + t_2^2 + t_3^2)^{\frac{1}{2}}$, then we have

$$r_{T_E} = r.$$

Proof of Lemma 10.12. By direct computation, we have

$$\begin{aligned}
 r_{T_E}^2 &= t_1^2 + t_2^2 + t_3^2 \\
 &= [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z]^2 + [(-\sin v)x + (\cos v)y]^2 \\
 &\quad + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z]^2 \\
 &= (\cos^2 u \cos^2 v)x^2 + 2(\cos u \cos v)[(\cos u \sin v)y + (-\sin u)z]x + [(\cos u \sin v)y + (-\sin u)z]^2 \\
 &\quad + (\sin^2 v)x^2 - (2 \sin v \cos v)xy + (\cos^2 v)y^2 + (\sin^2 u \cos^2 v)x^2 \\
 &\quad + (2 \sin u \cos v)[(\sin u \sin v)y + (\cos u)z]x + [(\sin u \sin v)y + (\cos u)z]^2 \\
 &= x^2 + 2(\cos u \cos v)[(\cos u \sin v)y + (-\sin u)z]x + [(\cos u \sin v)^2 y^2 - (2 \cos u \sin u \sin v)yz \\
 &\quad + (-\sin u)^2 z^2] - (2 \sin v \cos v)xy + (\cos^2 v)y^2 + (\sin^2 u \cos^2 v)x^2 \\
 &\quad + (2 \sin u \cos v)[(\sin u \sin v)y + (\cos u)z]x + [(\sin u \sin v)^2 y^2 \\
 &\quad + (2 \sin u \cos u \sin v)yz + (\cos u)^2 z^2] \\
 &= x^2 + y^2 + z^2 + (2 \cos^2 u \sin v \cos v - 2 \sin v \cos v + 2 \sin^2 u \cos v \sin v)xy \\
 &\quad (-2 \cos u \sin u \sin v + 2 \cos u \sin u \sin v)yz + (-2 \sin u \cos u \cos v + 2 \sin u \cos u \cos v)zx \\
 &= x^2 + y^2 + z^2 \\
 &= r^2
 \end{aligned}$$

which certainly gives

$$r_{T_E} = r$$

and this completes the proof of the lemma. ■

Now we return to the proof of the problem. After putting the identities (10.135), (10.136) and (10.137) into the 2-form (10.134) and using Lemma 10.11, we have

$$\begin{aligned}
 r_{T_E}^3 \zeta_{T_E} &= [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z] \times [-(\sin u) dx \wedge dy + (\sin v \cos u) dz \wedge dx \\
 &\quad + (\cos u \cos v) dy \wedge dz] \\
 &\quad + [(-\sin v)x + (\cos v)y] \times [(\cos v) dz \wedge dx + (-\sin v) dy \wedge dz] \\
 &\quad + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z] \times [(\cos u) dx \wedge dy + (\sin u \sin v) dz \wedge dx \\
 &\quad + (\sin u \cos v) dy \wedge dz] \\
 &= \left\{ [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z] \times (-\sin u) \right. \\
 &\quad \left. + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z] \times (\cos u) \right\} dx \wedge dy \\
 &\quad + \left\{ [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z] \times (\cos u \cos v) \right. \\
 &\quad \left. + [(-\sin v)x + (\cos v)y] \times (-\sin v) \right. \\
 &\quad \left. + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z] \times (\sin u \cos v) \right\} dy \wedge dz \\
 &\quad + \left\{ [(\cos u \cos v)x + (\cos u \sin v)y + (-\sin u)z] \times (\sin v \cos u) \right. \\
 &\quad \left. + [(-\sin v)x + (\cos v)y] \times (\cos v) \right. \\
 &\quad \left. + [(\sin u \cos v)x + (\sin u \sin v)y + (\cos u)z] \times (\sin u \sin v) \right\} dz \wedge dx \\
 &= z dx \wedge dy + x dy \wedge dz + y dz \wedge dx \\
 &= r^3 \zeta
 \end{aligned}$$

which implies that

$$\zeta_{T_E} = \zeta. \quad (10.138)$$

As we have shown that $d(\lambda_{T_E}) = \zeta_{T_E}$ holds in $E = \mathbb{R}^3 \setminus L$, this and the identity (10.138) imply that

$$\zeta = d(\lambda_{T_E}),$$

i.e., ζ is exact E .

This completes the proof of the problem. ■

Problem 10.23

Rudin Chapter 10 Exercise 23.

Proof.

(a) By applying Theorem 10.20 repeatedly, we have

$$\begin{aligned}
 & d[(-1)^{i-1}(r_k)^{-k}x_i dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k] \\
 &= d[(-1)^{i-1}(r_k)^{-k}x_i] \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k \\
 &\quad + (-1)^0(-1)^{i-1}(r_k)^{-k}x_i \wedge d(dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k) \\
 &= d[(-1)^{i-1}(r_k)^{-k}x_i] \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k \\
 &\quad + (-1)^{i-1}(r_k)^{-k}x_i \underbrace{[(d^2x_1) \wedge dx_2 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k]}_{\text{It is 0.}} \\
 &\quad + (-1)^1 dx_1 \wedge d(dx_2 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k) \\
 &= d[(-1)^{i-1}(r_k)^{-k}x_i] \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k \\
 &\quad + (-1)^i(r_k)^{-k}x_i dx_1 \wedge d(dx_2 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k) \\
 &= \cdots \\
 &= d[(-1)^{i-1}(r_k)^{-k}x_i] \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k. \tag{10.139}
 \end{aligned}$$

By [21, Eqn. (59), p.260], we have

$$d[(-1)^{i-1}(r_k)^{-k}x_i] = (-1)^{i-1} \sum_{j=1}^k \frac{\partial}{\partial x_j} \left(\frac{x_i}{r_k^k} \right) dx_j. \tag{10.140}$$

It is clear that

$$\frac{\partial}{\partial x_j} \left(\frac{x_i}{r_k^k} \right) = \begin{cases} \frac{-x_i \frac{\partial r_k^k}{\partial x_j}}{r_k^{2k}} = \frac{-x_i k r_k^{k-1} \frac{\partial r_k}{\partial x_j}}{r_k^{2k}} = \frac{-x_i k r_k^{k-1} \frac{x_j}{r_k}}{r_k^{2k}} = \frac{-k x_i x_j}{r_k^{k+2}}, & \text{if } j \neq i; \\ \frac{r_k^k - x_i \frac{\partial r_k^k}{\partial x_i}}{r_k^{2k}} = \frac{r_k^k - k x_i^2 r_k^{k-2}}{r_k^{2k}} = \frac{r_k^2 - k x_i^2}{r_k^{k+2}}, & \text{if } j = i. \end{cases}$$

Therefore, it follows from the summation (10.140) that

$$d[(-1)^{i-1}(r_k)^{-k}x_i] = (-1)^{i-1} \left(\frac{r_k^2 - k x_i^2}{r_k^{k+2}} dx_i + \sum_{\substack{j=1 \\ j \neq i}}^k \frac{-k x_i x_j}{r_k^{k+2}} dx_j \right)$$

and then the expression (10.139) with an application of the anticommutative relation shows that

$$\begin{aligned}
 & d[(-1)^{i-1}(r_k)^{-k}x_i dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k] \\
 &= (-1)^{i-1} \left(\frac{r_k^2 - k x_i^2}{r_k^{k+2}} dx_i + \sum_{\substack{j=1 \\ j \neq i}}^k \frac{-k x_i x_j}{r_k^{k+2}} dx_j \right) \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k \\
 &= (-1)^{i-1} \frac{r_k^2 - k x_i^2}{r_k^{k+2}} dx_i \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k
 \end{aligned}$$

$$\begin{aligned}
&= (-1)^{2i-2} \frac{r_k^2 - kx_i^2}{r_k^{k+2}} dx_1 \wedge \cdots \wedge dx_k \\
&= \frac{r_k^2 - kx_i^2}{r_k^{k+2}} dx_1 \wedge \cdots \wedge dx_k
\end{aligned} \tag{10.141}$$

Hence we deduce from the definition of ω_k and the relation (10.141) that

$$\begin{aligned}
d\omega_k &= \sum_{i=1}^k d[(-1)^{i-1} (r_k)^{-k} x_i dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_k] \\
&= \sum_{i=1}^k \frac{r_k^2 - kx_i^2}{r_k^{k+2}} dx_1 \wedge \cdots \wedge dx_k \\
&= \frac{kr_k^2 - k(x_1^2 + \cdots + x_k^2)}{r_k^{k+2}} dx_1 \wedge \cdots \wedge dx_k \\
&= 0
\end{aligned}$$

in E_k .

(b) We have $f_k : E_k \rightarrow \mathbb{R}$. Note that the gradient of f_k at $\mathbf{x}$ is given by

$$(\nabla f_k)(\mathbf{x}) = \sum_{i=1}^k (D_i f_k)(\mathbf{x}) \mathbf{e}_i. \tag{10.142}$$

– **Step 1:** f_k satisfies the equations given in the hint. By Theorem 6.20 (First Fundamental Theorem of Calculus), if $i \neq k$, then we have

$$\begin{aligned}
(D_i f_k)(\mathbf{x}) &= (-1)^k \frac{\partial}{\partial x_i} \left(\int_{-1}^{\frac{x_k}{r_k}} (1 - s^2)^{\frac{k-3}{2}} ds \right) \\
&= (-1)^k \frac{\partial(\frac{x_k}{r_k})}{\partial x_i} \left(1 - \frac{x_k^2}{r_k^2} \right)^{\frac{k-3}{2}} \\
&= (-1)^k \cdot \frac{-x_i x_k}{r_k^3} \left(\frac{x_1^2 + \cdots + x_{k-1}^2}{r_k^2} \right)^{\frac{k-3}{2}} \\
&= (-1)^{k+1} \frac{x_i x_k r_{k-1}^{k-3}}{r_k^k}.
\end{aligned} \tag{10.143}$$

Similarly, if $i = k$, then we have

$$\begin{aligned}
(D_k f_k)(\mathbf{x}) &= (-1)^k \frac{r_k - x_k \frac{\partial r_k}{\partial x_k}}{r_k^2} \left(1 - \frac{x_k^2}{r_k^2} \right)^{\frac{k-3}{2}} \\
&= (-1)^k \frac{r_k^2 - x_k^2}{r_k^3} \cdot \frac{(r_k^2 - x_k^2)^{\frac{k-3}{2}}}{r_k^{k-3}} \\
&= (-1)^k \frac{r_{k-1}^{k-1}}{r_k^k}.
\end{aligned} \tag{10.144}$$

By substituting the relations (10.143) and (10.144) into the definition (10.142) and then consider its dot product with $\mathbf{x}$, we get

$$\begin{aligned}
\mathbf{x} \cdot (\nabla f_k)(\mathbf{x}) &= \sum_{i=1}^k x_i (D_i f_k)(\mathbf{x}) \\
&= \sum_{i=1}^{k-1} (-1)^{k+1} \frac{x_i^2 x_k r_{k-1}^{k-3}}{r_k^k} + (-1)^k \frac{x_k r_{k-1}^{k-1}}{r_k^k}
\end{aligned}$$

$$\begin{aligned}
&= (-1)^{k+1} \frac{x_k r_{k-1}^{k-1}}{r_k^k} + (-1)^k \frac{x_k r_{k-1}^{k-1}}{r_k^k} \\
&= 0.
\end{aligned} \tag{10.145}$$

– **Step 2:** $\omega_k = d(f_k \omega_{k-1})$ for $k = 2, \dots, n$. We notice from Theorem 10.20, part (a) and Definition 10.17 that

$$\begin{aligned}
d(f_k \omega_{k-1}) &= (df_k) \wedge \omega_{k-1} + (-1)^0 f_k \wedge d\omega_{k-1} \\
&= (df_k) \wedge \omega_{k-1} \\
&= \frac{1}{r_{k-1}^{k-1}} \left[\sum_{i=1}^k (D_i f_k)(\mathbf{x}) dx_i \right] \\
&\quad \wedge \left[\sum_{i=1}^{k-1} (-1)^{i-1} x_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_{k-1} \right] \\
&= \frac{1}{r_{k-1}^{k-1}} \left[(-1)^{1-1} x_1 (D_1 f_k)(\mathbf{x}) \underbrace{dx_1 \wedge dx_2 \wedge \dots \wedge dx_{k-1}}_{\text{There are } (k-2) \text{ terms.}} \right. \\
&\quad + (-1)^{2-1} x_2 (D_2 f_k)(\mathbf{x}) \underbrace{dx_2 \wedge dx_1 \wedge dx_3 \wedge \dots \wedge dx_{k-1}}_{\text{There are } (k-2) \text{ terms.}} \\
&\quad + \dots + (-1)^{k-2} x_{k-1} (D_{k-1} f_k)(\mathbf{x}) \underbrace{dx_{k-1} \wedge dx_1 \wedge \dots \wedge dx_{k-2}}_{\text{There are } (k-2) \text{ terms.}} \left. \right] \\
&\quad + \left\{ \frac{1}{r_{k-1}^{k-1}} (D_k f_k)(\mathbf{x}) dx_k \right. \\
&\quad \left. \wedge \left[\sum_{i=1}^{k-1} (-1)^{i-1} x_i \underbrace{dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_{k-1}}_{\text{There are } (k-2) \text{ terms.}} \right] \right\}. \tag{10.146}
\end{aligned}$$

By applying the anticommutative relation $(k-2)$ -times to the **red** brackets in the expression (10.146) and then the formula (10.144), it becomes

$$\begin{aligned}
&(-1)^{2k-2} \frac{1}{r_{k-1}^{k-1}} \cdot \frac{r_{k-1}^{k-1}}{r_k^k} \sum_{i=1}^{k-1} (-1)^{i-1} x_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_k \\
&= \frac{1}{r_k^k} \sum_{i=1}^{k-1} (-1)^{i-1} x_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_k. \tag{10.147}
\end{aligned}$$

Now it remains to simplify the **blue** brackets in the expression (10.146). In fact, by the equation (10.145), the **blue** brackets are equivalent to

$$-\frac{x_k}{r_{k-1}^{k-1}} (D_k f_k)(\mathbf{x}) dx_1 \wedge \dots \wedge dx_{k-1} = (-1)^{k+1} \frac{x_k}{r_k^k} dx_1 \wedge \dots \wedge dx_{k-1}. \tag{10.148}$$

Hence we follow from substituting the expressions (10.147) and (10.148) back into the expression (10.146) that

$$\begin{aligned}
d(f_k \omega_{k-1}) &= (-1)^{k+1} \frac{x_k}{r_k^k} dx_1 \wedge \dots \wedge dx_{k-1} \\
&\quad + \frac{1}{r_k^k} \sum_{i=1}^{k-1} (-1)^{i-1} x_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_k \\
&= \frac{1}{r_k^k} \sum_{i=1}^k (-1)^{i-1} x_i dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_k
\end{aligned}$$

$$= \omega_k$$

which is our desired property.

- (c) We know from Problems 10.22 and 10.23 that $\omega_2 = \eta$ and $\omega_3 = \zeta$. By Examples 10.36 and 10.37, we know that

$$\int_{\gamma} \eta = 2\pi \neq 0 \quad \text{and} \quad \int_{\Sigma} \zeta = 4\pi \neq 0,$$

where γ and Σ are parametrizations of the unit circle and the unit sphere in $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Furthermore, we conclude from the discussion parts of Examples 10.36 and 10.37 and the facts $\partial\gamma = \partial\Sigma = 0$ that

$$\eta \quad \text{and} \quad \zeta$$

are *not* exact in $\mathbb{R}^2 \setminus \{0\}$ and $\mathbb{R}^3 \setminus \{0\}$ respectively. Thus it is reasonable to say that the answer in this part is *negative*.

Let $n \geq 2$. We basically follow a part of the argument as in [9, §6.1]. Consider the $(n-1)$ -sphere S^{n-1} of radius 1 in $\mathbb{R}^n$ defined by

$$S^{n-1} = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| = 1\}.$$

Let $\omega = r_n^n \omega_n$. Then we have $\omega = \omega_n$ on S^{n-1} so that

$$\omega(S^{n-1}) = \omega_n(S^{n-1})$$

by Definition 10.11, i.e.,

$$\int_{S^{n-1}} \omega_n = \int_{S^{n-1}} \omega.$$

Since ω_n is, by definition, a $(n-1)$ -form defined in E_n , ω is also a $(n-1)$ -form in E_n . We know that

$$S^{n-1} = \partial D^n,$$

where $D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \|\mathbf{x}\| \leq 1\}$ is the closed unit ball in $\mathbb{R}^n$ (see, for example, [11, p. 253]). Since S^{n-1} can be treated as an k -simplex in $\mathbb{R}^n$ of class $\mathcal{C}''$ in E_n , we apply Theorem 10.33 (Stokes' Theorem) to S^{n-1} and ω to get

$$\int_{S^{n-1}} \omega_n = \int_{S^{n-1}} \omega = \int_{\partial D^n} \omega = \int_{D^n} d\omega. \quad (10.149)$$

Lemma 10.13

We have

$$d\omega = n dx_1 \wedge \cdots \wedge dx_n.$$

Proof of Lemma 10.13. By Definition 10.18, we have

$$\begin{aligned} d\omega &= d\left(\sum_{i=1}^n (-1)^{i-1} x_i dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_n\right) \\ &= \sum_{i=1}^n (-1)^{i-1} d\mathbf{x}_i \wedge dx_1 \wedge \cdots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \cdots \wedge dx_n \\ &= \sum_{i=1}^n (-1)^{i-1} (-1)^{i-1} dx_1 \wedge \cdots \wedge dx_{i-1} \wedge d\mathbf{x}_i \wedge dx_{i+1} \wedge \cdots \wedge dx_n \\ &= n dx_1 \wedge \cdots \wedge dx_n \end{aligned}$$

which is the required result. ■

Now we apply Lemma 10.13 to the relation (10.149) and then using Definition 10.44 (Volume elements), we obtain

$$\int_{S^{n-1}} \omega = n \int_{D^n} dx_1 \wedge \cdots \wedge dx_n = \text{Vol}(D^n) = \frac{\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2} + 1)} \neq 0, \quad (10.150)$$

where Γ is the Gamma function.

For the second assertion, assume that λ was a $(n-2)$ -form defined in $E_n \subseteq \mathbb{R}^n$ such that

$$\omega_n = d\lambda. \quad (10.151)$$

By Problem 10.16 or Remarks 10.35(c), we know that

$$\partial S^{n-1} = \partial^2 D^n = 0.$$

Therefore, we follow from this fact, Theorem 10.33 (Stokes' Theorem) and the exactness (10.151) that

$$\int_{S^{n-1}} \omega_n = \int_{S^{n-1}} d\lambda = \int_{\partial S^{n-1}} \lambda = \int_{\partial^2 D^n} \lambda = 0$$

which contradicts the result (10.150). Hence ω_n is not exact in E_n .[▼]

(d) We have the following generalizations:

- **A generalization of Problem 10.22(c).** Suppose that $g \in \mathcal{C}''([0, 1])$, $h_i \in \mathcal{C}''([0, 1]^{n-2})$ and $g > 0$. Suppose further that

$$\Phi(s_1, \dots, s_{n-2}, t) = (x_1, \dots, x_n)$$

is a $(n-1)$ -surface with parameter domain I^{n-1} given by

$$x_i = g(t)h_i(s_1, \dots, s_{n-2}) \quad (1 \leq i \leq n). \quad (10.152)$$

Then we have

$$\int_{\Phi} \omega_n = 0. \quad (10.153)$$

To prove the formula (10.153), we need to compute the Jacobians

$$\frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)}{\partial(s_1, \dots, s_{n-2}, t)} = \det \begin{pmatrix} \frac{\partial x_1}{\partial s_1} & \cdots & \frac{\partial x_1}{\partial s_{n-2}} & \frac{\partial x_1}{\partial t} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial x_{i-1}}{\partial s_1} & \cdots & \frac{\partial x_{i-1}}{\partial s_{n-2}} & \frac{\partial x_{i-1}}{\partial t} \\ \frac{\partial x_{i+1}}{\partial s_1} & \cdots & \frac{\partial x_{i+1}}{\partial s_{n-2}} & \frac{\partial x_{i+1}}{\partial t} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial x_n}{\partial s_1} & \cdots & \frac{\partial x_n}{\partial s_{n-2}} & \frac{\partial x_n}{\partial t} \end{pmatrix}.$$

[▼]One can prove the same result by generalizing the argument used in part (b), but the steps are very cumbersome.

By the definition (10.152), we have

$$\begin{aligned} \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)}{\partial(s_1, \dots, s_{n-2}, t)} &= \det \begin{pmatrix} g(t) \frac{\partial h_1}{\partial s_1} & \cdots & g(t) \frac{\partial h_1}{\partial s_{n-2}} & g'(t) h_1 \\ \vdots & \ddots & \vdots & \vdots \\ g(t) \frac{\partial h_{i-1}}{\partial s_1} & \cdots & g(t) \frac{\partial h_{i-1}}{\partial s_{n-2}} & g'(t) h_{i-1} \\ g(t) \frac{\partial h_{i+1}}{\partial s_1} & \cdots & g(t) \frac{\partial h_{i+1}}{\partial s_{n-2}} & g'(t) h_{i+1} \\ \vdots & \ddots & \vdots & \vdots \\ g(t) \frac{\partial h_n}{\partial s_1} & \cdots & g(t) \frac{\partial h_n}{\partial s_{n-2}} & g'(t) h_n \end{pmatrix} \\ &= g^{n-2}(t) g'(t) \det \begin{pmatrix} \frac{\partial h_1}{\partial s_1} & \cdots & \frac{\partial h_1}{\partial s_{n-2}} & h_1 \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial h_{i-1}}{\partial s_1} & \cdots & \frac{\partial h_{i-1}}{\partial s_{n-2}} & h_{i-1} \\ \frac{\partial h_{i+1}}{\partial s_1} & \cdots & \frac{\partial h_{i+1}}{\partial s_{n-2}} & h_{i+1} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial h_n}{\partial s_1} & \cdots & \frac{\partial h_n}{\partial s_{n-2}} & h_n \end{pmatrix} \end{aligned}$$

so that

$$\begin{aligned} \int_{\Phi} \omega_n &= \int_{I^{n-1}} \frac{g'(t)}{g(t) \left(\sum_{j=1}^n h_j^2 \right)^{\frac{n}{2}}} \sum_{i=1}^n (-1)^{i-1} h_i \\ &\quad \times \det \begin{pmatrix} \frac{\partial h_1}{\partial s_1} & \cdots & \frac{\partial h_1}{\partial s_{n-2}} & h_1 \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial h_{i-1}}{\partial s_1} & \cdots & \frac{\partial h_{i-1}}{\partial s_{n-2}} & h_{i-1} \\ \frac{\partial h_{i+1}}{\partial s_1} & \cdots & \frac{\partial h_{i+1}}{\partial s_{n-2}} & h_{i+1} \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial h_n}{\partial s_1} & \cdots & \frac{\partial h_n}{\partial s_{n-2}} & h_n \end{pmatrix} ds_1 \cdots ds_{n-2} dt. \end{aligned}$$

Since the summation in the above integral is just the expansion of the following determinant along the first column

$$\det \begin{pmatrix} h_1 & \frac{\partial h_1}{\partial s_1} & \cdots & \frac{\partial h_1}{\partial s_{n-2}} & h_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ h_n & \frac{\partial h_n}{\partial s_1} & \cdots & \frac{\partial h_n}{\partial s_{n-2}} & h_n \end{pmatrix}$$

and Theorem 9.34(d) implies that it is actually 0. Hence we have

$$\int_{\Phi} \omega_n = \int_{I^{n-1}} \frac{g'(t)}{g(t) \left(\sum_{j=1}^n h_j^2 \right)^{\frac{n}{2}}} \times (0) ds_1 \cdots ds_{n-2} dt = 0$$

which proves the formula (10.153).

- **A generalization of Problem 10.22(d).** Our first step is to construct a $(n-1)$ -surface in E_n whose role is similar to that of Σ in Example 10.32. Let D_n be the $(n-1)$ -cell

$$[0, \pi]^{n-2} \times [0, 2\pi].$$

Suppose that $\Sigma^{n-1} : D_n \rightarrow E_n \subseteq \mathbb{R}^n \setminus \{\mathbf{0}\}$ is the $(n-1)$ -surface defined by

$$\Sigma^{n-1}(\phi_1, \dots, \phi_{n-1}) = (x_1, \dots, x_{n-1}, x_n),$$

where

$$\begin{aligned} x_1 &= \cos \phi_1, \\ x_2 &= \sin \phi_1 \cos \phi_2, \\ x_3 &= \sin \phi_1 \sin \phi_2 \cos \phi_3, \\ &\vdots \\ x_{n-2} &= \sin \phi_1 \cdots \sin \phi_{n-3} \cos \phi_{n-2}, \\ x_{n-1} &= \sin \phi_1 \cdots \sin \phi_{n-2} \cos \phi_{n-1}, \\ x_n &= \sin \phi_1 \cdots \sin \phi_{n-2} \sin \phi_{n-1} \end{aligned} \tag{10.154}$$

and $0 \leq \phi_1, \dots, \phi_{n-2} \leq \pi$, $0 \leq \phi_{n-1} \leq 2\pi$. (See [4] for further details of the derivation of Σ^{n-1} .) By direct computation, we know from the definitions (10.154) that

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1.$$

Thus the range of Σ^{n-1} is the $(n-1)$ -sphere S^{n-1} , i.e.,

$$\Sigma^{n-1}(D_n) = S^{n-1}. \tag{10.155}$$

Next, suppose E is a closed rectangle in D_n with edges parallel to those of D_n . In other words, we have

$$E = [a_1, b_1] \times \cdots \times [a_{n-1}, b_{n-1}],$$

where a_i and b_i are some constants with $0 < a_i < b_i < \pi$ for $1 \leq i \leq n-2$ and $0 < a_{n-1} < b_{n-1} < 2\pi$. Let $f \in \mathcal{C}''(D_n)$ and $f > 0$ on D_n . Let, further, that Ω be the $(n-1)$ -surface with parameter domain E , defined by

$$\Omega(\phi_1, \dots, \phi_{n-1}) = f(\phi_1, \dots, \phi_{n-1}) \Sigma^{n-1}(\phi_1, \dots, \phi_{n-1}).$$

Now we want to prove

Lemma 10.14

We have

$$\int_{\Omega} \omega_n = \int_S \omega_n = A_{n-1}(S),$$

where S and $A_{n-1}(S)$ denote the restriction Σ_E^{n-1} and the “area” of S .

Since the proof of Lemma 10.14 is quite lengthy, we present its proof in Appendix A.

– **A special case of Lemma 10.14.** We claim that

Lemma 10.15

For $n \geq 2$, we have

$$\int_{\Sigma^{n-1}} \omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2})},$$

where $\frac{2\pi^{\frac{n}{2}}}{\Gamma(\frac{n}{2})}$ is the surface area of the $(n-1)$ -sphere of radius 1, see [4, p. 66].

Proof of Lemma 10.15. The case for $n = 2$ is done in Example 10.36. So we prove the case for $n \geq 3$ by induction. By the formula (A.16), we have

$$\begin{aligned} \int_{\Sigma^{n-1}} \omega_n &= A_{n-1}(\Sigma^{n-1}) \\ &= \int_0^\pi \cdots \int_0^\pi \int_0^{2\pi} \sin \phi_{n-2} \sin^2 \phi_{n-3} \cdots \sin^{n-2} \phi_1 \, d\phi_1 \cdots d\phi_{n-1} \\ &= (2\pi) I_1 \times \cdots \times I_{n-2}, \end{aligned} \tag{10.156}$$

where

$$I_{n-2} = \int_0^\pi \sin^{n-2} x \, dx.$$

When $n = 3$, we obtain from the formula (A.9) that

$$\int_{\Sigma^2} \omega_3 = 2\pi I_1 = 2\pi \int_0^\pi \sin x \, dx = 4\pi.$$

By Theorem 8.18(a) and the fact that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, we have

$$\frac{2\pi^{\frac{3}{2}}}{\Gamma(\frac{3}{2})} = \frac{2\pi^{\frac{3}{2}}}{\frac{1}{2}\Gamma(\frac{1}{2})} = \frac{4\pi^{\frac{3}{2}}}{\sqrt{\pi}} = 4\pi.$$

Thus the statement is true for $n = 3$. Assume that it is also true for $n = k \geq 3$, i.e.,

$$\int_{\Sigma^{k-1}} \omega_k = \frac{2\pi^{\frac{k}{2}}}{\Gamma(\frac{k}{2})}.$$

For $n = k + 1$, we follow from the formula (A.9), the assumption and the properties of $\Gamma(x)$ that

$$\begin{aligned} \int_{\Sigma^k} \omega_{k+1} &= [(2\pi) I_1 \times \cdots \times I_{k-2}] \times I_{k-1} \\ &= \left(\int_{\Sigma^{k-1}} \omega_k \right) \times I_{k-1} \\ &= \frac{2\pi^{\frac{k}{2}}}{\Gamma(\frac{k}{2})} \times I_{k-1} \end{aligned}$$

$$\begin{aligned}
&= \begin{cases} \frac{2\pi^m}{\Gamma(m)} \times I_{2m-1}, & \text{if } k = 2m; \\ \frac{2\pi^{m+\frac{1}{2}}}{\Gamma(m+\frac{1}{2})} \times I_{2m}, & \text{if } k = 2m+1, \end{cases} \\
&= \begin{cases} \frac{2\pi^m}{\Gamma(m)} \times \frac{2(m-1)!\sqrt{\pi}}{\Gamma(m+\frac{1}{2})}, & \text{if } k = 2m; \\ \frac{2\pi^{m+\frac{1}{2}}}{\Gamma(m+\frac{1}{2})} \times \frac{2^m \Gamma(m+\frac{1}{2})}{\sqrt{\pi}} \times \frac{\pi}{2^m m!}, & \text{if } k = 2m+1, \end{cases} \\
&= \begin{cases} \frac{2\pi^{m+\frac{1}{2}}}{\Gamma(m+\frac{1}{2})}, & \text{if } k = 2m; \\ \frac{2\pi^{m+1}}{m!}, & \text{if } k = 2m+1, \end{cases} \\
&= \begin{cases} \frac{2\pi^{m+\frac{1}{2}}}{\Gamma(m+\frac{1}{2})}, & \text{if } k = 2m; \\ \frac{2\pi^{m+1}}{\Gamma(m+1)}, & \text{if } k = 2m+1, \end{cases} \\
&= \frac{2\pi^{\frac{k+1}{2}}}{\Gamma(\frac{k+1}{2})}.
\end{aligned}$$

Hence the statement is still true for $n = k + 1$ and this finishes the proof of the lemma. ■

This completes the proof of the problem. ■

Problem 10.24

Rudin Chapter 10 Exercise 24.

Proof. Let $\mathbf{x}, \mathbf{y} \in E$ and $\mathbf{x} \neq \mathbf{y}$. Since E is convex, the affine-oriented 2-simplex $\sigma = [\mathbf{p}, \mathbf{x}, \mathbf{y}]$ is in E . Furthermore, we know from Definition 10.30 that σ is of class $\mathcal{C}''$ because the identity mapping is of class $\mathcal{C}''$. By Theorem 10.33 (Stokes' Theorem) and the fact that $d\omega = 0$, we have

$$0 = \int_{\sigma} d\omega = \int_{\partial\sigma} \omega. \quad (10.157)$$

By Theorem 10.29, we have

$$\partial\sigma = [\mathbf{x}, \mathbf{y}] - [\mathbf{p}, \mathbf{y}] + [\mathbf{p}, \mathbf{x}],$$

so the integral (10.157) implies that

$$\begin{aligned}
\int_{[\mathbf{x}, \mathbf{y}]} \omega - \int_{[\mathbf{p}, \mathbf{y}]} \omega + \int_{[\mathbf{p}, \mathbf{x}]} \omega &= 0 \\
f(\mathbf{y}) - f(\mathbf{x}) &= \int_{[\mathbf{x}, \mathbf{y}]} \omega.
\end{aligned} \quad (10.158)$$

By definition, $[\mathbf{x}, \mathbf{y}]$ is the straight line segment in E joining the points $\mathbf{x}$ and $\mathbf{y}$. Let $\gamma : [0, 1] \rightarrow E$ be the 1-surface in $E \subseteq \mathbb{R}^n$ defined by

$$\gamma(t) = (1-t)\mathbf{x} + t\mathbf{y} = ((1-t)x_1 + ty_1, \dots, (1-t)x_n + ty_n).$$

By Definition 10.11 (or Example 10.12(a)), we have

$$\begin{aligned}
\int_{[\mathbf{x}, \mathbf{y}]} \omega &= \int_0^1 \sum_{i=1}^n a_i(\gamma(t)) \frac{\partial}{\partial t} ((1-t)x_i + ty_i) dt \\
&= \sum_{i=1}^n \int_0^1 a_i((1-t)\mathbf{x} + t\mathbf{y}) (y_i - x_i) dt
\end{aligned}$$

$$= \sum_{i=1}^n (y_i - x_i) \int_0^1 a_i((1-t)\mathbf{x} + t\mathbf{y}) dt. \quad (10.159)$$

Combining the expressions (10.158) and (10.159), we have

$$f(\mathbf{y}) - f(\mathbf{x}) = \sum_{i=1}^n (y_i - x_i) \int_0^1 a_i((1-t)\mathbf{x} + t\mathbf{y}) dt. \quad (10.160)$$

Next, by [21, Eqn. (25), p. 215] and the expression (10.160), we have

$$\begin{aligned} (D_j f)(\mathbf{x}) &= \lim_{s \rightarrow 0} \frac{f(\mathbf{x} + s\mathbf{e}_j) - f(\mathbf{x})}{s} \\ &= \lim_{s \rightarrow 0} (x_j + s - x_j) \frac{1}{s} \int_0^1 a_j((1-t)\mathbf{x} + t(\mathbf{x} + s\mathbf{e}_j)) dt \\ &= \lim_{s \rightarrow 0} \int_0^1 a_j(\mathbf{x} + ts\mathbf{e}_j) dt \\ &= \int_0^1 a_j(\mathbf{x}) dt \\ &= a_j(\mathbf{x}) \end{aligned}$$

for $j = 1, \dots, n$. Hence if we define the real function $f : E \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ by

$$f(\mathbf{x}) = \int_{[\mathbf{p}, \mathbf{x}]} \omega,$$

then it is of class $\mathcal{C}'$ in E and

$$df = \sum_{i=1}^n (D_i f)(\mathbf{x}) dx_i = \sum_{i=1}^n a_i(\mathbf{x}) dx_i = \omega$$

in E . This completes our proof of the problem. ■

Problem 10.25

Rudin Chapter 10 Exercise 25.

Proof. This problem relates the concepts “exactness” and “independence of path” of 1-forms. Suppose that

$$\omega = \sum a_i(\mathbf{x}) dx_i.$$

We need some results from topology. Given a space X , define an equivalence relation (Definition 2.3) on X by setting $x \sim y$ if there exists a connected subspace of X containing x and y . The equivalent classes are called the *connected component* of X . Let’s recall some well-known results about connected components of a space X ([18, Theorem 25.1, p. 159]):

Lemma 10.16

Let $\{X_\alpha\}$ be the collection of all connected components of X . Then we have

$$X = \bigcup_{\alpha} X_{\alpha}.$$

Furthermore, a space X is said to be *locally connected* at x if for every neighborhood U of x , there exists a connected neighborhood V such that

$$x \in V \subseteq U.$$

If X is locally connected at each x , then X is called a locally connected space. We have the following result about locally connected spaces ([18, Theorem 25.3, p. 161]):

Lemma 10.17

If X is a locally connected space, then each component of an open set U of X is also open in X .

We know, by checking the definition of locally connectedness directly, that $\mathbb{R}^n$ is locally connected. Since the set E in our question is supposed to be open in $\mathbb{R}^n$, Lemmas 10.16 and 10.17 imply that

$$E = \bigcup_{\alpha} E_{\alpha},$$

where each E_{α} is an open (in $\mathbb{R}^n$) connected component of E . We claim that ω is exact in E_{α} . To this end, fix $\mathbf{p}_{\alpha} \in E_{\alpha}$. Similar to Problem 10.24, we define a function $f_{\alpha} : E_{\alpha} \rightarrow \mathbb{R}$ by

$$f_{\alpha}(\mathbf{x}) = \int_{[\mathbf{p}_{\alpha}, \mathbf{x}]} \omega.$$

For any $\mathbf{x}, \mathbf{y} \in E_{\alpha}$, let $\gamma = [\mathbf{x}, \mathbf{y}] - [\mathbf{p}_{\alpha}, \mathbf{y}] + [\mathbf{p}_{\alpha}, \mathbf{x}]$. Then γ is a closed curve in E_{α} and the hypothesis shows that

$$0 = \int_{\gamma} \omega = \int_{[\mathbf{x}, \mathbf{y}] - [\mathbf{p}_{\alpha}, \mathbf{y}] + [\mathbf{p}_{\alpha}, \mathbf{x}]} \omega = \int_{[\mathbf{x}, \mathbf{y}]} \omega - \int_{[\mathbf{p}_{\alpha}, \mathbf{y}]} \omega + \int_{[\mathbf{p}_{\alpha}, \mathbf{x}]} \omega.$$

Thus we have

$$f_{\alpha}(\mathbf{x}) - f_{\alpha}(\mathbf{y}) = \int_{[\mathbf{x}, \mathbf{y}]} \omega$$

which is exactly the relation (10.158). By imitating the remaining part of the argument in Problem 10.24, we conclude that

$$\omega = df_{\alpha}$$

in E_{α} . Finally, if we define the function $f : E \rightarrow \mathbb{R}$ such that the restriction of f to E_{α} is f_{α} , i.e., $f|_{E_{\alpha}} = f_{\alpha}$, then we have

$$\omega = df$$

in E . This ends the proof of the problem. ■

Problem 10.26

Rudin Chapter 10 Exercise 26.

Proof. We follow the given hint. Let $E = \mathbb{R}^3 \setminus \{\mathbf{0}\}$. Then E is obviously open in $\mathbb{R}^3$. Define $\gamma : [0, 1] \rightarrow E$ to be a closed curve in E , of class $\mathcal{C}'$. Therefore, there is a 2-surface $\Phi : D \rightarrow E$ such that

$$\partial\Phi = \gamma,$$

where D is a compact subset of $\mathbb{R}^2$. Since ω is a 1-form in E of class $\mathcal{C}'$ and Φ is of class $\mathcal{C}''$ in E , it follows from Theorem 10.33 (Stokes' Theorem) and then the hypothesis $d\omega = 0$ that

$$\int_{\gamma} \omega = \int_{\partial\Phi} \omega = \int_{\Phi} d\omega = 0.$$

By Problem 10.25, we establish that ω is exact in $E = \mathbb{R}^3 \setminus \{\mathbf{0}\}$, finishing the proof of the problem. ■

Problem 10.27

Rudin Chapter 10 Exercise 27.

Proof. Let $E = (p_1, q_1) \times (p_2, q_2) \times (p_3, q_3)$ be the open 3-cell in $\mathbb{R}^3$. By Theorem 10.20 and then [21, Eqn. (59), p. 260], we have

$$\begin{aligned} d\lambda &= d(g_1 dx + g_2 dy) \\ &= (dg_1) \wedge dx + (-1)^0 g_1 \wedge d^2x + (dg_2) \wedge dy + (-1)^0 g_2 \wedge d^2y \\ &= \left(\frac{\partial g_1}{\partial x} dx + \frac{\partial g_1}{\partial y} dy + \frac{\partial g_1}{\partial z} dz \right) \wedge dx + \left(\frac{\partial g_2}{\partial x} dx + \frac{\partial g_2}{\partial y} dy + \frac{\partial g_2}{\partial z} dz \right) \wedge dy. \end{aligned} \quad (10.161)$$

Since the anticommutative relation and $dx \wedge dx = dy \wedge dy = 0$ (see [21, Eqn. (42) & (43), p.256]), we obtain from the expression (10.161) that

$$\begin{aligned} d\lambda &= \frac{\partial g_1}{\partial y} dy \wedge dx + \frac{\partial g_2}{\partial x} dx \wedge dy + \frac{\partial g_1}{\partial z} dz \wedge dx + \frac{\partial g_2}{\partial z} dz \wedge dy \\ &= \left(\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) dx \wedge dy + \frac{\partial g_1}{\partial z} dz \wedge dx + \frac{\partial g_2}{\partial z} dz \wedge dy. \end{aligned} \quad (10.162)$$

When x and y are fixed, since $f_2 \in \mathcal{C}'(E)$, we have $f_2 \in \mathcal{C}'((p_3, q_3))$. In particular, $f_2 \in \mathcal{R}$ on $[c, z]$ and f_2 is continuous at z , so Theorem 6.20 (First Fundamental Theorem of Calculus) implies that

$$\frac{\partial g_1}{\partial z} = \frac{\partial}{\partial z} \int_c^z f_2(x, y, s) ds - \frac{\partial}{\partial z} \int_b^y f_3(x, t, c) dt = f_2(x, y, z) - 0 = f_2(x, y, z). \quad (10.163)$$

Similarly, we have

$$\frac{\partial g_2}{\partial z} = -\frac{\partial}{\partial z} \int_c^z f_1(x, y, s) ds = -f_1(x, y, z) \quad (10.164)$$

and

$$\frac{\partial}{\partial y} \left(-\int_b^y f_3(x, t, c) dt \right) = -f_3(x, y, c). \quad (10.165)$$

Finally, we have to evaluate

$$\frac{\partial}{\partial x} \left(-\int_c^z f_1(x, y, s) ds \right) \quad \text{and} \quad \frac{\partial}{\partial y} \left(\int_c^z f_2(x, y, s) ds \right).$$

To do this, we need Theorem 9.42, so we have to check its hypotheses. For the first integral, we fix y and let $F_1(x, s) = f_1(x, y, s)$.

- Since $F_1(x, s)$ is defined for $a \leq x \leq r_1, c \leq s \leq r_3$ for some $r_1 < q_1$ and $r_3 < q_3$. (This is the hypothesis (a) in Theorem 9.42.)
- For every (fixed) $x \in [a, r_1]$, the condition $f_1 \in \mathcal{C}'(E)$ implies that $F_1(x, s) \in \mathcal{R}$ on $[c, r_3]$. (This is the hypothesis (c) in Theorem 9.42.)
- Since $f_1 \in \mathcal{C}'(E)$, $\frac{\partial F_1}{\partial s}$ is a uniformly continuous function on $[a, r_1] \times [c, r_3]$. By Definition 4.18, for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$\left| \frac{\partial}{\partial x} F_1(x, s) \Big|_{(p,q)} - \frac{\partial}{\partial x} F_1(x, s) \Big|_{(u,v)} \right| < \epsilon$$

for all $(p, q), (u, v) \in [a, r_1] \times [c, r_3]$ with $\sqrt{(p-u)^2 + (q-v)^2} < \delta$. In particular, if we fix u such that $a < u < r_1$ and put $q = v = s$, then the last inequality gives

$$\left| \frac{\partial}{\partial x} F_1(x, s) \Big|_{x=p} - \frac{\partial}{\partial x} F_1(x, s) \Big|_{x=u} \right| < \epsilon$$

for all $s \in [c, r_3]$ with $|p - u| < \delta$. This is exactly the hypothesis (d) in Theorem 9.42.

Hence, by Theorem 9.42, we have

$$\frac{\partial}{\partial x} \left(- \int_c^z f_1(x, y, s) \, ds \right) = - \int_c^z \frac{\partial}{\partial x} f_1(x, y, s) \, ds. \quad (10.166)$$

Similarly, we have

$$\frac{\partial}{\partial y} \left(\int_c^z f_2(x, y, s) \, ds \right) = \int_c^z \frac{\partial}{\partial y} f_2(x, y, s) \, ds. \quad (10.167)$$

Therefore, we follow from the partial derivatives (10.165) - (10.167) that

$$\frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} = f_3(x, y, c) - \int_c^z \left[\frac{\partial}{\partial x} f_1(x, y, s) + \frac{\partial}{\partial y} f_2(x, y, s) \right] \, ds. \quad (10.168)$$

Now we derive from Theorem 10.20 and [21, Eqn. (45), p. 257] that

$$\begin{aligned} d\omega &= d(f_1 \, dy \wedge dz) + d(f_2 \, dz \wedge dx) + d(f_3 \, dx \wedge dy) \\ &= d f_1 \wedge dy \wedge dz + f_1 \underbrace{d(dy \wedge dz)}_{\text{It is 0.}} + d f_2 \wedge dz \wedge dx + f_2 \underbrace{d(dz \wedge dx)}_{\text{It is 0.}} \\ &\quad + d f_3 \wedge dx \wedge dy + f_3 \underbrace{d(dx \wedge dy)}_{\text{It is 0.}} \\ &= \frac{\partial f_1}{\partial x} dx \wedge dy \wedge dz + \frac{\partial f_2}{\partial y} dy \wedge dz \wedge dx + \frac{\partial f_3}{\partial z} dz \wedge dx \wedge dy \\ &= \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) dx \wedge dy \wedge dz. \end{aligned} \quad (10.169)$$

By the assumption $d\omega = 0$, we obtain from the relation (10.169) that

$$\frac{\partial f_3}{\partial z} = - \frac{\partial f_1}{\partial x} - \frac{\partial f_2}{\partial y}$$

so that the relation (10.168) implies that

$$\begin{aligned} \frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} &= f_3(x, y, c) + \int_c^z \frac{\partial f_3}{\partial s} \, ds \\ &= f_3(x, y, c) + f_3(x, y, z) - f_3(x, y, c) \\ &= f_3(x, y, z). \end{aligned} \quad (10.170)$$

By substituting the relations (10.163), (10.164) and (10.170) into the relation (10.162), we reach the desired result that

$$d\lambda = \omega$$

in E . This proves our first assertion.

For the second assertion, if $\omega = \zeta$ and $E = \mathbb{R}^3 \setminus \{\mathbf{0}\}$, then, by Problem 10.22, we have

$$f_1(x, y, z) = \frac{x}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}, \quad f_2(x, y, z) = \frac{y}{(x^2 + y^2 + z^2)^{\frac{3}{2}}} \quad \text{and} \quad f_3(x, y, z) = \frac{z}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}.$$

To evaluate the integral

$$\int_c^z \frac{y}{(x^2 + y^2 + s^2)^{\frac{3}{2}}} \, ds, \quad (10.171)$$

we apply the substitution $s = \sqrt{x^2 + y^2} \tan \theta$, so the integral (10.171) becomes

$$\int_c^z \frac{y}{(x^2 + y^2 + s^2)^{\frac{3}{2}}} \, ds = y \int_{\arctan \frac{c}{\sqrt{x^2 + y^2}}}^{\arctan \frac{z}{\sqrt{x^2 + y^2}}} \frac{(x^2 + y^2)^{\frac{1}{2}} \sec^2 \theta \, d\theta}{(x^2 + y^2)^{\frac{3}{2}} \sec^3 \theta}$$

$$\begin{aligned}
&= y \int_{\arctan \frac{z}{\sqrt{x^2+y^2}}}^{\arctan \frac{z}{\sqrt{x^2+y^2}}} \frac{d\theta}{(x^2+y^2) \sec \theta} \\
&= \frac{y}{x^2+y^2} \int_{\arctan \frac{z}{\sqrt{x^2+y^2}}}^{\arctan \frac{z}{\sqrt{x^2+y^2}}} \cos \theta d\theta \\
&= \frac{y}{x^2+y^2} \sin \theta \Big|_{\arctan \frac{z}{\sqrt{x^2+y^2}}}^{\arctan \frac{z}{\sqrt{x^2+y^2}}} \\
&= \frac{y}{x^2+y^2} \left(\sin \arctan \frac{z}{\sqrt{x^2+y^2}} - \sin \arctan \frac{c}{\sqrt{x^2+y^2}} \right) \\
&= \frac{y}{x^2+y^2} \left[\frac{z}{(x^2+y^2+z^2)^{\frac{1}{2}}} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right]. \tag{10.172}
\end{aligned}$$

Similarly, we have

$$\int_b^y \frac{c}{(x^2+t^2+c^2)^{\frac{3}{2}}} dt = \frac{c}{x^2+c^2} \left[\frac{y}{(x^2+y^2+c^2)^{\frac{1}{2}}} - \frac{b}{(x^2+b^2+c^2)^{\frac{1}{2}}} \right] \tag{10.173}$$

and

$$\int_c^z \frac{x}{(x^2+y^2+s^2)^{\frac{3}{2}}} ds = \frac{x}{x^2+y^2} \left[\frac{z}{(x^2+y^2+z^2)^{\frac{1}{2}}} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right]. \tag{10.174}$$

Now we put the integrals (10.172) - (10.174) into the definitions of g_1 and g_2 to acquire

$$g_1(x, y, z) = \frac{y}{x^2+y^2} \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right] - \frac{c}{x^2+c^2} \left[\frac{y}{(x^2+y^2+c^2)^{\frac{1}{2}}} - \frac{b}{(x^2+b^2+c^2)^{\frac{1}{2}}} \right]$$

and

$$g_2(x, y, z) = -\frac{x}{x^2+y^2} \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right]$$

so that

$$\begin{aligned}
\lambda &= g_1 dx + g_2 dy \\
&= \frac{y}{x^2+y^2} \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right] dx - \frac{c}{x^2+c^2} \left[\frac{y}{(x^2+y^2+c^2)^{\frac{1}{2}}} - \frac{b}{(x^2+b^2+c^2)^{\frac{1}{2}}} \right] dx \\
&\quad - \frac{x}{x^2+y^2} \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right] dy \\
&= - \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right] \left(\frac{x dy - y dx}{x^2+y^2} \right) \\
&\quad - \frac{c}{x^2+c^2} \left[\frac{y}{(x^2+y^2+c^2)^{\frac{1}{2}}} - \frac{b}{(x^2+b^2+c^2)^{\frac{1}{2}}} \right] dx \\
&= - \left[\frac{z}{r} - \frac{c}{(x^2+y^2+c^2)^{\frac{1}{2}}} \right] \eta - \frac{c}{x^2+c^2} \left[\frac{y}{(x^2+y^2+c^2)^{\frac{1}{2}}} - \frac{b}{(x^2+b^2+c^2)^{\frac{1}{2}}} \right] dx. \tag{10.175}
\end{aligned}$$

Recall that (a, b, c) is a fixed point in $E = \mathbb{R}^3 \setminus \{\mathbf{0}\}$, so we put $b = c = 0$ into the expression (10.175) to get

$$\lambda = -\frac{z}{r} \eta$$

which is exactly the form that occurs in part (e) of Problem 10.22. This completes the proof of the problem. ■

Problem 10.28*Rudin Chapter 10 Exercise 28.*

Proof. We have $\Phi : [a, b] \times [0, 2\pi] \rightarrow \mathbb{R}^2$. Since $\omega = x^3 dy$, we follow from Theorem 10.20 and then [21, Eqn. (59), p. 260] that $d\omega = d(x^3 dy) = (dx^3) \wedge dy - x^3 d^2y = 3x^2 dx \wedge dy$. Therefore, it follows from Definition 10.11 that

$$\begin{aligned} \int_{\Phi} d\omega &= \int_{[a,b] \times [0,2\pi]} 3(r \cos \theta)^2 \frac{\partial(x, y)}{\partial(r, \theta)} dr d\theta \\ &= \int_a^b \int_0^{2\pi} 3r^2 \cos^2 \theta \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} dr d\theta \\ &= \int_a^b \int_0^{2\pi} 3r^3 \cos^2 \theta dr d\theta \\ &= \frac{3\pi}{4}(b^4 - a^4). \end{aligned}$$

By the hint in Problem 10.21(b), we know that $\partial\Phi = \Gamma - \gamma$, where $\Gamma : [0, 2\pi] \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ and $\gamma : [0, 2\pi] \rightarrow \mathbb{R}^2 \setminus \{\mathbf{0}\}$ are given by $\Gamma(t) = (b \cos \theta, b \sin \theta)$ and $\gamma(t) = (a \cos \theta, a \sin \theta)$. By Definition 10.11 again, we have

$$\begin{aligned} \int_{\partial\Phi} \omega &= \int_{\Gamma} \omega - \int_{\gamma} \omega \\ &= \int_0^{2\pi} (b \cos \theta)^3 \frac{\partial(b \sin \theta)}{\partial \theta} d\theta - \int_0^{2\pi} (a \cos \theta)^3 \frac{\partial(a \sin \theta)}{\partial \theta} d\theta \\ &= (b^4 - a^4) \int_0^{2\pi} \cos^4 \theta d\theta \\ &= (b^4 - a^4) \int_0^{2\pi} \frac{\cos 4\theta + 4 \cos 2\theta + 3}{8} d\theta \\ &= \frac{3\pi}{4}(b^4 - a^4). \end{aligned}$$

Hence we see that the two integrals are equal and this finishes the proof of the problem. ■

Problem 10.29*Rudin Chapter 10 Exercise 29.*

Proof. Let's recall the hypotheses of the problem first: Suppose that E is a convex open set in $\mathbb{R}^n$, $f \in \mathcal{C}'(E)$, p is an integer such that $1 \leq p \leq n$, and

$$(D_j f)(\mathbf{x}) = 0 \quad (p < j \leq n, \widehat{\mathbf{x}} \in E). \quad (10.176)$$

Denote $\widehat{\mathbf{x}} = (\mathbf{x}, x_p, \mathbf{x}')$, where $\mathbf{x} = (x_1, \dots, x_{p-1})$ and $\mathbf{x}' = (x_{p+1}, \dots, x_n)$. Furthermore, we define

$$V = \{(\mathbf{x}, x_p) \in \mathbb{R}^p \mid (\mathbf{x}, x_p, \mathbf{x}') \in E \text{ for some } \mathbf{x}'\}. \quad (10.177)$$

Now V is a convex open subset in $\mathbb{R}^p$. If we define U to be the projection of V onto its first $(p-1)$ -coordinates, i.e.,

$$U = \{\mathbf{x} = (x_1, \dots, x_{p-1}) \in \mathbb{R}^{p-1} \mid \mathbf{y} = (\mathbf{x}, x_p) \in V \text{ for some } x_p \in \mathbb{R}\}, \quad (10.178)$$

then U is also a convex open subset in $\mathbb{R}^{p-1}$.

Claim 1: $\text{graph}(\alpha) \subseteq V$. We show that there is a function $\alpha \in \mathcal{C}'(U)$ such that its *graph* lies in V , i.e.,

$$\text{graph}(\alpha) \subseteq V. \quad (10.179)$$

- **Case (i):** $E = B_n(\mathbf{a}, r)$ for some $\mathbf{a} = (a_1, \dots, a_n)$ and $r > 0$. For any $(x_1, \dots, x_p)$ satisfying

$$(x_1 - a_1)^2 + \dots + (x_p - a_p)^2 < r^2,$$

we may choose $(x_{p+1}, \dots, x_n)$ with the property that

$$(x_{p+1} - a_{p+1})^2 + \dots + (x_n - a_n)^2 < \frac{1}{2}\{r^2 - [(x_1 - a_1)^2 + \dots + (x_p - a_p)^2]\}$$

and this implies that

$$(x_1 - a_1)^2 + \dots + (x_n - a_n)^2 < r^2.$$

By the definition (10.177), we have

$$V = \{(x_1, \dots, x_{p-1}, x_p) \in \mathbb{R}^p \mid (x_1 - a_1)^2 + \dots + (x_p - a_p)^2 < r^2\} = B_p(\mathbf{a}', r) \quad (10.180)$$

which is the ball centered at the *fixed* point $\mathbf{a}' = (a_1, \dots, a_p)$ and of radius r . If we put $x_p = a_p$, then we get from the set (10.180) and the definition (10.178) that

$$U = \{(x_1, \dots, x_{p-1}) \in \mathbb{R}^p \mid (x_1 - a_1)^2 + \dots + (x_{p-1} - a_{p-1})^2 < r^2\}$$

We define $\alpha : U \rightarrow \mathbb{R}$ to be $\alpha(\mathbf{x}) = a_p$ for all $\mathbf{x} \in U$. It is clear that $\alpha \in \mathcal{C}'(U)$ and we have

$$\begin{aligned} \text{graph}(\alpha) &= \{(\mathbf{x}, \alpha(\mathbf{x})) \mid \mathbf{x} \in U\} \\ &= \{(\mathbf{x}, a_p) \mid \mathbf{x} \in U\} \\ &= \{(x_1, \dots, x_{p-1}, a_p) \mid (x_1 - a_1)^2 + \dots + (x_{p-1} - a_{p-1})^2 < r^2\} \\ &\subset V. \end{aligned}$$

To have a better understanding between V and its projection U , we consider the example that $n \geq 4, p = 3, \mathbf{a} = \mathbf{0}$ and $r = 1$, then the sets E, V and U are the unit balls in $\mathbb{R}^4, \mathbb{R}^3$ and $\mathbb{R}^2$ respectively, see Figure 10.16 below:

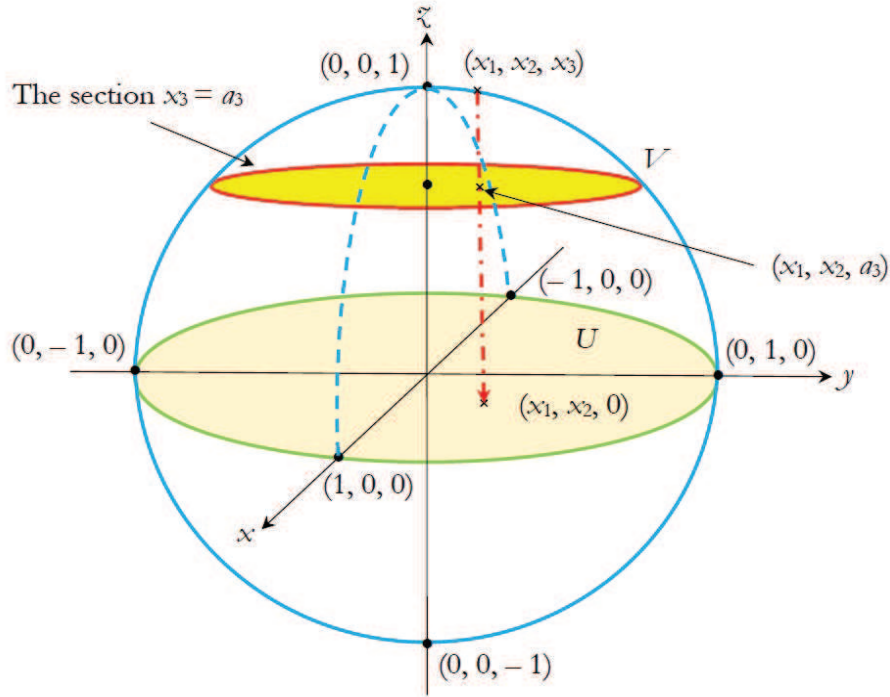


Figure 10.16: The unit disk U as the projection of the unit ball V .

Notice that $x_1^2 + x_2^2 + x_3^2 = 1$ and the point (x_1, x_2, a_3) on the section $x_3 = a_3$ is projected onto the point $(x_1, x_2, 0)$.

- **Case (ii):** $E = (a_1, b_1) \times \cdots \times (a_n, b_n)$. Then we follow from the definitions (10.177) and (10.178) that

$$\begin{aligned} V &= \{(x_1, \dots, x_{p-1}, x_p) \in \mathbb{R}^p \mid (x_1, \dots, x_{p-1}, x_p, x_{p+1}, \dots, x_n) \in E \text{ for some } (x_{p+1}, \dots, x_n)\} \\ &= \{(x_1, \dots, x_{p-1}, x_p) \in \mathbb{R}^p \mid x_i \in (a_i, b_i), \text{ where } 1 \leq i \leq p\} \\ &= (a_1, b_1) \times \cdots \times (a_p, b_p) \end{aligned}$$

and

$$U = (a_1, b_1) \times \cdots \times (a_{p-1}, b_{p-1}).$$

In this case, we define $\alpha : U \rightarrow \mathbb{R}$ by

$$\alpha(\mathbf{x}) = c$$

for some $c \in (a_p, b_p)$ and for all $\mathbf{x} \in U$. The definition shows clearly that $\alpha \in \mathcal{C}'(U)$ and

$$\begin{aligned} \text{graph}(\alpha) &= \{(\mathbf{x}, \alpha(\mathbf{x})) \mid \mathbf{x} \in U\} \\ &= \{(\mathbf{x}, c) \mid \mathbf{x} \in U\} \\ &= \{(x_1, \dots, x_{p-1}, c) \mid x_i \in (a_i, b_i), \text{ where } 1 \leq i \leq p-1\} \\ &= (a_1, b_1) \times \cdots \times (a_{p-1}, b_{p-1}) \times \{c\} \\ &\subset V. \end{aligned}$$

For instance, we take $n \geq 4, p = 3$ and $E = (0, 1) \times (0, 1) \times (0, 1) \times (0, 1)$. Then we have

$$V = (0, 1) \times (0, 1) \times (0, 1) \quad \text{and} \quad U = (0, 1) \times (0, 1),$$

see Figure 10.17 below:

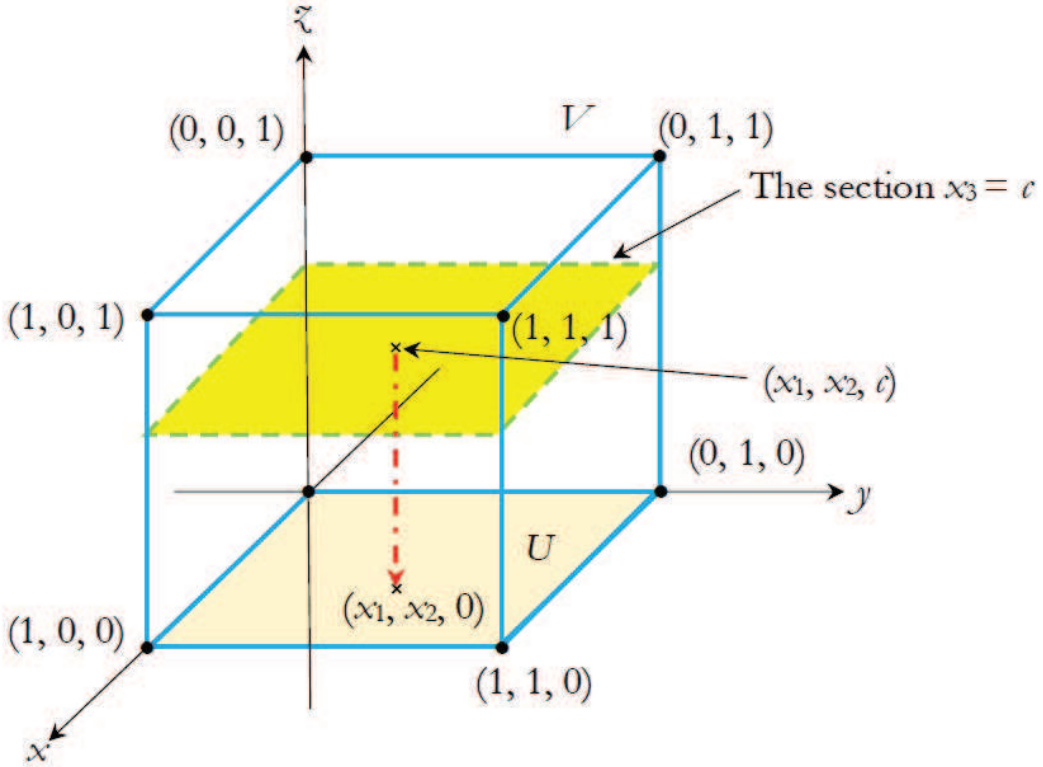


Figure 10.17: The open cells U and V .

- **Case (iii): E is an arbitrary open convex set in $\mathbb{R}^n$.** Since the set E is not compact, we can't apply Problem 10.6 directly to show our result. In fact, we need a variation of partitions of unity which is suitable for arbitrary sets in $\mathbb{R}^n$ and the following result (see [24, pp. 63, 64]) serves this purpose:

Lemma 10.18

Let $A \subseteq \mathbb{R}^n$ and let $\mathcal{O}$ be an open cover of A . Then there is a collection $\Phi = \{\varphi_\mu\}$ of infinitely differentiable functions $\varphi_\mu : W \rightarrow \mathbb{R}$, where W is an open set containing A , with the following properties:

- (a) For each $x \in A$, we have $0 \leq \varphi_\mu(x) \leq 1$.
- (b) For each $x \in A$, there is an open set V containing x (V may not be an element in the open cover $\mathcal{O}$) such that all but finitely many $\varphi_\mu \in \Phi$ are 0 on V .
- (c) For each $x \in A$, we have

$$\sum_{\varphi_\mu \in \Phi} \varphi_\mu(x) = 1.$$

- (d) For each $\varphi_\mu \in \Phi$, there is an open set $U \in \mathcal{O}$ such that $\varphi_\mu = 0$ outside of some closed set contained in U .

Now we are ready for the construction:

- **Step 1: Construction of an open cover of U .** Since V is open, for every $\mathbf{x} \in U$, we may take $\mathbf{y} \in V$ and a $\epsilon(\mathbf{y}) > 0$ such that the projection of $\mathbf{y}$ onto its first $(p-1)$ -coordinates is $\mathbf{x}$ (i.e., $\mathbf{y} = (\mathbf{x}, a_p)$ for some $a_p \in \mathbb{R}$) and $B_p(\mathbf{y}, \epsilon(\mathbf{y})) \subseteq V$. Now we see that

$$U = \bigcup_{\mathbf{x} \in U} B_{p-1}(\mathbf{x}, \epsilon(\mathbf{y})).$$

In other words, $\mathcal{O} = \{B_{p-1}(\mathbf{x}, \epsilon(\mathbf{y}))\}_{\mathbf{x} \in U}$ is an open cover of U .

- **Step 2: Construction of the required function α .** By Lemma 10.18, we can find a collection $\Phi = \{\varphi_\mu\}$ of $\mathcal{C}^\infty$ functions $\varphi_\mu : U \rightarrow \mathbb{R}$ (assume $W = U$ for simplicity) subordinate to the cover $\mathcal{O}$, i.e., each φ_μ satisfies the condition

$$\text{supp}(\varphi_\mu) \subseteq U_\mu = B_{p-1}(\mathbf{x}_\mu, \epsilon(\mathbf{y}_\mu))$$

for some $\mathbf{x}_\mu \in U$, where $\mathbf{y}_\mu = (\mathbf{x}_\mu, a_{\mu p})$ for some $a_{\mu p} \in \mathbb{R}$.

Lemma 10.19

Suppose that

$$V_\mu = B_p(\mathbf{y}_\mu, \epsilon(\mathbf{y}_\mu)) \quad \text{and} \quad U_\mu = B_{p-1}(\mathbf{x}_\mu, \epsilon(\mathbf{y}_\mu)).$$

Then the projection of V_μ is U_μ .

Proof of Lemma 10.19. Recall from the definition (10.178) that

$$U_\mu = B_{p-1}(\mathbf{x}_\mu, \epsilon(\mathbf{y}_\mu)) = \{\mathbf{x} \in \mathbb{R}^{p-1} \mid |\mathbf{x} - \mathbf{x}_\mu| < \epsilon(\mathbf{y}_\mu)\}.$$

By using the analysis as in proving the set relation (10.180) in **Case (i)**, we know that the projection of the ball $V_\mu = B_p(\mathbf{y}_\mu, \epsilon(\mathbf{y}_\mu))$ is exactly $U_\mu = B_{p-1}(\mathbf{x}_\mu, \epsilon(\mathbf{y}_\mu))$, see Figure 10.16 for an example in a lower dimension. This completes the proof of the lemma. ■

We return to the proof of Problem 10.29. Suppose that the constant $a_{\mu p} \in \mathbb{R}$ is chosen to be *fixed*. Then we have

$$\mathbf{y} = (\mathbf{x}, a_{\mu p}) \in V_{\mu} \quad \text{for every } \mathbf{x} \in U_{\mu}. \quad (10.181)$$

In fact, for $\mathbf{x} \in U_{\mu}$, we have $|\mathbf{x} - \mathbf{x}_{\mu}| < \epsilon(\mathbf{y})$ so that Lemma 10.19 shows that

$$|\mathbf{y} - \mathbf{y}_{\mu}| = |(\mathbf{x}, a_{\mu p}) - (\mathbf{x}_{\mu}, a_{\mu p})| = |\mathbf{x} - \mathbf{x}_{\mu}| < \epsilon(\mathbf{y}_{\mu}).$$

This proves the relation (10.181).

We define the function $\alpha : U \rightarrow \mathbb{R}$ by

$$\alpha(\mathbf{x}) = \sum_{\mu} \varphi_{\mu}(\mathbf{x}) a_{\mu p}$$

for every $\mathbf{x} \in U$. Now, for each $\mathbf{x} \in U$, Lemma 10.18(b) implies that there is an open ball

$$B_{p-1}(\mathbf{x}, \delta) \subseteq U$$

such that *all but finitely many* $\varphi_{\mu} \in \Phi$ are 0 on $B_{p-1}(\mathbf{x}, \delta)$, where $\delta > 0$. Let m be a finite positive integer such that $\varphi_{\mu_j}(\mathbf{x}) \neq 0$ on $B_{p-1}(\mathbf{x}, \delta)$ for $1 \leq j \leq m$. Thus we have

$$\alpha(\mathbf{x}) = \sum_{\mu} \varphi_{\mu}(\mathbf{x}) a_{\mu p} = \sum_{j=1}^m \varphi_{\mu_j}(\mathbf{x}) a_{\mu_j p}. \quad (10.182)$$

Since $\varphi_{\mu_j} \in \mathcal{C}^{\infty}(U)$ for every $1 \leq j \leq m$, each φ_{μ_j} is continuously differentiable at $\mathbf{x}$ and we conclude from the expression (10.182) that α is continuously differentiable at $\mathbf{x}$. Since $\mathbf{x}$ is arbitrary, we have $\alpha \in \mathcal{C}'(U)$ (in fact, $\mathcal{C}^{\infty}(U)$). Finally, by the convexity of the set V and Lemma 10.19(c), we have

$$(\mathbf{x}, \alpha(\mathbf{x})) = \left(\underbrace{\sum_{j=1}^m \varphi_{\mu_j}(\mathbf{x}) \mathbf{x}}_{\text{Its value is 1.}}, \sum_{j=1}^m \varphi_{\mu_j}(\mathbf{x}) a_{\mu_j p} \right) = \sum_{j=1}^m \varphi_{\mu_j}(\mathbf{x}) (\mathbf{x}, a_{\mu_j p}) \in V.$$

Hence we have $\text{graph}(\alpha) \subseteq V$ in this case. This completes the proof of **Claim 1**.

Claim 2: $F \in \mathcal{C}'(E)$. When $p = 1$, we have

$$F(\mathbf{x}) = \int_c^{x_1} \varphi(x) dx \quad (\mathbf{x} \in E). \quad (10.183)$$

Since V is convex in $\mathbb{R}$, Problem 2.21(c) implies that it is connected and then it follows from Theorem 2.47 that V is an interval. Pick $c \in V$. We know that

$$\varphi : [c, x_1] \rightarrow \mathbb{R}, \quad \varphi(x) = f(x, x_2, \dots, x_n) = f(\mathbf{x}) \quad \text{for } x \in [c, x_1] \quad \text{and} \quad f \in \mathcal{C}'(E),$$

so φ is continuous at x_1 . By Theorem 6.20 (First Fundamental Theorem of Calculus), F is differentiable at x_1 and

$$(D_1 F)(\mathbf{x}) = \frac{\partial}{\partial x_1} \int_c^{x_1} \varphi(x) dx = \varphi(x_1) = f(\mathbf{x}). \quad (10.184)$$

Now the continuity of $(D_1 F)(\mathbf{x})$ follows from the relation (10.184). By the definition (10.183), we see that

$$(D_j F)(\mathbf{x}) = 0$$

for $j = 2, \dots, n$ and for all $\mathbf{x} \in E$. Hence we follow from Theorem 9.21 that $F \in \mathcal{C}'(E)$.

Next, we suppose that $p > 1$. By the definition of the real function α , we have

$$(\mathbf{x}', \alpha(\mathbf{x}')) \in V.$$

Since $(\mathbf{x}', x_p) \in V$ and V is convex, we must have

$$(\mathbf{x}', \lambda\alpha(\mathbf{x}') + (1-\lambda)x_p) = \lambda(\mathbf{x}', \alpha(\mathbf{x}')) + (1-\lambda)(\mathbf{x}', x_p) \in V$$

for every $\lambda \in (0, 1)$. By this, instead of the definition (10.183), the integral

$$F(\mathbf{x}) = \int_{\alpha(\mathbf{x}')}^{x_p} \varphi(\mathbf{x}', x) dx \quad (\mathbf{x} \in E) \quad (10.185)$$

is well-defined. Similarly, $(D_j F)(\mathbf{x}) = 0$ for $j = p+1, \dots, n$ and for all $\mathbf{x} \in E$. If $\alpha(\mathbf{x}') > x_p$, then we can rewrite the integral (10.185) as

$$F(\mathbf{x}) = - \int_{x_p}^{\alpha(\mathbf{x}')} \varphi(\mathbf{x}', x) dx \quad (\mathbf{x} \in E).$$

Thus, without loss of generality, we may assume that $\alpha(\mathbf{x}') < x_p$.

We recall from the proof of Theorem 10.38 that since E is convex and $(D_j f)(\mathbf{x}) = 0$ for all $p < j \leq n$ and $\mathbf{x} \in E$, $f(\mathbf{x})$ does not depend on $\mathbf{x}'' = (x_{p+1}, \dots, x_n)$. Thus the function $\varphi : V \rightarrow \mathbb{R}$ is given by

$$\varphi(\mathbf{x}', x_p) = f(\mathbf{x}) = f(x_1, \dots, x_p, \underbrace{x_{p+1}, \dots, x_n}_{\text{They can be taken any values.}}).$$

For $1 \leq i \leq p-1$, we let x_i vary and keep other components of $\mathbf{x}'$ fix so that φ can be thought as a function of the two variables x_i and x_p . In this case, we define $\psi : [\alpha(\mathbf{x}'), x_p] \times [a_i, b_i] \rightarrow \mathbb{R}$ to be given by

$$\psi(x, x_i) = \varphi(\mathbf{x}', x) = f(\mathbf{x}), \quad (10.186)$$

where $\mathbf{x} = (x_1, \dots, x_{p-1}, x, x_{p+1}, \dots, x_n) \in E$.

Lemma 10.20

The function ψ defined in (10.186) satisfies the hypotheses of Theorem 9.42.

Proof of Lemma 10.20. By the definition of φ , we yield from the relation (10.186) that ψ is well-defined on $[\alpha(\mathbf{x}'), x_p] \times [a_i, b_i]$. For each fixed x_i in $[a_i, b_i]$, we have

$$\psi^{x_i}(x) = \psi(x, x_i) = f(\mathbf{x})$$

is a function of one variable. Since $f \in \mathcal{C}'(E)$, it is continuous on E and, in particular, on $[\alpha(\mathbf{x}'), x_p]$. Thus, by Theorem 6.8, we have

$$\psi^{x_i} \in \mathcal{R}$$

on $[\alpha(\mathbf{x}'), x_p]$ for every $x_i \in [a_i, b_i]$. Furthermore, by Theorem 9.21, $D_2 f$ is continuous on E and, in particular, on the compact set $[\alpha(\mathbf{x}'), x_p] \times [a_i, b_i]$. Hence, condition (d) of Theorem 9.42 is satisfied and this completes the proof of lemma. ■

Let's return to the proof of the problem. Now we apply Theorem 9.42 to the function ψ to get

$$(D_i F)(\mathbf{x}) = \frac{\partial}{\partial x_i} \int_{\alpha(\mathbf{x}')}^{x_p} \varphi(\mathbf{x}', x) dx = \frac{\partial}{\partial x_i} \int_{\alpha(\mathbf{x}')}^{x_p} \psi(x, x_i) dx = \int_{\alpha(\mathbf{x}')}^{x_p} \frac{\partial}{\partial x_i} \psi(x, x_i) dx. \quad (10.187)$$

In other words, the integral (10.187) shows that $(D_i F)(\mathbf{x})$ exists for all $\mathbf{x} \in E$ and $1 \leq i \leq p-1$. Finally, we are going to verify

$$(D_p F)(\mathbf{x}) = f(\mathbf{x}) \quad (10.188)$$

for all $\mathbf{x} \in E$. Since $\varphi(\mathbf{x}', x)$ is continuous at x_p , we deduce from Theorem 6.20 (First Fundamental Theorem of Calculus) that, for every $\mathbf{x} \in E$,

$$(D_p F)(\mathbf{x}) = \frac{\partial}{\partial x_p} \int_{\alpha(\mathbf{x}')}^{x_p} \varphi(\mathbf{x}', x) dx = \varphi(\mathbf{x}', x_p) = f(\mathbf{x})$$

which is exactly the relation (10.188). This proves **Claim 2** and so completes the proof of the problem. ■

10.6 Problems on vector fields

Problem 10.30*Rudin Chapter 10 Exercise 30.*

Proof. Let $\mathbf{A}$ be the matrix in the question. As $\mathbf{N} = (\alpha_2\beta_3 - \alpha_3\beta_2)\mathbf{e}_1 + (\alpha_3\beta_1 - \alpha_1\beta_3)\mathbf{e}_2 + (\alpha_1\beta_2 - \alpha_2\beta_1)\mathbf{e}_3$, we have

$$|\mathbf{N}|^2 = (\alpha_2\beta_3 - \alpha_3\beta_2)^2 + (\alpha_3\beta_1 - \alpha_1\beta_3)^2 + (\alpha_1\beta_2 - \alpha_2\beta_1)^2. \quad (10.189)$$

On the other hand, direct computation shows that

$$\begin{aligned} \det \mathbf{A} &= \alpha_1[\beta_2(\alpha_1\beta_2 - \alpha_2\beta_1) - \beta_3(\alpha_3\beta_1 - \alpha_1\beta_3)] - \alpha_2[\beta_1(\alpha_1\beta_2 - \alpha_2\beta_1) - \beta_3(\alpha_2\beta_3 - \alpha_3\beta_2)] \\ &\quad + \alpha_3[\beta_1(\alpha_3\beta_1 - \alpha_1\beta_3) - \beta_2(\alpha_2\beta_3 - \alpha_3\beta_2)] \\ &= (\alpha_1\beta_2 - \alpha_2\beta_1)(\alpha_1\beta_2 - \alpha_2\beta_1) + (\alpha_3\beta_1 - \alpha_1\beta_3)(\alpha_3\beta_1 - \alpha_1\beta_3) + (\alpha_2\beta_3 - \alpha_3\beta_2)(\alpha_2\beta_3 - \alpha_3\beta_2) \\ &= (\alpha_1\beta_2 - \alpha_2\beta_1)^2 + (\alpha_3\beta_1 - \alpha_1\beta_3)^2 + (\alpha_2\beta_3 - \alpha_3\beta_2)^2. \end{aligned} \quad (10.190)$$

Thus the first assertion follows immediately from the expressions (10.189) and (10.190).

Since $T(\mathbf{e}_1) = \alpha_1\mathbf{e}_1 + \alpha_2\mathbf{e}_2 + \alpha_3\mathbf{e}_3$ and $T(\mathbf{e}_2) = \beta_1\mathbf{e}_1 + \beta_2\mathbf{e}_2 + \beta_3\mathbf{e}_3$, it yields

$$\mathbf{N} \cdot T(\mathbf{e}_1) = \alpha_1\alpha_2\beta_3 - \alpha_1\alpha_3\beta_2 + \alpha_2\alpha_3\beta_1 - \alpha_2\alpha_1\beta_3 + \alpha_3\alpha_1\beta_2 - \alpha_3\alpha_2\beta_1 = 0$$

and

$$\mathbf{N} \cdot T(\mathbf{e}_2) = \beta_1\alpha_2\beta_3 - \beta_1\alpha_3\beta_2 + \beta_2\alpha_3\beta_1 - \beta_2\alpha_1\beta_3 + \beta_3\alpha_1\beta_2 - \beta_3\alpha_2\beta_1 = 0.$$

Hence we have $\mathbf{N} \cdot T(\mathbf{e}_1) = \mathbf{N} \cdot T(\mathbf{e}_2) = 0$. We complete the proof of the problem. ■

Problem 10.31*Rudin Chapter 10 Exercise 31.*

Proof.

(a) Let $\nabla h = (D_1h)\mathbf{e}_1 + (D_2h)\mathbf{e}_2 + (D_3h)\mathbf{e}_3$ so that

$$\mathbf{F} = g\nabla h = g(D_1h)\mathbf{e}_1 + g(D_2h)\mathbf{e}_2 + g(D_3h)\mathbf{e}_3.$$

Thus the divergence of $\mathbf{F}$ is given by

$$\begin{aligned} \nabla \cdot \mathbf{F} &= D_1F_1 + D_2F_2 + D_3F_3 \\ &= D_1(gD_1h) + D_2(gD_2h) + D_3(gD_3h) \\ &= (D_1gD_1h + gD_1^2h) + (D_2gD_2h + gD_2^2h) + (D_3gD_3h + gD_3^2h) \\ &= g(D_1^2h + D_2^2h + D_3^2h) + (D_1gD_1h + D_2gD_2h + D_3gD_3h) \\ &= g\nabla^2h + (\nabla g) \cdot (\nabla h). \end{aligned}$$

(b) As a remark, we call the first formula as the Green's first identity and the second one as the Green's second identity. Besides, there is a third identity, namely the Green's third identity, see [14, Eqn. (5.198), p. 362].

Since $\mathbf{F}$ is a vector field of class $\mathcal{C}'$ in the open set E and Ω is a closed subset of E with positively oriented boundary, Theorem 10.51 (Divergence Theorem) together with part (a) imply that

$$\int_{\Omega} [g\nabla^2h + (\nabla g) \cdot (\nabla h)] dV = \int_{\partial\Omega} (\mathbf{F} \cdot \mathbf{n}) dA = \int_{\partial\Omega} [(g\nabla h) \cdot \mathbf{n}] dA = \int_{\partial\Omega} g \frac{\partial h}{\partial \mathbf{n}} dA. \quad (10.191)$$

By interchanging the roles of g and h in the formula (10.191), we obtain

$$\int_{\Omega} [h \nabla^2 g + (\nabla h) \cdot (\nabla g)] dV = \int_{\partial\Omega} h \frac{\partial g}{\partial \mathbf{n}} dA. \quad (10.192)$$

Next, by subtracting the formula (10.192) from the formula (10.191), we yield

$$\int_{\Omega} (g \nabla^2 h - h \nabla^2 g) dV = \int_{\partial\Omega} \left(g \frac{\partial h}{\partial \mathbf{n}} - h \frac{\partial g}{\partial \mathbf{n}} \right) dA. \quad (10.193)$$

(c) Suppose that $\nabla^2 h = 0$. If $g = 1$, then it follows from the Green identity (10.193) that

$$\begin{aligned} \int_{\partial\Omega} \left(1 \times \frac{\partial h}{\partial \mathbf{n}} - h \frac{\partial(1)}{\partial \mathbf{n}} \right) dA &= \int_{\Omega} [1 \times \nabla^2 h - h \nabla^2(1)] dV \\ \int_{\partial\Omega} \frac{\partial h}{\partial \mathbf{n}} dA &= \int_{\Omega} \nabla^2 h dV \\ &= 0. \end{aligned}$$

To complete the last part, put $g = h$ into the Green's first identity (10.192), we have

$$\begin{aligned} \int_{\Omega} [h \nabla^2 h + \|\nabla h\|^2] dV &= \int_{\partial\Omega} h \frac{\partial h}{\partial \mathbf{n}} dA \\ \int_{\Omega} \|\nabla h\|^2 dV &= \int_{\partial\Omega} h \frac{\partial h}{\partial \mathbf{n}} dA - \int_{\Omega} h \nabla^2 h dV. \end{aligned}$$

Since $\nabla^2 h = 0$ in E and $h = 0$ on $\partial\Omega$, we have

$$\int_{\Omega} \|\nabla h\|^2 dV = 0$$

which implies that $\|\nabla^2 h\| = 0$ on Ω . Thus we obtain that

$$(Dh_1)(\mathbf{x}) = (D_2 h)(\mathbf{x}) = (D_3 h)(\mathbf{x}) = 0 \quad (10.194)$$

for every $\mathbf{x} \in \Omega$.

so every component of the interior Ω° of Ω is also open in $\mathbb{R}^3$ by Lemma 10.17. By the relations (10.194) and Theorem 9.17, we know that

$$h'(\mathbf{x}) = \mathbf{0}$$

on each connected component Ω_α° of Ω° . Then we follow from Problem 9.9 that

$$h(\mathbf{x}) = c_\alpha$$

for every $\mathbf{x} \in \Omega_\alpha^\circ$ for some constant c_α . Since $h \in \mathcal{C}''(E)$, the continuity of h and the hypothesis that $h = 0$ on $\partial\Omega$ ensures that the constant c_α must be 0. Hence we have

$$h(\mathbf{x}) = 0$$

for every $\mathbf{x} \in \Omega$.

(d) Let $E \subseteq \mathbb{R}^2$ be open and Ω be a closed subset of E with positively oriented boundary $\partial\Omega$ in $\mathbb{R}^2$. Suppose that $\mathbf{F} : E \rightarrow \mathbb{R}^3$ is a vector field defined by $\mathbf{F} = \alpha \mathbf{e}_1 + \beta \mathbf{e}_2$. We have to find a two-dimensional version of the divergence theorem. In fact,

Lemma 10.21

We have

$$\int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} ds = \int_{\Omega} \nabla \cdot \mathbf{F} dA,$$

where $\mathbf{n}$ is the outward normal vector to $\partial\Omega$ and ds is the element of arc length. See [20, pp. 422, 423].

Proof of Lemma 10.21. Now we apply Theorem 10.45 (Green's Theorem) with $\alpha = -f_2$, $\beta = f_1$ and $\mathbf{F} = f_1\mathbf{e}_1 + f_2\mathbf{e}_2$, we see that

$$\begin{aligned}\int_{\Omega} \nabla \cdot \mathbf{F} \, dA &= \int_{\Omega} \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} \right) dA \\ &= \int_{\partial\Omega} (-f_2 \, dx + f_1 \, dy) \\ &= \int_{\partial\Omega} (f_1, f_2) \cdot (dy, -dx) \\ &= \int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n} \, ds.\end{aligned}$$

This completes the proof of Lemma 10.21. ■

Let $f_1 = gD_1h$ and $f_2 = gD_2h$. Then $\mathbf{F} = g\nabla h$ and we have

$$\begin{aligned}\nabla \cdot \mathbf{F} &= D_1(gD_1h) + D_2(gD_2h) \\ &= (D_1gD_1h + gD_1^2h) + (D_2gD_2h + gD_2^2h) \\ &= g\nabla^2h + (\nabla g) \cdot (\nabla h).\end{aligned}$$

By Lemma 10.21, we have

$$\int_{\Omega} [g\nabla^2h + (\nabla g) \cdot (\nabla h)] \, dA = \int_{\partial\Omega} g\nabla h \cdot \mathbf{n} \, ds = \int_{\partial\Omega} g \frac{\partial h}{\partial \mathbf{n}} \, ds \quad (10.195)$$

which is the Green's first identity in the plane. Similar as the analysis in part (b), the Green's second identity in the plane can be derived from the formula (10.195):

$$\int_{\Omega} (g\nabla^2h - h\nabla^2g) \, dA = \int_{\partial\Omega} \left(g \frac{\partial h}{\partial \mathbf{n}} - h \frac{\partial g}{\partial \mathbf{n}} \right) \, dA.$$

We have completed the proof of the problem. ■

Problem 10.32

Rudin Chapter 10 Exercise 32.

Proof. We remark that for a detailed deduction of a parametrization a Möbius band, readers please refer to [7, pp. 108-110]. Besides, we also emphasize again that $-\Gamma_3(\theta, t)$ means that reverse of the curve $\Gamma_3(\theta, t)$, not the vector $-\Gamma_3(\theta, t)$, see the last two paragraphs in [21, pp. 268, 269].

Now we can start to prove the assertions. The figure of the parameter domain D is given in Figure 10.18:

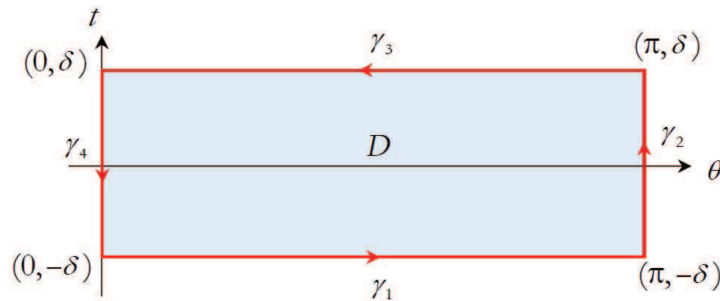


Figure 10.18: The parameter domain D .

Furthermore, the figure of a Möbius band is shown in 10.19.^w

^wThe figure is taken from <http://brookstonbeerbulletin.com/mobius-beer/>.



Figure 10.19: The figure of the Möbius band.

We prove the assertions as follows:

- **Computation of $\partial\Phi$.** By definition, D is a 2-cell with boundary $\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4$. By the analysis on the bottom on [21, p. 271], we know that

$$\partial\Phi = \Phi(\gamma_1) + \Phi(\gamma_2) + \Phi(\gamma_3) + \Phi(\gamma_4) = \Gamma_1 + \Gamma_2 + \Gamma_3 + \Gamma_4. \quad (10.196)$$

- **Computation of $\Phi(\mathbf{p}_1)$, $\Phi(\mathbf{p}_2)$, $\Phi(\mathbf{p}_3)$ and $\Phi(\mathbf{p}_4)$.** By direct computation, we have

$$\begin{aligned} \Phi(\mathbf{p}_1) &= \Phi(0, -\delta) = ((1 + \delta \sin 0) \cos 0, (1 + \delta \sin 0) \sin 0, -\delta) = (1, 0, -\delta) = \mathbf{a}, \\ \Phi(\mathbf{p}_2) &= \Phi(\pi, -\delta) = ((1 + \delta \sin \pi) \cos 2\pi, (1 + \delta \sin \pi) \sin 2\pi, -\delta \cos \pi) = (1, 0, \delta) = \mathbf{b}, \\ \Phi(\mathbf{p}_3) &= \Phi(\pi, \delta) = ((1 - \delta \sin \pi) \cos 2\pi, (1 - \delta \sin \pi) \sin 2\pi, \delta \cos \pi) = (1, 0, -\delta) = \mathbf{a}, \\ \Phi(\mathbf{p}_4) &= \Phi(0, \delta) = ((1 - \delta \sin 0) \cos 0, (1 - \delta \sin 0) \sin 0, \delta \cos 0) = (1, 0, \delta) = \mathbf{b}. \end{aligned}$$

- **Description of $\partial\Phi$.** Since $\gamma_1 = [\mathbf{p}_1, \mathbf{p}_2]$, we have $\gamma_1 = \{(\theta, -\delta) \mid 0 \leq \theta \leq \pi\}$. Therefore, if $(x, y, z) \in \Gamma_1$, then we have

$$x = (1 + \delta \sin \theta) \cos 2\theta, \quad y = (1 + \delta \sin \theta) \sin 2\theta, \quad z = -\delta \cos \theta$$

and so the projection of Γ_1 , denoted by $\lambda_1(\theta, \delta)$ whose domain is $[0, \pi]$, into the (x, y) -plane is given by

$$\lambda_1(\theta, \delta) = (1 + \delta \sin \theta) \cos 2\theta + i(1 + \delta \sin \theta) \sin 2\theta = (1 + \delta \sin \theta)(\cos 2\theta + i \sin 2\theta).$$

Now λ_1 is obviously a continuously differentiable closed curve in $\mathbb{C}$, so Problem 8.23 implies that

$$\begin{aligned} \text{Ind}(\lambda_1) &= \frac{1}{2\pi i} \int_0^\pi \frac{\lambda_1'(\theta, \delta)}{\lambda_1(\theta, \delta)} d\theta \\ &= \frac{1}{2\pi i} \int_0^\pi \frac{2(1 + \delta \sin \theta)(-\sin 2\theta + i \cos 2\theta) + \delta \cos \theta(\cos 2\theta + i \sin 2\theta)}{(1 + \delta \sin \theta)(\cos 2\theta + i \sin 2\theta)} d\theta \\ &= \frac{1}{\pi i} \int_0^\pi \frac{-\sin 2\theta + i \cos 2\theta}{\cos 2\theta + i \sin 2\theta} d\theta + \frac{1}{2\pi i} \int_0^\pi \frac{\delta \cos \theta}{1 + \delta \sin \theta} d\theta \\ &= \frac{1}{\pi i} \int_0^\pi i d\theta + \frac{1}{2\pi i} \int_0^\pi d(\ln(1 + \delta \sin \theta)) \\ &= 1 + \ln(1 + \delta \sin \theta) \Big|_0^\pi \\ &= 1. \end{aligned}$$

Since $\gamma_2 = [\mathbf{p}_2, \mathbf{p}_3]$, $\gamma_2 = \{(\pi, t) \mid -\delta \leq t \leq \delta\}$. Thus we have

$$\Gamma_2 = \Phi(\gamma_2)$$

$$\begin{aligned}
&= \Phi(\pi, t) \\
&= \{((1 - t \sin \pi) \cos 2\pi, (1 - t \sin \pi) \sin 2\pi, t \cos \pi) \mid -\delta \leq t \leq \delta\} \\
&= \{(1, 0, -t) \mid -\delta \leq t \leq \delta\} \\
&= [\mathbf{b}, \mathbf{a}].
\end{aligned}$$

For Γ_3 , we note that $\gamma_3 = [\mathbf{p}_3, \mathbf{p}_4]$, so

$$\gamma_3 = \{(\theta, \delta) \mid \theta \text{ decreases from } \pi \text{ to } 0\} = \{(\pi - \varphi, \delta) \mid \varphi \in [0, \pi]\}.$$

Therefore, if $(x, y, z) \in \Gamma_3$, then we have

$$x = [1 - \delta \sin(\pi - \varphi)] \cos 2(\pi - \varphi), \quad y = [1 - \delta \sin(\pi - \varphi)] \sin 2(\pi - \varphi), \quad z = \delta \cos(\pi - \varphi)$$

and so the projection of Γ_3 , denoted by $\lambda_3(\varphi, \delta)$ with domain $[0, \pi]$, into the (x, y) -plane is given by

$$\begin{aligned}
\lambda_3(\varphi, \delta) &= [1 - \delta \sin(\pi - \varphi)] \cos 2(\pi - \varphi) + i[1 - \delta \sin(\pi - \varphi)] \sin 2(\pi - \varphi) \\
&= [1 - \delta \sin(\pi - \varphi)] [\cos 2(\pi - \varphi) + i \sin 2(\pi - \varphi)] \\
&= (1 - \delta \sin \varphi)(\cos 2\varphi - i \sin 2\varphi).
\end{aligned}$$

Now λ_3 is obviously a continuously differentiable closed curve in $\mathbb{C}$, so Problem 8.23 implies that

$$\begin{aligned}
\text{Ind}(\lambda_3) &= \frac{1}{2\pi i} \int_0^\pi \frac{\lambda_3'(\varphi, \delta)}{\lambda_3(\varphi, \delta)} d\varphi \\
&= \frac{1}{2\pi i} \int_0^\pi \frac{2(1 - \delta \sin \varphi)(-\sin 2\varphi - i \cos 2\varphi) - \delta \cos \varphi(\cos 2\varphi - i \sin 2\varphi)}{(1 - \delta \sin \varphi)(\cos 2\varphi - i \sin 2\varphi)} d\varphi \\
&= \frac{1}{\pi i} \int_0^\pi \frac{-\sin 2\varphi - i \cos 2\varphi}{\cos 2\varphi - i \sin 2\varphi} d\varphi - \frac{1}{2\pi i} \int_0^\pi \frac{\delta \cos \varphi}{1 - \delta \sin \varphi} d\varphi \\
&= \frac{1}{\pi i} \int_0^\pi (-i) d\varphi + \frac{1}{2\pi i} \int_0^\pi d(\ln(1 - \delta \sin \varphi)) \\
&= -1 + \ln(1 - \delta \sin \varphi) \Big|_0^\pi \\
&= -1.
\end{aligned}$$

Since $\gamma_4 = [\mathbf{p}_4, \mathbf{p}_1]$, $\gamma_4 = \{(0, -t) \mid -\delta \leq t \leq \delta\}$. Thus we have

$$\begin{aligned}
\Gamma_4 &= \Phi(\gamma_4) \\
&= \Phi(0, -t) \\
&= \{((1 + t \sin 0) \cos 0, (1 + t \sin 0) \sin 0, -t \cos 0) \mid -\delta \leq t \leq \delta\} \\
&= \{(1, 0, -t) \mid -\delta \leq t \leq \delta\} \\
&= [\mathbf{b}, \mathbf{a}].
\end{aligned}$$

Hence we deduce from the above analysis and the expression (10.196) that

$$\partial\Phi = \Gamma_1 + \Gamma_3 + 2\Gamma_2.$$

- **The “geometric” boundary of M .** Recall that $\gamma_1 = \{(\theta, -\delta) \mid 0 \leq \theta \leq \pi\}$, so

$$\Gamma_1(0, -\delta) = \Phi(0, -\delta) = (1, 0, -\delta) = \mathbf{a} \quad \text{and} \quad \Gamma_1(\pi, -\delta) = \Phi(\pi, -\delta) = (1, 0, \delta) = \mathbf{b}.$$

By the previous analysis, we notice that Γ_1 and Γ_3 spiral up from $\mathbf{a}$ to $\mathbf{b}$. Furthermore, we have

$$\Gamma_1(\theta, -\delta) = ((1 + \delta \sin \theta) \cos 2\theta, (1 + \delta \sin \theta) \sin 2\theta, -\delta \cos \theta) \quad (10.197)$$

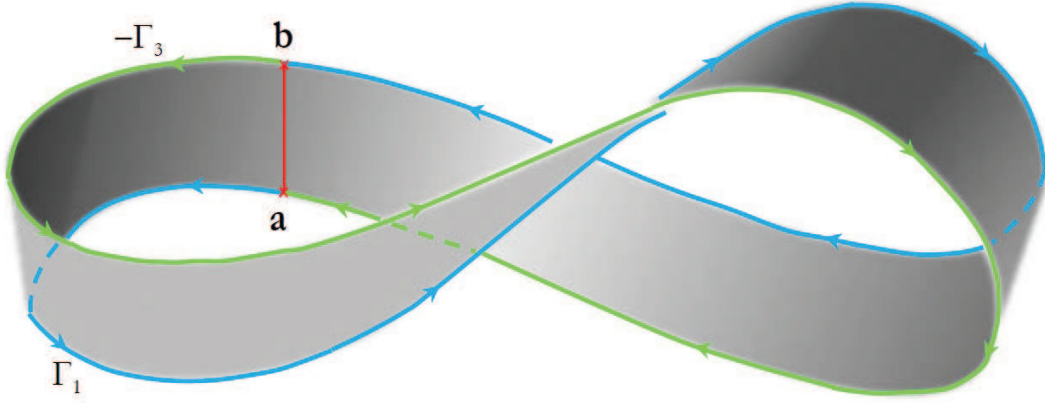
for $\theta \in [0, \pi]$ and

$$\Gamma_3(\theta, \delta) = ((1 - \delta \sin \theta) \cos 2\theta, (1 - \delta \sin \theta) \sin 2\theta, \delta \cos \theta), \quad (10.198)$$

where θ decreases from π to 0. Therefore,

$$\Gamma = \Gamma_1 - \Gamma_3$$

is a curve that spirals up from $\mathbf{a}$ to $\mathbf{b}$ and then back to $\mathbf{a}$, see Figure 10.20 below where the blue and green loci refer to Γ_1 and $-\Gamma_3$ respectively.

Figure 10.20: The “geometric” boundary of M .

- **A representation of Γ .** We claim that a parametrization of $\Gamma : [0, 2\pi] \rightarrow \mathbb{R}^3 \setminus \{\mathbf{0}\}$ is given by

$$\begin{aligned} x &= (1 + \delta \sin \theta) \cos 2\theta, \\ y &= (1 + \delta \sin \theta) \sin 2\theta, \\ z &= -\delta \cos \theta. \end{aligned} \quad (10.199)$$

In fact, when $\theta \in [0, \pi]$, the parametrization (10.199) is exactly the trace of Γ_1 (see the parametrization (10.197)). To see why the loci of the parametrization (10.199) for $\theta \in [\pi, 2\pi]$ is the “reverse” of Γ_3 , we put $\theta = \pi + \varphi$ into the parametrization (10.199) to get

$$\begin{aligned} x &= (1 - \delta \sin \varphi) \cos 2\varphi, \\ y &= (1 - \delta \sin \varphi) \sin 2\varphi, \\ z &= \delta \cos \varphi, \end{aligned} \quad (10.200)$$

where φ increases from 0 to π . Thus the parametrization (10.200) is exactly the reverse of the parametrization (10.198).

- **Proof of $\Gamma \neq \partial\Phi$.** If η is the 1-form discussed in Problems 10.21 and 10.22, then since η is closed in $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ by Example 10.36 (i.e., $d\eta = 0$), we get from Theorem 10.33 (Stokes’ Theorem) that

$$\int_{\partial\Phi} \eta = \int_{\Phi} d\eta = 0. \quad (10.201)$$

Since $-\delta \leq t \leq \delta$ and $-1 \leq \sin \theta \leq 1$, we have

$$-\delta \leq -t \sin \theta \leq \delta$$

and then

$$x^2 + y^2 = [(1 - t \sin \theta) \cos 2\theta]^2 + [(1 - t \sin \theta) \sin 2\theta]^2 = (1 - t \sin \theta)^2 \geq (1 - \delta)^2 > 0.$$

Thus η is well-defined on the projection of M into the (x, y) -plane. In this case, we follow from the previous calculation of the winding numbers of the projections of Γ_1 and Γ_3 into the (x, y) -plane around the origin that

$$\int_{\Gamma} \eta = \int_{\Gamma_1 - \Gamma_3} \eta = \int_{\Gamma_1} \eta \overset{\substack{\text{usual} \\ \text{subtrac-} \\ \text{tion}}}{=} \int_{\Gamma_3} \eta = 2\pi - (-2\pi) = 4\pi. \quad (10.202)$$

Hence integrals (10.201) and (10.202) shows that $\Gamma \neq \partial\Phi$.

This completes the proof of the problem. ■

CHAPTER 11

The Lebesgue Theory

11.1 Further properties of integrable functions

Problem 11.1

Rudin Chapter 11 Exercise 1.

Proof. We follow the hint. Let $E_n \subseteq E$ and $E_n = \{x \in E \mid f(x) > \frac{1}{n}\}$ for all $n = 1, 2, \dots$. We write

$$A = \bigcup_{n=1}^{\infty} E_n. \quad (11.1)$$

Suppose that $\mu(A) = 0$. Since $E_n \subset A$ for every n , we follow from [21, Eqn. (18), p. 304] that $0 \leq \mu(E_n) \leq \mu(A) = 0$ for every n . In other words, $\mu(E_n) = 0$ for every n . Conversely, suppose that $\mu(E_n) = 0$ for every positive integer n . We apply Theorem 11.8(b) to the definition (11.1) to get

$$0 \leq \mu(A) \leq \sum_{n=1}^{\infty} \mu(E_n) = 0.$$

Thus we have $\mu(A) = 0$. To prove our result, we need the following lemma:

Lemma 11.1

Let X be a measurable space and E be a measurable set. Let K_F be the *characteristic function* of F , where $F \subseteq E$ is measurable. Then we have $I_E(K_F) = \mu(E \cap F)$ and

$$\int_E K_F \, d\mu = \mu(F).$$

Furthermore, if f is a nonnegative simple measurable function defined on E and given by

$$f(x) = \sum_{k=1}^m d_k K_{F_k}(x),$$

where $d_k > 0$ and $F_k \subseteq E$ are measurable sets for $k = 1, \dots, m$, then we have

$$\int_E f \, d\mu = \sum_{k=1}^m d_k \mu(F_k).$$

Lemma of 11.1. We remark that some books call the K_F the *indicator function* of F . By Definition 11.19, K_F is also a simple function so that Definition 11.21 gives

$$I_E(K_F) = \mu(F). \quad (11.2)$$

Suppose that

$$s(x) = \sum_{i=1}^n c_i K_{E_i}(x),$$

where $x \in X$ and $c_1, \dots, c_n > 0$. By Definition 11.19, we know that $E_i \cap E_j = \emptyset$ for all $i \neq j$. Let

$$0 \leq s \leq K_F.$$

If $x \in E_i$ but $x \notin F$, then we have $0 < c_i \leq 0$, a contradiction. Thus $x \in F$ so that $E_i \subseteq F$ for every $i = 1, \dots, n$ which imply that

$$\bigcup_{i=1}^n E_i \subseteq F \subseteq E.$$

In addition, for every $x \in E_i \subseteq F$, we have $0 < c_i \leq 1$. Since each E_i is measurable by Proposition A.7, we follow from this, the additive property of μ , the inequality [21, Eqn. (8), p. 302] and the relation (11.2) that

$$I_E(s) = \sum_{i=1}^n c_i \mu(E \cap E_i) = \sum_{i=1}^n c_i \mu(E_i) \leq \sum_{i=1}^n \mu(E_i) = \mu\left(\bigcup_{i=1}^n E_i\right) \leq \mu(F) = I_E(K_F).$$

Hence, by Definition 11.21, we obtain

$$\int_E K_F d\mu = \sup I_E(s) = I_E(K_F) = \mu(F). \quad (11.3)$$

This proves the first assertion.

For the second assertion, we let f be the simple function given by the lemma. By the first assertion, we have

$$K_{F_i} \in \mathcal{L}(\mu)$$

on E . Then we deduce from Theorem 11.29, Remark 11.23(c) and the identity (11.3) that

$$\int_E f d\mu = \int_E \sum_{k=1}^m d_k K_{F_k} d\mu = \sum_{k=1}^m d_k \int_E K_{F_k} d\mu = \sum_{k=1}^m d_k \mu(F_k).$$

This proves the second assertion, completing the proof of the lemma. ■

We go back to the proof of Problem 11.1. By the definition (11.1), we have

$$A = \{x \in E \mid f(x) > 0\}.$$

We have to show that

$$\mu(A) = 0.$$

Since $E_n \subset E$ for every n , we must have $A \subset E$. Assume that $\mu(E_n) > 0$ for some n . By the result of the first part, we have $\mu(A) > 0$. By Definition 11.21 and Remark 11.23(b), we note the following

$$\int_E f d\mu = \sup I_E(s) \geq \sup I_{E_n}(s) = \int_{E_n} f d\mu \geq \frac{\mu(E_n)}{n} > 0$$

which contradicts the hypothesis. This completes the analysis of the problem. ■

Problem 11.2

Rudin Chapter 11 Exercise 2.

Proof. Let

$$A_+ = \{x \in E \mid f(x) \geq 0\} \quad \text{and} \quad A_- = \{x \in E \mid f(x) < 0\}.$$

It is clear that $E = A_+ \cup A_-$. Since f is measurable, Theorem 11.15 implies that A_+ and A_- are measurable. By the hypothesis, we have

$$\int_{A_+} f \, d\mu = \int_{A_-} (-f) \, d\mu = 0.$$

Since $f \geq 0$ on A_+ and $-f > 0$ on A_- , Problem 11.1 shows that $f(x) = 0$ almost everywhere on A_+ and $-f(x) = 0$ almost everywhere on A_- . Hence, these facts imply that

$$f(x) = 0$$

almost everywhere on E , completing the proof of the problem. ■

Problem 11.3

Rudin Chapter 11 Exercise 3.

Proof. Suppose that X is the measurable space we are considering. Let A be the set of points x at which the sequence $\{f_n(x)\}$ converges. Recall from the *Cauchy criterion for convergence* (see [21, p. 54]) that a sequence converges in $\mathbb{R}$ if and only if it is a Cauchy sequence. Therefore, if $x \in A$, then for every $n \in \mathbb{N}$, there exists an integer N such that

$$|f_p(x) - f_q(x)| < \frac{1}{n}$$

for all $p, q \geq N$ so that the set A is equal to

$$\bigcap_{n=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{p, q \geq N} \left\{ x \in X \mid |f_p(x) - f_q(x)| < \frac{1}{n} \right\}. \quad (11.4)$$

By the proof of Theorem 11.16, we have

$$\left\{ x \in X \mid |f_p(x) - f_q(x)| < \frac{1}{n} \right\} = \left\{ x \in X \mid f_p(x) - f_q(x) < \frac{1}{n} \right\} \cap \left\{ x \in X \mid f_p(x) - f_q(x) > -\frac{1}{n} \right\}. \quad (11.5)$$

Since f_p and f_q are measurable, $f_p - f_q$ is also measurable by Theorem 11.18. Thus the two sets in the right-hand side of (11.5) are measurable too. Since there exists a σ -ring $\mathfrak{M}$ of subsets of X , we see from Definition 11.1 that the set (11.4) is measurable. This completes the proof of the problem. ■

Problem 11.4

Rudin Chapter 11 Exercise 4.

Proof. Suppose that M is a positive constant such that

$$|g(x)| \leq M$$

for all $x \in E$. Since $f \in \mathcal{L}(\mu)$ on E , Remark 11.23(d) and Theorem 11.26 imply that

$$Mf \in \mathcal{L}(\mu) \quad \text{and} \quad |f| \in \mathcal{L}(\mu)$$

on E respectively. Since f and g are measurable, fg is also measurable by Theorem 11.18. It is obvious that

$$|fg| \leq M|f|$$

on E , so we follow from Theorem 11.27 that

$$fg \in \mathcal{L}(\mu)$$

on E . This completes the proof of the problem. ■

Problem 11.5*Rudin Chapter 11 Exercise 5.*

Proof. For the first assertion, we note that if $x \in [0, \frac{1}{2}]$, then $\{f_{2k}\}$ is a subsequence of $\{f_n\}$ such that

$$\lim_{k \rightarrow \infty} f_{2k}(x) = g(x) = 0.$$

If $x \in (\frac{1}{2}, 1]$, then $\{f_{2k+1}\}$ is a subsequence of $\{f_n\}$ such that

$$\lim_{k \rightarrow \infty} f_{2k+1}(x) = g(1-x) = 0.$$

By definition, we have the desired result that

$$\liminf_{n \rightarrow \infty} f_n(x) = 0 \quad (0 \leq x \leq 1).$$

For the second assertion, when $n = 2k$, we have

$$\int_0^1 f_n(x) dx = \int_0^1 f_{2k}(x) dx = \int_0^1 g(x) dx = \int_0^{\frac{1}{2}} 0 dx + \int_{\frac{1}{2}}^1 1 dx = \frac{1}{2}.$$

Similarly, when $n = 2k + 1$, we have

$$\int_0^1 f_n(x) dx = \int_0^1 f_{2k+1}(x) dx = \int_0^1 g(1-x) dx = \int_0^{\frac{1}{2}} 1 dx + \int_{\frac{1}{2}}^1 0 dx = \frac{1}{2}.$$

This completes the proof of the problem.^a ■

Problem 11.6*Rudin Chapter 11 Exercise 6.*

Proof. Given that $\epsilon > 0$. Let N be a positive integer such that $\frac{1}{N} < \epsilon$. On the one hand, for all $x \in \mathbb{R}$ and all integers $n \geq N$, we have

$$|f_n(x) - 0| = |f_n(x)| \leq \frac{1}{n} \leq \frac{1}{N} < \epsilon.$$

By Definition 7.7, $f_n(x) \rightarrow 0$ uniformly on $\mathbb{R}$. On the other hand, we have

$$\int_{-\infty}^{\infty} f_n dx = \int_{-\infty}^n f_n dx + \int_{-n}^n f_n dx + \int_n^{\infty} f_n dx = \int_{-n}^n \frac{1}{n} dx = 2.$$

This completes the proof of the problem.^b ■

11.2 The Riemann integrals and the Lebesgue integrals**Problem 11.7***Rudin Chapter 11 Exercise 7.*

^aThis problem says that the strict inequality in Theorem 11.31 (Fatou's Theorem) can happen.

^bAs the problem mentioned, the uniform convergence *does not* imply the dominated convergence in the case of *infinite* measure. Compare with [21, Eqn. (84), p.321].

Proof. Suppose that f is bounded on $[a, b]$ and α is a monotonically increasing function on $[a, b]$. As the hint says, we consider the regular measure μ_α induced by α in Example 11.6(b):

$$\begin{aligned}\mu_\alpha([a, b)) &= \alpha(b-) - \alpha(a-), \\ \mu_\alpha([a, b]) &= \alpha(b+) - \alpha(a-), \\ \mu_\alpha((a, b]) &= \alpha(b+) - \alpha(a+), \\ \mu_\alpha((a, b)) &= \alpha(b-) - \alpha(a+).\end{aligned}\tag{11.6}$$

However, the domain of the monotonically increasing function considered there is $\mathbb{R}$, so we have to extend the function α in question as follows:

$$\alpha(x) = \begin{cases} \alpha(a), & \text{if } x < a; \\ \alpha(x), & \text{if } a \leq x \leq b; \\ \alpha(b), & \text{if } x > b. \end{cases}$$

Then such extended $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ is now monotonically increasing on $\mathbb{R}$ and therefore, the regular measure μ_α is well-defined as in (11.6).

Motivating from Theorem 6.10, we assume that α is continuous on $[a, b]$.^c By this, the measure in the definition (11.6) unifies to the same value:

$$\mu_\alpha([a, b)) = \mu_\alpha([a, b]) = \mu_\alpha((a, b]) = \mu_\alpha((a, b)) = \alpha(b) - \alpha(a).$$

Now we claim that

“ $f \in \mathcal{R}(\alpha)$ on $[a, b]$ if and only if f is continuous a.e. on $[a, b]$ with respect to μ_α .”

To prove this claim, we mainly modify the proof of Theorem 11.33 and divide our proof into the following steps:

- **Step 1: Upper and lower Riemann-Stieltjes integrals.** By Theorem 3.17 and Theorem 6.4, there exists a sequence $\{P_k\}$ of partitions of $[a, b]$ such that P_{k+1} is a refinement of P_k , the distance between adjacent points of P_k is less than $\frac{1}{k}$ and

$$\lim_{k \rightarrow \infty} L(P_k, f, \alpha) = \mathcal{R} \int_a^b f d\alpha \quad \text{and} \quad \lim_{k \rightarrow \infty} U(P_k, f, \alpha) = \mathcal{R} \int_a^b f d\alpha.\tag{11.7}$$

- **Step 2: Representations of $\lim_{k \rightarrow \infty} L(P_k, f, \alpha)$ and $\lim_{k \rightarrow \infty} U(P_k, f, \alpha)$.** Let $P_k = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$, where $x_0 = a$ and $x_n = b$. For $1 \leq i \leq n$, let $E_i = (x_{i-1}, x_i]$ and

$$M_i = \sup_{x \in E_i} f(x) \quad \text{and} \quad m_i = \inf_{x \in E_i} f(x).$$

Next, we define the functions $L_k, U_k : [a, b] \rightarrow \mathbb{R}$ by

$$\begin{aligned}L_k(x) &= \begin{cases} 0, & \text{if } x = a; \\ m_i, & \text{if } x \in E_i \text{ for } 1 \leq i \leq n, \end{cases} \\ &= \sum_{i=1}^n m_i K_{E_i}(x)\end{aligned}\tag{11.8}$$

and

$$\begin{aligned}U_k(x) &= \begin{cases} 0, & \text{if } x = a; \\ M_i, & \text{if } x \in E_i \text{ for } 1 \leq i \leq n, \end{cases} \\ &= \sum_{i=1}^n M_i K_{E_i}(x).\end{aligned}\tag{11.9}$$

^cWe remark that this is a very restrictive requirement and one can find more general and less restrictive necessary and sufficient conditions for $f \in \mathcal{R}(\alpha)$ on $[a, b]$ in [13].

Thus the representations (11.8) and (11.9) of L_k and U_k imply that

$$\mathcal{R} \int_a^b L_k d\alpha = \mathcal{R} \int_a^a 0 d\alpha + \sum_{i=1}^n \mathcal{R} \int_{x_{i-1}}^{x_i} L_k d\alpha = \sum_{i=1}^n m_i \mathcal{R} \int_{x_{i-1}}^{x_i} d\alpha = \sum_{i=1}^n m_i \Delta\alpha_i = L(P_k, f, \alpha)$$

and

$$\mathcal{R} \int_a^b U_k d\alpha = \mathcal{R} \int_a^a 0 d\alpha + \sum_{i=1}^n \mathcal{R} \int_{x_{i-1}}^{x_i} U_k d\alpha = \sum_{i=1}^n M_i \mathcal{R} \int_{x_{i-1}}^{x_i} d\alpha = \sum_{i=1}^n M_i \Delta\alpha_i = U(P_k, f, \alpha),$$

where $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$ (since α is continuous on $[a, b]$) for $1 \leq i \leq n$. In conclusion, we have

$$L(P_k, f, \alpha) = \mathcal{R} \int_a^b L_k d\alpha \quad \text{and} \quad U(P_k, f, \alpha) = \mathcal{R} \int_a^b U_k d\alpha.$$

Putting these into the limits (11.7), we have

$$\lim_{k \rightarrow \infty} \mathcal{R} \int_a^b L_k d\alpha = \mathcal{R} \int_a^b f d\alpha \quad \text{and} \quad \lim_{k \rightarrow \infty} \mathcal{R} \int_a^b U_k d\alpha = \mathcal{R} \int_a^b f d\alpha.$$

- **Step 3: Computations of $\lim_{k \rightarrow \infty} L_k$ and $\lim_{k \rightarrow \infty} U_k$.** By the construction of the sequence of partitions $\{P_k\}$, we know that P_{k+1} refines P_k , so it leads from the definitions (11.8) and (11.9) that

$$L_1(x) \leq L_2(x) \leq \cdots \leq f(x) \leq \cdots \leq U_2(x) \leq U_1(x) \quad (11.10)$$

for all $x \in [a, b]$. Apply Theorem 3.14 (Monotone Convergence Theorem) to the inequalities (11.10), there exist functions $L, U : [a, b] \rightarrow \mathbb{R}$ such that

$$L(x) = \lim_{k \rightarrow \infty} L_k(x) \quad \text{and} \quad U(x) = \lim_{k \rightarrow \infty} U_k(x)$$

and

$$L(x) \leq f(x) \leq U(x) \quad (11.11)$$

for all $x \in [a, b]$.

- **Step 4: Computations of $\lim_{k \rightarrow \infty} \int_{[a,b]} L_k d\mu_\alpha$ and $\lim_{k \rightarrow \infty} \int_{[a,b]} U_k d\mu_\alpha$.** Since each E_i is a Borel set of $[a, b]$ (see [22, Sec. 1.11, p. 12]), it is measurable. By Definition 11.19, all L_k and U_k are simple functions. Thus they are measurable by Proposition A.7 and so the functions L and U are also (bounded) measurable by Theorem 11.17. By Theorem 11.28 (Lebesgue's Monotone Convergence Theorem)^d, we have

$$\lim_{k \rightarrow \infty} \int_{[a,b]} L_k d\mu_\alpha = \int_{[a,b]} L d\mu_\alpha \quad \text{and} \quad \lim_{k \rightarrow \infty} \int_{[a,b]} U_k d\mu_\alpha = \int_{[a,b]} U d\mu_\alpha.$$

- **Step 5: Representations of $U(P_k, f, \alpha)$ and $L(P_k, f, \alpha)$ in terms of Lebesgue integrals.** Since L_k and U_k are simple functions, it follows from the representation (11.8) and [21, Eqn. (52) & (54), p. 314] that^e

$$\int_{[a,b]} L_k d\mu_\alpha = \int_{\{a\}} L_k d\mu_\alpha + \sum_{i=1}^n \int_{E_i} L_k d\mu_\alpha = \sum_{i=1}^n I_{E_i}(L_k) = \sum_{i=1}^n \sum_{j=1}^n m_j \mu(E_i \cap E_j). \quad (11.12)$$

Since $E_i \cap E_j = \emptyset$ if $i \neq j$, we further deduce from the expression (11.12) that

$$\int_{[a,b]} L_k d\mu_\alpha = \sum_{i=1}^n m_i \mu(E_i) = \sum_{i=1}^n m_i \Delta\alpha = L(P_k, f, \alpha). \quad (11.13)$$

Similarly, we can show from the representation (11.9) of U_k that

$$\int_{[a,b]} U_k d\mu_\alpha = \sum_{i=1}^n M_i \Delta\alpha = U(P_k, f, \alpha). \quad (11.14)$$

^dThe constant “zero” in [21, Eqn. (64), p.318] can be replaced by any constant.

^eNotice that $\mu_\alpha(a) = \alpha(a) - \alpha(a) = 0$.

- **Step 6: Relations between Riemann-Stieltjes integrals and Lebesgue integrals.** Now by substituting the Lebesgue integrals (11.13) and (11.14) into the limits (11.7), we obtain

$$\lim_{k \rightarrow \infty} \int_{[a,b]} L_k d\mu_\alpha = \mathcal{R} \int_a^b f d\alpha \quad \text{and} \quad \lim_{k \rightarrow \infty} \int_{[a,b]} U_k d\mu_\alpha = \mathcal{R} \int_a^b f d\alpha. \quad (11.15)$$

By the result of **Step 3**, the expressions (11.15) can be written as

$$\int_{[a,b]} L d\mu_\alpha = \mathcal{R} \int_a^b f d\alpha \quad \text{and} \quad \int_{[a,b]} U d\mu_\alpha = \mathcal{R} \int_a^b f d\alpha. \quad (11.16)$$

- **Step 7: Equivalent statement of $f \in \mathcal{R}(\alpha)$ on $[a, b]$.** We note that $f \in \mathcal{R}(\alpha)$ on $[a, b]$ if and only if

$$\int_{[a,b]} (U - L) d\mu_\alpha = 0 \quad (11.17)$$

by the integrals (11.16). Since $U(x) \geq L(x)$ on $[a, b]$, the integral (11.17) holds if and only if $U(x) = L(x)$ a.e. on $[a, b]$ respect to μ_α by Problem 11.1. In other words, we have

$$“f \in \mathcal{R}(\alpha) \text{ on } [a, b] \text{ if and only if } U(x) = L(x) \text{ a.e. on } [a, b] \text{ with respect to } \mu_\alpha.” \quad (11.18)$$

To finish our proof, we have to show that

$$“U(x) = L(x) \text{ a.e. on } [a, b] \text{ w.r.t. } \mu_\alpha \text{ if and only if } f \text{ is continuous a.e. on } [a, b] \text{ w.r.t. } \mu_\alpha.” \quad (11.19)$$

This equivalent relation will be shown in the remaining three steps.

- **Step 8: A lemma.** We need the following lemma:^f

Lemma 11.2

Suppose that x_0 is chosen such that $x_0 \notin P_k$ for every $k \in \mathbb{N}$ and $U(x_0) = L(x_0)$. Then f is *continuous* at x_0 .

Proof of Lemma 11.2. Assume that f was discontinuous at x_0 . Then there exists a $\epsilon > 0$ and a sequence $\{x_p\} \subset [a, b]$ such that $|x_p - x_0| < \frac{1}{p}$ but

$$|f(x_p) - f(x_0)| \geq \epsilon$$

for all $p \in \mathbb{N}$. Now for every $k \in \mathbb{N}$, we can find an i such that $x_0 \in E_i^\circ$. (Recall that E_i° denotes the interior of E_i , see Problem 2.9.) Thus it implies that $x_p \in E_i$ for all large enough p . Hence we have

$$U_k(x_0) - L_k(x_0) = \sum_{i=1}^n (M_i - m_i) K_{E_i}(x_0) = M_i - m_i \geq \sup_{a, b \in E_i} |f(a) - f(b)| \geq |f(x_p) - f(x)| \geq \epsilon,$$

but this implies that

$$U(x_0) - L(x_0) \geq \epsilon$$

which contradicts the hypothesis $U(x_0) = L(x_0)$. This completes the proof of our lemma. ■

By Lemma 11.2, if f is discontinuous at $x \in [a, b]$, then $x \in S_1 \cup S_2$, where

$$S_1 = \{x \in [a, b] \mid U(x) \neq L(x)\} \quad \text{and} \quad S_2 = \{x \in [a, b] \mid x \in P_k \text{ for some } k \in \mathbb{N}\}.$$

Therefore, f is continuous a.e. on $[a, b]$ with respect to μ_α if and only if

$$\mu_\alpha(S_1) = \mu_\alpha(S_2) = 0.$$

^fLemma 11.2 and its proof are basically motivated by **Step 7** of the proof of Theorem C in [13].

- **Step 9: Proofs of $\mu_\alpha(S_1) = \mu_\alpha(S_2) = 0$.** Since α is continuous on $[a, b]$, we always have

$$\mu_\alpha(\{x\}) = 0 \quad (11.20)$$

for every $x \in [a, b]$. Since each P_k is at most countable, their union

$$\bigcup_{k=1}^{\infty} P_k$$

is countable by Theorem 2.12. Since $S_2 \subseteq \bigcup_{k=1}^{\infty} P_k$, we know from this, the measure (11.20), Definition 11.7 and Theorem 11.8(b) that

$$0 \leq \mu_\alpha(S_2) \leq \mu_\alpha\left(\bigcup_{k=1}^{\infty} P_k\right) \leq \sum_{k=1}^{\infty} \mu_\alpha(P_k) = 0.$$

In other words, we always have $\mu_\alpha(S_2) = 0$.

By the definition of S_1 , $U(x) = L(x)$ a.e. on $[a, b]$ with respect to μ_α if and only if $\mu_\alpha(S_1) = 0$.

- **Step 10: Final step.** By the first part of **Step 9**, we always have $\mu_\alpha(S_2) = 0$. Then, by the result of **Step 8** and the second part of **Step 9**, we know that f is continuous a.e. on $[a, b]$ with respect to μ_α if and only if $U(x) = L(x)$ a.e. on $[a, b]$ with respect to μ_α . Hence our claim follows from the relations (11.18) and (11.19) in **Step 7**.

This completes the proof of the problem. ■

Problem 11.8

Rudin Chapter 11 Exercise 8.

Proof. Since $f \in \mathcal{R}$ on $[a, b]$, we know from Definition 6.1 that f is supposed to be bounded on $[a, b]$. By Theorem 11.33(b), f is continuous almost everywhere on $[a, b]$. By Theorem 6.20, F is differentiable almost everywhere on $[a, b]$ and

$$F'(x) = f(x),$$

where f is continuous at $x \in [a, b]$, finishing the proof of the problem. ■

Problem 11.9

Rudin Chapter 11 Exercise 9.

Proof. Given that $f \in \mathcal{L}$ on $[a, b]$. Let $x_0 \in [a, b]$ and $\{x_n\}$ be a sequence in $[a, b]$ converging to x_0 , where $x_n \neq x_0$ for all $n \in \mathbb{N}$. We want to show that

$$\lim_{n \rightarrow \infty} F(x_n) = F(x_0).$$

Now for each $n \in \mathbb{N}$, we define the sequences of intervals $\{E_n\}$ and functions $\{f_n\}$ given by $E_n = [a, x_n]$ and

$$f_n(x) = \begin{cases} f(x), & x \in E_n; \\ 0, & x \in [a, b] \setminus E_n \end{cases} \quad (11.21)$$

respectively. By the definition (11.21), we can write

$$f_n = f \times K_{E_n}$$

on $[a, b]$. Since f and K_{E_n} are real-valued measurable functions defined on $[a, b]$, it follows from Theorem 11.18 that f_n is also measurable on $[a, b]$. Furthermore, we have

$$\lim_{n \rightarrow \infty} f_n(x) = f(x)K_{[a, x_0]}(x)$$

for all $x \in [a, b]$. Therefore, this shows that

$$|f_n(x)| \leq |f(x)|$$

for all $n = 1, 2, 3, \dots$ and $x \in [a, b]$. Hence, we deduce easily from Theorem 11.32 (Lebesgue's dominated convergence theorem) that

$$\lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} \int_a^{x_n} f(x) dx = \lim_{n \rightarrow \infty} \int_{[a, b]} f_n dm = \int_{[a, b]} f K_{[a, x_0]} dm = \int_a^{x_0} f(x) dx = F(x_0)$$

as desired. Hence F is continuous at x_0 and since x_0 is arbitrary, F is continuous on $[a, b]$. This completes the proof of the problem. ■

11.3 Functions of classes $\mathcal{L}$ and $\mathcal{L}^2$

Problem 11.10

Rudin Chapter 11 Exercise 10.

Proof. Suppose that $g(x) = 1$ for every $x \in X$. Then we have

$$E = \{x \in X \mid g(x) > a\} = \begin{cases} X, & \text{if } a \leq 1; \\ \emptyset, & \text{if } a > 1. \end{cases} \quad (11.22)$$

Since X is a measurable space, we know from Definition 11.12 that $X \in \mathfrak{M}$. Since $\mathfrak{M}$ is a σ -ring, we follow from Definition 11.1 that $\emptyset = X \setminus X \in \mathfrak{M}$. Thus X and $\emptyset$ are measurable and then the set E defined in (11.22) is also measurable. Since

$$\int_X |1|^2 d\mu = \int_X d\mu = \mu(X) < +\infty,$$

we get from Definition 11.34 that $1 \in \mathcal{L}^2(\mu)$ on X . By Theorem 11.35, we have

$$f = f \cdot 1 \in \mathcal{L}(\mu)$$

on X .

On the other hand, this is false if $\mu(X) = +\infty$. For example, we consider $X = \mathbb{R}$ and the function

$$f(x) = \frac{1}{1 + |x|}.$$

Then we have $\mu(\mathbb{R}) = +\infty$ and

$$\int_{\mathbb{R}} |f|^2 dm = \int_{-\infty}^{\infty} \frac{dx}{(1 + |x|)^2} \leq \int_{-\infty}^{\infty} \frac{dx}{x^2} = 0$$

so that $f \in \mathcal{L}$ on $\mathbb{R}$. However, since

$$\int_{\mathbb{R}} f^+ dm = \int_0^{\infty} \frac{dx}{1 + x} = \lim_{N \rightarrow \infty} \int_0^N \frac{dx}{1 + x} = \lim_{N \rightarrow \infty} \ln(1 + N) \rightarrow \infty,$$

it follows from Definition 11.22 that $f \notin \mathcal{L}$ on $\mathbb{R}$. This ends the proof of the problem. ■

Problem 11.11*Rudin Chapter 11 Exercise 11.*

Proof. By definition, we have

$$d(f, g) = \int_X |f - g| d\mu.$$

Recall from Remarks 11.25 that two functions $f, g \in \mathcal{L}(\mu)$ on X are supposed to be the *same* if $f = g$ almost everywhere on X . We first show that $\mathcal{L}(\mu)$ is a metric space by checking Definition 2.15.

- (a) If $f \neq g$ on $E \subseteq X$ with $\mu(E) > 0$, then we have from Remark 11.23(f) that $f, g \in \mathcal{L}(\mu)$ on E and $X \setminus E$. By Theorem 11.29, we have $f - g \in \mathcal{L}(\mu)$ on E and $X \setminus E$. Thus Theorem 11.26 implies that

$$|f - g| \in \mathcal{L}(\mu)$$

on E and $X \setminus E$. Now we get from Theorem 11.24(b) that

$$d(f, g) = \int_X |f - g| d\mu = \int_{X \setminus E} |f - g| d\mu + \int_E |f - g| d\mu. \quad (11.23)$$

Since $|f - g| \geq 0$ on X and $|f - g| \geq \delta > 0$ on E for some fixed δ , we apply Remark 11.23(c) to the relation (11.23) to get

$$d(f, g) \geq \int_E |f - g| d\mu > \delta \mu(E) > 0.$$

Furthermore, if $f = \hat{f}$ almost everywhere on X , then we deduce from Remark 11.23(e) that

$$d(f, \hat{f}) = \int_X |f - \hat{f}| d\mu = \int_{X \setminus E} |f - \hat{f}| d\mu + \int_E |f - \hat{f}| d\mu = 0,$$

where $E = \{x \in X \mid f(x) \neq \hat{f}(x)\}$ which has measure zero.

- (b) Now $d(f, g) = d(g, f)$ is obvious because of the identity $|f - g| = |g - f|$.

- (c) Suppose that $f, g, h \in \mathcal{L}(\mu)$ on X . Then it follows from the triangle inequality

$$|f - g| \leq |f - h| + |h - g|$$

that the inequality $d(f, g) \leq d(f, h) + d(h, g)$ always holds.

Hence $\mathcal{L}(\mu)$ is a metric space.

Next, we want to show that $\mathcal{L}(\mu)$ is complete. By Definition 3.12, we have to show that every Cauchy sequence converges in $\mathcal{L}(\mu)$: Suppose that $\{f_n\}$ is a Cauchy sequence in $\mathcal{L}(\mu)$. We must show that $\{f_n\}$ converges to a function f in $\mathcal{L}(\mu)$. We mainly follow the proof of Theorem 11.42 here and we prove our result in two steps.

- **Step 1: Construction of the function f .** Now we can find a sequence $\{n_k\}, k = 1, 2, \dots$, such that

$$\int_X |f_{n_k} - f_{n_{k+1}}| d\mu = d(f_{n_k}, f_{n_{k+1}}) < \frac{1}{2^k} \quad (k = 1, 2, 3, \dots)$$

which implies that

$$\sum_{k=1}^{\infty} \int_X |f_{n_k} - f_{n_{k+1}}| d\mu \leq \sum_{k=1}^{\infty} \frac{1}{2^k} = 1. \quad (11.24)$$

By Theorem 11.30, we may interchange the integration and summation on the left-hand side in the inequality (11.24) to get

$$\int_X \sum_{k=1}^{\infty} |f_{n_k} - f_{n_{k+1}}| d\mu \leq 1.$$

Therefore, we must have

$$\sum_{k=1}^{\infty} |f_{n_k} - f_{n_{k+1}}| < +\infty \quad (11.25)$$

almost everywhere on X . Apply Theorem 3.45 to the inequality (11.25), we know that the series

$$\sum_{k=1}^{\infty} (f_{n_k} - f_{n_{k+1}})$$

converges almost everywhere on X . Since the k th partial sum of the series

$$\sum_{k=1}^{\infty} (f_{n_k} - f_{n_{k+1}})$$

is $f_{n_{k+1}}(x) - f_{n_1}(x)$, we see that the limit

$$f(x) = \lim_{k \rightarrow \infty} f_{n_k}(x)$$

defines $f(x)$ for almost all $x \in X$.

- **Step 2: Convergence of the sequence of functions $\{f_n\}$.** We claim that $\{f_n\}$ converges to f in $\mathcal{L}(\mu)$. Given that $\epsilon > 0$. Since $\{f_n\}$ is Cauchy, there exists a positive integer N such that

$$d(f_n, f_m) < \epsilon \quad (11.26)$$

for all $m, n \geq N$. If $n_k > N$, then we may apply Theorem 11.31 (Fatou's Theorem) to the sequence

$$\{F_j = |f_{n_j} - f_{n_k}|\}$$

of nonnegative measurable functions to get

$$|f - f_{n_k}| = \liminf_{j \rightarrow \infty} |f_{n_j} - f_{n_k}|$$

and

$$d(f, f_{n_k}) \leq \liminf_{j \rightarrow \infty} d(f_{n_j}, f_{n_k}). \quad (11.27)$$

By the assumption (11.26), we deduce from the inequality (11.27) that

$$d(f, f_{n_k}) = \int_X |f - f_{n_k}| d\mu \leq \epsilon. \quad (11.28)$$

By the result of the inequality (11.28), we have

$$|f - f_{n_k}| \in \mathcal{L}(\mu).$$

Since $\{f_{n_k}\}$ is a sequence of measurable functions, f is also measurable by the corollary of Theorem 11.17 and then Theorem 11.18 ensures that

$$f - f_{n_k}$$

is also a measurable function. Thus $(f - f_{n_k})^+$ and $(f - f_{n_k})^-$ are measurable by the corollary of Theorem 11.17 again. Furthermore, we see that

$$|(f - f_{n_k})^+| = (f - f_{n_k})^+ \leq |f - f_{n_k}| \quad \text{and} \quad |(f - f_{n_k})^-| = (f - f_{n_k})^- \leq |f - f_{n_k}| \quad (11.29)$$

hold almost everywhere on X . If we apply Theorem 11.27 to the inequalities (11.29), then we can conclude that

$$(f - f_{n_k})^+, (f - f_{n_k})^- \in \mathcal{L}(\mu).$$

Therefore, since

$$f - f_{n_k} = (f - f_{n_k})^+ - (f - f_{n_k})^-,$$

it follows from Theorem 11.29 that

$$f - f_{n_k} \in \mathcal{L}(\mu).$$

By this and the fact that $f = (f - f_{n_k}) + f_{n_k}$, we see that

$$f \in \mathcal{L}(\mu).$$

Since ϵ is arbitrary, we can derive from the inequality (11.28) that

$$\lim_{k \rightarrow \infty} d(f, f_{n_k}) = 0.$$

Now we obtain from the triangle inequality

$$d(f, f_n) \leq d(f, f_{n_k}) + d(f_{n_k}, f_n) \quad (11.30)$$

that the sequence $\{f_n\}$ converges to f in $\mathcal{L}(\mu)$ if we take both n and n_k become large enough on the right-hand side in the equality (11.30).

This completes the proof of the problem. ■

Problem 11.12

Rudin Chapter 11 Exercise 12.

Proof. The answer is affirmative. It is clear that the set $[0, 1]$ is Lebesgue measurable. Let $\alpha \in [0, 1]$ and $\{x_n\} \subseteq [0, 1]$ be a sequence such that $x_n \rightarrow \alpha$ as $n \rightarrow \infty$. By condition (b), for each *fixed* $x_n \in [0, 1]$, the function f_n defined by

$$f_n(y) = f(x_n, y)$$

is continuous in y . Thus, by Example 11.14, each $f_n(y)$ is Lebesgue measurable.

Furthermore, the continuity of $f(x, y)$ for *fixed* $y \in [0, 1]$ (condition (c)) shows that

$$\lim_{n \rightarrow \infty} f_n(y) = \lim_{n \rightarrow \infty} f(x_n, y) = f(\alpha, y).$$

By condition (a), we know that

$$|f_n(y)| = |f(x_n, y)| \leq 1$$

for all $x_n, y \in [0, 1]$. Since $1 \in \mathcal{L}$ on $[0, 1]$, we follow from Theorem 11.32 (Lebesgue's dominated convergence theorem) that

$$\lim_{n \rightarrow \infty} g(x_n) = \lim_{n \rightarrow \infty} \int_0^1 f(x_n, y) dy = \lim_{n \rightarrow \infty} \int_0^1 f_n(y) dy = \int_0^1 f(\alpha, y) dy = g(\alpha).$$

Since the sequence $\{x_n\}$ is arbitrary, Theorem 4.6 implies that g is continuous at α . This completes the proof of the problem. ■

Problem 11.13

Rudin Chapter 11 Exercise 13.

Proof. Let E be the set of these points. By Definition 11.34,

$$\|f\| = \left\{ \int_{-\pi}^{\pi} |f(x)|^2 dx \right\}^2$$

defines a norm in $\mathcal{L}^2$. By direct computation, we have

$$\|f_n\|^2 = \int_{-\pi}^{\pi} |f(x)|^2 dx = \int_{-\pi}^{\pi} \sin^2 nx dx = \int_{-\pi}^{\pi} \left(\frac{1}{2} - \frac{\cos 2x}{2}\right) dx = \pi$$

for all $n = 1, 2, 3, \dots$. In other words, we have

$$\|f_n\| = \sqrt{\pi}$$

for all $n = 1, 2, 3, \dots$. Therefore, E is bounded. Furthermore, it is also easy to see that if $n \neq m$, then we have

$$\begin{aligned} \|f_n - f_m\|^2 &= \int_{-\pi}^{\pi} |f_n(x) - f_m(x)|^2 dx \\ &= \int_{-\pi}^{\pi} (\sin nx - \sin mx)^2 dx \\ &= \int_{-\pi}^{\pi} (\sin^2 nx - 2 \sin nx \sin mx + \sin^2 mx) dx \\ &= 2\pi - 2 \int_{-\pi}^{\pi} \sin nx \sin mx dx \\ &= 2\pi. \end{aligned} \tag{11.31}$$

This means that the neighborhood $N_{\sqrt{\pi}}(f_n)$ contains only the point f_n itself, so any point f_n of E is *not* a limit point of E by Theorem 2.20. In other words, $E' = \emptyset$. By Problem 2.1, $\emptyset \subset E$. Hence E is closed by Definition 2.18(d).

Assume that E was compact. By Theorem 11.42, $\mathcal{L}^2$ is a complete metric space. By the identity (11.31), E has *no* Cauchy sequence. Thus E cannot be complete by Definition 3.12. However, since E is assumed to be compact, we deduce from Theorem 3.11(b) that E is complete, a contradiction. Hence E is not compact and this ends the proof of the problem. ■

Problem 11.14

Rudin Chapter 11 Exercise 14.

Proof. Let $f : X \rightarrow \mathbb{C}$ and V be an open set in $\mathbb{C}$. By definition, we have $f = u + iv$, where u and v are real. Recall that f is measurable if and only if u and v are measurable.

Suppose that $f^{-1}(V)$ is measurable for every open set V in $\mathbb{C}$. It is easy to see that $V_x = (a, \infty) \times \mathbb{R}$ and $V_y = \mathbb{R} \times (a, \infty)$ are open sets in $\mathbb{C}$ for every $a \in \mathbb{R}$, so $f^{-1}(V_x)$ and $f^{-1}(V_y)$ are measurable. By direct checking, we know that

$$f^{-1}(V_x) = f^{-1}((a, \infty) \times \mathbb{R}) = \{x \in X \mid f(x) = u(x) + iv(x) \in (a, \infty) \times \mathbb{R}\} = \{x \in X \mid u(x) > a\}$$

and

$$f^{-1}(V_y) = f^{-1}(\mathbb{R} \times (a, \infty)) = \{x \in X \mid f(x) = u(x) + iv(x) \in \mathbb{R} \times (a, \infty)\} = \{x \in X \mid v(x) > a\}.$$

Since a is arbitrary, we obtain from Definition 11.13 that u and v are measurable. Hence f is measurable too.

Conversely, suppose that f is measurable so that u and v are measurable. Let $(p, q) \in V$. Since V is open in $\mathbb{C}$, there exists a $\delta > 0$ such that $N_{\sqrt{2}\delta}((p, q)) \subseteq V$.[§] By definition, we have

$$N_{\sqrt{2}\delta}((p, q)) = \{(x, y) \in \mathbb{C} \mid |(x, y) - (p, q)| < \sqrt{2}\delta\}.$$

Now the open square

$$R_\delta((p, q)) = \{(x, y) \in X \mid x \in (p - \delta, p + \delta) \text{ and } y \in (q - \delta, q + \delta)\}$$

[§]Notice that the number δ depends on the point (p, q) .

is obviously a subset of $N_{\sqrt{2}\delta}((p, q))$ so that $R_\delta((p, q)) \subseteq V$ and then

$$V = \bigcup_{(p, q) \in V} R_\delta((p, q)), \quad (11.32)$$

see Figure 11.1 for details.

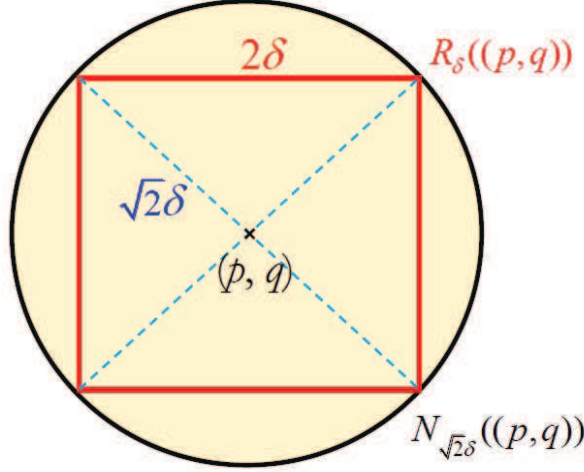


Figure 11.1: The open square $R_\delta((p, q))$ and the neighborhood $N_{\sqrt{2}\delta}((p, q))$.

By the definition in Problem 2.22, we see that the set of points

$$Q = \{p_n + iq_n \mid p_n, q_n \in \mathbb{Q} \text{ and } n \in \mathbb{N}\}$$

is clearly a *countable dense subset* of $\mathbb{C}$ and so $\mathbb{C}$ is separable. Thus, by Problem 2.23, it has a countable base $\{V_m\}$. Since $R_\delta((p, q))$ is an open subset of V , there exists a $m \in \mathbb{N}$ such that

$$(p, q) \in V_m \subseteq R_\delta((p, q)) \subseteq V. \quad (11.33)$$

Therefore, we know from the representation (11.32) that V is a union of subcollection of $\{V_m\}$. Since every infinite subset of a countable set is also countable (see Theorem 2.8), there exists a subsequence $\{m_k\}$ such that

$$V = \bigcup_{k=1}^{\infty} V_{m_k}$$

and furthermore, we deduce from the relation (11.33) that

$$V = \bigcup_{k=1}^{\infty} R_\delta((p_k, q_k)), \quad (11.34)$$

where $\{(p_k, q_k)\} \subseteq V$. Now we follow from the representation (11.34) that

$$\begin{aligned} f^{-1}(V) &= f^{-1}\left(\bigcup_{k=1}^{\infty} R_\delta((p_k, q_k))\right) \\ &= \bigcup_{k=1}^{\infty} \{x \in X \mid f(x) \in R_\delta((p_k, q_k))\} \\ &= \bigcup_{k=1}^{\infty} \{x \in X \mid u(x) \in (p_k - \delta, p_k + \delta) \text{ and } v(x) \in (q_k - \delta, q_k + \delta)\} \end{aligned}$$

$$= \bigcup_{k=1}^{\infty} \{x \in X \mid p_k - \delta < u(x) < p_k + \delta\} \cap \{x \in X \mid q_k - \delta < v(x) < q_k + \delta\}.$$

Since u and v are measurable, the set

$$\{x \in X \mid p_k - \delta < u(x) < p_k + \delta\} \cap \{x \in X \mid q_k - \delta < v(x) < q_k + \delta\}$$

is measurable for each $k \in \mathbb{N}$. Since the set of all measurable sets of $\mathbb{R}$ is a σ -ring, Definition 11.1 implies that $f^{-1}(V)$ is also measurable. This completes the proof of the problem. ■

Problem 11.15

Rudin Chapter 11 Exercise 15.

Proof. By Definition 11.4, an interval in $(0, 1]$ is one of the following forms:

$$(0, b), (0, b], [a, b], [a, b), (a, b), (a, b), \quad (11.35)$$

where $0 < a \leq b \leq 1$. If $A \in \mathcal{R}$, then by a similar argument as in the proof of Proposition A.2(c), A can be written as a finite union of *disjoint* intervals of $(0, 1]$. By a similar argument as in the proof of Proposition A.3(d), any two different decompositions of A into disjoint intervals of $(0, 1]$ give the same value of $\phi(A)$.

Suppose that $A, B \in \mathcal{R}$. By the previous paragraph, we have

$$A = I_1 \cup \cdots \cup I_m \quad \text{and} \quad B = J_1 \cup \cdots \cup J_n,$$

where $\{I_1, \dots, I_m\}$ and $\{J_1, \dots, J_n\}$ are collections of disjoint intervals of $(0, 1]$. Let $A \cap B = \emptyset$. Since

$$A \cap B = \bigcup_{p=1}^m \bigcup_{q=1}^n (I_p \cap J_q),$$

we must have

$$I_p \cap J_q = \emptyset$$

for all $p = 1, \dots, m$ and $q = 1, \dots, n$. In other words, $\{I_1, \dots, I_m, J_1, \dots, J_n\}$ is a set of disjoint intervals of $(0, 1]$. Therefore, we have

$$\begin{aligned} \phi(A \cup B) &= \phi(I_1 \cup \cdots \cup I_m \cup J_1 \cup \cdots \cup J_n) \\ &= [\phi(I_1) + \cdots + \phi(I_m)] + [\phi(J_1) + \cdots + \phi(J_n)] \\ &= \phi(A) + \phi(B). \end{aligned}$$

Thus ϕ is an additive set function on $\mathcal{R}$.

Let $A = (0, \frac{1}{2}) \in \mathcal{R}$ so that $\phi(A) = 1 + \frac{1}{2} = \frac{3}{2}$. Given that $\epsilon = \frac{1}{3}$. If $F \subset A$ and F is closed, then F does not contain the end point 0 and thus

$$\phi(F) + \epsilon \leq 1 + \frac{1}{3} < \phi(A).$$

By Definition 11.5, ϕ cannot be regular. For each $k \in \mathbb{N}$, we suppose that $A_k = (\frac{1}{2^{k+1}}, \frac{1}{2^k}]$. Then we have

$$A_i \cap A_j = \emptyset$$

for $i \neq j$ and

$$\left(0, \frac{1}{2}\right] = A_1 \cup A_2 \cup \cdots.$$

We notice that

$$\phi(A_k) = \frac{1}{2^k} - \frac{1}{2^{k+1}}$$

so that

$$\sum_{k=1}^{\infty} \phi(A_k) = \sum_{k=1}^{\infty} \left(\frac{1}{2^k} - \frac{1}{2^{k+1}} \right) = \frac{1}{2},$$

but $\phi((0, \frac{1}{2}]) = 1 + \frac{1}{2} = \frac{3}{2}$. Therefore, we have

$$\phi\left(\bigcup_{k=1}^{\infty} A_k\right) \neq \sum_{n=1}^{\infty} \phi(A_k),$$

i.e., ϕ cannot be extended to a countably additive set function on a σ -ring, completing the proof of the problem. ■

Problem 11.16

Rudin Chapter 11 Exercise 16.

Proof. Let $f_k(x) = \sin n_k x$. By definition, we have

$$E = \{x \in (-\pi, \pi) \mid \{f_k(x)\} \text{ converges}\}.$$

By Example 11.14, since each f_k is continuous, $\{f_k\}$ is a sequence of measurable functions. Thus, it follows from Problem 11.3 that E is measurable. Let A be a measurable subset of E . By the hypothesis, we let $f : E \rightarrow \mathbb{R}$ be the function such that

$$\lim_{k \rightarrow \infty} f_k(x) = f(x)$$

on E . Since each f_k is continuous on $(-\pi, \pi)$, Example 11.14 says that each f_k is measurable on $(-\pi, \pi)$. By Remark 11.23(f), f_k is measurable on E and also on A . Since

$$|f_k(x)| = |\sin n_k x| \leq 1$$

for all $k \in \mathbb{N}$ and $x \in A$ and $1 \in \mathcal{L}$ on A , we deduce from Theorem 11.32 (Lebesgue's dominated Convergence Theorem) and the first part of the given hint that

$$\int_A f \, dx = \lim_{k \rightarrow \infty} \int_A f_k \, dx = \lim_{k \rightarrow \infty} \int_A \sin n_k x \, dx = 0. \quad (11.36)$$

By Problem 11.2, the relation (11.36) implies that $f(x) = 0$ almost everywhere on E and then

$$f^2(x) = 0$$

almost everywhere on E . Now, on the one hand, we apply Theorem 11.32 (Lebesgue's dominated Convergence Theorem) to the sequence of measurable functions $\{f_k^2\}$ (the measurability of f_k^2 follows from Theorem 11.18) to obtain that

$$\lim_{k \rightarrow \infty} \int_A f_k^2 \, dx = \int_A f^2 \, dx = 0. \quad (11.37)$$

On the other hand, since $2 \sin^2 n_k x = 1 - \cos 2n_k x$, we obtain from the second part of the hint that

$$\lim_{k \rightarrow \infty} \int_A f_k^2 \, dx = \lim_{k \rightarrow \infty} \int_A \sin^2 n_k \, dx = \frac{1}{2} \lim_{k \rightarrow \infty} \int_A (1 - \cos 2n_k x) \, dx = \frac{m(A)}{2}. \quad (11.38)$$

Combining the limits (11.37) and (11.38), we derive

$$\frac{m(A)}{2} = \lim_{k \rightarrow \infty} \int_A f_k^2 \, dx = \int_A f^2 \, dx = 0$$

which is equivalent to $m(A) = 0$. Since A is arbitrary, we have $m(E) = 0$ in particular. This finishes the proof of the problem. ■

Problem 11.17*Rudin Chapter 11 Exercise 17.*

Proof. Here we must have Theorem 8.12 (Bessel inequality) in terms of Lebesgue's sense. If $\{\phi_n\}$ is orthonormal on a measurable set $X \subseteq \mathbb{R}$ and $f \in \mathcal{L}^2$ on X , then we have

$$f(x) \sim \sum_{n=1}^{\infty} c_n \phi_n(x) \quad \text{and} \quad \sum_{n=1}^{\infty} |c_n|^2 \leq \int_X |f(x)|^2 dx = \|f\|^2,$$

where

$$c_n = \int_X f \bar{\phi}_n dx \quad (n = 1, 2, 3, \dots). \quad (11.39)$$

In particular, we have

$$\lim_{n \rightarrow \infty} c_n = 0. \quad (11.40)$$

Assume that there were infinitely many integers n such that

$$\sin nx \geq \delta$$

for all $x \in E$. Consider the function K_E which is clearly measurable. By Remark 11.23(c), since $|K_E(x)|^2 \leq 1$ on $[-\pi, \pi]$, we have

$$0 < m(E) = \int_{-\pi}^{\pi} |K_E(x)|^2 dx \leq \int_{-\pi}^{\pi} dx = 2\pi.$$

Thus we must have $K_E \in \mathcal{L}^2$ on $[-\pi, \pi]$ by Definition 11.34.

For each $n \in \mathbb{N}$, we define

$$\phi_n(x) = \frac{1}{\sqrt{\pi}} \sin nx.$$

Then we can see easily that $\{\phi_n\}$ forms an orthonormal set of functions on $[-\pi, \pi]$. Now we apply the definition (11.39) to the function $f = K_E$ and the orthonormal system $\{\phi_n\}$ to obtain

$$c_n = \frac{1}{\sqrt{\pi}} \int_X K_E \sin nx dx = \frac{1}{\sqrt{\pi}} \int_E \sin nx dx. \quad (11.41)$$

By our hypothesis, we can deduce from the identity (11.41) that

$$c_n \geq \frac{1}{\sqrt{\pi}} \int_E \delta dx = \frac{\delta m(E)}{\sqrt{\pi}} > 0$$

for infinitely many integers n , but this obviously contradicts the special case (11.40) of the Bessel inequality. Hence there are *at most* finitely many integers n such that $\sin nx \geq \delta$ on E , completing the proof of the problem. ■

Problem 11.18*Rudin Chapter 11 Exercise 18.*

Proof. We remark that there is a mistake in the statement. It still holds if we take $f(x) \equiv 0$ and $g(x) \equiv 1$ on X . Furthermore, the equality holds trivially if $\mu(X) = 0$ by Remark 11.23(e). Therefore, we suppose that $f \not\equiv 0$ almost everywhere and $\mu(X) > 0$ so that

$$\int |f|^2 d\mu \neq 0.$$

We consider the (complex) constant given by

$$\bar{c} = \frac{\int f \bar{g} d\mu}{\int |f|^2 d\mu}. \quad (11.42)$$

Now it is easy to see that

$$\begin{aligned} \int |g - cf|^2 d\mu &= \int (g - cf)(\bar{g} - \overline{cf}) d\mu \\ &= \int (|g|^2 - \overline{cf}g - cf\bar{g} + |c|^2|f|^2) d\mu \\ &= \int |g|^2 d\mu + \frac{|\int f \bar{g} d\mu|^2}{(\int |f|^2 d\mu)^2} \cdot \int |f|^2 d\mu - 2\operatorname{Re} \left(\int cf\bar{g} d\mu \right) \\ &= \int |g|^2 d\mu + \frac{|\int f \bar{g} d\mu|^2}{\int |f|^2 d\mu} - 2\operatorname{Re} \left(\int cf\bar{g} d\mu \right). \end{aligned} \quad (11.43)$$

If $g(x) = cf(x)$ almost everywhere, then $g(x) - cf(x) = 0$ and thus $cf\bar{g} = |g|^2$ almost everywhere. By the identity (11.43), we establish that

$$\int |g|^2 d\mu + \frac{|\int f \bar{g} d\mu|^2}{\int |f|^2 d\mu} = 2\operatorname{Re} \left(\int cf\bar{g} d\mu \right) = 2\operatorname{Re} \left(\int |g|^2 d\mu \right) = 2 \int |g|^2 d\mu$$

which implies that

$$\begin{aligned} \frac{|\int f \bar{g} d\mu|^2}{\int |f|^2 d\mu} &= \int |g|^2 d\mu \\ \left| \int f \bar{g} d\mu \right|^2 &= \int |f|^2 d\mu \int |g|^2 d\mu. \end{aligned} \quad (11.44)$$

On the other hand, if the identity (11.44) holds, then we derive from the definition (11.42) and the identity (11.43) that

$$\begin{aligned} \int |g - cf|^2 d\mu &= 2 \int |g|^2 d\mu - 2\operatorname{Re} \left(c \int f \bar{g} d\mu \right) \\ &= 2 \int |g|^2 d\mu - \frac{2}{\int |f|^2 d\mu} \operatorname{Re} \left(\overline{\int f \bar{g} d\mu} \cdot \int f \bar{g} d\mu \right) \\ &= 2 \int |g|^2 d\mu - \frac{2}{\int |f|^2 d\mu} \cdot \operatorname{Re} \left(\left| \int f \bar{g} d\mu \right|^2 \right) \\ &= 2 \int |g|^2 d\mu - \frac{2}{\int |f|^2 d\mu} \cdot \left| \int f \bar{g} d\mu \right|^2 \\ &= 2 \int |g|^2 d\mu - 2 \int |g|^2 d\mu \\ &= 0. \end{aligned}$$

Since $|g - cf|^2 \geq 0$, we find from Problem 11.1 that

$$|g - cf|^2 = 0$$

almost everywhere. Hence we have our desired result that $g(x) = cf(x)$ almost everywhere. This completes the proof of the problem. ■

APPENDIX A

A proof of Lemma 10.14

We see from the definition (10.154) that Σ^{n-1} is of class $\mathcal{C}''$, so is S . We consider the n -surface $\Psi : [0, 1] \times E \rightarrow \mathbb{R}^n \setminus \{\mathbf{0}\}$ defined by

$$\Psi(t, \phi_1, \dots, \phi_{n-1}) = [1 - t + tf(\phi_1, \dots, \phi_{n-1})]\Sigma^{n-1}(\phi_1, \dots, \phi_{n-1}), \quad (\text{A.1})$$

where $(\phi_1, \dots, \phi_{n-1}) \in E$ and $0 \leq t \leq 1$. For each fixed ϕ_i , define

$$\begin{aligned} & E_{(\phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1})} \\ &= \{(\phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) \in [0, \pi]^{n-2} \times [0, 2\pi] \mid (\phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) \in E\} \\ &= \prod_{\substack{j=1 \\ j \neq i}}^{n-1} [a_j, b_j] \end{aligned}$$

and the mapping

$$\Phi : [0, 1] \times \prod_{\substack{j=1 \\ j \neq i}}^{n-1} [a_j, b_j] \rightarrow \mathbb{R}^n \setminus \{\mathbf{0}\}$$

defined by

$$\Phi(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) = \Psi(t, \phi_1, \dots, \overset{\text{fixed}}{\downarrow} \phi_i, \dots, \phi_{n-1}).$$

Following the idea as in the proof of “**A generalization of Problem 10.22(c)**”, if we write

$$\Phi(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) = (x_1, \dots, x_n),$$

then we put

$$g(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) = 1 - t + tf(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1})$$

into the right-hand side of the definition (A.1) to get

$$x_i = g(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1})h_i(\phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}),$$

where $1 \leq i \leq n$. Since $f \in \mathcal{C}''(D_n)$ and $f > 0$, we must have $g \in \mathcal{C}''(D_n)$ and $g > 0$. Furthermore, since

$$[0, 1] \times \prod_{\substack{j=1 \\ j \neq i}}^{n-1} [a_j, b_j] \quad (\text{A.2})$$

is clearly a $(n-1)$ -cell, it is compact by Theorem 2.40 and then the mapping Φ is a $(n-1)$ -surface of class $\mathcal{C}''$ with (A.2) as the parameter domain to which the formula (10.153) holds. Since $f \in \mathcal{C}''(D_n)$,

Ψ is a n -surface of class $\mathcal{C}''$ in $\mathbb{R}^n \setminus \{\mathbf{0}\}$. Since each $\frac{x_i}{r^n}$ is of class $\mathcal{C}'$ in $\mathbb{R}^n \setminus \{\mathbf{0}\}$, the same is true for the $(n-1)$ -form ω_n by Definition 10.11 and thus it follows from Theorem 10.33 (Stokes' Theorem) and Problem 10.23(a) that

$$\int_{\partial\Psi} \omega_n = \int_{\Psi} d\omega_n = 0. \quad (\text{A.3})$$

To finish the proof of the lemma, we need to compute $\partial\Psi$ by generalizing the relation (10.124). In fact, the definition of Ψ shows that

$$\begin{aligned} \partial\Psi &= \Phi(0, \phi_1, \dots, \phi_{n-1}) - \Phi(1, \phi_1, \dots, \phi_{n-1}) + \sum_{i=1}^{n-1} [\Psi(t, \phi_1, \dots, \textcolor{red}{a}_i, \phi_{i+1}, \dots, \phi_{n-1}) \\ &\quad - \Psi(t, \phi_1, \dots, \textcolor{red}{b}_i, \phi_{i+1}, \dots, \phi_{n-1})] \\ &= S(\phi_1, \dots, \phi_{n-1}) - \Omega(\phi_1, \dots, \phi_{n-1}) + \sum_{i=1}^{n-1} [\Phi_{a_i}(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1}) \\ &\quad - \Phi_{b_i}(t, \phi_1, \dots, \phi_{i-1}, \phi_{i+1}, \dots, \phi_{n-1})]. \end{aligned}$$

The above analysis implies that

$$\int_{\Phi_{a_i}} \omega_n = \int_{\Phi_{b_i}} \omega_n = 0 \quad (1 \leq i \leq n)$$

so that we derive from these and the formula (A.3) that

$$\begin{aligned} \int_S \omega_n - \int_{\Omega} \omega_n &= 0 \\ \int_{\Omega} \omega_n &= \int_S \omega_n. \end{aligned} \quad (\text{A.4})$$

To compute the integral on the right-hand side in the identity (A.4), we consider the following steps:

- **Step 1: Observations from Examples 10.36 and 10.37.** By Examples 10.36 and 10.37 (see also Problem 10.21(a) and Problem 10.22(b)), we have

$$\int_{\gamma} \omega_2 = \int_{\gamma} \eta = \int_{\gamma} \textcolor{red}{(1)} \, dt = \int_0^{2\pi} dt = 2\pi$$

and

$$\int_S \omega_3 = \int_S \zeta = \int_S \textcolor{red}{\sin u} \, du \wedge dv = \int_0^{\pi} \int_0^{2\pi} \sin u \, du \, dv = 4\pi.$$

Thus we have

$$(\omega_2)_{\gamma} = dt \quad \text{and} \quad (\omega_3)_{\Sigma} = \sin u \, du \wedge dv.$$

In other words, we need to find a way to express ω_n in terms of the parameters of a *parametrization of the $(n-1)$ -sphere*. Fortunately, this has been done in (10.154) for the $(n-1)$ -surface (don't forget it is also a $\mathcal{C}'$ -mapping) $\Sigma^{n-1} : D_n \rightarrow E_n$ already.

By Definition 10.21 with carefully chosen n, m, t_i and x_j or the deduction up to [21, Eqn. (72), pp. 264, 265], we have^a

$$\begin{aligned} (\omega_n)_{\Sigma^{n-1}} &= \left[\sum_{i=1}^n (-1)^{i-1} x_i \, dx_1 \wedge \dots \wedge dx_{i-1} \wedge dx_{i+1} \wedge \dots \wedge dx_n \right]_{\Sigma^{n-1}} \\ &= \left[\sum_{i=1}^n (-1)^{i-1} x_i(\phi_1, \dots, \phi_{n-1}) \det \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)}{\partial(\phi_1, \dots, \phi_{n-1})} \right] \times \\ &\quad d\phi_1 \wedge \dots \wedge d\phi_{n-1}. \end{aligned} \quad (\text{A.5})$$

Therefore, if we can compute the Jacobians in the formula (A.5), then an explicit formula of $(\omega_n)_{\Sigma^{n-1}}$ in terms of $d\phi_1, \dots, d\phi_{n-1}$ can be derived.

^aThe formula (A.5) can also be found in [25, Eqn. (12.31), p. 206].

- **Step 2: Statement of the formula** $(\omega_n)_{\Sigma^{n-1}}$.

Lemma A.1

For every $n \geq 2$, we have

$$(\omega_n)_{\Sigma^{n-1}} = \sin \phi_{n-2} \sin^2 \phi_{n-3} \cdots \sin^{n-2} \phi_1 d\phi_1 \wedge \cdots \wedge d\phi_{n-1}.$$

To prove this lemma, we have to compute the Jacobians in the formula (A.5), so we need to consider the following $n \times (n-1)$ Jacobian matrix

$$\begin{aligned} \mathbf{M} &= \frac{\partial(x_1, \dots, x_n)}{\partial(\phi_1, \dots, \phi_{n-1})} \\ &= \begin{pmatrix} -\sin \phi_1 & 0 & \cdots & 0 \\ \cos \phi_1 \cos \phi_2 & -\sin \phi_1 \sin \phi_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \cos \phi_1 \prod_{\substack{j=1 \\ j \neq 1}}^{n-2} \sin \phi_j \cos \phi_{n-1} & \cos \phi_2 \prod_{\substack{j=1 \\ j \neq 2}}^{n-2} \sin \phi_j \cos \phi_{n-1} & \cdots & \prod_{j=1}^{n-2} \sin \phi_j (-\sin \phi_{n-1}) \\ \cos \phi_1 \prod_{\substack{j=1 \\ j \neq 1}}^{n-1} \sin \phi_j & \cos \phi_2 \prod_{\substack{j=1 \\ j \neq 2}}^{n-1} \sin \phi_j & \cdots & \cos \phi_{n-1} \prod_{\substack{j=1 \\ j \neq n-1}}^{n-1} \sin \phi_j \end{pmatrix} \end{aligned}$$

so that

$$\det \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)}{\partial(\phi_1, \dots, \phi_{n-1})} = \det \mathbf{M}_i \quad (1 \leq i \leq n),$$

where $\mathbf{M}_i$ is the lower triangular $(n-1) \times (n-1)$ matrix obtain by deleting the i row from $\mathbf{M}$.

To get the idea of the computation of $\det \mathbf{M}_i$, we first consider a special case in **Step 3**. As a result, a “guess” of the formula of $\det \mathbf{M}_i$ can be obtained in the expression (A.8). A proof of such formula is provided in **Step 4** and finally the proof of Lemma A.1 is presented in **Step 5**.

- **Step 3: The special case when $n = 5$.** In this step, we consider the special case that $n = 5$. Then we have

$$\begin{aligned} x_1 &= \cos \phi_1, \\ x_2 &= \sin \phi_1 \cos \phi_2, \\ x_3 &= \sin \phi_1 \sin \phi_2 \cos \phi_3, \\ x_4 &= \sin \phi_1 \sin \phi_2 \sin \phi_3 \cos \phi_4, \\ x_5 &= \sin \phi_1 \sin \phi_2 \sin \phi_3 \sin \phi_4 \end{aligned} \tag{A.6}$$

and^b

$$\begin{aligned}
\det \frac{\partial(x_2, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} &= \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \cos \phi_1 \\
&= (-1)^{1-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \mathbf{x}_1, \\
\det \frac{\partial(x_1, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} &= -\sin^4 \phi_1 \sin^2 \phi_2 \sin \phi_3 \cos \phi_2 \\
&= (-1)^{2-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \times \cos \phi_2 \prod_{j=1}^{2-1} \sin \phi_j \\
&= (-1)^{2-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \mathbf{x}_2, \\
\det \frac{\partial(x_1, x_2, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} &= \sin^4 \phi_1 \sin^3 \phi_2 \sin \phi_3 \cos \phi_3 \\
&= (-1)^{3-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \times \cos \phi_3 \prod_{j=1}^{3-1} \sin \phi_j \\
&= (-1)^{3-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \mathbf{x}_3, \\
\det \frac{\partial(x_1, x_2, x_3, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} &= -\sin^4 \phi_1 \sin^3 \phi_2 \sin^2 \phi_3 \cos \phi_4 \\
&= (-1)^{4-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \times \cos \phi_4 \prod_{j=1}^{4-1} \sin \phi_j \\
&= (-1)^{4-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \mathbf{x}_4, \\
\det \frac{\partial(x_1, x_2, x_3, x_4)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} &= (-1)^{5-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \times \prod_{j=1}^{5-1} \sin \phi_j \\
&= (-1)^{5-1} \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 \mathbf{x}_5.
\end{aligned} \tag{A.7}$$

Substituting the definitions (A.6) and the determinants (A.7) into the formula (A.5) (with $n = 5$) and using the fact that

$$(x_1, x_2, x_3, x_4, x_5)$$

is a point of the 4-sphere, we obtain that

$$\begin{aligned}
(\omega_5)_{\Sigma^4} &= \left[x_1 \det \frac{\partial(x_2, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} - x_2 \det \frac{\partial(x_1, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} \right. \\
&\quad + x_3 \det \frac{\partial(x_1, x_2, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} - x_4 \det \frac{\partial(x_1, x_2, x_3, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} \\
&\quad \left. + x_5 \det \frac{\partial(x_1, x_2, x_3, x_4)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} \right] \times d\phi_1 \wedge d\phi_2 \wedge d\phi_3 \wedge d\phi_4 \\
&= \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 (x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2) d\phi_1 \wedge d\phi_2 \wedge d\phi_3 \wedge d\phi_4 \\
&= \sin^3 \phi_1 \sin^2 \phi_2 \sin \phi_3 d\phi_1 \wedge d\phi_2 \wedge d\phi_3 \wedge d\phi_4
\end{aligned}$$

which is consistent with Lemma A.1. Hence, it is reasonable to make the following claim:

$$\det \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)}{\partial(\phi_1, \dots, \phi_{n-1})} = (-1)^{i-1} \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \cdots \sin \phi_{n-2} \mathbf{x}_i \tag{A.8}$$

where $1 \leq i \leq n$ and $n \geq 2$.

^bThe explicit form of the Jacobian matrix $\frac{\partial(x_1, x_2, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)}$ can be found on the next page.

$$\begin{aligned}
& \frac{\partial(x_1, x_2, x_3, x_4, x_5)}{\partial(\phi_1, \phi_2, \phi_3, \phi_4)} \\
&= \begin{pmatrix} \frac{\partial x_1}{\partial \phi_1} & \frac{\partial x_1}{\partial \phi_2} & \frac{\partial x_1}{\partial \phi_3} & \frac{\partial x_1}{\partial \phi_4} \\ \frac{\partial x_2}{\partial \phi_1} & \frac{\partial x_2}{\partial \phi_2} & \frac{\partial x_2}{\partial \phi_3} & \frac{\partial x_2}{\partial \phi_4} \\ \frac{\partial x_3}{\partial \phi_1} & \frac{\partial x_3}{\partial \phi_2} & \frac{\partial x_3}{\partial \phi_3} & \frac{\partial x_3}{\partial \phi_4} \\ \frac{\partial x_4}{\partial \phi_1} & \frac{\partial x_4}{\partial \phi_2} & \frac{\partial x_4}{\partial \phi_3} & \frac{\partial x_4}{\partial \phi_4} \\ \frac{\partial x_5}{\partial \phi_1} & \frac{\partial x_5}{\partial \phi_2} & \frac{\partial x_5}{\partial \phi_3} & \frac{\partial x_5}{\partial \phi_4} \end{pmatrix} \\
&= \begin{pmatrix} -\sin \phi_1 & 0 & 0 & 0 \\ \cos \phi_1 \cos \phi_2 & -\sin \phi_1 \sin \phi_2 & 0 & 0 \\ \cos \phi_1 \sin \phi_2 \cos \phi_3 & \sin \phi_1 \cos \phi_2 \cos \phi_3 & -\sin \phi_1 \sin \phi_2 \sin \phi_3 & 0 \\ \cos \phi_1 \sin \phi_2 \sin \phi_3 \cos \phi_4 & \sin \phi_1 \cos \phi_2 \sin \phi_3 \cos \phi_4 & \sin \phi_1 \sin \phi_2 \cos \phi_3 \cos \phi_4 & -\sin \phi_1 \sin \phi_2 \sin \phi_3 \sin \phi_4 \\ \cos \phi_1 \sin \phi_2 \sin \phi_3 \sin \phi_4 & \sin \phi_1 \cos \phi_2 \sin \phi_3 \sin \phi_4 & \sin \phi_1 \sin \phi_2 \cos \phi_3 \sin \phi_4 & \sin \phi_1 \sin \phi_2 \sin \phi_3 \cos \phi_4 \end{pmatrix}.
\end{aligned}$$

- **Step 4: A proof of the formula (A.8).** We apply induction to prove the formula (A.8). To do so, we rewrite the equations (10.154) as follows:

$$\begin{aligned}
x_1 &= \cos \phi_1, \\
\frac{x_2}{\sin \phi_1} &= \cos \phi_2, \\
\frac{x_3}{\sin \phi_1} &= \sin \phi_2 \cos \phi_3, \\
&\vdots \\
\frac{x_{k-1}}{\sin \phi_1} &= \sin \phi_2 \cdots \sin \phi_{k-2} \cos \phi_{k-1}, \\
\frac{x_k}{\sin \phi_1} &= \sin \phi_2 \cdots \sin \phi_{k-2} \sin \phi_{k-1},
\end{aligned} \tag{A.9}$$

where $0 \leq \phi_1, \dots, \phi_{k-2} \leq \pi$, $0 \leq \phi_{k-1} \leq 2\pi$. For $j = 1, \dots, k-2$, let $\varphi_j = \phi_{j+1}$ so that

$$0 \leq \varphi_1, \dots, \varphi_{k-3} \leq \pi \quad \text{and} \quad 0 \leq \varphi_{k-2} \leq 2\pi.$$

Furthermore, for $j = 1, \dots, k-2$, we let

$$X_j = \frac{x_{j+1}}{\sin \phi_1} = \sin \phi_2 \cdots \sin \phi_j \cos \phi_{j+1} = \sin \varphi_1 \cdots \sin \varphi_{j-1} \cos \varphi_j$$

and

$$X_{k-1} = \frac{x_k}{\sin \phi_1} = \sin \phi_2 \cdots \sin \phi_{k-2} \sin \phi_{k-1} = \sin \varphi_1 \cdots \sin \varphi_{k-3} \sin \varphi_{k-2}.$$

Thus the last $(k-1)$ equations of (10.154) can be reduced to

$$\begin{aligned}
X_1 &= \cos \varphi_1, \\
X_2 &= \sin \varphi_1 \cos \varphi_2, \\
&\vdots \\
X_{k-2} &= \sin \varphi_1 \cdots \sin \varphi_{k-3} \cos \varphi_{k-2}, \\
X_{k-1} &= \sin \varphi_1 \cdots \sin \varphi_{k-3} \sin \varphi_{k-2}.
\end{aligned} \tag{A.10}$$

Note that

$$X_1^2 + \cdots + X_{k-1}^2 = 1.$$

Assume the induction is true for $n = k-1$ for some $k \geq 3$, i.e.,

$$\det \frac{\partial(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{k-1})}{\partial(\varphi_1, \dots, \varphi_{k-2})} = (-1)^{i-1} \sin^{k-3} \varphi_1 \sin^{k-4} \varphi_2 \cdots \sin \varphi_{k-3} X_i, \tag{A.11}$$

where $1 \leq i \leq k-1$. Since

$$x_1 = \cos \phi_1 \quad \text{and} \quad x_{j+1} = X_j \sin \phi_1 \quad (1 \leq j \leq k-1),$$

we have

$$\frac{\partial(x_1, \dots, x_k)}{\partial(\phi_1, \dots, \phi_{k-1})} = \begin{pmatrix} \frac{\partial x_1}{\partial \phi_1} & \frac{\partial x_1}{\partial \phi_2} & \cdots & \frac{\partial x_1}{\partial \phi_{k-1}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_{k-1}}{\partial \phi_1} & \frac{\partial x_{k-1}}{\partial \phi_2} & \cdots & \frac{\partial x_{k-1}}{\partial \phi_{k-1}} \\ \frac{\partial x_k}{\partial \phi_1} & \frac{\partial x_k}{\partial \phi_2} & \cdots & \frac{\partial x_k}{\partial \phi_{k-1}} \end{pmatrix}$$

$$\begin{aligned}
&= \begin{pmatrix} -\sin \phi_1 & 0 & \cdots & 0 \\ X_1 x_1 & \frac{\partial X_1}{\partial \phi_2} \sin \phi_1 & \cdots & \frac{\partial X_1}{\partial \phi_{k-1}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-2} x_1 & \frac{\partial X_{k-2}}{\partial \phi_2} \sin \phi_1 & \cdots & \frac{\partial X_{k-2}}{\partial \phi_{k-1}} \sin \phi_1 \\ X_{k-1} x_1 & \frac{\partial X_{k-1}}{\partial \phi_2} \sin \phi_1 & \cdots & \frac{\partial X_{k-1}}{\partial \phi_{k-1}} \sin \phi_1 \end{pmatrix} \\
&= \begin{pmatrix} -\sin \phi_1 & 0 & \cdots & 0 \\ X_1 x_1 & \frac{\partial X_1}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_1}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-2} x_1 & \frac{\partial X_{k-2}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{k-2}}{\partial \varphi_{k-2}} \sin \phi_1 \\ X_{k-1} x_1 & \frac{\partial X_{k-1}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{k-1}}{\partial \varphi_{k-2}} \sin \phi_1 \end{pmatrix}_{k \times (k-1)}. \quad (\text{A.12})
\end{aligned}$$

By deleting the first row of the matrix (A.12), we obtain

$$\begin{aligned}
\det \frac{\partial(x_2, \dots, x_k)}{\partial(\phi_1, \dots, \phi_{k-1})} &= \begin{vmatrix} \textcolor{red}{X_1} x_1 & \frac{\partial X_1}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_1}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ \textcolor{red}{X_{k-2}} x_1 & \frac{\partial X_{k-2}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{k-2}}{\partial \varphi_{k-2}} \sin \phi_1 \\ \textcolor{red}{X_{k-1}} x_1 & \frac{\partial X_{k-1}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{k-1}}{\partial \varphi_{k-2}} \sin \phi_1 \end{vmatrix} \\
&= x_1 \sin^{k-2} \phi_1 \begin{vmatrix} \textcolor{red}{X_1} & \frac{\partial X_1}{\partial \varphi_1} & \cdots & \frac{\partial X_1}{\partial \varphi_{k-2}} \\ \vdots & \vdots & \ddots & \vdots \\ \textcolor{red}{X_{k-2}} & \frac{\partial X_{k-2}}{\partial \varphi_1} & \cdots & \frac{\partial X_{k-2}}{\partial \varphi_{k-2}} \\ \textcolor{red}{X_{k-1}} & \frac{\partial X_{k-1}}{\partial \varphi_1} & \cdots & \frac{\partial X_{k-1}}{\partial \varphi_{k-2}} \end{vmatrix}. \quad (\text{A.13})
\end{aligned}$$

Now, by expanding the determinant (A.13) along the first column and then using the assumption (A.11), we have

$$\begin{aligned}
 \det \frac{\partial(x_2, \dots, x_k)}{\partial(\phi_1, \dots, \phi_{k-1})} &= x_1 \sin^{k-2} \phi_1 \sum_{i=1}^{k-1} (-1)^{i-1} \det \frac{\partial(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_{k-1})}{\partial(\varphi_1, \dots, \varphi_{k-2})} X_i \\
 &= x_1 \sin^{k-2} \phi_1 \sum_{i=1}^{k-1} \sin^{k-3} \varphi_1 \sin^{k-4} \varphi_2 \cdots \sin \varphi_{k-3} X_i^2 \\
 &= x_1 \sin^{k-2} \phi_1 \sin^{k-3} \varphi_1 \sin^{k-4} \varphi_2 \cdots \sin \varphi_{k-3} \sum_{i=1}^{k-1} X_i^2 \\
 &= \sin^{k-2} \phi_1 \sin^{k-3} \varphi_2 \cdots \sin \varphi_{k-2} x_1.
 \end{aligned} \tag{A.14}$$

Next, for $2 \leq i \leq k$, we first delete the i row from the matrix (A.12), then expand its corresponding determinant along the first row and finally apply the assumption (A.11) to obtain that

$$\begin{aligned}
 \det \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k)}{\partial(\phi_1, \dots, \phi_{k-1})} &= \begin{vmatrix} -\sin \phi_1 & 0 & \cdots & 0 \\ X_1 x_1 & \frac{\partial X_1}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_1}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ X_{i-2} x_1 & \frac{\partial X_{i-2}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{i-2}}{\partial \varphi_{k-2}} \sin \phi_1 \\ X_i x_1 & \frac{\partial X_i}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_i}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ X_{k-1} x_1 & \frac{\partial X_{k-1}}{\partial \varphi_1} \sin \phi_1 & \cdots & \frac{\partial X_{k-1}}{\partial \varphi_{k-2}} \sin \phi_1 \end{vmatrix} \\
 &= -\sin \phi_1 \begin{vmatrix} \frac{\partial X_1}{\partial \varphi_1} \sin \phi_1 & \cdots & \cdots & \frac{\partial X_1}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial X_{i-2}}{\partial \varphi_1} \sin \phi_1 & \cdots & \cdots & \frac{\partial X_{i-2}}{\partial \varphi_{k-2}} \sin \phi_1 \\ \frac{\partial X_i}{\partial \varphi_1} \sin \phi_1 & \cdots & \cdots & \frac{\partial X_i}{\partial \varphi_{k-2}} \sin \phi_1 \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial X_{k-1}}{\partial \varphi_1} \sin \phi_1 & \cdots & \cdots & \frac{\partial X_{k-1}}{\partial \varphi_{k-2}} \sin \phi_1 \end{vmatrix} \\
 &= -\sin^{k-1} \phi_1 \det \frac{\partial(X_1, \dots, X_{i-2}, X_i, \dots, X_{k-1})}{\partial(\varphi_1, \dots, \varphi_{k-2})} \\
 &= (-1)^{i-1} \sin^{k-1} \phi_1 \sin^{k-3} \varphi_1 \sin^{k-4} \varphi_2 \cdots \sin \varphi_{k-3} X_{i-1}
 \end{aligned}$$

$$= (-1)^{i-1} \sin^{k-2} \phi_1 \sin^{k-3} \phi_2 \cdots \sin \phi_{k-2} x_i. \quad (\text{A.15})$$

Combining the results (A.14) and (A.15), we have shown that

$$\det \frac{\partial(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k)}{\partial(\phi_1, \dots, \phi_{k-1})} = (-1)^{i-1} \sin^{k-2} \phi_1 \sin^{k-3} \phi_2 \cdots \sin \phi_{k-2} x_i$$

for $1 \leq i \leq k$. Hence, by induction, our claim (A.8) holds for every positive integer $n \geq 2$.

- **Step 5: Completion of the proof of Lemma A.1.** By substituting the determinant (A.8) into the formula (A.5), we establish easily that

$$\begin{aligned} (\omega_n)_{\Sigma^{n-1}} &= \left(\sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \cdots \sin \phi_{n-2} \sum_{i=1}^n x_i^2 \right) d\phi_1 \wedge \cdots \wedge d\phi_{n-1} \\ &= \sin^{n-2} \phi_1 \sin^{n-3} \phi_2 \cdots \sin \phi_{n-2} d\phi_1 \wedge \cdots \wedge d\phi_{n-1}, \end{aligned}$$

completing the proof of Lemma A.1.

If we apply Lemma A.1 to the integral on the right-hand side in (A.4), then we acquire

$$\int_S \omega_n = \int_S \sin \phi_{n-2} \sin^2 \phi_{n-3} \cdots \sin^{n-2} \phi_1 d\phi_1 \wedge \cdots \wedge d\phi_{n-1} = A_{n-1}(S), \quad (\text{A.16})$$

where $A_{n-1}(S)$ is the “area” of the restriction $S = \Sigma_E^{n-1}$. This completes the proof of Lemma 10.14.

APPENDIX *B*

Solid angle subtended by a surface at the origin

Roughly speaking, a *solid angle* is the two-dimensional “angle” in three-dimensional space. In fact, it is an analogy to the case that a plane angle θ measured in radians equals its corresponding arc length s of a circle of radius r . Recall that the angle θ can be found by the formula

$$\theta = \frac{s}{r}. \quad (\text{B.1})$$

See Figure B.1 for an illustration.

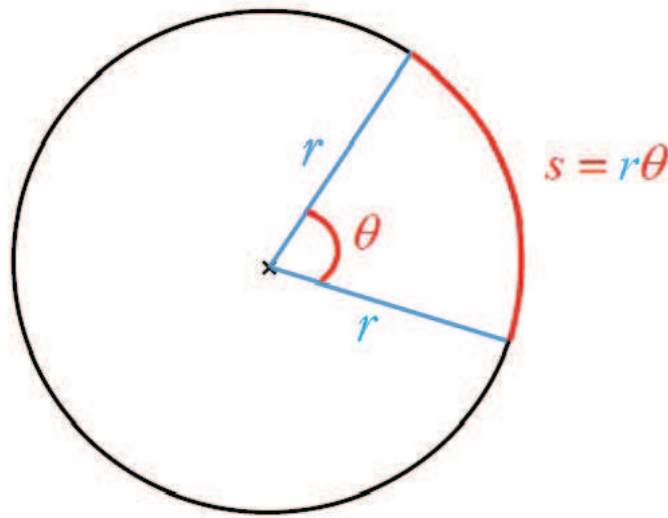
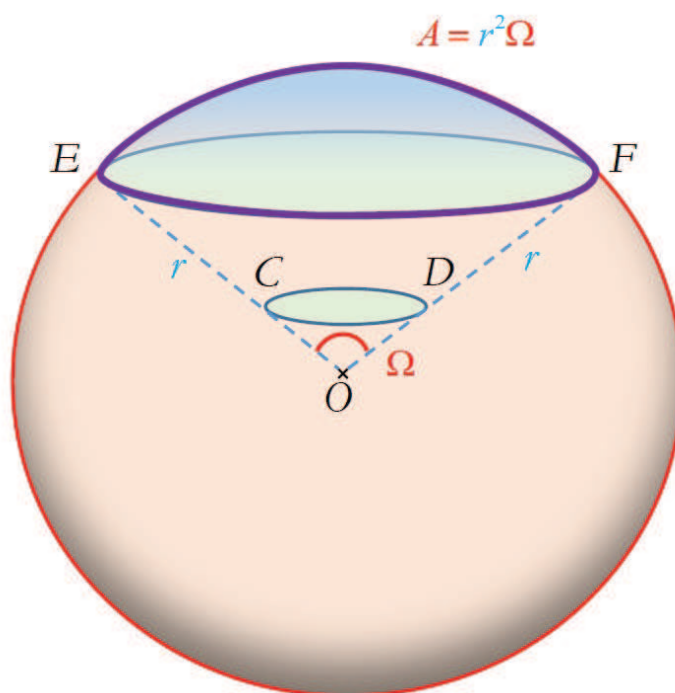


Figure B.1: The plane angle θ measured in radians.

To extend this concept to our three dimensional space, we consider the solid angle Ω subtended by an object to be the surface area A of its radial projection onto the unit sphere (the unit is *steradians*). In other words, you have an object and you project it onto the the unit sphere. After this, you compute the surface area of the projection and this number is called the solid angle. Similar to the definition (B.1), we define the solid angle to be

$$\Omega = \frac{A}{r^2}, \quad (\text{B.2})$$

where A is the surface area of the projection and r is the radius of the sphere. The idea can be seen in Figure B.2:


 Figure B.2: The solid angle Ω measured in steradians.

To find the formula (B.2), we know from the spherical coordinates system that $dA = r^2 \sin \theta d\theta d\phi$ so that

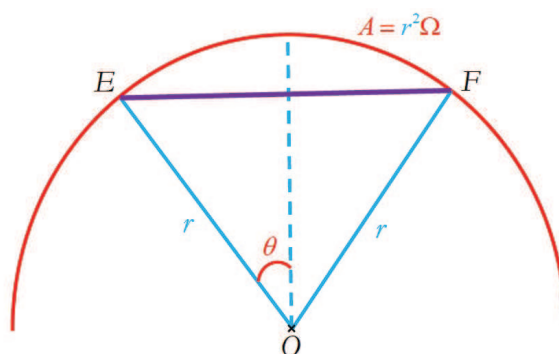
$$d\Omega = \frac{dA}{r^2} = \sin \theta d\theta d\phi. \quad (\text{B.3})$$

By integration, we deduce from the expression (B.3) that

$$\Omega = \int \int_S \sin \theta d\theta d\phi, \quad (\text{B.4})$$

where S denotes the projection of the object onto the sphere. Here the variables θ and ϕ in the formula (B.4) correspond to the variables u and v in Problem 10.22 (see Figure 10.13).

When we look at Figure B.2 again, it may happen that two similar objects (two discs in Figure B.2) may have the same solid angle. This coincides with our daily experience that a small nearby object and a large far away object may “look” the same. Since the objects OCD or OEF are similar cones, let’s calculate the solid angle of a cone with apex angle 2θ . First, we have a section of the cone in Figure B.3:


 Figure B.3: A section of the cone with apex angle 2θ .

In this case, our formula (B.4) becomes

$$\Omega = \int_0^{2\pi} \int_0^\theta \sin \varphi \, d\varphi \, d\phi = 2\pi \int_0^\theta \sin \varphi \, d\varphi = 2\pi(1 - \cos \theta). \quad (\text{B.5})$$

Now the formula (B.5) says that the solid angle Ω of a cone (a three-dimensional angle) can be found by the plane angle θ (a two-dimensional angle). For instance, we know that the sun is much larger than the moon and also much far away from the earth. However, the “sizes” of both objects (the sun and the moon) are approximately the same because they have approximately the same solid angle. In fact, the *angular diameters* of the sun and the moon seen from earth are approximately 9.35×10^{-3} and 9.2×10^{-3} in radians respectively. By substituting these values into the formula (B.5), we obtain the solid angle subtended by the sun and the moon are given by

$$\Omega_{\text{sun}} = 6.87 \times 10^{-5} \text{ sr} \quad \text{and} \quad \Omega_{\text{moon}} = 6.67 \times 10^{-5} \text{ sr}$$

respectively. Hence, these values explain why the moon can almost cover the whole sun in the sky during a solar eclipse.

APPENDIX C

Proofs of some basic properties of a measure

Let p be a positive integer and $\mathbb{R}^p$ denote the p -dimensional Euclidean space. We call I an *interval*^a in $\mathbb{R}^p$ if

$$\begin{aligned} I &= \{\mathbf{x} = (x_1, \dots, x_p) \mid -\infty \leq a_i \leq x_i \leq b_i \leq +\infty \quad (i = 1, \dots, p)\} \\ &= [a_1, b_1] \times \cdots \times [a_p, b_p], \end{aligned} \quad (\text{C.1})$$

where any or all of the “ $\leq$ ” signs can be replaced by “ $<$ ” and the notation “ $\leq b_i$ ” (resp. “ $a_i \leq$ ”) is replaced by “ $< b_i$ ” if $b_i = +\infty$ (resp. “ $a_i <$ ” if $a_i = -\infty$) in (C.1). Let $I_1, \dots, I_n$ be intervals for some positive integer n . If

$$A = \bigcup_{i=1}^n I_i,$$

then A is called an *elementary set*.

We obtain some fundamental results about intervals in the following proposition, see [19, pp. 4-7] for reference.

Proposition C.1

Suppose that I and J are intervals in the form (C.1).

(a) If $I \subseteq J$, then there exist intervals $I_1, \dots, I_n$ such that

$$I = I_0 \subseteq I_1 \subseteq \cdots \subseteq I_n = J. \quad (\text{C.2})$$

(b) If $I \subseteq J$, then for such a sequence of intervals in the relation (C.2), the set-theoretic differences

$$I_m \setminus I_{m-1} \quad (m = 1, \dots, n)$$

form a sequence of intervals of type (C.1).

(c) The difference $J \setminus I$ is a union of finite number of intervals.

Proof.

(a) We consider the special case $J = \mathbb{R}^p$ first, i.e., $a_1 = \cdots = a_p = -\infty$ and $b_1 = \cdots = b_p = +\infty$ in the

^aSome books call it a *rectangle* or a *box*.

definition (C.1). Suppose that $I_0 = I$. Then we define I_1 and I_2 as follows:

$$\begin{aligned} I_1 &= \{(x_1, \dots, x_p) \mid x_1 \in [a_1, \infty), a_i \leq x_i \leq b_i \quad (i = 2, \dots, p)\}, \\ I_2 &= \{(x_1, \dots, x_p) \mid x_1 \in \mathbb{R}, a_i \leq x_i \leq b_i \quad (i = 2, \dots, p)\}, \\ I_{2k-1} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{k-1} \in \mathbb{R}, x_k \in [a_k, \infty), a_i \leq x_i \leq b_i \quad (i = k+1, \dots, p)\}, \\ I_{2k} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_k \in \mathbb{R}, a_i \leq x_i \leq b_i \quad (i = k+1, \dots, p)\}, \\ I_{2p-1} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{p-1} \in \mathbb{R}, x_p \in [a_p, \infty)\}, \\ I_{2p} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_p \in \mathbb{R}\} = \mathbb{R}^p, \end{aligned} \quad (\text{C.3})$$

where $k = 2, \dots, p-1$. Hence it is easy to see from our construction that

$$I = I_0 \subseteq I_1 \subseteq \dots \subseteq I_{2p} = \mathbb{R}^p. \quad (\text{C.4})$$

For the general case (i.e., $J \subset \mathbb{R}^p$), our special case shows that there exists intervals $I_0, I_1, \dots, I_{2p}$ such that the relation (C.3) holds. Let

$$I'_i = I_i \cap J,$$

where $i = 0, 1, \dots, 2p$. It is obvious from the definition (C.1) that the intersection of two intervals must be an interval. Thus each I'_i is an interval and then the relation (C.3) implies that

$$I = I'_0 \subseteq I'_1 \subseteq \dots \subseteq I'_{2p} = J.$$

(b) Similar as part (a), we prove the result for the special case $J = \mathbb{R}^p$ first. Note that we have^b

$$\begin{aligned} I_{2k} \setminus I_{2k-1} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{k-1} \in \mathbb{R}, x_k \in (-\infty, a_k), \\ &\quad a_i \leq x_i \leq b_i \quad (i = k+1, \dots, p)\}, \\ I_{2k-1} \setminus I_{2k-2} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{k-1} \in \mathbb{R}, x_k \in (b_k, \infty), \\ &\quad a_i \leq x_i \leq b_i \quad (i = k+1, \dots, p)\}, \\ I_{2p-1} \setminus I_{2p-2} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{p-1} \in \mathbb{R}, x_p \in (b_p, +\infty)\}, \\ I_{2p} \setminus I_{2p-1} &= \{(x_1, \dots, x_p) \mid x_1, \dots, x_{p-1} \in \mathbb{R}, x_p \in (-\infty, a_p)\}, \end{aligned} \quad (\text{C.5})$$

where $k = 1, \dots, p-1$. By the definition (C.1), all the above differences are certainly intervals and so our result follows in this special case.

For the general case (i.e., $J \subset \mathbb{R}^p$), we deduce from the identity

$$A \setminus B = A \cap B^c \quad (\text{C.6})$$

that

$$\begin{aligned} I'_m \setminus I'_{m-1} &= I'_m \cap (I'_{m-1})^c \\ &= (I_m \cap J) \cap (I_{m-1} \cap J)^c \\ &= (I_m \cap J) \cap (I_{m-1}^c \cup J^c) \\ &= (I_m \cap J \cap I_{m-1}^c) \cup (I_m \cap J \cap J^c) \\ &= [(I_m \cap I_{m-1}^c) \cap J] \cup (I_m \cap \emptyset) \\ &= (I_m \setminus I_{m-1}) \cap J, \end{aligned}$$

where $m = 1, \dots, n$. Since both $I_m \setminus I_{m-1}$ and J are intervals, the difference $I'_m \setminus I'_{m-1}$ must be an interval too.

(c) If $J \setminus I$ is an interval, then there is nothing to prove. Suppose that $J \setminus I$ is not an interval. We consider the set $\hat{I} = I \cap J$. Then we have

$$\hat{I} \subset J \quad \text{and} \quad J \setminus I = J \setminus \hat{I}.$$

^bNotice that the intervals $(-\infty, a_k)$ and $(b_k, +\infty)$ are replaced by $(-\infty, a_k]$ and $[b_k, +\infty)$ if the interval $[a_k, +\infty)$ and the inequality $x_k \leq b_k$ in (C.3) are replaced by $(a_k, +\infty)$ and $x_k < b_k$ respectively.

By part (a), there exist intervals $I_1, \dots, I_n$ such that

$$\widehat{I} = I_0 \subseteq I_1 \subseteq \dots \subseteq I_n = J. \quad (\text{C.7})$$

By part (b), the sequence in the relation (C.7) satisfies the condition that

$$J_k = I_k \setminus I_{k-1} \quad (k = 1, \dots, n)$$

form a set of intervals and

$$\bigcup_{k=1}^n J_k = \bigcup_{k=1}^n (I_k \setminus I_{k-1}) = I_n \setminus I_0 = J \setminus \widehat{I} = J \setminus I$$

which is our desired result. ■

Proposition C.2

Denote $\mathcal{E}$ to be the collection of all elementary sets of $\mathbb{R}^p$.

- (a) The set $\mathcal{E}$ is a ring, but not a σ -ring.
- (b) If $A \in \mathcal{E}$, then $A^c \in \mathcal{E}$.
- (c) If $A \in \mathcal{E}$, then A is the union of a finite number of disjoint intervals.
- (d) If $A \in \mathcal{E}$ and

$$A = I_1 \cup \dots \cup I_m = J_1 \cup \dots \cup J_n,$$

where $\{I_1, \dots, I_m\}$ and $\{J_1, \dots, J_n\}$ are disjoint intervals, then we have

$$m(A) = m(I_1) + \dots + m(I_m) = m(J_1) + \dots + m(J_n).$$

Proof.

- (a) Suppose that $A, B \in \mathcal{E}$. Then we have

$$A = I_1 \cup \dots \cup I_m \quad \text{and} \quad B = J_1 \cup \dots \cup J_n,$$

where $I_1, \dots, I_m, J_1, \dots, J_n$ are intervals. Since

$$A \cup B = I_1 \cup \dots \cup I_m \cup J_1 \cup \dots \cup J_n$$

is also a union of a finite number of intervals, we have $A \cup B \in \mathcal{E}$. Furthermore, we have

$$\begin{aligned} A \setminus B &= \bigcup_{s=1}^n I_s \setminus \bigcup_{t=1}^m J_t \\ &= \left(\bigcup_{s=1}^n I_s \right) \cap \left(\bigcup_{t=1}^m J_t \right)^c \\ &= \left(\bigcup_{s=1}^n I_s \right) \cap \left(\bigcap_{t=1}^m J_t^c \right) \\ &= \bigcup_{s=1}^n \left(\bigcap_{t=1}^m I_s \cap J_t^c \right) \\ &= \bigcup_{s=1}^n \bigcap_{t=1}^m (I_s \setminus J_t). \end{aligned}$$

By Proposition A.1(c), each $I_s \setminus J_t$ is a union of finite number of intervals. As we note in the proof of Proposition A.1(a), intersections of intervals is still an interval, so the intersection

$$\bigcap_{t=1}^m (I_s \setminus J_t)$$

is a finite union of intervals. Hence the difference $A \setminus B$ is a finite union of intervals. In other words, we have $A \setminus B \in \mathcal{E}$. By Definition 11.1, $\mathcal{E}$ is a ring.

However, $\mathcal{E}$ is not a σ -ring. Let $p = 1$ and $A_n = (n - \frac{1}{2}, n + \frac{1}{2}) \subset \mathbb{R}$ for $n = 1, 2, \dots$. Then we have $A_n \in \mathcal{E}$ for $n = 1, 2, \dots$ and

$$A = \bigcup_{n=1}^{\infty} A_n = \left(\frac{1}{2}, +\infty\right) \setminus \left\{\frac{1}{2}, \frac{3}{2}, \dots\right\}$$

is not elementary. Otherwise, we have

$$A = \bigcup_{k=1}^N I_k,$$

where $I_1, \dots, I_N$ are some intervals. Since A is unbounded, I_k is unbounded for some $k \in \{1, \dots, N\}$ and this implies that such I_k contains $\frac{n}{2}$ for some positive integer n , a contradiction.

(b) By definition, we have $A^c = \mathbb{R}^p \setminus A$. Since $A, \mathbb{R}^p \in \mathcal{E}$ and $\mathcal{E}$ is a ring, part (a) implies that $A^c \in \mathcal{E}$.

(c) Let $A = \bigcup_{k=1}^n I_k$ with $I_1, \dots, I_n \in \mathcal{E}$. Write $A_1 = I_1$ and

$$A_k = (I_1 \cup \dots \cup I_k) \setminus (I_1 \cup \dots \cup I_{k-1}), \quad (\text{C.8})$$

where $k = 1, \dots, n$. Then we have

$$A = \bigcup_{k=1}^n A_k.$$

We claim that $A_1, \dots, A_n$ are pairwise disjoint. Let $s > t$. Apply the identity (C.6) to the definition (C.8), we get

$$A_s = I_s \setminus (I_1 \cup \dots \cup I_t \cup \dots \cup I_{s-1}) \quad \text{and} \quad A_t = I_t \setminus (I_1 \cup \dots \cup I_{t-1}) \subset I_t$$

which shows that

$$A_s \cap A_t = \emptyset.$$

(d) For each $p = 1, \dots, m$, we have

$$m(I_p) = m(I_p \cap J_1) + \dots + m(I_p \cap J_n).$$

Similarly, for each $q = 1, \dots, n$, we have

$$m(J_q) = m(J_q \cap I_1) + \dots + m(J_q \cap I_m).$$

Since

$$\sum_{p=1}^m m(I_p) = \sum_{p=1}^m \sum_{q=1}^n m(I_p \cap J_q) = \sum_{q=1}^n \sum_{p=1}^m m(J_q \cap I_p) = \sum_{q=1}^n m(J_q),$$

this means that $m(A)$ is well-defined.

This finishes the proof of the proposition. ■

Proposition C.3

Suppose that $A \in \mathfrak{M}(\mu)$ and $\epsilon > 0$. Then there exist a closed set F and an open set G such that $F \subseteq A \subseteq G$, where

$$\mu(G \setminus A) < \epsilon \quad \text{and} \quad \mu(A \setminus F) < \epsilon.$$

Proof. Since $A \in \mathfrak{M}(\mu)$, we deduce from Definition 11.9 that A can be represented as the union of a countable collection of μ -measurable sets $\{A_n\}$ of $\mathfrak{M}_F(\mu)$, i.e.,

$$A = \bigcup_{n=1}^{\infty} A_n.$$

By the definition of $\mathfrak{M}_F(\mu)$, for each $n \in \mathbb{N}$, there exists a sequence $\{A_{nk}\} \subseteq \mathcal{E}$ such that $A_{nk} \rightarrow A_n$ as $k \rightarrow \infty$. Now for each k , since μ^* is regular, there exist $F_{nk}, G_{nk} \in \mathcal{E}$ such that F_{nk} is closed and G_{nk} is open,

$$F_{nk} \subseteq A_{nk} \subseteq G_{nk} \quad (\text{C.9})$$

and

$$\mu^*(G_{nk}) - 2^{-n}2^{-k}2^{-1}\epsilon \leq \mu^*(A_{nk}) \leq \mu^*(F_{nk}) + 2^{-n}2^{-k}2^{-1}\epsilon. \quad (\text{C.10})$$

By the left-hand side of the inequality (C.10), we have

$$\mu^*(G_{nk}) - \mu^*(A_{nk}) \leq 2^{-n}2^{-k}2^{-1}\epsilon. \quad (\text{C.11})$$

Since each G_{nk} is open, Theorem 2.24(a) shows that the sets

$$G_n = \bigcup_{k=1}^{\infty} G_{nk} \quad \text{and} \quad G = \bigcup_{n=1}^{\infty} G_n \quad (\text{C.12})$$

are open. Now, by Remark 11.11(a), we have $G_{nk}, G_n, G \in \mathfrak{M}(\mu)$.^c We define

$$A_n = \bigcup_{k=1}^{\infty} A_{nk} \quad \text{and} \quad A = \bigcup_{n=1}^{\infty} A_n. \quad (\text{C.13})$$

Combining the relations (C.9), the definitions (C.12) and (C.13), we obtain

$$A_n \subseteq G_n \quad \text{and} \quad A \subseteq G.$$

For each n , if $x \in G_n \setminus A_n$, then $x \in G_n$ but $x \notin A_n$ so that $x \in G_{nk}$ for some k and $x \notin A_{ni}$ for all i . Therefore, we have $x \in G_{nk} \setminus A_{nk}$ for some k which implies that

$$G_n \setminus A_n \subseteq \bigcup_{k=1}^{\infty} (G_{nk} \setminus A_{nk}). \quad (\text{C.14})$$

Similarly, we can show that

$$G \setminus A \subseteq \bigcup_{n=1}^{\infty} (G_n \setminus A_n). \quad (\text{C.15})$$

By applying [21, Eqn. (18), p. 304] and Theorem 11.8(b) to the relation (C.15) first and then the relation (C.14), we conclude that

$$\begin{aligned} \mu^*(G \setminus A) &\leq \mu^*\left(\bigcup_{n=1}^{\infty} (G_n \setminus A_n)\right) \\ &\leq \sum_{n=1}^{\infty} \mu^*(G_n \setminus A_n) \\ &\leq \sum_{n=1}^{\infty} \mu^*\left(\bigcup_{k=1}^{\infty} (G_{nk} \setminus A_{nk})\right) \\ &\leq \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} \mu^*(G_{nk} \setminus A_{nk}). \end{aligned} \quad (\text{C.16})$$

^cNote that these also follow directly from the fact that $\mathfrak{M}(\mu)$ is a σ -ring.

Since μ^* is a nonnegative additive set function and $G_{nk}, A_{nk} \in \mathcal{E}$, we further apply the monotonic property [21, Eqn. (8), p. 302] and the inequality (C.11) to the inequality (C.16) to obtain

$$\mu^*(G \setminus A) \leq \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} [\mu^*(G_{nk}) - \mu^*(A_{nk})] \leq \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} 2^{-n} 2^{-k} 2^{-1} \epsilon = 2^{-1} \epsilon < \epsilon. \quad (\text{C.17})$$

As Rudin commented on [21, p. 308], $\mu^*(A)$ can be replaced by $\mu(A)$ if $A \in \mathfrak{M}(\mu)$. Recall that $G \setminus A \in \mathfrak{M}(\mu)$, so our first half of the proposition follows from the inequality (C.17).

For the second half of the proposition, we note that F is a closed set contained in A if and only if its open complement F^c contained in A^c , i.e., $A^c \subseteq F^c$. Next, we follow from the identity (C.2) that

$$A \setminus F = A \cap F^c = F^c \cap (A^c)^c = F^c \setminus A^c.$$

Therefore, we have $\mu(A \setminus F) < \epsilon$ if and only if $\mu(F^c \setminus A^c) < \epsilon$. If we let

$$A_n = A_{nk} = \underbrace{(n, n) \times \cdots \times (n, n)}_{p \text{ terms}}$$

for every $n, k \in \mathbb{N}$, then we have $A_{nk} \in \mathcal{E}$. By Eqn. (5) and (24) [21, pp. 301, 306], we obtain $A_{nk} \rightarrow A_n$ as $k \rightarrow \infty$ and thus $A_n \in \mathfrak{M}_F(\mu)$. Since $\mathbb{R}^p = \bigcup_{n=1}^{\infty} A_n$, we have $\mathbb{R}^p \in \mathfrak{M}(\mu)$. Since $A^c = \mathbb{R}^p \setminus A$ and $\mathfrak{M}(\mu)$ is a ring, Definition 11.1 implies that $A^c \in \mathfrak{M}(\mu)$. Now we apply the previous analysis to the set A^c to get an open set G containing A^c such that $\mu(G \setminus A^c) < \epsilon$ and this implies that

$$\mu(A \setminus G^c) < \epsilon,$$

where $F = G^c$ is our desired closed set contained in A . This completes the proof of the proposition. ■

Proposition C.4

The collection $\mathcal{B}$ of all Borel sets in $\mathbb{R}^p$ is a σ -ring and it is the smallest σ -ring containing all open sets of $\mathbb{R}^p$.

Proof. Let $\mathcal{O}$ be the collection of all open subsets of $\mathbb{R}^p$ and Ω be the family of all σ -rings $\mathcal{R}$ in $\mathbb{R}^p$ which contain the collection $\mathcal{O}$. Since the power set of $\mathbb{R}^p$ (denoted by $2^{\mathbb{R}^p}$) is a σ -ring, $\Omega \neq \emptyset$. Suppose that

$$\mathcal{R}^* = \bigcap_{\mathcal{R} \in \Omega} \mathcal{R}.$$

By definition, it is clear that $\mathcal{O} \subset \mathcal{R}^*$ and that $\mathcal{R}^*$ must lie in every σ -ring containing $\mathcal{O}$. If $A, B \in \mathcal{R}^*$, then $A, B \in \mathcal{R}$ for every $\mathcal{R} \in \Omega$. Since $\mathcal{R}$ is a σ -ring, we have $A \cup B \in \mathcal{R}$ and $A \setminus B \in \mathcal{R}$ which imply that $A \cup B \in \mathcal{R}^*$ and $A \setminus B \in \mathcal{R}^*$. By Definition 11.1, $\mathcal{R}^*$ is a σ -ring and thus it is the smallest σ -ring containing the collection of all open subsets of $\mathbb{R}^p$. Hence we have $\mathcal{R}^* = \mathcal{B}$. ■

Proposition C.5

For every measure μ , the sets of measure zero form a σ -ring.

Proof. Suppose that $\mathfrak{M}_0$ is the class of all sets of measure zero under the measure μ . If $A, B \in \mathfrak{M}_0$, then $\mu(A) = \mu(B) = 0$ and we follow from [21, Eqn. (7), p. 302] that

$$0 \leq \mu(A \cup B) \leq \mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B) = 0.$$

Thus we have $A \cup B \in \mathfrak{M}_0$. By the identity (C.6), we have

$$0 \leq \mu(A \setminus B) = \mu(A \cap B^c) \leq \mu(A) = 0$$

which means that $A \setminus B \in \mathfrak{M}_0$. Let $A_n \in \mathfrak{M}_0$ for $n = 1, 2, \dots$. By Theorem 11.8(b), we have

$$0 \leq \mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n) = 0$$

which means that $\bigcup_{n=1}^{\infty} A_n \in \mathfrak{M}_0$. By Definition 11.1, $\mathfrak{M}_0$ is a σ -ring. ■

Proposition C.6

Every countable subset of $\mathbb{R}^p$ has Lebesgue measure zero.

Proof. Let $\mathbf{x} = (x_1, \dots, x_p) \in \mathbb{R}^p$. Given $\epsilon > 0$. Since $\{x_i\} \subset (x_i - \epsilon, x_i + \epsilon)$ for $i = 1, \dots, p$, we have

$$\{\mathbf{x}\} \subset I = \prod_{i=1}^p (x_i - \epsilon, x_i + \epsilon).$$

By Definition 11.4, we have

$$m(\mathbf{x}) \leq \prod_{i=1}^p (2\epsilon) = 2^p \epsilon^p \rightarrow 0$$

as $\epsilon \rightarrow 0$. In other words, we obtain $m(\mathbf{x}) = 0$. Let $E = \{\mathbf{x}_1, \mathbf{x}_2, \dots\}$ be a countable subset of $\mathbb{R}^p$. By Proposition A.5, we see that $E \in \mathfrak{M}_0$ which is a subset of $\mathfrak{M}(m)$. By Theorem 11.10, the Lebesgue measure m is countably additive on $\mathfrak{M}(\mu)$ and in particular, on $\mathfrak{M}_0$. Hence we have

$$m(E) = m\left(\bigcup_{n=1}^{\infty} \mathbf{x}_n\right) = \sum_{n=1}^{\infty} m(\mathbf{x}_n) = 0$$

which is our desired result. ■

Proposition C.7

Let s be a real-valued function defined on the measurable space X , the range of s consist of the distinct numbers $c_1, \dots, c_n$ and

$$E_i = \{x \mid s(x) = c_i\} \quad (i = 1, \dots, n).$$

Then every simple function s is measurable if and only if the sets $E_1, \dots, E_n$ are measurable.

Proof. We claim that the characteristic function K_E is measurable if and only if E is measurable. Suppose that E is measurable. It is obvious that $x \notin \{0, 1\}$ implies that $K_E^{-1}(x) = \emptyset$. Let $a \in \mathbb{R}$. By Theorem 11.15, we have

$$\{x \in X \mid K_E(x) \geq a\} = \begin{cases} \emptyset, & \text{if } a > 1; \\ E, & \text{if } 0 < a < 1; \\ X, & \text{if } a \leq 0. \end{cases} \quad (\text{C.18})$$

Since X is a measurable space, we have, by Definition 11.12, X is measurable. Since $\emptyset$ and E are measurable, it follows from (C.18) that $\{x \in X \mid K_E(x) \geq a\}$ is measurable. By Definition 11.13, the characteristic function K_E is measurable. Conversely, for every $x \in X$, we have

$$\{x \in X \mid K_E(x) > 0\} = E.$$

Since K_E is measurable, the set $\{x \mid K_E(x) > 0\}$ is measurable by Definition 11.13. This proves the claim.

Suppose that $E_1, \dots, E_n$ are measurable. Then each characteristic function is measurable. Since

$$s = \sum_{i=1}^n c_i K_{E_i},$$

Theorem 11.18 implies that s is also measurable. Conversely, suppose that s is measurable. Then the set $s^{-1}(c_i) = \{x \in X \mid s(x) = c_i\}$ is measurable, where $i = 1, \dots, n$. Since $E_i = \{x \in X \mid s(x) = c_i\}$, E_i is measurable for $i = 1, \dots, n$. ■

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